# BRILLIANT PUBLIC SCHOOL, SITAMARHI

**(Affiliated up to +2 level to C.B.S.E., New Delhi)**

## Class-XI

## IIT-JEE Advanced Chemistry

## Study Package

## Session: 2014-15

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# STUDY PACKAGE

## Target: IIT-JEE (Advanced)

## SUBJECT: CHEMISTRY-XI

**Chapters:**


**1. Mole Concept**

**2. Stoichiometry**

**3. Atomic Structure**

**4. Chemical Bonding**

**5. Periodic Table and Representative Element**

**6. Gaseous State**

**7. Chemical Equilibrium**

**8. Ionic Equilibrium**

**9. Thermodynamics**

**10. IUPAC Nomenclature**

**11. Hydrocarbons**

**12. Nitrogen Family**

# STUDY PACKAGE

# Target: IIT-JEE(Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 1. MOLE CONCEPT

**Index:**


**1. Key Concepts**

**2. Exercise I**

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**4. Exercise III**

**5. Exercise IV**

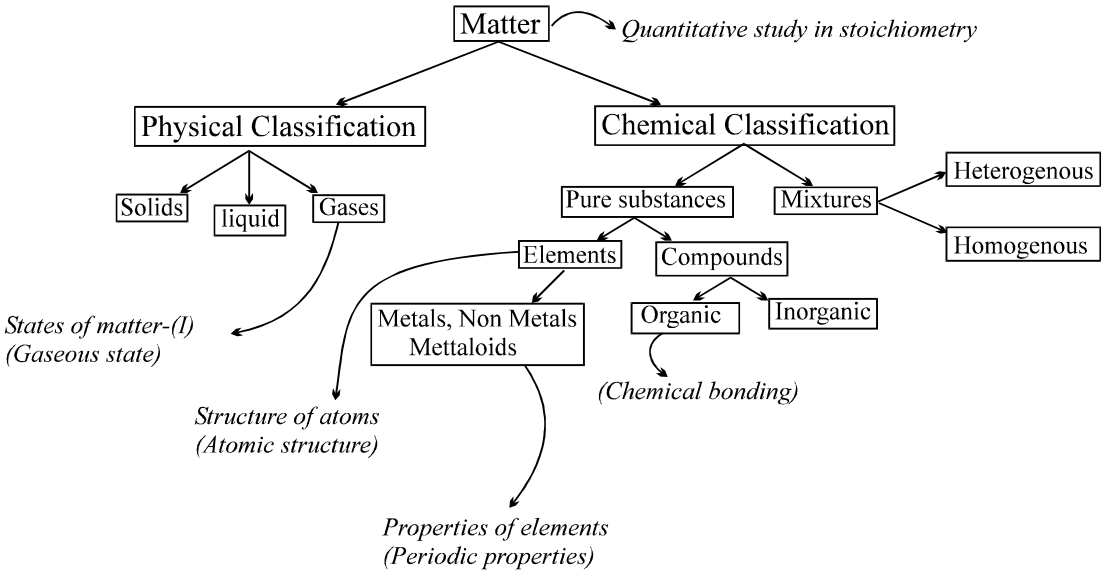
**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

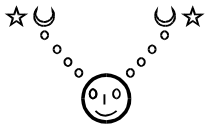
# *CHEMISTRY – STUDY OF MATTER*

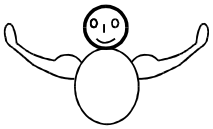
## *Overview of Chemistry*




**Friends for you used in the sheet.**

1. Teacher's advice ⟶  : Tips which can enhance your performance.

2. Student's query ⟶  : Arbit doubts which are generally developed among students.

3. Boost your confidence ⟶  : Some additional information.

4. Dangers ⟶  Take care of the general mistakes and crucial points.

# *KEY CONCEPTS*

## 1. LAWS OF CHEMICAL COMBINATION

***1.1 Law of conservation of mass [Lavoisier]***

***1.2 Law of constant composition [Proust]***

***1.3 Law of multiple proportions [Dalton]***

***1.4 Law of reciprocal proportions [Richter]***

***1.5 Gay Lussac law of combining volumes [Guess Who??]***



"*Wonder these laws are useful?*"



"*These are no longer useful in chemical calculations now but gives an idea of earlier methods of analysing and relating compounds by mass.*"

## 2. MOLE CONCEPT

***2.1 Definition of mole :*** One mole is a collection of that many entities as there are number of atoms exactly in 12 gm of C-12 isotope.

or 1 mole = collection of $6.02 \times 10^{23}$ species

$6.02 \times 10^{23} = N_A$ = Avogadro's No.



1 mole of atoms is also termed as 1 gm-atom, 1 mole of ions is termed as 1 gm-ion and 1 mole of molecule termed as 1 gm-molecule.

***2.2 Methods of Calculations of mole :***

(a) If no. of some species is given, then no. of moles = $\frac{\text{Given no.}}{N_A}$

(b) If weight of a given species is given, then no of moles = $\frac{\text{Given wt.}}{\text{Atomic wt.}}$ (for atoms),

or $= \frac{\text{Given wt.}}{\text{Molecular wt.}}$ (for molecules)

(c) If volume of a gas is given along with its temperature (T) and pressure (P)

use $n = \frac{PV}{RT}$

where R = 0.0821 lit-atm/mol-K (when P is in atmosphere and V is in litre.)



1 mole of any gas at STP occupies 22.4 litre.



Gases do not have volume. What is meant by "Volume of gas"?



Do not use this expression (PV = nRT) for solids/liquids.



How would I calculate moles if volume of a solid is given?

***2.3*** ***Atomic weight:*** It is the weight of an atom relative to *one twelvth of weight of 1 atom of C-12*



Be clear in the difference between 1 amu and 1 gm.

(a) Average atomic weight = $\sum$ % of isotope X molar mass of isotope.



The % obtained by above expression (used in above expression) is by number (i.e. its a mole%)

***2.4*** ***Molecular weight :*** It is the sum of the atomic weight of all the constituent atom.

(a) Average molecular weight = $\dfrac{\sum n_i M_i}{\sum n_i}$

where $n_i$ = no. of moles of any compound and $m_i$ = molecular mass of any compound.



Make yourselves clear in the difference between mole% and mass% in question related to above.



Shortcut for % determination if average atomic weight is given for X having isotopes $X^A$ & $X^B$.

$$\% \text{ of } X^A = \left| \frac{\text{Average atomic weight} - \text{wt of } X^B}{\text{difference in weight of } X^A \text{ \& } X^B} \right| \times 100$$



Try working out of such a shortcut for $X^A$, $X^B$, $X^C$

## 3. EMPIRICAL FORMULA, MOLECULAR FORMULA :

***3.1*** ***Empirical formula :*** Formula depicting constituent atom in their simplest ratio.
***Molecular formula :*** Formula depicting actual number of atoms in one molecule of the compound

***3.2*** ***Relation between the two :*** Molecular formula = Empirical formula × n

$$n = \frac{Molecular\ mass}{Empirical\ \ Formula\ mass}$$



Check out the importance of each step involved in calculations of *empirical formula*.

***3.3*** ***Vapour density :***

*Vapour density :* Ratio of density of vapour to the density of hydrogen at similar pressure and temperature.

$$\text{Vapour density} = \frac{Molecular\ mass}{2}$$



Can you prove the above expression?



Is the above parameter temperature dependent?

**4. STOICHIOMETRY :** Stoichiometry pronounced ("stoy – key – om – e – tree") is the calculations of the quantities of reactants and products involved in a chemical reaction.

This can be divided into two category.

(A) Gravimetric analysis

(B) Volumetric analysis (to be discussed later)

***4.1 Gravimetric Analysis :***

***4.1.1 Methods for solving :***

(a) Mole Method
(b) Factor Label Method } Balance reaction required

(c) POAC method } Balancing not required but common sense ☺ *use it with slight care*.

(d) Equivalent concept } to be discussed later

**5. CONCEPT OF LIMITING REAGENT.**

***5.1 Limiting Reagent :*** It is very important concept in chemical calculation. It refers to reactant which is present in minimum stoichiometry quantity for a chemical reaction. It is reactant consumed fully in a chemical reaction. So all calculations related to various products or in sequence of reactions are made on the basis of limiting reagent.

☠ It comes into picture when reaction involves two or more reactants. For solving any such reactions, first step is to calculate L.R.

***5.2 Calculation of Limiting Reagent :***

(a) By calculating the required amount by the equation and comparing it with given amount. [Useful when only two reactant are there]

(b) By calculating amount of any one product obtained taking each reactant one by one irrespective of other reactants. The one giving least product is *limiting reagent*.

(c) Divide given moles of each reactant by their stoichiometric coefficient, the one with least ratio is *limiting reagent*. [Useful when number of reactants are more than two.]

**6. PERCENTAGE YIELD :** The percentage yield of product $= \frac{actual\ yield}{the\ theoretical\ maximum\ yield} \times 100$

☺ The actual amount of any limiting reagent consumed in such incomplete reactions is given by [% yield × given moles of limiting reagent] [For reversible reactions]

☺ For irreversible reaction with % yield less than 100, the reactants is converted to product (desired) and waste.

**7. CONCENTRATION TERMS :**

***7.1 General concentraction term :***

(a) $\text{Density} = \frac{\text{Mass}}{\text{Volume}}$, Unit : gm/cc

(b) $\text{Relative density} = \frac{Density\ \ of\ \ any\ \ substance}{Density\ of\ \ refrence\ substance}$

(c) Specific gravity = $\frac{Density\ of\ any\ substance}{Density\ of\ water\ at\ 4°C}$

(d) Vapour density = $\frac{Density\ of\ vapour\ at\ some\ temperature\ and\ pressure}{Density\ of\ H_2\ gas\ at\ same\ temperature\ and\ pressure}$



(1) Which of these are temperature dependent.

(2) Classify them as w/w, w/v, v/v ratio.

***7.2 For solutions (homogeneous mixture) :***



What is solute and solvent in a solution.



If the mixture is not homogeneous, then none of them is applicable.



Classify each given ratio as w/w, w/v, v/v and comment on their temperature dependence.

(a) **% by mass** $\left(\frac{w}{W}\right)$ **: =** $\frac{\text{wt .of solute}}{\text{wt .of solution}} \times 100$

[X % by mass means 100 gm solution contains X gm solute ; $\therefore$ (100 – X) gm solvent ]

(b) **%** $\left(\frac{w}{V}\right)$ **: =** $\frac{\text{wt.of solute}}{\text{volumeof solution}} \times 100$ [for liq. solution]

[X % $\left(\frac{w}{V}\right)$ means 100 ml solution contains X gm solute ]



for gases **% by volume** is same as **mole %**

(c) **%** $\left(\frac{v}{V}\right)$ **: =** $\frac{\text{volumeof solute}}{\text{volumeof solution}} \times 100$

(d) **Mole % : =** $\frac{\text{Moles of solute}}{\text{Total moles}} \times 100$

(e) **Mole fraction ($X_a$) : =** $\frac{\text{Moles of solute}}{\text{Total moles}}$

(f) **Molarity (M) : =** $\frac{\text{Mole of solute}}{\text{volume of solution in litre}}$

(g) **Molality (m) : =** $\frac{\text{Moles of solute}}{\text{Mass of solvent(in kg)}}$

(h) **Parts per million (ppm) : =** $\frac{\text{Mass of solute}}{\text{Mass of solvent}} \times 10^6 \cong \frac{Mass\ of\ solute}{Mass\ of\ solution} \times 10^6$



Get yourselves very much confortable in their interconversion. It is very handy.

### *7.3 Some typical concentration terms :*

(a) **Oleum :** Labelled as '% oleum' (for e.g. 102% oleum), it means maximum amount of $H_2SO_4$ that can be obtained from 100 gm of such oleum (mix of $H_2SO_4$ and $SO_3$) by adding sufficient water.

Work out what are the maximum and minimum value of the %

(b) **$H_2O_2$ :** Labelled as 'volume $H_2O_2$ (for e.g. 20V $H_2O_2$), it means volume of $O_2$ (in litre) at STP that can be obtained from 1 litre of such a sample when it decomposes according to

$$H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2$$

Work out a relationship between M and volume $H_2O_2$ and remember it

## 8. SOME EXPERIMENTAL METHODS :

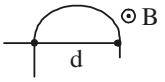
### *8.1 For determination of atomic mass :*

(a) Dulong's and Petit's Law :

Atomic weight × specific heat (cal/gm°C) $\propto\cong 6.4$

Gives approximate atomic weight and is applicable for metals only. Take care of units of specific heat.

(b) Mass spectrometry : $\frac{mv^2}{r} = qvB$




B is the magnitude of magnetic field r = d/2

*m* is mass of ion, *v* is velocity of ion, *r* is the distance where the ions strikes, *q* is the charge on the ion.

### *8.2 For molecular mass determination :*

(a) **Victor Maeyer's process :** (*for volatile substance*)

*Procedure :* Some known weight of a volatile substance (w) is taken, converted to vapour and collected over water. The volume of air displaced over water is given (V) and the following expressions are used.

$M = \frac{w}{PV}RT$ or $M = \frac{w}{(P-P')V}RT$

If aq. tension is not given If aq. tension is P'

**Aqueous tension** : Pressure exerted due to water vapours at any given temperature.

This comes in picture when any gas is collected over water. Can you guess why?

(b) **Silver salt method :** (*for organic acids*)

*Basicity of an acid :* No. of replacable $H^+$ atoms in an acid (H contained to more electronegative atom is acidic)

*Procedure :* Some known amount of silver salt ($w_1$ gm) is heated to obtain $w_2$ gm of while shining residue of silver. Then if the basicity of acid is n, molecular weight of acid would be

$$\left(\frac{w_2}{108}\times\frac{1}{n}\right)\times M_{salt} = w_1 \text{ and molecular weight of acid} = M_{salt} - n(107)$$

This is one good practical application of POAC.

(c) **Chloroplatinate salt method :** (*for organic bases*)

Lewis acid : electron pair acceptor

Lewis base :electron pair donor

Procedure : Some amount of organic base is reacted with $H_2PtCl_6$ and forms salt known as chloroplatinate. If base is denoted by B then salt formed

(i) with monoacidic base = $B_2H_2PtCl_6$

(ii) with diacidic base = $B_2(H_2PtCl_6)_2$

(iii) with triacidic base = $B_2(H_2PtCl_6)_3$

The known amount ($w_1$ gm) of salt is heated and Pt residue is measured. ($w_2$ gm). If acidity of base is 'n'

then $\left(\frac{w_2}{195}\times\frac{1}{n}\right)\times M_{salt} = w_1$ and $M_{base} = \frac{M_{salt} - n(410)}{2}$

***8.3 For % determination of elements in organic compounds :***

☺ All these methods are applications of POAC

☺ Do not remember the formulas, derive them using the concept, its easy.

(a) **Liebig's method :** (*Carbon and hydrogen*)

$$(w)\,\text{Organic Compound} \xrightarrow[CuO]{\Delta} (w_1)CO_2 + H_2O\,(w_2)$$

$$\%\ \text{of C} = \frac{w_1}{44}\times\frac{12}{w}\times 100$$

$$\%\ \text{of H} = \frac{w_2}{18}\times\frac{1}{w}\times 100$$

where $w_1$ = wt. of $CO_2$ produced, $w_2$ = wt. of $H_2O$ produced,
w = wt. of organic compound taken

(b) **Duma's method :** (*for nitrogen*)

$$(w)\ \text{Organic Compound} \xrightarrow[CuO]{\Delta} N_2 \rightarrow (P, V, T \text{ given})$$

use PV = nRT to calculate moles of $N_2$, n.

$$\therefore \quad \%\ \text{of N} = \frac{n\times 28}{w}\times 100$$

w = wt of organic compound taken

(c) **Kjeldahl's method :** (*for nitrogen*)

$$(w)\text{O.C.} + H_2SO_4 \rightarrow (NH_4)_2SO_4 \xrightarrow{NaOH} NH_3 + H_2SO_4 \rightarrow (\text{molarity and volume } (V_1) \text{ consumed given})$$

$$\Rightarrow \quad \%\ \text{of N} = \frac{MV_1\times 2\times 14}{w}\times 100$$

where M = molarity of $H_2SO_4$.

Some N containing compounds do not give the above set of reaction as in Kjeldahl's method.

(d) **Sulphur :**

(w) O.C. + $HNO_3 \rightarrow H_2SO_4 + BaCl_2 \rightarrow (w_1)\ BaSO_4$

$\Rightarrow \quad \%\ \text{of S} = \frac{w_1}{233} \times 1 \times 32 \times 100\%$ .

where $w_1$ = wt. of $BaSO_4$, w = wt. of organic compound

(e) **Phosphorus :**

O.C + $HNO_3 \rightarrow H_3PO_4$ + [$NH_3$ + magnesia mixture ammonium molybdate] $\rightarrow MgNH_4PO_4$

$\xrightarrow{\Delta} Mg_2P_2O_7$

$$\%\ \text{of P} = \frac{w_1}{222} \times \frac{2 \times 31}{w} \times 100$$

(f) **Carius method :** (*Halogens*)

O.C. + $HNO_3$ + $AgNO_3 \rightarrow AgX$

If X is Cl then colour = white

If X is Br then colour = dull yellow

If X is I then colour = bright yellow

Flourine can't be estimated by this

$$\%\ \text{of X} = \frac{w_1}{(\text{M. weight of AgX})} \times \frac{1 \times (\text{At. wt. of X})}{w} \times 100$$

**9. EUDIOMETRY :** [For reactions involving gaseous reactants and products]

The stoichiometric coefficient of a balanced chemical reactions also gives the ratio of volumes in which gasesous reactants are reacting and products are formed at same temperature and pressure. The volume of gases produced is often given by mentioning certain solvent which absorb contain gases.

| **Solvent** | **gas (es) absorb** |
|---|---|
| KOH | $CO_2$, $SO_2$, $Cl_2$ |
| Ammon $Cu_2Cl_2$ | CO |
| Turpentine oil | $O_3$ |
| Alkaline pyrogallol | $O_2$ |
| water | $NH_3$, HCl |
| $CuSO_4$ | $H_2O$ |

Check out for certain assumption which are to be used for solving problem related to this.

## *EXERCISE # I*

### *LAWS OF CHEMICAL COMBINATION*

Q.10 What mass of sodium chloride would be decomposed by 9.8 gm of sulphuric acid, if 12 gm of sodium bisulphate and 2.75 gm of hydrogen chloride were produced in a reaction assuming that the law of conservation of mass is true?[Assume none of the reactants are remaining] [Ans. 4.95 gm]

Q.12 Zinc sulphate crystals contain 22.6% of zinc and 43.9% of water. Assuming the law of constant proportions to be true, how much zinc should be used to produce 13.7 gm of zinc sulphate crystal and how much water will they contain?

Q.13 Carbon combines with hydrogen to form three compounds A, B and C. The percentage of hydrogen in A, B and C are 25, 14.3 and 7.7 respectively. Which law of chemical combination is illustrated?
[Ans. law of multiple proportions]

Q.14 Illustrate the law of reciprocal proportions from the following data : KCl contains 52.0% potassium, KI contains 23.6% potassium and ICl contains 78.2% iodine.

### *ATOMIC MASS & MOLECULAR MASS*

Q.1 The average mass of one gold atom in a sample of naturally occuring gold is $3.2707 \times 10^{-22}$g. Use this to calculate the molar mass of gold.

Q.2 A plant virus is found to consist of uniform symmetrical particles of 150 Å in diameter and 5000 Å long. The specific volume of the virus is 0.75 $cm^3/g$. If the virus is considered to be a single particle, find its molecular weight.

Q.3 Density of a gas relative to air is 1.17. Find the mol. mass of the gas. [$M_{air}$ = 29g/mol]

### *MOLE*

Q.4 If all 1 billion ($10^9$) people in India were put to work counting the atoms in a mole of gold and if each person could count one atom per second day and night for 365 days a year, how many years would it take to finish the count ?

Q.5 Vitamin C, ascorbic acid, has the formula $C_6H_8O_6$.

(a) The recommended daily dose of vitamin C is 60 milligrams. How many moles are you consuming if you ingest 60 mg of the vitamin ?

(b) A typical tablet contains 1.00 g of vitamin C. How many moles of vitamin C does this represent ?

(c) When you consume 1.00 gram of vitamin C, how many oxygen atoms are you eating ?

Q.6 Precious metals such as gold and platinum are sold in units of "troy ounces",where 1 troy ounce is 31.1 grams. If you have a block of platinum with a mass of 15.0 troy ounces, how many mole of the metal do you have ? What is the size of the block in cubic centimeters ? (The density of platinum is 21.45 $g/cm^3$at 20°C) (Atomic wt.of Pt. = 195)

Q.7 One type of artifical diamond (commonly called YAG for yttrium aluminium garnet) can be represented by the formula $Y_3Al_5O_{12}$.

(a) Calculate the weight percentage composition of this compound.

(b) What is the weight of yttrium present in a 200 – carat YAG if 1 carat - 200 mg ? (Y = 89, Al = 27)

Q.8 A chemical commonly called "dioxin" has been very much in the news in the past few years. (It is the by – product of herbicide manufacture and is thought to be quite toxic.) Its formula is $C_{12}H_4Cl_4O_2$. If you have a sample of dirt (28.3 g) that contains $1.0 \times 10^{-4}$ % dioxin, how many moles of dioxin are in the dirt

sample ?

## *LIMITING REACTANT*

Q.9 Titanium, which is used to make air plane engines and frames, can be obtained from titanium tetrachloride, which in turn is obtained from titanium oxide by the following process :

$$3\,TiO_2(s) + 4C\,(s) + 6Cl_2\,(g) \longrightarrow 3TiCl_4(g) + 2CO_2(g) + 2CO\,(g)$$

A vessel contains 4.15 g $TiO_2$, 5.67 g C and; 6.78 g $Cl_2$, suppose the reaction goes to completion as written, how many gram of $TiCl_4$ can be produced ? (Ti = 48)

Q.10 A chemist wants to prepare diborane by the reaction

$6\,LiH + 8BF_3 \longrightarrow 6Li\,BF_4 + B_2H_6$

If he starts with 2.0 moles each of LiH & $BF_3$. How many moles of $B_2H_6$ can be prepared.

Q.11 When you see the tip of a match fire, the chemical reaction is likely to be

$P_4S_3 + 8O_2 \longrightarrow P_4O_{10} + 3SO_2$

What is the minimum amount of $P_4S_3$ that would have to be burned to produce at least 1.0 g of $P_4O_{10}$ and at least 1.0 g of $SO_2$

## *GRAVIMETRIC ANALYSIS*

Q.12 1 gm sample of $KClO_3$ was heated under such conditions that a part of it decomposed according to the equation (1) $2KClO_3 \longrightarrow 2\,KCl + 3O_2$

and remaining underwent change according to the equation.

(2) $4KClO_3 \longrightarrow 3\,KClO_4 + KCl$

If the amount of $O_2$ evolved was 146.8 ml at S.T.P., calculate the % by weight of $KClO_4$ in the residue.

Q.13 A sample of calcium carbonate contains impurities which do not react with a mineral acid. When 2 grams of the sample were reacted with the mineral acid, 375 ml of carbon dioxide were obtained at 27°C and 760 mm pressure. Calculate the % purity of the sample of $CaCO_3$?

Q.14 One gram of an alloy of aluminium and magnesium when heated with excess of dil. HCl forms magnesium chloride, aluminium chloride and hydrogen. The evolved hydrogen collected over mercury at 0°C has a volume of 1.2 litres at 0.92 atm pressure. Calculate the composition of the alloy.

Q.15 A sample containing only $CaCO_3$ and $MgCO_3$ is ignited to CaO and MgO. The mixture of oxides produced weight exactly half as much as the original sample. Calculate the percentages of $CaCO_3$ and $MgCO_3$ in the sample.

Q.16 Determine the percentage composition of a mixture of anhydrous sodium carbonate and sodium bicarbonate from the following data:

wt. of the mixture taken = 2g

Loss in weight on heating = 0.124 g.

Q.17 A 10 g sample of a mixture of calcium chloride and sodium chloride is treated with $Na_2CO_3$ to precipitate calcium as calcium carbonate. This $CaCO_3$ is heated to convert all the calcium to CaO and the final mass of CaO is 1.62g. Calculate % by mass of NaCl in the original mixture.

Q.18 In a gravimetric determination of P an aqueous solution of $NaH_2PO_4$ is treated with a mixture of ammonium and magnesium ions to precipitate magnesium ammonium phosphate $Mg(NH_4)PO_4.\ 6H_2O$. This is heated and decomposed to magnesium pyrophosphate, $Mg_2P_2O_7$ which is weighed. A solution of $NaH_2PO_4$ yielded 1.054 g of $Mg_2P_2O_7$. What weight of $NaH_2PO_4$ was present originally ?

Q.19 By the reaction of carbon and oxygen, a mixture of CO and $CO_2$ is obtained. What is the composition of the mixture obtained when 20 grams of $O_2$ reacts with 12 grams of carbon ?

Q.20 A mixture of nitrogen and hydrogen. In the ratio of one mole of nitrogen to three moles of hydrogen, was partially converted into $NH_3$ so that the final product was a mixture of all these three gases. The mixture was to have a density of 0.497 g per litre at 25°C and 1.00 atm. What would be the mass of gas in 22.4 litres at S.T.P? Calculate the % composition of this gaseous mixture by volume.

Q.21 Direct reaction of iodine ($I_2$) and chlorine ($Cl_2$) produces an iodine chloride, $I_xCI_y$ , a bright yellow solid. If you completely used up 0.508 g of iodine and produced 0.934 g of $I_xCI_y$, what is the empirical formula of the compound? Later experiment showed the molar mass, of $I_xCI_y$ was 467 g/mol. What is the molecular formnula of the compound ? (I = 127)

Q.22 Equal weights of mercury and $I_2$ are allowed to react completely to form a mixture of mercurous and mercuric iodide leaving none of the reactants. Calculate the ratio of the wts of $Hg_2I_2$ and $HgI_2$ formed.

## ***EMPIRICAL & MOLECULAR FORMULA***

Q.23 Tha action of bacteria on meat and fish produces a poisonous compound called cadaverine. As its name and origin imply, it stinks! It is 58.77% C, 13.81% H, and 27.42% N. Its molar mass is 102 g/mol. Determine the molecular formula of cadaverine.

Q.24 Polychlorinated biphenyls, PCBs, known to be dangerous environmental pollutants, are a group of compounds with the general empirical formula $C_{12}H_mCl_{10-m}$, where m is an integer. What is the value of m, and hence the empirical formula of the PCB that contains 58.9% chlorine by mass ?

Q.25 Given the following empirical formulae and molecular weights, compute the true molecular formulae :

| | Empirical Formula | Molecular weight | | Empirical Formula | Molecular weight |
|---|---|---|---|---|---|
| (a) | $CH_2$ | 84 | (b) | $CH_2O$ | 150 |
| (c) | HO | 34 | (d) | HgCl | 472 |
| (e) | HF | 80 | | | |

Q.26 Hexachlorophene, $C_{13}H_6CI_6O_2$, is a germicide in soaps. Calculate weight percent of each element in the compound.

0.9482 gm of carbon dioxide and 0.1939 gm of water ?

Q.28 What is the percentage of nitrogen in an organic compound 0.14 gm of which gave by Dumas method 82.1 c.c. of nitrogen collected over water at 27°C and at a barometric pressure of 774.5 mm? (aqueous tension of water at 27°C is 14.5 mm)

Q.29 0.2000 gm of an organic compound was treated by Kjeldahl's method and the resulting ammonia was passed into 50 cc of M/4 $H_2SO_4$. The residual acid was then found to require 36.6 cc of M/2 NaOH for neutralisation. What is the percentage of nitrogen in the compound?

Q.27 What is the empirical formula of a compound 0.2801 gm of which gave on complete combustion

Q.30 0.275 gm of an organic compound gave on complete combustion 0.22 gm of carbon dioxide and 0.135 gm of water. 0.275 gm of the same compound gave by Carius method 0.7175 gm of silver chloride. What is the empirical formula of the compound ?

Q.31 0.6872 gm of an organic compound gave on complete combustion 1.466 gm of carbon dioxide and 0.4283 gm of water. A given weight of the compound when heated with nitric acid and silver nitrate gave an equal weight of silver chloride. 0.3178 gm of the compound gave 26.0cc of nitrogen at 15°C and 765 mm pressure. Deduce the empirical formula of the compound?

Q.32 0.80g of the chloroplatinate of a mono acid base on ignition gave 0.262g of Pt. Calculate the mol wt of the base.

Q.33 A compound which contains one atom of X and two atoms of Y for each three atoms of Z is made by mixing 5.00 g of X, $1.15\times10^{23}$ atoms of Y, 0.03 mole of Z atoms. Given that only 4.40 g of compound results. Calculate the atomic weight of Y if the atomic weight of X and Z are 60 and 80 a.m.u. respectively.

## ***CONCENTRATION TERMS***

Q.34 Calculate the molarity of the following solutions :

(a) 4g of caustic soda is dissolved in 200 mL of the solution.

(b) 5.3 g of anhydrous sodium carbonate is dissolved in 100 mL of solution.

(c) 0.365 g of pure HCl gas is dissolved in 50 mL of solution.

Q.35 The density of a solution containing 13% by mass of sulphuric acid is 1.09 g/mL. Calculate the molarity of the solution.

Q.36 The mole fraction of $CH_3OH$ in an aqueous solution is 0.02 and its density is 0.994 g $cm^{-3}$. Determine its molarity and molality.

Q.37 The density of a solution containing 40% by mass of HCl is 1.2 g/mL. Calculate the molarity of the solution.

Q.38 A mixture of ethanol and water contains 54% water by mass. Calculate the mole fraction of alcohol in this solution.

Q.39 15 g of methyl alcohol is present in 100 mL of solution. If density of solution is 0.90 g $mL^{-1}$. Calculate the mass percentage of methyl alcohol in solution.

Q.40 Units of parts per million (ppm) or per billion (ppb) are often used to describe the concentrations of solutes in very dilute solutions. The units are defined as the number of grams of solute per million or per billion grams of solvent. Bay of Bengal has 1.9 ppm of lithium ions. What is the molality of $Li^+$ in this water ?

Q.41 A 6.90 M solution of KOH in water contains 30% by mass of KOH. Calculate the density of the solution.

Q.42 Fill in the blanks in the following table.

| Compound | Grams Compd | Grams Water | Molality of Compd | Mole Fraction of Compd |
|---|---|---|---|---|
| $Na_2CO_3$ | _____ | 250 | 0.0125 | _____ |
| $CH_3OH$ | 13.5 | 150 | _____ | _____ |
| $KNO_3$ | _____ | 555 | _____ | 0.0934 |

Q.43 A solution of specific gravity 1.6 is 67% by weight. What will be the % by weight of the solution of same acid if it is diluted to specific gravity 1.2 ?

Q.44 Find out the volume of 98% w/w $H_2SO_4$ (density = 1.8 gm/ ml) must be diluted to prepare 12.5 litres of 2.5 M sulphuric acid solution.

Q.45 Determine the volume of diluted nitric acid (d = 1.11 g $mL^{-1}$, 19% w/v $HNO_3$) That can be prepared by diluting with water 50 mL of conc. $HNO_3$ (d =1.42 g $mL^{-1}$, 69.8% w /v).

Q.46 A mixture of Xe and $F_2$ was heated. A sample of white solid thus formed reacted with $H_2$, to give 112 ml of Xe at STP and HF formed required 30 ml of 1 M NaOH for complete neutralization. Determine empirical formula.

Q.47 A certain oxide of iron contains 2.5 grams of oxygen for every 7.0 grams of iron. If it is regarded as a mixture of FeO and $Fe_2O_3$ in the weight ratio x : y, what is x : y, (atomic weight of iron = 56).

Q.48 In what ratio shoudl you mix 0.2M $NaNO_3$ and 0.1M $Ca(NO_3)_2$ solution so that in resulting solution, the concentration of negative ion is 50% greater than conc. of positive ion.

Q.49 Sulfur dioxide is an atmospheric pollutant that is converted to sulfuric acid when it reacts with water vapour. This is one source of acid rain, one of our most pressing environmental problems. The sulfur dioxide content of an air sample can be determined as follows. A sample of air is bubbled through an aqueous solution of hydrogen peroxide to convert all of the $SO_2$ to $H_2SO_4$

$$H_2O_2 + SO_2 \longrightarrow H_2SO_4$$

Titration of the resulting solution completes the analysis. In one such case, analysis of 1550 L of Los Angeles air gave a solution that required 5.70 ml of 5.96 x $10^{-3}$M NaOH to complete the titration. Determine the number of grams of $SO_2$ present in the air sample.

## ***SOME TYPICAL CONCENTRATION TERMS***

Q.50 Calculate the St. of "20V " of $H_2O_2$ in terms of

(i) gm/L (ii) M (iii) % by volume

Q.51 Calculate composition of the final solution if 100 gm oleum labelled as 109% is added with
(a) 9 gm water (b) 18 gm water (c) 120 gm water

## ***EUDIOMETRY***

Q.52 10 ml of a mixture of CO, $CH_4$ and $N_2$ exploded with excess of oxygen gave a contraction of 6.5 ml. There was a further contraction of 7 ml, when the residual gas treated with KOH. What is the composition of the original mixture?

Q.53 When 100 ml of a $O_2 - O_3$ mixture was passed through turpentine, there was reduction of volume by 20 ml. If 100 ml of such a mixture is heated, what will be the increase in volume?

Q.54 9 volumes of a gaseous mixture consisting of a gaseous organic compound A and just sufficient amount of oxygen required for complete combustion yielded on burning 4 volumes of $CO_2$, 6 volumes of water vapour and 2 volumes of $N_2$, all volumes measured at the same temperature and pressure. If the compound A contained only C, H and N (i) how many volumes of oxygen are required for complete combustion and (ii) what is the molecular formula of the compound A?

Q.55 60 ml of a mixture of nitrous oxide and nitric oxide was exploded with excess of hydrogen. If 38 ml of $N_2$ was formed, calculate the volume of each gas in the mixture.

Q.56 When a certain quantity of oxygen was ozonised in a suitable apparatus, the volume decreased by 4 ml. On addition of turpentine the volume further decreased by 8 ml. All volumes were measured at the same temperature and pressure. From these data, establish the formula of ozone.

Q.57 10 ml of ammonia were enclosed in an eudiometer and subjected to electric sparks. The sparks were continued till there was no further increase in volume. The volume after sparking measured 20 ml. Now 30 ml of $O_2$ were added and sparking was continued again. The new volume then measured 27.5 ml. All volumes were measured under identical conditions of temperature and pressure. V.D. of ammonia is 8.5. Calculate the molecular formula of ammonia. Nitrogen and Hydrogen are diatomic.

## ***EXERCISE # II***

Q.1 Nitrogen (N), phosporus (P), and potassium (K) are the main nutrients in plant fertilizers. According to an industry convention, the numbers on the label refer to the mass % of N, $P_2O_5$, and $K_2O$, in that order. Calculate the N : P : K ratio of a 30 : 10 : 10 fertilizer in terms of moles of each elements, and express it as x : y : 1.0.

Q.2 One mole of a mixture of $N_2$, $NO_2$ and $N_2O_4$ has a mean molar mass of 55.4. On heating to a temperature at which $N_2O_4$ may be dissociated : $N_2O_4 \longrightarrow 2NO_2$, the mean molar mass tends to the lower value of 39.6. What is the mole ratio of $N_2$ : $NO_2$ : $N_2O_4$ in the original mixture?

Q.3 10 mL of gaseous organic compound contain C, H and O only was mixed with 100 mL of $O_2$ and exploded under identical conditions and then cooled. The volume left after cooling was 90 mL. On treatment with KOH a contraction of 20 mL was observed. if vapour density of compound is 23 derive molecular formula of the compound.

Q.4 Fluorocarbon polymers can be made by fluorinating polyethylene according to the reaction $(CH_2)n + 4nCoF_3 \longrightarrow (CF_2)_n + 2nHF + 4nCoF_2$, where n is a large integer. The $CoF_3$ can be regenerated by the reaction $2\,CoF_2 + F_2 \longrightarrow 2CoF_3$. If the HF formed in the first reaction cannot be reused, how many kg of fluorine are consumed per kg of fluorocarbon produced, $(CF_2)_n$ ? If HF can be recovered and electrolyzed to hydrogen and fluorine, and if this fluorine is used for regenerating $CoF_3$,

what is the net consumption of fluorine per kg of fluorocarbon ?

Q.5 $A_2 + 2B_2 \longrightarrow A_2B_4$

$\frac{3}{2} A_2 + 2B_2 \longrightarrow A_3B_4$

Two substance $A_2$ & $B_2$ react in the above manner when $A_2$ is limited it gives $A_2B_4$ in excess gives $A_3B_4$. $A_2B_4$ can be converted to $A_3B_4$ when reacted with $A_2$. Using this information calculate the composition of the final mixture when the mentioned amount of A & B are taken

(a) 4 moles $A_2$ & 4 moles $B_2$ (b) $\frac{1}{2}$ moles $A_2$ & 2 moles $B_2$

(c) 1.25 moles $A_2$ & 2 moles $B_2$

Q.6 Exchange of ions in a solution by two compounds is known as metathesis reaction. How much minimum volume of 0.1 M aluminium sulphate solution should be added to excess calcium nitrate to obtain atleast 1 gm of each salt in the metathesis reaction.

$$Al_2(SO_4)_3 + 3Ca(NO_3)_2 \longrightarrow 2Al(NO_3)_3 + 3CaSO_4$$

Q.7 In a water treatment plant, $Cl_2$ used for the treatment of water is produced from the following reaction $2KMnO_4 + 16\ HCl \longrightarrow 2KCl + 2MnCl_2 + 8H_2O + Cl_2$. If during each feed 1 *l* $KMnO_4$ having 79% (w/v) $KMnO_4$ & 9 *l* HCl with d = 1.825 gm/ ml & 10% (w/w) HCl are entered & if that percent yield is 80% then calculate

(a) amount of $Cl_2$ produced.

(b) amount of water that can be treated by $Cl_2$ if 1 litres consumes 28.4 g of $Cl_2$ for treatment.

(c) Calculate efficiency $\eta$ of the process if $\eta = \frac{\text{vol. of water treated}}{\text{vol of total feed}}$

Q.8 Hexane ($C_6H_{14}$) & aniline ($C_6H_7N$) are partially miscible. At 25°C, 0.5 mole of hexane & 0.5 mol of aniline are shaken together & allowed to settle. Two liquid layers are formed.On analysis, the layer A rich in aniline has 10 mol% of hexane while the layer B, rich in hexane has 70 mole% of hexane. What is the weight ratio of layers A & B?

Q.9 The molecular mass of an organic acid was determined by the study of its barium salt. 2.562 g of salt was quantitatively converted to free acid by the reaction 30 ml of 0.2 M $H_2SO_4$, the barium salt was found to have two moles of water of hydration per $Ba^{+2}$ ion and the acid is mono basic. What is molecularweight of anhydyous acid ? (At. mass of Ba = 137)

Q.10 Three different brands of liquid chlorine are available in the market for the use in purifying water of swimming pools. All are sold at the same rate of Rs 10 per litre and all are water solutions. Brand A contains 10% hypochlorite (ClO), brand B contains 7% available chlorine (Cl) and brand C contains 14% sodium hypochlorite (NaClO). All percentage are (w/v) ratios. Which of the three would you buy?

Q.11 A complex compound cobalt has : Co = 22.58%, H = 5.79%, N = 32.2%, O = 12.26% and Cl=27.17%. When the compound is heated it lost $NH_3$ to the extent of 32.63% of its original weight. How many molecules of $NH_3$ are present in the complex compound ? Derive empirical formula of the compound. (Co = 59)

Q.12 A sea water sample has a density of 1.03 g/cm$^3$ and 2.8% NaCl by mass. A saturated solution of NaCl in water is 5.45 M NaCl. How much water would have to be evaporated from $1.00 \times 10^6$ L of the sea water before NaCl would precipitate ?

Q.13 A mixture of formic acid (HCOOH) and oxalic acid ($H_2C_2O_4$) is heated with conc. $H_2SO_4$. The gas

produced is collected and on its treatment with KOH solution the volume of the gas decreases by one sixth. Calculate the molar ratio of the two acid in the original mixture. The reactions are

$HCOOH\ (l) + H_2SO_4\ (l) \longrightarrow CO(g) + H_2SO_4.\ H_2O$ (liq.)

$H_2C_2O_4\ (l) + H_2SO_4\ (l) \longrightarrow CO(g) + CO_2(g) + H_2SO_4\ .\ H_2O$ (liq.)

Q.14 A sample of oleum is such that ratio of "free $SO_3$" by "combined $SO_3$" is equal to unity. Calculate its labelling in terms of percentage oleum.

Q.15 One litre of milk weighs 1.035 kg. The butter fat is 4% (v/v) of milk has density of 875 kg/m$^3$. Find the density of fat free skimed milk.

Q.16 A sample of fuming sulphuric acid containing $H_2SO_4$, $SO_3$ and $SO_2$ weighing 1.00 g is found to require 23.47 mL of 1.00 M alkali (NaOH) for neutralisation. A separate sample shows the presence of 1.50% $SO_2$. Find the percentage of "free" $SO_3$, $H_2SO_4$ and "combined" $SO_3$ in the sample.

Q.17 Chloride samples are prepared for analysis by using NaCl, KCl and $NH_4Cl$ separately or as mixture. What minimum volume of 5 % by weight $AgNO_3$ solution(sp.gr, 1.04 g ml$^{-1}$) must be added to a sample of 0.3 g in order to ensure complete precipitation of chloride in every possible case?

Q.18 In one process for waterproofing, a fabric is exposed to $(CH_3)_2SiCl_2$ vapour. The vapour reacts with hydroxyl groups on the surface of the fabric or with traces of water to form the waterproofing film $[(CH_3)_2SiO]_n$, by the reaction

$$n(CH_3)_2SiCl_2 + 2nOH^- \longrightarrow 2nCl^- + nH_2O + [(CH_3)_2SiO]_n$$

where n stands for a large integer. The waterproofing film is deposited on the fabric layer upon layer. Each layer is 6.0 Å thick [ the thickness of the $(CH_3)_2SiO$ group]. How much $(CH_3)_2\ SiCl_2$ is needed to waterproof one side of a piece of fabric, 1.00 m by 3.00 m, with a film 300 layers thick ? The density of the film is 1.0 g/cm$^3$.

Q.19 Diatoms, microscopic organism, produce carbohydrates from carbon dioxide and water by normal photosynthesis :

$6\ CO_2 + 6\ H_2O$ + solar energy $\longrightarrow C_6H_{12}O_6 + 6\ O_2$. During the first five years of life whales gain 75 kg of mass per day.

(a) Assuming that the mass gain in the first five years of a whale's life is due to the production of carbohydrates, calculated the volume of $CO_2$ per day at 0°C and 101 kPa that must be used by the diatoms to produce the carbohydrates.

(b) There is 0.23 mL of dissolved $CO_2$ per *l* sea water (at 24°C and 101 kPa). If diatoms can completely remove carbon dioxide from the water they process, what volume of water would they process to produced the carbohydrates required by a blue whale per day?

(c) 3% of the mass of a $9.1 \times 10^4$ kg adult whale is nitrogen. What is the maximum mass of $NH_4^+$ that can become available for other marine organisms if one adult whale dies ?

(d) 18% of a adult whale's mass is carbon which can be returned to the atmosphere as $CO_2$ being removed from there by weathering of rocks containing calcium silicate.

$$CaSiO_3(s) + 2CO_2 + 3H_2O(l) \longrightarrow Ca^{2+}(aq) + 2HCO_3^-(aq) + H_4SiO_4(aq)$$

What are the maximum grams of $CaSiO_3$ that can be weathered by the carbon dioxide produced from the decomposition of 1000 blue whales, the number estimated to die annually ?

Q.20 20 ml of a mixture of methane and a gaseous compound of Acetylene series were mixed with 100 ml of oxygen and exploded. The volume of the products after cooling to original room temperature and pressure, was 80 ml and on treatment with potash solution a further contracting of 40 ml was observed. Calculate (a) the molecular formula of the hydrocarbon, (b) the percentage composition of the mixture.

Q.21 In a solution the concentrations of $CaCl_2$ is 5M & that of $MgCl_2$ is 5m. The specific gravity of solution is 1.05, calculate the concentration of $Cl^-$ in the solution in terms of Molarlity.

Q.22 3.6 g of Mg is burnt in limited supply of oxygen. The residue was treated with 100 mL of $H_2SO_4$ (35% by mass, 1.26 g $mL^{-1}$ density). When 2.463 L of $H_2$ at 760 mm Hg at $27^0C$ was evolved. After the reaction, $H_2SO_4$ was found to have a density of 1.05 g $mL^{-1}$. Assuming no volume change in $H_2SO_4$ solution. Find

(i) % by mass of final $H_2SO_4$

(ii) % by mass of Mg converted to oxide

(iii) mass of oxygen used. (Mg = 24, S= 32)

Q.23 A mixture of $H_2$, $N_2$ & $O_2$ occupying 100 ml underwent reaction so as to form $H_2O_2$ (*l*) and $N_2H_2$ (g) as the only products, causing the volume to contract by 60 ml. The remaining mixture was passed through pyrogallol causing a contraction of 10 ml. To the remaining mixture excess $H_2$ was added and the above reaction was repeated, causing a reduction in volume of 10 ml. Identify the composition of the initial mixture in mol %. (No other products are formed)

Q.24 For a gas $A_2B_6$ dissociating like $A_2B_6(g) \longrightarrow A_2(g) + 3B_2(g)$, Vapour densities of the mixture at various time is observed. From the data & informations given.

**Informations**

(1) At t = 0, reaction starts with 1 mole of $A_2B_6$ only & observed V.D. = 50.

(2) Density of $A_2B_6$ relative to $A_2$ is 2.5.

(3) Reaction is complete at time t = 40 min.

**Observations**

(i) time t = 0 , V.D. = 50

(ii) time t = 10 min., V.D. = 25

(iii) time t = 20 min., V.D.=20

Calculate

(a) Molecular weight of $A_2B_6$, Atomic weight of A, Atomic weight of B.

(b) Mole percent of $A_2B_6$, $A_2B_6$, $A_2$ & $B_2$ at t =10 min.

(c) Mass percent of $A_2B_6$, $A_2$ & $B_2$ at t = 20 min.

(d) Rate of disappearance of $A_2B_6$ between t =10 to t = 20, if it is assumed that it disappears uniformly during this time interval. [Rate of disappearance = $\frac{Mole\ \ dissociated}{Time\ \ taken}$]

(e) Vap. density of mixture at t = 40 min.

Q.25 An **impure** sample of $CH_3COONa$, $Na_2SO_4$ & $NaHCO_3$ containing equal moles of each component was heated to cause liberation of $CO_2$ gas [Assume no dissociation of $CH_3COONa$ to give $CO_2$ gas]. If 7.389 *l* of $CO_2$ gas at 1 atm pressure & 300 K is evolved & it is known that the sample contains 50% by mass inert impurities (which are not involved in any reactions) then calculate

(a) moles of each component

(b) wt. of total impure sample

(c) Volume of 0.2 M HCl required for complete neutralisation of that wt. of fresh impure sample as obtained in (b) part. [Assume no interference by weaker acid (if formed) in neutralization process in presence of strong acid]

Q.26 An **impure** sample of $CuSO_4.5H_2O$ (having 40% purity) undergoes following sequence of reactions in a reaction flask having large amount of KCN

$CuSO_4.5H_2O \longrightarrow CuSO_4 + 5H_2O$ ....(1)

$CuSO_4 + KCN \longrightarrow Cu(CN)_2 + K_2SO_4$ ....(2)

$Cu(CN)_2 \longrightarrow Cu_2(CN)_2 + (CN)_2\uparrow$ ....(3)

$Cu_2(CN)_2 + KCN \longrightarrow K_3[Cu(CN)_4]$ ....(4)

If % yield of react. (1) is 100% (2) is 80% (3) is 60% & (4) is 50%. Calculate

(i) wt. of impure sample of $CuSO_4.5H_2O$ required for producing 28.5 gm of complex compound $K_3[Cu(CN)_4]$

(ii) vol. of $(CN)_2$ gas produced at STP if wt. of impure sample of $CuSO_4.5H_2O$ as obtained in 'a' is reacted

## *EXERCISE # III*

Q.1 Equal volumes of 10% (v/v) of HCl is mixed with 10% (v/v) NaOH solution. If density of pure NaOH is 1.5 times that of pure HCl then the resultant solution be.
(A) basic (B) neutral (C) acidic (D) can't be predicted.

Q.2 A definite amount of gaseous hydrocarbon having (carbon atoms less than 5) was burnt with sufficient amount of $O_2$. The volume of all reactants was 600 ml, after the explosion the volume of the products [$CO_2$(g) and $H_2O$(g)] was found to be 700 ml under the similar conditions. The molecular formula of the compound is
(A) $C_3H_8$ (B) $C_3H_6$ (C) $C_3H_4$ (D) $C_4H_{10}$

Q.3 A mixture (15 mL) of CO and $CO_2$ is mixed with V mL (excess) of oxygen and electrically sparked. The volume after explosion was (V + 12) mL. What would be the residual volume if 25 mL of the original mixture is exposed to KOH. All volume measurements were made at the same temperature and pressure
(A) 7 mL (B) 12 mL (C) 10 mL (D) 9 mL

Q.4 One gram of the silver salt of an organic dibasic acid yields, on strong heating, 0.5934 g of silver. If the weight percentage of carbon in it 8 times the weight percentage of hydrogen and one-half the weight percentage of oxygen, determine the molecular formula of the acid. [Atomic weight of Ag = 108]
(A) $C_4H_6O_4$ (B) $C_4H_6O_6$ (C) $C_2H_6O_2$ (D) $C_5H_{10}O_5$

Q.5 The density of vapours of a particular volatile specie was found to be 10 miligram / ml at STP. Its atomic weight in amu is
(A) 20 amu (B) 112 amu (C) 224 amu (D) data insufficient

Q.6 A mixture of $C_3H_8$ (g) & $O_2$ having total volume 100 ml in an Eudiometry tube is sparked & it is observed that a contraction of 45 ml is observed what can be the composition of reacting mixture.
(A) 15 ml $C_3H_8$ & 85 ml $O_2$ (B) 25 ml $C_3H_8$ & 75 ml $O_2$
(C) 45 ml $C_3H_8$ & 55 ml $O_2$ (D) 55 ml $C_3H_8$ & 45 ml $O_2$

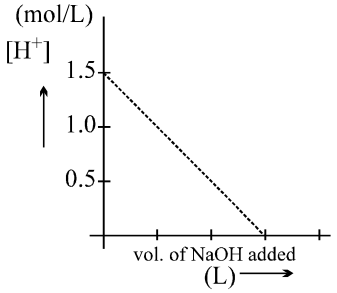
Q.7 Carbon can react with $O_2$ to form CO & $CO_2$ depending upon amount of substances taken. If each option is written in an order like (x, y, z, p) where x represents moles of C taken, y represents moles of $O_2$ taken z represents moles of CO formed & p represents moles of $CO_2$ formed, then which options are correct.
(A) (1, 0.75, 0.5, 0.5) (B) (1, 0.5, 0, 0.5)
(C) (1, 0.5, 0.5, 0) (D) (1, 2, 1, 1)

Q.8 One mole mixture of $CH_4$ & air (containing 80% $N_2$ 20% $O_2$ by volume) of a composition such that when underwent combustion gave maximum heat (assume combustion of only $CH_4$). Then which of the statements are correct, regarding composition of initial mixture.(X presents mole fraction)

(A) $X_{CH_4} = \frac{1}{11}, X_{O_2} = \frac{2}{11}, X_{N_2} = \frac{8}{11}$ (B) $X_{CH_4} = \frac{3}{8}, X_{O_2} = \frac{1}{8}, X_{N_2} = \frac{1}{2}$

(C) $X_{CH_4} = \frac{1}{6}, X_{O_2} = \frac{1}{6}, X_{N_2} = \frac{2}{3}$ (D) Data insufficient

Q.9 To 500 ml of 2 M impure $H_2SO_4$ sample, NaOH soluton 1 M was slowly added & the followng plot was obtained. The percentage purity of $H_2SO_4$ sample and slope of the curve respectively are:




(A) 50%, $-\frac{1}{3}$ (B) 75%, $-\frac{1}{2}$
(C) 75% , –1 (D) none of these

Q.10 Two gaseous ions of unknown charge & mass initially at rest are subjected to same potential difference for accelerating the charges & then subjected to same magnetic field (placed perpendicular to the velocity) & following observations were made.

Obs 1. Before entering the magnetic field zone both ions had same kinetic energy.

Obs 2. The radius of curvature of ion A was greater than that of B.

Stat 1: The magnitude of charge on both the ions should be same

Stat 2: Particle A is more massive than particle B

Stat 3: The e/m ratio of A is higher than that of B

(A) Only Stat 1 & Stat 2 are correct (B) Only Stat 3 is correct

(C) Only Stat 2 is incorrect (D) Only Stat 1 is incorrect

Q.11 Two gases A and B which react according to the equation

$$aA_{(g)} + bB_{(g)} \longrightarrow cC_{(g)} + dD_{(g)}$$

to give two gases C and D are taken (amount not known) in an Eudiometer tube (operating at a constant Pressure and temperature) to cause the above.

If on causing the reaction there is no volume change observed then which of the following statement is/ are correct.

(A) (a + b) = (c + d)

(B) average molecular mass may increase or decrease if either of A or B is present in limited amount.

(C) Vapour Density of the mixture will remain same throughout the course of reaction.

(D) Total moles of all the component of taken mixture will change.

***Question No. 12 to 13 (2 questions)***

A mixture of $H_2$ and Acetylene ($C_2H_2$) was collected in a Eudiometer tube. Then, 60 ml of oxygen were also introduced. The resulting mixture of all the gases was exploded. **After** cooling **a resulting gaseous mixture passes through** Caustic potash solution a contraction of 32 ml occurred and 13 ml of oxygen alone were left behind.

Q.12 After explosion, on cooling of resulting mixture, contraction in volume will be

(A) 21 ml (B) 30 ml (C) 45 ml (D) none

Q.13 Percentage composition of the gaseous mixture of $H_2$ & acetylene are

(A) 53.3, 46.7 (B) 46.7, 53.3 (C) 15.7, 84.3 (D) 84.3, 15.7

***Question No. 14 to 17 (4 questions)***

A 4.925 g sample of a mixture of $CuCl_2$ and $CuBr_2$ was dissolved in water and mixed thoroughly with a 5.74 g portion of AgCl. After the reaction the solid, a mixture of AgCl and AgBr, was filtered, washed, and dried. Its mass was found to be 6.63 g.

Q.14 % By mass of $CuBr_2$ in original mixture is

(A) 2.24 (B) 74.5 (C) 45.3 (D) None

Q.15 % By mass of Cu in original mixture is

(A) 38.68 (B) 19.05 (C) 3.86 (D) None

Q.16 % by mole of AgBr in dried precipate is

(A) 25 (B) 50 (C) 75 (D) 60

Q.17 No. of moles of $Cl^-$ ion present in the solution after precipitation are

(A) 0.06 (B) 0.02 (C) 0.04 (D) None

***Question No. 18 to 20 (3 questions)***

NaBr, used to produce AgBr for use in photography can be self prepared as follows :

$Fe + Br_2 \longrightarrow FeBr_2$ ....(1)

$FeBr_2 + Br_2 \longrightarrow Fe_3Br_8$ ....(2) (not balanced)

$Fe_3Br_8 + Na_2CO_3 \longrightarrow NaBr + CO_2 + Fe_3O_4$ ....(3) (not balanced)

How much Fe in kg is consumed to produce $2.06 \times 10^3$ kg NaBr. ....(4)

Q.18 Mass of iron required to produce $2.06 \times 10^3$ kg NaBr

(A) 420 gm (B) 420 kg (C) $4.2 \times 10^5$ kg (D) $4.2 \times 10^8$ gm

Q.19 If the yield of (ii) is 60% & (iii) reaction is 70% then mass of iron required to produce $2.06 \times 10^3$ kg NaBr

(A) $10^5$ kg (B) $10^5$ gm (C) $10^3$ kg (D) None

Q.20 If yield of (iii) reaction is 90% then mole of $CO_2$ formed when $2.06 \times 10^3$ gm NaBr is formed

(A) 20 (B) 10 (C) 40 (D) None

***Question No. 21 to 23 (3 questions)***

In the gravimetric determination of sulfur the ignited precipitate of $BaSO_4$ sometimes partially reduces to BaS. This cause an error, of course, if the analyst does not realize this and convert the BaS back to $BaSO_4$. Suppose a sample which contains 32.3% $SO_3$ is analyzed and 20.0% of the final precipitate that is weighed is BaS. (80.0% is $BaSO_4$). What percentage of $SO_3$ in the sample would the analyst calculate if he assume the entire precipitate as $BaSO_4$? Repeat the question if BaS was 20% by mole.

Q.21 Calculate the mass of sample, assuming 100 gm precipitate is formed

(A) 106.3 gm (B) 114.35 gm (C) 110.5 gm (D) None

Q.22 Percentage of $SO_3$ in the sample, calculated by analyst is (if the assume the entire precipiate as $BaSO_4$)

(A) 30 (B) 30.5 (C) 32 (D) 32.3

Q.23 If BaS was 20% by mole in precipitate, then percentage of $SO_3$ in the sample calculated by analyst is (if he assume the entire precipitate as $BaSO_4$)

(A) 30 (B) 30.5 (C) 32 (D) 32.3

***Question No. 24 to 25 are based on the following Passage. Read it carefully & answer the questions that follow***

A monobasic acid of weight 15.5 gms is heated with excess of oxygen & evolved gases when passed through KOH solution increased its weight by 22 gms and when passed through anhydrous $CaCl_2$, increased its weight by 13.5 gms. When the same mass of this organic acid is reacted with excess of silver nitrate solution form 41.75 gm silver salt of the acid which on ignition gave the residue of weight 27 gm.

Q.24 The molecular formula of the organic acid is

(A) $C_2H_6$ (B) $C_2H_5O_2$ (C) $C_2H_6O_2$ (D) $C_2H_4O$

Q.25 The molar masses of the acid & its silver salt respectively are

(A) 60, 168 (B) 167, 60 (C) 60, 167 (D) 168, 60

***Question No. 26 to 28 (3 questions)***

$N_2O_5$ and $H_2O$ can react to form $HNO_3$, according to given reaction

$$N_2O_5 + H_2O \longrightarrow 2HNO_3$$

the concentration of a mixture of $HNO_3$ and $N_2O_5$ (g) can be expressed similar to oleum. Then answer the following question.

Q.26 Find the percentage labelling of a mixture containing 23 gm $HNO_3$ and 27 gm $N_2O_5$.

(A) 104.5% (B) 109% (C) 113.5% (D) 118%

Q.27 Find the maxmum and minimum value of percentage labelling

(A) 133.3 % (B) 116.66%, 0% (C) 116.66%, 100% (D) None

Q.28 Find the new labelling if 100 gm of this mixture (original) is mixed with 4.5 gm water

(A) $100 + \frac{4.5}{1}$ (B) $100 + \frac{4.5}{1.045}$ (C) $100 + \frac{4.5}{104.5}$ (D) $100 + \frac{4.5}{1.09}$

***Question No. 29 and 30 are based on the following piece of information. Mark the appropriate options on the basis of information.***

342 gm of 20% by mass of $Ba(OH)_2$ solution (sp. gr. 0.57) is reacted with 200 ml of 2M $HNO_3$ according to given balanced reaction.

$$\mathbf{Ba(OH)_2 + 2HNO_3 \longrightarrow Ba(NO_3)_2 + 2H_2O}$$

Q.29 The nature of the final solution is

(A) acidic (B) neutral (C) basic (D) can't say

Q.30 If density of final solution is 1.01 gm /ml then find the molarity of the ion in resulting solution by which nature of the above solution is identified, is

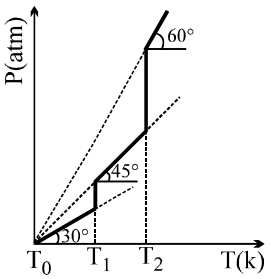
(A) 0.5 M (B) 0.8 M (C) 0.4 M (D) 1 M

***Question No. 31 & 32 are based on the piece of information.***

For a gaseous reaction,

$$2A(g) \longrightarrow 3B(g) + C(g)$$

Whose extent of dissociation depends on temperature is performed in a closed container, it is known that extent of dissociation of A is different in different temperature range. With in a temperature range it is constant. (Temperature range $T_0 - T_1, T_1 - T_2, T_2 - T_\infty$). A plot of P v/s T is drawn under the given condition.




Q.31 If $\alpha_{T_i - T_{i+1}}$ is the degree of dissociation of A then in the temperature range $T_i \rightarrow T_{i+1}$

(A) $\alpha_{T_0 - T_1}$ is lowest (B) $\alpha_{T_0 - T_1}$ is highest

(C) $\alpha_{T_2 - T_\infty} = 1$ (D) $\alpha_{T_2 - T_\infty} = 0$

Q.32 If initially 1 mole of A is taken in a 1 *l* container then [R = 0.0821 atm lit / k]

(A) $\alpha_{T_0 - T_1} = \frac{1}{2\sqrt{3R}} + \frac{1}{2}$ (B) $\alpha_{T_0 - T_1} = \frac{1}{2\sqrt{3R}} - \frac{1}{2}$

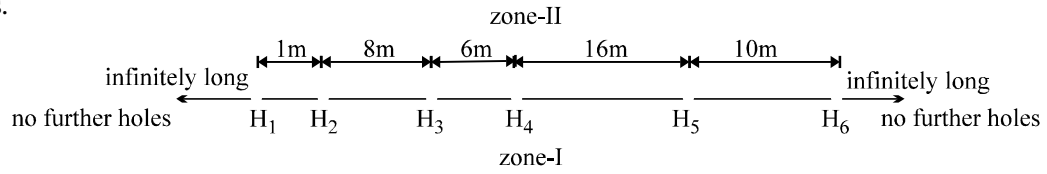
(C) $\alpha_{T_1 - T_2} = \frac{1}{R} + 1$ (D) $\alpha_{T_1 - T_2} = \frac{1}{R} - 1$

Q.33 Which has maximum number of atoms of oxygen

(A) 10 ml $H_2O(l)$ (B) 0.1 mole of $V_2O_5$

(C) 12 gm $O_3(g)$ (D) $12.044 \times 10^{22}$ molecules of $CO_2$

Q.34 In a mass spectrometry experiment, various ions $H^+$, $Li^+$, $O^{2+}$ & $N_3^+$ were projected with a same velocity into a same magnetic field zone (alligned perpendicular to the direction of velocity). The sheet on which they are striking is pierced at certain points (marked as $H_1$, $H_2$ etc.) as shown in the diagram. **It is known that $H^+$ comes back to zone-I from $H_2$ when projected from $H_1$**. Mark out the correct options.




(A) Out of all remaining ions when projected from $H_1$, only $N_3^+$ will come back to zone-I.

(B) When all the remaining ions were projected from $H_2$, only $O^{2+}$ will come back in zone-I.

(C) When all the remaining ions were projected from $H_3$, none of the them will come back to zone-I.

(D) When all the remaining ions were projected from $H_4$, none of the them will come back to zone-I.

## ***EXERCISE # IV***

Q.1 An evacuated glass vessel weighs 50 gm when empty , 148.0 g when filled with liquid of density 0.98 $gml^{-1}$ and 50.5 g when filled with an ideal gas at 760 mm at 300 K . Determine the molecular weight of the gas . **[JEE '98,3]**

Q.2 At 100° C and 1 atmp , if the density of liquid water is 1.0 g $cm^{-3}$ and that of water vapour is 0.0006 g $cm^{-3}$ , then the volume occupied by water molecules in 1 L of steam at that temperature is :

(A) 6 $cm^3$ (B) 60 $cm^3$ (C) 0.6 $cm^3$ (D) 0.06 $cm^3$

**[JEE '2001 (Scr), 1]**

Q.3 How many moles of $e^-$ weigh one Kg **[JEE'2002 (Scr), 1]**

(A) $6.023 \times 10^{23}$ (B) $\frac{1}{9.108} \times 10^{31}$ (C) $\frac{6.023}{9.108} \times 10^{54}$ (D) $\frac{1}{9.108 \times 6.023} \times 10^{8}$

Q.4 Calculate the molarity of pure water using its density to be 1000 kg $m^{-3}$.

**[JEE'2003]**

Q.5 One gm of charcoal absorbs 100 ml 0.5 M $CH_3COOH$ to form a monolayer, and thereby the molarity of $CH_3COOH$ reduces to 0.49. Calculate the surface area of the charcoal adsorbed by each molecule of acetic acid. Surface area of charcoal = $3.01 \times 10^2$ $m^2$/gm. **[JEE'2003]**

Q.6 Calculate the amount of Calcium oxide required when it reacts with 852 gm of $P_4O_{10}$. **[JEE 2005]**

$$6CaO + P_4O_{10} \longrightarrow 2Ca_3(PO_4)_2$$

Q.7 20% surface sites have adsorbed $N_2$. On heating $N_2$ gas evolved from sites and were collected at 0.001 atm and 298 K in a container of volume is 2.46 $cm^3$. Density of surface sites is $6.023 \times 10^{14}$/$cm^2$ and surface area is 1000 $cm^2$, find out the no. of surface sites occupied per molecule of $N_2$.

**[JEE 2005]**

# ANSWER KEY

## *EXERCISE # I*

Q.1 196.2 Q.2 $7.09 \times 10^7$ Q.3 33.9 Q.4 $19.09 \times 10^6$ years Q.5

(a) $3.41 \times 10^{-4}$mole, (b) $5.68 \times 10^{-3}$ mole, (c) $2.05 \times 10^{22}$ atoms

Q.6 2.39 mole Pt, 21.7 $cm^3$

Q.7 (a) Y = 44.95%, Al = 22.73%, O = 32.32%, (b) 17.98 gm Q.8 $8.8 \times 10^{-8}$ mole Q.9

9.063 gm Q.10 0.250 Q.11 1.14 g Q.12 49.9%

Q.13 76.15% Q.14 Al = 0.546 g; Mg = 0.454 g Q.15 28.4%, 71.6%Q.16

%$NaHCO_3$ = 16.8, % $Na_2CO_3$= 83.2 Q.17 67.9% Q.18 1.14 gm Q.19 CO : $CO_2$ =

21 : 11 Q.20 12.15 gm; 14.28%, $H_2$ 42.86%, $NH_3$ 42.86%

Q.21 $ICl_3$, $I_2Cl_6$ Q.22 0.532 : 1 Q.23 $C_5H_{14}N_2$ Q.24 m = 4, $C_6H_2Cl_3$ Q.25

(a) $C_6H_{12}$, (b) $C_5H_{10}O_5$, (c) $H_2O_2$, (d) $Hg_2Cl_2$, (e) $H_4F_4$

Q.26 H = 1.486%, C = 38.37%, O = 7.87%, Cl = 52.28% Q.27 CH

Q.28 6.67% Q.29 46.9% Q.30 $CH_3Cl$ Q.31 $C_7H_{10}NCl$ Q.32

92.70 Q.33 70 Q.34 (a) 0.5 M, (b) 0.5 M, (c) 0.2 M Q.35 1.445 M

Q.36 1.088 M, 1.13 m Q.37 13.15 M Q.38 0.25 Q.39

16.67% Q.40 $2.7 \times 10^{-4}$ m Q.41 1.288 gm/ml Q.42 0.331, $2.25 \times 10^{-4}$, 2.81, 0.0482, 321,

5.72 Q.43 29.77% Q.44 1736.1 ml Q.45 183.68 ml Q.46 $XeF_6$

Q.47 9 : 10 Q.48 1 : 2

Q.49 $1.09 \times 10^{-3}$ gm Q.50 (i) 60.71 gm/l, (ii) 1.78 M, (iii) 6.071%

Q.51 (a) pure $H_2SO_4$(109 gm); (b) 109 gm $H_2SO_4$, 9 gm $H_2O$; (c) 109 gm $H_2SO_4$, 111 gm $H_2O$ Q.52

CO = 5 ml ; $CH_4$ = 2 ml ; $N_2$ = 3 ml Q.53 10 ml

Q.54 (i) 7 volumes, (ii) $C_2H_6N_2$ Q.55 NO = 44 ml ; $N_2O$ = 16 ml

Q.56 $O_3$ Q.57 $NH_3$

## *EXERCISE # II*

Q.1 10.07 : 0.662 : 1 Q.2 0.5 : 0.1 : 0.4 Q.3 $C_2H_6O$ Q.4 1.52 kg, 0.76 kg

Q.5 (a) $A_3B_4 = 2$ & $A_2 = 1$ ; (b) $A_2B_4 = \frac{1}{2}$ & $B_2 = 1$ (c) $A_2B_4 = 0.5$ & $A_3B_4 = 0.5$

Q.6 24. 51 ml Q.7 (a)10 mol, (b) 25 *lit.*, (c) 2.5 Q.8 $W_A : W_B = 0.524$

Q.9 122.6 Q.10 Brand B Q.11 5 molecule, $[Co(NH_3)_5(NO_2)]Cl_2$

Q.12 $9.095 \times 10^5$ lit Q.13 $\frac{HCOOH}{H_2C_2O_4} = \frac{4}{1}$ Q.14 110.11% Q.15 1.041 gm/ml

Q.16 $H_2SO_4 = 35.38\%$, Free $SO_3 = 63.1\%$, combined $SO_3 = 28.89\%$ Q.17 18.38 ml

Q.18 0.9413 gram Q.19 (a) 56000 lit/day, (b) $2.6 \times 10^8$ litres, (c) 3510 kg/day, (d) $7.917 \times 10^7$ kg

Q.20 (b) 50 Q.21 $[Cl^-] = 13.36$ M Q.22 (i) 28%, (ii) 33.33%, (iii) 0.8g

Q.23 $N_2 = 30$ ml, $H_2 = 40$ ml

Q.24 (a) 100, 20, 10; (b) 33.33%, 16.67%, 50%; (c) 50%, 20%, 30%; (d) 0.0167 mol/min; (e) 12.5

Q.25 (a) 0.6, (b) 369.6 gm (c)6 (*l*) Q.26 (i) 521.25 gm, (ii) 2.24 *l*

Q.27 In initial gaseous mixture gases of same molar mass are present.
∴ Avg. molar mass of the mixture will be 28.
After the appropriate reactions , the gas that will remain will be $N_2$ only since both $C_2H_4$ and CO will get oxidised to $CO_2$ which is then removed from KOH.
∴ Average Molar mass of final gaseous mixture is 28

## *EXERCISE # III*

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Q.1 | A | Q.2 | A | Q.3 | C | Q.4 | B |
| Q.5 | D | Q.6 | A,B | Q.7 | A | Q.8 | A |
| Q.9 | D | Q.10 | A | Q.11 | A,C | Q.12 | C |
| Q.13 | B | Q.14 | C | Q.15 | A | Q.16 | B |
| Q.17 | C | Q.18 | B,C,D | Q.19 | C | Q.20 | B |
| Q.21 | B | Q.22 | A | Q.23 | B | Q.24 | C |
| Q.25 | C | Q.26 | B | Q.27 | C | Q.28 | B |
| Q.29 | C | Q.30 | A | Q.31 | A | Q.32 | D |
| Q.33 | C | Q.34 | B,C,D | | | | |

## *EXERCISE # IV*

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Q.1 | 123 g/mol | Q.2 | C | Q.3 | D | Q.4 | 55.5 mol $L^{-1}$ |
| Q.5 | $5 \times 10^{-19}$ $m^2$ | Q.6 | 1008 gm | Q.7 | 2 | | |

# STUDY PACKAGE

# Target: IIT-JEE(Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 2. Stoichiometry

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

**4. Exercise III**

**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

# OXIDATION & REDUCTION

***Molecular Equations:***

$BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4\downarrow + 2\ NaCl$ .

***Ionic Equations :***

$Ba^{2+} + SO_4^{2-} \longrightarrow BaSO_4\downarrow$ .

***Spectator Ions :***

Ions which do not undergo change during a reaction , they are not included in the final balanced equation .

***Rules For Writing Ionic Equations :***

**(i)** All soluble electrolytes involved in a chemical change are expresses in ionic symbols and covalent substances are written in molecular form .

**(ii)** The electrolyte which is highly insoluble , is expressed in molecular form .

**(iii)** The ions which are common and equal in number on both sides (spectator ions) are cancelled .

**(iv)** Besides the atoms , the ionic charges must also balance on both the sides .

**Valency *:***

Valency of an element is defined as the number of hydrogen atoms that combine with or are displaced by one atom of the element . Cl, monovalent; O, divalent; N, trivalent; tetravalent C; variable valency P (3 , 5). It is never a useful concept despite of physical reality, so more common & artificial concept is oxidation state (oxidation number) .

***Oxidation Number :***

It is the charge (real or imaginary) which an atom appears to have when it is in combination. It may be a whole no. or fractional. For an element may have different values. It depends on nature of compound in which it is present. There are some operational rules to determine oxidation number.

***Stock's Notation :***

Generally used for naming compounds of metals , some non-metals also. eg. $Cr_2O_3$ Chromium (III) oxide and $P_2O_5$ Phosphorous (V) oxide .

***Oxidation :***

Addition of oxygen , removal of hydrogen , addition of electro- negative element , removal of electro-positive element, loss of electrons, increase in oxidation number (de-electronation) .

***Reduction :***

Removal of oxygen, addition of hydrogen, removal of electronegative element, addition of electro-positive element, gain of electrons, decrease in oxid. no. (electronation).

***Redox Reactions :*** A reaction in which oxidation & reduction occur simultaneously .

***Oxidising Agents (oxidants / oxidisors):***

They oxidise others, themselves are reduced & gain electrons. eg. $O_2$, $O_3$, $HNO_3$, $MnO_2$, $H_2O_2$, halogens, $KMnO_4$, $K_2Cr_2O_7$, $KIO_3$, $Cl(SO_4)_3$, $FeCl_3$, NaOCl, hydrogen ions.
[Atoms present in their higher oxidation state.]

***Reducing Agents ( reductants / reducers):***

They reduce others, themselves get oxidised & lose electrons. $H_2$, molecule is weak but Nascent hydrogen is powerful . C, CO, $H_2S$, $SO_2$, $SnCl_2$, Sodium thio Sulphate, Al, Na, $CaH_2$, $NaBH_4$, $LiAlH_4$
[Atoms present in their lower oxidation state].

***Both Oxidising & Reducing Agents :*** $SO_2$ , $H_2O_2$ , $O_3$ , $NO_2$ , etc .

***Balancing Of Equations :***

**(i)** Ion - electron method **(ii)** Oxidation number method

[Concept involved that in any chemical reaction electrons cannot be produced so no. of electrons in both the sides should be same]

***Oxidation Half Reaction :*** $Na \longrightarrow Na^{+} + e^{-}$ .

***Reduction Half Reaction :*** $F_2 + 2e^{-} \longrightarrow 2\,F^{-}$ .

## COMMON OXIDATION AND REDUCTION PARTS

| OXIDATION PARTS | REDUCTION PARTS |
|---|---|
| $Fe^{2+} \longrightarrow Fe^{3+}$ | $Fe^{3+} \longrightarrow Fe^{2+}$ |
| $Zn \longrightarrow Zn^{2+}$ | |
| $X^{-} \longrightarrow X_2$ | $X_2 \longrightarrow X^{-}$ |
| $S^{2-} \longrightarrow S$ | $Cr_2O_7^{2-} \longrightarrow Cr^{3+}$ |
| $H_2O_2 \longrightarrow O_2$ | $NO_3^{-} \longrightarrow NO$ |
| $SO_3^{2-} \longrightarrow SO_4^{2-}$ | $MnO_4^{-} \longrightarrow Mn^{2+}$ (neutral med.) |
| $C_2O_4^{2-} \longrightarrow CO_2$ | $MnO_4^{-} \longrightarrow MnO_2$ (Basic med.) |
| $S_2O_3^{2-} \longrightarrow S_4O_6^{2-}$ | $SO_4^{2-} \longrightarrow SO_2$ |
| $I_2 \longrightarrow IO_3^{-}$ | $MnO_2 \longrightarrow Mn^{2+}$ |

***Types Of Redox Reduction :***

Intermolecular redox, disproportion, Intra molecular redox .

***Disproportion:***

In such reactions the oxidising and reducing agents(atom) are the same .

$$\overset{\text{oxidation}}{H_2O_2} + \overset{\text{reduction}}{H_2O_2} \longrightarrow 2\,H_2O + O_2 \ .$$

To identify whether a reaction is redox or not , find change in oxidation number or loss and gain of electrons. If there is no change in oxidation number , the reaction is not a redox reaction .

**NOTE :** To predict the product of reaction remember :

**(a)** Free halogen on reduction gives halide ion $(F_2 \rightarrow F^{-})$

**(b)** Alkali metals on oxidation give metallic ion with + 1 oxidation state.

**(c)** Conc. $HNO_3$ on reduction gives $NO_2$ , while dilute $HNO_3$ can give $NO, N_2, NH_4^{+}$ or other products depending on the nature of reducing agent and on dilution.

**(d)** In acid solution $KMnO_4$ is reduced to $Mn^{2+}$ while in neutral or alkaline , it gives $MnO_2$ or $K_2MnO_4$ .

**(e)** $H_2O_2$ on reduction gives water and on oxidation gives oxygen.

**(f)** Dichromate ion in acid solution is reduced to $Cr^{3+}$.

***Nature of oxides based on oxidation number :***

| | | | |
|---|---|---|---|
| Lowest oxidation state | $\longrightarrow$ | Basic | $(MnO)$ |
| Intermediate oxidation state | $\longrightarrow$ | Amphoteric | $(Mn_3O_4 , MnO_2)$ |
| Highest oxidation state | $\longrightarrow$ | Acidic | $(Mn_2O_7)$ |

***Metathesis Reactions :***

Never redox reactions . In these two compounds react to form two new compounds and no change in oxidation number occur . eg.

**(i)** $Pb(NO_3)_2 + K_2CrO_4 \longrightarrow PbCrO_4 + 2\,KNO_3$ **(ii)** $HCl + NaOH \longrightarrow NaCl + H_2O$

***Rules For Assigning Oxidation Number :***

**(i)** Oxidation number of free elements or atoms is zero .

**(ii)** Oxidation number of allotropes is zero .

**(iii)** Oxidation number of atoms in homo-nuclear molecules is zero .

**(iv)** Oxidation number of mono-atomic ions is equal to the algebric charge on them .

**(v)** Oxidation number of F in compounds is - 1 .

**(vi)** Oxidation number of H in its compounds is + 1 , except in metalhydrides where it is - 1 .

**(vii)** Oxidation number of O is - 2 in its compounds , but in $F_2O$ it is + 2 and in peroxides it is - 1 and - 0.5 in $KO_2$ .

**(viii)** Oxidation number of alkali metals in their compounds + 1 .

**(ix)** Oxidation number of alkaline earth metals in their compounds is + 2 .

**(x)** Oxidation number of an ion is equal to its charge .

**(xi)** Oxidation number of a molecule as a whole is zero .

**(xii)** The sum of oxidation number of all the atoms in a molecule should be zero and in an ion equal to its charge .

***Average Oxidation Number :*** Find Oxidation Number of Fe in $Fe_3O_4$

$Fe_3O_4$ is FeO. $Fe_2O_3$.

O. N. of Fe in FeO is + 2 ; O. N. of Fe in $Fe_2O_3$ is + 3 .

Therefore average O. N. of three Fe atoms $= \frac{+2+2x(+3)}{3} = +\frac{8}{3}$ .

# EQUIVALENT CONCEPT

***(A) Volumetric analysis:*** This mainly involve titrations based chemistry. It can be divided into two major category. (I) Non-redox system (II) Redox system

***(I) Non – redox system:*** This involve following kind of titrations:

1. Acid-Base titrations
2. Back titration
3. Precipitation titration
4. Double indicator acid base titration

***Titrimetric Method of Analysis :*** A titrimetric method of analysis is based on chemical reaction such as.

$$aA + tT \longrightarrow \text{Product}$$

Where 'a' molecules of "analysis", A, reacts with t molecules of reagent T.

T is called Titrant normally taken in buret in form of solution of known concentration. The solution of titrant is called "standard solution".

The addition of titrant is added till the amount of T, chemically equivalent to that of 'A' has been added. It is said equivalent point of titration has been reached. In order to know when to stop addition of titrant, a chemical substance is used called indicator, which respond to appearance of excess of titrant by changing colour precisely at the equivalence point. The point in the titration where the indicator changes colour is termed the 'end point'. It is possible that end point be as close as possible to the equivalence point.

The term titration refer's to process of measuring the volume of titrant required to reach the end point. For many years the term volumetric analysis was used rather than titrimetric analysis. However from a rigorons stand point the term titrimetric is preferable because volume measurement may not be confirmed to titration. In certain analysis, for example one might measure the volume of a gas.

We can adopt mole method in balanced chemical reactions to relate reactant and products but it is more easier to apply law of equivalents in volumetric calculations because it does not require knowledge of balanced chemical reactions involved in sequence.

Law of equivalents refers to that, equivalents of a limiting reactant is equal to equivalent of other reactant reacting in a chemical reaction or equal to equivalents of products formed. *n* factor here we mean a conversion factor by which we divide molar mass of substance to get equivalent mass and it depends on nature of substance which vary from one condition to another condition.

We can divide n-factor calculations in two category.

(A) when compound is not reacting.

(B) when compound is reacting.

***Acid-Base titration***

To find out strength or concentration of unknown acid or base it is titrated against base or acid of known strength. At the equivalence point we can know amount of acid or base used and then with the help of law of equivalents we can find strength of unknown.

Meq of acid at equivalence point = Meq of base at equivalence point

***Back titration***

Back titration is used in volumetric analysis to find out excess of reagent added by titrating it with suitable reagent. It is also used to find out percentage purity of sample. For example in acid-base titration suppose we have added excess base in acid mixture. To find excess base we can titrate the solution with another acid of known strength.

***Precipitation titration :***

In ionic reaction we can know strength of unknown solution of salt by titrating it against a reagent with which it can form precipitate. For example NaCl strength can be known by titrating it against $AgNO_3$ solution with which it form white ppt. of AgCl.

Meq. of NaCl at equivalence point = meq of $AgNO_3$ used = meq of AgCl formed

***Double indicator acid-base titration:***

In the acid-base titration the equivalence point is known with the help of indicator which changes its colour at the end point. In the titration of polyacidic base or polybasic acid there are more than one end point for each step neutralization. Sometimes one indicator is not able to give colour change at every end point. So to find out end point we have to use more than one indicator. For example in the titration of $Na_2CO_3$ against HCl there are two end points.

$$Na_2CO_3 + HCl \longrightarrow NaHCO_3 + NaCl$$

$$NaHCO_3 + HCl \longrightarrow H_2CO_3 + NaCl$$

When we use phenophthalein in the above titration it changes its colour at first end point when $NaHCO_3$ is formed and with it we can not know second end point. Similarly with methyl orange it changes its colour at second end point only and we can not know first end point. It is because all indicator changes colour on the basis of pH of medium. So in titration of $NaHCO_3$, $KHCO_3$ against acid phenolphthalein can not be used.

So we can write with phenolpthalein, if total meq of $Na_2CO_3$ = 1 then

½ meq of $Na_2CO_3$ = meq of HCl

with methyl orange,

meq of $Na_2CO_3$ = meq of HCl

***n-factor in non-redox system***

| Titration | Indicator | pH Range | n factor |
|---|---|---|---|
| $Na_2CO_3$ against acid | Phenolphthalein | 8.3 – 10 | 1 |
| $K_2CO_3$ of products formed in reaction. | Methyl orange | 3.1 – 4.4 | 2 |

***Note:*** When we carry out dilution of solution, meq eq, milli mole or mole of substance does not change because they represent amount of substance, however molar concentration may change.

***Solubilities of some important salt's :***

| | | | |
|---|---|---|---|
| 1. | **Chloride :** | $AgCl$ – White ppt.<br>$Hg_2Cl_2$ – White ppt.<br>$PbCl_2$ – White ppt.<br>$CuCl$ – Insolution ppt.<br>$BiOCl$ – White ppt.<br>$SbOCl$ – White ppt.<br>$Hg_2OCl_2$ – White ppt. | All other chlorides are soluble in water. |
| 2. | **Bromide :** | $AgBr$ – Pate yellow ppt.<br>$PbBr_2$ – White ppt.<br>$Hg_2Br_2$ – White ppt.<br>$CuBr$ – White ppt. | All other bromides are soluble in water |
| 3. | **Iodide :** | $AgI$ – Yellow ppt.<br>$PbI_2$ – Yellow ppt.<br>$Hg_2I_2$ – Green ppt.<br>$HgI_2$ – Red ppt.<br>$CuI$ – White ppt.<br>$BI_3$ – Black ppt. | |
| 4. | **Some important oxides and hydroxises :** | $Ag_2O$ – Brown ppt.<br>$Pb(OH)_2$ – White ppt.<br>$Pb(OH)_4$ – White ppt.<br>$Hg_2O$ – Black ppt.<br>$HgO$ – Yellow ppt.<br>$Cu_2O$ – Red ppt.<br>$CuO$ – Black ppt.<br>$Cu(OH)_2$ – Blue ppt.<br>$Cd(OH)_2$ – White ppt.<br>$Fe(OH)_2$ – White ppt.<br>$Fe(OH)_3$ – Red ppt.<br>$Sn(OH)_2$ – White ppt.<br>$Sn(OH)_4$ – White ppt.<br>$Al(OH)_3$ – White gelatenons<br>$Cr(OH)_3$ – Grey-Green<br>$Co(OH)_2$ – Pink<br>$Co(OH)_3$ – Brownish black<br>$Ni(OH)_2$ – Green<br>$Ni(OH)_3$ – Black<br>$Mn(OH)_2$ – White<br>$MnO(OH)_2$ – Brown | |

$Zn(OH)_2$ – White
$Mg(OH)_2$ – White

***Carbonates :*** Except Alkali metals and $NH_4^+$ all other carbonates are insoluble.

$Ag_2CO_3 \longrightarrow$ White ppt. $\longrightarrow Ag_2O + CO_2$
$3HgO.HgCO_3 \longrightarrow$ basic murcuric carbonate White ppt.
$CuCO_3 \longrightarrow$ Green ppt.
$CaCO_3 \longrightarrow$ White ppt.

***Sulphites ( $SO_3^{2-}$ ) :*** Except Alkali metal and Ammonium, all other sulphite are generally insoluble.

Examples : $Ag_2SO_3$, $PbSO_3$, $BaSO_3$, $CaSO_3$ $\longrightarrow$ White ppt.

***Thiosulphates :*** Mostly soluble except

$Ag_2S_2O_3 \longrightarrow$ White ppt. $[Ag(S_2O_3)_2]^{3-}$ soluble
$PbS_2O_3 \longrightarrow$ White ppt.
$BaS_2O_3 \longrightarrow$ White ppt.

***Thiocynate ($SCN^-$) :*** $Hg(SCN)_2$ – White ppt. (Pharaoh's serpent)
$Ag(SCN)$ – White ppt.
$Cu(SCN)_2$ – Black ppt.
$Cu(SCN)$ – White ppt.
$Fe(SCN)_3$ – Red complex.
$[Co(SCN)_4]^{2-}$ – Blue complex
$Co[Hg(SCN)_4]$ – Blue ppt.

***Cynaides($CN^-$) :*** Except Alkali metal Alkaline earth metal cyanides are soluble in water.

$Hg(CN)_2$ – soluble in water in undissociated form
$Ag(CN)$ – White ppt. $[Ag(CN)_2]^-$ soluble
$Pb(CN)_2$ – White ppt.
$Fe(CN)_3$ – Brown ppt. $[Fe(CN)_6]^{3-}$ soluble
$Co(CN)_2$ – Brown ppt. $[Co(CN)_6]^{4-}$ soluble
$Ni(CN)_2$ – Green $[Ni(CN)_4]^{2-}$ soluble

***Sulphides :*** Except Alkali metals and ammonium salt's all other sulphides are insoluble. Some insoluble sulphides with unusual colour are

$CdS \longrightarrow$ Yellow
$MnS \longrightarrow$ Pink
$ZnS \longrightarrow$ White
$SnS \longrightarrow$ Brown
$SnS_2 \longrightarrow$ Yellow
$As_2S_3 \longrightarrow$ Yellow
$Sb_2S_3 \longrightarrow$ Orange

***Chromates :***

$Ag_2CrO_4 \longrightarrow$ Red ppt.
$PbCrO_4 \longrightarrow$ Yellow ppt.
$BaCrO_4 \longrightarrow$ Yellow ppt.
$FeCrO_4 \longrightarrow$ Green ppt.
Dichromates are generally soluble.
$MnO_4$ – Permangnates are generally soluble.

***Phosphates:*** Are generally insoluble :

$Ag_3PO_4 \longrightarrow$ Yellow ppt.

$AlPO_4 \longrightarrow$ Yellow ppt.

$ZrO(HPO_4) \longrightarrow$ White ppt.

$Mg(NH_4)PO_4 \longrightarrow$ White ppt.

$(NH_4)_3[P\,Mo_{12}O_{40}] \longrightarrow$ Canary yellow ppt.

***Phosphite ( $HPO_4^{2-}$ ):*** Except Alkali metals all other phosphites are insoluble

***Hypo phosphite:*** All hypophosphites are soluble in water.

All Acetate are soluble except $Ag(CH_3COO)$

All formates are soluble except $Ag(HCOO)$

Tatarate,Citrate,Salicylate,Succinate of Silver-are all insoluble white ppt.

***Some Important ppt.:*** KH (Tartarate) $\longrightarrow$ White ppt.

$NH_4H$(Tartarate) $\longrightarrow$ White ppt.

$K_2[PtCl_6] \longrightarrow$ White ppt.

$K_3[Co(NO_2)_6] \longrightarrow$ Yellow ppt.

$(NH_4)_3[Co(NO_2)_6] \longrightarrow$ Yellow ppt.

$(NH_4)_2[PtCl_6] \longrightarrow$ Yellow ppt.

## ***HEATING EFFECTS***

**Heating effect of carbonate & bicarbonate salts**

All carbonates except (Na, K, Rb, Cs) decompose on heating giving $CO_2$

$Li_2CO_3 \longrightarrow Li_2O + CO_2$

$MCO_3 \longrightarrow MO + CO_2$ [M = Be, Mg, Ca, Sr, Ba]

$$\underbrace{Cu(OH)_2.CuCO_3}_{\text{Basic Cu(II) carbonate}} \xrightarrow{\Delta} 2CuO + CO_2 + H_2O$$

$$\underset{\text{white}}{ZnCO_3} \longrightarrow \underset{\substack{\text{Yellow(hot)}\\ \text{white(cold)}}}{ZnO} + CO_2; \qquad PbCO_3 \longrightarrow \underset{\text{Yellow}}{PbO} + CO_2$$

$$\underset{\text{Yellow}}{Ag_2CO_3} \longrightarrow \underset{\text{Black}}{2Ag} + CO_2 + \frac{1}{2}O_2; \qquad HgCO_3 \longrightarrow Hg + \frac{1}{2}O_2\uparrow + CO_2$$

$(NH_4)_2CO_3 \longrightarrow 2NH_3 + H_2O + CO_2$

All bicarbonates decompose to give carbonates and $CO_2$. eg.

$2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$

[General reaction: $2HCO_3^- \longrightarrow CO_3^{2-} + H_2O + CO_2$ ]

**Heating effect of ammonium salts**

$NH_4NO_2 \longrightarrow N_2 + 2H_2O$; $\qquad NH_4NO_3 \longrightarrow N_2O + 2H_2O$

$(NH_4)_2Cr_2O_7 \longrightarrow N_2 + Cr_2O_3 + 4H_2O$

$2NH_4ClO_4 \longrightarrow N_2 + Cl_2 + 2O_2 + 4H_2O$

$2NH_4IO_3 \longrightarrow N_2 + I_2 + O_2 + 4H_2O$

[If anionic part is oxdising in nature, then $N_2$ will be the product (some times $N_2O$).]

$(NH_4)_2HPO_4 \longrightarrow HPO_3 + H_2O + 2NH_3$

$(NH_4)_2SO_4 \longrightarrow NH_3 + H_2SO_4$

$2(NH_3)_3PO_4 \longrightarrow 2NH_3 + P_2O_5 + 3H_2O$

$(NH_4)_2CO_3 \longrightarrow 2NH_3 + H_2O + CO_2$

[If anionic part weakly oxidising or non oxidising in nature then $NH_3$ will be the product.]

**Heating effect of nitrate salts**

$MNO_3 \longrightarrow KNO_2 + \frac{1}{2}O_2$ [M = Na, K, Rb, Cs]

$2LiNO_3 \longrightarrow Li_2O + 2NO_2 + \frac{1}{2}O_2$

$2M(NO_3)_2 \longrightarrow 2MO + 4NO_2 + O_2$

[M = all bivalent metal's ions eg. $Zn^{+2}$, $Mg^{+2}$, $Sr^{+2}$, $Ca^{+2}$, $Ba^{+2}$, $Cu^{+2}$, $Pb^{+2}$]

$Hg(NO_3)_2 \longrightarrow Hg + 2NO_2 + O_2$ ; $2AgNO_3 \longrightarrow 2Ag + 2NO_2 + O_2$

**Heating effect of Halides salts**

$2FeCl_3 \longrightarrow 2FeCl_2 + Cl_2$ ; $AuCl_3 \longrightarrow AuCl + Cl_2$

$Hg_2Cl_2 \longrightarrow HgCl_2 + Hg$ ; $NH_4Cl \longrightarrow NH_3 + HCl$

$PbX_4 \longrightarrow PbX_2 + X_2$ [ X = Cl, Br, $SCN^-$ ]

**Heating effect of hydrated chloride salts**

$MgCl_2 . 6H_2O \xrightarrow{\Delta} MgO + 2HCl + 5H_2O$

$2FeCl_3 . 6H_2O \longrightarrow Fe_2O_3 + 6HCl + 9H_2O$

$2AlCl_3 . 6H_2O \longrightarrow Al_2O_3 + 6HCl + 9H_2O$

$\underset{\text{Pink}}{CoCl_2.6H_2O} \xrightarrow[-2H_2O]{50°C} \underset{\text{Pink}}{CoCl_2.4H_2O} \xrightarrow[-2H_2O]{58°C} \underset{\text{Red violet}}{CoCl_2.2H_2O} \xrightarrow[-2H_2O]{140°C} \underset{\text{Blue}}{CoCl_2}$

**Heating effect of hydrated Sulphate salts**

$\underset{\text{Blue vitriol}}{CuSO_4.5H_2O} \xrightarrow[-4H_2O]{100°C} \underset{\text{Bluish White}}{CuSO_4.H_2O} \xrightarrow[-H_2O]{220°C} \underset{\text{White}}{CuSO_4} \xrightarrow{>800°C} CuO + SO_2 + \frac{1}{2}O_2$

$\underset{\text{White}}{CuSO_4} \xrightarrow{750°C} \underset{\text{Black}}{CuO} + SO_3$

$\underset{\text{Green Vitriol}}{FeSO_4.7H_2O} \xrightarrow[-7H_2O]{300°C} FeSO_4 \xrightarrow{\Delta} Fe_2SO_3 + SO_2 + SO_3$ (very important)

$Fe_2(SO_4)_3 \xrightarrow{\Delta} Fe_2O_3 + 3SO_3$

$\underset{\text{epsom salt}}{MgSO_4.7H_2O} \xrightarrow[-7H_2O]{\Delta} MgSO_4 \downarrow$ [Same as $ZnSO_4$ ]

$\underset{\text{gypsum}}{CaSO_4.2H_2O} \xrightarrow{120°C} (CaSO_4.\frac{1}{2}H_2O) + 1\frac{1}{2}H_2O$

Plaster of pairs

$CaSO_4.2H_2O \xleftarrow{+1\frac{1}{2}H_2O}$ Plaster of pairs $\xrightarrow{\Delta} \underset{\text{Dead burnt}}{CaSO_4} + \frac{1}{2}H_2O$

$Na_2S_2O_3.5H_2O \xrightarrow{220°C} Na_2S_2O_3 + 5H_2O$

$Na_2S_2O_3 \xrightarrow[\text{(From 4 moles)}]{\Delta} 3Na_2SO_4 + Na_2S_5.$

$ZnSO_4.7H_2O \xrightarrow{70°C} ZnSO_4.6H_2O \xrightarrow{70°–200°C} ZnSO_4.H_2O \xrightarrow{>200°C} ZnSO_4 \xrightarrow[\Delta]{>800°C} ZnO + SO_2 + \frac{1}{2}O_2$

$2NaHSO_3 \xrightarrow{\Delta} Na_2S_2O_5 + H_2O$ ; $2NaHSO_3 \xrightarrow{\Delta} Na_2SO_3 + SO_2$, $Na_2SO_3 \longrightarrow Na_2S + Na_2SO_4$

$2NaHSO_4 \xrightarrow{\Delta} Na_2SO_4 + H_2O + SO_3$

**Heating effect of Oxide salts**

$2Ag_2O \xrightarrow[300°C]{\Delta} 4Ag + O_2$ ; $\underset{\text{yellow}}{HgO} \xrightarrow[400°C]{\Delta} \underset{\text{red}}{HgO} \xrightarrow[\text{strong heating}]{\Delta} Hg + \frac{1}{2}O_2$

$PbO_2 \xrightarrow{\Delta} PbO + \frac{1}{2}O_2$ ; $\underset{\text{Red lead}}{Pb_3O_4} \underset{340°C}{\overset{>470°C}{\rightleftharpoons}} 6PbO + O_2$ Litharge

$3MnO_2 \xrightarrow{900°C} Mn_3O_4 + O_2$ ; $\underset{\text{white}}{ZnO} \underset{\text{Cold}}{\overset{\text{Hot}}{\rightleftharpoons}} \underset{\text{yellow}}{ZnO}$

$2CrO_5 \longrightarrow Cr_2O_3 + \frac{1}{2}O_2$ ; $2CrO_3 \xrightarrow{420°C} Cr_2O_3 + \frac{3}{2}O_2$

$I_2O_5 \longrightarrow I_2 + \frac{5}{2}O_2$

**Heating effect of dichromate & chromate salts**

$(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} N_2 + Cr_2O_3 + 4H_2O$; $K_2Cr_2O_7 \xrightarrow{\Delta} 2K_2CrO_4 + Cr_2O_3 + \frac{7}{2}O_2$

**Heating effect of phosphate salts**

$\underset{1^o \text{ phosphate salt}}{NaH_2PO_4} \xrightarrow{\Delta} H_2O + NaPO_3$; $\underset{2^o \text{ phosphate salt}}{Na_2HPO_4} \xrightarrow{\Delta} H_2O + Na_2P_2O_7$

$\underset{3^o \text{ phosphate salt}}{Na_3PO_4} \xrightarrow{\Delta}$ No effect

$Na(NH_4)HPO_4.4H_2O \xrightarrow[-4H_2O]{\Delta} NaNH_4HPO_4 \xrightarrow{\text{High temp.}} NaPO_3 + NH_3 + H_2O$

$2Mg(NH_4)PO_4 \xrightarrow{\Delta} Mg_2P_2O_7 + 2NH_3 + H_2O$

***Heating effects of acetate, formate, oxalate salts***

$CH_3CO_2K \xrightarrow{\Delta} K_2CO_3 + CH_3COCH_3$

$Pb(OAc)_2 \xrightarrow{\Delta} PbO + CO_2 + CH_3COCH_3$

$Mg(OAc)_2 \longrightarrow MgO + CO_2 + CH_3COCH_3$

$Be(OAc)_2 \longrightarrow BeO + CO_2 + CH_3COCH_3$

$Ca(OAc)_2 \longrightarrow CaCO_3 + CH_3COCH_3$

$Ba(OAc)_2 \longrightarrow BaCO_3 + CH_3COCH_3$

$HCO_2Na \xrightarrow{350°C} Na_2C_2O_4 + H_2\uparrow$

$2HCOOAg \xrightarrow{\Delta} HCOOH + 2Ag + CO_2$

$Na_2C_2O_4 \longrightarrow Na_2CO_3 + CO$

$SnC_2O_4 \xrightarrow{\Delta} SnO + CO_2 + CO$

$FeC_2O_4 \longrightarrow FeO + CO + CO_2$

$Ag_2C_2O_4 \xrightarrow{\Delta} 2Ag + 2CO_2$

$HgC_2O_4 \longrightarrow Hg + 2CO_2$

$(HCOO)_2Hg \longrightarrow HCOOH + Hg + CO_2$

**Heating effect of Acids**

$2HNO_3 \xrightarrow{\Delta} H_2O + 2NO_2 + \frac{2}{4}O_2$

$H_2SO_4 \xrightarrow{444°C} H_2O + SO_3$;

$H_2SO_4 \xrightarrow{>800°C} H_2O + SO_2 + \frac{1}{2}O_2$

$3H_2SO_3 \longrightarrow 2H_2SO_4 + S\downarrow + H_2O$

$3HNO_2 \longrightarrow HNO_3 + 2NO + H_2O$

$HClO_3 \longrightarrow HClO_4 + ClO_2 + H_2O$

$3HOCl \longrightarrow 2HCl + HClO_3$

$4H_3PO_3 \xrightarrow[200°C]{\Delta} 3H_3PO_4 + PH_3$

$2H_3PO_2 \longrightarrow H_3PO_4 + PH_3$

$2NaH_2PO_2 \longrightarrow Na_2HPO_4 + PH_3$

} Undergoes disproportionation reaction

$H_3PO_4 \xrightarrow{220°C} H_4P_2O_7 \xrightarrow{320°C} 4HPO_3 \xrightarrow{>320°C} 2P_2O_5 + 2H_2O$

$$H_3BO_3 \xrightarrow{100°C} 4HBO_2 \xrightarrow{140°C} H_2B_4O_7 \xrightarrow[hot]{Red} 2B_2O_3 + H_2O$$

$$H_2C_2O_4 \xrightarrow{\Delta} H_2O + CO + CO_2$$

***Some reactions of important oxidising agents***

**(I)** ***Potassium dichromate ($K_2Cr_2O_7$) :***

$Cr_2O_7^{2-}$ ion takes electrons in the acidic medium and is reduced to $Cr^{3+}$ ion. Thus $Cr_2O_7^{2-}$ acts as an oxidising agent in acidic medium.

$$K_2Cr_2O_7 + 4H_2SO_4(dil.) \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 4H_2O + 3O$$

or $$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$$

(1) $$K_2Cr_2O_7 + 7H_2SO_4 + 6KI \longrightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3I_2$$

or $$Cr_2O_7^{2-} + 14H^+ + 6I^- \longrightarrow 2Cr^{3+} + 7H_2O + 3I_2$$

(2) $$K_2Cr_2O_7 + 7H_2SO_4 + 6FeSO_4 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3Fe_2(SO_4)_3$$

or $$Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \longrightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$$

(3) $$K_2Cr_2O_7 + 4H_2SO_4 + 3H_2S \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 7H_2O + 3S$$

or $$Cr_2O_7^{2-} + 8H^+ + 3H_2S \longrightarrow 2Cr^{3+} + 7H_2O + 3S$$

(4) $$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$$

or $$Cr_2O_7^{2-} + 2H^+ + 3SO_2 \longrightarrow 2Cr^{3+} + 3MnO_4^- + H_2O$$

(5) $$K_2Cr_2O_7 + 4H_2SO_4 + 3Na_2SO_3 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3Na_2SO_3 + 4H_2O$$

or $$Cr_2O_7^{2-} + 8H^+ + 3SO_3^{2-} \longrightarrow 2Cr^{3+} + 3SO_4^{2-} + 4H_2O$$

(6) $$K_2Cr_2O_7 + 14HCl \longrightarrow 2KCl + 2CrCl_3 + 7H_2O + 3Cl_2$$

or $$Cr_2O_7^{2-} + 14H^+ + 6Cl^- \longrightarrow 2Cr^{3+} + 7H_2O + 3Cl_2$$

(7) $$K_2Cr_2O_7 + H_2SO_4 + 4H_2O_2 \xrightarrow{ether} K_2SO_4 + 2CrO_5 + 5H_2O$$

or $$Cr_2O_7^{2-} + 2H^+ + 4H_2O_2 \xrightarrow{ether} 2CrO_5 + 5H_2O$$

**(II)** ***Manganese dioxide ($MnO_2$) :***

In presence of excess of $H^+$ ions, $MnO_2$ acts as a stronge oxidising agent. In showing this behaviour $Mn^{4+}$ changes to $Mn^{2+}$ ion.

$$MnO_2 + 4H^+ + 4e^- \longrightarrow Mn^{2+} + 2H_2O$$

(1) $$MnO_2 + 4H^+ + C_2O_4^- \longrightarrow Mn^{2+} + 2H_2O + 2CO_2$$

(2) $$MnO_2 + 4H^+ + 2Fe^{2+} \longrightarrow Mn^{2+} + 2H_2O + 2Fe^{3+}$$

(3) $$MnO_2 + 4H^+ + 2Cl^- \longrightarrow Mn^{2+} + 2H_2O + Cl_2$$

**(III)** ***Potassium permangate ($KMnO_4$) :***

**(A)** **In acidic medium:** The reduction of $MnO_4^-$ ion into $Mn^{2+}$ ion san be represented by the following ionic equation :

$$2MnO_4^- + 4H^+ \longrightarrow Mn^{2+} + 2H_2O + 5O$$

or $$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$$

*(1)* *KI to $I_2$* ($I^- \rightarrow I_2$)

$$2KMnO_4 + 8H_2SO_4 + 10KI \longrightarrow 6K_2SO_4 + 2MnSO_4 + 8H_2O + 5I_2$$

or $$2MnO_4^- + 16H^+ + 10I^- \longrightarrow 2Mn^{2+} + 8H_2O + 5I_2$$

*(2)* *Ferrous salts to ferric salts* ($Fe^{2+} \rightarrow Fe^{3+}$)

$$2KMnO_4 + 8H_2SO_4 + 10FeSO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5Fe_2(SO_4)_3$$

or $$MnO_4^- + 8H^+ + 5Fe^{2+} \longrightarrow 2Mn^{2+} + 4H_2O + 5Fe^{3+}$$

*(3)* *Oxalic acid* ($H_2C_2O_4$) *or oxalate* ($C_2O_4^{2-}$) *to* $CO_2$

$$2KMnO_4 + 8H_2SO_4 + 5H_2C_2O_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 10CO_2$$

or $$2\,MnO_4^- + 16H^+ + 5C_2O_4^- \longrightarrow 2Mn^{2+} + 8H_2O + 5CO_2$$

*(4)* *$H_2S$ to S* ($S^{2-} \rightarrow S°$)

$$2KMnO_4 + 3H_2SO_4 + 5H_2S \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5S$$

or $$2\,MnO_4^- + 16H^+ + 5S^{2-} \longrightarrow 2Mn^{2+} + 8H_2O + 5S$$

*(5)* *Nitrite to nitrate* ($NO_2^- \rightarrow NO_3^-$)

$$2KMnO_4 + 3H_2SO_4 + 5KNO_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5KNO_3$$

or $$2\,MnO_4^- + 6H^+ + 5NO_2^- \longrightarrow 2Mn^{2+} + 3H_2O + 5NO_3^-$$

*(6)* *Arsenite $AsO_3^{3-}$* (As = +3) *to arsenate, $AsO_4^{3-}$* (As = +5)

$$2KMnO_4 + 3H_2SO_4 + 5Na_3AsO_3 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5Na_3AsO_4$$

or $$2\,MnO_4^- + 6H^+ + 5AsO_3^{3-} \longrightarrow 2Mn^{2+} + 3H_2O + 5AsO_4^{3-}$$

*(7)* *Sulphite, $SO_3^{2-}$ (S = +4) to sulphate, $SO_4^{2-}$* (S = +6)

$$2KMnO_4 + 3H_2SO_4 + 5Na_2SO_3 \longrightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5Na_2SO_4$$

or $$2\,MnO_4^- + 6H^+ + 5SO_3^{2-} \longrightarrow 2Mn^{2+} + 3H_2O + 5SO_4^{2-}$$

*(8)* *Halogen acid, HX* (X = –1) *to the corresponding halogen,* $X_2$(X = 0) ($X^- \rightarrow X_2$)

$$2KMnO_4 + 3H_2SO_4 + 10HCl \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5Cl_2$$

or $$2\,MnO_4^- + 16H^+ + 10Cl^- \longrightarrow 2Mn^{2+} + 3H_2O + 5Cl_2$$

*(9)* *$H_2O_2$* (O = –1) *to $O_2$* (O = 0)

$$2KMnO_4 + 3H_2SO_4 + 5H_2O_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5O_2$$

or $$2\,MnO_4^- + 6H^+ + 5H_2O_2 \longrightarrow 2Mn^{2+} + 8H_2O + 5H_2O$$

*(10)* *$SO_2$* (S = +4) *to $H_2SO_4$* (S = +6) ($SO_2 \rightarrow SO_4^{2-}$)

$$2KMnO_4 + 2H_2O + 5SO_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 2H_2SO_4$$

or $$2\,MnO_4^- + 2H_2O + 5SO_2 \longrightarrow 2Mn^{2+} + 5SO_4^{2-}$$

*(11)* *Hydrazine, $N_2H_4$* (N = –2) *to $N_2$*(N = 0)

$$4KMnO_4 + 6H_2SO_4 + 5N_2H_4 \longrightarrow 2K_2SO_4 + 4MnSO_4 + 16H_2O + 5N_2$$

or $$4\,MnO_4^- + 12H^+ + 5N_2H_4 \longrightarrow 2Mn^{2+} + 16H_2O + 5N_2$$

*(12)* *Hydrazoic acid, $HN_3$* (N = –1/3) *to $N_2$*(N = 0)

$$2KMnO_4 + 3H_2SO_4 + 10HN_3 \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 15N_2$$

or $$2\,MnO_4^- + 6H^+ + 5HN_3 \longrightarrow 2Mn^{2+} + 8H_2O + 15N_2$$

*(13)* *Nitric oxide, NO* (N = +2) *to $HNO_3$*(N = +5)

$$6KMnO_4 + 9H_2SO_4 + 10NO \longrightarrow 3K_2SO_4 + 6MnSO_4 + 4H_2O + 10HNO_3$$

or $$3\,MnO_4^- + 9H^+ + 5NO \longrightarrow 3Mn^{2+} + 2H_2O + 5HNO_3$$

*(14)* *Potassium ferrocyanide,* $K_4[Fe(CN)_6]$ *to potassium ferricyanide,* $K_3[Fe(CN)_6]$

$$2KMnO_4 + 3H_2SO_4 + 10K_4[Fe(CN)_6] \longrightarrow K_2SO_4 + 2MnSO_4 + 2H_2O + 10K_3[Fe(CN)_6] + 10KOH$$

or $$MnO_4^- + 8H^+ + 5[Fe(CN)_6]^{4-} \longrightarrow Mn^{2+} + 4H_2O + 5[Fe(CN)_6]^{3-}$$

*(15)* *Sodium thiosulphate,* $Na_2S_2O_3$ (S = +3) *to sodium dithionate,* $Na_2S_2O_6$ (S = +5)

$$6KMnO_4 + 9H_2SO_4 + 5Na_2S_2O_3 \longrightarrow 3K_2SO_4 + 6MnSO_4 + 9H_2O + 5Na_2S_2O_6$$

or $$6\,MnO_4^- + 18H^+ + 5S_2O_3^{2-} \longrightarrow 6Mn^{2+} + 9H_2O + 5S_2O_6^{2-}$$

**(B)** **In alkaline medium:** In alkaline solution $MnO_4^-$ ion is reduced to colourless & insoluble $MnO_2$ according to the following equations:

$$2\,MnO_4^- + H_2O \xrightarrow{\text{alkali}} 2MnO_2 + 2OH^- + 3O$$

or $$MnO_4^- + 2H_2O + 3e^- \longrightarrow MnO_2 + 4OH^-$$

*(1)* *Iodides($I^-$) to iodates ($IO_3^-$)*

$$2KMnO_4 + H_2O + KI \longrightarrow 2MnO_2 + 2KOH + KIO_3$$

or $$2\,MnO_4^- + H_2O + I^- \longrightarrow 2MnO_2 + 2OH^- + IO_3^-$$

*(2)* *$NH_3$(N = –3) to $N_2$ (N = 0)*

$$2KMnO_4 + 2NH_3 \longrightarrow 2MnO_2 + 2KOH + N_2 + 2\,H_2O$$

or $$2\,MnO_4^- + 2NH_3 \longrightarrow 2H_2O + 2OH^- + N_2 + 2\,MnO_2$$

*(3)* *Nitrotoluene to nitrobenzoic acid*

$$2KMnO_4 + C_6H_4(CH_3)(NO_2) \longrightarrow 2MnO_2 + 2KOH + C_6H_4(COOH)(NO_2)$$

Nitrotoluene — Nitrobenzoic acid

or $$2\,MnO_4^- + C_6H_4(CH_3)(NO_2) \longrightarrow MnO_2 + C_6H_4(COOH)(NO_2) + OH^-$$

*(4)* *Ethylene ($H_2C = CH_2$) to ethylene glycol ($HO–CH_2–CH_2–OH$)*

$$2KMnO_4 + 4H_2O + 3H_2C{=}CH_2 \longrightarrow 2MnO_2 + 2KOH + 3HO–CH_2–CH_2–OH$$

or $$2\,MnO_4^- + 4H_2O + 3H_2C{=}CH_2 \longrightarrow 2MnO_2 + 2OH^- + 3HO–CH_2–CH_2–OH$$

**(C)** **In neutral medium:** In neutral solution, $KMnO_4$ is directly reduced to $MnO_2$

$$2KMnO_4 + H_2O \longrightarrow 2KOH + 2MnO_2 + 3O$$

or $$2\,MnO_4^- + H_2O \longrightarrow 2OH^- + 2MnO_2 + 3O$$

or $$MnO_4^- + 2H_2O + 3e^- \longrightarrow 2MnO_2 + 4OH^-$$

*(1)* *Manganous salt (e.g. $MnSO_4$) to insoluble $MnO_2$ ($Mn^{2+} \longrightarrow Mn^{4+}O_2$)*

$$2KMnO_4 + 4H_2O + 3MnSO_4 + H_2SO_4 \longrightarrow 5MnO_2 + 3H_2SO_4 + K_2SO_4 + 2H_2O$$

or $$2\,MnO_4^- + 10H_2O + 3Mn^{2+} \longrightarrow 5MnO_2 + 8H_2O + 4H^+$$

*(2)* *Sodium thiosulphate,* $Na_2S_2O_3$ (S = +2) *to* $Na_2SO_4$ (S = +6) [$S_2O_4^{2-} \longrightarrow SO_4^{2-}$]

$$8KMnO_4 + H_2O + 3Na_2S_2O_3 \longrightarrow 8MnO_2 + 2KOH + 3Na_2SO_4 + 3K_2SO_4$$

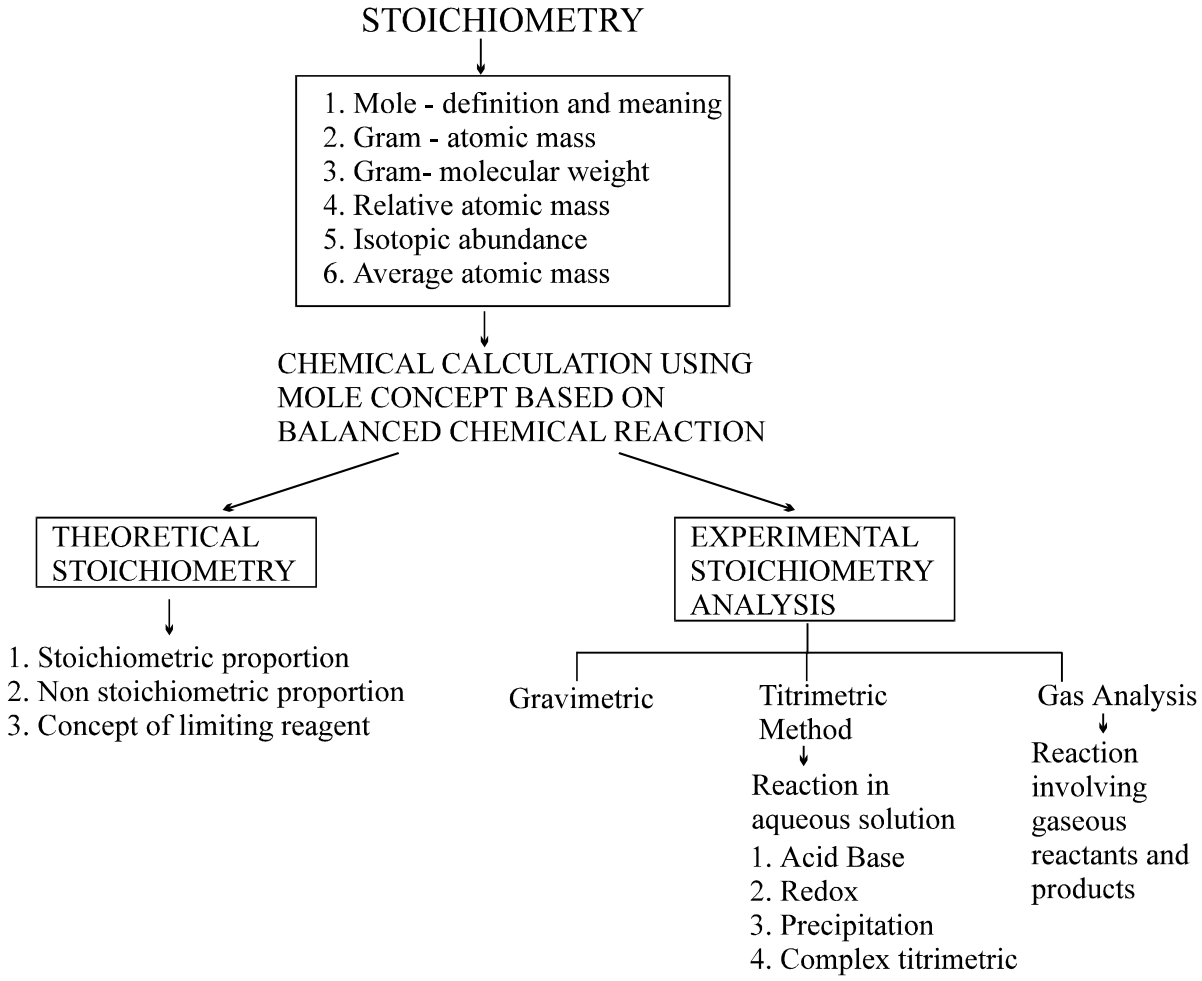
or $$8\,MnO_4^- + H_2O + 3S_2O_3^{2-} \longrightarrow 8MnO_2 + 2OH^- + 6SO_4^{2-}$$

*(3)* *Nitrogen dioxide,* $NO_2$ (N = +4) *to* $HNO_3$ (N = +5) [$NO_2 \longrightarrow NO_3^-$]

$$2KMnO_4 + 4H_2O + 6NO_2 \longrightarrow 2KOH + 2MnO_2 + 2MnO_2 + 6HNO_3$$

or $$MnO_4^- + H_2O + 3NO_2 \longrightarrow MnO_2 + 3NO_3^- + 2H^+$$

## ***THE ATLAS***


STOICHIOMETRY
1. Mole - definition and meaning
2. Gram - atomic mass
3. Gram- molecular weight
4. Relative atomic mass
5. Isotopic abundance
6. Average atomic mass
CHEMICAL CALCULATION USING MOLE CONCEPT BASED ON BALANCED CHEMICAL REACTION
THEORETICAL STOICHIOMETRY
1. Stoichiometric proportion
2. Non stoichiometric proportion
3. Concept of limiting reagent
EXPERIMENTAL STOICHIOMETRY ANALYSIS
Gravimetric
Titrimetric Method
Reaction in aqueous solution
1. Acid Base
2. Redox
3. Precipitation
4. Complex titrimetric
Gas Analysis
Reaction involving gaseous reactants and products

# *GLOSSARY*

**Aliquot.** A portion of the whole, usually a simple fraction. A portion of a sample withdraw from a volumetric flask with a pipet is called an aliquot.

**Analytical concentration.** The total number of moles per litre of a solute regardless of any reactions that might occur when the solute dissolves. Used synonymously with formality.

**Equivalent.** The amount of a substance which furnishes or reacts with 1 mol of $H^+$ (acid-base), 1 mol of electrons (redox), or 1 mol of a univalent cation (precipitation and complex formation).

**Equivalent weight.** The weight in grams of one equivalent of a substance.

**Equivalence point.** The point in a titration where the number of equivalents of titrant is the same as the number of equivalents of analyte.

**End point.** The point in a titration where an indicator changes color.

**Formula weight.** The number of formula weights of all the atoms in the chemical formula of a substance.

**Formality.** The number of formula weights of solute per litre of solution; synonymous with analytical concentration.

**Indicator.** A chemical substance which exhibits different colors in the presence of excess analyte or titrant.

**Normality.** The number of equivalents of solute per litre of solution.

**Primary standard.** A substance available in a pure form or state of known purity which is used in standardizing a solution.

**Standardization.** The process by which the concentration of a solution is accurately ascertained.

**Standard solution.** A solution whose concentration has been accurately determined.

**Titrant.** The reagent (a standard solution) which is added from a buret to react with the analyte.

# ***EASY RIDE***

## ***Acid Base Titration***

Q1. A small amount of $CaCO_3$ completely neutralized 52.5 mL of N/10 HCl and no acid is left at the end. After converting all calcium chloride to $CaSO_4$, how much plaster of paris can be obtained?

Q2. How many ml of 0.1 N HCl are required to react completely with 1 g mixture of $Na_2CO_3$ and $NaHCO_3$ containing equimolar amounts of two?

Q3. 10 g $CaCO_3$ were dissolved in 250 ml of M HCl and the solution was boiled. What volume of 2 M KOH would be required to equivalence point after boiling? Assume no change in volume during boiling.

Q4. 125 mL of a solution of tribasic acid (molecular weight = 210) was neutralized by 118mL of decinormal NaOH solution and the trisodium salt was formed. Calculate the concentration of the acid in grams per litre.

Q5. Upon heating one litre of N/2 HCl solution, 2.675g of hydrogen chloride is lost and the volume of solution shrinks to 750 ml. Calculate (i) the normality of the resultant solution (ii) the number of milli-equivalents of HCl in 100 mL of the original solution.

Q6. For the standardization of a $Ba(OH)_2$ solution, 0.2g of potassium acid phthalate (m.wt. 204.2g) weighed which was then titrated with $Ba(OH)_2$ solution. The titration requires 27.80mL $Ba(OH)_2$ solution. What is the molarity of base? The reaction products include $BaC_8H_4O_4$ as only Ba containing species.

Q7. A definite amount of $NH_4Cl$ was boiled with 100mL of 0.8N NaOH for complete reaction. After the reaction, the reactant mixture containing excess of NaOH was neutralized with 12.5mL of 0.75N $H_2SO_4$. Calculate the amount of $NH_4Cl$ taken.

Q8. $H_3PO_4$ is a tri basic acid and one of its salt is $NaH_2PO_4$. What volume of 1 M NaOH solution should be added to 12 g of $NaH_2PO_4$ to convert it into $Na_3PO_4$?

Q9. Calculate the number of gm. of borax, $Na_2B_4O_7.10H_2O$, per litre of a solution of which 25cc required 15.6 cc of N/10 hydrochloric acid for naturalization, methyl orange being used as indicator. In aqueous solution, borax hydrolyses according to the equation:

$$Na_2B_4O_7 + 7H_2O = 2NaOH + 4H_3BO_3$$

The liberated boric acid is a weak acid and is without effect on methyl orange.

Q10. 25mL of a solution of $Na_2CO_3$ having a specific gravity of 1.25g $ml^{-1}$ required 32.9 mL of a solution of HCl containing 109.5g of the acid per litre for complete neutralization. Calculate the volume of 0.84N $H_2SO_4$ that will be completely neutralized by 125g of $Na_2CO_3$ solution.

Q11. A solution containing 4.2 g of KOH and $Ca(OH)_2$ is neutralized by an acid. It consumes 0.1 equivalent of acid, calculate the percentage composition of the sample.

Q12. 5gm of a double sulphate of iron and ammonia was boiled with an excess of sodium hydroxide solution and the liberated ammonia was passed into 50cc of normal sulphuric acid. The excess of acid was found to require 24.5cc of normal sodium hydroxide for naturalization. Calculate the percentage of ammonia (expressed as $NH_3$) in the double salt.

Q13. 0.5 g of fuming $H_2SO_4$ (oleum) is diluted with water. The solution requires 26.7 ml of 0.4 N NaOH for complete neutralization. Find the % of free $SO_3$ in the sample of oleum.

Q14. 1.64 g of a mixture of $CaCO_3$ and $MgCO_3$ was dissolved in 50 mL of 0.8 M HCl. The excess of acid required 16 mL of 0.25 M NaOH for neutralization. Calculate the percentage of $CaCO_3$ and $MgCO_3$ in the sample.

Q15. 1.5 g of chalk were treated with 10 ml of 4N – HCl. The chalk was dissolved and the solution made to 100 ml 25 ml of this solution required 18.75 ml of 0.2 N – NaOH solution for complete neutralisation. Calculate the percentage of pure $CaCO_3$ in the sample of chalk?

Q16. 2.013g of a commercial sample of NaOH containing $Na_2CO_3$ as an impurity was dissolved to give 250ml solution. A 10ml portion of this solution required 20ml of 0.1N $H_2SO_4$ for complete neutralization. Calculate % by weight of $Na_2CO_3$.

Q17. Exactly 50 ml of $Na_2CO_3$ solution is equivalent to 56.3 ml of 0.102 N HCl in an acid-base neutralisation. How many gram $CaCO_3$ would be precipitated if an excess of $CaCl_2$ solution were added to 100 ml of this $Na_2CO_3$ solution.

Q18. 6g mixture of $NH_4Cl$ and NaCl is treated with 110mL of a solution of caustic soda of 0.63N. The solution was then boiled to remove $NH_3$. The resulting solution required 48.1mL of a solution of 0.1N HCl. What is % composition of mixture?

Q19. Calculate the number of gm(a) of hydrochloric acid, (b) of potassium chloride in 1 litre of a solution, 25cc of which required 21.9cc of N/10 sodium hydroxide for naturalization and another 25cc after the addition of an excess of powdered chalk, required 45.3cc of N/10 silver nitrate for the complete precipitation of the chloride ion.

Q20. 2.5 gm of a mixture containing $NaHCO_3$, $Na_2CO_3$ and NaCl is dissolved in 100 ml water and its 50 ml portion required 13.33 ml 1.0 N HCl solution to reach the equivalence point. On the other hand its other 50 ml portion required 19 ml 0.25 M NaOH solution to reach the equivalence point. Determine mass % of each component? ($Na_2CO_3$ = 36.38%, $NaHCO_3$ = 31.92%, NaCl = 31.7%)

## ***Redox Titration***

Q21. It requires 40.05 ml of 1M $Ce^{4+}$ to titrate 20ml of 1M $Sn^{2+}$ to $Sn^{4+}$. What is the oxidation state of the cerium in the product.

Q22. A volume of 12.53 ml of 0.05093 M $SeO_2$ reacted with exactly 25.52 ml of 0.1M $CrSO_4$. In the reaction, $Cr^{2+}$ was oxidized to $Cr^{3+}$. To what oxidation state was selenium converted by the reaction.

Q23. A 1.0g sample of $H_2O_2$ solution containing x % $H_2O_2$ by mass requires *x* $cm^3$ of a $KMnO_4$ solution for complete oxidation under acidic conditions. Calculate the normality of $KMnO_4$ solution.

Q24. Metallic tin in the presence of HCI is oxidized by $K_2Cr_2O_7$ to stannic chloride, $SnCl_4$. What volume of deci-normal dichromate solution would be reduced by 1g of tin.

Q25. Calculate the mass of oxalic acid which can be oxidized by 100ml of M $MnO_4^-$ solution, 10ml of which is capable of oxidizing 50ml of 1N $I^-$ of $I_2$.

Q26. Exactly 40ml of an acidified solution of 0.4M iron(II) ion of titrated with $KMnO_4$ solution. After addition of 32ml $KMnO_4$, one additional drop turns the iron solution purple. Calculate the concentration of permangnate solution.

Q27. The iodide content of a solution was determined by the titration with Cerium(IV) sulfate in the presence of HCl, in which $I^-$ is converted to ICl. A 250ml sample of the solution required 20ml of 0.058N $Ce^{4+}$ solution. What is the iodide concentration in the original solution in gm/lt.

Q28. Potassium acid oxalate $K_2C_2O_4 \cdot 3HC_2O_4 \cdot 4H_2O$ can be oxidized by $MnO_4^-$ in acid medium. Calculate the volume of 0.1M $KMnO_4$ reacting in acid solution with one gram of the acid oxalate.

Q29. 5g sample of brass was dissolved in one litre dil. $H_2SO_4$. 20 ml of this solution were mixed with KI, liberating $I_2$ and $Cu^+$ and the $I_2$ required 20 ml of 0.0327 N hypo solution for complete titration. Calculate the percentage of Cu in the alloy.

Q30. 1.44g pure $FeC_2O_4$ was dissolved in dil. HCl and solution diluted to 100 mL. Calculate volume of 0.01M $KMnO_4$ required to oxidize $FeC_2O_4$ solution completely.

Q31. 0.84 g iron ore containing x percent of iron was taken in a solution containing all the iron in ferrous condition. The solution required x ml of a dichromatic solution for oxidizing the iron content to ferric state. Calculate the strength of dichromatic solution.

Q32. 0.5M $KMnO_4$ solution completely reacts with 0.05M $FeC_2O_4$ solution under acidic conditions where the products are $Fe^{3+}$, $CO_2$ and $Mn^{2+}$. The volume of $FeC_2O_4$ used is 125 ml. What volume of $KMnO_4$ was used.

Q33. A solution is made by mixing 200 ml of 0.1M $FeSO_4$, 200 gm of 0.1M $KMnO_4$ and 600 ml 1M $HClO_4$. A reaction occurs in which $Fe^{2+}$ is converted to $Fe^{3+}$ & $MnO_4^-$ to $Mn^{2+}$ in acid solution. Calculate the concentration of each ion.

Q34. To 100ml of $KMnO_4$ solution containing 0.632 gm of $KMnO_4$, 200 ml of $SnCl_2$ solution containing 2.371 gm is added in presence of HCl. To the resulting solution excess of $HgCl_2$ solution is added all at once. How many gms of $Hg_2Cl_2$ will be precipitated. (Mn = 55; K = 39; Sn = 118.7; Hg = 201)

Q35. A 1.0 g sample of $Fe_2O_3$ solid of 55.2% purity is dissolved in acid and reduced by heating the solution with zinc dust. The resultant solution is cooled and made upto 100.0 mL. An aliquot of 25.0 mL of this solution requires 17.0 mL of 0.0167 M solution of an oxidant for titration. Calculate the number of moles of electrons taken up by the oxidant in the reaction of the above titration.

Q36. A mixture of FeO and $Fe_2O_3$ is reacted with acidified $KMnO_4$ solution having a concentration of 0.2278 M, 100 ml of which was used. The solution was then titrated with Zn dust which converted $Fe^{3+}$ of the solution to $Fe^{2+}$. The $Fe^{2+}$ required 1000 ml of 0.13 M $K_2Cr_2O_7$ solution. Find the % of FeO & $Fe_2O_3$

Q37. 2 gms of $FeC_2O_4$ are made to react in acid solution with 0.25 M $KMnO_4$ solution. What volume of $KMnO_4$ solution would be required. The resulting solution is treated with excess of $NH_4Cl$ and $NH_4OH$ solution. The precipitated $Fe(OH)_3$ is filtered off, washed and ignited. What is the mass of the product obtained. (Fe = 56 )

Q38. The neutralization of a solution of 1.2 g of a substance containing a mixture of $H_2C_2O_4.2H_2O$, $KHC_2O_4.H_2O$ and different impurities of a neutral salt consumed 18.9 ml of 0.5 N NaOH solution. On titration with $KMnO_4$ solution, 0.4 g of the same substance needed 21.55 ml of 0.25 N $KMnO_4$. Calculate the % composition of the substance.

Q39. A 1.0 g sample containing $BaCl_2 . 2H_2O$ was dissolved and an excess of $K_2CrO_4$ solution added. After a suitable period, the $BaCrO_4$ was filtered, washed and redissolved in HCl to convert $CrO_4^{2-}$ to $Cr_2O_7^{2}$. An excess of KI was added, and the liberated iodine was titrated with 84.7 mL of 0.137 M sodium thiosulphate. Calculate the percent purity of $BaCl_2 . 2H_2O$.

Q40. A sample of Mg was burnt in air to give a mix of MgO and $Mg_3N_2$. The ash was dissolved in 60meq HCl and the resulting solution was back titrated with NaOH. 12 meq of NaOH were required to reach end point. An excess of NaOH was then added and the solution distilled. The $NH_3$ released was then trapped in 10 meq of second acid solution. Back titration of this solution required 6 meq of the base. Calculate the % of Mg burnt to the nitride.

## ***Double titration***

Q41. A solution contains $Na_2CO_3$ and $NaHCO_3$. 20ml of this solution required 4ml of 1N – HCl for titration with Ph indicator. The titration was repeated with the same volume of the solution but with MeOH. 10.5 ml of 1 – N HCl was required this time. Calculate the amount of $Na_2CO_3$ & $NaHCO_3$.

Q42. A solution contains a mix of $Na_2CO_3$ and NaOH. Using Ph as indicator 25ml of mix required 19.5 ml of 0.995 N HCl for the end point. With MeOH, 25 ml of the solution required 25ml of the same HCl for the end point. Calculate gms/L of each substance in the mix .

Q43. 200ml of a solution of mixture of NaOH and $Na_2CO_3$ was first titrated with Ph and $\frac{N}{10}$ HCl. 17.5 ml of HCl was required for end point. After this MeOH was added and 2.5 ml of some HCl was again required for next end point. Find out amounts of NaOH and $Na_2CO_3$ in the mix.

Q44. What is the concentration of a solution of orthophosphoric acid(gm $H_3PO_4$ per litre), 25cc of which required 18.8cc of N sodium hydroxide for naturalization in the presence of phenolphthalein as indicator?

Q45. 2gm of mixture of hydrated sodium carbonate $Na_2CO_3 . 10H_2O$, and sodium bicarbonate was dissolved in water and made up to 250 cc. 25 cc of this solution was titrated, using methyl orange as indicator, and 22.5cc of 0.087N HCl were required for naturalization. Calculate the percentage of sodium bicarbonate in the mixture.

Q46. A solution contains $Na_2CO_3$ and $NaHCO_3$. 10ml of this requires 2ml of 0.1M $H_2SO_4$ for neutralisation using Ph indicator. MeOH is then added when a further 2.5 ml of 0.2 M $H_2SO_4$ was needed. Calculate strength of $Na_2CO_3$ and $NaHCO_3$.

Q47. A sample containing $Na_2CO_3$ & NaOH is dissolved in 100ml solution. 10ml of this solution requires 25ml of 0.1N HCl when Ph is used as indicator. If MeOH is used as indicator 10ml of same solution requires 30ml of same HCl. Calculate % of $Na_2CO_3$ and NaOH in the sample.

Q48. What is the concentration of a solution of sodium carbonate (expressed as gm. of anhydrous sodium carbonate per litre), 25cc of which required 18.3cc of 0.12N sulphuric acid for neutralization, phenolphthalein being used as an indicator?

Q49. When the salt, $KNaC_4H_4O_6 . 4H_2O$ (molecular weight 282) is ignited, there is a residue of sodium carbonate and potassium carbonate. A gram of this salt gave a residue which required 63.8cc of N/10 hydrochloric acid for neutralization, methyl orange being used as indicator. Calculate the percentage purity of the salt.

Q50. Calculate (i) the number of gm. of anhydrous sodium carbonate, (ii) the number of gm. of sodium bicarbonate, present together in one litre of a solution. 25cc of this solution required 11.8cc of N/10 hydrochloric acid for naturalization when phenolphthalein was used as indicator and 31.0cc of N/10 hydrochloric acid when methyl orange was used as indicator.

## ***Back Titration***

Q51. 50gm of a sample of $Ca(OH)_2$ is dissolved in 50ml of 0.5N HCl solution. The excess of HCl was titrated with 0.3N – NaOH. The volume of NaOH used was 20cc. Calculate % purity of $Ca(OH)_2$.

Q52. One gm of impure sodium carbonate is dissolved in water and the solution is made up to 250ml. To 50ml of this made up solution, 50ml of 0.1N – HCl is added and the mix after shaking well required 10ml of 0.16N – NaOH solution for complete titration. Calculate the % purity of the sample.

Q53. What amount of substance containing 60% NaCl, 37% KCl should be weighed out for analysis so that after the action of 25 ml of 0.1N $AgNO_3$ solution, excess of $Ag^+$ is back titrated with 5 ml of $NH_4SCN$ solution? Given that 1 ml of $NH_4SCN$ = 1.1 ml of $AgNO_3$.

Q54. 5g of pyrolusite (impure $MnO_2$) were heated with conc. HCl and $Cl_2$ evolved was passed through excess of KI solution. The iodine liberated required 40 mL of $\frac{N}{10}$ hypo solution. Find the % of $MnO_2$ in the pyrolusite.

Q55. 1.64 g of a mixture of $CaCO_3$ and $MgCO_3$ was dissolved in 50 mL of 0.8 M HCl. The excess of acid required 16 mL of 0.25 M NaOH for neutralization. Calculate the percentage of $CaCO_3$ and $MgCO_3$ in the sample.

Q56. A mixture of $CaCl_2$ and NaCl weighing 2.385g was dissolved in water and treated with a solution of sodium oxalate which produces a precipitate of calcium oxalate. The precipitate was filtered from the mixture and then dissolved in HCl to give oxalic acid which when titrated against 0.2M $KMnO_4$ consumed 19.64 mL of the latter. What was percentage by mass of $CaCl_2$ in the original sample?

Q57. An acid solution of a $KReO_4$ sample containing 26.83 mg of combined rhenium was reduced by passage through a column of granulated zinc. The effluent solution, including the washings from the column, was then titrated with 0.10 N $KMnO_4$. 11.45 mL of the standard permanganate was required for the re-oxidation of all the rhenium to the perrhenate ion, $ReO_4^-$. Assuming that rhenium was only element reduced. What is the oxidation state to which rhenium was reduced by the Zn column.

(Atomic mass of Re = 186.2)

Q58. $H_2O_2$ is reduced rapidly by $Sn^{2+}$, the products being $Sn^{4+}$ & water. $H_2O_2$ decomposes slowly at room temperature to yield $O_2$ & water. Calculate the volume of $O_2$ produced at 20°C & 1.00 atm when 200 g of 10.0 % by mass $H_2O_2$ in water is treated with 100.0 ml of 2.00 M $Sn^{2+}$ & then the mixture is allowed to stand until no further reaction occurs.

Q59. A mixture containing $As_2O_3$ and $As_2O_5$ required 20.1 ml of 0.05N iodine for titration. The resulting solution is then acidified and excess of KI was added. The liberated iodine required 1.1113g hypo ($Na_2S_2O_3 . 5H_2O$) for complete reaction. Calculate the mass of the mixture. The reactions are

$$As_2O_3 + 2I_2 + 2H_2O \rightarrow As_2O_5 + 4H^+ + 4I^-$$
$$As_2O_5 + 4H^+ + 4I^- \rightarrow As_2O_3 + 2I_2 + 2H_2O$$

Q60. A sample of $MnSO_4 . 4H_2O$ is strongly heated in air which gives $Mn_3O_4$ as residue.
(i) The residue is dissolved in 100 ml of 0.1N $FeSO_4$ containing $H_2SO_4$.
(ii) The solution reacts completely with 50ml of $KMnO_4$ solution.
(iii) 25 ml of $KMnO_4$ solution used in step (ii) requires 30 ml of 0.1N $FeSO_4$ solution for the complete reaction.
Find the weight of $MnSO_4.4H_2O$ in the sample.

# *PROFICIENCY TEST*

**Q.1** ***Fill in the blanks with appropriate items :***

1. The number of water molecules in 0.5 mol of barium chloride dihydrate is _____.
2. 20ml of 0.1 M $H_2C_2O_4 \cdot 2H_2O$ (oxalic acid) solution contains oxalic acid equal to _____ moles.
3. The volume of $1.204 \times 10^{24}$ molecules of water at 4°C is _____.
4. 0.2 mol of ozone ($O_3$) at N.T.P. will occupy volume _____ L.
5. The balancing of chemical equation is based upon _____.
6. 2 gm of hydrogen will have same number of H atoms as are there in _____ g hydrazine ($NH_2$–$NH_2$).
7. The mass of $x$ atoms of element = $\frac{.....x}{N_A}$.
8. The moles of $x$ atoms of a triatomic gas = $\frac{x}{N_A} \times$ _____.
9. The amount of $Na_2SO_4$ which gives 9.6 gm of $SO_4^{2-}$ is _____.
10. The 44 mg of certain substance contain $6.02 \times 10^{20}$ molecules. The molecular mass of the substance is _____.
11. The mass of $1 \times 10^{22}$ molecules of $CuSO_4 . 5H_2O$ is _____.
12. The atomic mass of iron is 56. The equivalent mass of the metal in $FeCl_2$ is _____ and that in $FeCl_3$ is _____.
13. The sulphate of a metal M contains 9.87% of M. The sulphate is isomorphous with $ZnSO_4.7H_2O$. The atomic mass of M is _____.
14. A binary compound contains 50% of A (at. mass = 16) & 50% B (at. mass = 32). The empirical formula of the compound is _____.
15. 10.6 g of $Na_2CO_3$ react with 9.8 g of $H_2SO_4$ to form 16 g of $Na_2SO_4$ & 4.4 g $CO_2$. This is in accordance with the law of _____.
16. 3 g of a salt (m. wt. 30) are dissolved in 250 ml of water. The molarity of solution is _____.
17. 0.5 mole of $BaCl_2$ are mixed with 0.2 mole of $Na_3PO_4$ the maximum number of mole of $Ba_3(PO_4)_2$ formed are _____.
18. The Eq. weight of $Na_2HPO_4$ when it reacts with excess of HCl is _____.
19. The mole fraction of solute in 20% (by weight) aqueous $H_2O_2$ solution is _____.
20. A metallic oxide contains 60% of the metal. The Eq. weight of the metal is _____.
21. The number of gm of anhydrous $Na_2CO_3$ present in 250 ml of 0.25 N solution is_____.
22. _____ ml of 0.1 M $H_2SO_4$ is required to neutralize 50 ml of 0.2 M NaOH solution.
23. The number of mole of water present in 90 g $H_2O$ are _____.
24. The concentration of $K^+$ ion in 0.2 M $K_2Cr_2O_7$ solution would be_____.
25. 280 ml of sulphur vapour at NTP weight 3.2 g . The Mol. formula of the sulphur vapour is _____.

**Q.2** ***True or False Statements :***

1. Equal volumes of helium and nitrogen under similar conditions have equal number of atoms.
2. The smallest particle is a substance which is capable in independent existence is called an atom.
3. The number of formula units in 0.5 mole of KCl is $6.02 \times 10^{23}$.
4. 22.4 L of ethane gas at S.T.P. contains H atoms as are present in 3 gram molecules of dihydrogen.
5. Molarity of pure water is 55.5.
6. A 20% solution of KOH (density = 1.02 g/ml) has molarity = 3.64.
7. In a mixture of 1 g $C_6H_6$ & 1 g $C_7H_8$, the mole fraction of both are same.
8. 1 mole of $C_{12}H_{22}O_{11}$ contains 22 hydrogen atoms.
9. $KClO_4$ & $KMnO_4$ are isomorphous in nature.
10. Mass of $3.01 \times 10^{23}$ molecules of of methane is 8 gm.
11. A hydrocarbon contains 86% C. 448 ml of the hydrocarbon weighs 1.68 g at STP. Then the hydrocarbon is an alkene.
12. $6.023 \times 10^{54}$ $e^{-s}$ weigh one kg.
13. An oxide of metal M has 40% by mass of oxygen. Metal M has relative atomic mass of 24. The empirical formula of the oxide is MO.
14. 5 g of a crystalline salt when rendered anhydrous lost 1.8 g of water. The formula weight of the anhydrous salt is 160. The number of molecules of water of crystallisation in the salt is 5.
15. Number of valence $e^{-s}$ in 4.2 g of $N^{3-}$ is 24 $N_A$.
16. The equivalent mass of $KMnO_4$ in alkaline medium is molar mass divided by five.
17. The equivalent mass of $Na_2S_2O_3$ in its reaction with $I_2$ is molar mass divided by two.
18. In a reaction, $H_2MoO_4$ is changed to $MoO_2^+$. In this case, $H_2MoO_4$ acts as an oxidising agent.
19. $KBrO_3$ acts as a strong oxidising agent. It accepts 6 electrons to give KBr.
20. 0.1 M sulphuric acid has normality of 0.05 N.
21. The reaction, $2H_2O_2 \longrightarrow 2H_2O + O_2$ is not an example of a redox reaction.
22. The disproportionation reaction,
    $2Mn^{3+} + 2H_2O \longrightarrow MnO_2 + Mn^{+2} + 4H^+$
    is an example of a redox reaction.
23. The oxidation number of hydrogen is always taken as + 1 in its all compounds.
24. The increase in oxidation number of an element implies that the element has undergone reduction.
25. The oxidation state of oxygen atom in potassium super oxide is $-\frac{1}{2}$.

## *MIDDLE GAME*

Q1. A sample of calcium carbonate contains impurities which do not react with a mineral acid. When 2 grams of the sample were reacted with the mineral acid, 375 ml of carbon dioxide were obtained at 27°C and 760 mm pressure. Calculate the % purity of the sample of $CaCO_3$?

Q2. One gram of an alloy of aluminium and magnesium when heated with excess of dil. HCl forms magnesium chloride, aluminium chloride and hydrogen. The evolved hydrogen collected over mercury at 0°C has a volume of 1.2 litres at 0.92 atm pressure. Calculate the composition of the alloy.

Q3. 10 gm of a mixture of anhydrous nitrates of two metal A & B were heated to a constant weight & gave 5.531 gm of a mixture of the corresponding oxides. The equivalent weights of A & B are 103.6 & 31.8 respectively. What was the percentage of A in the mixture.

Q4. 50ml of a solution, containing 0.01 mole each $Na_2CO_3$, $NaHCO_3$ and NaOH was titrated with N-HCl. What will be the titre readings if
(a) only Ph is used as indicator.
(b) only MeOH is used as indicator from the beginning.
(c) MeOH is added after the first end point with Ph.

Q5. Chrome alum $K_2SO_4 . Cr_2(SO_4)_3 . 24\ H_2O$ is prepared by passing $SO_2$ gas through an aqueous solution of $K_2Cr_2O_7$ acidified with dilute sulphuric acid till the reduction is complete. The alum is crystallized followed by filtration/centrifugation. If only 90% of the alum can be recovered from the above process, how much alum can be prepared from 10kg of $K_2Cr_2O_7$? Give the number of moles of electrons supplied by $SO_2$ for reducing one mole of $K_2Cr_2O_7$.

Q6. 25 mL of a solution containing HCl was treated with excess of M/5 $KIO_3$ and KI solution of unknown concentration where $I_2$ liberated is titrated against a standard solution of 0.021M $Na_2S_2O_3$ solution whose 24 mL were used up. Find the strength of HCl and volume of $KIO_3$ solution consumed.

Q7. A 10g sample of only CuS and $Cu_2S$ was treated with 100 mL of 1.25 M $K_2Cr_2O_7$. The products obtained were $Cr^{3+}$, $Cu^{2+}$ and $SO_2$. The excess oxidant was reacted with 50 mL of $Fe^{2+}$ solution. 25 ml of the same $Fe^{2+}$ solution required 0.875M acidic $KMnO_4$ the volume of which used was 20 mL. Find the % of CuS and $Cu_2S$ in the sample.

Q8. A substance of crude copper is boiled in $H_2SO_4$ till all the copper has reacted. The impurities are inert to the acid. The $SO_2$ liberated in the reaction is passed into 100 mL of 0.4 M acidified $KMnO_4$. The solution of $KMnO_4$ after passage of $SO_2$ is allowed to react with oxalic acid and requires 23.6 mL of 1.2 M oxalic acid. If the purity of copper is 91%, what was the weight of the sample.

Q9.. A 1.87gm. sample of chromite ore($FeO.Cr_2O_3$) was completely oxidized by the fusion of peroxide. The fused mass was treated with water and boiled to destroy the excess of peroxide. After acidification the sample was treated with 50ml. of 0.16M $Fe^{2+}$. In back titration 2.97 ml of 0.005 M barium dichromate was required to oxidize the excess iron (II). What is the percentage of chromite in the sample?

Q10. 0.6213 g of sample contains an unknown amount of $As_2O_3$. The sample was treated with HCl resulting in formation of $AsCl_3(g)$ which was distilled into a beaker of water. The hydrolysis reaction is as follows

$$AsCl_3 + 2H_2O \rightarrow HAsO_2 + 3H^+ + 3Cl^-.$$

The amount of $HAsO_2$ was determined by titration with 0.04134 M $I_2$, requiring 23.04 mL to reach the equivalence point. The redox products in the titration were $H_3AsO_4$ and $I^-$. Find the amount of $KMnO_4$ needed to oxidize As in $As_2O_3$ to its maximum possible oxidation state in acidic medium.

Q11. A sample of steel weighing 0.6 gm and containing S as an impurity was burnt in a stream of $O_2$, when S was converted to its oxide $SO_2$. $SO_2$ was then oxidized to $SO_4^{--}$ by using $H_2O_2$ solution containing 30ml of 0.04 M NaOH. 22.48 ml of 0.024 M HCl was required to neutralize the base remaining after oxidation. Calculate the % of S in the sample.

Q12. Sulfur dioxide is an atmospheric pollutant that is converted to sulfuric acid when it reacts with water vapour. This is one source of acid rain, one of our most pressing environmental problems. The sulfur dioxide content of an air sample can be determined as follows. A sample of air is bubbled through an aqueous solution of hydrogen peroxide to convert all of the $SO_2$ to $H_2SO_4$

$$H_2O_2 + SO_2 \longrightarrow H_2SO_4$$

Titration of the resulting solution completes the analysis. In one such case, analysis of 1550 L of Los Angeles air gave a solution that required 5.70 ml of $5.96 \times 10^{-3}$M NaOH to complete the titration. Determine the number of grams of $SO_2$ present in the air sample.

Q13. 1.4 g of a complex $[Co(NH_3)_x]Cl_3$ was treated with 50 mL of 2N NaOH solution and boiled. Ammonia gas evolved was passed through 50 mL of 1N $H_2SO_4$. After the reaction was over, excess acid required 37.2 mL of 0.5 N NaOH. Calculate

(i) The percentage of ammonia in the sample. (ii) The value of x in the formula.

Q14. 3.3 gm of a sample of Anhydrous $CuSO_4$ was dissolved in water and made to 250ml. 25 ml of this solution after taking usual precautions was treated with a little excess of KI solution. A white ppt. of $Cu_2I_2$ and iodine was evolved. The iodine so evolved required 24.6 ml of hypo solution containing 20gm of $(Na_2S_2O_3 \cdot 5H_2O)$ per litre. What is the purity of $CuSO_4$ solution.

Q15. A certain sample of coal contained some iron pyrite ($FeS_2$) – a pollution causing impurity. When the coal was burned iron(II) was oxidised and $SO_2$ was formed. The $SO_2$ was reacted with NaOH when sodium sulphite and water was formed. On a particular fay $10^3$ kg of coal was burned and it required 4 litres of 5M NaOH for the treatment of $SO_2$. What was the percentage of pyrite in the coal. What was the percentage of sulphur in the coal.

Q16. Calculate the % of $MnO_2$ in a sample of pyrolusite ore, 1.5 g which was made to react with 10 g. of Mohr's salt ($FeSO_4.(NH_4)_2SO_4.6H_2O$) and dilute $H_2SO_4$. $MnO_2$ was converted $Mn^{2+}$. After the reaction the solution was diluted to 250 ml and 50 ml of this solution, when titrated with 0.1 N $K_2Cr_2O_7$, required 10 ml of the dichromate solution.

Q17. Chlorine dioxide ($ClO_2$), has been used as a disinfectant in air conditioning systems. It reacts with water according to the reaction: $ClO_2 + H_2O \longrightarrow HClO_3 + HCl$

In an experiment, a 10.0 L sealed flask containing $ClO_2$ and some inert gas at 300 K and 1.0 atmosphere pressure is opened in a bath containing excess of water and all $ClO_2$ is reacted quantitatively. The resulting solution requried 200 mL 0.9 M NaOH solution for naturalization. Determine mole fraction of $ClO_2$ in the flask.

Q18. Consider the following reactions: $XeF_2 + F_2 \longrightarrow XeF_6$

and $XeF_6 + (-CH_2-CH_2)_n \longrightarrow (-CH_2-CH_2-)_n \longrightarrow (-CH_2-CH_2-) + HF + XeF_4$

Determine mass of $F_2$ (g) required for preparation of 1.0 kg fluorinated polymer.

Q19. 2.0 g of a sample containing NaCl, NaBr and some inert impurity is dissolved in enough water and treated with excess of $AgNO_3$ solution. A 2.0 g of precipitate was formed. Precipitate on shaking with aqueous NaBr gain 0.76 g of weight. Determine mass percentage of NaCl in the original sample.

Q20. 2.725 g of a mixture of $K_2C_2O_4$, $KHC_2O_4$ and $H_2C_2O_4 \cdot 2H_2O$ is dissolved in 100 mL $H_2O$ and its 10mL portion is titrated with 0.1 N HCl solution. 20 mL acid was required to reach the equivalence point. In another experiment, 10mL portion of the same stock solution is titrated with 0.1 N KOH solution. 20 mL of base was required to reach the equivalence point. Determine mass percentage of each component in the mixture.

Q21. A 0.127 g of an unsaturated oil was treated with 25 mL of 0.1 M ICl solution. The unreacted ICl was then treated with excess of KI. Liberated iodine required 40 mL 0.1 M hypo solution. Determine mass of $I_2$ that would have been required with 100.0 g oil if $I_2$ were used in place of ICl.

Q22. The CO in a 20.3 L sample of gas was converted to $CO_2$ by passing the gas over iodine pentoxide heated to $150^0$C. $I_2O_5(s) + 5CO(g) \longrightarrow 5CO_2(g) + I_2(g)$. The iodine distilled at this temperature and collected in a vessel containing 8.25 ml of 0.011 M $Na_2S_2O_3$. The excess $Na_2S_2O_3$ was back titrated with 2.16 ml of 0.00947 M $I_2$ solution. Calculate the number of milligrams of CO per litre of the sample.

Q23. The chromate ion may be present in waste from a chrome plating plant. It is reduced to insoluble chromium hydroxide by dithionate ion in basic medium $S_2O_4^{2-} + Cr_2O_4^{2-} \longrightarrow SO_3^{2-} + Cr(OH)_3$. 100 ml of water require 387 gm of $Na_2S_2O_4$. Calculate molarity and normality of $CrO_4^{2-}$ in waste water. Also express concentration in ppm of $Na_2CrO_4$.

Q24. A gas mixture was passed at the rate of 2.5 L/min. through a solution of NaOH for a total of 64 minutes. The $SO_2$ in the mixture was retained as sulphite ion: $SO_2(g) + 2OH^- \longrightarrow SO_3^{2-} + H_2O$. After acidification with HCl, the sulphite was titrated with 4.98 mL of 0.003125 M $KIO_3$.
$IO_3^- + SO_3^- + HCl \longrightarrow ICl_2^- + SO_4^{2-} + H_2O$. If density of the mixture is 1.2 gm/lt, calculate concentration of $SO_2$ in ppm.

Q25. The arsenic in a 1.223 gm sample of a pesticide was converted to $H_3AsO_4$ by suitable treatment. The acid was then neutralized and exactly 40 ml of 0.08 M $AgNO_3$ was added to precipitate the arsenic quantitavely as $Ag_3AsO_4$. The excess $Ag^+$ in the filterate required 11.27 ml of 0.1 M KSCN as
$Ag^+ + SCN^- \longrightarrow AgSCN$ (s) Calculate the percent $As_2O_3$ in the sample.($As_2O_3 = 198$)

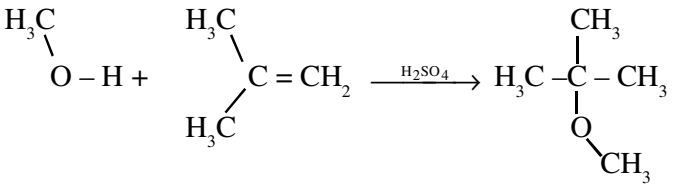
Q26. 5 gm of bleaching powder was suspended in water and volume made up to half a litre. 20 ml of this suspension when acidified with acetic acid and treated with excess of potassium iodide solution liberated iodine which required 20 ml of a decinormal hypo solution for titration. Calculate percentage of available chlorine in bleaching powder.

Q27. 25 mL of a 0.107M $H_3PO_4$ was titrated with a 0.115M solution of a NaOH solution to the end point identified by the colour change of the indicator, bromocresol green. This required 23.1 mL. The titration was repeated using phenolphthalein indicator. This time, 25 mL of same $H_3PO_4$ solution required 46.8 mL of same NaOH solution. What is the coefficient 'n' in the equation

$$H_3PO_4 + nOH^- \rightarrow nH_2O + [H_{(3-n)}PO_4]^{n-} \text{ for each reaction?}$$

Q28. 1 gm sample of $KClO_3$ was heated under such conditions that a part of it decomposed according to the equation (1) $2KClO_3 \longrightarrow 2KCl + 3O_2$
and remaining underwent change according to the equation. (2) $4KClO_3 \longrightarrow 3KClO_4 + KCl$
If the amount of $O_2$ evolved was 146.8 ml at S.T.P., calculate the % of weight of $KClO_4$ in the residue.

Q29. Methyl t-butyl ether (MTBE) is a carbon-based compound that has replaced lead - containing materials as the principal antiknock ingredient in gasoline. Today's gasoline contains about 7% MTBE by mass. MTBE is produced from isobutene by the following reaction:




| Methanol | Isobutene | MTBE |
|---|---|---|
| $CH_4O$ | $C_4H_8$ | $C_5H_{12}O$ |

Approximately 2 billion pounds of MTBE are produced each year at a cost of about 10¢ per pound. Assume that you are a chemist working for a company that sold 750 million pounds of MTBE last year.
(a) If the synthesis has a reaction yield of 86%, how much isobutene was used to produce the MTBE?

(b) You have improved the synthesis of MTBE so that the yield of the reaction increases from 86% to 93%. If the company uses the same mass of isobutene for next year's production, how many pounds of MTBE will the company sell if it uses your new process?
(c) Assuming that the price of MTBE does not change, how much more money will the company make next year because of your work?

Q30. A mixture of NaCl and NaBr weighing 3.5084 gm was dissolved and treated with enough $AgNO_3$ to precipitate all of the chloride and bromide as AgCl and AgBr. The washed precipitate was treated with KCN to solubilize the silver and the resulting solution was electrolyzed. The equations are :

$$NaCl + AgNO_3 \longrightarrow AgCl + NaNO_3$$
$$NaBr + AgNO_3 \longrightarrow AgBr + NaNO_3$$
$$AgCl + 2KCN \longrightarrow KAg(CN)_2 + KCl$$
$$AgBr + 2KCN \longrightarrow KAg(CN)_2 + KBr$$
$$4KAg(CN)_2 + 4KOH \longrightarrow 4Ag + 8KCN + O_2 + 2H_2O$$

After the final step was complete, the deposit of metallic silver weighed 5.5028 gm. What was the composition of the initial mixture.

Q31. Phosphorus is essential for plant growth, and it is often the limiting nutrient in aqueous ecosystems. However, too much phosphorus can cause algae to grow at an explosive rate. This process, known as eutrophication, robs the rest of the ecosystem of essential oxygen, often destroying all other aquatic life. One source of aquatic phosphorus pollution is the $HPO_4^{2-}$ used in detergents in sewage plants. The simplest way to remove $HPO_4^{2-}$ is to treat the contaminated water with lime, CaO, which generates $Ca^{2+}$ and $OH^-$ ions in water. The phosphorus precipitates as $Ca_5(PO_4)_3OH$.
(a) Write the balanced equation for CaO dissolving in water.
(b) Write the balanced equation for the precipitation reaction.
(c) How many kilograms of lime are required to remove all the phosphorus from a $1.00 \times 10^4$ L holding tank filled with contaminated water that is 0.0156 M in $HPO_4^{2-}$?

Q32. It was desired to neutralize a certain solution prepared by mixing KCl and hydrobromic acid. Titration of 10ml of this solution with 0.1M $AgNO_3$ solution required 50ml of the latter for the complete precipitation of the halides. The resulting precipitate when filtered, washed and dried weighed 0.771 gm. How much 0.1M NaOH must have been used for the neutralization of 10ml of the solution.

Q33. The element Se, dispersed in a 5.0 ml sample of detergent for dandruff control, was determined by suspending the sample in warm, ammonical solution that contain 45.0 ml of 0.020 M $AgNO_3$.
$6Ag^+ + 3Se(s) + 6NH_3 + 3H_2O \longrightarrow 2Ag_2Se(s) + Ag_2SeO_3(s) + 6NH_4^+$
The mixture was next treated with excess nitric acid which dissolves the $Ag_2SeO_3$ but not the $Ag_2Se$. The $Ag^+$ from the $Ag_2SeO_3$ and excess $AgNO_3$ consumed 16.74 ml of 0.0137 N KSCN in a Volhard titration. How many milligrams of Se were contained per millilitre of sample.

Q34. In the presence of fluoride ion, $Mn^{2+}$ can be titrated with $MnO_4^-$, both reactants being converted to a complex of Mn(III). A 0.545 g sample containing $Mn_3O_4$ was dissolved and all manganese was converted to $Mn^{2+}$. Titration in the presence of fluoride ion consumed 31.1 ml of $KMnO_4$ that was 0.117 N against oxalate.
(a) write a balanced chemical equation for the reaction, assuming that the complex is $MnF_4^-$.
(b) what was the % of $Mn_3O_4$ in the sample?

Q35. $CuSO_4$ reacts with KI in an acidic medium to liberate $I_2$ $2CuSO_4 + 4KI \longrightarrow Cu_2I_2 + 2K_2SO_4 + I_2$.
Mercuric periodate $Hg_5(IO_6)_2$ reacts with a mixture of KI & HCl according to the following equation:
$Hg_5(IO_6)_2 + 34KI + 24\ HCl \longrightarrow 4K_2HgI_4 + 8I_2 + 24\ KCl + 12\ H_2O$
The liberated iodine is titrated against $Na_2S_2O_3$ solution; 1 ml of which is equivalent to 0.0499 gm of $CuSO_4 \cdot 5H_2O$. What volume in ml of $Na_2S_2O_3$ solution will be required to react with the $I_2$ liberated from 0.7245 gm of $Hg_5(IO_6)_2$?
Given Mol. wt. of $Hg_5(IO_6)_2$ = 1448.5 gm/mol; Mol. wt. of $CuSO_4 \cdot 5H_2O$ = 249.5 gm/mol

# ANSWER KEY

## EASY RIDE

### Acid Base Titration

Q1. 0.381 g Q2. V = 157.8 ml Q3. V = 25 mL Q4. 6.608 g/litre Q5. (i) 0.569N, (ii) 50
Q6. 0.0176M Q7. 3.78g Q8.200 mL Q9. 11.92 g/litre Q10. 470 mL
Q11. KOH = 35%, $Ca(OH)_2$ = 65% Q12. 8.67 Q13. 20.72 %
Q14. $MgCO_3$ = 52.02%, $CaCO_3$ = 47.98 % Q15. 83.33 Q16. 2.63% Q17. 0.575 gm
Q18. % of $NH_4Cl$ = 57.5%, % of NaCl = 42.5%
Q19. 3.198 g HCl/litre, 6.974 g KCl/litre
Q20. 0.06gm; 0.0265gm

### Redox Titration

Q21. + 3 Q22. zero Q23. 0.588 N Q24. 337 mL Q25. 22.5gm Q26. 0.1M
Q27. 0.254gm/lt Q28. V = 31.68 ml Q29. 41.53% Q30. 600 L $MnO_4^-$ solution
Q31. 0.15 N Q32. 7.5 ml Q33. $Fe^{3+}$ = 0.02M; $MnO_4^-$ = 0.016 M; $H^+$ = 0.568 M; $Mn^{2+}$ = 0.004M; $SO_4^{2-}$ = 0.02M; $K^+$ = 0.02M, $ClO_4^-$ = 0.6M
Q34. 1.176 gm Q35. 6.07 ≈ 6 Q36. FeO = 13.34%; $Fe_2O_3$ = 86.66%
Q37. 33.33 ml ; 1.486 gm Q38. $H_2C_2O_4 . 2H_2O$ = 14.35%, $KHC_2O_4 . H_2O$ = 81.71%
Q39. 94.38% Q40. 27.27%

### Double titration

Q41. 0.424 gm; 0.21gm Q42. 23.2 gm, 22.28gm Q43. 0.06gm; .0265gm
Q44. 36.85 g/litre Q45. 56.7% Q46. 4.24 g/L; 5.04 g/L Q47. 39.85%; 60.15%
Q48. 9.31 g/litre Q49. 90.0% Q50. (i) 5.003 g/litre, (ii) 2.486 g/litre

### Back Titration

Q51.1.406% Q52. 90.1% Q53. 0.1281 g Q54. 0.174g; 3.48%
Q55. $MgCO_3$= 52.02% , $CaCO_3$= 47.98% Q56. 45.7% $CaCl_2$ Q57. –1
Q58. 4.67L Q59. 0.25g Q60. 1.338gm

## PROFICIENCY TEST

**Q.1**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | $6.02 \times 10^{23}$ | 2. | $2 \times 10^{-3}$ mol | 3. | 36 ml | 4. | 4.48 L |
| 5. | Laws of conservation of mass | | | 6. | 16 gm | 7. | GAM |
| 8. | 1/3 | 9. | 14.2 gm | 10. | 44 g $mol^{-1}$ | 11. | 4.13 g |
| 12. | 44.8 L | 13. | 24.3 | 14. | $A_2B$ | 15. | Conservation of mass |
| 16. | 0.4 | 17. | 0.1 | 18. | M/2 | 19. | 0.1168 |
| 20. | 12 | 21. | 3.3125 g | 22. | 50 | 23. | 5 |
| 24. | 0.4 M | 25. | $S_8$ | | | | |

**Q.2**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | False | 2. | False | 3. | False | 4. | True |
| 5. | True | 6. | True | 7. | False | 8. | False |
| 9. | True | 10. | True | 11. | True | 12. | False |
| 13. | True | 14. | True | 15. | False | 16. | False |
| 17. | False | 18. | True | 19. | True | 20. | False |
| 21. | False | 22. | True | 23. | False | 24. | False |
| 25. | True | | | | | | |

## ***MIDDLE GAME***

Q1. 76.15% Q2. Al = 0.546 g; Mg = 0.454 g Q3. 51.6% Q4. 20ml, 40ml, 20ml

Q5. 30.55kg, 6 electrons Q6. $V_{KIO_3}$ = 0.42 mL, [HCl] = 0.02N, 0.73 gm/lt

Q7. 57.4% CuS, 42.6% $Cu_2S$ Q8. 5 gm Q9. 15.8% Q10. 0.06 gm

Q11. 1.7613% Q12. $1.087 \times 10^{-3}$gm Q13. (a) 38.13%; (b) x = 6

Q14. 95.9% Q15. 60%, 320gm Q16. 59.48% Q17. 44.335%

Q18. 3.04 kg Q19. 33.15% Q20. 27.9%, 50.92%, 21.18% Q21. 100g

Q22. 0.2424 mg/lt Q23. 0.0445 N; 0.0148 M; 2400 ppm Q24. 10.375ppm

Q25. 5.594% $As_2O_3$ Q26. 35.5% available $Cl_2$

Q27. For Bromocresol, n = 1; For Phenolphthalein n = 2 Q28. 49.8%

Q29. (a) $2.52 \times 10^{11}$ gm; (b) 811 million pounds; (c) 6.1 million dollars

Q30. NaCl = 67%; NaBr = 33% Q31. 14.56 Kg Q32. 12.02 ml Q33. 7.95mg/ml

Q34. 40.77% Q35. 40 ml $Na_2S_2O_3$ solution.

## ***ANSWERS OF STOICHIOMETRY: MOLE II***
## ***EX: MIDDLE GAME***

Q1.76.15% Q2. Al = 0.546 g; Mg = 0.454 g Q3. 51.6% Q4. 20ml, 40ml, 20ml

Q5. 30.55kg, 6 electrons Q6. $V_{KIO_3}$ = 0.42 mL, [HCl] = 0.02N, 0.73 gm/lt

Q7. 57.4% CuS, 42.6% $Cu_2S$ Q8. 5 gm Q9. 15.8% Q10. 0.06 gm
Q11. 1.7613% Q12. $1.087 \times 10^{-3}$gm Q13. (a) 38.13%; (b) x = 6
Q14. 95.9% Q15. 60%, 320gm Q16. 59.48% Q17. 44.335%
Q18. 3.04 kg Q19. 33.15% Q20. 27.9%, 50.92%, 21.18% Q21.100g
Q22. 0.2424 mg/lt Q23. 0.0445 N; 0.0148 M; 2400 ppm Q24. 10.375ppm
Q25. 5.594% $As_2O_3$ Q26. 35.5% available $Cl_2$
Q27. For Bromocresol, n = 1; For Phenolphthalein n = 2 Q28. 49.8%
Q29. (a) $2.52 \times 10^{11}$ gm; (b) 811 million pounds; (c) 6.1 million dollars
Q30. NaCl = 67%; NaBr = 33% Q31. 14.56 Kg Q32. 12.02 ml Q33. 7.95mg/ml
Q34. 40.77% Q35. 40 ml $Na_2S_2O_3$ solution.


## ***ANSWERS OF STOICHIOMETRY: MOLE II***
## ***EX: MIDDLE GAME***

Q1.76.15% Q2. Al = 0.546 g; Mg = 0.454 g Q3. 51.6% Q4. 20ml, 40ml, 20ml

Q5. 30.55kg, 6 electrons Q6. $V_{KIO_3}$ = 0.42 mL, [HCl] = 0.02N, 0.73 gm/lt

Q7. 57.4% CuS, 42.6% $Cu_2S$ Q8. 5 gm Q9. 15.8% Q10. 0.06 gm
Q11. 1.7613% Q12. $1.087 \times 10^{-3}$gm Q13. (a) 38.13%; (b) x = 6
Q14. 95.9% Q15. 60%, 320gm Q16. 59.48% Q17. 44.335%
Q18. 3.04 kg Q19. 33.15% Q20. 27.9%, 50.92%, 21.18% Q21.100g
Q22. 0.2424 mg/lt Q23. 0.0445 N; 0.0148 M; 2400 ppm Q24. 10.375ppm
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# STUDY PACKAGE

# Target: IIT-JEE(Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 3. Atomic Structure

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

**4. Exercise III**

**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

## Physical Constants[a]

| Constant and Symbol[b] | | SI Value | Gaussian Value |
|---|---|---|---|
| Speed of light in vaccum | c | $2.99 \times 10^{8}$ m/s | $2.99 \times 10^{10}$ cm/s |
| Proton & electron charge | e | $1.60 \times 10^{-19}$ C | $4.8 \times 10^{-10}$ statC |
| Permittivity of vaccum | $\varepsilon_0$ | $8.85 \times 10^{-12}$ $C^2/N\text{-}m^2$ | |
| Avogadro constant | $N_A$ | $6.02 \times 10^{23}$ $mol^{-1}$ | $6.02 \times 10^{23}$ $mol^{-1}$ |
| Electron rest mass (0.000548 amu) | $m_e$ | $9.10 \times 10^{-31}$ kg | $9.10 \times 10^{-28}$ g |
| Proton rest mass (1.00757 amu) | $m_P$ | $1.67 \times 10^{-27}$ kg | $1.67 \times 10^{-24}$ g |
| Neutron rest mass (1.00893 amu) | $m_n$ | $1.67 \times 10^{-27}$ kg | $1.67 \times 10^{-24}$ g |
| Planck constant | h | $6.62 \times 10^{-34}$ J s | $6.62 \times 10^{-27}$ erg s |
| Permeability of vaccums | $\mu_0$ | $4\pi \times 10^{-7}$ $NC^{-2}$ $s^2$ | |
| Bohr radius | $a_0$ | $5.29 \times 10^{-11}$ m | $0.529 \times 10^{-8}$ cm |
| Bohr's velocity | | $2.188 \times 10^{6} \times \frac{Z}{n}$ m/sec. | $2.188 \times 10^{8} \times \frac{Z}{n}$ cm/sec. |
| Bohr's energy (–13.6 eV/atom) | | $-21.8 \times 10^{-19} \frac{Z^2}{n^2}$ J/atom | $-21.8 \times 10^{-12}$ erg/atom |
| Bohr magneton (BM) | $\beta_e$ | $9.27 \times 10^{-24}$ J/T | |
| Gas constant | R | 8.3145 J/mol-K | $8.3145 \times 10^{7}$ erg/mol-K |
| Boltzmann constant | k | $1.38 \times 10^{-23}$ J/K | $1.30 \times 10^{-16}$ erg/K |
| Gravitional constant | G | $6.67 \times 10^{-11}$ $m^3/kg\text{-}s^2$ | $6.67 \times 10^{-8}$ $cm^3/g\text{-}s^2$ |

## Energy Conversion Factors[a]

1 erg = $10^{-7}$ J
1 cal = 4.184 J
1 eV = $1.602177 \times 10^{-19}$ J = $1.602177 \times 10^{-12}$ erg = 23.0605 kcal/mol

## Greek Alphabet

| | | | | | |
|---|---|---|---|---|---|
| Alpha | A | α | Beta | B | β |
| Gamma | Γ | γ | Delta | Δ | δ |
| Epsilon | E | ε | Zeta | Z | ζ |
| Eta | H | η | Theta | Θ | θ |
| Iota | I | ι | Kappa | K | κ |
| Lambda | Λ | λ | Mu | M | μ |
| Nu | N | ν | Xi | Ξ | ξ |
| Omicron | O | ο | Pi | Π | π |
| Rho | P | ρ | Sigma | Σ | σ |
| Tau | T | τ | Upsilon | Y | υ |
| Phi | Φ | ϕ | Chi | X | χ |
| Psi | Ψ | ψ | Omega | Ω | ω |

# ***KEY CONCEPT***

**STRUCTURE OF ATOM**

Rutherford's Model
Bohr's Model
Wave mechanical model

**EXTRA NUCLEAR PART ($e^-$)**

Electrons,protons & neutrons are the most important fundamental particles of atoms of all elements (Except hydrogen)

Some uncommon Fundamental particles :

**1.** ${}_ZX^A$ , $A = Z + n$

**2.** Reduced mass $\frac{1}{\mu}=\frac{1}{M}+\frac{1}{m}=\frac{mM}{m+M}$ m = mass of $e^-$ ; M = Mass of nucleus

**3.** Photon is considered massless bundle of energy. But to find its mass use $m = \frac{h}{\lambda c}$

**4.** $E = mc^2$ , $E = h\nu = hc/\lambda = hc\,\bar{\nu}$

**5.** Quantum efficiency or Quantum Yield $= \frac{\text{no. of molecules reacting}}{\text{no. of quanta absorbed}}$

**6.** $R_n = R_1 (A)^{1/3}$ , $R_1 = 1.33 \times 10^{-13}$ cm A = mass number

**7.** $\frac{1}{2} m_\alpha v_\alpha^2 = K\frac{Z_e . 2e}{r}$ ; $\text{Tan}\frac{\theta}{2} \alpha \frac{1}{b}$

number of a particles at $\theta = \frac{K}{\sin^4 \theta/2}$ ; b = impact parameter

**8.** Rydberg's Equation $\frac{1}{\lambda}=\bar{\nu}=R_H\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]\times Z^2$

**9.** Limiting spectral line (series limit) means $n_2 = \infty$

**10.** $H_\alpha$ line means we know $n_1$ , $n_2$ (longest $\lambda$ , shortest $\nu$ , least E) [ $H_\alpha$ , $H_\beta$ , $H_\gamma$ , $H_\delta$ ]

**11.** No. of wavelengths observed in the spectrum $= \frac{n(n-1)}{2}$

when $e^-$ deexcites to ground state , n = no. of higher orbit

**12.** $1/2\, mv^2 = h\nu - h\nu^0(w)$ (work function or B.E.)

$\nu^0$ = Threshhold frequency $W = h\nu_0 = \frac{hc}{\lambda_0}$

**13.** Accelerating potential = eV = KE = $\frac{1}{2} mv^2$

**14.** $\lambda = hc/E = 1240$ ev. nm

**15.** $K=\frac{1}{4\pi\varepsilon_0}$ ; P.E. $= \frac{K q_1 q_2}{r}$ centrifugal force = $\mathbf{mv^2/r}$

**16.** $mvr = n\cdot \frac{h}{2\pi} = n.\hbar$

**17.** $E_n = \frac{E_1}{n^2} z^2 = -\frac{2\pi^2 me^4}{n^2 h^2} z^2$ ; $E_1 = \frac{-2\pi^2 me^4}{h^2}$

**18.** $r_n = \frac{n^2}{Z} x \frac{h^2}{4\pi^2 e^2 m}$ **19.** $v = \frac{z}{n} \times \frac{2\pi e^2}{h}$

**20.** revolutions per sec = $v/2\pi r$ **21.** Time for one revolution = $2\pi r/v$

**22.** Separation energy = $E_{n_\infty = \infty} - E_{n \text{ given } = 2,3,4,.....}$

**23.** No. of waves = n = no. of shells **24.** I.E. = $E_{n=\infty} - E_{\text{ground state of e-}}$ (K, L, M, N)

**25.** $\lambda = h/mv = h/p$ **26.** $\lambda = \sqrt{\frac{150}{V \text{in volts}}}$ Å

**27.** $E_n \neq KE$ $KE = 1/2\ mv^2$, $E = h\nu$ **28.** $\Delta x.\Delta p > h/4\pi$

**29.** $\nu^{1/2} = a(z-b)$ b = screening constant

**30.** Nucleons **31.** Isotopes, Isobars, Isotones (A – Z)

**32.** Isoelectronic **33.** Isosters

**34.** Isodiaphers (A – 2Z) **35.** paramagnetic

**36.** Diamagnetic **37.** $S = \frac{h}{2\pi}\sqrt{S(S+1)}$

**38.** $\mu = \sqrt{n(n+2)}$ B.M. n = number of unpaired $e^-$ ;

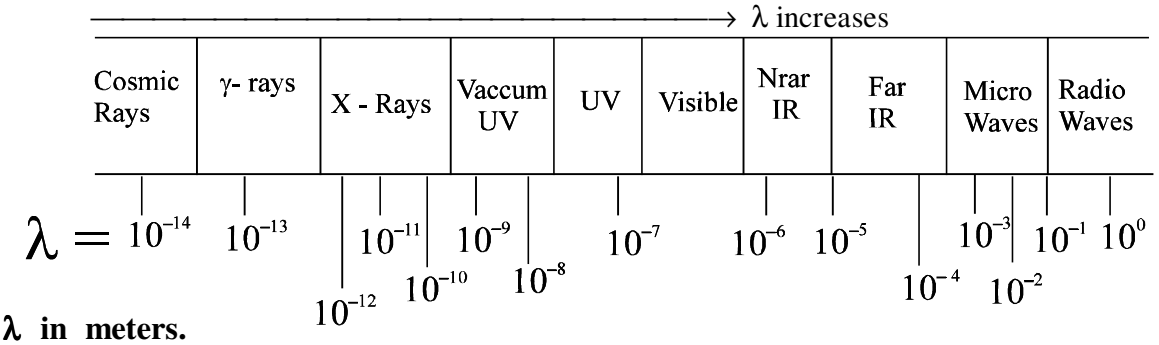
**39.**

| Radial Nodes | ; | Angular nodes ; | Total nodes |
|---|---|---|---|
| $(n - l - 1)$ | | $l$ | $(n-1)$ |

**40.** Total no. of $e^-$ in an energy level $= 2n^2$
Total no. of $e^-$ in a sublevel $= 2(2l+1)$
Maximum no. of $e^-$ in an orbital $= 2$
Total no. of orbitals in a sublevel $= (2l+1)$
No. of subshells in main energyshell $= n$
No. of orbitals in a main energy shell $= n^2$

| $l$ = | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| | s | p | d | f | g |

**41. ELEECTROMEGNETIC SPECTRUM**



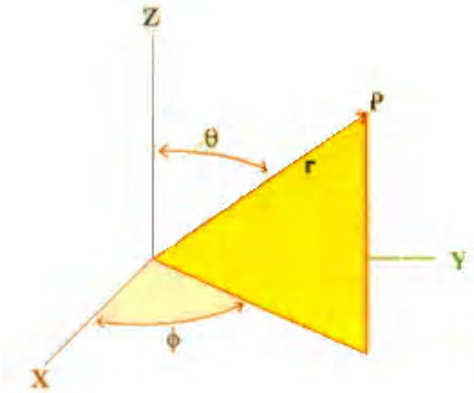

**λ in meters.**

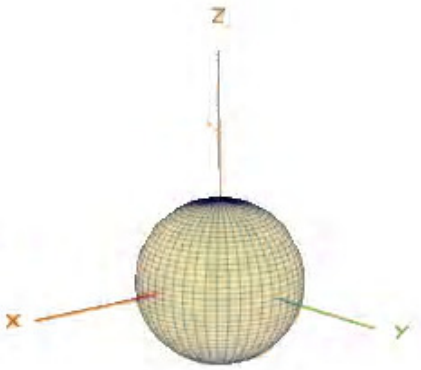
Distinction between the wave – particle nature of a photon and the particle–wave nature of sub-atomic particle.

| | **PHOTON** | **SUB ATOMIC PARTICLE** |
|---|---|---|
| 1. | Energy = $h\nu$ | Energy = $\frac{1}{2}mv^2$ |
| 2. | Wavelength = $\frac{c}{\nu}$ | Wavelength = $\frac{h}{mv}$ |

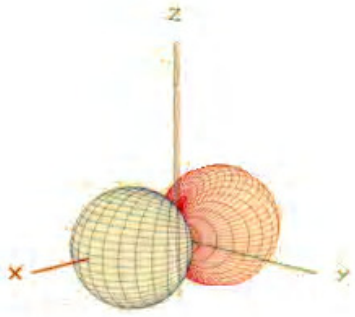
***Note**: We should never interchange any of the above and to write electronic conf. of Cation first write for neutral atom & then remove $e^-$ from outermost shell.*

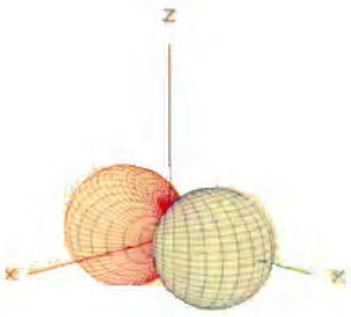
# ***SHAPES OF ATOMIC ORBITALS***

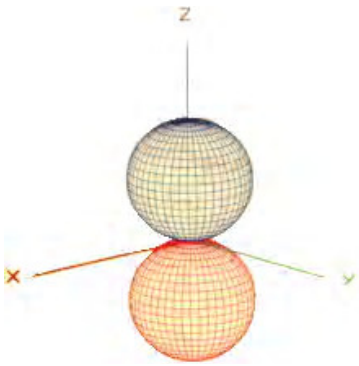


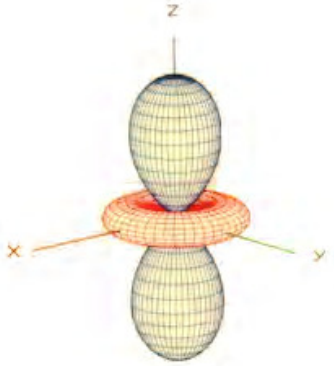

The spherical Polar Coordinates S

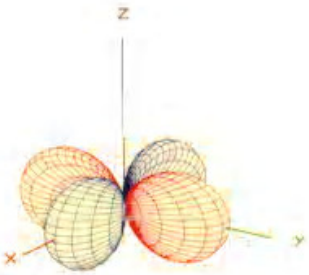


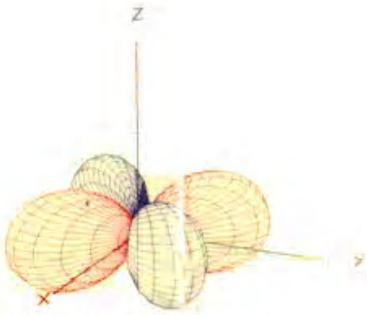



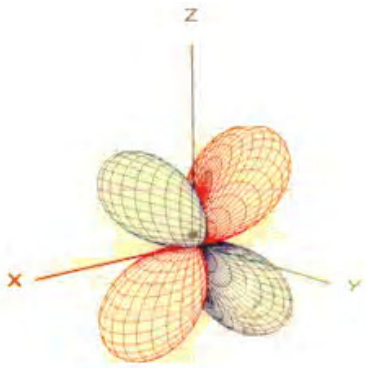

$p_X$ $p_y$ $p_z$



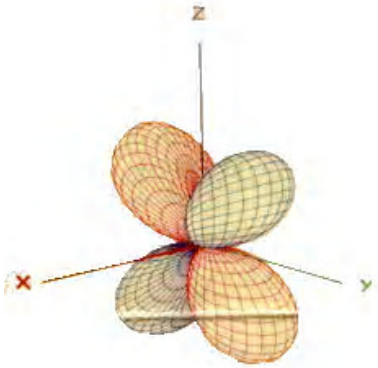



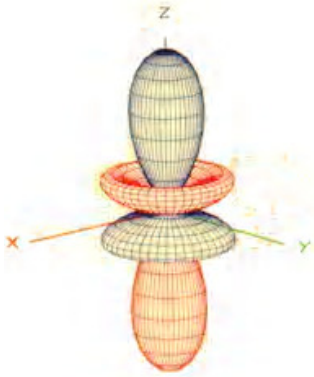

$d_{z^2}$ $d_{x^2-y^2}$ $d_{xy}$



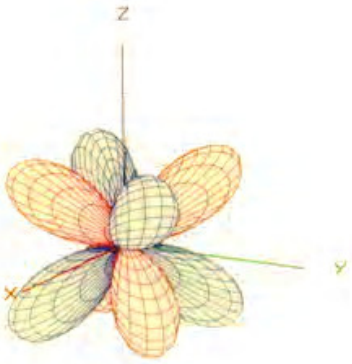

$d_{xz}$



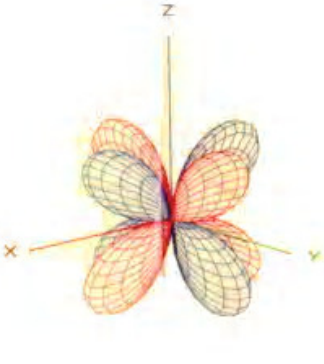

$d_{yz}$



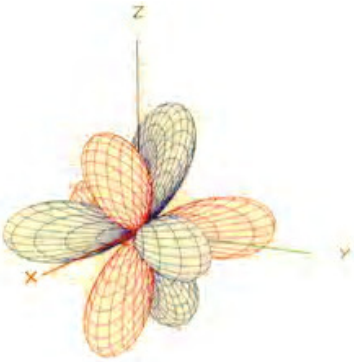

$f_{z^3}$



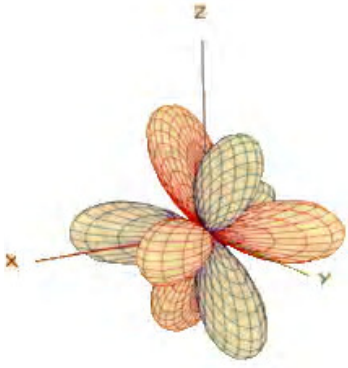

$f_{xyz}$



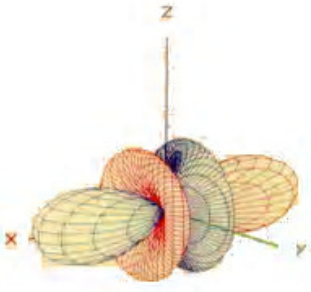

$f_{z(x^2-y^2)}$



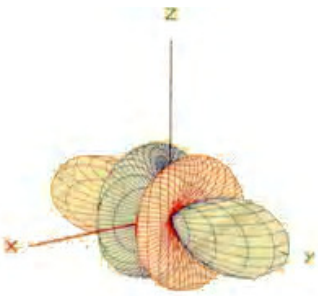

$f_{x(y^2-z^2)}$




$f_{y(z^2-x^2)}$




$f_{x^3}$




$f_{y^3}$

## ***EXERCISE -I***

### *LIGHT*

Q.1 H- atom is exposed to electromagnetic radiation of 1028 Å and gives out induced radiations. Calculate $\lambda$ of induced radiations.

Q.2 The wavelength of a certain line in the Paschen series in 1093.6 nm. What is the value of $n_{high}$ for this line. $[R_H = 1.0973 \times 10^{+7}\ m^{-1}]$

Q.3 A certain dye absorbs 4530 Å and fluoresces at 5080 Å these being wavelengths of maximum absorption that under given conditions 47% of the absorbed energy is emitted. Calculate the ratio of the no. of quanta emitted to the number absorbed.

Q.4 The reaction between $H_2$ and $Br_2$ to form HBr in presence of light is initiated by the photo decomposition of $Br_2$ into free Br atoms (free radicals) by absorption of light. The bond dissociation energy of $Br_2$ is 192 KJ/mole. What is the longest wavelength of the photon that would initiate the reaction.

Q.5 Wavelength of the Balmer $H_\alpha$ line (first line) is 6565 Å. Calculate the wavelength of $H_\beta$ (second line).

Q.6 Calculate the Rydberg constant R if $He^+$ ions are known to have the wavelength difference between the first (of the longest wavelength) lines of Balmer and Lyman series equal to 133.7nm.

Q.7 The quantum yield for decomposition of HI is 2. In an experiment 0.01 moles of HI are decomposed. Find the number of photons absorbed.

Q.8 The light radiations with discrete quantities of energy are called _____.

Q.9 What transition in the hydrogen spectrum would have the same wavelength as the Balmer transition, n=4 to n=2 of $He^+$ spectrum.

Q.10 Calculate the energy emitted when electrons of 1.0 g atom of hydrogen undergo transition giving the spectral line of lowest energy in the visible region of its atomic spectrum.

### PLANCK'S QUANTUM THEORY

Q.11 Calculate the wavelength of the radiation that would cause photo dissociation of chlorine molecule if the Cl- Cl bond energy is 243 KJ/mol.

Q.12 Suppose $10^{-17}$ J of light energy is needed by the interior of the human eye to see an object. How many photons of green light ($\lambda = 550$ nm) are needed to generate this minimum amount of energy.

Q.13 A photon having $\lambda = 854$ Å causes the ionization of a nitrogen atom. Give the I.E. per mole of nitrogen in KJ.

Q.14 Calculate the threshold frequency of metal if the binding energy is 180.69 KJ $mol^{-1}$ of electron.

Q.15 Calculate the binding energy per mole when threshold wavelength of photon is 240 nm.

Q.16 A metal was irriadated by light of frequency $3.2 \times 10^{15}\ S^{-1}$. The photoelectron produced had its KE, 2 times the KE of the photoelectron which was produced when the same metal was irriadated with a light of frequency $2.0 \times 10^{15}\ S^{-1}$. What is work function.

Q.17 U.V. light of wavelength 800 Å & 700 Å falls on hydrogen atoms in their ground state & liberates electrons with kinetic energy 1.8 eV and 4 eV respectively. Calculate planck's constant.

Q.18 The dissociation energy of $H_2$ is 430.53 KJ/mol. If $H_2$ is exposed to radiant energy of wavelength 253.7 nm, what % of radiant energy will be converted into K.E.

Q.19 A potential difference of 20 KV is applied across an X-ray tube. Find the minimum wavelength of X-ray generated.

Q.20 The K.E. of an electron emitted from tungstan surface is 3.06 eV. What voltage would be required to bring the electron to rest.

## *BOHR'S MODEL*

Q.21 Calculate energy of electron which is moving in the orbit that has its rad. sixteen times the rad. of first Bohr orbit for H–atom.

Q.22 The electron energy in hydrogen atom is given by, $E_n = \frac{-21.7\times10^{-12}}{n^2}$ ergs. Calculate the energy required to remove an $e^-$ completely from n = 2 orbit . What is the largest wavelength in cm of light that can be used to cause this transition.

Q.23 Calculate the wavelength in angstrom of photon that is emitted when an $e^-$ in Bohr orbit n=2 returns to the orbit n=1. The ionization potential of the ground state of hydrogen atom is $2.17\times10^{-11}$ erg/atom.

Q.24 The radius of the fourth orbit of hydrogen atom is 0.85 nm. Calculate the velocity of electron in this orbit.

Q.25 The velocity of $e^-$ in a certain Bohr orbit of the hydrogen atom bears the ratio 1:275 to the velocity of light. What is the quantum no. "n" of the orbit and the wave no. of the radiation emitted for the transition from the quatum state (n+1) to the ground state.

Q.26 Electrons of energy 12.09 eV can excite hydrogen atoms. To which orbit is the electron in the hydrogen atom raised and what are the wavelengths of the radiations emitted as it drops back to the ground state.

Q.27 A doubly ionised lithium atom is hydrogen like with atomic number z = 3. Find the wavelength of the radiation required to excite the electron in $Li^{2+}$ from the first to the third Bohr orbit.

Q.28 Estimate the difference in energy between I and II Bohr Orbit for a hydrogen atom. At what minimum at no. a transition from n=2 to n=1 energy level would result in the emission of X–rays with $\lambda = 3.0\times10^{-8}$ m? Which hydrogen like species does this at no correspond to.

Q.29 Find out the no. of waves made by a Bohr electron in one complete revolution in its 3rd orbit.

Q.30 Iodine molecule dissociates into atoms after absorbing light of $4500A^0$. If one quantum of radiation is absorbed by each molecule, calculate the K.E. of iodine atoms
(Bond energy of $I_2$ = 240 KJ/mol)

Q.31 Calculate the wavelength of radiation emitted, producing a line in Lyman series, when an electron falls from fourth stationary state in hydrogen atom.

Q.32 Calculate the wave no. for the shortest wavelength transition in the Balmer series of atomic hydrogen.

## *GENERAL*

Q.33 What is de-Broglie wavelength of a He-atom in a container at room temperature.(Use $U_{avg}$)

Q.34 Through what potential difference must an electron pass to have a wavelength of 500 Å.

Q.35 A proton is accelerated to one- tenth of the velocity of light. If its velocity can be measured with a precision $\pm$ 1%. What must be its uncertainity in position.

Q.36 To what effective potential a proton beam be subjected to give its protons a wavelength of $1\times10^{-10}$ m.

Q.37 Calculate magnitude of angular momentum of an $e^-$ that occupies 1s, 2s , 2p , 3d , 3p.

Q.38 Calculate the number of exchange pairs of electrons present in configuration of Cu according to Aufbau Principle considering 3d & 4s orbitals.

Q.39 He atom can be excited to $1s^1\ 2p^1$ by $\lambda$ = 58.44 nm. If lowest excited state for He lies $4857cm^{-1}$ below the above. Calculate the energy for the lower excitation state.

Q.40 Wave functions of electrons in atoms & molecules are called_____.

Q.41 The outermost electronic conf. of Cr is_____.

## ***EXERCISE-II***

Q.1 X-rays emitted from a copper target and a molybdenum target are found to contain a line of wavelength 22.85 nm attributed to the $K_\alpha$ line of an impurity element. The $K_\alpha$ lines of copper (Z = 29) and molybdenum ( Z = 42) have wavelength 15.42 nm and 7.12 nm respectively. Using Moseley's law, $\gamma^{1/2} = a\,(Z - b)$ calculate the atomic number of the impurity element.

Q.2 1.8 g hydrogen atoms are excited to radiations. The study of spectra indicates that 27% of the atoms are in 3rd energy level and 15% of atoms in 2nd energy level and the rest in ground state. If I.P. of H is $21.7 \times 10^{-12}$ erg. Calculate –

(i) No. of atoms present in III & II energy level.

(ii) Total energy evolved when all the atoms return to ground state.

Q.3 One mole $He^+$ ions are excited. Spectral analysis showed existence of 50% ions in $3^{rd}$ orbit, 25% in $2^{nd}$ and rest in ground state. Calculate total energy evolved when all the ions return to the ground state.

Q.4 The energy of an excited H-atom is –3.4 eV. Calculate angular momentum of $e^-$.

Q.5 The vapours of Hg absorb some electrons accelerated by a potential diff. of 4.5 volt as a result of which light is emitted. If the full energy of single incident $e^-$ is supposed to be converted into light emitted by single Hg atom, find the wave no. of the light.

Q.6 The hydrogen atom in the ground state is excited by means of monochromatic radiation of wavelength x $A^0$. The resulting spectrum consists of 15 different lines . Calculate the value of x.

Q.7 The eyes of certain member of the reptile family pass a single visual signal to the brain when the visual receptors are struck by photons of wavelength 850 nm . If a total energy of $3.15 \times 10^{-14}$ J is required to trip the signal, what is the minimum number of photons that must strike the receptor.

Q.8 If the average life time of an excited state of H atom is of order $10^{-8}$ sec, estimate how many orbits an $e^-$ makes when it is in the state n = 2 and before it suffers a transition to n =1 state.

Q.9 Calculate the frequency of $e^-$ in the first Bohr orbit in a H-atom.

Q.10 A single electron orbits around a stationary nucleus o f charge +Ze where Z is a constant from the nucleus and e is the magnitude of the electric charge. The hydrogen like species required 47.2 eV to excite the electron from the second Bohr orbit to the third Bohr orbit. Find

(i) the value of Z and give the hydrogen like species formed.

(ii) the kinetic energy and potential energy of the electron in the first Bohr orbit.

Q.11 A stationary $He^+$ ion emitted a photon corresponding to a first line of the Lyman series. The photon liberated a photon electron from a stationary H atom in ground state. What is the velocity of photoelectron.

Q.12 To what series does the spectral lines of atomic hydrogen belong if its wave number is equal to the difference between the wave numbers of the following two lines of the Balmer series 486.1 and 410.2 nm. What is the wavelength of this.

Q.13 A particle of charge equal to that of an electron and mass 208 times the mass of the electron moves in a circular orbit around a nucleus of charge +3e. Assuming that the Bohr model of the atom is applicable to this system, (a) derive an expression for the radius of the nth bohr orbit, (b) find the value of n for which the radius of the orbit is approximately the same as that of the first Bohr orbit for th ehydrogen atom, and (c) find the wavelength of the radiation emitted when the revolving particle jumps from the third orbit to the first.

Q.14 A neutrons breaks into a proton and an electron. This decay of neutron is accompanied by release of energy. Assuming that 50% of the energy is produced in the form of electromagentic radiation, what will be the frequency of radiation produced. Will this photon be sufficient to cause ionization of Aluminium. In case it is able to do so what will be the energy of the electron ejected from the Aluminium atom. $IE_1$ = of Al = 577 KJ/mol

Q.15 Find the number of photons of radiation of frequency $5 \times 10^{13}\ s^{-1}$ that must be absorbed in order to melt one gm ice when the latent heat of fusion of ice is 330 J/g.

Q.16 The dye acriflavine, when dissolved in water, has its maximum light absorption at 4530 Å and its maximum fluorescence emission at 5080 Å. The number of fluorescence quanta is, on the average, 53% of the number of quanta absorbed. Using the wavelengths of maximum absorption and emission, what % of absorbed energy is emitted as fluorescence?

Q.17 Hydrogen atom in its ground state is excited by means of monochromatic radiation of wavelength 975Å. How many different lines are possible in the resulting spectrum? Calculate the longest wavelength amongst them.

Q.18 An alpha particle after passing through a potential difference of $2 \times 10^6$ volt falls on a silver foil. The atomic number of silver is 47. Calculate (i) the K.E. of the alpha-particle at the time of falling on the foil. (ii) K.E. of the $\alpha$ – particle at a distance of $5 \times 10^{-14}$m from the nucleus, (iii) the shortest distance from the nucleus of silver to which the $\alpha$–particle reaches.

Q.19 Suppose the potential energy between electron and proton at a distance r is given by $-\frac{ke^2}{3r^3}$. Use Bohr's theory to obtain energy of such a hypothetical atom.

Q.20 An energy of 68 eV is required to excite a hydrogen like atom from its second Bohr orbit to the third. The nuclear charge is Ze. Find the value of Z, the kinetic energy of the electron in the first Bohr orbit and the wavelength of the radiation required to eject the electrons from the first Bohr orbit to infinity.

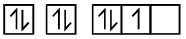
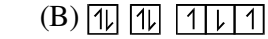
Q.21 A proton captures a free electron whose K.E. is zero & forms a hydrogen atom of lowest energy-level (n = 1). If a photon is emitted in this process, what will be the wavelength of radiation? In which region of electromagnetic spectrum, will this radiation fall? (Ionisation potential of hydrogen = 13.6 volt, $h = 6.6 \times 10^{-34}$K/s, $C = 3.0 \times 10^8$ m/s)

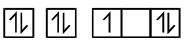
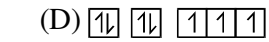
Q.22 The ionisation energy of the hydrogen atom is given to be 13.6 eV. A photon falls on a hydrogen atom which is initially in the ground state and excites it to the (n = 4)state.

(a) show this transition in the energy-level diagram &

(b) calculate the wavelength of the photon.

Q.23 Calculate Total spin and the multiplicity for each possible configuration of N-atom.

(A)  (B) 

(C)  (D) 

Q.24 Find the wavelength of the first line of $He^+$ ion spectral series whose interval between extreme line is

$$\left[\frac{1}{\lambda_1}-\frac{1}{\lambda_2}=2.7451\times10^4\ cm^{-1}\right]$$

Q.25 The ionisation energy of a H-like Bohr atom is 4 Rydbergs

(i) What is the wavelength of radiation emitted when the $e^-$ jumps from the first excited state to the ground state.

(ii) What is the radius of first Bohr orbit for this atom. [ 1 Rydberg = $2.18 \times 10^{-18}$ J]

## *DE-BROGLIE*

Q.26 What is de Broglie wavelength associated with an $e^-$ accelerated through P.D. = 100 KV.

Q.27 Calculate the de-broglie wavelength associated with motion of earth (mass $6 \times 10^{24}$ Kg) orbiting around the sun at a speed of $3 \times 10^6$ m/s.

## *HEISENBERG*

Q.28 A base ball of mass 200 g is moving with velocity $30 \times 10^2$ cm/s. If we can locate the base ball with an error equal in magnitude to the $\lambda$ of the light used (5000 Å), how will the uncertainty in momentum be compared with the total momentum of base ball.

Q.29 An electron has a speed of 40 m/s, accurate up to 99.99%. What is the uncertainity in locating its position.

## ***EXERCISE-III***

Q.1 The ratio of the energy of a photon of 2000 Å wavelength radiation to that of 4000 Å radiation is
(A) 1 / 4 (B) 4 (C) 1 / 2 (D) 2

Q.2 The maximum energy is present in any electron at
(A) Nucleus (B) Ground state
(C) First excited state (D) Infinite distance from the nucleus

Q.3 Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon
(A) 3s (B) 2p (C) 2s (D) 1s

Q.4 The third line in Balmer series corresponds to an electronic transition between which Bohr's orbits in hydrogen
(A) $5 \rightarrow 3$ (B) $5 \rightarrow 2$ (C) $4 \rightarrow 3$ (D) $4 \rightarrow 2$

Q.5 Correct set of four quantum numbers for valence electron of rubidium( $Z = 37$) is

(A) $5, 0, 0, +\frac{1}{2}$ (B) $5, 1, 0, +\frac{1}{2}$ (C) $5, 1, 1, +\frac{1}{2}$ (D) $6, 0, 0, +\frac{1}{2}$

Q.6 The correct set of quantum numbers for the unpaired electron of chlorine atom is

| | n | $l$ | m | | n | $l$ | m |
|---|---|---|---|---|---|---|---|
| (A) | 2 | 1 | 0 | (B) | 2 | 1 | 1 |
| (C) | 3 | 1 | 1 | (D) | 3 | 0 | 0 |

Q.7 The orbital diagram in which the Aufbau's principle is violated is

| | 2s | $2p_x$ | $2p_y$ | $2p_z$ | | 2s | $2p_x$ | $2p_y$ | $2p_z$ |
|---|---|---|---|---|---|---|---|---|---|
| (A) | ↑↓ | ↑↓ | ↑ | | (B) | ↑ | ↑↓ | ↑ | ↑ |
| (C) | ↑↓ | ↑ | ↑ | ↑ | (D) | ↑↓ | ↑↓ | ↑↓ | ↑ |

Q.8 The total number of neutrons in dipositive zinc ions with mass number 70 is
(A) 34 (B) 40 (C) 36 (D) 38

Q.9 Principal quantum number of an atom represents
(A) Size of the orbital (B) Spin angular momentum
(C) Orbital angular momentum (D) Space orientation of the orbital

Q.10 Which of the following sets of quantum numbers represent an impossible arrangement

| | n | $l$ | m | $m_s$ | | n | $l$ | m | $m_s$ |
|---|---|---|---|---|---|---|---|---|---|
| (A) | 3 | 2 | –2 | $\frac{1}{2}$ | (B) | 4 | 0 | 0 | $\frac{1}{2}$ |
| (C) | 3 | 2 | –3 | $\frac{1}{2}$ | (D) | 5 | 3 | 0 | $\frac{1}{2}$ |

Q.11 The orbital angular momentum of an electron in 2s orbital is:

(A) $+\frac{1}{2}.\frac{\hbar}{2\pi}$ (B) Zero (C) $\frac{\hbar}{2\pi}$ (D) $\sqrt{2}.\frac{\hbar}{2\pi}$

Q.12 The explanation for the presence of three unpaired electrons in the nitrogen atom can be given by
(A) Pauli's exclusion principle (B) Hund's rule
(C) Aufbau's principle (D) Uncertainty principle

Q.13 The maximum number of electrons that can be accommodated in the $M^{th}$ shell is
(A) 2 (B) 8 (C) 18 (D) 32

Q.14 Which quantum number will determine the shape of the subshell
(A) Principal quantum number (B) Azimuthal quantum number
(C) Magnetic quantum number (D) Spin quantum number

Q.15 The electronic configuration of an element is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$. This represents its
(A) Excited state (B) Ground state (C) Cationic form (D) None

Q.16 Which of the following has maximum number of unpaired electron (atomic number of Fe 26)
(A) Fe (B) Fe (II) (C) Fe (III) (D) Fe (IV)

Q.17 Which quantum number is not related with Schrodinger equation
(A) Principal (B) Azimuthal (C) Magnetic (D) Spin

Q.18 According to Bohr's atomic theory, which of the following is/are correct:

(I) $\text{Kinetic energy of electron} \propto \frac{Z^2}{n^2}$

(II) The product of velocity of electron and principle quantum number 'n' $\propto Z^2$

(III) Frequency of revolution of electron in an orbit $\propto \frac{Z^2}{n^3}$

(IV) Coulombic force of attraction on the electron $\propto \frac{Z^3}{n^4}$

(A) I, III, IV (B) I, IV (C) II (D) I

Q.19 If $\lambda_0$ is the threshold wavelength for photoelectric emission, $\lambda$ wavelength of light falling on the surface of metal, and m, mass of electron, then de Broglie wavelength of emitted electron is

(A) $\left[\frac{h(\lambda\lambda_0)}{2mc(\lambda_0-\lambda)}\right]^{\frac{1}{2}}$ (B) $\left[\frac{h(\lambda_0-\lambda)}{2mc\lambda\lambda_0}\right]^{\frac{1}{2}}$ (C) $\left[\frac{h(\lambda-\lambda_0)}{2mc\lambda\lambda_0}\right]^{\frac{1}{2}}$ (D) $\left[\frac{h\lambda\lambda_0}{2mc}\right]^{\frac{1}{2}}$

Q.20 It is known that atom contain protons, neutrons and electrons. If the mass of neutron is assumed to half of its original value where as that of proton is assumed to be twice of its original value then the atomic mass of ${}^{14}_{6}C$ will be
(A) same (B) 25% more (C) 14.28 % more (D) 28.5% less

Q.21 Give the correct order of initials **T** (true) or **F** (false) for following statements.
(I) If an ion has 2 electrons in K shell, 8 electrons in L shell and 6 electrons in M shell, then number of S electrons present in that element is 6.
(II) The maximum number of electrons in a subshell is given by $2n^2$.
(III) If electron has magnetic number –1, then it cannot be present in s-orbital.
(IV) Only one radial node is present in 3p orbital.
(A) TTFF (B) FFTF (C) TFTT (D) FFTF

Q.22 Predict the magnetic moment for $S^{2-}$, $Co^{3+}$ [At. no. of S = 16, Co = 27]

Q.23 The critical wavelength for producing the photoelectric effect in tungsten is 2600Å. What wavelength would be necessary to produce photoelectrons from tungston having twice the kinetic energy of these produced at 2200Å ?

Q.24 The shortest wavelength of He atom in Balmer series is x, then longest wavelength in the Paschene series of $Li^{+2}$ is

(A) $\frac{36x}{5}$ (B) $\frac{16x}{7}$ (C) $\frac{9x}{5}$ (D) $\frac{5x}{9}$

Q.25 An electron in a hydrogen atom in its ground state absorbs energy equal to the ionisation energy of $Li^{+2}$. The wavelength of the emitted electron is:

(A) $3.32 \times 10^{-10}$ m (B) 1.17 Å (C) $2.32 \times 10^{-9}$ nm (D) 3.33 pm

Q.26 In compound $FeCl_2$ the orbital angular momentum of last electron in its cation & magnetic moment (in Bohr Magneton) of this compound are

(A) $(\sqrt{6})\hbar, \sqrt{35}$ (B) $(\sqrt{6})\hbar, \sqrt{24}$ (C) 0, $\sqrt{35}$ (D) none of these

Q.27 An electron, a proton and an alpha particle have kinetic energies of 16E, 4E and E respectively. What is the qualitative order of their de Broglie wavelengths?

(A) $\lambda_e > \lambda_p = \lambda_\alpha$ (B) $\lambda_p = \lambda_\alpha > \lambda_e$ (C) $\lambda_p > \lambda_e > \lambda_\alpha$ (D) $\lambda_\alpha < \lambda_e \gg \lambda_p$

Q.28 **Question**: Is the specie paramagnetic?

STAT-1: The atomic number of specie is 29.

STAT-2: The charge on the specie is + 1.

(A) Statements (1) alone is sufficient but statement (2) is not sufficient.

(B) Statement (2) alone is sufficient but statement (1) is not sufficient.

(C) Both statement together are sufficient but neither statement alone is sufficient.

(D) Statement (1) & (2) together are not sufficient.

Q.29 **Question :** Are the rays in discharge tube cathode rays?

STAT1 : Rays are deflected towards – ve electrode kept externally.

STAT2 : Rays are produced at low pressure and high voltage.

(A) Statements (1) alone is sufficient but statement (2) is not sufficient.

(B) Statement (2) alone is sufficient but statement (1) is not sufficient.

(C) Both statement together are sufficient but neither statement alone is sufficient.

(D) Any one of them is sufficient.

Q.30 Given $\Delta H$ for the process $Li(g) \longrightarrow Li^{+3}(g) + 3e^-$ is 19800 kJ/mole & $IE_1$ for Li is 520 then $IE_2$ & $IE_1$ of $Li^+$ are respectively (approx, value)

(A) 11775, 7505 (B) 19280, 520 (C) 11775, 19280 (D) Data insufficient

Q.31 The ratio of difference in wavelengths of 1st and 2nd lines of Lyman series in H–like atom to difference in wavelength for 2nd and 3rd lines of same series is:

(A) 2.5 : 1 (B) 3.5 : 1 (C) 4.5 : 1 (D) 5.5 : 1

Q.32 Which of the following statement is INCORRECT.

(A) $\frac{e}{m}$ ratio for canal rays is maximum for hydrogen ion.

(B) $\frac{e}{m}$ ratio for cathode rays us independent of the gas taken.

(C) The nature of canal rays is dependent on the electrode material.

(D) The $\frac{e}{m}$ ratio for electron is expressed as $\frac{E^2}{2B^2V}$, when the cathode rays go undeflected under the influence of electric field E, magnetic field B and V is potential difference applied across electrodes.

Q.33 The quantum numbers of four electrons (e1 to e4) are given below

| | n | $l$ | m | s | | n | $l$ | m | s |
|---|---|---|---|---|---|---|---|---|---|
| e1 | 3 | 0 | 0 | +1/2 | e2 | 4 | 0 | 1 | 1/2 |
| e3 | 3 | 2 | 2 | –1/2 | e4 | 3 | 1 | –1 | 1/2 |

The correct order of decreasing energy of these electrons is:
(A) e4 > e3 > e2 > e1 (B) e2 > e3 > e4 > e1 (C) e3 > e2 > e4 > e1(D) none

Q.34 If radius of second stationary orbit (in Bohr's atom) is R. Then radius of third orbit will be
(A) R/3 (B) 9R (C) R/9 (D) 2.25R

Q.35 An electron in a hydrogen atom in its ground state absorbs 1.5 times as much energy as the minimum required for it to escape from the atom. What is the velocity of the emitted electron?
(Given mass of $e^- = 9.1 \times 10^{-28}$ gm)

***PROBLEM ON DE-BROGLIE, HEISENBERG & SCHRODINGER EQUATIONS***

Q.36 An electron can undergo diffraction by crystals . Through what potential should a beam of electron be accelerated so that its wavelength become equal to 1.54 A° .

Q.37 The first use of quantum theory to explain the structure of atom was made by :
(A) Heisenburg (B) Bohr (C) Planck (D) Einstein

Q.38 The wavelength associated with a golf weighing 200g and moving at a speed of 5m/h is of the order
(A) $10^{-10}$m (B) $10^{-20}$m (C) $10^{-30}$m (D) $10^{-40}$m

Q.39 If the nitrogen atom had electronic configuration $1s^7$, it would have energy lower that of normal ground state configuration $1s^2\ 2s^2\ 2p^3$ , because the electrons would be closer to the nucleus. Yet $1s^7$ is not observed because it violates :–
(A) Heisenberg uncertainity principle (B) Hunds rule
(C) Pauli's exclusion principle (D) Bohr postulate of stationary orbits

Q.40 Wavelength of high energy transition of H-atoms is 91.2 nm. Calculate the corresponding wavelength of He atoms.

Q.41(i) The wave function of 2s electron is given by

$$\Psi 2s = \frac{1}{4\sqrt{2\pi}}\left(\frac{1}{a_o}\right)^{3/2}\left(2-\frac{r}{a_o}\right)e^{\left(2-\frac{r}{a_o}\right)}$$

It has a node at $r = r_0$, find relation between $r_0$ and $a_0$.

(ii) Find wavelength for 100 g particle moving with velocity 100 $ms^{-1}$.

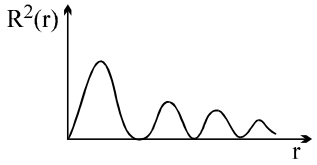
Q.42 The electron in the first excited state of H-atom absorbs a proton and is further excited. the Debroglie wavelength of the electron in this excited state is 1340 pm. Calculate the wavelength of photon absorbed by the atom and also longest wavelength radiation emitted when this electron de-excited to ground state.

Q.43 The uncertainity principle may be stated mathematically

$$\Delta p.\Delta x \approx \frac{h}{4\pi}$$

where $\Delta p$ represents the uncertainity in the momentum of a particle and $\Delta x$ represnts the uncertainity in its position. If an electron is traveling at 200 m/s within 1 m/s uncertainity, what is the theoretical uncertainity in its position in μm (micrometer)?

Q.44 From the following observations predict the type of orbital:
Observation 1: x y plane acts as a nodal plane
Observation 2: The angular function of the orbital intersect the three axis at origin only.
Observation 3: $R^2(r)$ / v/s r curve is obtained for the orbital is




(A) $5p_z$ (B) $6d_{xy}$ (C) $6\ dx^2–y^2$ (D) $6\ d_{yz}$

Q.45 **Question :** Is the orbital of hydrogen atom $3p_x$?

STAT 1: The radial function of the orbital is $R(r) = \frac{1}{9\sqrt{6}\,a_0^{3/2}}(4-\sigma)\sigma e^{-\sigma/2}, \sigma = \frac{r}{2}$

STAT 2: The orbital has 1 radial node & 0 angular node.

(A) Statement (1) alone is sufficient. (B) Statement (2) alone is sufficient
(C) Both together is sufficient. (D) Neither is sufficient

Q.46 What is uncertainity in location of a photon of wavelength 5000Å if wavelength is known to an accuracy of 1 pm?

(A) $7.96 \times 10^{-14}$ m (B) 0.02 m (C) $3.9 \times 10^{-8}$ m (D) none

## ***EXERCISE-IV***

Q.1 With what velocity should an $\alpha$–particle travel towards the nucleus of a Cu atom so as to arrive at a distance $10^{-13}$ m. **[JEE 1997]**

Q.2 A compound of Vanadium has magnetic moment of 1.73 BM work out electronic configuration of Vanadium Ion in the compound. **[JEE 1997]**

Q.3 The energy of an electron in the first Bohr orbit of H atom is – 13.6 eV . The possible energy value(s) of the excited state(s) for electrons in Bohr orbits of hydrogen is/are :
(A) – 3.4 eV (B) – 4.2 eV (C) – 6.8 eV (D) + 6.8 eV **[JEE 1998]**

Q.4 The electrons, identified by n & $l$ ; (i) n = 4 , $l$ = 1 (ii) n = 4 , $l$ = 0
(iii) n = 3 , $l$ = 2 (iv) n = 3 , $l$ = 1 can be placed in order of increasing energy, from the lowest to highest as :
(A) (iv) < (ii) < (iii) < (i) (B) (ii) < (iv) < (i)
(C) (i) < (iii) < (ii) < (iv) (D) (iii) < (i) < (iv) < (ii) **[JEE 1999]**

Q.5 Gaseous state electronic configuration of nitrogen atom can be represented as:

(A) ↑↓ ↑↓ ↑ ↑ ↑ (B) ↑↓ ↑↓ ↑ ↓ ↑

(C) ↑↓ ↑↓ ↑ ↓ ↓ (D) ↑↓ ↑↓ ↓ ↓ ↓ **[JEE 1999]**

Q.6 The electronic configuration of an element is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$. This represents its:
(A) excited state (B) ground state (C) cationic form (D) none **[JEE 2000]**

Q.7 The number of nodal planes in a $p_x$ orbital is:
(A) one (B) two (C) three (D) zero **[JEE 2000]**

Q.8 Calculate the energy required to excite one litre of hydrogen gas at 1 atmp and 298K to the first excited state of atomic hydrogen. The energy for the dissociation of H – H is 436 KJ $mol^{-1}$.

Q.9 The quantum numbers +1/2 and –1/2 for the electron spin represent:
(A) rotation of the electron in clockwise and anticlockwise direction respectively.
(B) rotation of the electron in anticlockwise and clockwise direction respectively.
(C) magnetic moment of the electron pointing up and down respectively.
(D) two quantum mechanical spin states which have no classical analogue. **[JEE 2001]**

Q.10 Rutherfords experiment , which established the nuclear model of atom, used a beam of :–
(A) $\beta$ - particles, which impinged on a metal foil and get absorbed.
(B) $\gamma$ - rays, which impinged on a metal foil and ejected electron.
(C) Helium atoms, which impinged on a metal foil and got scattered.
(D) Helium nuclie, which impinged on a metal foil and got scattered. **[JEE 2002]**

Q.11 The spin magnetic moment of cobalt of the compund $Hg[Co(SCN)_4]$ is [Given : $Co^{+2}$]
(A) $\sqrt{3}$ (B) $\sqrt{8}$ (C) $\sqrt{15}$ (D) $\sqrt{24}$ **[JEE 2004]**

Q.12 The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom?
(A) $He^+$ (n = 2) (B) $Li^{2+}$ (n = 2) (C) $Li^{2+}$ (n = 3) (D) $Be^{3+}$ (n = 2)
**[JEE 2004]**

Q.13 Given in hydrogenic atom $r_n$, $V_n$, E, $K_n$ stand for radius, potential energy, total energy and kinetic energy in $n^{th}$ orbit. Find the value of U,v,x,y. **[JEE 2006]**

| | | | |
|---|---|---|---|
| (A) | $U = \frac{V_n}{K_n}$ | (P) | 1 |
| (B) | $\frac{1}{r_n} \propto E^x$ | (Q) | –2 |
| (C) | $r_n \propto Z^y$ (Z = Atomic number) | (R) | –1 |
| (D) | v = (Angular momentum of electron in its lowest energy ) | (S) | 0 |

# ANSWER KEY

## EXERCISE -I

### LIGHT

Q.1 6563 Å ; 1216 Å ; 1026 Å  Q.2 6  Q.3 0.527
Q.4 6235 Å  Q.5 4863 Å  Q.6 $1.096 \times 10^7\ m^{-1}$
Q.7 $3 \times 10^{21}$  Q.8 photons  Q.9 $n_1 = 1, n_2 = 2$  Q.10 $1.827 \times 10^5$ J/mol

### PLANCK'S QUANTUM THEORY

Q.11 $4.9 \times 10^{-7}$ m  Q.12 28 photons  Q.13 1403 KJ/mol  Q.14 $4.5 \times 10^{14}\ s^{-1}$
Q.15 497 KJ/mol  Q.16 319.2 KJ/mol  Q.17 $6.57 \times 10^{-34}$ Js
Q.18 8.68 %  Q.19 0.62 Å  Q.20 3.06 V

### BOHR'S MODEL

Q.21 $-1.36 \times 10^{-19}$ Joules  Q.22 $5.425 \times 10^{-12}$ ergs, $3.7 \times 10^{-5}$ cm
Q.23 1220 Å  Q.24 $5.44 \times 10^5$ m/s  Q.25 2 ; $9.75 \times 10^4\ cm^{-1}$
Q.26 3 , 6563 Å , 1215 Å , 1026 Å  Q.27 113.74 Å
Q.28 10.2 eV , z = 2  Q.29 3  Q.30 $2.186 \times 10^{-20}$ Joules
Q.31 $9.7 \times 10^{-8}$ m  Q.32 27419.25 $cm^{-1}$

### GENERAL

Q.33 0.79 Å  Q.34 $6.03 \times 10^{-4}$ volt  Q.35 $1.05 \times 10^{-13}$ m
Q.36 0.0826 volts  Q.37 0 ; 0 ; $\sqrt{2}\frac{h}{2\pi}$ ; $\sqrt{6}\frac{h}{2\pi}$ ; $\sqrt{2}\frac{h}{2\pi}$  Q.38 25
Q.39 $3.3 \times 10^{-18}$ J  Q.40 orbitals  Q.41 $3s^2\ 3p^6\ 3d^5\ 4s^1$

## EXERCISE-II

Q.1 24
Q.2 $292.68 \times 10^{21}$ atoms, $162.60 \times 10^{21}$ atoms, 832.50 KJ  Q.3 $331.13 \times 10^4$ J
Q.4 $h/\pi$  Q.5 $3.63 \times 10^6\ m^{-1}$  Q.6 938 Å  Q.7 $1.35 \times 10^5$
Q.8 $8 \times 10^6$  Q.9 $6530 \times 10^{12}$ Hz  Q.10 5 ; 340 ev , – 680 eV
Q.11 $3.09 \times 10^8$ cm/sec  Q.12 Brackett ; $2.63 \times 10^{-4}$ cm
Q.13 $r_n = \frac{n^2 h^2}{4K\pi^2 \times 3e^2 \times 208\, m_e}$  n = 25 ; 55.2 pm  Q.14 $9.15 \times 10^{19}$ Hz , yes, $58.5 \times 10^{-15}$ J
Q.15 $10^{22}$  Q.16 47.26%  Q.17 six , 18800 Å
Q.18 $6.4 \times 10^{-13}$ J, $2.1 \times 10^{-13}$J, $3.4 \times 10^{-14}$m  Q.19 $E = \frac{n^6 h^6}{384\, m^3\ K^2\ e^4\ \pi^6}$
Q.20 6 ; 489.6 eV , 25.28 Å  Q.21 910 Å ;U.V.  Q.22 973.5 Å
Q.23 +1/2 , +1/2 , +1/2 , +3/2 and 2,2,2,4  Q.24 4689 Å
Q.25 303.89 Å , $2.645 \times 10^{-9}$ cm

### DE-BROGLIE

Q.26 3.88 pm  Q.27 $3.68 \times 10^{-65}$ m

### HEISENBERG

Q.28 $1.75 \times 10^{-29}$  Q.29 0.0144 m

## ***EXERCISE-III***

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Q.1 | D | Q.2 | D | Q.3 | D | Q.4 | B | Q.5 | A | Q.6 | C | Q.7 | B |
| Q.8 | B | Q.9 | A | Q.10 | C | Q.11 | B | Q.12 | B | Q.13 | C | Q.14 | B |
| Q.15 | B | Q.16 | C | Q.17 | D | Q.18 | A | Q.19 | A | Q.20 | C | Q.21 | C |
| Q.22 | zero, 4.9 B.M. | | | Q.23 | $\lambda = 1900\text{Å}$ | | | Q.24 | B | Q.25 | B | Q.26 | B |
| Q.27 | A | Q.28 | C | Q.29 | A | Q.30 | A | Q.31 | B | Q.32 | C | Q.33 | C |
| Q.34 | D | Q.35 | $1.54 \times 10^6$m/s | | | | | | | | | | |

***PROBLEM ON DE-BROGLIE, HEISENBERG & SCHRODINGER EQUATIONS***

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Q.36 | 63.12 volts | Q.37 | B | Q.38 | C | | | Q.39 | C |
| Q.40 | 22.8 nm | Q.41 | (i) $r_0 = 2a_0$, (ii) $6.626 \times 10^{-35}$ m | | | | | Q.42 | 4860 Å, 18788 Å |
| Q.43 | $\approx 2.9 \times 2 \times 10^{-5}$ m $\approx$ 58 μm | | | Q.44 | D | Q.45 | B | Q.46 | B |

## ***EXERCISE-IV***

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Q.1 | $6.3 \times 10^6$ m/s | Q.2 | [Ar] $3d^1$ | Q.3 | A | Q.4 | A |
| Q.5 | A,D | Q.6 | B,C | Q.7 | A | Q.8 | 97.819 KJ |
| Q.9 | D | Q.10 | D | Q.11 | C | Q.12 | D |
| Q.13 | (A) Q, (B) P, (C) R, (D) S | | | | | | |

# STUDY PACKAGE

# Target: IIT-JEE (Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 4. Chemical Bonding

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

**4. Exercise III**

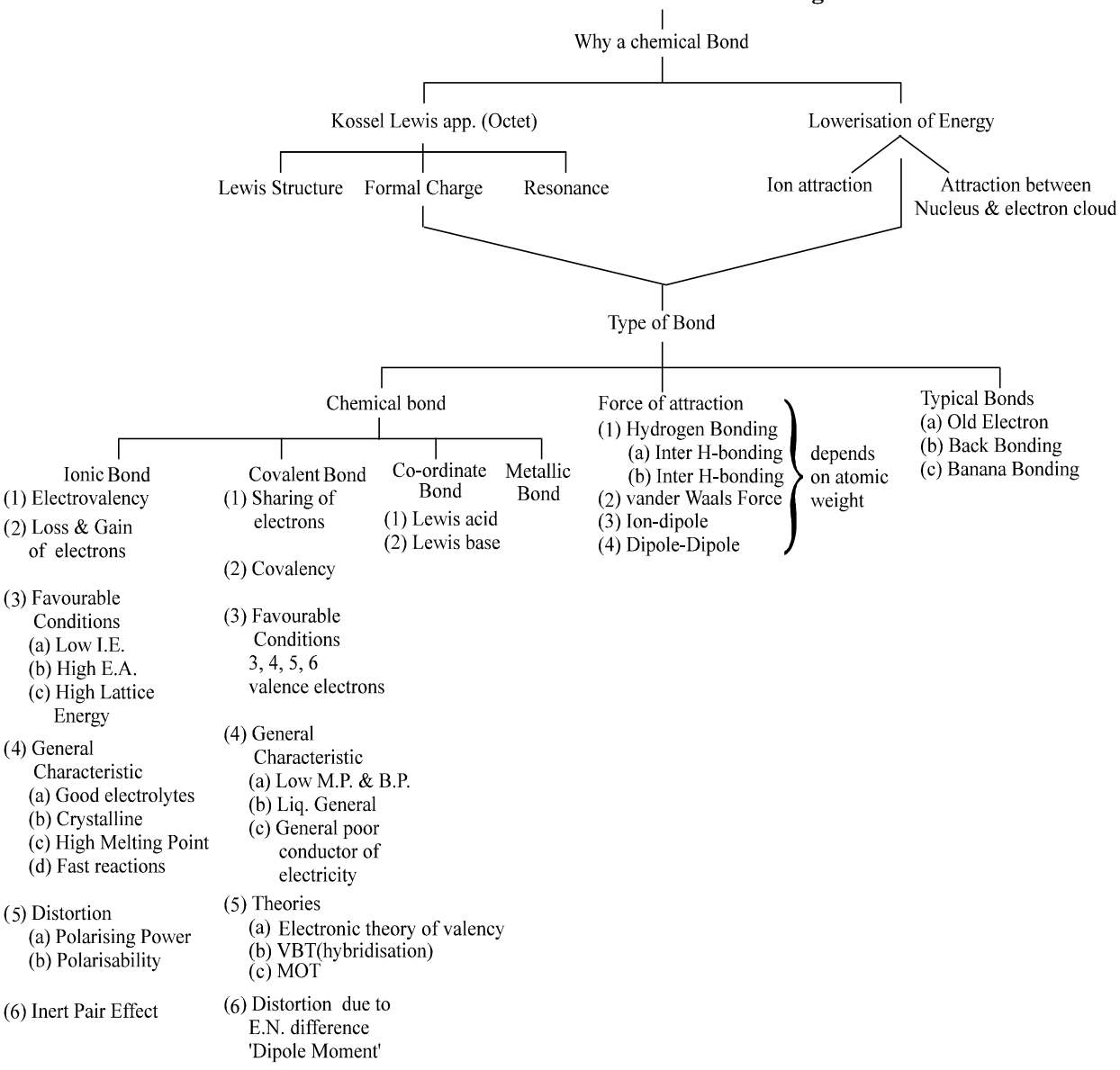
**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

## Flowchart to Chemical Bonding



# ***KEY CONCEPT***

***Reasons for Bond formation:***

- ✦ Lowerisation of energy due to attractions.
- ✦ Attainment of Octet [$ns^2\ np^6$], assumed to be most stable.

***Types of bonds*** : Ionic, covalent, co-ordinate

## *IONIC BOND* [ELECTROVALENT]

- ✦ Complete loss of $e^-$ to form ions.
- ✦ Electrostatic attraction between ions.
- ✦ Elements of 'p' & 'd' block may show variable electrovalency due to

(a) **Inert Pair effect (for p block):** The reluctance of 's' electron pair to take part in bond formation on moving down a group in 'P' block elements.

***Finds application in***

- ✦ Stability of oxidation state of a particular metal atom.
- ✦ Oxidizing & reducing power of compounds.

(b) **Unstability of core:** For 'd' block elements the core may either have pseudo inert configuration (having 18 electrons in outermost shell )or any other.

***Properties of Ionic compounds***

- ✦ Ionic bonds are Non directional in nature
- ✦ High Melting points / Boiling points.
- ✦ In solid state they are conductor ( due to absence of charge carrier) while in aqueous & molten state they are good conductor of electricity.
- ✦ Soluble in Polar solvents
- ✦ Show isomorphism.

☹ No sp. theories to understand bond formation.

☺ characteristics like various crystal lattices to be done in solid state.

## *COVALENT BOND*

- ✦ Sharing of electrons
- ✦ Overlapping of orbitals
- ✦ Types : single, double, triple, polar, non–polar bonds.
- ✦ Variable covalency : Shown by elements having vacant 'd' orbitals (caused due to excitation of the electron.)

***Properties:***

- ✦ Covalent bonds are directional in nature
- ✦ Low melting point & boiling point. (except Diamond / Graphite, due to their peculiear structure)
- ✦ Electrical conductivity due to auto-protolysis or self ionisation.
- ✦ Show isomerism.

## *CO-ORDINATE BOND*

Bonding between lewis acid & lewis base or electron deficient & electron rich species.

**Lewis base:** Species with lone pair on 'central atom' available for donation. eg. $NH_3$, $H_2O$

**Lewis acid:** Electron deficient due to incomplete octal, vacant p or d orbital & high + ve $\frac{\text{charge}}{\text{size}}$ ratio.

### *Lewis Dot structures:*

- ✦ Arrangement of various atoms & types of bonding present but no idea of geometry.
- ✦ Selection of central atom [least E.N. of all elements excluding hydrogen]
- ✦ In hydrogen containing proton donor oxy acids all 'H' atoms are attached to oxygen as –OH groups except in $H_3PO_3$(dibasic), $H_3PO_2$(monobasic) & $H_4P_2O_5$ (dibasic).

***Applications:***

- ✦ To know various linkages present
- ✦ To calculate O.S. of various elements.

### *Various Theories For Explaining Bonding*

- ✦ Electronic theory of valency (Kossel, Lewis) ;Singlat linkages
- ✦ Valence bond theory (Heitler, London, Pauling, Slater).
- ✦ M.O.T. (Hund , Mulliken). ***(will be discussed in class XII th)***

## *DIPOLE MOMENT*

Dipole moment is a vector quantity = $\mu = q \times d$. Units = col m (S.I.) or esu cm(cgs) or Debey(common unit) 1 D = $10^{-18}$ esu cm = $3.33 \times 10^{-30}$ col. m

$$\% \text{ ionic character} = \frac{\text{observed D.M.}}{\text{calculated D.M. for 100\% ionic}} \times 100\ \%$$

$$\% \text{ ionic character} = [\,16\,(X_A - X_B) + 3.5\,(X_A - X_B)^2\,]\ \%$$ [Hanny & Smyth equation]

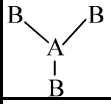
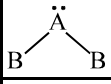
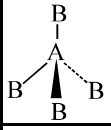
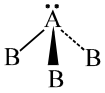
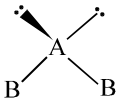
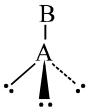
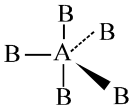
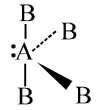
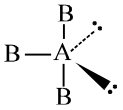
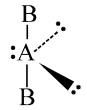
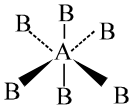
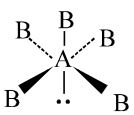
***Dipole moment depends on***

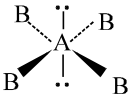
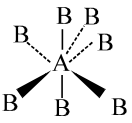
- ✦ Electronegativity difference between atoms
- ✦ Angle between various bonds
- ✦ Magnetic of polarity of the molecule
- ✦ Direction of bond dipole moment
- ✦ Influence of unshared $e^-$ pairs
- ✦ Symmetrical / Unsymmetrical shape.

**Bond Moments:**

| | | | |
|---|---|---|---|
| H–F (1.9 D) | H–O (1.5 D) | C – C (0 D) | C–F (1.4 D) |
| H–Cl (1.1 D) | H–N (1.3 D) | C = O (2.3 D) | C–Cl (1.5 D) |
| H – Br (0.8 D) | H– C (0.4 D) | | C – Br (1.4 D) |
| H – I (0.4 D) | | | C – I (1.2 D) |

## *SHAPES OF MOLECULES BASED ON VSEPR THEORY*

| **Total no. of hybrid orbitals** | **No. of b.p. (bond pairs)** | **No. of unshared pair i.e. $l_p$** | **General formula** | **Type of hybridisations** | **Stereo chemical formula** | **Shape** | **Exam.** |
|---|---|---|---|---|---|---|---|
| 2 | 2 | 0 | $AB_2$ | sp | B–A–B | linear | $BeCl_2$ |
| 3 | 3 | 0 | $AB_3$ | $sp^2$ |  | Trigonal planar | $BCl_3$, $GaF_3$ |
| 3 | 2 | 1 | $AB_2$ | $sp^2$ |  | Bent or angular | $GeF_2$, $O_3$ |
| 4 | 4 | 0 | $AB_4$ | $sp^3$ |  | Tetrahedral | $CH_4$ |
| 4 | 3 | 1 | $AB_3$ | $sp^3$ |  | Trigonal pyramid | $NH_3$ |
| 4 | 2 | 2 | $AB_2$ | $sp^3$ |  | Bent or angular | $H_2O$ |
| 4 | 1 | 3 | AB | $sp^3$ |  | linear | HF |
| 5 | 5 | 0 | $AB_5$ | $sp^3d$ |  | Trigonal bipyramidal | $PF_5$, $NbBr_5$ |
| 5 | 4 | 1 | $AB_4$ | $sp^3d$ |  | Seesaw | $SF_4$ |
| 5 | 3 | 2 | $AB_3$ | $sp^3d$ |  | T-shaped | $ClF_3$, $BrF_3$ |
| 5 | 2 | 3 | $AB_2$ | $sp^3d$ |  | Linear | $ICl_2^-$ $XeF_2$ |
| 6 | 6 | 0 | $AB_6$ | $sp^3d^2$ |  | Octahedral | $SF_6$ |
| 6 | 5 | 1 | $AB_5$ | $sp^3d^2$ |  | Square pyramidal | $IF_5$ |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6 | 4 | 2 | $AB_4$ | $sp^3d^2$ |  | Square planar | $IF_4$ $XeF_4$ |
| 7 | 7 | 0 | $AB_7$ | $sp^3d^3$ |  | Pentagonal bipyramidal | $IF_7$ |

**How To Decide The Type Of Hybridisation :**

Type of hybridisation = (number of σ bonds + number of lone pairs)

## *RESONANCE*

- ✦ Delocalisations of π electron cloud in between orbitals of various atoms in a molecule (provided all the atoms are in the same plane)
- ✦ Exists where more than one Lewis dot structure are possible for a molecule.
- ✦ Resonance causes stablisation of the molecule & difference in the energies of hybrid & other structure is termed as Resonance energy.
- ✦ R.E. → Experimental heat of formation-Theoretical heat of formation.
- ✦ The properties of the actual structure (Resonance hybrid) are decided by the weighed average (depending on stability) of the contributing molecule.
- ✦ More the resonating structure more stable the molecule becomes.

## *FORCES OF ATTRACTION (WEAKER BONDS)*

- ✦ **Hydrogen bonding:** When a hydrogen atom is linked to a highly electronegative atom (like F, O or N) comes under the influence of another strongly electronegative atom, then a weak bond is developed between them, which is called as hydrogen bond.

***Types of H-bonding:***

- ✦ Intermolecular
- ✦ Intramolecular

***Applications in:***

(a) Abnormal behaviour of water.
(b) Association of a molecule as in carboxylic acid.
(c) Dissociation of a polar species.
(d) Abnormal melting point & boiling point.
(e) Enhanced solubility in water.

- ✦ **Ion dipole attraction**
- ✦ **Dipole-dipole attraction**
- ✦ **Ion-induced dipole attraction**
- ✦ **Dipole-Induced Dipole attraction**
- ✦ **Induced -dipole Induced Dipole attraction**
- ✦ **Metallic bonds:** Electron gas model or sea model, with metal atom existing as kernels along with less firmly held valence $e^{-s}$ & bonds between various kernels ( at the lattice site) & valence $e^{-s}$ is known as metallic bonds.

## *SOME TYPICAL BOND*

1. ***Back bonding:***

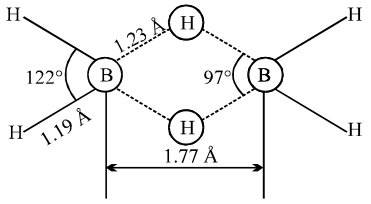
- ✦ If among the bonded atoms, one atom has a vacant orbital & another has excess of $e^{-s}$ then a sort of π bonding takes place between he two. If this is between 'P' orbitals of the two, this is known as pπ-pπ back bonding.

✦ Most efficient when the atoms are very small & the orbitals involved of the two are of same energy level.

2. ***Banana bond:***

✦ This type of bonding is present in $B_2H_6$.

✦ This structure shows that there are two types of hydrogen atom-Terminals and bridging.

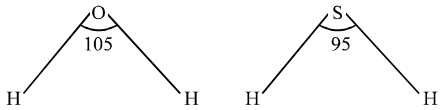



## *MISCELLANEOUS CONCEPT*

1. ***Comparison of bond angles.***

(a) In case central atoms are having different hybridisation then it can be compared.

(b) If same hybridisation but different central atom then bond angle would be more of the molecule in which C.A. is more E.N. eg. $H_2S$ & $H_2O$.




(c) If C.A. is same & bonded atoms different then bond angle increases as the attached atom size increases.

2. ***Bond strength , Bond length & Paramagnetic nature***

(a) Using concepts of resonance.

(b) Use of Lewis dot structure for the rest.

## *EXERCISE - I*

### *IONIC BOND*

Q.1 The combination of atoms take place so that
(A) They can gain two electrons in the outermost shell
(B) They get eight electrons in the outermost shell
(C) They acquire stability by lowering of energy
(D) They get eighteen electrons in the outermost shell.

Q.2 An ionic bond $A^+B^-$ is most likely to be formed when :
(A) the ionization energy of *A* is high and the electron affinity of *B* is low
(B) the ionization energy of *A* is low and the electron affinity of *B* is high
(C) the ionization energy of *A* and the electron affinity of *B* is high
(D) the ionization energy of *A* and the electron affinity of *B* is low

Q.3 Which of the following compounds of elements in group *IV* is expected to be most ionic ?
(A) $PbCl_2$ (B) $PbCl_4$ (C) $CCl_4$ (D) $SiCl_4$

Q.4 The compound which contains ionic as well as covalent bonds is
(A) $C_2H_4Cl_2$ (B) $CH_3I$ (C) KCN (D) $H_2O_2$

Q.5 The hydration of ionic compounds involves :
(A) Evolution of heat (B) Weakening of attractive forces
(C) Dissociation into ions (D) All of these

Q.6 In which of the following species the bonds are Non-directional ?
(A) $NCl_3$ (B) RbCl (C) $BeCl_2$ (D) $BCl_3$

Q.7 Which has the lowest anion to cation size ratio :
(A) *LiF* (B) *NaF* (C) *CsI* (D) *CsF*

Q.8 Which of the following statement(s) is/are correct regarding ionic compounds?
(A) They are good conductors at room temperature in aqueous solution.
(B) They are generally soluble in polar solvents.
(C) They consist of ions.
(D) They generally have high melting and boiling points.

Q.9 Which of the following compounds contain/s both ionic and covalent bonds?
(A) $NH_4Cl$ (B) KCN (C) $CuSO_4 \cdot 5H_2O$ (D) NaOH

Q.10 Among the following isostructural compounds, identify the compound, which has the highest Lattice energy
(A) LiF (B) LiCl (C) NaCl (D) MgO

Q.11 A bond formed between two like atoms cannot be
(A) ionic (B) covalent (C) coordinate (D) metallic

Q.12 Which of the following, when dissolved in water forms a solution, which is Non-conductivity?
(A) Green Vitriol (B) Indian salt Petre
(C) Alcohol (D) Potash alum

Q.13 Most ionic compounds have :
(A) high melting points and low boiling points
(B) high melting points and nondirectional bonds
(C) high solubilities in polar solvents and low solubilities in nonpolar solvents
(D) three-dimensional network structures, and are good conductors of electricity in the molten state

Q.14 An electrovalent compound does not exhibit space isomerism because of
(A) Presence of oppositively charged ions
(B) High melting points
(C) Non-directional nature of the bond
(D) Crystalline nature

Q.15 Which of the following have an (18 + 2) electron configuration ?
(A) $Pb^{2+}$ (B) $Cd^{2+}$ (C) $Bi^{3+}$ (D) $SO_4^{2-}$

Q.16 Which of the following contains (electrovalent) and non-polar (covalent) bonds ?
(A) $CH_4$ (B) $H_2O_2$ (C) $NH_4Cl$ (D) $HCN$

***COVALENT BOND***

Q.17 A sigma bond may be formed by the overlap of 2 atomic orbitals of atoms *A* and *B*. If the bond is formed along as the *x*-axis, which of the following overlaps is acceptable ?
(A) *s* orbital of *A* and $p_z$ orbital of *B* (B) $p_x$ orbital of *A* and $p_y$ orbital of *B*
(C) $p_z$ orbital of *A* and $p_x$ orbital of *B* (D) $p_x$ orbital of *A* and *s* orbital of *B*

Q.18 The maximum covalency is equal to
(A) the number of unpaired *p*-electrons
(B) the number of paired *d*-electrons
(C) the number of unpaired *s* and *p*-electrons
(D) the actual number of *s* and *p*-electrons in the outermost shell.

Q.19 How many bonded electron pairs are present in $IF_7$ molecule :
(A) 6 (B) 7 (C) 5 (D) 8

Q.20 $PCl_5$ exists but $NCl_5$ does not because :
(A) Nitrogen has no vacant 2*d*-orbitals (B) $NCl_5$ is unstable
(C) Nitrogen atom is much smaller than P (D) Nitrogen is highly inert

Q.21 Which of the following has/have a strong covalent bond?
(A) Cl-F (B) F-F (C) C-Cl (D) C-F

Q.22 Which of the following statements is/are true?
(A) Covalent bonds are directional
(B) Ionic bonds are nondirectional
(C) A polar bond is formed between two atoms which have the same electronegativity value.
(D) The presence of polar bonds in a polyatomic molecule suggests that it has zero dipole moment

Q.23 Rotation around the bond (between the underlined atoms) is restricted in :
(A) $\underline{C}_2H_4$ (B) $H_2\underline{O}_2$ (C) $\underline{Al}_2Cl_6$ (D) $\underline{C}_2H_6$

Q.24 The octet rule is not obeyed in :
(A) $CO_2$ (B) $BCl_3$ (C) $PCl_5$ (D) $SiF_4$

Q.25 Which of the following two substances are expected to be more covalent :
(A) $BeCl_2$ (B) $SnCl_4$ (C) $ZnS$ (D) $ZnCl_2$

Q.26 To which of the following species octet rule is not applicable :
(A) $BrF_5$ (B) $SF_6$ (C) $IF_7$ (D) $CO$

Q.27 Which of the following species are hypervalent?

1. $ClO_4^-$, 2. $BF_3$, 3. $SO_4^{2-}$, 4. $CO_3^{2-}$

(A) 1, 2, 3 (B) 1, 3 (C) 3, 4 (D) 1, 2

Q.28 The types of bond present in $N_2O_5$ are

(A) only covalent (B) only ionic

(C) ionic and covalent (D) covalent & coordinate

***CO-ORDINATE BOND***

Q.29 $NH_3$ and $BF_3$ combine readily because of the formation of :

(A) a covalent bond (B) a hydrogen bond (C) a coordinate bond (D) an ionic bond

Q.30 Which of the following species contain covalent coordinate bond :

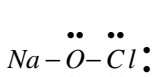
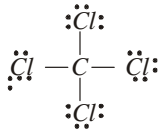
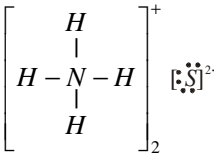
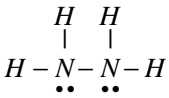
(A) $AlCl_3$ (B) $CO$ (C) $[Fe(CN)_6]^{4-}$ (D) $N_3^-$

Q.31 Which of the following molecules does not have coordinate bonds?

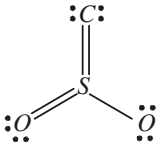
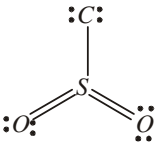
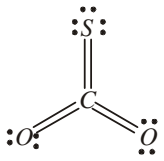
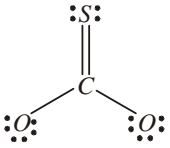
(A) $CH_3$–NC (B) CO (C) $O_3$ (D) $CO_3^{2-}$

***LEWIS STRUCTURE***

Q.32 Which of the following Lewis diagrams is(are) incorrect ?

(A)  (B)  (C)  (D) 

Q.33 The possible structure(s) of monothiocarbonate ion is :

(A)  (B)  (C)  (D) 

Q.34 The valency of sulphur in sulphuric acid is :

(A) 2 (B) 8 (C) 4 (D) 6

Q.35 The total number of valence electrons in 4.2g of $N_3^-$ ion are :

(A) 2.2 $N$ (B) 4.2 $N$ (C) 1.6 $N$ (D) 3.2 $N$

Q.36 No $X-X$ bond exists in which of the following compounds having general form of $X_2H_6$ ?

(A) $B_2H_6$ (B) $C_2H_6$ (C) $Al_2H_6$ (D) $Si_2H_6$

Q.37 Pick out among the following species isoelectronic with $CO_2$ :

(A) $N_3^-$ (B) $(CNO)^-$ (C) $(NCN)^{2-}$ (D) $NO_2^-$

Q.38 Which of the following have a three dimensional network structure ?

(A) $SiO_2$ (B) $(BN)_x$ (C) $P_4$(white) (D) $CCl_4$

Q.39 Which of the following oxyacids of sulphur contain $S-S$ bonds ?

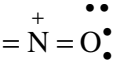
(A) $H_2S_2O_8$ (B) $H_2S_2O_6$ (C) $H_2S_2O_4$ (D) $H_2S_2O_5$

***RESONANCE***

Q.40 Resonating structures of a molecule should have:
(A) identical bonding
(B) identical arrangement of atoms
(C) nearly the same energy content
(D) the same number of paired electrons

Q.41 Which of the following conditions apply to resonating structures ?
(A) The contributing structures should have similar energies
(B) The contributing structures should be represented such that unlike formal charges reside on atoms that are far apart
(C) The more electropositive element should preferably have positive formal charge and the more electronegative element have negative formal charge
(D) The contributing structures must have the same number of unpaired electrons

Q.42 $N_2O$ has a linear, unsymmetrical structure that may be thought of as a hybrid of two resonance forms. If a resonance form must have a satisfactory Lewis structure, which of the five structures shown below are the resonance forms of $N_2O$ ?

(A)  (B)  (C)  (D)  (E) 

Q.43 Resonance occurs due to the
(A) delocalization of a lone pair of electrons
(B) delocalization of sigma electrons
(C) delocalization of pi electrons
(D) migration of protons

***V.B.T. & HYBRIDISATION***

Q.44 The strength of bonds by $s-s,\ p-p,\ s-p$ overlap is in the order :
(A) $s-s < s-p < p-p$
(B) $s-s < p-p < s-p$
(C) $s-p < s-s < p-p$
(D) $p-p < s-s < s-p$

Q.45 In the following compound $\overset{1}{C}H_2=\overset{2}{C}H-\overset{3}{C}CH_2-C\equiv CH$, the $C_2-C_3$ bond is of the type :
(A) $sp-sp^2$
(B) $sp^3-sp^3$
(C) $sp-sp^3$
(D) $sp^2-sp^3$

Q.46 Which of the following has a geometry different from the other three species (having the same geometry)?
(A) $BF_4^-$
(B) $SO_4^{2-}$
(C) $XeF_4$
(D) $PH_4^+$

Q.47 Maximum bond energy is in :
(A) $F_2$
(B) $N_2$
(C) $O_2$
(D) equal

Q.48 Among the following species, identify the isostructural pairs : $NF_3, NO_3^-, BF_3, H_3O^+, HN_3$
(A) $[NF_3, NO_3^-]$ and $[BF_3, H_3O^+]$
(B) $[NF_3, HN_3]$ and $[NO_3^-, BF_3]$
(C) $[NF_3, H_3O^+]$ *and* $[NO_3^-, BF_3]$
(D) $[NF_3, H_3O^+]$ and $[HN_3, BF_3]$

Q.49 Number and type of bonds between two carbon atoms in $CaC_2$ are :
(A) one sigma ($\sigma$) and one pi ($\pi$) bond
(B) one $\sigma$ and two $\pi$ bonds
(C) one $\sigma$ and one and a half $\pi$ bond
(D) one $\sigma$ bond

Q.50 In $C-C$ bond is $C_2H_6$ undergoes heterolytic fission, the hybridisation of two resulting carbon atoms is/are
(A) $sp^2$ both
(B) $sp^3$ both
(C) $sp^2, sp^3$
(D) $sp, sp^2$

Q.51 The hybridisation and geometry of $BrF_3$ molecules are :

(A) $sp^3d$ and *T* shaped (B) $sp^2d^2$ and tetragonal

(C) $sp^3d$ and bent (D) none of these

Q.52 The shape of methyl cation $(CH_3^+)$ is likely to be:

(A) linear (B) pyramidal (C) planar (D) spherical

Q.53 The structure of $XeF_2$ involves hybridization of the type :

(A) $sp^3$ (B) $dsp^2$ (C) $sp^3d$ (D) $sp^3d^2$

Q.54 In the $XeF_4$ molecule, the Xe atom is in the

(A) $sp^2$-hybridized state (B) $sp^3$-hybridised state (C) $sp^2d$-hybridized state (D) $sp^3d^2$-hybridized state

Q.55 How many σ- and π- bonds are there in salicyclic acid?

(A) 10σ, 4π (B) 16σ, 4π (C) 18σ, 2π (D) 16σ, 2π

Q.56 Which of the following statements are not correct?

(A) Hybridization is the mixing of atomic orbitals of large energy difference.

(B) $sp^2$ – hybrid orbitals are formed from two *p* - atomic orbitals and one *s*- atomic orbitals

(C) $dsp^2$ – hybrid orbitals are all at 90º to one another

(D) $d^2sp^3$ – hybrid orbitals are directed towards the corners of a regular octahedron

Q.57 Which of the following has been arranged in increasing order of size of the hybrid orbitals ?

(A) $sp < sp^2 < sp^3$ (B) $sp^3 < sp^2 < sp$ (C) $sp^2 < sp^3 < sp$ (D) $sp^2 < sp < sp^3$

Q.58 In the context of carbon, which of the following is arranged in the correct order of electronegativity :

(A) $sp > sp^2 > sp^3$ (B) $sp^3 > sp^2 > sp$ (C) $sp^2 > sp > sp^3$ (D) $sp^3 > sp > sp^2$

Q.59 When $2s-2s, 2p-2p$ and $2p-2s$ orbitals overlap, the bond strength decreases in the order :

(A) $p-p > s-s > p-s$ (B) $p-p > p-s > s-s$ (C) $s-s > p-p > p-s$ (D) $s-s > p-s > p-p$

Q.60 The shapes of $IF_5$ and $IF_7$ are respectively :

(A) square pyramidal and pentagonal bipyramidal (B) octahedral and pyramidal

(C) trigonal bipyramidal and square antiprismatic (D) distorted square planar and distorted octahedral

Q.61 Carbon atoms in $C_2(CN)_4$ are :

(A) *sp*-hybridized (B) $sp^2$-hybridized

(C) *sp*- and $sp^2$ hybridized (D) *sp*, $sp^2$ and $sp^3$ - hybridized

Q.62 $CO_2$ has the same geometry as :

(I) $HgCl_2$ (II) $NO_2$ (III) $SnCl_4$ (IV) $C_2H_2$

(A) I and III (B) II and IV (C) I and IV (D) III and IV

Q.63 Strongest bond is formed by the head on overlapping of :

(A) 2*s*- and 2*p*- orbitals (B) 2*p*- and 2*p*- orbitals

(C) 2*s*- and 2*s*- orbitals (D) All

Q.64 The ratio of σ and π bonds in benzene is :
(A) 2 (B) 6 (C) 4 (D) 8

Q.65 The bond angle and hybridization in ether $(CH_3OCH_3)$ is :
(A) 106°51′, $sp^3$ (B) 104°31′, $sp^3$ (C) 109° 28' $sp^3$ (D) None of these

Q.66 The enolic form of acetone contains :
(A) 9 sigma, 1 pi bond and 2 lone pairs (B) 8 sigma, 2 pi bond and 2 lone pairs
(C) 10 sigma, 1 pi bond and 1 lone pairs (D) 9 sigma, 2 pi bond and 1 lone pairs

Q.67 The shape of a molecule which has 3 bond pairs and one lone pair is :
(A) Octahedral (B) Pyramidal (C) Triangular planar (D) Tetrahedral

Q.68 [illegible] *T* shaped :
(A) $BeF_2$ (B) $BCl_3$ (C) $NH_3$ (D) $ClF_3$

Q.69 Maximum s-character is in bonds formed by () atom:
(A) $\overset{*}{C}H_4$ (B) $\overset{*}{Xe}O_3$ (C) $XeO_6^{4-}$ (D) $SF_4$

Q.70 Which of the following species is (are) isostructural with $XeF_4$ ?
(A) $ICl_4^-$ (B) $I_5^-$ (C) $BrF_4^-$ (D) $XeO_4$

Q.71 A hydrazine molecule is split in $NH_2^+$ and $NH_2^-$ ions. Which of the following statements is/are correct ?
(A) $NH_2^+$ shows $sp^2$ – hybridisation whereas $NH_2^-$ shows $sp^3$ – hybridisation
(B) $Al(OH)_4^-$ has a regular tetrahedral geometry
(C) $sp^2$ – hybridized orbitals have equal *s*- and *p*- character
(D) Hybridized orbitals always form σ - bonds

Q.72 There is change in the type of hybridisation when:
(A) $NH_3$ combines with $H^+$ (B) $AlH_3$ combines with $H^-$
(C) $NH_3$ forms $NH_2^-$ (D) $SiF_4$ forms $SiF_6^{2-}$

Q.73 Which of the following statement is/are correct
(A) Hybridisation is the mixing of atomic orbitals prior to their combining into molecular orbitals :
(B) $sp^3d^2$ – hybrid orbitals are at 90° to one another
(C) $sp^3d$ – hybrid orbitals are directed towards the corners of a regular tetrahedron
(D) $sp^3d^2$ – hybrid orbitals are directed towards the corners of a regular octahedron

Q.74 A σ-bond may between two $p_x$ orbitals containing one unpaired electron each when they approach each other appropriately along :
(A) *x* - axis (B) *y* - axis (C) *z* - axis (D) any direction

Q.75 Indicate the wrong statement :
(A) A sigma bond has no free rotation around its axis
(B) *p*-orbitals always have only sideways overlap
(C) *s*-orbitals never form π - bonds
(D) There can be more than one sigma bond between two atoms

Q.76 $sp^3$ hybridisation is in :

(A) $AlH_4^-$ (B) $CH_3^-$ (C) $ClO_2^-$ (D) $NH_2^-$

Q.77 Which of the following pairs is (are) isostructural?

(A) $SF_4$ and $SiF_4$ (B) $SF_6$ and $SiF_6^{2-}$ (C) $SiF_6^{2-}$ and $SeF_6^{2-}$ (D) $XeO_6^{4-}$ and $TeF_6^{2-}$

Q.78 Which of the following has (have) octahedral geometry :

(A) $SbCl_6^-$ (B) $SnCl_6^{2-}$ (C) $XeF_6$ (D) $IO_6^{5-}$

Q.79 Shape of $NH_3$ is very similar to :

(A) $SeO_3^{2-}$ (B) $CH_3^-$ (C) $BH_3$ (D) $CH_3^+$

Q.80 Which of the following have same shape as $NH_2^+$ ?

(A) $CO_2$ (B) $SnCl_2$ (C) $SO_2$ (D) $BeCl_2$

Q.81 Which of the following is (are) linear ?

(A) $I_3^-$ (B) $I_3^+$ (C) $PbCl_2$ (D) $XeF_2$

Q.82 Which of the following species are linear ?

(A) $ICl^{2-}$ (B) $I^{3-}$ (C) $N_3^-$ (D) $ClO_2$

Q.83 The structure of $XeF_6$ is :

(A) pentagonal bipyramidal (B) distorted octahedral (C) capped octahedral (D) square pyramidal

***OTHER FORCES***

Q.84 Which of the following models best describes the bonding within a layer of the graphite structure ?

(A) metallic bonding (B) ionic bonding
(C) non-metallic covalent bonding (D) van der Waals forces

Q.85 The critical temperature of water is higher than that of $O_2$ because the $H_2O$ molecule has :

(A) fewer electrons than $O_2$ (B) two covalent bonds
(C) $V$ - shape (D) dipole moment

Q.86 Ethanol has a higher boiling point than dimethyl ether though they have the same molecular weight. This is due to :

(A) resonance (B) coordinate bonding (C) hydrogen bonding (D) ionic bonding

Q.87 Arrange the following in order of decreasing boiling point :

(I) *n*-Butane (II) *n*-Butanol (III) *n*-Butyl chloride (IV) Isobutane

(A) $IV > III > II > I$ (B) $IV > II > III > I$ (C) $I > II > III > IV$ (D) $II > III > I > IV$

Q.88 Which of the following compounds would have significant intermolecular hydrogen bonding ?

$HF, CH_3OH, N_2O_4, CH_4$

(A) $HF, N_2O_4$ (B) $HF, CH_4, CH_3OH$ (C) $HF, CH_3OH$ (D) $CH_3OH, CH_4$

Q.89 For $H_2O_2, H_2S, H_2O$ and $HF$, the correct order of increasing extent of hydrogen bonding is :

(A) $H_2O > HF > H_2O_2 > H_2S$ (B) $H_2O > HF > H_2S > H_2O_2$
(C) $HF > H_2O > H_2O_2 > H_2S$ (D) $H_2O_2 > H_2O > HF > H_2S$

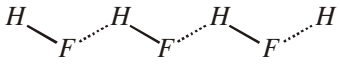
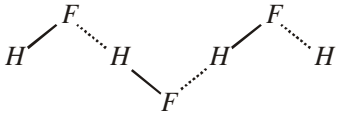
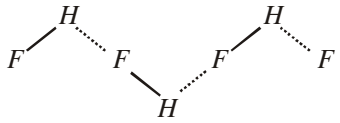
Q.90 Iron is harder than sodium because
(A) iron atoms are smaller (B) iron atoms are more closely packed
(C) metallic bonds are stronger in sodium (D) metallic bonds are stronger in iron

Q.91 Which one of the following does not have intermolecular H-bonding?
(A) $H_2O$ (B) *o*-nitro phenol (C) HF (D) $CH_3COOH$

Q.92 The order of strength of hydrogen bonds is:
(A) $ClH...Cl > NH...N > OH...O > FH...F$ (B) $ClH...Cl < NH...N < OH...O < FH...F$
(C) $ClH...Cl < NH...N > OH...O > FH...F$ (D) $ClH...Cl < NH...N < OH...O > FH...F$

Q.93 Which of the following exhibit/s H-bonding?
(A) $CH_4$ (B) $H_2Se$ (C) $N_2H_4$ (D) $H_2S$

Q.94 Among the following, van der Waals forces are maximum in
(A) HBr (B) LiBr (C) LiCl (D) AgBr

Q.95 The *H* bond in solid *HF* can be best represented as:
(A) $H-F....H-F....H-F$ (B) 
(C)  (D) 

Q.96 The volatility of *HF* is low because of :
(A) its low polarizability (B) the weak dispersion interaction between the molecules
(C) its small molecular mass (D) its strong hydrogen bonding

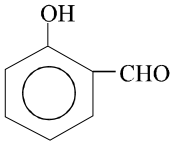
Q.97 The melting point of $AlF_3$ is 104° *C* and that of $SiF_4$ is - 77° *C* (it sublimes) because :
(A) there is a very large difference in the ionic character of the $Al-F$ and $Si-F$ bonds
(B) in $AlF_3$, $Al^{3+}$ interacts very strongly with the neighbouring $F^-$ ions to give a three dimensional structure but in $SiF_4$ no interaction is possible
(C) the silicon ion in the tetrahedral $SiF_4$ molecule is not shielded effectively from the fluoride ions whereas in $AlF_3$, the $Al^{3+}$ ion is shielded on all sides
(D) the attractive forces between the $SiF_4$ molecules are strong whereas those between the $AlF_3$ molecules are weak

Q.98 Two ice cubes are pressed over each other and unite to form one cube. Which force is responsible for holding them together :
(A) van der Waal's forces (B) Covalent attraction
(C) Hydrogen bond formation (D) Dipole-dipole attraction

Q.99 Intramolecular hydrogen bonding is found in :
(A) Salicylaldehyde (B) Water (C) Acetaldehyde (D) Phenol

Q.100 The pairs of bases in DNA are held together by :
(A) Hydrogen bonds (B) Ionic bonds (C) Phosphate groups (D) Deoxyribose groups

Q.101 In dry ice there are :
(A) Ionic bond (B) Covalent bond (C) Hydrogen bond (D) None of these

Q.102 


(A) has intermolecular H - bonding (B) has intramolecular H- bonding
(C) has low boiling point (D) is steam-volatile

Q.103 Which of the following bonds/forces is/are weakest?
(A) covalent bond (B) vander Waals force (C) hydrogen bond (D) london force

Q.104 Compare O–O bond energy among $O_2$, $H_2O_2$ and $O_3$ with reasons.

Q.105 Which of the following is/are observed in metallic bonds ?
(A) Mobile valence electrons (B) Overlapping valence orbitals
(C) Highly directed bond (D) Delocalized electrons

Q.106 Which of the following factors are responsible for van der Waals forces ?
(A) Instantaneous dipole-induced dipole interaction
(B) Dipole-induced dipole interaction and ion-induced dipole interaction
(C) Dipole-dipole interaction and ion-induced dipole interaction
(D) Small size of molecule

Q.107 Which of the following are true ?
(A) Van der Waals forces are responsible for the formation of molecular crystals
(B) Branching lowers the boiling points of isomeric organic compounds due to van der Waals forces of attraction
(C) In graphite, van der Waals forces act between the carbon layers
(D) In diamond, van der Waals forces act between the carbon layers

Q.108 Intermolecular hydrogen bonding increases the enthalpy of vapourization of a liquid due to the:
(A) decrease in the attraction between molecules
(B) increase in the attraction between molecules
(C) decrease in the molar mass of unassociated liquid molecules
(D) increase in the effective molar mass of hydrogen - bonded molecules

Q.109 Which of the following molecules have intermolecular hydrogen bonds ?
(A) $KH_2PO_4$ (B) $H_3BO_3$ (C) $C_6H_5CO_2H$ (D) $CH_3OH$

Q.110 Which of the following have dipole moment ?
(A) nitrobenzene (B) *p*-chloronitrobenzene
(C) *m*-dichlorobenzene (D) *o*-dichlorobenzene

Q.111 In which of the following compounds, breaking of covalent bond takes place?
(A) Boiling of $H_2O$ (B) Melting of KCN (C) Boiling of $CF_4$ (D) Melting of $SiO_2$

***MISCELLEANEOUS***

Q.112 Among $KO_2$, $AlO_2^-$, $BaO_2$ and $NO_2^+$ unpaired electron is present in :
(A) $KO_2$ only (B) $NO_2^+$ and $BaO_2$ (C) $KO_2$ and $AlO_2^-$ (D) $BaO_2$ only

Q.113 Cyanogen, $(CN)_2$, has a ____ shape/structure :
(A) Linear (B) Zig-zag (C) Square (D) Cyclic

Q.114 In which of the following sovents, KI has highest solubility? The dielectric constant ( $\in$ ) of each liquid is given in parentheses.
(A) $C_6H_6(\in = 0)$ (B) $(CH_3)_2CO$ $(\in = 2)$ (C) $CH_3OH$ $(\in = 32)$ (D) $CCl_4(\in = 0)$

Q.115 The formal charges on the three O-atoms in $O_3$ molecule are
(A) 0, 0, 0 (B) 0, 0, –1 (C) 0, 0, +1 (D) 0, +1, –1

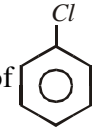
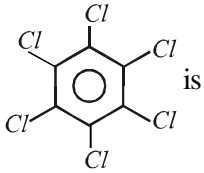
Q.116 The types of bonds present in $CuSO_4 \cdot 5H_2O$ are
(A) electrovalent and covalent (B) electrovalent and coordinate covalent
(C) covalent and coordinate covalent (D) electrovalent, covalent and coordinate covalent

Q.117 For which of the following crystalline substances does the solubility in water increase upto 32° *C* and then decrease rapidly ?
(A) $CaCl_2.2H_2O$ (B) $Na_2SO_4.10H_2O$ (C) $FeSO_4.7H_2O$ (D) Alums

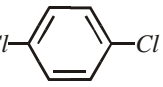
Q.118 Which of the following has been arranged in order of decreasing dipole moment ?
(A) $CH_3Cl > CH_3F > CH_3Br > CH_3I$ (B) $CH_3F > CH_3Cl > CH_3Br > CH_3I$
(C) $CH_3Cl > CH_3Br > CH_3I > CH_3F$ (D) $CH_3F > CH_3Cl > CH_3I > CH_3Br$

Q.119 Which of the following has the least dipole moment
(A) $NF_3$ (B) $CO_2$ (C) $SO_2$ (D) $NH_3$

Q.120 The experimental value of the dipole moment of *HCl* is 1.03 *D*. The length of the $H-Cl$ bond is $1.275\,\mathring{A}$ . The percentage of ionic character in *HCl* is :
(A) 43 (B) 21 (C) 17 (D) 7

Q.121 The dipole moment of  is 1.5 *D*. The dipole moment of  is :
(A) 0 *D* (B) 1.5 *D* (C) 2.86 *D* (D) 2.25 *D*

Q.122 In the cyanide ion the formal negative charge is on
(A) *C* (B) *N*
(C) Both *C* and *N* (D) Resonate between *C* and *N*

Q.123 Which has (have) zero value of dipole moment?
(A) $[Ni(CN)_4]^{2-}$ square planner (B) $CHCl_3$
(C) $CO_2$ (D) 

Q.124 Which of the following compounds possesses zero dipole moment?
(A) Water (B) Benzene (C) Carbon tetrachloride (D) Boron trifluoride

Q.125 Hypervalent compound is (are) :
(A) $SO_3^{2-}$ (B) $PO_4^{3-}$ (C) $SO_4^{2-}$ (D) $ClO_4^-$

Q.126 Which of the following statements are correct?
(A) The crystal lattice of ice is mostly formed by covalent as well as hydrogen bonds
(B) The density of water increases when heated from 0° *C* to 4° *C* due to the change in the structure of the cluster of water molecules
(C) Above 4° *C* the thermal agitation of water molecules increases. Therefore, intermolecular distance increases and water starts expanding
(D) The density of water increases from 0° *C* to a maximum at 4° *C* because the entropy of the system increases

***BONDS ANGLES & BOND LENGTH***

Q.127 The correct order of increasing $X-O-X$ bond angle is $(X = H, F \text{ or } Cl)$ :
(A) $H_2O > Cl_2O > F_2O$ (B) $Cl_2O > H_2O > F_2O$
(C) $F_2O > Cl_2O > H_2O$ (D) $F_2O > H_2O > Cl_2O$

Q.128 Which of the following is true ?
(A) $\text{Bond order} \propto \frac{1}{\text{bond length}} \propto \text{bond energy}$ (B) $\text{Bond order} \propto \text{bond length} \propto \frac{1}{\text{bond energy}}$
(C) $\text{Bond order} \propto \frac{1}{\text{bond length}} \propto \frac{1}{\text{bond energy}}$ (D) $\text{Bond order} \propto \text{bond length} \propto \text{bond energy}$

Q.129 Which of the following has been arranged in order of decreasing bond length ?
(A) $P-O > Cl-O > S-O$ (B) $P-O > S-O > Cl-O$
(C) $S-O > Cl-O > P-O$ (D) $Cl-O > S-O > P-O$

Q.130 If a molecule $MX_3$ has zero dipole moment, the sigma bonding orbitals used by *M* (atm. no. < 21) are :
(A) pure *p* (B) *sp* hybrid (C) $sp^2$ hybrid (D) $sp^3$ hybrid

Q.131 How many sigma and pi bonds are present in tetracyanoethylene ?
(A) Nine σ and nine π (B) Five π and nine σ (C) Nine σ and seven π (D) Eight σ and eight π

Q.132 Among the following species, which has the minimum bond length ?
(A) $B_2$ (B) $C_2$ (C) $F_2$ (D) $O_2^-$

Q.133 Which has higher bond energy :
(A) $F_2$ (B) $Cl_2$ (C) $Br_2$ (D) $I_2$

Q.134 The bond angle in $PH_3$ is :
(A) Much lesser than $NH_3$ (B) Equal to that in $NH_3$
(C) Much greater than in $NH_3$ (D) Slightly more than in $NH_3$

Q.135 $H-B-H$ bond angle in $BH_4^-$ is :
(A) 180° (B) 120° (C) 109° (D) 90°

Q.136 In the series ethane, ethylene and acetylene, the $C-H$ bond energy is :
(A) The same in all the three compounds (B) Greatest in ethane
(C) Greatest in ethylene (D) Greatest in acetylene

Q.137 Which one of the following compounds has bond angle as nearly 90° ?
(A) $NH_3$ (B) $H_2S$ (C) $H_2O$ (D) $SF_6$

Q.138 **State whether each statement is true or false. If false, write the correct statement.**

(i) The polarising power of a cation is directly proportional to its charge.
(ii) The polarising power of a cation is directly proportional to its size.
(iii) The polarisability of an anion is directly proportional to its charge.
(iv) The polarisability of an anion is directly proportional to its size.
(v) For a given anion, greater the polarising power of the cation, more the ionic character.
(vi) For a given cation, greater the polarisability of the anion, more the covalent character.
(vii) An element with low ionization potential is most likely to form a covalent bond with an other element having a high electron affinity.
(viii) Ionic interactions are stronger than covalent bonds.
(ix) Two non-metal atoms are likely to form covalent bonds on combination.
(x) Ionic interactions are directional.

Q.139 **State whether each statements is T or F, if F rectify.**

(i) All diatomic molecules are non-polar.
(ii) All molecules having polar bonds are polar (i.e., have a net dipole)
(iii) The lone pairs of electrons do not contribute to the net dipole of a molecule.
(iv) The $CH_2Cl_2$ molecule may be polar or nonpolar depending on its geometry.
(v) The net dipole in the water molecule is the resultant of its bond dipoles.
(vi) $SO_2$ is polar whereas $CO_2$ is non-polar.
(vii) $NH_3$ is less polar than $NF_3$
(viii) If all bonds in a molecule are polar, the molecule as a whole must be polar.

Q.140 **Fill in the blanks.**

(i) $\pi$–bonds are formed by the lateral overlap of a p-orbital with another ____ orbital.
(ii) Free rotation is possible if two atoms are bonded together only by a_____ bond.
(iii) The maximum number of $\sigma$ bonds that can be formed between two atoms is_____.
(iv) The repulsion between ____ is greater than the repulsion between two bonded pairs
(v) A lone pair is _____ polarisable compared to a $\sigma$ bonded pair which in turn is _____ polarisable compared to a $\pi$- bonded pair.
(vi) In nitro benzene the total number of bonded electrons equals _____.

Q.141 $AgNO_3$ gives a white precipitate with *NaCl* but not with $CCl_4$. Why ?

Q.142 Using *VSEPR* theory identify the type of hybridisation and draw the structure of $OF_2$.

Q.143 What should be the structure of the following as per *VSEPR* theory ?
(a) $XeF_2$ (b) $XeF_4$ (c) $PBr_5$ (d) $OF_2$ (e) $I_3^-$ and (f) $I_3^+$

Q.144 The percent ionic character in *HCl* is 18.08. The observed dipole moment is 1.08 *D*. Find the inter-nuclear distance in *HCl*.

Q.145 In the hydrides of group VI elements the central atoms involve $sp^3$ hybridisation but the bond angles decrease in the order, $H_2O, H_2S, H_2Si, H_2Te$. How would you account for this ?

Q.146 Assuming that all the four valency of carbon atom in propane pointing towards the corners of a regular tetrahedron. Calculate the distance between the terminal carbon atoms in propane. Given, $C-C$ single bond length is 1.54 Å.

Q.147 The dipole moment of *HBr* is 7.95 debye and the intermolecular separation is $1.94\times10^{-10}\,m$ Find the % ionic character in *HBr* molecule.

Q.148 *HBr* has dipole moment $2.6\times10^{-30}\ cm$ . If the ionic character of the bond is 11.5 %, calculate the interatomic spacing.

Q.149 Dipole moment of *LiF* was experimentally determined and was found to be 6.32 *D*. Calculate percentage ionic character in *LiF* molecule $Li-F$ bond length is 0.156 pm.

Q.150 A diatomic molecule has a dipole moment of 1.2 *D*. If bond length is 1.0 Å, what percentage of an electronic charge exists on each atom.

## *EXERCISE - II*

**Choose the correct alternative (only one correct answer).**

Q.1 The bond between carbon atom (1) & carbon atom (2) in compound $N \equiv \underset{1}{C} - \underset{2}{C} H = CH_2$ involves the hybrids as : **[ JEE '87]**

(A) $sp^2$ & $sp^2$ (B) $sp^3$ & sp (C) sp & $sp^2$ (D) sp & sp

Q.2 Hydrogen bonding is maximum in **[ JEE '87]**

(A) Ethanol (B) Diethylether (C) Ethyl chloride (D) Triethylamine

Q.3 The species which the central atom uses $sp^2$ hybrid orbitals in its bonding is **[ JEE '88]**

(A) $PH_3$ (B) $NH_3$ (C) $CH_3^+$ (D) $SbH_3$

Q.4 The molecule that has linear stucture is **[ JEE '88]**

(A) $CO_2$ (B) $NO_2$ (C) $SO_2$ (D) $SiO_2$

Q.5 The compound which has zero dipole moment is **[ JEE '89]**

(A) $CH_2Cl_2$ (B) $BF_3$ (C) $NF_3$ (D) $ClO_2$

Q.6 Which of the following is paramagnetic **[ JEE '89]**

(A) $O_2^-$ (B) $CN^-$ (C) CO (D) $NO^+$

Q.7 The molecule which has pyramidal shape is **[ JEE '89]**

(A) $PCl_3$ (B) $SO_3$ (C) $CO_3^{2-}$ (D) $NO_3^-$

Q.8 The compound in which $\overset{*}{C}$ uses its $sp^3$ hybrid orbitals for bond formation is : **[ JEE '89]**

(A) $H\overset{*}{C}OOH$ (B) $(H_2N)\overset{*}{C}O$ (C) $(CH_3)_3\overset{*}{C}OH$ (D) $CH_3\overset{*}{C}HO$

Q.9 The C - H bond distance is the longest in **[ JEE '89]**

(A) $C_2H_2$ (B) $C_2H_4$ (C) $C_2H_6$ (D) $C_2H_2Br_2$

Q.10 Which one of the following is the smallest in size **[ JEE '89]**

(A) $N^{3-}$ (B) $O^{2-}$ (C) $F^-$ (D) $Na^+$

Q.11 The number of sigma and pi bonds in 1-butene-3-yne are **[ JEE '89]**

(A) 5 sigma 5 pi (B) 7 sigma 3 pi (C) 8 sigma 2 pi (D) 6 sigma 4 pi

Q.12 Amongst the following the one having highest I.E. is **[ JEE '90]**

(A) [Ne] $3s^2 3p^1$ (B) [Ne] $3s^2 3p^3$ (C) [Ne] $3s^2 3p^2$ (D) [Ar] $3d^\circ 4s^2 4p^3$

Q.13 The hybridisation of C atoms in C - C single bond of $HC \equiv C - CH = CH_2$ is **[ JEE '91]**

(A) $sp^3$ - $sp^3$ (B) $sp^2$ - $sp^3$ (C) sp - $sp^2$ (D) $sp^3$ - sp

Q.14 The type of hybrid orbitals used by the chlorine atom in $ClO_2^-$ is **[ JEE '92]**

(A) $sp^3$ (B) $sp^2$ (C) sp (D) none

Q.15 The $CN^-$ & $N_2$ are isoelectronic. But in contrast to $CN^-$, $N_2$ is chemically inert because of

(A) Low bond energy **[ JEE '92]**

(B) Absence of bond polarity

(C) Unsymmetrical electron distribution

(D) Presence of more number of electron in bonding orbitals.

Q.16 The maximum possible number of hydrogen bonds a water molecule can form is **[ JEE '92]**
(A) 2 (B) 4 (C) 3 (D) 1

Q.17 Pick out the isoelectronic structures from the following **[ JEE '93]**
I. $CH_3^+$ II. $H_3O^+$ III. $NH_3$ IV. $CH_3^-$
(A) I and II (B) III and IV (C) I and III (D) II, III and IV

Q.18 The number of electrons that are paired in oxygen molecule is **[ JEE '95]**
(A) 7 (B) 8 (C) 16 (D) 14

Q.19 Allyl isocyanide has **[ JEE '95]**
(A) 9s, 4p bonds (B) 9s, 3p bonds and 2 non-bonding electrons
(C) 8s, 5p bonds (D) 8s, 3p bonds and 4 non- bonding electrons

Q.20 The order of increasing thermal stabilities of $K_2CO_3$(I) , $MgCO_3$(II) , $CaCO_3$(III) , $BaCO_3$(IV) is **[ JEE '96]**
(A) II < III < IV < I (B) IV < II < III < I (C) IV < II < I < III (D) II < IV < III < I

Q.21 Identify isostructural pairs from $NF_3$(I) , $NO_3^-$(II) , $BF_3$(III) , $H_3O^+$(IV), $HN_3$(V) **[ JEE '96]**
(A) I & II, III & IV (B) I & V, II & III (C) I & IV, II & III (D) I & IV, III & V

Q.22 (i)The number and type of bonds between two C - atom in $CaC_2$ are **[ JEE '96]**
(A) 1 sigma 1 pi (B) 1 sigma 2 pi (C) 1 sigma , ½ pi (D) 1 sigma

Q.23 Which is correct for $CsBr_3$ ? **[ JEE '96]**
(A) it is a covalent compound (B) it contains $Cs^{3+}$ & $Br^-$ ions
(C) it contains $Cs^+$ & $Br_3^-$ ions (D) it contains $Cs^+$, $Br^-$ & lattice $Br_2$ molecule

Q.24 Among $KO_2$ , $AlO_2^-$ , $BaO_2$ & $NO_2^+$ unpaired electron is present in **[ JEE '97]**
(A) $NO_2^+$ & $BaO_2$ (B) $KO_2$ & $AlO_2^-$ (C) $KO_2$ only (D) $BaO_2$ only

Q.25 Which of the following has maximum number of unpaired electrons? **[ JEE '96]**
(A) $Mg^{2+}$ (B) $Ti^{3+}$ (C) $V^{3+}$ (D) $Fe^{2+}$

Q.26 KF combines with HF to form $KHF_2$. The compound contains the species **[ JEE '97]**
(A) $K^+$, $F^-$ and $H^+$ (B) $K^+$, $F^-$ and HF (C) $K^+$ and $[HF_2]^-$ (D) $[KHF]^+$ and $F^-$

Q.27 Among the following compounds the one that is polar and has the central atom with $sp^2$ hybridisation is **[ JEE '97]**
(A) $H_2CO_3$ (B) $SiF_4$ (C) $BF_3$ (D) $HClO_2$

Q.28 Which contains both polar & non polar covalent bonds **[ JEE '97]**
(A) $NH_4Cl$ (B) HCN (C) $H_2O_2$ (D) $CH_4$

Q.29 The type of hybrid orbitals used by the chlorine atom in $ClO_3^-$ is **[ JEE '97]**
(A) $sp^3$ (B) $sp^3d$ (C) $sp^3d^2$ (D) $sp^2$

Q.30 Hybridisation seen in cation of solid $PCl_5$ **[ JEE '97]**
(A) $sp^3d$ (B) $sp^3$ (C) $sp^3d^2$ (D) sp

Q.31 What type of hybridisation and how many lone pair of electrons are present in the species $I_3^-$ on the central atom. **[ JEE '97]**

(A) $sp^2$ one lone pair (B) $sp^3d$ three lone pair (C) sp three lone pair (D) sp no lone pair

Q.32 In which of the following the central atom does not use $sp^3$ hybrid orbitals in its bonding?

(A) $BeF_3^-$ (B) $OH_3^+$ (C) $NH_2^-$ (D) $NF_3$ **[ JEE '97]**

Q.33 The structure of $IBr_2^-$ involves hybridisation of the type.

(A) $sp^3d$ (B) $sp^3d^2$ (C) $dsp^3$ (D) $d^2sp^3$

Q.34 The maximum angle around the central atom H-M-H is present in

(A) $AsH_3$ (B) $PH_3$ (C) $NH_3$ (D) $SbH_3$

Q.35 Which one of the following molecules is planar : **[ JEE '97]**

(A) $NF_3$ (B) $NCl_3$ (C) $PH_3$ (D) $BF_3$

Q.36 Which one has $sp^2$ hybridisation **[ JEE '97]**

(A) $CO_2$ (B) $SO_2$ (C) $N_2O$ (D) CO

Q.37 The geometry & the type of hybrid orbitals present about the central atom in $BF_3$ is : **[ JEE '98]**

(A) linear, sp (B) trigonal planar, $sp^2$ (C) tetrahedra $sp^3$ (D) pyramidal, $sp^3$

Q.38 The correct order of increasing C - O bond length of, CO, $CO_3^{2-}$, $CO_2$ is **[ JEE '99]**

(A) $CO_3^{2-} < CO_2 < CO$ (B) $CO_2 < CO_3^{2-} < CO$

(C) $CO < CO_3^{2-} < CO_2$ (D) $CO < CO_2 < CO_3^{2-}$

Q.39 In the dichromate anion **[ JEE '99]**

(A) 4 Cr - O bonds are equivalent (B) 6 Cr - O bonds are equivalent

(C) all Cr - O bonds are equivalent (D) all Cr - O bonds are non equivalent

Q.40 The geometry of $H_2S$ and its dipole moment are **[ JEE '99]**

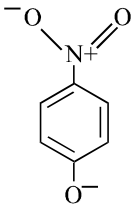
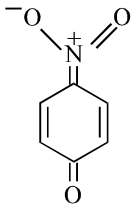
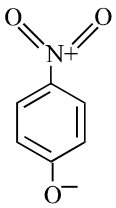
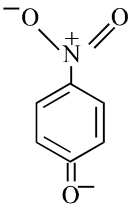
(A) angular & non zero (B) angular & zero

(C) linear & non zero (D) linear & zero

Q.41 In compounds type $ECl_3$, where E = B, P, As or Bi, the angles Cl - E - Cl for different E are in the order

(A) B > P = As = Bi (B) B > P > As > Bi (C) B < P = As = Bi (D) B < P < As < Bi **[ JEE '99]**

Q.42 The most unlikely representation of resonance structure of p–nitrophenoxide is:

(A)  (B)  (C)  (D) 

Q.43 Amongst $H_2O$, $H_2S$ , $H_2Se$ and $H_2Te$, the one with the highest boiling point is **[JEE 2000]**

(A) $H_2O$ because of hydrogen bonding (B) $H_2Te$ because of higher molecular weight

(C) $H_2S$ because of hydrogen bonding (D) $H_2Se$ because of lower molecular weight

Q.44 The hybridization of atomic orbitals of nitrogen in $NO_2^+$, $NO_3^-$ and $NH_4^+$ are **[JEE 2000]**

(A) $sp^2$, $sp^3$ and $sp^2$ respectively (B) sp, $sp^2$ and $sp^3$ respectively
(C) $sp^2$, sp and $sp^3$ respectively (D) $sp^2$, $sp^3$ and sp respectively

Q.45 Specify the coordination geometry around and hybridization of N and B atoms in a 1 : 1 complex of $BF_3$ and $NH_3$ **[JEE 2002]**

(A) N : tetrahedral, $sp^3$ ; B : tetrahedral, $sp^3$ (B) N : pyramidal, $sp^3$; B : pyramidal, $sp^3$
(C) N : pyramidal, $sp^3$ ; B : planar, $sp^2$ (D) N : pyramidal, $sp^3$; B : tetrahedral, $sp^3$

Q.46 The nodal plane in the $\pi$-bond of ethene is located in **[JEE 2002]**

(A) the molecular plane
(B) a plane parallel to the molecular plane
(C) a plane perpendicular to the molecular plane which bisects, the carbon-carbon $\sigma$ bond at right angle.
(D) a plane perpendicular to the molecular plane which contains, the carbon-carbon bond.

Q.47 Which of the following molecular species has unpaired electron(s)? **[JEE 2002]**

(A) $N_2$ (B) $F_2$ (C) $O_2^-$ (D) $O_2^{2-}$

Q.48 Which of the following are isoelectronic and isostructural ? $NO_3^-, CO_3^{2-}, ClO_3^-, SO_3$ **[JEE 2003]**

(A) $NO_3^-, CO_3^{2-}$ (B) $SO_3$, $NO_3^-$ (C) $ClO_3^-, CO_3^{2-}$ (D) $CO_3^{2-}, SO_3$

Q.49 Which species has the maximum number of lone pair of electrons on the central atom? **[JEE 2005]**

(A) $ClO_3^-$ (B) $XeF_4$ (C) $SF_4$ (D) $I_3^-$

**Fill in the blanks.** **[ 12 × 2 = 24]**

Q.1 Silver chloride is sparingly soluble in water because its lattice energy is greater than _____ energy. **[ JEE '87]**

Q.2 _____ phosphorous is reactive because of its highly strained tetrahedral structure. **[ JEE '87]**

Q.3 The shape of $CH_3^+$ is _____. **[ JEE '90]**

Q.4 The valence atomic orbitals on C in silver acetylide is _____ hybridised. **[ JEE '90]**

Q.5 Amongst the three isomers of nitrophenol , the one that is least soluble in water is _____. **[ JEE '94]**

Q.6 The kind of delocalization involving sigma bond orbitals are called _____. **[ JEE '94]**

Q.7 The two types of bonds present in $B_2H_6$ are covalent & _____. **[ JEE '94]**

Q.8 When $N_2$ goes to $N_2^+$, the N - N distance _____ & when $O_2$ goes to $O_2^+$, the O - O bond distance _____ . **[ JEE '96]**

Q.9 Among $N_2O$, $SO_2$, $I_3^+$ & $I_3^-$, the linear species are _____ & _____ . **[ JEE '97]**

Q.10 Among $PCl_3$, $CH_3^+$, $NH_2^-$ & $NF_3$, _____ is least relative towards water. **[ JEE '97]**

Q.11 The P - P - P angle in $P_4$ molecule is _____. **[ JEE '97]**

Q.12 Compounds that formally contain $Pb^{4+}$ are easily reduced to $Pb^{2+}$. The stability of lower oxidation state is due to _____ . **[ JEE '97]**

**State whether true or false.** **[ 16 × 2 = 32]**

Q.1 In benzene carbon uses all the three p-orbitals for hybridisation. **[ JEE '87]**

Q.2 $sp^2$ hybrid orbitals have equal S & P character . **[ JEE '87]**

Q.3 In group I A of alkali metals, the ionisation potential decreases down the group. Therefore lithium is a poor reducing agent . **[ JEE '87]**

Q.4 All the Al - Cl bond in $Al_2Cl_6$ are equivalent . **[ JEE '88]**

Q.5 Both potassium ferrocyanide & potassium ferricyanide are diamagnetic. **[ JEE '88]**

Q.6 The presence of polar bonds in a polyatomic molecule suggests that the molecule has non - zero dipole moment . **[ JEE '90]**

Q.7 Nitric oxide, though an odd electron molecule, is diamagnetic in liquid state. **[ JEE '91]**

Q.8 The decreasing order of E A of F, Cl, Br is F > Cl > Br . **[ JEE '93]**

Q.9 Diamond is harder than graphite . **[ JEE '93]**

Q.10 The basic nature of hydroxides of group 13 (III B) decreases progressively down the group. **[ JEE '93]**

Q.11 The tendency for catenation is much higher for C than Si. **[ JEE '93]**

Q.12 The dipolemoment of $CH_3F$ is greater than $CH_3Cl$. **[ JEE '93]**

Q.13 HBr is stronger acid than HI because of H - bonding. **[ JEE '97]**

Q.14 F atom has less negative E A than Cl atom. **[ JEE '97]**

Q.15 LiCl is predominantly a covalent compound. **[ JEE '97]**

Q.16 $Al(OH)_3$ is amphoteric in nature. **[ JEE '97]**

**Explain the following.** **[ 10 × 3 = 30]**

Q.1 Explain the molecule of magnesium chloride is linear whereas that of stannous chloride is angular. **[ JEE '87]**

Q.2 Give reason carbon oxygen bond lengths in formic acid are 1.23 A° & 1.36 A° and both the carbon oxygen bonds in sodium formate have the same value i.e. 1.27 A°. **[ JEE '88]**

Q.3 Give reason that valency of oxygen is generally two whereas sulphur shows of 2 , 4, & 6. **[ JEE '88]**

Q.4 Explain the first I.E. of carbon atom is greater than that of boron atom whereas the reverse is true for the second I.E. **[ JEE '89]**

Q.5 Explain why the dipolemoment of $NH_3$ is more than that of $NF_3$. **[ JEE '95]**

Q.6 The experimentally determined N - F bond length in $NF_3$ is greater than the sum of single bond covalent radii of N & F . Explain. **[ JEE '95]**

Q.7 Explain the difference in the nature of bonding in LiF & LiI. **[ JEE '96]**

Q.8 Explain $PCl_5$ is formed but $NCl_5$ cannot. **[JEE '97]**

Q.9 Give reasons for the following in one or two sentences only. **[ JEE '99]**

(a) $BeCl_2$ can be easily hydrolyed (b) $CrO_3$ is an acid anhydride .

Q.10 Explain why o-hydroxybenzaldehyde is a liquid at room temperature, while p-hydroxybenzaldehyde is a high melting solid. **[ JEE '99]**

**Arrange as directed.** **[ 9 × 2 = 18]**

Q.1 $N_2$ , $O_2$ , $F_2$ , $Cl_2$ in increasing order of bond dissociation energy. **[ JEE '88]**

Q.2 $CO_2$ , $N_2O_5$ , $SiO_2$ , $SO_3$ is the increasing order of acidic character. **[ JEE '88]**

Q.3 HOCl , $HOClO_2$ , $HOClO_3$ , HOClO in increasing order of thermal stability. **[ JEE '88]**

Q.4 Increasing order of ionic size : $N^{3-}$ , $Na^+$ , $F^-$ , $O^{2-}$ , $Mg^{2+}$

Q.5 Increasing strength of H - bonding . (X ..... H - X) O , S , F , Cl , N .

Q.6 Increasing order of extent of hydrolysis $CCl_4$ , $MgCl_2$ , $AlCl_3$ , $PCl_5$ , $SiCl_4$

Q.7 Arrange in increasing order of dipole moment . **[ JEE '96]**
Toluene , m - dichcorobenzene , O - dichlorobenzene , p - dichlorobenzene .

Q.8 The decreasing order of acid strength of ClOH , BrOH , IOH. **[ JEE '97]**

Q.9 Arrange in order of increasing radii , $Li^+$ , $Mg^{2+}$ , $K^+$ , $Al^{3+}$ . **[ JEE '97]**

**Miscellaneous.**

Q.1 Write two resonance structures of ozone which satisfy the octet rule. **[JEE '91]**

Q.2 Using VSEPR theory , identify the type of hybridisation & draw the structure of $OF_2$. What are oxidation states of O & F. **[JEE '94]**

Q.3 What are the types of bond present in $B_2H_6$? **[IIT 1994]**

Q.4 Arrange toluene, m–dichlorobenzene, o–dicholorobenzene and p–dichlorobenzene in order of increasing dipole moment. **[IIT 1996]**

Q.5 Draw the structures of **[JEE '97]**
(i) $XeF_2$ (ii) $XeO_3$ (iii) $XeF_4$ (iv) $BrF_5$ (v) $SO_3^{2-}$

Q.6 Interpret the non-linear shape of $H_2S$ molecule & non planar shape of $PCl_3$ using VSEPR theory. **[JEE '98]**

Q.7 Discuss the hybridisation of C - atoms in allene ($C_3H_4$) and show the $\pi$ – orbital overlaps. **[JEE '99]**

Q.8 Using VSEPR theory, draw the shape of $PCl_5$ and $BrF_5$. **[JEE 2003]**

Q.9 Draw the structure of $XeF_4$ and $OSF_4$ according to VSEPR theory, clearly indicating the state of hybridisation of the central atom and lone pair of electrons (if any) on the central atom. **[JEE 2004]**

# *ANSWER KEY*

## ***EXERCISE - I***

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Q.1 | C | Q.2 | B | Q.3 | A | Q.4 | C | Q.5 | D |
| Q.6 | B | Q.7 | D | Q.8 | A,B,C,D | Q.9 | A,B,C,D | Q.10 | D |
| Q.11 | A | Q.12 | C | Q.13 | B,C,D | Q.14 | C | Q.15 | A,C |
| Q.16 | C | Q.17 | D | Q.18 | D | Q.19 | B | Q.20 | A |
| Q.21 | D | Q.22 | A,B | Q.23 | A,C | Q.24 | B,C | Q.25 | A,B |
| Q.26 | A,B,C | Q.27 | B | Q.28 | D | Q.29 | C | Q.30 | B,C,D |
| Q.31 | D | Q.32 | A | Q.33 | D | Q.34 | D | Q.35 | C |
| Q.36 | A,C | Q.37 | A,B,C | Q.38 | A,B | Q.39 | B,C,D | Q.40 | B,C,D |
| Q.41 | A,B,C,D | Q.42 | A,E | Q.43 | A,C | Q.44 | A | Q.45 | D |
| Q.46 | C | Q.47 | B | Q.48 | C | Q.49 | B | Q.50 | C |
| Q.51 | A | Q.52 | C | Q.53 | C | Q.54 | D | Q.55 | B |
| Q.56 | A | Q.57 | A | Q.58 | A | Q.59 | B | Q.60 | A |
| Q.61 | C | Q.62 | C | Q.63 | B | Q.64 | C | Q.65 | C |
| Q.66 | A | Q.67 | B | Q.68 | D | Q.69 | A | Q.70 | A,C,D |
| Q.71 | A,B,D | Q.72 | B,D | Q.73 | A,B | Q.74 | A | Q.75 | A,B |
| Q.76 | A,B,C,D | Q.77 | B | Q.78 | A,B,D | Q.79 | A,B | Q.80 | B,C |
| Q.81 | A,D | Q.82 | A,B,C | Q.83 | C | Q.84 | C | Q.85 | D |
| Q.86 | C | Q.87 | D | Q.88 | C | Q.89 | C | Q.90 | D |
| Q.91 | B | Q.92 | B | Q.93 | C | Q.94 | D | Q.95 | C |
| Q.96 | D | Q.97 | B | Q.98 | C | Q.99 | A | Q.100 | A |
| Q.101 | B | Q.102 | B,C,D | Q.103 | B,D | Q.104 | $O_2>O_3>H_2O_2$ | | |
| Q.105 | A,D | Q.106 | A,B,C | Q.107 | A,B | Q.108 | B | | |
| Q.109 | A,B,C,D | Q.110 | A,B,C,D | Q.111 | D | Q.112 | A | Q.113 | A |
| Q.114 | C | Q.115 | D | Q.116 | D | Q.117 | B | Q.118 | A |
| Q.119 | B | Q.120 | C | Q.121 | A | Q.122 | D | Q.123 | A,C,D |
| Q.124 | B,C,D | Q.125 | B,C,D | Q.126 | A,B,C,D | Q.127 | B | Q.128 | A |
| Q.129 | B | Q.130 | C | Q.131 | A | Q.132 | B | Q.133 | B |
| Q.134 | A | Q.135 | C | Q.136 | D | Q.137 | B,D | | |
| Q.138 | T, F, T, T, F, F, F, T, T, F | | | Q.139 | F, F, F, F, T, T, F, F | | | | |

Q.140 (i) p-orbital, (ii) σ–bond, (iii) 1 , (iv) LP–LP & LP–BP, (v) more, less, (vi) 36

Q.143 (a) Linear, (b) square planar, (c) T.B.P. (d) bent, (e) linear, (f) bent

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Q.144 | 1.2Å | Q.146 | 2.33 Å | Q.147 | 85% | Q.148 | 1.4 Å |
| Q.149 | 84.5% | Q.150 | 25% | | | | |

## ***EXERCISE - II***

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Q.1 | C | Q.2 | A | Q.3 | C | Q.4 | A | Q.5 | B | Q.6 | A | Q.7 | A |
| Q.8 | C | Q.9 | C | Q.10 | D | Q.11 | B | Q.12 | B | Q.13 | C | Q.14 | A |
| Q.15 | B | Q.16 | B | Q.17 | D | Q.18 | D | Q.19 | A | Q.21 | C | Q.22 | B |
| Q.23 | C | Q.24 | C | Q.25 | D | Q.26 | C | Q.27 | A | Q.28 | A | Q.29 | A |
| Q.30 | B | Q.31 | B | Q.32 | A | Q.33 | B | Q.34 | C | Q.35 | D | Q.36 | B |
| Q.37 | B | Q.38 | D | Q.39 | B | Q.40 | A | Q.41 | B | Q.42 | C | Q.43 | A |
| Q.44 | B | Q.45 | A | Q.46 | A | Q.47 | C | Q.48 | A | Q.49 | D | | |

**Fill in the blanks.**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Q.1 | hydration | Q.2 | white | Q.3 | trigonal planar | Q.4 | sp |
| Q.5 | ortho | Q.7 | banana | Q.8 | increases, decreases | | |
| Q.9 | $N_2O$, $I_3^-$ | Q.10 | $NH_2^-$ | Q.11 | 60° | Q.12 | inert pair effect |

**State whether true or false.**

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Q.1 | F | Q.2 | F | Q.3 | F | Q.4 | F | Q.5 | F | Q.6 | F | Q.7 | T |
| Q.8 | F | Q.9 | T | Q.10 | F | Q.11 | T | Q.12 | F | Q.13 | F | Q.14 | T |
| Q.15 | T | Q.16 | T | | | | | | | | | | |

**Explain the following.**

| | | | | | |
|---|---|---|---|---|---|
| Q.1 | Lone pair | Q.2 | Resonance | Q.3 | expansion of octet |
| Q.5 | Lone pair contribution | Q.7 | LiF → Ionic charge, LiI → covalent charge | | |
| Q.8 | d-orbitals | Q.10 | Intra-H-bonding in o-hydroxybenzaldehyde | | |

**Arrange as directed.**

Q.1 $F_2 < Cl_2 < O_2 < N_2$

Q.2 $SiO_2 < CO_2 < SO_3 < N_2O_5$

Q.3 $HClO < HClO_2 < HClO_3 < HClO_4$

Q.4 $Mg^{2+} < Na^{+} < F^{-} < O^{2-} < N^{-3}$

Q.5 S < Cl < N < O < F

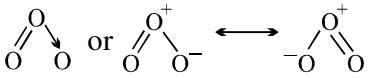
Q.6 $CCl_4 < MgCl_2 < AlCl_3 < SiCl_4 < PCl_5$

Q.7 p - dichlorobenzene < Toluene < m-dichcorobenzene < o-dichlorobenzene

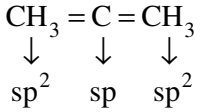
Q.8 ClOH < BrOH < IOH

Q.9 $LI^{+} < Al^{3+} < Mg^{2+} < K^{+}$

**Miscellaneous.**

Q.1 

Q.5 (i) Linear, (ii) Pyramidal, (iii) Square planar, (iv) Square pyramidal, (v) pyramidal

Q.7 

# STUDY PACKAGE

# Target: IIT-JEE (Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 5. Periodic Table and Representative Element

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

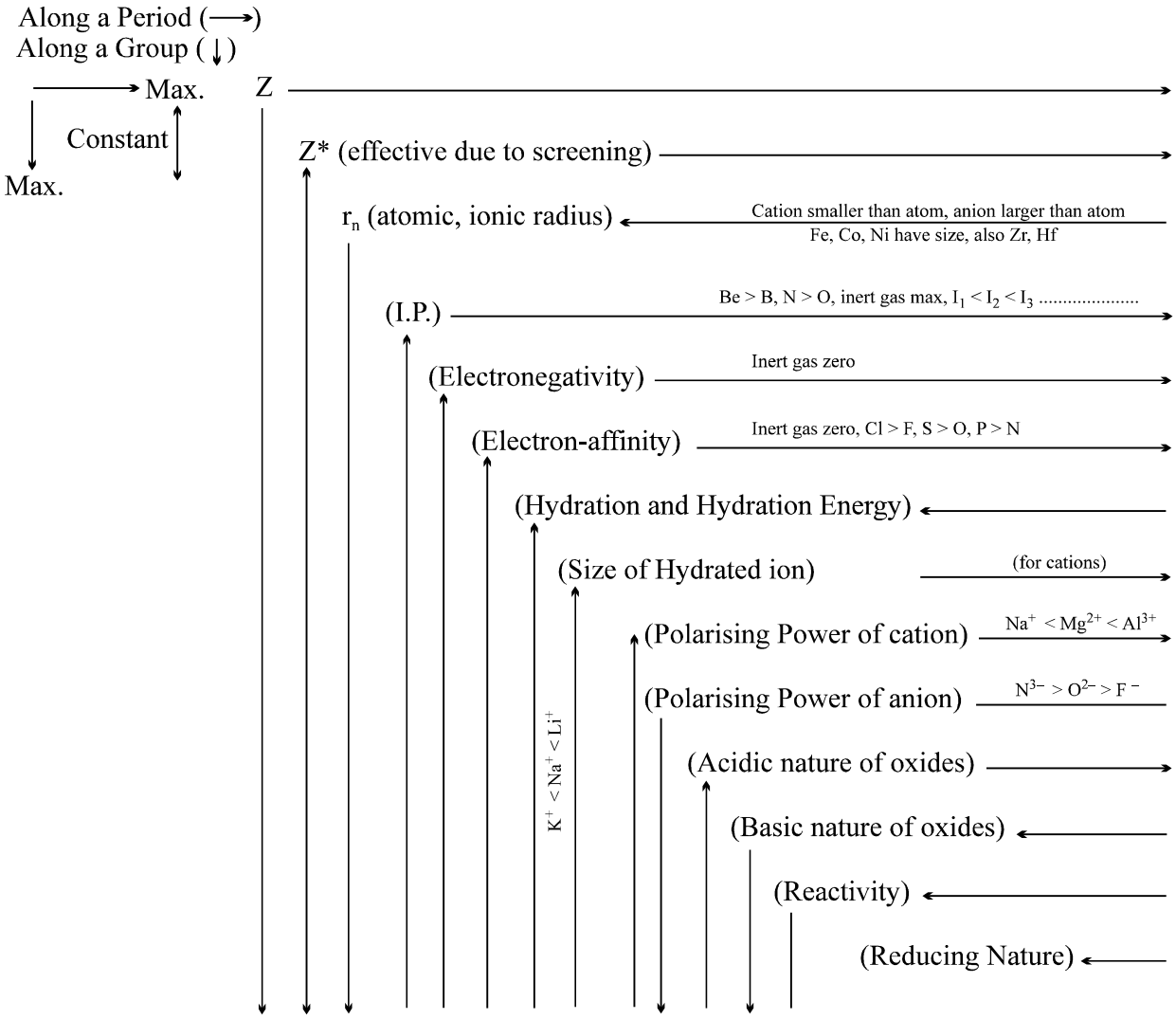
**4. Exercise III**

**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

# PERIODICITY




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- Isoelectric ions have different size.
- Inert pair effect is in p-block. Stability of higher state decreases and that of lower state increases going along a group. $Ge^{2+} < Sn^{2+} < Pb^{2+}$
- Reducing nature of hydride increases in a group and decreases in a period.

# Part–A (Periodic Table)

***INTRODUCTION :***

Many attempts were made to classify the known elements from time to time . These are :

**(i)** Proust Hypothesis **(ii)** Doberniers Triad law **(iii)** Newlands Octave law

**(iv)** Lother Meyer's curve **(v)** Mendeleev Periodic law **(vi)** Modern periodic law

**PERIODIC LAW (1869) :**

The physical and chemical properties of elements are periodic functions of their **atomic weight**.

**MODERN PERIODIC LAW :**

The physical and chemical properties of elements are periodic functions of their **atomic number**.

**LONG FORM OF PERIODIC TABLE : [ BOHR'S TABLE ]**

based on modern periodic law and Bohr Burry Scheme of E.C.

**CLASSIFICATION OF ELEMENT INTO GROUPS AND PERIODS :**

Group A : s and p block elements, representative elements. IA to VII A and O group.

Group B : d and f block elements, transition and inner transition elements IB to VII B and VIII groups.

Total 16 Groups

Period 1 to 7 classified as short, shortest, long, longest and incomplete period.

**CLASSIFICATION OF ELEMENTS INTO s , p , d & f BLOCK ELEMENTS :**

**s – block :**

**(i)** configuration $ns^{1-2}$ **(ii)** last $e^-$ enters in s orbital

**(iii)** two groups I A or 1 ; II A or 2

**p – block :**

**(i)** configuration $ns^2\ np^{1-6}$ **(ii)** last $e^-$ enters in p orbital

**(iii)** six groups III A , IV A , V A , VI A , VII A , zero or 13 , 14 , 15 , 16 , 17 , 18

**d – block : [ Transition Elements ]**

**(i)** configuration $ns^{1-2}\ (n-1)\ d^{1-10}$ **(ii)** last $e^-$ enters in d orbital

**(iii)** their two outermost shell are incomplete

**(iv)** 10 groups III B , IV B , V B , VI B , VII B , VIII (Triad) , I B , II B or 3 , 4 , 5 , 6 , 7 , (8 , 9 , 10) , 11 , 12 .

**(v)** four series 3 d , 4 d , 5 d , 6 d .

**f – block : [ Inner Transition ]**

**(i)** configuration $ns^2\ (n-1)\ d^{0-1}\ (n-2)\ f^{1-14}$

**(ii)** last $e^-$ enters in f orbital

**(iii)** two series 4 f Lanthanides & 5 f Actinides

**ESTIMATING POSITION OF AN ELEMENT FROM ITS ELECTRONIC CONFIGURATION:**

The last electron enters which subshell gives idea of its block.

[ ***Think*** : $1s^1$ and $1s^2$ belongs to which block]

Period no. is equal to the valence shell present in the configuration. Also for s and p block elements.

Period no. = valence shell, for 'd' block = shell + 1, for f block = shell + 2 and so on.

Group no. for s and p block = valence shell electron (A)

for d block = $d^1$ to $d^5$ [no. of (s + d) electron (B)]

$d^6$, $d^7$, $d^8$ (VIII)

$s^1\ d^9$, $d^{10}$ (IB, IIB)

**[ Use these carefully while locating the position.]**

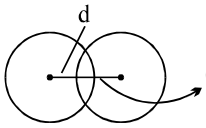
**COMMONLY ASKED PROPERTIES :**

**1.** ***Atomic Volume*** **:** Volume occupied by one gm atom of an element .

$$\text{Atomic volume} = \frac{\text{Atomic weight}}{\text{density}}$$

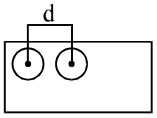
Lower atomic volume leads to higher density , increased hardness higher melting point, boiling point, less malleability & ductility.

**2.** ***Atomic Radius :*** Problem in calculating actual size of atom and hence distance between nuclei is calculated giving rise to three type of radii for atoms.

(a) ***Covalent radius :*** 

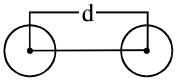
$C_r = \frac{d}{2}$ $C_r <$ actual atom size [Slight difference]

[Used for $H_2$, $Cl_2$ and such molecules]

(b) ***Metallic Radius :*** 

$M_r = \frac{d}{2}$ $M_r >$ actual size [Slight difference]

[Used for metals]

(c) ***Vanderwaal radius :*** 

$V_r = \frac{d}{2}$ $V_r >>$ actual size [very large difference]

In general $V_R > M_r > C_R$

(d) ***Ionic Radius*** **:** A cation is smaller than parent atom . An anion is larger than parent atom.

**FACTORS AFFECTING ATOMIC SIZE :**

(a) 'n' increase size increases

(b) $Z_{eff}$ increase size decrease $[Z_{eff} = Z - \sigma]$

(c) Type of measurement of radii.

Calculation of $Z_{eff}$

$Z_{eff} = \sqrt{\frac{En^2}{1312}}$ where E is I.E. in kJ/mole ; $Z_{eff} = \sqrt{\frac{E \times n^2}{13.6}}$ E is I.E. in eV per atom.

or Calculated by *Slaters rule*.

**SLATER'S RULE :**

(a) For calculating $\sigma$ on a (s or p) block (other than on 1s)

Rule-1 : Each (ns, nP) electron contribute to a screening factor of 0.35.

Rule-2 : Each $(n-1)^{th}$ shell electron contribute to a screening factor of 0.85.

Rule-3 : Each $(n-2)^{nd}$ and deeper shell electron contribute to a screening factor of 1.

[* On 1s, the screening factor due to other electron is taken as 0.3]

(b) For calculating $\sigma$ on (d or f) block.

Rule-1 : Each screening causing electron (d or *f* ) of same shell has factor of 0.35.

Rule-2 : Each electron other than *Rule*-1 have screening factor of 1.

***General Trend :***

Along a period, size decrease ['n' constant, $Z_{eff} \uparrow$ ]

Along a group, size increase ['n' increasing, $Z_{eff}$ constant ]

***Exceptions :***

(1) Noble gases have largest atomic sizes [Vander waal radii]. However, their covalent radii are smaller *e.g.* Xe.

(2) Size of Ga and Al are same, [$Z_{eff}$ increasing]

**ISOELECTRONIC SPECIES [Size depends upon Z, more Z less size]:**

**(i)** $S^{2-}$, $Cl^-$, $K^+$, $Ca^{2+}$, $Sc^{2+}$ **(ii)** $SO_3$, $NO_3^-$, $CO_3^{2-}$, $COCl_2$

**(iii)** $N_2$, CO, $CN^-$ **(iv)** $NH_3$, $H_3O^+$

**(v)** $H^-$, He, $Li^+$ **(vi)** $CH_4$, $NH_4^+$

**(vii)** $NCS^-$, $CS_2$ **(viii)** $\frac{\text{radius of cation}}{\text{radius of anion}} = \frac{Z_{eff} \text{ of Anion}}{Z_{eff} \text{ of Cation}}$

[ Check out for size for an isoelectronic noble gas.]

**IONISATION ENERGY :**

Amount of energy required to remove the most loosely bounded electron from an isolated gaseous atom.

Units : kJ $mol^{-1}$, k cal $mol^{-1}$, eV per atom.

Ionisation is endothermic (endoergic) i.e. requires energy hence

$\Delta H$ is +ve

$M + \text{Energy } (IE_1) \longrightarrow M^+ + e^-$

$M \longrightarrow M^+ + e^- \quad \Delta H = IE_1$

$M^+ \longrightarrow M^{+2} + e^- \quad \Delta H = IE_2$

$M^{+2} \longrightarrow M^{+3} + e^- \quad \Delta H = IE_3$

$IE_3 > IE_2 > IE_1$ always

**FACTORS AFFECTING IONISATION ENERGY :**

(1) Atomic size : Varies inversely

(2) Screening effect : varies inversely

(3) Nuclear charge : varies directly

(4) Sp Elect. config of outermost electron (half filled / fully filled)

(5) Type of orbital involved in Ionisation : s > p > d > f.

Half fillness and full fillness of inner orbitals. [affects d block and f block trends]

***General Trend:*** Along period I.E. increases [with some exception] [$Z_{eff} \uparrow$]

Along a group I.E. decrease [$Z_{eff}$ constant, $n \uparrow$]

***Exception :***

(1) Along a period, half filled and fully filled have higher I.E. e.g. Be > B and N > O.

(2) along a group, Ga > Al

**PROPERTIES AFFECTED BY IONISATION ENERGY:**

(1) Metallic character

(2) Tending to stay in which state $A^{+1}$. $A^{+2}$ or $A^{+3}$

(3) Other properties based on (1)

**ELECTRON AFFINITY :**

Amount of energy released when an electron is added to an isolated gaseous atom.

*Units :* k J $mol^{-1}$, k Cal $mol^{-1}$ and eV per atom.

Addition of electron results in release of energy in most of the cases but 2nd E. A. is always energy required. The sum of $EA_1$ & $EA_2$ is energy required.

$EA \propto \frac{1}{\text{atomic size}} \propto Z_{eff}$. Cl has the highest E.A.

**ELECTRON GAIN ENTHALPY :**

When expressed in terms of enthalpy change ($\Delta H$) then it is termed as E.G.E. Remember that $\Delta H$ = – ve for exothermic change.

For $EA_1$, energy is released $\therefore \Delta H_{EA1}$ = – ve

For $EA_2$, onwards is required $\therefore \Delta H_{EA2}$ = + ve

$EA_1 + EA_2$ , energy is always required.

**FACTORS AFFECTING ELECTRON AFFINITY :**

(1) ***Atomic size :*** varies inversely

(2) ***Nuclear change :*** varies directly

(3) Sp E.C. of half filled and fully filled shells.

***General Trend :*** Along a period, electron affinity increases [with exception] as $Z_{eff} \uparrow$.

Along a group, electron affinity decreases after 3rd period. Between 2nd and 3rd period in p block electron affinity of 2nd period is lesser due to high electron density.

***Exception :***

(1) A fully filled and half filled which have low values or even sometimes energy is required rather than getting released.

(2) 2nd period has lower value than 3rd owing to repulsion between electrons.

**ELECTRO NEGATIVITY : [ Properties of an atom in a molecule]**

F has highest. Decreasing order $\rightarrow$ F > O > Cl = N > Br > S = C > I > H.

**Pauling Scale:** $X_A - X_B = 0.208\sqrt{\Delta}$ E in kcal/mol

$\Delta = E_{A-B} - (E_{A-A} \times E_{B-B})^{1/2}$

**Mulliken's Scale :** $X_A = \frac{I_p + E_A}{2}$ (e v) .

Mulliken's values of E N are about 2.8 times as large as Pauling .

**Allred–Rochows :** $X_A = \frac{0.359\ Z_{eff}}{r^2} + 0.744$

**FACTOR AFFECTING ELECTRO NEGATIVITY :**

(1) ***Nuclear attraction :*** varies directly

(2) ***Atomic radius :*** varies inversely

(3) ***Change on ions :*** More positive charge more electronegativity and more –ve change less electronegativity.

(4) ***Hybridisation :*** to be discussed later in bonding.

***General Trends :*** Along a period, electronegativity increases

Along a group, electronegativity decreases

***Exceptions :*** None noteworthy.

**FACTORS DEPENDENT ON ELECTRO NEGATIVITY :**

(1) % ionic character varies directly.

(2) Strength of bond varies directly.

(3) B.L. : varies inversely.

(4) Nature of hydrides

(5) Nature of hydroxide.

**MISCELLANEOUS CHEMICAL PROPERTIES :**

***1. Periodicity of hydra acids :***

(a) Acidic character of hydra acid increases from left to right in a period.

(b) Acidic character of hydra acid increases from top to bottom in a group.

***2. Periodicity of oxy acids :***

(a) Acidic character of oxy acid increases from left in a period.

(b) Acidic character of oxy acid decreases from top to bottom in a group.

***3. Periodicity of nature of oxide :***

(a) On moving from left to right in a period acidic nature of oxide generally increases.
e.g. $CO_2 < P_2O_3 < SO_2 < ClO_2$

(b) On moving from top to bottom in a group acidic nature of oxide generally decreases.

***4. Solubility of salt in water :***

(a) ***Hydration energy*** decreases along a group.

(b) ***Lattice energy*** decreases along a group.

**TRENDS IN PHYSICAL PROPERTIES :**

Physical properties are mostly dependent on. Atomic weight and so not regular trend. Mark out exception in the graph and think out of the reasons?

**SOME COMMONLY USED TERMS :**

1. ***Noble Gases :*** Element of group 18 are called noble gases. These are also called as inert gases because their outermost ns and np orbitals are completely filled (except He and $1s^2$) and these gaseous are non-reactive in nature under ordinary conditions.

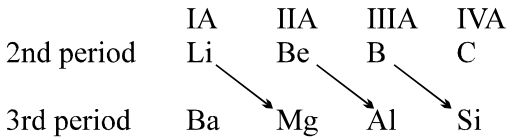
2. ***Representative elements :*** All the s and p block elements are known as representative elements except zero group.

3. ***Transition elements :*** All the d-block elements (except IIB group) are called transition element. It comprises into $4^{th}$, $5^{th}$, $6^{th}$ and $7^{th}$ period. They lie between s and p block elements.

4. ***Inner transition elements :*** All the f-block elements or 4f and 5f block elements are called inner transition element. Total number of these elements is 28. They lie in IIIB and placed at the bottom of periodic table.

5. ***Typical elements :*** Elements second and third period are known as typical elements.

6. ***Diagonal relationship :*** Properties of elements of second period resemble with the element of third period. These resembled properties between two periods or this type of relation between two periods are called diagonal relationship.




**<u>Increase your I bank</u>**

**(i) Resemblance between Li and Mg :**

(a) Unlike the other members of the group, lithium reacts with $N_2$ to form a nitride in the same way as magnesium does.

(b) Lithium hydroxide, carbonate and nitrate decomposes on heating to give $Li_2O$ as like Mg but other alkali hydroxides and carbonates are unchanged on heating where as the nitrate decompose to give nitrite.

(c) Lithium hydroxide carbonate and fluoride are much soluble than the corresponding sodium or potassium compounds. The solubilities are comparable to those of corresponding magnesium compound.

**(ii) Resemblance between Be and Al :**

(a) The ionic radius of $Be^{2+}$ is nearly same as that for the $Al^{3+}$.
(b) Like aluminium, beryllium is not readily attacked by acids because of the pressure of an oxide film.
(c) Beryllium dissolved in alkali to give the beryllate ion $[B(OH)_4]^{2-}$ just as aluminium does to give $(Al(OH)_6)^3$.
(d) The oxides BeO and $Al_2O_3$ are hard high melting insoluble solids. The oxides as well as their hydroxides amphoteric and dissolve in sodium hydroxide solution.
(e) Beryllium and aluminium from fluoro complex anion, $BeF_4^{2-}$ and $AlF_6^{3-}$ in solution, the other group II metals do not form stable fluoro complexes in solution.
(f) Beryllium chloride ($BeCl_2$) is essentially covalent and has a bridged polymeric structure just as aluminium trichloride is covalent forming a bridged dimer, $Al_2Cl_6$. Both the chlorides are soluble in organic solvent and are strong Lewis acid.

**(iii) Resemblance between B and Si :**

(a) Boron and silicon form numerous hydride which spontaneously catch fire on exposure to air and are easily hydrolysed.
(b) Boron halide like silicon halides hydrolysed by water. Aluminium halides are only partially hydrolysed by water.
(c) Boron forms binary compounds with several metals known as borides just as silicon forms metal silicides some of the borides and silicides under go hydrolysis to yield boron and silicon respectively.
(d) $B_2O_3$ and $SiO_2$ are acidic in nature, Borates and silicates have tetrahedral $BO_4$ and $SiO_4$ structural units respectively. Boro silicates are known in which form can replace silicon in three dimensional lattice. However boron can also form planer $BO_3$ unit.
(e) Both B and Al are semiconductors.

***Bridge Elements :*** Typical elements of II period.

**NOMENCLATURE OF THE ELEMENT :**

The names are derived by using roots for the three digits in the atomic number of the element and adding the ending –ium. The roots for the number are

| Digit | Name | Abbreviation |
|---|---|---|
| 0 | nil | n |
| 1 | un | u |
| 2 | bi | b |
| 3 | tri | t |
| 4 | quad | q |
| 5 | pent | p |
| 6 | hex | h |
| 7 | sept | s |
| 8 | oct | o |
| 9 | enn | e |

Thus element with atomic number 109 will be named as **une** (**u** for 1, **n** for 0 and **e** for 9). Table summarises the names of the elements with atomic number above 100.

# ***EXERCISE # I***

## ***General Info about periodic table***

Q.1 Which of the following does not reflect the periodicity of element
(A) Bonding behaviour (B) Electronegativity (C) Ionisation potential (D) Neutron/ Proton ratio

Q.2 Choose the s-block element in the following:
(A) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^5, 4s^1$ (B) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}, 4s^1$
(C) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$ (D) all of the above

Q.3 False statement for periodic classification of elements is
(A) The properties of the elements are periodic function of their atomic numbers.
(B) No. of non-metallic elements is less than the no. of metallic elements.
(C) First ionization energy of elements does change continuously with increasing of atomic no. in a period.
(D) d-subshell is filled by final electron with increasing atomic no. of inner transition elements.

Q.4 Pick out the isoelectronic structure from the following:

| I. $^+CH_3$ | II. $H_3O^+$ | III. $NH_3$ | IV. $CH_3^-$ |
|---|---|---|---|
| (A) I and II | (B) III and IV | (C) I and III | (D) II, III and IV |

Q.5 If there were 10 periods in the periodic table then how many elements would this period can maximum comprise of.

Q.6 If $(n + l)$ rule for energy is not followed, what are the blocks of the following elements if they are filled according to increasing shell number
(a) K(19) (b) Fe(26) (c) Ga(31) (d) Sn(50)

Q.7 Use the following system of naming elements in which first alphabets of the digits are written collectively,

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| nil | uni | bi | tri | quad | pent | hex | sept | oct | enn |

to write three-letter symbols for the elements with atomic number 101 to 109.
[Example : 101 is Unu....]

## ***Properties and Periodic trends***

Q.8 The size of the following species increases in the order:
(A) $Mg^{2+} < Na^+ < F^- < Ar$ (B) $F^- < Ar < Na^+ < Mg^{2+}$
(C) $Ar < Mg < F^- < Na^+$ (D) $Na^+ < Ar < F^- < Mg^{2+}$

Q.9 Element in which maximum ionization energy of following electronic configuration would be
(A) [Ne] $3s^2\ 3p^1$ (B) [Ne] $3s^2\ 3p^2$ (C) [Ne] $3s^2\ 3p^3$ (D) [Ar] $3d^{10}\ 4s^2\ 4p^3$

Q.10 The outermost electronic configuration of most electronegative element is:
(A) $ns^2\ np^1$ (B) $ns^2\ np^4$ (C) $ns^2\ np^5$ (D) $ns^2\ np^6$

Q.11 The electron affinity of the members of oxygen of the periodic table, follows the sequence
(A) O > S > Se (B) S > O < Se (C) O < S > Se (D) Se > O > S

Q.12 The process of requiring absorption of energy is
(A) $F \rightarrow F^-$ (B) $Cl \rightarrow Cl^-$ (C) $O^- \rightarrow O^{2-}$ (D) $H \rightarrow H^-$

Q.13 In the following which configuration of element has maximum electronegativity.
(A) $1s^2, 2s^2\ 2p^5$ (B) $1s^2, 2s^2\ 2p^6$ (C) $1s^2, 2s^2\ 2p^4$ (D) $1s^2, 2s^2\ 2p^6, 3s^2\ 3p^3$

Q.14 Highest size will be of
(A) $Br^-$ (B) I (C) $I^-$ (D) $I^+$

Q.15 Atomic radii of flourine and neon in Å units are respectively given by
(A) 0.72, 1.60 (B) 1.60, 1.60 (C) 0.72, 0.72 (D) 1.60, 0.72

Q.16 The correct order of second ionisation potential of C, N, O and F is:
(A) C > N > O > F (B) O > N > F >C (C) O > F > N > C (D) F > O > N > C

Q.17 Decreasing ionization potential for K, Ca & Ba is
(A) Ba> K > Ca (B) Ca > Ba > K (C) K > Ba > Ca (D) K > Ca > Ba

Q.18 Element Hg has two oxidation states $Hg^{+1}$ & $Hg^{+2}$. the right order of radii of these ions.
(A) $Hg^{+1} > Hg^{+2}$ (B) $Hg^{+2} > Hg^{+1}$ (C) $Hg^{+1} = Hg^{+2}$ (D) $Hg^{+2} \geq Hg^{+1}$

Q.19 The ionization energy will be maximum for the process.
(A) $Ba \rightarrow Ba^{++}$ (B) $Be \rightarrow Be^{++}$ (C) $Cs \rightarrow Cs^+$ (D) $Li \rightarrow Li^+$

Q.20 Why the first ionisation energy of carbon atom is greater than that of boron atom whereas, the reverse is true for the second ionisation energy.

Q.21 On the Pauling's electronegativity scale, which element is next to F.

Q.22 $Mg^{2+}$, $O^{2-}$, $Na^+$, $F^-$, $N^{3-}$ (Arrange in decreasing order of ionic size)

Q.23 Why $Ca^{2+}$ has a smaller ionic radius than $K^+$.

Q.24 Which of the ions are paramagnetic $Sr^{2+}$, $Fe^{3+}$, $Co^{2+}$, $S^{2-}$, $Pb^{2+}$

Q.25 Why do alkaline earth metals always form dipositive ions.

Q.26 State giving reasons which one have higher value :
(a) $IE_1$ of F or Cl (b) E A of O or $O^-$ (c) ionic radius of $K^+$ or $Cl^-$

Q.27 Explain why a few elements such as Be (+0.6), N(+0.3) & He(+0.6) have positive electron gain enthalpies while majority of elements do have negative values.

Q.28 From among the elements, choose the following: Cl, Br, F, Al, C, Li, Cs & Xe.
(i) The element with highest electron affinity. (ii) The element with lowest ionisation potential.
(iii) The element whose oxide is amphoteric. (iv) The element which has smallest radii.
(v) The element whose atom has 8 electrons in the outermost shell.

Q.29 Which property will increase and which will decrease for IA group as we go down the group.
(a) Atomic size (g) E N
(b) Ionic radii (h) At. mass
(c) I E (i) Valance $e^-$
(d) Density (j) Metallic ch
(e) Melting point (k) Chemical reactivity
(f) Boiling point

Q.30 The IE do not follow a regular trend in II & III periods with increasing atomic number. Why?

Q.31 Arrange in decreasing order of atomic size : Na, Cs, Mg, Si, Cl.

Q.32 In the ionic compound KF, the $K^+$ and $F^-$ ions are found to have practically radii, about 1.34 Å each. What do you predict about the relative covalent radii of K and F?

Q.33 Does $Na_2(g)$ molecule exhibit metallic properties.

Q.34 Which will have a higher boiling point, $Br_2$ or ICl , & why?

Q.35 Which bond in each pair is more polar
(a) P – Cl or P – Br (b) S – Cl or S – O (c) N – O or N – F

Q.36 Arrange noble gases , in the increasing order of b.p.

## ENERGY BASED CALCULATIONS

Q.37 Calculate E.N. of flourine if $(r_F)_{covalent} = 0.72$ Å.

Q.38 Calculate E.N. of chlorine atom on Pauling scale if I.E. of $Cl^-$ is 4eV & of E.A. of $Cl^+$ is + 13.0 eV.

Q.39 Calculate the electronegativity of fluorine from the following data :
$E_{H-H} = 104.2$ kcal $mol^{-1}$ $E_{F-F} = 36.2$ kcal $mol^{-1}$
$E_{H-F} = 134.6$ kcal $mol^{-1}$ $X_H = 2.1$

Q.40 Calculate the E.N. of Cl from the bond energy of ClF (61 KCal/mol). Given that bond energies of $F_2$ and $Cl_2$ are 38 and 58 KCal/mol respectively.

Q.41 The IE values of $Al_{(g)} = Al^+ + e$ is 577.5 kJ $mol^{-1}$ and $\Delta H$ for $Al_{(g)} = Al^{3+} + 3e$ is 5140 kJ $mol^{-1}$. If second and third IE values are in the ratio 2 : 3. Calculate $IE_2$ and $IE_3$.

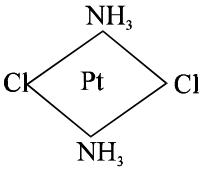
Q.42 How many chlorine atoms will be ionised $Cl \longrightarrow Cl^+ + e^{-1}$ by the energy released from the process $Cl + e^{-1} \longrightarrow Cl^-$ for $6.02 \times 10^{23}$ atoms (I.P. for Cl = 1250 kJ $mol^{-1}$ and E.A. = 350 kJ $mole^{-1}$)

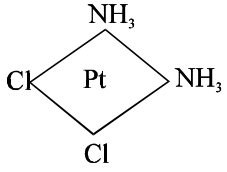
Q.43 For the gaseous reaction,
$K + F \longrightarrow K^+ F^-$, $\Delta H$ was calculated to be 19 kcal under conditions where the cations and anions were prevented by electrostatic separation from combining with each other. The ionisation potential of K is 4.3 eV. What is the electron affinity of F?

Q.44 The ionisation potentials of atoms A and B are 400 and 300 kcal $mol^{-1}$ respectively. The electron affinities of these atoms are 80.0 and 85.0 k cal $mol^{-1}$ respectively. Prove that which of the atoms has higher electronegativity.

Q.45 The As-Cl bond distance in $AsCl_3$ is 2.20 Å. Estimate the SBCR (single bond covalent radius) of As. (Assume EN of both to be same and radius of Cl = 0.99 Å.)

Q.46 The Pt-Cl distance has been found to be 2.32 Å in several crystalline compounds. If this value applies to both of the compounds shown in figure. What is Cl - Cl distance in (a) and (b)

(a) 


(b) 


### ***Effective nuclear charge and screening***

Q.47 Calculate the screening constant of Ca . (atomic number 20)

Q.48 Calculate the effective nuclear charge on–
(i) 4s valency $e^-$ in Bromine atom. and (ii) 3d electron in Bromine atom.

Q.49 I.P. of $Be^{+x}$ is found to be 217.6 electron volt. What is the value of x.

Q.50 Calculate $Z_{eff}$ from slater's rule & from Bohr's model. Take I.E. of K from graph. IE of K is 4.3 eV.

## ***Miscellaneous Properties***

Q.51 Arrange following oxides in increasing acidic nature
$Li_2O$, BeO, $B_2O_3$

Q.52 Which oxide is more basic, MgO or BaO? Why?

Q.53 The basic nature of hydroxides of group 13 (III-A) decreases progressively down the group. Comment.

Q.54 Based on location in P.T., which of the following would you expect to be acidic & which basic.
(a) CsOH (b) IOH (c) $Sr(OH)_2$ (d) $SeO_3(OH)_2$ (e) FrOH (f) BrOH

# ***EXERCISE # II***

***Question No. 1 and 2 are based on the following information.***
Four elements P, Q, R & S have ground state electronic configuration as:
P $\rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$ Q $\rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^1$
R $\rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^3$ S $\rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^1$

Q.1 Comment which of the following option represent the correct order of true (T) & false (F) statement.
I size of P < size of Q
II size of R < size of S
III size of P < size of R (appreciable difference)
IV size of Q < size of S (appreciable difference)
(A) TTTT (B) TTTF (C) FFTT (D) TTFF

Q.2 Order of $IE_1$ values among the following is
(A) P > R > S > Q (B) P < R < S < Q (C) R > S > P > Q (D) P > S > R > Q

In *following question* a statement S and an explanation E is given. Choose the correct answers from the codes A, B, C, D given for given question.
(A) S is correct but E is wrong.
(B) S is wrong but E is correct.
(C) Both S and E are correct and E is correct explanation of S.
(D) Both S and E are correct but E is not correct explanation of S.

Q.3 S : Lithium is a better reducing agent than Cs.
E : Sublimation energy and Ionisation energy of lithium is less than that of Cs.

Q.4 S : The first ionization energy of Be is greater than that of B.
E : 2p orbital is lower in energy than 2s.

Q.5 Bond distance C–F in ($CF_4$) & Si–F in ($SiF_4$) are respective 1.33Å & 1.54 Å. C–Si bond is 1.87 Å. Calculation the covalent radius of F atom ignoring the electronegativity differences.
(A) 0.64 Å (B) $\frac{1.33+1.54+1.8}{3}$ Å (C) 0.5 Å (D) $\frac{1.54}{2}$ Å

Q.6 Two elements A & B are such that B. E. of A–A, B–B & A–B are respectively 81 Kcal / mole, 64 Kcal / mole, 76 Kcal / mole & if electronegativity of B is 2.4 then the electronegativity of A may be approximately
(A) 2.81 (B) 1.8 (C) 1.99 (D) 3.0

## ***EXERCISE # III***

Q.1 Moving from right to left in a periodic table, the atomic size is: **[JEE 1995]**
(A) increased (B) decreased (C) remains constant (D) none of these

Q.2 The increasing order of electronegativity in the following elements: **[JEE 1995]**
(A) C, N, Si, P (B) N, Si, C, P (C) Si, P, C, N (D) P, Si, N, C

Q.3 One element has atomic weight 39. Its electronic configuration is $1s^2$, $2s^2$ $2p^6$, $3s^2$ $3p^6$ $4s^1$. The true statement for that element is:
(A) Hight value of IE (B) Transition element
(C) Isotone with $_{18}Ar^{38}$ (D) None **[JEE 1995]**

Q.4 The number of paired electrons in oxygen is: **[JEE 1995]**
(A) 6 (B) 16 (C) 8 (D) 32

Q.5 Fluorine is the most reactive among all the halogens, becuase of its: **[JEE 1995]**
(A) small size (B) low dissociation energy of F-F bond.
(C) large size (D) high dissociation energy of F-F bond.

Q.6 The decreasing size of $K^+$, $Ca^{2+}$, $Cl^-$ & $S^{2-}$ follows the order: **[REE 1995]**
(A) $K^+ > Ca^{+2} > S^{-2} > Cl^-$ (B) $K^+ > Ca^{+2} > Cl^- > S^{-2}$
(C) $Ca^{+2} > K^+ > Cl^- > S^{-2}$ (D) $S^{-2} > Cl^- > K^+ > Ca^{+2}$

Q.7 Which of the following oxide is neutral? **[JEE 1996]**
(A) CO (B) $SnO_2$ (C) ZnO (D) $SiO_2$

Q.8 Which of the following has the maximum number of unpaired electrons **[JEE 1996]**
(A) $Mg^{2+}$ (B) $Ti^{3+}$ (C) $V^{3+}$ (D) $Fe^{2+}$

Q.9 The following acids have been arranged in the order of decreasing acid strength. Identify the correct order **[JEE 1996]**

ClOH(I) BrOH(II) IOH(III)

(A) I > II > III (B) II > I > III (C) III > II > I (D) I > III > II

Q.10 The incorrect statement among the following is **[JEE 1997]**
(A) $IE_1$ of Al is less than $IE_1$ of Mg (B) $IE_2$ of Mg is greater than $IE_2$ of Na
(C) $IE_1$ of Na is less than $IE_1$ of Mg (D) $IE_3$ of Mg is greater than $IE_3$ of Al

Q.11 The incorrect statement among the following is: **[JEE 1997]**
(A) the first ionisation potential of Al is less that the first ionisation potential of Mg
(B) the second ionisation potential of Mg is greater that the second ionisation potential of Na
(C) the first ionisation potential of Na is less than the first ionisation potential of Mg
(D) the third ionisation potential of Mg is greater than the third ionisation potential of Al

Q.12 Which of the following are amphoteric? **[REE 1997]**
(A) $Be(OH)_2$ (B) $Sr(OH)_2$ (C) $Ca(OH)_2$ (D) $Al(OH)_3$

Q.13 $Li^+$, $Mg^{2+}$, $K^+$, $Al^{3+}$ (Arrange in increasing order of radii) **[JEE 1997]**

Q.14 Ionic radii of: **[JEE 1999]**
(A) $Ti^{4+} < Mn^{7+}$ (B) $^{35}Cl^- > ^{37}Cl^-$ (C) $K^+ > Cl^-$ (D) $P^{3+} > P^{5+}$

***Directions:*** The questions below to consist of an 'assertion in column-1 and the 'reason' in column-2. Against the specific question number, write in the appropriate space.
(A) If both assertion and reason are CORRECT, and reason is the CORRECT explanation of the assertion. (B) If both assertion and reason are CORRECT, but reason is not the CORRECT explanation of the assertion. (C) If assertion if CORRECT but reason is INCORRECT (D) If assertion is INCORRECT reason in CORRECT.

Q.15 **Assertion**: F atom has a less negative electron gain enthalpy than Cl atom. **[JEE 2000]**
**Reason**: Additional electron is repelled more efficiently by 3p electron in Cl atom than by 2p electron in F atom.

Q.16 **Assertion**: $Al(OH)_3$ is amphoteric in nature. **[JEE 2000]**
**Reason**: Al –O and O – H bonds can be broken with equal case in $Al(OH)_3$.

Q.17 The correct order of radii is: **[JEE 2000]**
(A) N < Be < B (B) $F^- < O^{2-} < N^{3-}$ (C) Na < Li < K (D) $Fe^{3+} < Fe^{2+} < Fe^{4+}$

Q.18 The correct order of acidic strength is: **[JEE 2000]**
(A) $Cl_2O_7 > SO_2 > P_4O_{10}$ (B) $CO_2 > N_2O_5 > SO_3$
(C) $Na_2O > MgO > Al_2O_3$ (D) $K_2O > CaO > MgO$

Q.19 The $IE_1$ of Be is greater than that of B. [**T/F**] **[JEE 2001]**

Q.20 The set representing correct order of $IP_1$ is **[JEE 2001]**
(A) K > Na > Li (B) Be > Mg > Ca (C) B > C > N (D) Fe > Si > C

Q.21 Identify the least stable ion amongst the following: **[JEE 2002]**
(A) $Li^-$ (B) $Be^-$ (C) $B^-$ (D) $C^-$

Q.22 Identify the correct order of acidic strengths of $CO_2$, CuO, CaO, $H_2O$: **[JEE 2002]**
(A) $CaO < CuO < H_2O < CO_2$ (B) $H_2O < CuO < CaO < CO_2$
(C) $CaO < H_2O < CuO < CO_2$ (D) $H_2O < CO_2 < CaO < CuO$

# ANSWER KEY

## EXERCISE # I

Q.1 D Q.2 C Q.3 D Q.4 D

Q.5 72 Q.6 (a) d block, (b) d block, (c) p block, (e) f block

| Q.7 | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 |
|---|---|---|---|---|---|---|---|---|---|
| | Unu | Unb | Unt | Unq | Unp | Unh | Uns | Uno | Une |

Q.8 A Q.9 C Q.10 C Q.11 C

Q.12 C Q.13 A Q.14 C Q.15 A

Q.16 C Q.17 B Q.18 A Q.19 B

Q.20 Zeff & half filled config. Q.21 O

Q.22 $N^{3-} > O^{2-} > F^{-} > Na^{+} > Mg^{2+}$ Q.23 Isolelectronic $Ca^{+2}$(higher)

Q.24 $Fe^{3+}$ , $CO^{2+}$ Q.25 difference in $1E_1$ & $1E_2$ is less than 10ev.

Q.26 (a) F (b) O (c) $Cl^{-}$ Q.27 half filled and fully filled orbitals

Q.28 (i) Cl (ii) Cs (iii) Al (iv) F (v) Xe

Q.29 Increases $\rightarrow$ a , b , d, h , j , k , Decrease $\rightarrow$ c , e , f , g , Same $\rightarrow$ i

Q.30 half filled & fully filled orbitals Q.31 Cs > Na > Mg > Si > Cl

Q.32 $r_k > 1.34\text{Å} > r_F$ Q.33 No Q.34 ICl

Q.35 (a) P–Cl (b) S–O, (C) N–F Q.36 He < Ne < Ar < Kr < Xe < Rn

Q.37 4, 4.3 Q.38 3.03 (Pauling) Q.39 3.8752 Q.40 3.2

Q.41 $IE_2 = 1825$ kJ/mole, $IE_3 = 2737.5$ kJ/mol Q.42 $1.686 \times 10^{23}$ atom

Q.43 3.476 eV Q.44 $EN_1 > EN_2$

Q.45 1.21 Å Q.46 4.64 Å ; b = 3.28 Å Q.47 17.15 Q.48 (i) 7.6 (ii) 13.85

Q.49 $Be^{+3}$ Q.50 2.2, (Slater's rule) 2.25 (Bohr's model)

| Q.51 | $Li_2O$ | < | BeO | < | $B_2O_3$ |
|---|---|---|---|---|---|
| | basic | | amphoteric | | acidic |

Q.52 BaO Q.53 False

Q.54 (a) basic (b) acidic (c) basic (d) acidic (e) basic (f) acidic

## EXERCISE # II

Q.1 B Q.2 A Q.3 A Q.4 C Q.5 C Q.6 A, C

## EXERCISE # III

Q.1 A Q.2 C Q.3 C Q.4 A

Q.5 A B Q.6 D Q.7 A Q.8 D

Q.9 A Q.10 B Q.11 B Q.12 AD

Q.13 $Al^{+3} < Li^{+} < Mg^{2+} < K^{+}$ Q.14 D Q.15 C Q.16 C

Q.17 B Q.18 A Q.19 True Q.20 B

Q.21 A Q.22 A

## MODERN (MODIFIED) MENDELEEFF'S PERIODIC TABLE

| GROUP / PERIOD | I A | I B | II A | II B | III A | III B | IV A | IV B | V A | V B | VI A | VI B | VII A | VII B | VIII | | | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. | | | | | | | | | | | | | | H 1 | | | | He 2 |
| 2. | Li 3 | | Be 4 | | | B 5 | | C 6 | | N 7 | | O 8 | | F 9 | | | | Ne 10 |
| 3. | Na 11 | | Mg 12 | | | Al 13 | | Si 14 | | P 15 | | S 16 | | Cl 17 | | | | Ar 18 |
| 4. | K 19 | Cu 29 | Ca 20 | Zn 30 | Sc 21 | Ga 31 | Ti 22 | Ge 32 | V 23 | As 33 | Cr 24 | Se 34 | Mn 25 | Br 35 | Fe 26 | Co 27 | Ni 28 | Kr 36 |
| 5. | Rb 37 | Ag 47 | Sr 38 | Cd 48 | Y 39 | In 49 | Zr 40 | Sn 50 | Nb 41 | Sb 51 | Mo 42 | Te 52 | Tc 43 | I 53 | Ru 44 | Rh 45 | Pd 46 | Xe 54 |
| 6. | Cs 55 | Au 79 | Ba 56 | Hg 80 | La* 57-71 | Tl 81 | Hf 72 | Pb 82 | Ta 73 | Bi 83 | W 74 | Po 84 | Re 75 | At 85 | Os 76 | Ir 77 | Pt 78 | Rn 86 |
| 7. | Fr 87 | Uuu 111 | Ra 88 | Uub 112 | Ac†. 89-103 | —— | Ku(Rf) 104 | Uuq 114 | Ha 105 | —— | Unh(Sg) 106 | Uuh 116 | Uns(Bh) 107 | —— | Uno (Hs) 108 | Une (Mt) 109 | Uun 110 | Uuo 118 |

## INNER TRANSITION ELEMENTS

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *LANTHANIDE SERIES | Ce 58 | Pr 59 | Nd 60 | Pm 61 | Sm 62 | Eu 63 | Gd 64 | Tb 65 | Dy 66 | Ho 67 | Er 68 | Tm 69 | Yb 70 | Lu 71 |
| †ACTINIDE SERIES | Th 90 | Pa 91 | U 92 | Np 93 | Pu 94 | Am 95 | Cm 96 | Bk 97 | Cf 98 | Es 99 | Fm 100 | Md 101 | No 102 | Lr 103 |

**Fig. 1.3** Modern Mendeleeff's Periodic Table

# STUDY PACKAGE

# Target: IIT-JEE (Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 6. Gaseous State

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

**4. Exercise III**

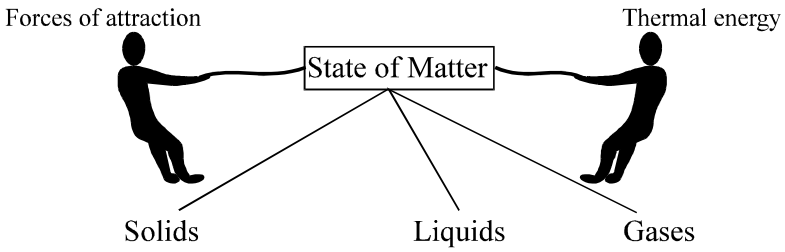
**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

## *KEY CONCEPTS*




**Parameters:**

| | |  |  |  |
|---|---|---|---|---|
| (i) | Forces of attraction | | | |
| (ii) | Thermal energy | down | moderate | up |
| (iii) | Space | down | moderate | up |
| (iv) | Shape | up | down | down |
| (v) | Volume | up | up | down |
| (vi) | Density | up | up | down |

**GASEOUS STATE :** The state is characterized by sensitivity of volume change with change of pressure and temperature. It is due to large distance between molecules as compared to their own dimensions. There exists weak Vander Waal's forces, molecules move independent of each other with speed about 400 $m\,s^{-1}$ .




***Are volume of solids & liquid totally independent of pressure??***

**IDEAL GAS :** A gas with no intermolecular attractions & having very negligible volume occupied by molecules when compared with volume of gas is termed as ideal gas. A theoretical concept which for gases present can be obtained only under certain condition.

**REAL GAS** : Considerable forces of attraction & appreciable size of molecules. These under " certain conditions" behalve like ideal. [Refer : section of real gas behaviour]

**Parameter associated with the gas :** P , V , T , n

where

**P** represents pressure exerted by the gas molecules on the walls of the container assuming negilible intermolecular attractions,

**V** represents free volume available for motion (equal to the volume of the container),

**T** represents absolute temperature, **n** represents no of moles.

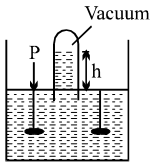
**Conversion factors :** Pressure $\rightarrow$ 1 **atm** = $1.013 \times 10^5$ **Pa** = 760 **mm of Hg** = 760 **torr** = 1.013 **bar**

Volume $\rightarrow 1\,\boldsymbol{l} = 1\,\mathbf{dm^3} = 10^{-3}\,\mathbf{m^3} = 1000\,\mathbf{ml} = 1000\,\mathbf{cm^3}$

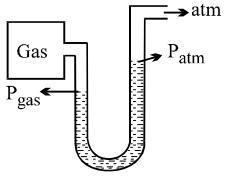
Temperature $\rightarrow \mathbf{T_K} = \mathbf{T_C^\circ} + 273 = \frac{5}{9}\mathbf{T_F^\circ} + 255.22$

## INSTRUMENTS FOR PRESSURE CALCULATIONS :

**Barometer :** $P = \dfrac{A \times h \times d \times g}{A}$ where d = density of fluid
h = vertical height
g = acceleration due to



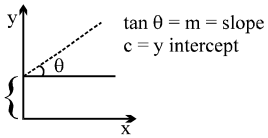

**Manometer :** $P_{gas} = P_{atm} + hdg$



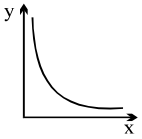

## EQUATION & GRAPHS OF EXTENSIVE UTILITY IN GASEOUS STATE :

***Two Parameters 'y' & 'x' if are related as***

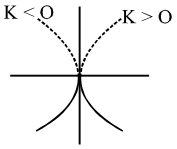
(a) $y = m x + C$ (where m & C are constants) [then there will be a <u>direct relationship</u> between them & graph will be a straight line as shown ]
$y = mx + C$ (straight line)



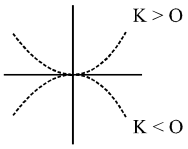

(b) $xy$ = constant (rect hyperbola) / $y = \dfrac{m}{x} + C$ (where m & C are constant)




(c) $y^2 = Kx$ ( K is a constant)




(d) $x^2 = Ky$ ( K is a constant)

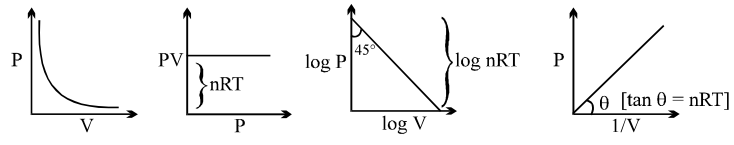



**Experimental Gas laws** ⟶ Relationship between various parameter of the gas. Gaseous state is the only state that allows a quantitative descriptive between the four parameters, P, V, T & n. The relationship which connects the four variables is known as equation of state, which can be obtained experimentally from the following gas laws.

# All are based on experimental data.
# All are applicable for ideal gases only.
# Get yourselves comfortable with all the various types of graphs to get a 'feel' of them.

I. **Boyle's law** $V \propto \dfrac{1}{P}$ $\left(\begin{matrix} T \text{ , constant} \\ n \text{ constant} \end{matrix}\right)$ $P_1 V_1 = P_2 V_2$

**graphs are Isotherms**

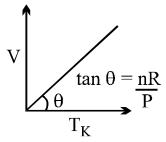


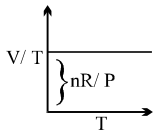


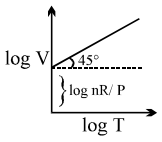
Plot the different curves for difference values of n & V to compare.

II. **Charle's law** $V \propto T$ $\begin{pmatrix} P \text{ , constant} \\ n \text{ constant} \end{pmatrix}$ $\frac{V_1}{T_1} = \frac{V_2}{T_2}$

**graphs are Isobars**



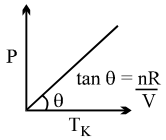



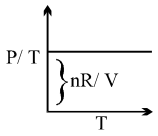


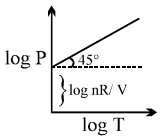

Plot graphs of V vs $T_c$ & V vs $T_F$

III. **Gay Lussac's law** $P \propto T$ $\begin{pmatrix} V \text{ , constant} \\ n \text{ constant} \end{pmatrix}$ $\frac{P_1}{T_1} = \frac{P_2}{T_2}$

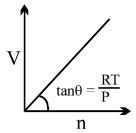
**graphs are Isochor**

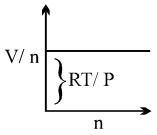


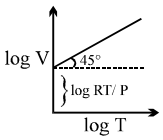




IV. **Avogadro's law** $V \propto n$ $(T, P \text{ constant})$ $\frac{V_1}{n_1} = \frac{V_2}{n_2}$




**Combined Gas Law :** $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$

**Equation Of State :** $PV = nRT$

$d$ = density of gas

$\Rightarrow$ $PV = \frac{W}{M} RT$ $R$ = Universal Gas constant = 0.0821 atm litres /kelvin/mol

$\Rightarrow$ $PM = dRT$ = 8.314 joule/kelvin = 2cal / kelvin/mol

**Dalton's law of partial pressure :**

$P_{total} = P_A + P_B + .....$ ; $P_{wet\ gas} = P_{dry\ gas} + P_{H_2O\ vapour}$ i.e. aq. tension.

$P_A$, $P_B$ are partial pressures . ; $P_A$ = mole fraction$_A$ × Total pressure

and % of gas in mixture = $\frac{\text{Partial pressure}}{\text{Total pressure}} \times 100$ .

**Amagat's Law :**

The total volume of a mixture of gases is equal to the sum of the partial volumes of the constituent gases, at same Temperature & Pressure.

**Graham's law of Diffusion & Effusion :**

$r \propto \frac{1}{\sqrt{d}}$ or $r \propto \frac{1}{\sqrt{M}}$ $r \propto \frac{P}{\sqrt{M}}$ [For gases effusing at different pressures]

**r** is rate of diffusion of any gas.

$\frac{r_1}{r_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{M_2}{M_1}}$ ; $\frac{\text{volume / time}}{\text{volume / time}} = \sqrt{\frac{M_2}{M_1}}$ **d** is density at some temperature.

$r = \frac{\text{moles diffused}}{\text{time taken}} = \frac{\text{distance travelled in a narrow tube}}{\text{time taken}} = \frac{\text{Pressure drop I}}{\text{Pressure drop II}}$



It should be noted that the rate of diffusion or effusion actually depends on pressure difference of the gas and not simply on its pressure. Moreover the pressure difference is to be measured for this gas only i.e. if a container holds [He] at a pressure of 0.1 atm and if a small pin-hole is made in the container and if the container is placed in a room, then the rate of effusion of He gas from the container to outside depends only on its pressure difference , which is 0.1 - 0 (as there is no He in the atmosphere). This implies that the diffusion of a gas is not dependent on the diffusion of any other gas.



Whenever we consider the diffusion of a gas under experimental conditions, we always assume that the gas diffuses in vaccum and during the time period for which the diffusion is studied the rate of diffusion (or the composition of diffusing or effusing mixture of gases) remains constant.

**Miscellaneous concepts used in Gaseous State:**

(a) **Bursting of containers :** two concepts used depending upon type of container.

(i) **Bubble type** (very thin skin) cannot tolerate difference in pressure on the skin

outside pressure = inside pressure



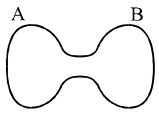
Any change in these cause change in volume & the container burst due to maximum stretching.

(ii) **Cylinder type** (thick skin) can withstand pressure difference till a limit but cannot have volume change.



Any change cause a change in pressure & when it exceeds the limits the container burst.

(b) **Connecting containers having gases**






On removal of nozzel the gas from higher pressure will travel so as to have equal pressure at both the containers.... from idea of total moles & final temperature each parameter can be calculated.

(c) **Changes in Open vessel :** Pressure of gas remains constant & so is the volume.

$\therefore\ n_1 T_1 = n_2 T_2$

(e) **Changes in closed vessel :** $\frac{P_1}{n_1} = \frac{P_2}{n_2}$

(f) **Barometric pressure distributor in a gas** [To calculate pressure at various height in a gas]

$$\ln \frac{P_2}{P_1} = \frac{-Mg}{RT}[H_2 - H_1] \qquad P_2 = P_1\, e^{-\frac{Mg}{RT}[H_2 - H_1]}$$

(g) **I separation**

For separating lighter gas from a mixture.

**Separation Factor :**

$$f = \frac{n_1^1/n_2^1}{n_1/n_2} = f.$$

$n_1$, $n_2$ and $n_1^1$, $n_2^1$ are the concentration of two isotopes before and after processing .

Theoretical separation factor $f' = \sqrt{\frac{M_2}{M_1}}$

If required enrichment of species (1) is attained after 'x' times, then :

$$(f')^x = \frac{n_1^1/n_2^1}{n_1/n_2} = f.$$

Solving using Graham's law , $x = \frac{2 \log f}{\log\left(\frac{M_2}{M_1}\right)}$ .

(h) **Payload / lifting power** [based on Buoyancy]

L.P. of balloon = $V(d - d_1)g - Mg$

V = Volume of balloon

d = density of outside gas

d = density of gas in the balloon

M = Mass of balloon

(i) **Analysis of a reaction involving gaseous**

$A(g) + B(g) \longrightarrow C(g)$

$\rightarrow$ What happens to pressure as reaction proceeds (in a closed container)

(j) **Vapour density and degree of dissociation** $\alpha = \frac{D - d}{(n-1)d}$

**Kinetic theory of gases :**

$$PV = \frac{1}{3} mNu^2 = \frac{1}{3} Mu^2 \text{ (For 1 mole)}$$

**Types of velocities :**

$u^2 = \frac{u_1^2 + u_2^2 + ..... + u_N^2}{N}$ ; $u =$ **root mean square speed** .

??

**Finds applications in K.E.**

$$u = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3PV}{M}} = \sqrt{\frac{3P}{d}} \quad ;$$

$$\textbf{Average speed} = \frac{u_1 + u_2 + ..... + u_N}{N} = \sqrt{\frac{8RT}{\pi M}}$$

??

**Finds application in Collision theroy**

$$\textbf{most probable speed} = \sqrt{\frac{2RT}{M}}$$

most probable : average : r. m. s. = $1 : 1.13 : 1.22$

$$= \sqrt{2} : \sqrt{\frac{8}{\pi}} : \sqrt{3}$$

*Relationship between three types of speeds*

$$u_{rms} > u_{av} > u_{mp}$$

Average kinetic energy of a single molecule $= \frac{3}{2} \cdot \frac{R}{N} \cdot T = \frac{3}{2} kT$

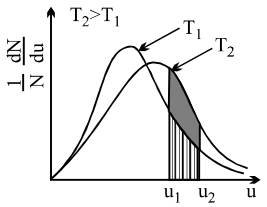
k = Boltzman constant = $1.3806 \times 10^{-16}$ erg $deg^{-1}$ .

Total kinetic energy for **one mole** of a gas $= \frac{3}{2} RT$ .

kinetic energy of **n moles** of a gas $= n \times \frac{3}{2} RT$

**Maxwell distribution Laws:** $dN_u = 4\pi N\left(\frac{M}{2\pi RT}\right)^{3/2} \exp(-Mu^2 / 2RT)\, u^2\, du$

$$= 4\pi N\left(\frac{m}{2\pi kT}\right)^{3/2} \exp(-mu^2 / 2kT)\, u^2\, du$$




**Collision frequency & Mean Free Path :**

Mean free path $\lambda = \frac{d_1 + d_2 + ..... + d_n}{n}$

$$\lambda = \frac{\text{Average velocity / RMS velocity}}{\text{collision number or frequency}} = \frac{kT}{\sqrt{2}\,\pi\,\sigma^2\,P}$$

k = Boltzman constant ; $\sigma$ = collision diameter .

$Z_1 = \sqrt{2}\,\pi\sigma^2\bar{u}\,N$ * [collisions made by one molecule

$$Z_{11} = \frac{1}{\sqrt{2}}\,\pi\sigma^2\bar{u}\,N^{*2}$$

## *THE REAL PATH*

**Vander Waals equation of state :**

$$\left(P+\frac{a}{V^2}.n^2\right)(V-nb) = nRT.$$

a , b are Vander Waals constants ; different for each gas

unit of a $\longrightarrow$ atm $L^2$ $mol^{-2}$ ; S.I. unit $\longrightarrow$ Pa $m^6$ $mol^{-2}$

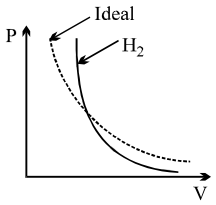
unit of b $\longrightarrow$ L $mol^{-1}$ ; S.I. unit $\longrightarrow$ $m^3$ $mol^{-1}$

greater the value of 'a' more easily the gas is liquefiable ;

greater the value of 'b' greater the molecular size ,

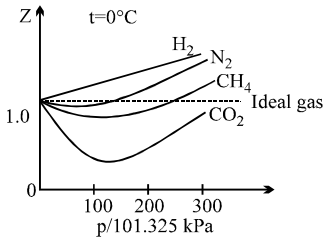
**Real gases : Deviation from ideal behaviour**

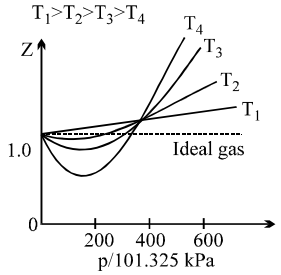
The curve for the real gas has a tendency to coincide with that of an ideal gas at low pressures when the volume is large. At higher pressures, however deviations are observed.




**Compressibility factor :**

$$z = \frac{PV}{nRT} = \frac{\text{volume observed}}{\text{volume ideal}}$$




**Boyle Temperature :**

$$T_B = \frac{a}{bR}$$

**Inversion Temperature :**

$$T_i = \frac{2a}{bR}$$

**Interpretation Of Deviation From Vander Waals Equation :**

**(i)** At low pressure $z = \frac{PV}{RT} = 1 - \frac{a}{VRT}$

**(ii)** At high pressure $z = \frac{PV}{RT} = 1 + \frac{Pb}{RT}$

**(iii)** At extremely low pressure $z = \frac{PV}{RT} = 1$ ; $Pb = \frac{a}{V}$

## HEAT CAPACITIES

$C_P$ = Molar heat capacity at constant pressure
$C_V$ = Molar heat capacity at constant volume
$c_p$ = specific heat capacity at constant pressure
$c_v$ = specific heat capacity at constant volume

$$\Rightarrow C_P - C_V = R$$

$$\Rightarrow \frac{C_P}{C_V} = r \qquad r = 1.66 \text{ (monoatomic); } 1.4 \text{ (diatomic)}$$

**Molar Specific Heat :**

= specific heat × molecular mass

$C_P - C_V = R/J$ ; $\frac{C_P}{C_V} = 1.66$ for monoatomic ; $\frac{C_P}{C_V} = 1.4$ for diatomic

**Degree Of Freedom :**

Three for monoatomic gas ; Five for diatomic gas .

For a molecule having N atoms, total are 3N

***Translational*** : 3 for all types [at all temp.]

***Rotational*** : 2 for linear
3 for N-linear

] *Each contributing* $\frac{1}{2}KT$

***Vibrational :*** 3N – 5 for linear
3N – 6 for N-linear

] *Each contributing KT*

**Law Of Equipartition Of Energy :**

***Translational Motion*** : $\overline{E}_{trans} = \frac{1}{2} mv^2 = \frac{1}{2} m v_x^2 + \frac{1}{2} m v_y^2 + \frac{1}{2} m v_z^2 \Rightarrow \frac{3}{2} KT$

***Rotational Motion*** : $\overline{E}_{rot} = \frac{1}{2} I_x \omega_x^2, \frac{1}{2} I_y \omega_y^2$ (linear)

$$= \frac{1}{2} I_x \omega_x^2 + \frac{1}{2} I_y \omega_y^2 + \frac{1}{2} I_z \omega_z^2 \text{ (N.L.)}$$

$\therefore$ linear = KT ; N.L. = $\frac{3}{2} KT$

***Vibrational Motion*** : $\overline{E}_{vib} = \frac{1}{2} KT + \frac{1}{2} KT = KT$

### *SOME OTHER EQUATION OF STATE*

**Dieterici Equation :**

$P e^{na/VRT} \times (V - nb) = nRT$

**Berthelot Equation :**

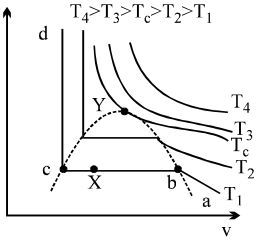
$\left(P + \frac{n^2 a}{T V^2}\right)(V - nb) = nRT$ (a & b are Berthlot's constant different from vander Waal's constant)

**Virial Equation Of State For 1 Mole Of Gas :**

$$z = \frac{P\,V}{R\,T} = 1 + B\,\frac{1}{V} + C\,\frac{1}{V^2} + D\,\frac{1}{V^3} + \ldots\ldots \text{ (B, C, D... are temp. dependent constant)}$$

$$\left.\begin{aligned} &B = \text{second virial co–efficient} = b - \frac{a}{R\,T} \\ &C = \text{third virial co–efficient} = b^2 . \end{aligned}\right] \textit{gas dependent}$$

**Critical Phenomenon : Critical Temp , Critical pressure , Critical volume**




$$T_c = \frac{8\,a}{27\,R\,b} \quad ; \quad P_c = \frac{a}{27\,b^2} \quad ; \quad V_c = 3\,b$$

**The law of corresponding states :**

$$p_r = \frac{p}{p_c}, \qquad T_r = \frac{T}{T_c} \qquad \text{and} \qquad V_r = \frac{V_m}{V_c}$$

$$p = p_r p_c, \qquad T = T_r T_c \qquad \text{and} \qquad V_m = V_r V_c$$

Substituting these expression in the van der Waals equation

$$\left(p + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

we obtain $\left(p_r p_c + \dfrac{a}{V_r^2 V_c^2}\right)(V_r V_c - b) = RT_r T_c$

Replacing $p_c$, $V_c$ and $T_c$ in terms of a, b and R, we get

$$\left\{p_r \frac{a}{27b^2} + \frac{a}{V_r^2 (3b)^2}\right\}\{V_r (3b) - b\} = RT_r \left(\frac{8a}{27Rb}\right)$$

i.e. $\quad (p_r + 3/V_r^2)\,(3V_r - 1) = 8T_r$

$$Z = \frac{pV_m}{RT} = \frac{(p_r p_c)(V_r V_c)}{R(T_r T_c)} = \frac{p_c V_c}{T_c}\left(\frac{p_r V_r}{T_r}\right) = \frac{3}{8}\frac{p_r V_r}{T_r}$$

# EXERCISE # I

## EXPERIMENTAL GAS LAW AND APPLICATION OF IDEAL GAS EQUATION.

Q.1 3.6 gm of an ideal gas was injected into a bulb of internal volume of 8L at pressure P atmp and temp T-K. The bulb was then placed in a thermostat maintained at (T+ 15) K.0.6 gm of the gas was let off to keep the original pressure. Find P and T if mol weight of gas is 44.

Q.2 A toy balloon originally held 1.0 gm of He gas and had a radius 10 cm. During the night, 0.25 gm of the gas effused from the balloon. Assuming ideal gas behaivour, under these constant P and T conditions, what was the radius of the balloon the next morning.

Q.3 If a scuba diver is to remain submerged for 1 hr, what pressure must be applied to force sufficient air into the tank to be used . Assume 0.5 $dm^3$ of air per breath at standard atmospheric pressure, a respiration rate of 38 breaths per minute, and a tank capacity of 30 $dm^3$.

Q.4 While resting, the average human male use 0.2 $dm^3$ of $O_2$ per hour at S T P for each kg of body mass. Assume that all this $O_2$ is used to produce energy by oxidising glucose in the body . What is the mass of glucose required per hour by a resting male having mass 60 kg . What volume, at S T P of $CO_2$ would be produced.

Q.5 In a basal metabolism measurement timed at 6.00 min, a patient exhaled 52.5 L of air, measured over water at $20^0$C. The vapour pressure of water at $20^0$C is 17.5 torr. The barometric pressure was 750 torr. The exhaled air analyzed 16.75 vol% oxygen and the inhaled air 20.32 vol% oxygen, both on dry basis. Neglecting any solubility of the gases in water and any difference in the total volumes of inhaled and exhaled air, calculate the rate of oxygen consumption by the patient in ml (S.T.P) per minute.

Q.6 One mole of $NH_4Cl$ is kept in an open container & then covered with a lid. The container is now heated to 600 K where all $NH_4Cl$ dissociates into $NH_3$ & HCl. If volume of the container is 24.63 litres, calculate what will be the final pressure of gases inside the container. Also find whether the lid would stay or bounce off if it can with stand a pressure difference of 5.5 atm. Assume that outside air is at 300 K and 1 atm pressure.

## DALTON'S LAW OF PARTIAL PRESSURE

Q.7 12 g $N_2$, 4 gm $H_2$ and 9 gm $O_2$ are put into a one litre container at 27°C. What is the total pressure.

Q.8 $1.0\times10^{-2}$ kg of hydrogen and $6.4\times10^{-2}$ kg of oxygen are contained in a $10\times10^{-3}$ $m^3$ flask at 473 K. Calculate the total pressure of the mixture. If a spark ignities the mixture. What will be the final pressure.

## GRAHAM'S LAW OF DIFFUSION AND EFFUSION

Q.9 At room temp , $NH_3$ gas at one atmp & HCl gas at "P" atmp are allowed to effuse through identical pin holes to the opposite ends of a glass tube 1m long & uniform cross-section. A white deposit is observed at a distance of 60 cm from the HCl end. What is "P".

Q.10 A gas mixture contains equal number of molecules of $N_2$ and $SF_6$ , some of it is passed through a gaseous effusion apparatus . Calculate how many molecules of $N_2$ are present in the product gas for every 100 molecules of $SF_6$.

Q.11 Two gases NO and $O_2$ were introduced at the two ends of a one metre long tube simultaneously (tube of uniform cross- section). At what distance from NO gas end , Brown fumes will be seen.

Q.12 At 20 ºC two balloons of equal volume and porosity are filled to a pressure of 2 atm, one with 14 kg $N_2$ & other with 1 kg $H_2$ . The $N_2$ balloon leaks to a pressure of $\frac{1}{2}$ atm in one hour. How long will it take for $H_2$ balloon to leaks to a pressure of $\frac{1}{2}$ atm.

Q.13 Naturally occurring Fluorine is entirely $^{19}F$, but suppose that it were 50% $F^{19}$ and 50% $F^{20}$ whether gaseous diffusion of $UF_6$ would then work to separate $U^{235}$ from $U^{238}$.

Q.14 Pure $O_2$ diffuses through an aperture in 224 sec, whereas mixture of $O_2$ and another gas containing 80 % $O_2$ diffuses from the same in 234 sec. What is molecular weight of the gas?

Q.15 A space capsule is filled with neon gas at 1.00 atm and 290 K. The gas effuses through a pin-hole into outer space at such a rate that the pressure drops by 0.3 torr/sec

(a) If the capsule were filled with ammonia at the same temperature and pressure, what would be the rate of pressure drop.

(b) If the capsule were filled with 30.0 mol % helium, 20.0 mol % oxygen & 50.0 mol % nitrogen at a total pressure of 1.00 atm & a temp. of 290 K, what would be the corresponding rate of pressure drop.

Q.16 The composition of the equilibrium mixture ($Cl_2 \Leftrightarrow 2Cl$) which is attained at 1200 ºC is determined by measuring the rate of effusion through a pin hole . It is observed that at 1.8 mm Hg pressure, the mixture effuses 1.16 times as fast as Kr effuses under the same conditions. Calculate the fraction of chlorine molecules dissociated into atoms. [ Kr = 84 a. m. u.]

## ***APPLICATION OF CONCEPT OF GASOUS STATE***

Q.17 Show that the height at which the atmospheric pressure is reduced to half its value is given by

$$h = \frac{0.6909\,RT}{Mg}$$

Q.18(a) How much $H_2$ (in mol) is needed to inflate a balloon of radius 3m to a pressure of 1 atmp in an ambient temp at $25^0$ C at sea level.

(b) What mass can the balloon lift at sea level, where the density of air is $1{\cdot}22\ Kg\ m^{-3}$ .

(c) What would be the pay load if He were used instead of $H_2$.

Q.19 Calculate the pressure of a barometer on an aeroplane which is at an altitude of 10 Km. Assume the pressure to be 101.325 Kpa at sea level & the mean temperature 243 K. Use the average molar mass of air (80% $N_2$ , 20 % $O_2$)

Q.20 Automobile air bags are inflated with $N_2$ gas which is formed by the decomposition of solid sodium azide ($NaN_3$). The other product is Na - metal. Calculate the volume of $N_2$ gas at 27°C and 756 Torr formed by the decomposing of 125 gm of sod azide.

Q.21 What will be the temperature difference needed in a hot air balloon to lift 1.0 kg of mass ? Assume that the volume of balloon is 100 $m^3$ , the temperature of ambient air is 25 ºC , the pressure is 1 bar , and air is an ideal gas with an average molar mass of 29 g $mol^{-1}$ ( hot and cold both).

Q.22 An iron cylinder contains helium at a pressure of 250 k pa and 27°C. The cylinder can withstand a pressure of $1 \times 10^6$ pa . The room in which cylinder is placed catches fire. Predict whether the cylinder will blow up before it metls or not. [melting point of cylinder = 1800 k ]

Q.23 Determine the molar mass of a gas if its pressure is to fall to one-half of its value in a vertical distance of one meter at 298 K.

## *KINETIC THEORY OF GASEOUS LAW MAXWELL DISTRIBUTION OF SPEEDS*

Q.24 The time taken for a given volume of gas E to effuse through a hole is 75 sec. Under identical conditions the same volume of a mix of CO & $N_2$ ( containing 40% of $N_2$ by volume ) effused in 70 seconds. Calculate

(i) the relative mol mass of E , and

(ii) the RMS velocity ( in $ms^{-1}$ units ) of E at $0^0$C.

Q.25 At what temperature in °C, the $U_{rms}$ of $SO_2$ is equal to the average velocity of $O_2$ at 27°C.

Q.26 Calculate $U_{rms}$ of molecules of $H_2$ at 1 atmp density of $H_2$ is 0.00009 g/cc.

Q.27 A bulb of capacity 1 $dm^3$ contains $1.03 \times 10^{23}$ $H_2$ molecules & pressure exerted by these molecules is 101.325 kPa. Calculate the average square molecular speed and the temperature.

Q.28 The mean kinetic energy of a molecule at $0^0$C is $5.621 \times 10^{-14}$ ergs. Calculate the number of molecules in gm molecule of gas.

Q.29 The density of CO at 273 K and 1 atm is 1.2504 kg $m^{-3}$. Calculate (a) root mean square speed (b) the average speed and (c) most probable speed.

Q.30 Calculate the temperature values at which the molecules of the first two members of the homologous series, $C_nH_{2n+2}$ will have the same rms speed as $CO_2$ gas at 770 K. The normal b.p. of n-butane is 273 K. Assuming ideal gas behaviour of n-butane upto this temperature, calculate the mean velocity and the most probable velocity of its molecules at this temperature.

Q.31 Calculate the temperature at which the root mean square velocity, average velocity and most probable velocity of oxygen gas are all equal to 1500 $ms^{-1}$.

Q.32 Calculate the fraction of $N_2$ molecules at 101.325 kPa and 300 K whose speeds are in the range of $u_{mp} - 0.005\ u_{mp}$ to $u_{mp} + 0.005\ u_{mp}$.

Q.33 What is the ratio of the number of molecules having speeds in the range of $2u_{mp}$ and $2u_{mp} + du$ to the number of molecules having speeds in the range of $u_{mp}$ and $u_{mp} + du$?

## *COLLISIONS AMONG GASEOUS MOLECULES*

Q.34 At low pressure and high temperature, the vander Waals equation is reduced to

(A) $PV_m = RT$ (B) $(P + a/V_m^2)(V_m - b) = RT$

(C) $P(V_m - b) = RT$ (D) $(P + a/V_m^2)(V_m) = RT$

Q.35 Calculate the mean free path in $CO_2$ at 27°C and a pressure of $10^{-6}$ mm Hg. (molecular diameter = 460 pm)

Q.36 Calculate the value of $\sigma$, $\lambda$, $Z_1$ and $Z_{11}$ for nitrogen molecules at 25°C and at pressure of $10^{-3}$ mm Hg. Given that b for nitrogen is 39.1 $cm^3\ mol^{-1}$.

Q.37 A mixture of hydrogen and helium is prepared such that the number of wall collisions per unit time by molecules of each gas is the same. Which gas has the higher concentration?

Q.38 The mean free path of the molecule of a certain gas at 300 K is $2.6 \times 10^{-5}$ m. The collision diameter of the molecule is 0.26 nm. Calculate
(a) pressure of the gas, and (b) number of molecules per unit volume of the gas.

## ***REAL GAS AND VANDER WAALS EQUATION OF STATE***

Q.39 The density of mercury is 13.6 $g/cm^3$. Estimate the b value.

Q.40 Calculate the pressure exerted by 22 g of carbon dioxide in 0.5 $dm^3$ at 298.15 K using:
(a) the ideal gas law and (b) vander waals equation. Given:
[a = 363.76 kPa $dm^6\ mol^{-2}$ and b = 42.67 $cm^3\ mol^{-1}$]

## ***COMPRESSIBILITY FACTOR***

Q.41 The compressibility factor for $N_2$ at – 50°C and 800 atmp pressure is 1.95 and at 100°C and 200 atmp, it is 1.10. A certain mass of nitrogen occupied one litre at – 50°C and 800 atmp. Calculate the volume occupied by the same quantity of $N_2$ at 100°C and 200 atmp.

Q.42 At 273.15 K and under a pressure of 10.1325 MPa, the compressibility factor of $O_2$ is 0.927. Calculate the mass of $O_2$ necessary to fill a gas cylinder of 100 $dm^3$ capacity under the given conditions.

## ***BOYLE'S TEMPERATURE CRITICAL PHENOMENON AND INVERSION TEMPERATURE***

Q.43 The vander waals constant for $O_2$ are a = 1.36 atm $L^2\ mol^{-2}$ and b = 0.0318 L $mol^{-1}$. Calculate the temperature at which $O_2$ gas behaves, ideally for longer range of pressure.

Q.44 The vander Waals constants for gases A, B and C are as follows

| Gas | a/$dm^6$ kPa $mol^{-2}$ | b/$dm^3\ mol^{-1}$ |
|---|---|---|
| A | 405.3 | 0.027 |
| B | 1215.9 | 0.030 |
| C | 607.95 | 0.032 |

Which gas has (i) the highest critical temperature, (ii) the largest molecular volume, and (iii) most ideal behaviour around STP?

## ***HEAT CAPACITY***

Q.45 One mole of a non linear triatomic gas is heated in a closed rigid container from 500°C to 1500°C. Calculate the amount of energy required if vibrational degree of freedom become effective only above 1000°C.

## *OTHER EQUATION OF STATE*

Q.46 A commercial cylinder contains 6.91 $m^3$ of $O_2$ at 15.18 M Pa and $21^0$C. the critical constants for $O_2$ are $T_C = -118.4^0$C , $P_C = 50.1$ atmp. Determine the reduced pressure and reduced temperature for $O_2$ under these conditions.

Q.47 Show that at low densities, the vander waals equation

$$\left(p + \frac{a}{V_m^2}\right)(V_m - b) = RT$$

and the Dieterici's equation

$$p(V_m - b) = RT \exp(-a/RTV_m)$$

give essentially the same value of p.

## *SOME PROBLEMS ON REAL GAS & VANDER WAALS*

Q.48 Calculate from the vander waal's equation, the temperature at which 192 gm of $SO_2$ would occupy a vol. of 10 $dm^3$ at 15 atm pressure.[a = 6.7 atm $lit^2$ $mol^2$, b = 0.0564 lit $mol^{-1}$]

Q.49 Calculate the pressure of 15 mol neon at 30° C in a 12 lit container using
(i) the ideal gas equation (ii) the vander waals equation
[a = 0.2107 atm $lit^2$ $mol^{-2}$, b = 0.0171 lit $mol^{-1}$]

Q.50 The molar volume of He at 10.1325 MPa and 273 K is 0.011075 of its molar volume at 101.325 KPa at 273 K.Calculate the radius of helium atom. The gas is assumed to show real gas nature. Neglect the value of a for He.

Q.51 The density of water vapour at 327.6 atm and 776.4 K is 133.2 gm/$dm^3$.
Determine the molar volume, $V_m$ of water and the compression factor.

Q.52 $N_2$ molecule is spherical of radius 100 pm.
(a) What is the volume of molecules is one mole of a gas?
(b) What is the value of vander waal's constant b?

## *EXERCISE # II*

Q.1 A 50 litre vessel is equally divided into three parts with the help of two stationary semi permeable membrane (SPM). The vessel contains 60 g $H_2$ gas in the left chamber, 160 g $O_2$ in the middle & 140 g $N_2$ in the right one. The left SPM allows transfer of only $H_2$ gas while the right one allows the transfer of both $H_2$ & $N_2$. Calculate the final ratio of pressure in the three chambers.

Q.2 Militants are hiding at the top of the kargil peak which is 7000 m above the plains. Major of a troop of soldiers wants to send few soldiers to the peak to kill the enemies by balloons, then find the minimum volume of each balloon (volume remain constant throughout the mission) if he attach 10 balloons to each soldier. Given

(i) Change in density in atmosphere is $d = d_0 e^{-Mgh/RT}$
(where $d_0$ is density at plain and d is density at height 'h')

(ii) M = 29 gm/mole (constant)

(iii) T = 27°C (constant)

(iv) g = 10 m/sec$^2$

(v) Each balloon contains 10 moles of $H_2$

(vi) weigth of each soldier is 75 kg.

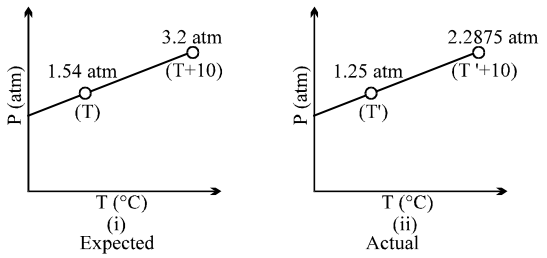
Q.3 There are two vessels of same volume consisting same no of moles of two different gases at same temperature. One of the gas is $CH_4$ & the other is unknown X. Assuming that all the molecules of X are under random motion whereas in $CH_4$ except one all are stationary. Calculate $Z_1$ for X in terms of $Z_1$ of $CH_4$. Given that the collision diameter for both the gases are same & $(U_{rms})_x = \frac{1}{\sqrt{6}}(Uav)_{CH_4}$.

Q.4 A mixture of $CH_4$ & $O_2$ is used as an optimal fuel if $O_2$ is present in thrice the amount required theoretically for combustion of $CH_4$. Calculate number of effusions steps required to convert a mixture containing 1 part of $CH_4$ in 193 parts mixture (parts by volume). If calorific value (heat evolved when 1 mole is burnt) of $CH_4$ is 100 cal/mole & if after each effusion 90% of $CH_4$ is collected, find out what initial mole of each gas in initial mixture required for producing 1000 cal of energy after processing.
[Given $(0.9)^5 = 0.6$]

Q.5 A closed vessel of known volume containing known amount of ideal gaseous substance 'A' was observed for variation of pressure with temperature. The expected graph was to be like as in (i) However actual observations revealed the graph to be like. (ii) The deviation was attributed to polymerisation of gas molecules as $nA(g) \rightleftharpoons A_n(g)$. If it is known that the above reaction gives only 50% yield

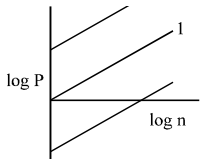
(a) Calculate the ratio of $\frac{n_{experiment}}{n_{theoritical}}$ (where $n_{exp.}$ = Total no. of gaseous mole acutally present
$n_{theoritical}$ = Total no. of mole original taken)

(b) Find the value of n to which the gas A is being polymerised into



Q.6 You are told to prepare a closed experimental environment (a box) for student mice. The box volume will be 294 liters (about 10 $ft^3$) and the entire air volume will be changed every minute. The relative humidity of the air entering the box is to be controlled at 40% at 21°C. What weight of $H_2O$ must be added to the flow of dry air per minute? (Equilibrium vapour pressure for $H_2O$ at $21^0C \simeq 19$ torr).
(R = 0.082 liter atm $mole^{-1}deg^{-1}$ mol wt: $H_2O$ = 18)

Q.7 Graph between log P (atm) v/s log n is plotted for an ideal gas enclosed in 24.63 litre container at three different temperatures. If $T_1 = \frac{T_2}{3} = 2T_3$ [where $T_1$, $T_2$, $T_3$ are temperature in kelvin of graph 1, 2 & 3] then




(a) Mention graph 2 & graph 3.
(b) Calculate $T_1$, $T_2$, $T_3$.
(c) Calculate slope of graphs 1, 2 & 3.
(d) Calculate intercept of graphs 2 & 3.

Q.8 During one of his adventure, Chacha chaudhary got trapped in an underground cave which was sealed two hundred year back. The air inside the cave was poisonous, having some amount of carbon monoxide in addition to $O_2$ and $N_2$. Sabu, being huge could not enter into the cave, so in order to save chacha choudhary be started sucking the poisonous air out of the cave by mouth. Each time, he filled his lunge with cave air and exhaled it out in the surroundings. In the mean time fresh air from surrounding effused into the cave till the pressure was again one atmosphere. Each time Sabu sucked out some air, the pressure in the cave dropped to half of its initial value of one atmosphere.
If the initial sample of air from the cave contain 5% by volume CO.
If the safe level of CO in the atmosphere is less than 0.001% by volume how many times does Sabu need to such out air in order to save Chacha chaudhary.

Q.9 A compound exists in the gaseous state both as a monomer (A) and dimer ($A_2$). The molecular weight of the monomer is 48. In an experiment, 96 g of the compound was confined in a vessel of volume 33.6 litres and heated to $273^0$ C. Calculate the pressure developed, if the compound exists as a dimer to the extent of 50 per cent by weight, under these conditions. (R = 0.082)

Q.10 The following reaction is carried out in a flask at 101325 pascal and 383 k with the initial concentration of $CH_4$ , $O_2$ as 0.01 & 0.03 mole.

$$2\,CH_4 + 3\,O_2 \longrightarrow 2\,CO + 4\,H_2O.$$

All reactants and products are gases at 383 k. A shortwhile after the completion of reaction the flask is cooled to 283 k at which $H_2O$ is completely condensed. Calculate :
(a) Volume of flask.
(b) Total pressure and partial pressure of various species at 383 k , 283 k.
(c) number of molecules of various substance before and after reaction.

Q.11 A closed vertical cylinder is divided into two parts by a frictionless piston, each part contains 1 mole of air . At 27 ºC the volume of the upper part is 4 times than that of the lower part. Calculate the temperature when volume of the upper part will be three times than that of the lower part.

Q.12 A water gas mixture has the compsition by volume of 50% $H_2$, 45% CO and 5% $CO_2$.
(i) Calculate the volume in litres at STP of the mixture which on treatment with excess steam will contain 5 litres of $H_2$. The stoichiometry for the water gas shift reaction is

$$CO + H_2O \rightarrow CO_2 + H_2$$

(ii) Find the density of the water gas mixture in $kg/m^3$.
(iii) Calculate the moles of the absorbants KOH, $Ca(OH)_2$ and ethanolamine.
$HO–CH_2–CH_2–NH_2$ required respectively to collect the $CO_2$ gas obtained.

Q.13 One mole of an ideal gas is subjected to a process in which $P = \frac{1}{8.21} V$ where P is in atm & V in litre. If the process is operating from 1 atm to finally 10 atm (no higher pressure achieved during the process) then what would be the maximum temperature obtained & at what instant will it occur in the process.

Q.14 A gas present in a container connected to frictionless, weightless piston operating always at one atmosphere pressure such that it permits flow of gas outside (with no adding of gas). The graph of $n$ vs T (Kelvin) was plotted & was found to be a straight line with co-ordinates of extreme points as (300, 2) & (200, 3). Calculate

(i) relationship between $n$ & T

(ii) relationship between V & T

(iii) Maxima or minima value of 'V'

Q.15 Find the critical constant ($P_c$, $V_c$ and $T_c$) in terms of A and B, also find compressibility factor (z) for the following equation of state.

$$PV = RT - \frac{A}{V} + \frac{2B}{V^2}$$

where A and B are constant, P = pressure and V = molar volume.

Q.16 Calculate the volume occupied by 14.0 g $N_2$ at 200 K and 8.21 atm pressure if $\frac{P_C V_C}{RT_C} = \frac{3}{8}$ and $\frac{P_r V_r}{T_r} = 2.2$.

## *EXERCISE # III*

***Question No. 1 to 3 are based on the following information. Read it carefully to answer the questions that follows.***

A gas undergoes dissociation as $A_4(g) \longrightarrow 4A(g)$ in a closed rigid container having volume 22.4 litres at 273 K. If the initial moles of $A_4$ taken before dissociation is 1 then

Q.1 The total pressure (in atm) after 50% completion of the reaction (assuming ideal behaviour)

(A) 1/2 (B) 2.5 (C) 2 (D) 4

Q.2 If the gases are not ideal & at the beginning total pressure observed is less than 1 atm then

(A) compressibility factor of $A_4 > 1$ (B) compressibility factor of $A_4 < 1$

(C) compressibility factor of $A_4 = 1$ (D) compressibility factor of $A > 1$

Q.3 If the gases are non–ideal & after 100% dissociation total pressure is greater than 4 atm, then

(A) The compression of A(g) will be easier than that of ideal gas

(B) The compression of A(g) will be difficult than that of ideal gas

(C) The compression of A(g) will be same as that of ideal gas

(D) A cannot be compressed

***Question No. 4 to 6 are based on the following Passage. Read it carefully & answer the questions that follow***

Two containers X & Y are present with container X consisting of some mass of He at some temperature while container Y having double the volume as that of container X & kept at same temperature containing same mass of $H_2$ gas as the mass of Helium gas. Based on this data & the following conditions answer the question that follows

**Assume sizes of $H_2$ molecule & He atom to be same & size of H–atom to be half to that of He–atom & only bimolecular collisions to be occuring.**

**Condition I:** all except one atom of He are stationary in cont. X & all molecules of $H_2$ are moving in container Y.

**Condition II:** both containers contain all moving molecules

Q.4 Assuming **condition I** to be applicable & if no. of *total collisions* occuring per unit time is 'A' in container X then no. of total collisions made by any one molecule in container Y will be:

(A) A (B) $\sqrt{2}$ A (C) $\frac{A}{\sqrt{2}}$ (D) none of these

Q.5 Assuming **condition II** then ratio of 'total no. of collisions per unit volume per unit time' in container X & container Y is (container X : container Y)

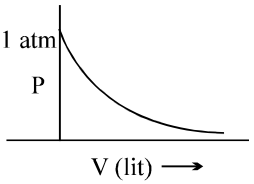
(A) 1:1 (B) $\sqrt{2}$ :1 (C) 1: $\sqrt{2}$ (D) 4:1

Q.6 Assuming **condition II** to be applicable, if temperature only of container Y is doubled to that of original (causing dissociation of all $H_2$ gas into H gaseous atoms) then, if no. of total collisions per unit volume per unit time in container X is A then, no. of 'total collisions made by all molecules per unit volume in container Y would be

(A) $2\sqrt{2}$ A (B) $\sqrt{2}$ A (C) $8\sqrt{2}$ A (D) none of these

***Question No. 7 to 9 are based on the following Passage. Read it carefully & answer the questions that follow***

On the recently discovered 10th planet it has been found that the gases follow the relationship $\mathbf{Pe^{V/2} = nCT}$ where C is constant other notation are as usual (V in lit., P in atm and T in Kelvin). A curve is plotted between P and V at 500 K & 2 moles of gas as shown in figure




Q.7 The value of constant C is

(A) 0.01 (B) 0.001 (C) 0.005 (D) 0.002

Q.8 Find the slope of the curve plotted between P Vs T for closed container of volume 2 lit. having same moles of gas

(A) $\frac{e}{2000}$ (B) 2000 e (C) 500 e (D) $\frac{2}{1000e}$

Q.9 If a closed container of volume 200 lit. of $O_2$ gas (ideal gas) at 1 atm & 200 K is taken to planet. Find the pressure of oxygen gas at the planet at 821 K in same container

(A) $\frac{10}{e^{100}}$ (B) $\frac{20}{e^{50}}$ (C) 1 atm (D) 2 atm

***Question No. 10 to 11 (2 questions)***

For a gaseous molecular system the probability of finding a molecule with velocity betwen v and v + dv is given by

$$\frac{dN_v}{N} = \left(\frac{m}{kT}\right) e^{-mv^2/2kT}\, v\, dv$$

where m = mass of gas molecule
k = Boltzmann constant
T = Temperature of gas
$N_v$ = No. of molecules with velocity between v and v + dv
N = Total No. of molecules

Q.10 At some temperature the fraction of molecules with kinetic energies between E and E + dE is given by

(A) $\frac{1}{kT} e^{-E/kT}\, E\, dE$ (B) $\left(\frac{m}{kT}\right) e^{-E/kT}\, dE$ (C) $\frac{1}{kT} e^{-E/kT}\, dE$ (D) $\left(\frac{m}{kT}\right) e^{-E/kT}\, E\, dE$

Q.11 Fraction of molecules with K.E. greater than E is given by

(A) $\int_0^E \frac{1}{KT} e^{-E/KT}\, dE$ (B) $\int_E^\infty \frac{1}{KT} e^{-E/KT}\, E\, dE$ (C) $\int_E^\infty \frac{1}{KT} e^{-E/KT}\, dE$ (D) $\int_0^E \frac{1}{KT} e^{-E/KT}\, E\, dE$

Q.12 Ratio of fraction of molecules with K.E. greater than and less than average K.E. is:

(A) $\frac{1}{e^{1/2}+1}$ (B) $\frac{1}{e^{3/2}-1}$ (C) $\frac{e^{1/2}}{1-e^{1/2}}$ (D) $\frac{e^{3/2}}{e^{3/2}-1}$

***Question No. 13 to 15 (3 questions)***

The rate of change of pressure (p) of a gas at constant temperature and constant external pressure due to effusion of gas from a vessel of constant volume is related to rate of change of number of molecules present by

$$\frac{dp}{dt} = \frac{kT}{V}\frac{dN}{dt}$$

where k = Boltzmann constant, T = temperature, V = volume of vessel & N = No. of molecules and

$\frac{dN}{dt} = \frac{-pA_0}{(2\pi mkT)^{1/2}}$, where $A_0$ = area of orifice and m = mass of molecule

Q.13 Time required for pressure inside vessel to reduce to 1/e of its initial value is (ln e = 1)

(A) $\left(\frac{2\pi m}{kT}\right)^{1/2}\frac{V}{A_0}$ (B) $\left(\frac{kT}{2\pi m}\right)^{1/2}\frac{V}{A_0}$ (C) $\left(\frac{2\pi mkT}{A_0}\right)^{1/2}$ (D) $\frac{2\pi m}{kT}\frac{V}{A_0}$

Q.14 If the gas inside the vessel had molecular weight 9 times the gas in previous example and area of orifice was doubled and temperature maintained at 4T, time required for pressure to fall to 1/e times of its initial value would be (t = answer of previous option)

(A) 1.33 t (B) 4.24 t (C) 0.75 t (D) 1.125 t

Q.15 The incorrect statement(s) is/are

[I] Pressure will not fall to zero in finite time

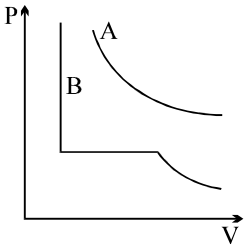
[II] Time required for pressure to decrease to half its initial value is independent of initial pressure

[III] The relations given above are true for real gases also

(A) I (B) II (C) III (D) I and III

***Question No. 16 to 17 (2 questions)***

For two gases A and B, P v/s V isotherms are drawn at T K as shown. $T_A$ & $T_B$ are critical temperatures of A & B respectively




Q.16 Which of following is true?

(A) $T_A < T < T_B$ (B) $T_A > T > T_B$

(C) $T_A > T_B > T$ (D) none of above

Q.17 The correct statement(s) is/are

(I) Pressure correction term will be more negligible for gas B at T K.

(II) The curve for gas 'B' will be of same shape as for gas A if $T > T_B$

(III) Gas 'A' will show same P v/s V curve as of gas 'B' if $T > T_A$

(A) III only (B) II and III (C) II only (D) All

Q.18 n moles of Helium gas are placed in a vessel of volume V Liter. at T K. If $V_I$ is ideal volume of Helium then diameter of He atom is

(A) $\left[\frac{3}{2}\frac{V_I}{\pi N_A n}\right]^{\frac{1}{3}}$ (B) $\left[\frac{3(V-V_I)}{2\ \pi N_A n}\right]^{\frac{1}{3}}$ (C) $\left[\frac{6(V-V_I)}{\pi N_A n}\right]^{\frac{1}{3}}$ (D) $\left[\frac{6V_I}{\pi N_A n}\right]^{\frac{1}{3}}$

***Q.19 and Q.20*** **are based on the following passage.**

Under a given condition, it is found that two separate gases effuse out of two separate container in such a way that they follows the equation $\frac{dN}{dt} = -K_1N$ & $\frac{dN}{dt} = -K_2N$, $K_1 = 6.93 \times 10^{-3} sec^{-1}$, $K_2 = 6.93 \times 10^{-5} sec^{-1}$, where N is no. of molecules remaining in the container.

Q.19 Which one of the following may represent fraction of no. of molecules present after the given interval for gas-I?

(A) t = 0 t = 100sec t = 200 sec

$1 \quad \frac{1}{2} \quad \frac{1}{8}$

(B) t = 0 t = 100 sec t = 200 sec

$1 \quad \frac{1}{8} \quad \frac{1}{16}$

(C) t = 0 t = 100 sec t = 200 sec

$1 \quad \frac{1}{2} \quad \frac{1}{4}$

(D) t = 0 t = 100 sec t = 200 sec

$1 \quad \frac{1}{4} \quad \frac{1}{16}$

Q.20 Identify the correct option regarding sequence of (True) & (False) statements

(i) The time required for moles of gas I to get reduced to half of original & that of gas II to reduced to half of original is independent of initial moles of gas I & gas II.

(ii) The rate at which **initially** molecules will come out in gas I as compared to gas II will be greater **in gas II** if initial no. of molecules are same.

(iii) The time required for moles to get reduced from 1 to 0.8 in gas I and 2 to 1.6 in gas II will be same

(iv) For the two gases, moles remaining on the container after same interval should be in Geometrical Progression.

(A) TFFT (B) TFTT (C) FTFT (D) TTFF

## *EXERCISE # IV*

Q.1 A mixture of ideal gases is cooled upto liquid He temperature (4.22 K) to form an ideal solution. Is this statement **true** or **false**. Justify your answer in not more than two lines. **[JEE 1996]**

Q.2 The ratio between the r. m. s. velocity of $H_2$ at 50 K and that of $O_2$ at 800 K is :
(A) 4 (B) 2 (C) 1 (D) 1/4 **[JEE 1996]**

Q.3 X ml of $H_2$ gas effuses through a hole in a container in 5 sec. The time taken for the effusion of the same volume of the gas specified below under identical conditions is : **[JEE 1996]**
(A) 10 sec, He (B) 20 sec, $O_2$ (C) 25 sec, CO (D) 55 sec, $CO_2$

Q.4 One mole of $N_2O_4$ (g) at 300 k is kept in a closed container under one atmp. It is heated to 600 k when 20 % by mass of $N_2O_4$ (g) decomposes to $NO_2$ (g) . The resultant pressure is : **[JEE 1996]**
(A) 1.2 atm (B) 2.4 atm (C) 2.0 atm (D) 1.0 atm

Q.5 The absolute temperature of an ideal gas is _____ to/than the average kinetic energy of the gas molecules. **[JEE 1997]**

Q.6 One way of writing the equation for state for real gases is,

$$P\bar{V} = RT\left[1 + \frac{B}{\bar{V}} + .....\right] \quad \text{where B is a constant.}$$

Derive an approximate expression for 'B' in terms of Vander Waals constant 'a' & 'b'. **[JEE 1997]**

Q.7 Calculate the total pressure in a 10 litre cylinder which contains 0.4 g He, 1.6 g oxygen and 1.4 g of nitrogen at 27 ºC. Also calculate the partial pressure of He gas in the cylinder. Assume ideal behavious for gases. **[JEE 1997]**

Q.8 According to Graham's law , at a given temperature the ratio of the rates of diffusion $\frac{r_A}{r_B}$ of gases A and B is given by : **[JEE 1998]**

(A) $\frac{P_A}{P_B}\left(\frac{M_A}{M_B}\right)^{1/2}$ (B) $\left(\frac{M_A}{M_B}\right)\left(\frac{P_A}{P_B}\right)^{1/2}$ (C) $\frac{P_A}{P_B}\left(\frac{M_B}{M_A}\right)^{1/2}$ (D) $\frac{M_A}{M_B}\left(\frac{P_B}{P_A}\right)^{1/2}$

Q.9 An evacuated glass vessel weighs 50.0 g when empty, 148.0 gm when filled with a liquid of density 0.98 g /mL and 50.5 g when filled with an ideal gas at 760 mm Hg at 300 k . Determine the molecular weight of the gas . **[JEE 1998]**

Q.10 Using Vander Waals equation, calculate the constant "a" when 2 moles of a gas confined in a 4 litre flask exerts a pressure of 11.0 atmp at a temperature of 300 k. The value of "b" is 0.05 litre $mol^{-1}$. **[JEE 1998]**

Q.11 The pressure exerted by 12 g of an ideal gas at temperature t ºC in a vessel of volume V is one atmp . When the temperature is increased by 10 degrees at the same volume, the pressure increases by 10 %. Calculate the temperature 't' and volume 'V'. [molecular weight of gas = 120] **[JEE 1999]**

Q.12 One mole of $N_2$ gas at 0.8 atmp takes 38 sec to diffuse through a pin hole, whereas one mole of an unknown compound of Xenon with F at 1.6 atmp takes 57 sec to diffuse through the same hole . Calculate the molecular formula of the compound.(At. wt. Xe = 138, F = 19) **[JEE 1999]**

Q.13 A gas will approach ideal behaviour at : **[JEE 1999]**

(A) low temperature and low pressure (B) low temperature and high pressure

(C) low pressure and high temperature (D) high temperature and high pressure .

Q.14 The compressibility of a gas is less than unity at STP. Therefore **[JEE 2000]**

(A) $V_m > 22.4$ L (B) $V_m < 22.4$ L (C) $V_m = 22.4$ L (D) $V_m = 44.8$ L

Q.15 The r.m.s. velocity of hydrogen is $\sqrt{7}$ times the r.m.s. velocity of nitrogen. If T is the temperature of the gas : **[JEE 2000]**

(A) $T(H_2) = T(N_2)$ (B) $T(H_2) > T(N_2)$

(C) $T(H_2) < T(N_2)$ (D) $T(H_2) = \sqrt{7}\ T(N_2)$

Q.16 The pressure of a fixed amount of an ideal gas is proportional to its temperature. Frequency of collision and their impact both increase in proportion to the square root of temperature. True/False. **[JEE 2000]**

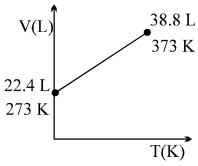
Q.17 Calculate the pressure exerted by one mole of $CO_2$ gas at 273 k, if the Vander Waals constant $a = 3.592$ $dm^6$ atm $mol^{-2}$. Assume that the volume occupied by $CO_2$ molecules is negligible. **[JEE 2000]**

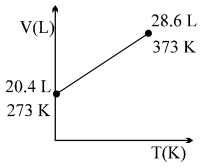
Q.18 The root mean square velocity of an ideal gas at constant pressure varies with density as

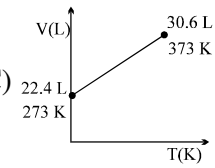
(A) $d^2$ (B) $d$ (C) $d^{1/2}$ (D) $1/d^{1/2}$ **[JEE 2001]**

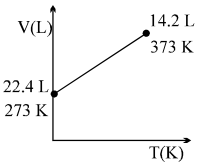
Q.19 The compression factor (compressibility factor) for one mole of a vander Waals gas at 0° C and 100 atmosphere pressure is found to be 0.5. Assuming that the volume of a gas molecule is negligible, calculate the vander waals constant 'a'. **[JEE 2001]**

Q.20 Which one of the following V, T plots represents the behaviour of one mole of an ideal gas at one atmp?

(A) 


(B) 


(C) 


(D) 


**[JEE 2002]**

Q.21 The density of the vapour of a substance at 1 atm pressure and 500 K is 0.36 Kg $m^{-3}$. The vapour effuses through a small hole at a rate of 1.33 times faster than oxygen under the same condition.

(a) Determine

(i) mol. wt.; (ii) molar volume; (iii) compression factor (z) of the vapour and

(iv) which forces among the gas molecules are dominating, the attractive or the repulsive

(b) If the vapour behaves ideally at 1000K , determine the average translational K.E. of a molecule. **[JEE 2002]**

Q.22 The average velocity of gas molecules is 400 m/sec. Calculate its (rms) velocity at the same temperature. **[JEE 2003]**

Q.23 $C_V$ value of He is always 3R/2 but $C_V$ value of $H_2$ is 3R/2 at low temperature and 5R/2 at moderate temperature and more than 5R/2 at higher temperature explain in two to three lines. **[JEE 2003]**

Q.24 Positive deviation from ideal behaviour takes place because of **[JEE 2003]**

(A) molecular interaction between atoms and $\frac{PV}{nRT} > 1$

(B) molecular interation between atoms and $\frac{PV}{nRT} < 1$

(C) finite size of atoms and $\frac{PV}{nRT} > 1$

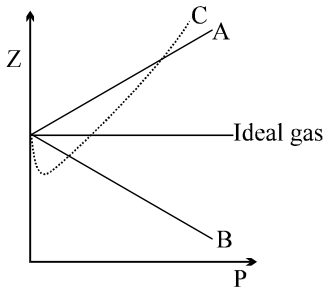
(D) finite size of atoms and $\frac{PV}{nRT} < 1$

Q.25 For a real gas obeying van der Waal's equation a graph is plotted between $PV_m$ (y-axis) and P(x-axis) where $V_m$ is molar volume. Find y-intercept of the graph. **[JEE 2004]**

Q.26 The ratio of the rate of diffusion of helium and methane under identical condition of pressure and temperature will be

(A) 4 (B) 2 (C) 1 (D) 0.5 **[JEE 2005]**

Q.27




where $Z = \frac{PV}{nRT}$,

a = Van der Waal's constant for pressure correction

b = Van der Waal's constant for volume correction

Pick the only incorrect statement

(A) for gas A, if a = 0, the compressibility factor is directly proportional to pressure

(B) for gas B, if b = 0, the compressibility factor is directly proportional to pressure

(C) for gas C, $a \neq 0$, $b \neq 0$, it can be used to calculate a and b by giving lowest P value and its intercept with Z = 1.

(D) slope for all three gases at high pressure (not shown in graph) is positive. **[JEE 2006]**

# ANSWER KEY

## EXERCISE # I

Q.1 P = 0.062 atm , T = 75 K Q.2 9.08 cm Q.3 $3.8\times10^3$ kpa

Q.4 16.07 gm ; 12 $dm^3$ Q.5 280 ml/min Q.6 6 atm, No Q.7 66.74 atm

Q.8 $P_{total} = 27.54\times10^5$ N/$m^2$ , $P_{final} = 19.66\times10^5$N/$m^2$ Q.9 2.19 atmp Q.10 228

Q.11 50.8 cm Q.12 16 min Q.13 yes Q.14 46.6

Q.15 (a) 0.33 Torr/sec , (b) 0.29 Torr/sec Q.16 0.137 Q.18 $4.62\times10^3$ moles, 128.79 Kg, 119.55Kg

Q.19 25.027 Kpa Q.20 71.4 L Q.21 2.53°C Q.22 yes

Q.23 175.133 kg $mol^{-1}$ Q.24 32.14 g / mol , 460.28 m/s Q.25 236.3°C

Q.26 183,800 cm/sec Q.27 $8.88\times10^5$ $(m/s)^2$ ; 71.27 K

Q.28 $6.06\times10^{23}$ molecules $mol^{-1}$ Q.29 $U_{RMS} = 493$ m/s , $U_{mp} = 403$m/s , $U_{av}$=454.4 m/s

Q.30 280 K, 525 K , $3.157\times10^2$ m/sec, $2.798\times10^2$ m/sec

Q.31 $T_{RMS}$= 2886 K, $T_{av}$ = 3399 K, $T_{mp}$=4330K Q.32 $8.303\times10^{-3}$

Q.33 0.199 Q.34 A Q.35 $3.3\times10^3$ cm

Q.36 314 pm, 7.015 cm, 6742 $s^{-1}$, $1.09\times10^{17}$ $cm^{-3}s^{-1}$ Q.37 He

Q.38 (a) $1.281\times10^{23}$ $m^{-3}$, (b) $5.306\times10^2$ Pa Q.39 58.997 $cm^3$

Q.40 (a) $2.479\times10^3$ kPa, (b) 2225.55 kPa Q.41 3.77 L Q.42 15.40 kg

Q.43 521 K Q.44 (i) B, (ii) C, (iii) A Q.45 4500 RJ Q.46 $\pi = 2.99$ , $\theta = 1.90$

Q.48 350.5°C Q.49 (i) 31.1 atm, (ii) 31.4 atm Q.50 $r = 1.33\times10^{-8}$

Q.51 Molar vol = 0.1353 L/mol; Z = 0.6957

Q.52 (a) $2.52\times10^{-3}$ l $mol^{-1}$, (b) $10.08\times10^{-3}$ $dm^3$ $mol^{-1}$

## EXERCISE # II

Q.1 4 : 7 : 5 Q.2 14.41 $m^3$. Q.3 $\frac{2\sqrt{2}}{3\sqrt{\pi}}Z_1$

Q.4 10 Steps, 27.78 mol $CH_4$, 5333.3 mol $O_2$ Q.5(a) 0.625, (b) 4 Q.6 2.2 g

Q.7 $T_1$ = 300 K, $T_2$ = 900 K Q.8 13 Q.9 2 atmp

Q.10 (a) 1.257 L ; (b) At 383 K $P_T = 113.99$ kpa , $P_{O_2} = 38$ kpa , $P_{CO} = 25.33$ kpa , $P_{H_2O} = 50.66$ kpa,
At 283 K $P_T = 46.81$ kpa , $P_{O_2} = 28.086$ kpa , $P_{CO} = 18.724$ kpa , $P_{H,O} = 0$
(c) Before reaction : $CH_4 = 0.01\ N_A$ , $O_2 = 0.03\ N_A$
After reaction : $O_2 = 0.015\ N_A$ , $CO = 0.01\ N_A$ , $H_2O = 0.02\ N_A$

Q.11 421.9 K

Q.12 (i) 5.263 L ,(ii) 0.7 Kg/$m^3$ ,(iii) $K_{OH}$=0.2348 moles,$Ca(OH)_2$=0.1174 moles, ethanolamine=0.2348 moles

Q.13 10,000 K Q.14 $n = \frac{-T}{100} + 5$ , $V = \frac{-RT^2}{100} + 5RT$ , 51.3125 *l*

Q.15 $V_C = \frac{6B}{A}$ , $T_C = \frac{A^2}{6RB}$ , $P_C = \frac{A^3}{108B^2}$ , compressibility factor = $\frac{P_C V_C}{RT_C} = \frac{1}{3}$ Q.16 0.825 L

## ***EXERCISE # III***

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Q.1 | B | Q.2 | B | Q.3 | B | Q.4 | D | Q.5 | C | Q.6 | A | Q.7 | B |
| Q.8 | D | Q.9 | A | Q.10 | C | Q.11 | C | Q.12 | B | Q.13 | A | Q.14 | C |
| Q.15 | C | Q.16 | A | Q.17 | C | Q.18 | B | Q.19 | C | Q.20 | A | | |

## ***EXERCISE # IV***

Q.1 yes it is false statement Q.2 C Q.3 B Q.4 B Q.5 directly proportional

Q.6 $B = \left(b - \frac{\alpha}{RT}\right)$ Q.7 0.492 atmp ; 0.246 atmp Q.8 C Q.9 123

Q.10 6.46 atmp $L^2\ mol^{-2}$ Q.11 –173$^0$C , 0.82 L Q.12 $XeF_6$ Q.13 C

Q.14 B Q.15 C Q.16 Both statements are correct Q.17 34.8 atmp

Q.18 D Q.19 1.2544 atmp $L^2\ mol^{-2}$ Q.20 C

Q.21 (a) (i) 18.1 g/mol , (ii) 50.25 L $mol^{-1}$ , (iii) 1.224 , (iv) repulsive, (b) $2.07 \times 10^{-20}$ J

Q.22 434.17 m/sec

Q.23 Since $H_2$ is diatomic and He is monoatomic degree of freedom for mono is 3 and only translational but for diatomic, vibrational and rotational are also to be considered

Q.24 C Q.25 RT Q.26 B Q.27 C

# STUDY PACKAGE

# Target: IIT-JEE (Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 7. Chemical Equilibrium

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

**4. Exercise III**

**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

## *THE KEY*

### *CHEMICAL EQUILIBRIUM*

Most of the chemical reaction do not go to completion in a closed system and attain a state of equilibrium. Equilibrium is said to have reached in a physical or chemical system when rate of forward and reverse processes are equal. At equilibrium macroscopic properties of the system like concentration. Pressure ect. become constant at constant temperature.

State of chemical equilibrium is characterised by equilibrium constant. Equilibrium constant have constant value at a given temperature.

### *UNDERSTANDING EQUILIBRIUM*

There are two approaches to understand nature of equilibrium. One stems from kinetics as developed by Gulberg and Wagge (1863). The other approach comes from thermodynaics. Equilibrium criteria is explained on the basis of thermodynamic function like $\Delta H$ (change in enthalpy), $\Delta S$ (change in entropy) and $\Delta G$ (change in Gibb's function).

According to kinetic approaches -The state of equilibrium is characterised by equal rate of forward and backward process.
At equilibrium

| Rate of forward reaction = Rate of backward reaction. |
|---|

Example : Physical equilibria. Solid liquid equilibria
Solid $\rightleftharpoons$ liquid
Example: $H_2O$ (s) $\rightleftharpoons$ $H_2O(l)$ : 273 K ; 1 atm P.

Solid ice and liquid can coexist at 273 K and 1 atm. Solid form is said to be in equilibrium with liquid form. At equilibrium, if heat exchanged from surrounding is zero, amount of solid ice and liquid water will remain unchanged. However it must be noted that, the process of conversion of ice into water and vica-versa-never ceases. At equilibrium

| Net rate of conversion of ice into water = Net rate of conversion of water into ice. |
|---|

### *OTHER EXAMPLES OF PHYSICAL EQUILIBRIA*

The liquid vapour equilibria :

(i) Example : $H_2O$ ($l$) $\rightleftharpoons$ $H_2O$ (g) 373 K ; 1 atm pressure.
Equilibrium is characterized by constant value of vapour pressure of $H_2O$ ($l$) at 373 K (= 1 atm)

Net rate of condensation of $H_2O$ (g) = net rate of evaporation of $H_2O$ ($l$)

(ii) Sugar (s) $\rightleftharpoons$ sugar (aq)
This is example of dissolution equilibria. Equilibrium is characterised by constant molar concentration of sugar at specified temperature. At equilibrium, the solution of sugar in aqueous solution is called saturated solution.

(iii) The dissolution of gas in liquid.
Example $CO_2$ (g) + $H_2O$ $\rightleftharpoons$ $CO_2$(aq)
The concentration of gas in liquid is proportional to the pressure of gas over the liquid.

| *Process* | *Characteristic constant* |
|---|---|
| $H_2O$ (*l*) ⇌ $H_2O$ (g) | $P_{H_2O}$ constant at given temperature |
| $H_2O$ (s) ⇌ $H_2O$ (*l*) | $P_{H_2O}$ constant at given temperature |
| solute (s) ⇌ solute (sol$^n$) | concentration of solute is constant at given temperature |
| gas (g) ⇌ gas (aq) | [Gas(aq)]/[Gas(g)] = constant at given temperature |

## *IMPORTANT CHARACTERISTIC OF EQUILIBRIUM*

(i) Equilibrium is possible only in closed system.
(ii) The rate of forward process at equilibrium is equal to rate of backward process.
(iii) All measurable properties of system remain constant over time.
(iv) When equilibriums is attained for a physical process, it is characterised by constant value of one of its parameter.
(v) The constant value of these parameters indicate extent to which equilibria is shifted in forward direction.
(vi) Both, *Kinetic* and *Thermodynamics* theories can be invoked to understand the extent to which a reaction proceed to forward direction. *e.g.* If extent of reaction is too large for forward direction (equilibrium is tilted heavily to forward direction) than
  (a) Specific rate of forward reaction >>> specific rate of backward reaction
  (b) Product is thermodynamically very stable as compared to reactant.
  (c) Gibb's function of product is vary small as compared to Gibb's function of reactant.

## *EQUILIBRIUM IN CHEMICAL PROCESS*

A general equation for a reversible reaction may be written

$$mA + nB + ..... \rightleftharpoons xC + yD + .....$$

we can write the reaction quotient, Q for this equation as

$$Q = \frac{[C]^x[D]^y.....}{[A]^m[B]^n.....}$$

where we use bracket to indicate "molar concentration of." The reaction quotient is a ratio of the molar concentrations of the product of the chemical equation (multiplied together) and of the reactants (also multiplied together), each raised to a power equal to the coefficient preceding that substance in the balanced chemical equation. The reaction quotient for the reversible reaction.

$$2NO_2\,(g) \rightleftharpoons N_2O_4\,(g)$$

is given by the expression $Q = \frac{[N_2O_4]}{[NO_2]^2}$

The numerical value of Q for a given reaction varies ; it depends on the concentration of products and reactants present at the time when Q is determined.

When a mixture of reactants and products of a reaction reaches equilibrium at a given temperature, its reaction quotient always has the same value. This value is called the equilibrium constant, K, of the reaction at that temperature. When a reaction is at equilibrium at a given temperature, the concentration of reactants and products is such that the value of reaction quotient, Q is always equal to the equilibrium constant, K, for that reaction at that temperature.

The mathematical expression that indicates that a reaction quotient always assumes the same value at equilibriums

$$Q = K = \frac{[C]^x[D]^y.....}{[A]^m[B]^n.....}$$

is a mathematical statement of the *law of mass action*. When a reaction has attained equilibrium at a given temperature, the reaction quotient for the reaction always has the same value.

The magnitude of an equilibrium constant is a measure of the yield of a reaction when it reaches equilibrium.

A large value for K indicates that equilibrium is attained only after the reactants have been largely converted into products. A small value of K-much less than 1-indicates the equilibrium is attained when only a small proportion of the reactants have been converted into products.

Regardless of the initial mixture of reactants and products in a reversible reaction, the composition of a system will always adjust itself to a condition of equilibrium for which the value of the reaction quotient is equal to the equilibrium constant for the system, provided that the temperature does not change.

An equilibrium can be established either starting from reactants or starting from products. In fact, one technique that is used to determine whether a reaction it truly at equilibrium is to approach equilibrium starting with reactants in one experiment and starting with products in another. If the same value of the reaction quotient is observed when the concentrations stop changing in both experiments, then we may be certain that the system has reached equilibrium.

We should calculate the value of Q or K from the activities of the reactants and products rather than from their concentrations. However, the activity of a dilute solute is usefully approximated by its molar concentration, so we will use concentrations as approximated by its pressure (in atmospheres), so we use pressures for gases. However, we also can use molar concentrations of gases in our equilibrium calculations, because the molar concentration of a gas is directly proportional to its pressure. The activity of a pure solid or pure liquid is 1, and the activity of a solvent in a dilute solution is close to 1. Thus these species (solids, liquids, and solvents) are omitted from reactions quotients and equilibrium calculations.

Using concentrations and pressure instead of activities means that we calculate approximate values for reaction quotients and equilibrium constants. However, these approximations hold well for dilute solutions and for gases with pressures less than about 2 atmospheres.

## *CONCENTRATION VERSES TIME GRAPH FOR HABER PROCESS*

$$3H_2(g) + N_2(g) \longrightarrow 2NH_3(g)$$

Starting with pure $H_2$ and $N_2$ as reaction proceeds in forward direction. Ammonia is formed. At initially conc. of $H_2$ and $N_2$ drops and attain a steady value at equilibrium. On the others hand conc. of $NH_3$ increases and at equilibrium attains a constant value.

Concentration time graphs for $N_2 + 3H_2 \rightleftharpoons 2NH_3$

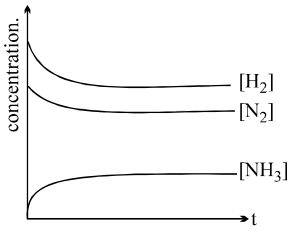



fig (1)

Concentration time graph for $H_2 + I_2 \rightleftharpoons 2HI$

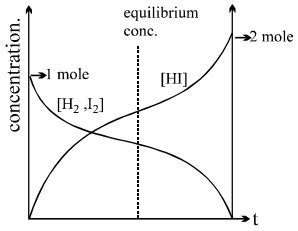



fig (2)

This graph shows how equilibrium state can be achieved from both direction.

## *HOMOGENEOUS CHEMICAL EQUILIBRIA*

A homogeneous equilibrium is equilibrium with in a single phase i.e. when physical state of all the reactants and product are same.

## *LIQUID PHASE HOMOGENEOUS EQUILIBRIUM*

Example : (i) $I_2\,(aq) + I^-\,(aq) \rightleftharpoons I_3^-(aq)$

(ii) $Hg_2^{2+}(aq) + NO_3^-(aq) + 3H_3O^+(aq) \rightleftharpoons 2Hg^{2+}(aq) + HNO_2(aq) + 4H_2O(l)$

(iii) $NH_3(aq) + H_2O\,(l) \rightleftharpoons NH_4^+(aq) + OH^-\,(aq)$

Eq. constants for

(i) $K = \dfrac{[I_3^-(aq)]}{[I_2(aq)][I^-(aq)]}$

(ii) $K = \dfrac{[Hg^{2+}]^2[HNO_2]}{[Hg_2^{2+}][NO_3^-][H_3O^+]^3}$

(iii) $K = \dfrac{[NH_4^+(aq)][OH^-(aq)]}{[NH_3(aq)]}$

The equilibrium constant in all theses cases can be called $K_C$. The subscrit 'C' denoting active masses of solute expressed in terms of molar concentration.

## *HOMOGENEOUS EQUILIBRIA IN GASES*

Example : (i) $C_2H_6(g) \rightleftharpoons C_2H_4() + H_2(g)$

(ii) $3O_2(g) \rightleftharpoons 2O_3(g)$

(iii) $C_3H_8(g) + 5O_2(g) \rightleftharpoons 3CO_2(g) + 4H_2O(g)$

Equilibrium constant expression for then are

(i) $K_C = \dfrac{[C_2H_4O)][H_2]}{[C_2H_6(g)]}$ $\qquad$ $K_P = \dfrac{[P_{C_2H_4}][P_{H_2}]}{[P_{C_2H_6}]}$

[ ] represents concentration in mol/ litre at equilibrium $\qquad$ $P_{C_2H_4}$ & other are partial pressure at equilibrium

(ii) $K_C = \dfrac{[O_3]^2}{[O_2]^3}$ $\qquad$ $K_P = \dfrac{P_{O_3}^2}{P_{O2}^3}$

(iii) $K_C = \dfrac{[CO_2]^2[H_2O]^4}{[C_3H_8][O_2]^5}$ $\qquad$ $K_P = \dfrac{P_{CO_2}^3 \cdot P_{H_2O}^4}{P_{C_3H_8} \cdot P_{O_2}^5}$

***Note*** : Equilibrium constant for gaseous homogeneous equilibrium can be expressed in two ways Vi2; $K_P$ and $K_C$. This means value of equilibrium constant depends upon choice of standard state in which concentration of reactant's and product are expressed.

## *HETROGENEOUS EQUILIBRIA*

If reactants and product are found in two or more phases, the equilibria describing them is called hetrogeneous equilibrium.

Example: (i) $PbCl_2(s) \rightleftharpoons Pb^{2+}(aq) + 2Cl^-\,(aq)$

(ii) $CaO(s) + CO_2(g) \rightleftharpoons CaCO_3(s)$

(iii) $Br_2\,(l) \rightleftharpoons Br_2(g)$

Equilibrium expression for them can be written as

(i) $K = [Pb^{2+}(aq)][Cl^-(aq)]^2$

(ii) $K_P = \dfrac{1}{P_{CO_2}}$ $\qquad$ $K_C = \dfrac{1}{[CO_2(g)]}$

(iii) $K_P = P_{Br_2}$ $\qquad$ $K_C = [Br_2(g)]$

***Note*** **:** Active masses of pure solid and liquid are taken as 'I'. It is because as pure solids and liquid took part in reaction, their concentration (or density) remain constant. In thermodynamic sense. We can say this is because Gibb's functions for pure solid and liquid is defined at stipulated pressure of 1.00 bar and as pressure of system changes, Gibb's function for pure solid and liquid remain constant and equal to their value at 1 bar.

### *UNIT OF EQUILIBRIUM CONSTANT*

We have already noted that the value of an equilibrium constant has meaning only when we give the corresponding balanced chemical equation. Its value changes for the new equation obtained by multiplying or dividing the original equation by a number. The value for equilibrium constant, $K_C$ is calculate substituting the concentration in mol/L and for $K_P$ by substituting partial pressure in Pa, kPa, etc. in atm. Thus, units of equilibrium constant will turn out to be units based on molarity or pressure, unless the sum of the exponents in the numerator is equal to the sum of the exponents in the denominator. Thus for the reaction:

$H_2(g) + I_2(g) \rightleftharpoons 2HI$, $K_C$ and $K_P$ do not have any unit

$N_2(g) + 3H_2(g) \rightleftharpoons 2\ NH_3$ , $K_C$ has unit $(mol/L)^{-2}$ and $K_P$ has unit $bar^{-2}$ or

$N_2O_4\ (g) \rightleftharpoons 2NO_2$, $K_C$ has unit mol / L and $K_P$ has unit bar

However, these days we express equilibrium constants in dimensionless quantities by specifying the standard state of the reactants and the products. The standard state for pure gas is 1 bar and now the partial pressure are measured with respect to this standard. Thus a pressure of 2 bar in term of this standard state is equal to 2 bar / 1 bar = 2, a dimensionless number. Similarly for a solute the standard state; $c_0$, is 1 molar solution and all concentrations are measured with respect to it. The numerical value of equilibrium constant depends on the standard state chosen.

### *FACTOR'S AFFECTING EQUILIBRIA*

Effect of change in concentration on equilibrium. A chemical system at equilibrium can be shifted out of equilibrium by adding or removing one more of reactants or products. Shifting out of equilibrium doesn't mean that value of equilibrium constant change. Any alteration of concentration of reactant or product will disturb the equilibrium and concentration of reactant and product one readjust to one again attain equilibrium concentration.

In other word, as we add or remove reactant (or product) the ratio of equilibrium concentration become 'Q' (reaction quotient) and depending upon.

$Q < K$ : equilibrium will shift in forward direction.

$Q > K$ : equilibrium will shift in backward direction.

Example : $Fe^{3+}(a) + SCN^{-}(aq) \rightleftharpoons Fe(SCN)^{2+}(aq)$

(i) adding $Fe^3$ or $SCN^-$ will more $\dfrac{[Fe(SCN)^{2+}]}{[Fe^{3+}][SCN^-]} = Q$ less then $K_C$ and equilibria will shift in forward direction.

(ii) Removing $Fe(SCN)^{2+}$ will have same effect

(iii) Adding $Fe(SCN)^{2+}$ from outside source in equilibrium mixture will have effect of increasing 'Q' hence reaction shift in backward direction.

### *EFFECT OF CHANGE IN PRESSURE*

Sometimes we can change the position of equilibrium by changing the pressure on a system. However, changes in pressure have a measurable effect only in system where gases are involved – and then only when the chemical reaction produces a change in the total number of gas molecules in the system.

As we increase the pressure of a gaseous system at equilibrium, either by decreasing the volume of the system or by adding more of the equilibrium mixture, we introduce a stress by increasing the number of molecules per unit of volume. In accordance with Le Chatelier's principle, a chemical reaction that reduces the total number of molecules per unit of volume will be favored because this relieves the stress. The reverse reaction would be favoured by a decrease in pressure.

Consider what happens when we increase the pressure on a system in which NO, $O_2$ and $NO_2$ are in equilibrium.

Example : $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

The formation of additional amounts of $NO_2$ decreases the total number of molecules in the system, because each time two molecules of $NO_2$ form, a total of three molecules of NO and $O_2$ react. This reduces the total pressure exerted by the system and reduces, but does not completely relieve, the stress of the increased pressure. On the other hand, a decrease in the pressure on the system favors decomposition of $NO_2$ into NO and $O_2$ which tends to restore the pressure.

Let us now consider the reaction

$$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$$

Because there is no change in the total number of molecules in the system during reaction, a change in pressure does not favor either formation or decomposition of gaseous nitric oxide.

### *EFFECT OF CHANGE IN TEMPERATURE ON EQUILIBRIUM*

Changing concentration or pressure upsets an equilibrium because the reaction quotient is shifted away from the equilibrium value. Changing the temperature of a system at equilibrium has a different effect: A change in temperature changes the value of the equilibrium constant. However, we can predict the effect of the temperature change by treating it as a stress on the system and applying Le Chatelier's principle. When hydrogen reacts with gaseous iodine, energy is released as heat is evolved.

$$H_2(g) + I_2(g) \rightleftharpoons 2I(g) \quad \Delta H = -9.4 \text{ kJ (exothermic)}$$

Because this reaction is exothermic, we can write it with heat as a product.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) + 9.4 \text{ kJ}$$

Increasing the temperature of the reaction increases the amount of energy present. Thus, increasing the temperature has the effect of increasing the amount of one of the products of this reaction. The reaction shifts to the left to relieve the stress, and there is an increase in the concentration of $H_2$ and $I_2$ and a reduction in the concentration of HI. When we change the temperature of a system at equilibrium, the equilibrium constant for the reaction changes. Lowering the temperature in the HI system increases the equilibrium constant from 50.0 at 400°C to 67.5 at 357°C. At equilibrium at the lower temperature, the concentration of HI has increased and the concentrations of $H_2$ and $I_2$ have decreased. Raising the temperature decreases the value of the equilibrium constant from 67.5 at 357°C to 50.0 at 400°C.

### *EFFECT OF TEMPERATURE : VAN'T HOFF EQUATION*

(a) $\dfrac{d(\ell nK)}{dT} = \dfrac{\Delta H^\circ}{RT^2}$ (b) $\dfrac{d(\ell nK)}{d\left(\frac{1}{T}\right)} = -\dfrac{\Delta H^\circ}{R}$ . **Integrated form** $\ell n\left(\dfrac{K_2}{K_1}\right) = -\dfrac{\Delta H^\circ}{R}\left(\dfrac{1}{T_2} - \dfrac{1}{T_1}\right)$

### *A THERMODYNAMIC RELATIONSHIP :*

$\Delta G^\circ = -RT\ell nK$ .

### *EFFECT OF CATALYST ON EQUILIBRIUM*

A catalyst has no effect on the value of an equilibrium constant or on equilibrium concentrations. The catalyst merely increase the rates of both the forward and the reverse reactions to the same extent so that equilibrium is reached more rapidly.

All of these effects change in concentration or pressure, change in temperature, and the effect of a catalyst on a chemical equilibrium play a role in the industrial synthesis of ammonia from nitrogen and hydrogen according to the equation.

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

One way to increase the yield of ammonia is to increase the pressure on the system in which $N_2$, $H_2$ and $NH_3$ are in equilibrium or are coming to equilibrium.

$$N_2(g)\ 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The formation of additional amounts of ammonia reduces the total pressure exerted by the system and somewhat reduces the stress of the increased pressure.

Although increasing the pressure of a mixture of $N_2$ , $H_2$ and $NH_3$ increase the yield ammonia, at low temperatures the rate of formation of ammonia is slow. At room temperature, for example, the reaction is so slow that if we prepared a mixture of $N_2$ and $H_2$, no detectable amount of ammonia would form during our lifetime. Attempts to increase the rate of the reaction by increasing the temperature are counterproductive. The formation of ammonia from hydrogen and nitrogen is an exothermic process:

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g) \qquad \Delta H = -92.2\ kJ$$

Thus increasing the temperature to increase the rate lowers the yield. If we lower the temperature to shift the equilibrium to the right to favor the formation of more ammonia, equilibrium is reached more slowly because of the large decrease of reaction rate with decreasing temperature.

Part of the rate of formation lost by operating at lower temperatures can be recovered by using a catalyst to increase the reaction rate. Iron powder is one catalyst used. However, as we have seen, a catalyst serves equally well to increase the rate of a reverse reaction in this case, the decomposition of ammonia into its constituent elements. Thus the net effect of the iron catalyst on the reaction is to cause equilibrium to be reached more rapidly.

In the commercial production of ammonia, conditions of about 500°C and 150–900 atmosphere are selected to give the best compromise among rate, yield and the cost of the equipment necessary to produce and contain gases at high pressure and high temperatures.

## *APPLICATION OF EQUILIBRIUM CONSTANT*

Before we consider the applications of equilibrium constants, let us consider its important features:

(i) the expression for equilibrium constant, K is applicable only when concentrations of the reactants and products have attained their equilibrium values and do not change with time.

(ii) The value of equilibrium constant is independent of initial concentration of the reactants and product. Equilibrium constant has one unique value for a particular reaction represented by a balanced equation at a given temperature.

(iii) The equilibrium constant for the reverse reaction is equal constant for the forward reaction.

(iv) The equilibrium constant, K for a reaction is related to the equilibrium constant of the corresponding reaction whose equation is obtained by multiplying or dividing the equation for the original reaction by a small integer.

Now we will consider some applications of equilibrium constant and use it to answer question like:

(i) predicting the extent of a reaction on the basis of its magnitude.

(ii) predicting the direction of the reaction, and

(iii) calculating equilibrium concentration.

### *Predicting the extent of a reaction*

The magnitude of equilibrium constant is very useful especially in reactions of industrial importance. An equilibrium constant tells us whether we can expect a reaction mixture to contain a high or low concentration of product(s) at equilibrium. (It is important to note that an equilibrium constant tells us nothing about the rate at which equilibrium is reached). In the expression of $K_C$ or $K_P$, product of the concentrations of

products is written in numerator and the product of the concentrations of reactants is written in denominator. High value of equilibrium constant indicates that product(s) concentration is high and its low value indicates that concentration of the product(s) in equilibrium mixture is low.

For reaction, $H_2(g) + Br_2(g) \rightleftharpoons 2HBr(g)$, the value of

$$K_P = \frac{(P_{HBr})^2}{(P_{H_2})(P_{Br_2})} = 5.4 \times 10^{18}$$

The large value of equilibrium constant indicates that concentration of the product, HBr is very high and reaction goes nearly to completion.

Similarly, equilibrium constant for the reaction $H_2(g) + Cl_2(g) \rightleftharpoons 2HCl(g)$ aty 300 K is very high and reaction goes virtually to completion.

$$K_C = \frac{[HCl]^2}{[H_2][Cl_2]} = 4.0 \times 10^{31}$$

Thus, large value of $K_P$ or $K_C$ (larger than about $10^3$), favour the products strongly. For intermedicate values of K (approximately in the range of $10^{-3}$ to $10^3$), the concentrations of reactants and products are comparable. Small values of equilibrium constant (smaller than $10^{-3}$), favour the reactants strongly.

At 298 K for reaction, $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

$$K_C = \frac{[NO]^2}{[N_2][O_2]} = 4.8 \times 10^{-31}$$

The very small value of $K_C$ implies that reactants $N_2$ and $O_2$ will be the predominant species in the reaction mixture at equilibrium.

### *Predicting the direction of the reaction.*

The equilibrium constant is also used to find in which direction an rabidity reaction mixture of reactants and products will proceed. For this purpose, we calculate the reaction quotient, Q. The reaction quotient is defined in the same way as the equilibrium constant ( with molar concentrations to give $Q_C$, or with partial pressure to give $Q_P$) at any stage of reaction. For a general reaction:

$$aA + bB \rightleftharpoons cC + dD$$

$$Q_C = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

Then, if $Q_C > K_c$ , the reaction will proceed in the direction of reactants (reverse reaction).

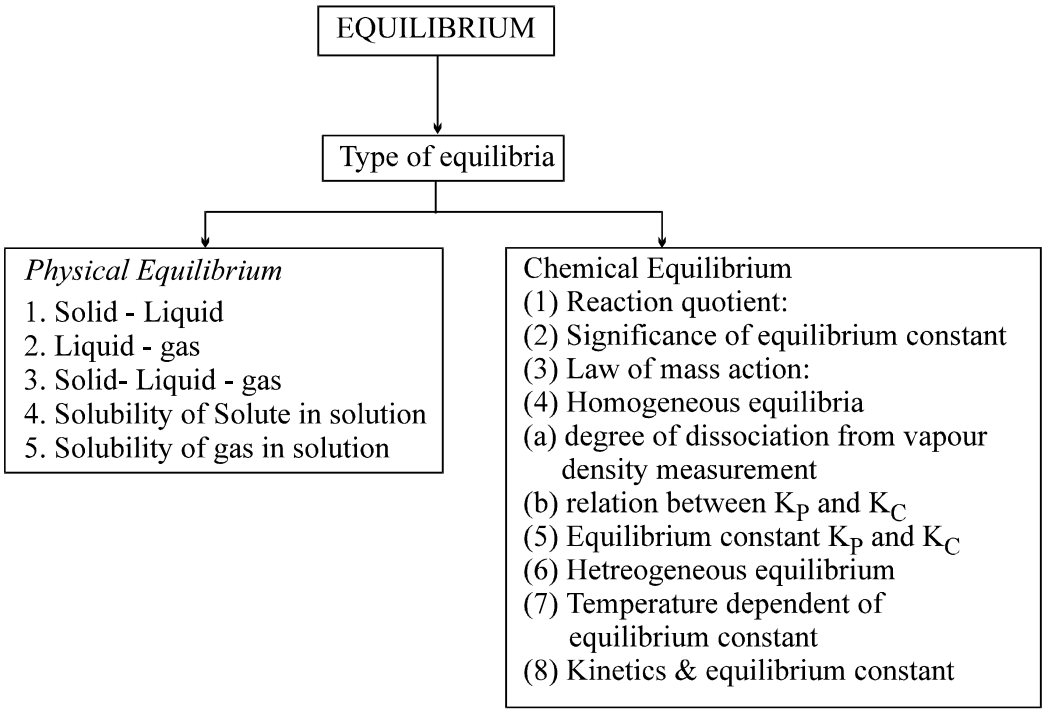
if $Q_C < K_c$, the reaction will move in the direction of the products

if $Q_C = K_c$, the reaction mixture is already at equilibrium.

In the reaction, $H_2(g) + I_2(g) \rightleftharpoons 2Hl(g)$, if the molar concentrations of $H_2$ , $I_2$ and HI are 0.1 mol $L^{-1}$ respectively at 783 K, then reaction quotient at this stage of the reaction is

$$Q_C = \frac{[HI]^2}{[H_2][I_2]} = \frac{(0.4)^2}{(0.1)(0.2)} = 8$$

$K_C$ for this reaction at 783 K is 46 and we find that $Q_C < K_C$. The reaction, therefore, will move to right i.e. more $H_2(g)$ and $I_2(g)$ will react to form more HI (g) and their concentration will decrease till $Q_C = K_C$.




Application of equilibrium constant

(i) Pridicting extent of reaction
(ii) Pridicting direction of change
(iii) Calculation of equilibrium concentration

Le Chatleir's principle

Effect of following factor's on equlibrium — Once pressure temperature catalyst inert gas

Temperature dependence- Von't Hoff's equation

Understanding equilibrium
(i) From kinetic's approach (Gulber & Wagge approach)
(ii) From thermodynamics approach – Criteria for equilibrium in terms of Gibb's function

# ***EXERCISE I***

### *Reaction quotient and equilibrium constant*

Q.1 The initial concentrations or pressure of reactants and products are given for each of the following systems. Calculate the reaction quotient and determine the directions in which each system will shift to reach equilibrium.

(a) $2NH_3\ (g) \rightleftharpoons N_2\ (g) + 3H_2\ (g)$ $K = 17$
$[NH_3] = 0.20\ M$ ; $[N_2] = 1.00\ M$ ; $[H_2] = 1.00\ M$

(b) $2NH_3\ (g) \rightleftharpoons N_2\ (g) + 3H_2\ (g)$ $K_p = 6.8 \times 10^4\ atm^2$
Initial pressure : $NH_3 = 3.0$ atm ; $N_2 = 2.0$ atm ; $H_2 = 1.0$ atm

(c) $2SO_3(g) \rightleftharpoons 2SO_2\ (g) + O_2\ (g)$ $K = 0.230$ atm
$[SO_3] = 0.00\ M$ ; $[SO_2] = 1.00\ M$ ; $[O_2] = 1.00\ M$

(d) $2SO_3(g) \rightleftharpoons 2SO_2\ (g) + O_2\ (g)$ $K_p = 16.5$ atm
Initial pressure : $SO_3 = 1.0$ atm ; $SO_2 = 1.0$ atm ; $O_2 = 1.0$ atm

(d) $2NO(g) + Cl_2\ (g) \rightleftharpoons 2NOCl\ (g)$ $K = 4.6 \times 10^4$
$[NO] = 1.00\ M$ ; $[Cl_2] = 1.00\ M$ ; $[NOCl] = 0\ M$

(f) $N_2\ (g) + O_2\ (g) \rightleftharpoons 2NO\ (g)$ $K_p = 0.050$
Initial pressure : $NO = 10.0$ atm ; $N_2 = O_2 = 5$ atm

Q.2 Among the solubility rules is the statement that all chlorides are soluble except $Hg_2Cl_2$, AgCl, $PbCl_2$, and CuCl.

(a) Write the expression for the equilibrium constant for the reaction represented by the equation.
$AgCl\ (s) \rightleftharpoons Ag^+(aq) + Cl^-\ (aq)$
Is K greater than 1, less than 1, or about equal to 1? Explain your answer

(b) Write the expression for the equilibrium constant for the reaction represented by the equation
$Pb^{2+}\ (aq) + 2Cl^-\ (aq) \rightleftharpoons PbCl_2\ (s)$
Is K greater than 1, less than 1, or about equal to 1? Explain your answer.

Q.3 Among the solubility rules is the statement that carbonates, phosphates, borates, arsenates, and arsenites, except those of the ammonium ion and the alkali metals are insoluble.

(a) Write the expression for the equilibrium constant for the reaction represented by the equation
$CaCO_3\ (s) \rightleftharpoons Ca^{2+}\ (aq) + CO_3^{2-}\ (aq)$
Is K greater than 1, less than 1, or about equal to 1? Explain your answer

(b) Write the expression for the equilibrium constant for the reaction represented by the equation.
$3Ba^{2+}\ (aq) + 2PO_4^{3-}\ (aq) \rightleftharpoons Ba_3(PO_4)_2\ (s)$
Is K greater than 1, less than 1, or about equal to 1? Explain your answer.

Q.4 Benzene is one of the compounds used as octane enhancers in unleaded gasoline. It is manufactured by the catalytic conversion of acetylene to benzene.
$3C_2H_2 \longrightarrow C_6H_6$
Would this reaction be most useful commercially if K were about 0.01, about 1, or about 10? Explain your answer.

Q.5 Show the complete chemical equation and the net ionic equation for the reaction represented by the equation
$KI\ (aq) + I_2\ (aq) \rightleftharpoons KI_3\ (aq)$
give the same expression for the reaction quotient. $KI_3$ is composed of the ions $K^+$ and $I_3^-$.

### *Using the equilibrium constant*

Q.6 Which of the following reactions goes almost all the way to completion, and which proceeds hardly at all?

(a) $N_2(g) + O_2(g) \rightleftharpoons 2NO\ (g)$; $K_c = 2.7 \times 10^{-18}$

(b) $2NO(g) + O_2(g) \rightleftharpoons 2NO_2\ (g)$; $K_c = 6.0 \times 10^{13}$

Q.7 For which of the following reactions will the equilibrium mixture contain an appreciable concentration of both reactants and products?

(a) $Cl_2(g) \rightleftharpoons 2Cl\ (g)$ ; $K_c = 6.4 \times 10^{-39}$

(b) $Cl_2(g) + 2NO\ (g) \rightleftharpoons 2NOCl\ (g)$ ; $K_c = 3.7 \times 10^{8}$

(c) $Cl_2(g) + 2NO_2\ (g) \rightleftharpoons 2NO_2Cl\ (g)$ ; $K_c = 1.8$

Q.8 The value of $K_c$ for the reaction $3O_2\ (g) \rightleftharpoons 2O_3\ (g)$ is $1.7 \times 10^{-56}$ at 25°C. Do you expect pure air at 25°C to contain much $O_3$ (ozone) when $O_2$ and $O_3$ are in equilibrium? If the equilibrium concentration of $O_2$ in air at 25°C is $8 \times 10^{-3}$ M, what is the equilibrium concentration of $O_3$?

Q.9 At 1400 K, $K_c = 2.5 \times 10^{-3}$ for the reaction $CH_4\ (g) + 2H_2S \rightleftharpoons CS_2(g) + 4H_2(g)$. A 10.0 L reaction vessel at 1400 K contains 2.0 mol of $CH_4$, 3.0 mol of $CS_2$, 3.0 mol of $H_2$ and 4.0 mol of $H_2S$. Is the reaction mixture at equilibrium? If not, in which direction does the reaction proceed to reach equilibrium?

Q.10 The first step in the industrial synthesis of hydrogen is the reaction of steam and methane to give water gas, a mixture of carbon monoxide and hydrogen.

$H_2O\ (g) + CH_4(g) \rightleftharpoons CO\ (g) + 3H_2(g)$ $K_c = 4.7$ at 1400 K

A mixture of reactants and product at 1400 K contains 0.035 M $H_2O$, 0.050M $CH_4$, 0.15 M CO, and 0.20 M $H_2$. In which direction does the reaction proceed to reach equilibrium?

Q.11 An equilibrium mixture of $N_2$, $H_2$, and $NH_3$ at 700 K contains 0.036 M $N_2$ and 0.15 M $H_2$. At this temperature, $K_c$ for the reaction $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is 0.29. What is the concentration of $NH_3$?

Q.12 The air pollutant NO is produced in automobile engines from the high temperature reaction $N_2(g) + O_2(g) \rightleftharpoons 2NO\ (g)$ ; $K_c = 1.7 \times 10^{-3}$ at 2300 K. If the initial concentrations of $N_2$ and $O_2$ at 2300 K are both 1.40 M, what are the concentrations of NO, $N_2$, and $O_2$ when the reaction mixture reaches equilibrium?

Q.13 At a certain temperature, the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ has an equilibrium constant $K_c = 5.8 \times 10^{-2}$. Calculate the equilibrium concentrations of $PCl_5$, $PCl_3$ and $Cl_2$ if only $PCl_5$ is present initially, at a concentration of 0.160 M.

Q.14 At 700 K, $K_p = 0.140$ for the reaction $ClF_3\ (g) \rightleftharpoons ClF\ (g) + F_2(g)$. Calculate the equilibrium partial pressure of $ClF_3$, ClF, and $F_2$ if only $ClF_3$ is present initially, at a partial pressure of 1.47 atm.

### *Homogeneous equilibria degree of dissociation, vapour density and equilibrium constant*

Q.15 The degree of dissociation of $N_2O_4$ into $NO_2$ at 1.5 atmosphere and 40°C is 0.25. Calculate its $K_p$ at 40°C.Also report degree of dissociation at 10 atmospheric pressure at same temperature.

Q.16 At 46°C, $K_p$ for the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ is 0.667 atm . Compute the percent dissociation of $N_2O_4$ at 46° C at a total pressure of 380 Torr .

Q.17 When 36.8g $N_2O_4$ (g) is introduced into a 1.0-litre flask at 27°C . The following equilibrium reaction occurs : $N_2O_4$ (g) $\rightleftharpoons$ $2NO_2$ (g) ; $K_p = 0.1642$ atm.

(a) Calculate $K_c$ of the equilibrium reaction.

(b) What are the number of moles of $N_2O_4$ and $NO_2$ at equilibrium?

(c) What is the total gas pressure in the flask at equilibrium?

(d) What is the percent dissociation of $N_2O_4$?

Q.18 At some temperature and under a pressure of 4 atm , $PCl_5$ is 10% dissociated . Calculate the pressure at which $PCl_5$ will be 20% dissociated , temperature remaining same.

Q.19 In a mixture of $N_2$ and $H_2$ in the ratio of 1:3 at 64 atmospheric pressure and 300°C, the percentage of ammonia under equlibrium is 33.33 by volume. Calculate the equilibrium constant of the reaction using the equation . $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$.

Q.20 The system $N_2O_4 \rightleftharpoons 2NO_2$ maintained in a closed vessel at 60° C & a pressure of 5 atm has an average (i.e. observed) molecular weight of 69, calculate $K_p$. At what pressure at the same temperature would the observed molecular weight be (230/3) ?

Q.21 The vapour density of $N_2O_4$ at a certain temperature is 30. Calculate the percentage dissociation of $N_2O_4$ at this temperature. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$.

Q.22 In the esterfication $C_2H_5OH$ (l) + $CH_3COOH$ (l) $\rightleftharpoons$ $CH_3COOC_2H_5$ (l) + $H_2O$ (l) an equimolar mixture of alcohol and acid taken initially yields under equilibrium, the water with mole fraction = 0.333. Calculate the equilibrium constant.

### *Hetrogeneous equilibrium*

Q.23 Solid Ammonium carbamate dissociates as: $NH_2COONH_4$ (s) $\rightleftharpoons$ $2NH_3(g) + CO_2(g)$. In a closed vessel solid ammonium carbamate is in equilibrium with its dissociation products. At equilibrium, ammonia is added such that the partial pressure of $NH_3$ at new equilibrium now equals the original total pressure. Calculate the ratio of total pressure at new equilibrium to that of original total pressure.

Q.24 A sample of $CaCO_3(s)$ is introduced into a sealed container of volume 0.821 litre & heated to 1000K until equilibrium is reached. The equilibrium constant for the reaction $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ is $4 \times 10^{-2}$ atm at this temperature. Calculate the mass of CaO present at equilibrium.

Q.25 Anhydrous calcium chloride is often used as a dessicant. In the presence of excess of $CaCl_2$, the amount of the water taken up is governed by $K_p = 6.4 \times 10^{85}$ for the following reaction at room temperature, $CaCl_2(s) + 6H_2O(g) \rightleftharpoons CaCl_2 .6H_2O(s)$ . What is the equilibrium vapour pressure of water in a closed vessel that contains $CaCl_2(s)$ ?

Q.26 20.0 grams of $CaCO_3(s)$ were placed in a closed vessel, heated & maintained at 727° C under equilibrium $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ and it is found that 75 % of $CaCO_3$ was decomposed. What is the value of $K_p$ ? The volume of the container was 15 litres.

### *Changes in concentration at equilibrium Le Chatelier's principle*

Q.27 Suggest four ways in which the concentration of hydrazine, $N_2H_4$, could be increased in an equilibrium described by the equation

$$N_2 (g) + 2H_2 (g) \rightleftharpoons N_2H_4 (g) \qquad \Delta H = 95 \text{ kJ}$$

Q.28 How will an increase in temperature affect each of the following equilibria? An increase in pressure?

(a) $2NH_3\ (g) \rightleftharpoons N_2\ (g) + 3H_2\ (g)$ $\Delta H = 92$ kJ

(b) $N_2\ (g) + O_2\ (g) \rightleftharpoons 2NO\ (g)$ $\Delta H = 181$ kJ

(c) $2O_3\ (g) \rightleftharpoons 3O_2\ (g)$ $\Delta H = -285$ kJ

(d) $CaO\ (s) + CO_2\ (g) \rightleftharpoons CaCO_3\ (s)$ $\Delta H = -176$ kJ

Q.29(a) Methanol, a liquid fuel that could possibly replace gasoline, can be prepared from water gas and additional hydrogen at high temperature and pressure in the presence of a suitable catalyst. Write the expression for the equilibrium constant for the reversible reaction.

$2H_2\ (g) + CO\ (g) \rightleftharpoons CH_3OH\ (g)$ $\Delta H = -90.2$ kJ

(b) Assume that equilibrium has been established and predict how the concentration of $H_2$, CO and $CH_3OH$ will differ at a new equilibrium if (1) more $H_2$ is added. (2) CO is removed. (3) $CH_3OH$ is added. (4) the pressure on the system is increased. (5) the temperature of the system is increased. (6) more catalyst is added.

Q.30(a) Water gas, a mixture of $H_2$ and CO, is an important industrial fuel produced by the reaction of steam with red-hot coke, essentially pure carbon. Write the expression for the equilibrium constant for the reversible reaction.

$C(s) + H_2O\ (g) \rightleftharpoons CO\ (g) + H_2\ (g)$ $\Delta H = 131.30$ kJ

(b) Assume that equilibrium has been established and predict how the concentration of each reactant and product will differ at a new equilibrium if (1) more C is added. (2) $H_2O$ is removed. (3) CO is added. (4) the pressure on the system is increased. (5) the temperature of the system is increased.

Q.31 Ammonia is a weak base that reacts with water according to the equation

$NH_3\ (aq) + H_2O\ (l) \rightleftharpoons NH_4^+ + OH^-\ (aq)$

Will any of the following increase the percent of ammonia that is converted to the ammonium ion in water? (a) Addition of NaOH. (b) Addition of HCl. (c) Addition of $NH_4Cl$.

Q.32 Suggest two ways in which the equilibrium concentration of $Ag^+$ can be reduced in a solution of $Na^+$, $Cl^-$, $Ag^+$ and $NO_3^-$, in contact with solid AgCl.

$Na^+\ (aq) + Cl^-\ (aq) + Ag^+\ (aq) + NO_3^-\ (aq) \rightleftharpoons AgCl\ (s) + Na^+\ (aq) + NO_3^-\ (aq)$ $\Delta H = -65.9$ kJ

Q.33 Additional solid silver sulfate, a slightly soluble solid, is added to a solution of silver ion and sulfate ion in equilibrium with solid silver sulfate. Which of the following will occur? (a) The $Ag^+$ and $SO_4^{2-}$ concentration will not change. (b) The added silver sulfate will dissolve. (c) Additional silver sulfate will form and precipitate from solution as $Ag^+$ ions and $SO_4^{2-}$ ions combine. (d) The $Ag^+$ ion concentration will increase and the $SO_4^{2-}$ ion concentration will decrease.

### *Kinetics and equilibrium constant*

Q.34 Consider a general, single-step reaction of the type $A + B \rightleftharpoons C$. Show that the equilibrium constant is equal to the ratio of the rate constant for the forward and reverse reaction, $K_c = k_f/k_r$.

Q.35 Which of the following relative values of $k_f$ and $k_r$ results in an equilibrium mixture that contains large amounts of reactants and small amounts of product?

(a) $k_f > k_r$ (b) $k_f = k_r$ (c) $k_f < k_r$

Q.36 Consider the gas-phase hydration of hexafluoroacetone, $(CF_3)_2CO$:

$$(CF_3)_2CO\ (g) + H_2O\ (g) \underset{k_r}{\overset{k_f}{\bullet}} (CF_3)_2C(OH)_2\ (g)$$

At 76°C, the forward and reverse rate constants are $k_f = 0.13\ M^{-1}s^{-1}$ and $k_r = 6.02 \times 10^{-4} s^{-1}$. What is the value of the equilibrium constant $K_c$?

Q.37 Consider the reaction of chloromethane with $OH^-$ in aqueous solution

$$CH_3Cl\,(aq) + OH^-\,(aq) \underset{k_r}{\overset{k_f}{\bullet}} CH_3OH\,(aq) + Cl^-\,(aq)$$

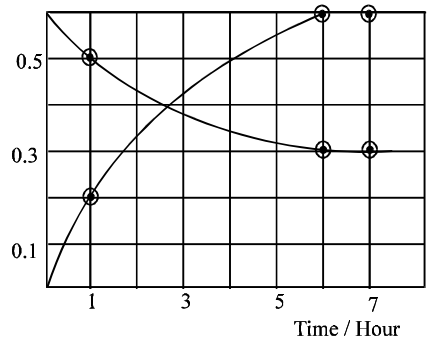
At 25°C, the rate constant for the forward reaction is $6 \times 10^{-6}\ M^{-1}s^{-1}$, and the equilibrium constant $K_c$ is $1 \times 10^{16}$. Calculate the rate constant for the reverse reaction at 25°C.

Q.38 The progress of the reaction A ⇌ nB with time, is presented in figure. Determine

(i) the value of n.

(ii) the equilibrium constant k.

(iii) the initial rate of conversion of A.




***Temperature dependence of equilibrium constant***

Q.39 Listed in the table are forward and reverse rate constants for the reaction $2NO\,(g) \rightleftharpoons N_2(g) + O_2(g)$

| **Temperature (K)** | $\mathbf{k_f\ (M^{-1}s^{-1})}$ | $\mathbf{k_r\ (M^{-1}s^{-1})}$ |
|---|---|---|
| 1400 | 0.29 | $1.1 \times 10^{-6}$ |
| 1500 | 1.3 | $1.4 \times 10^{-5}$ |

Is the reaction endothermic or exothermic? Explain in terms of kinetics.

Q.40 Forward and reverse rate constant for the reaction $CO_2(g) + N_2(g) \rightleftharpoons CO(g) + N_2O\,(g)$ exhibit the following temperature dependence.

| **Temperature (K)** | $\mathbf{k_f\ (M^{-1}s^{-1})}$ | $\mathbf{k_r\ (M^{-1}s^{-1})}$ |
|---|---|---|
| 1200 | $9.1 \times 10^{-11}$ | $1.5 \times 10^{5}$ |
| 1500 | $2.7 \times 10^{-9}$ | $2.6 \times 10^{5}$ |

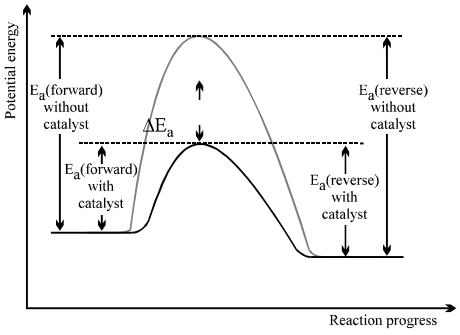
Is the reaction endothermic or exothermic? Explain in terms of kinetics.

Q.41 The equilibrium constant $K_p$ for the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ is $3.81 \times 10^2$ at 600 K and $2.69 \times 10^3$ at 700 K. Calculate $\Delta_r H$.

Q.42 As shown in figure a catalyst lowers the activation energy for the forward and reverse reactions by the same amount, $\Delta E_a$.

(a) Apply the Arrhenius equation, $K = Ae^{-E_a/RT}$ to the forward and reverse reactions, and show that a catalyst increases the rates of both reactions by the same factor.

(b) Use the relation between the equilibrium constant and the forward and reverse rate constants, $K_c = k_f/k_r$, to show that a catalyst does not affect the value of the equilibrium constant.



### *Temperature dependence of equilibrium constant*

Q.43 Variation of equilibrium constant 'K' with temperature 'T' is given by equation

$$\log K = \log A - \frac{\Delta H^\circ}{2.303\,RT}$$

A graph between log K and 1/T was a straight line with slope of 0.5 and intercept 10. Calculate

(a) $\Delta H^\circ$

(b) Pre exponential factor

(c) Equilibrium constant at 298 K

(d) Equilibrium constant at 798 K assuming $\Delta H^\circ$ to be independent of temperature.

Q.44 Rate of disappearance of the reactant A at two different temperature is given by A → B

$\frac{-d[A]}{dt} = (2\times10^{-2}\ S^{-1})\,[A] \qquad -4\times10^{-3}\ S^{-1}[B]$ ; 300K

$\frac{-d[A]}{dt} = (4\times10^{-2}\ S^{-1})\,[A] \qquad -16\times10^{-4}\ [B]$ ; 400K

Calculate heat of reaction in the given temperature range. When equilibrium is set up.

Q.45 The $K_P$ for reaction A + B → C + D is1.34 at 60°C and 6.64 at 100°C. Determine the free energy change of this reaction at each temperature and $\Delta H^\circ$ for the reaction over this range of temperature?

### *Equilibrium expressions and equilibrium constants*

Q.46 If $K_c = 7.5\times10^{-9}$ at 1000 K for the reaction $N_2\,(g) + O_2\,(g) \rightarrow 2NO\,(g)$, what is $K_c$ at 1000 K for the reaction $2NO\,(g) \rightarrow N_2\,(g) + O_2\,(g)$?

Q.47 An equilibrium mixture of $PCl_5$, $PCl_3$ and $Cl_2$ at a certain temperature contains $8.3\times10^{-3}$ M $PCl_5$, $1.5\times10^{-2}$ M $PCl_3$, and $3.2\times10^{-2}$ M $Cl_2$. Calculate the equilibrium constant $K_c$ for the reaction $PCl_5\,(g) \rightarrow PCl_3\,(g) + Cl_2\,(g)$.

Q.48 A sample of HI ($9.30\times10^{-3}$ mol) was placed in an empty 2.00 L container at 1000 K. After equilibrium was reached, the concentration of $I_2$ was $6.29\times10^{-4}$ M. Calculate the value of $K_c$ at 1000 K for the reaction $H_2\,(g) + I_2\,(g) \rightarrow 2HI\,(g)$.

Q.49 The vapour pressure of water at 25°C is 0.0313 atm. Calculate the values of $K_p$ and $K_c$ at 25°C for the equilibrium $H_2O\,(l) \rightarrow H_2O\,(g)$.

Q.50 For each of the following equilibria, write the equilibrium constant expression for $K_c$. Where appropriate, also write the equilibrium constant expression for $K_p$.

(a) $Fe_2O_3\,(s) + 3CO\,(g) \rightarrow 2Fe\,(l) + 3CO_2\,(g)$ (b) $4Fe\,(s) + 3O_2\,(g) \rightarrow 2Fe_2O_3\,(s)$

(c) $BaSO_4\,(s) \rightarrow BaO\,(s) + SO_3\,(g)$ (d) $BaSO_4\,(s) \rightarrow Ba^{2+}\,(aq) + SO_4^{2-}\,(aq)$

### *General problems*

Q.51 When 0.5 mol of $N_2O_4$ is placed in a 4.00 L reaction vessel and heated at 400 K, 79.3% of the $N_2O_4$ decomposes to $NO_2$.
Calculate $K_c$ and $K_p$ at 400 K for the reaction $N_2O_4\,(g) \rightarrow 2NO_2(g)$

Q.52 What concentration of $NH_3$ is in equilibrium with $1.0\times10^{-3}$ M $N_2$ and $2.0\times10^{-3}$ M $H_2$ at 700K? At this temperature $K_c = 0.291$ for the reaction $N_2(g) + 3H_2\,(g) \rightarrow 2NH_3\,(g)$.

Q.53 At 100 K, the value of $K_c$ for the reaction $C\,(s) + H_2O\,(g) \rightarrow CO\,(g) + H_2\,(g)$ is $3.0\times10^{-2}$. Calculate the equilibrium concentrations of $H_2O$, $CO_2$, and $H_2$ in the reaction mixture obtained by heating 6.0 mol of steam and an excess of solid carbon in a 5.0 L container. What is the molar composition of the equilibrium mixture?

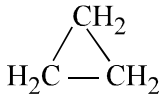
Q.54 When 1.0 mol of $PCl_5$ is introduced into a 5.0 L container at 500 K, 78.5 % of the $PCl_5$ dissociates to give an equilibrium mixture of $PCl_5$, $PCl_3$, and $Cl_2$.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

(a) Calculate the values of $K_c$ and $K_p$.

(b) If the initial concentrations in a particular mixture of reactants and products are $[PCl_5] = 0.5$ M, $[PCl_3] = 0.15$ M, and $[Cl_2] = 0.6$ M, in which direction does the reaction proceed to reach equilibrium? What are the concentrations when the mixture reaches equilibrium?

Q.55 The equilibrium constant $K_c$ for the gas-phase thermal decomposition of cyclopropane to propene is $1.0 \times 10^5$ at 500 K.



$$\rightleftharpoons CH_3{-}CH{=}CH_2 \qquad K_c = 1.0 \times 10^5$$

cyclopropane Propene

(a) What is the value of $K_p$ at 500 K?

(b) What is the equilibrium partial pressure of cyclopropane at 500 K when the partial pressure of propene is 5.0 atm?

(c) Can you alter the ratio of the two concentrations at equilibrium by adding cyclopropane or by decreasing the volume of the container? Explain.

(d) Which has the larger rate constant, the forward reaction or the reverse reaction?

(e) Why is cyclopropane so reactive?

***Thermodynamic and equilibrium constant***

Q.56 α-D-Glucose undergoes mutarotation to β-D-Glucose in aqueous solution. If at 298 K there is 60% conversion. Calculate $\Delta G^\circ$ of the reaction.

α-D-Glucose ⇌ β-D-Glucose

Q.57 For the reaction at 298 K

$$A(g) + B(g) \rightleftharpoons C(g) + D(g)$$

$\Delta H^\circ = -29.8$ kcal ; $\Delta S^\circ = -0.1$ kcal / K

Calculate $\Delta G^\circ$ and K.

Q.58 The equilibrium constant of the reaction $2C_3H_6(g) \rightleftharpoons C_2H_4(g) + C_4H_8(g)$ is found to fit the expression

$$\ln K = -1.04 - \frac{1088\,K}{T}$$

Calculate the standard reaction enthalpy and entropy at 400 K.

## ***PROFICIENCY TEST***

1. K for the reaction $2A + B \rightleftharpoons 2C$ is $1.5 \times 10^{12}$. This indicates that at equilibrium the concentration of _____ would be maximum.

2. The reaction $N_2 + O_2 \rightleftharpoons 2NO$ – Heat, would be favoured by _____ temperature.

3. K for the reaction $X_2 + Y_2 \rightleftharpoons 2XY$ is 100 K. For reaction $XY \rightleftharpoons \frac{1}{2}X_2 + \frac{1}{2}Y_2$ would be _____.

4. Compared to K for the dissociation, $2H_2S \rightleftharpoons 2H^+ + 2HS^-$, then K' for the $H^+ + HS^- \rightleftharpoons H_2S$ would have _____.

5. The equilibrium constant for a reaction decreases with increase in temperature, the reaction must be _____.

6. For the reaction, $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, $K_P$ and $K_C$ are related as _____.

7. For the reactions, $N_2O_4(g) \rightleftharpoons 2NO_2(g)$, at equilibrium, increase in pressure shifts the equilibrium in _____ direction.

8. $\Delta G^\circ$ is related to K by the relation_____.

9. Vant Hoff's equation is _____.

10. When the reaction is at equilibrium, the value of $\Delta G$ is _____.

11. Dimensions of equilibrium constant, $K_c$ for the reaction $2NH_3 \rightleftharpoons N_2 + 3H_2$, are _____.

12. The value of K for a reaction can be changed by changing _____.

13. The law of mass action was proposed by _____.

14. The degree of dissociation of $PCl_5$ [$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$], _____ with increase in pressure at equilibrium.

15. If concentration quotient, Q is greater than $K_C$, the net reaction in taking place in _____ direction.

16. The reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3$ would be favoured by ____ pressure.

17. $K_P$ is related to $K_C$ as _____.

18. Solubility of a gas in water ___ with increase in temperature.

19. Introduction of inert gas at constant volume to a gaseous reaction at equilibrium results in formation of _____ product.

20. The product is more stable than reactants in reaction having _____K.

21. Van't Hoff's equation gives the quantitative relation between change in value of K with change in temperature.

22. The larger value of K indicates that the product is more stable relative to reactants.

23. The value of equilibrium constant changes with change in the initial concentration of the reactants.

24. Extent of a reaction can always be increased by increasing the temperature.

25. $K_P$ is related to $K_C$ as $K_P = K_C (RT)^{\Delta n}$ .

26. Introduction of inert gas at a gaseous reaction ($\Delta n_g \neq 0$) at equilibrium keeping pressure constant has no effect on equilibrium state.

27. For the reaction, $N_2O_4 (g) \rightleftharpoons 2NO_2(g)$, $K_P = K_C (RT)$.

28. For a reaction the value of Q greater than K indicates that the net reaction is proceeding in backward direction.

29. Solubilities of all solids in water increase with increase in temperature.

30. Dissolution of all gases in water is accompanied by evolution of heat.

31. For the reaction, $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the equilibrium expression may be written as $K = \frac{[NH_3]^2}{[N_2][H_2]^3}$ .

32. For the reaction, $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$, $K_P = p_{CO_2}$ .

33. A catalyst increases the value of the equilibrium constant for a reaction.

34. If concentration quotient of reaction is less than K, the net reaction is proceeding in the backward direction.

35. In case of endothermic reactions, the equilibrium shifts in backward direction on increasing the temperature.

36. The value of K increases with increase in pressure.

37. For the reaction, $H_2 + I_2 \rightleftharpoons 2HI$, the equilibrium constant, K is dimensionless.

38. The reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$, $\Delta H = -X$ kJ, is favoured by high pressure and high temperature.

39. A very high value of K indicates that at equilibrium most of the reactants are converted into products.

40. The value of K for the reaction, $N_2 + 2H_2 \rightleftharpoons 2NH_3$, can be increased by applying high pressure or by using a catalyst.

## ***EXERCISE II***

Q.1 At high temperatures phosgene, $COCl_2$ decompose to give CO & $Cl_2$. In a typical experiment $9.9 \times 10^{-4}$ kg of $COCl_2$ is injected into a flask of volume 0.4105 $dm^3$ at 1000 K. When equilibrium is established it is found that the total pressure in the flask is $3.039 \times 10^5$ pascals. Calculate the equilibrium constant ($K_p$) for this reaction at 1000 K.

Q.2 2 moles of A & 3 moles of B are mixed in 1 litre vessel and the reaction is carried at 400°C according to the equation; A + B ⇌ 2 C. The equilibrium constant of the reaction is 4. Find the number of moles of C at equilibrium.

Q.3 $2\,NOBr\,(g) \rightleftharpoons 2\,NO\,(g) + Br_2\,(g)$. If nitrosyl bromide (NOBr) is 33.33% dissociated at 25° C & a total pressure of 0.28 atm. Calculate $K_p$ for the dissociation at this temperature.

Q.4 At 90°C, the following equilibrium is established :
$H_2(g) + S(s) \rightleftharpoons H_2S(g)$ $\quad K_p = 6.8 \times 10^{-2}$
If 0.2 mol of hydrogen and 1.0 mol of sulphur are heated to 90°C in a 1.0 litre vessel, what will be the partial pressure of $H_2S$ at equilibrium?

Q.5 The equilibrium constant for the reaction is 9.40 at 900°C $S_2(g) + C(s) \rightleftharpoons CS_2(g)$ . Calculate the pressure of two gases at equilibrium, when 1.42 atm of $S_2$ and excess of C(s) come to equilibrium.

Q.6 A mixture of 2 moles of $CH_4$ & 34 gms of $H_2S$ was placed in an evacuated container, which was then heated to & maintained at 727° C. When equilibrium was established in the gaseous reaction $CH_4 + 2\,H_2S \rightleftharpoons CS_2 + 4\,H_2$ the total pressure in the container was 0.92 atm & the partial pressure of hydrogen was 0.2 atm. What was the volume of the container ?

Q.7 At 817° C, $K_p$ for the reaction between pure $CO_2$ and excess hot graphite to form 2 CO(g) is 10 atm.
(a) What is the analysis of the gases at equilibrium at 817°C & a total pressure of 4.0 atm ? What is the partial pressure of $CO_2$ at equilibrium ?
(b) At what total pressure will the gas mixture analyze 6%, $CO_2$ by volume ?

Q.8 The equilibrium mixture $SO_2 + NO_2 \rightleftharpoons SO_3 + NO$ was found to contain 0.6 mol of $SO_3$, 0.40 mol of NO, 0.8 mol of $SO_2$ & 0.1 mol of $NO_2$ in a 1L vessel. One mole of NO was then forced into the reaction vessel with V & T constant. Calculate the amounts of each gas in the new equilibrium mixture.

Q.9 For the reaction $N_2O_4 \rightleftharpoons 2NO_2$, equilibrium mixture contains $NO_2$ at P = 1.1 atm & $N_2O_4$ at P = 0.28 atm at 350 K. The volume of the container is doubled. Calculate the equilibrium pressures of the two gases when the system reaches new equilibrium.

Q.10 In the preceding problem, calculate the degree of dissociation, $\alpha$ at both pressures corresponding to mean molar masses of 65 & 76.667. Use data from the preceding problem.

Q.11 $PCl_5$ dissociates according to the reaction $PCl_5 \rightleftharpoons PCl_3(g) + Cl_2(g)$ . At 523 K, $K_p = 1.78$ atm. Find the density of the equilibrium mixture at a total pressure of 1 atm.

Q.12 The reaction $3/2H_2(g) + 1/2\,N_2(g) \rightleftharpoons NH_3(g)$ was carried out at T = 620 K & P = 10 atm with an initial mixture of $H_2 : N_2 = 3 : 1$, the mixture at equilibrium contained 7.35 % $NH_3$. Find $K_p$ and $K_c$.

Q.13 For the reaction $SO_2(g) + 1/2\ O_2(g) \rightleftharpoons SO_3(g)$ $\Delta H^\circ_{298} = -98.32$ kJ/mole, $\Delta S^\circ_{298} = -95.0$ J/K/mole . Find the $K_p$ for this reaction at 298 K.

Q.14 The following data for the equilibrium composition of the reaction

$$2Na(g) \rightleftharpoons Na_2(g)$$

at 1.013 MPa pressure and 1482.53 K have been obtained.

mass % Na (monomer gas) = 71.3

mass % $Na_2$ (dimer gas) = 28.7

Calculate the equilirium constant $K_p$.

Q.15 The degree of dissociation of HI at a particular temperature is 0.8 . Find the volume of 1.5M sodium thiosulphate solution required to react completely with the iodine present at equilibrium in acidic conditions, when 0.135 mol each of $H_2$ and $I_2$ are heated at 440 K in a closed vessel of capacity 2.0 L.

Q.16 A reaction system in equilibrium according to the equation $2\,SO_2 + O_2 \rightleftharpoons 2\,SO_3$ in 1 litre reaction vessel at a given temperature was found to contain 0.11 mol of $SO_2$, 0.12 mol of $SO_3$ and 0.05 mol of $O_2$. Another 1 litre reaction vessel contains 64 g of $SO_2$ at the same temperature. What mass of $O_2$ must be added to this vessel in order that at equilibrium half of $SO_2$ is oxidised to $SO_3$ ?

Q.17 A mixture of hydrogen & iodine in the mole ratio 1.5 : 1 is maintained at 450°C. After the attainment of equilibrium $H_2(g) + I_2(g) \rightleftharpoons 2\,HI(g)$, it is found on analysis that the mole ratio of $I_2$ to HI is 1 : 18. Calculate the equilibrium constant & the number of moles of each species present under equilibrium, if initially, 127 grams of iodine were taken.

Q.18 In a closed container nitrogen and hydrogen mixture initially in a mole ratio of 1:4 reached equilibrium. It is found that the half hydrogen is converted to ammonia. If the original pressure was 180 atm, what will be the partial pressure of ammonia at equilibrium. (There is no change in temperature)

Q.19 The equilibrium constant for the reaction $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$ is 7.3 at 450°C & 1atm pressure . The initial concentration of water gas [$CO + H_2$] & steam are 2 moles & 5 moles respectively. Find the number of moles of CO, $H_2$, $CO_2$ & $H_2O$ (vapour) at equilibrium.

Q.20 At 1200°C, the following equilibrium is established between chlorine atoms & molecule:

$$Cl_2(g) \rightleftharpoons 2Cl\,(g)$$

The composition of the equilibrium mixture may be determined by measuring the rate of effusion of the mixture through a pin hole. It is found that at 1200°C and 1 atm pressure the mixtureeffuses 1.16 times as fast as krypton effuses under the same condition. Calculate the equilibrium constant $K_c$.

Q.21 Two solids X and Y disssociate into gaseous products at a certain temperature as follows: $X(s) \rightleftharpoons A(g) + C(g)$, and $Y(s) \rightleftharpoons B(g) + C(g)$. At a given temperature, pressure over excess solid X is 40 mm and total pressure over solid Y is 60 mm. Calculate:

(a) the values of $K_p$ for two reactions (in mm)

(b) the ratio of moles of A and B in the vapour state over a mixture of X and Y.

(c) the total pressure of gases over a mixture of X and Y.

Q.22 $SO_3$ decomposes at a temperature of 1000 K and at a total pressure of 1.642 atm. At equilibrium, the density of mixture is found to be 1.28 g/l in a vessel of 90 literes. Find the degree of dissociation of $SO_3$ for $SO_3 \rightleftharpoons SO_2 + 1/2O_2$.

Q.23 Consider the equilibrium: P(g) + 2Q(g) ⇌ R(g). When the reaction is carried out at a certain temperature, the equilibrium conceentration of P and Q are 3M and 4M respectively. When the volume of the vessel is doubled and the equilibrium is allowed to be reestablished, the concentration of Q is found to be 3M. Find (A) $K_c$ (B) concentration of R at two equilibrium stages.

Q.24 When $PCl_5$ is heated, it dissociates into $PCl_3$ and $Cl_2$. The vapor density of the gaseous mixture at 200°C and 250°C is 70.2 and 57.9 respectively. Find the % dissociation of $PCl_5$ at 200°C and 250°C.

Q.25 The density of an equilibrium mixture of $N_2O_4$ and $NO_2$ at 101.32 $KP_a$ is 3.62g $dm^{-3}$ at 288 K and 1.84 g $dm^{-3}$ at 348K. What is the heat of the reaction for $N_2O_4 \rightleftharpoons 2NO_2$ (g) .

Q.26 Two solid compounds A & C dissociates into gaseous productat temperature as follows

$$A(s) \overset{\Delta}{\rightleftharpoons} B(g) + E(g)$$

$$C(s) \overset{\Delta}{\rightleftharpoons} D(g) + E(g)$$

At 20° C pressure over excess solid A is 50atm & that over excess solid C is 68atm. Find the total pressure of gases over the solid mixture.

Q.27 The equilibrium constant for the following reaction at 1395 K.

$2H_2O \rightleftharpoons 2H_2 + O_2 \qquad K_1 = 2.1 \times 10^{-13}$

$2CO_2 \rightleftharpoons 2CO + O_2 \qquad K_2 = 1.4 \times 10^{-12}$

Calculate the value of K for the reaction : $H_2 + CO_2 \rightleftharpoons CO + H_2O$

Q.28 A saturated solution of iodine in water contains 0.33g $I_2$ / L. More than this can dissolve in a KI solution because of the following equilibrium : $I_2(aq) + I^-(aq) \rightleftharpoons I_3^-(aq)$. A 0.10 M KI solution (0.10 M $I^-$) actually dissolves 12.5 g of iodine/L, most of which is converted to $I_3^-$. Assuming that the concentration of $I_2$ in all saturated solutions is the same, calculate the equilibrium constant for the above reaction. What is the effect of adding water to a clear saturated of $I_2$ in the KI solution ?

Q.29 The equilibrium p–Xyloquinone + methylene white ⇌ p–Xylohydroquinone + methylene blue may be studied convinently by observing the difference in color methylene white and methylene blue. One mmol of methylene blue was added to 1L of solution that was 0.24 M in p–Xylohydroquinone and 0.012 M in p–Xyloquinone. It was then found that 4% of the added methylene blue was reduced to methylene white. What is the equilibrium constant of the above reaction? The equation is balanced with one mole each of 4 substances.

Q.30 A mixture of $N_2$ & $H_2$ are in equilibrium at 600 K at a total pressure of 80 atm. If the initial ratio of $N_2$ and $H_2$ are 3:1 and at equilibrium $NH_3$ is 10% by volume. Calculate $K_P$ of reaction at given temperature.

Q.31 $\Delta G^\circ$ (298 K) for the reaction $1/2\ N_2 + 3/2\ H_2 \overset{K_1}{\rightleftharpoons} NH_3$ is $-16.5$ kJ $mol^{-1}$ . Find the equilibrium constant ($K_1$) at 25°C .What will be the equilibrium constants $K_2$ and $K_3$ for the following reactions:

$N_2 + 3H_2 \overset{K_2}{\rightleftharpoons} 2NH_3$

$NH_3 \overset{K_3}{\rightleftharpoons} 1/2\ N_2 + 3/2\ H_2$

Q.32 A certain gas A polymerizes to a small extent at a given temperature & pressure, $nA \rightleftharpoons A_n$ . Show that the gas obeys the approx. equation $\frac{PV}{RT} = \left[1 - \frac{(n-1)K_c}{V^{n-1}}\right]$ where $K_c = \frac{[A_n]}{[A]^n}$ & V is the volume of the conatiner. Assume that initially one mole of A was taken in the container.

Q.33 $10^{-3}$ mol of $CuSO_4.5H_2O$ is introduced in a 1.9 L vessel maintained at a constant temperature of 27°C containing moist air at relative humidity of 12.5%. What is the final molar composition of solid mixture? For $CuSO_4.5H_2O(s) \rightleftharpoons CuSO_4(s) + 5H_2O(g)$, $K_p(atm) = 10^{-10}$. Take vapor pressure of water at 27°C as 28 torrs.

Q.34 When 1 mol of A(g) is introduced in a closed 1L vessel maintained at constant temperature, the following equilibria are established.

$A(g) \rightleftharpoons B(g) + 2C(g)$ ; $K_{C_1} = ?$

$C(g) \rightleftharpoons 2D(g) + 3B(g)$ ; $K_{C_2} = ?$

The pressure at equilibrium is $\left(\frac{13}{6}\right)$ times the initial pressure.

Calculate $K_{C_1}$ & $K_{C_2}$ if $\frac{[C]_{eq}}{[A]_{eq}} = \frac{4}{9}$.

Q.35 When NO & $NO_2$ are mixed, the following equilibria are readily obtained;

$2NO_2 \rightleftharpoons N_2O_4$ $K_p = 6.8\ atm^{-1}$

$NO + NO_2 \rightleftharpoons N_2O_3$ $K_p = ?$

In an experiment when NO & $NO_2$ are mixed in the ratio of 1 : 2, the total final pressure was 5.05 atm & the partial pressure of $N_2O_4$ was 1.7 atm. Calculate

(a) the equilibrium partial pressure of NO.

(b) $K_p$ for $NO + NO_2 \rightleftharpoons N_2O_3$

Q.36 Solid $NH_4I$ on rapid heating in a closed vessel at 357°C develops a constant pressure of 275 mm Hg owing to partial decomposition of $NH_4I$ into $NH_3$ and HI but the pressure gradually increases further (when the excess solid residue remains in the vessel) owing to the dissociation of HI. Calculate the final pressure developed at equilibrium.

$NH_4I(s) \rightleftharpoons NH_3(g) + HI(g)$

$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$, $K_c = 0.065$ at 357°C

Q.37 Given are the following standard free energies of formation at 298K.

| | CO(g) | $CO_2(g)$ | $H_2O(g)$ | $H_2O(l)$ |
|---|---|---|---|---|
| $\Delta_rG°$ / kJ $mol^{-1}$ | –137.17 | –394.36 | –228.57 | –237.13 |

(a) Find $\Delta_rG°$ and the standard equilibrium constant $K_p^0$ at 298 K for the reaction

$CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$

(b) If CO, $CO_2$ and $H_2$ are mixed so that the partial pressure of each is 101.325 kPa and the mixture is brought into contact with excess of liquid water, what will be the partial pressure of each gas when equilibrium is attained at 298K. The volume available to the gases is constant.

Q.38 For the reaction

$C_2H_6(g) \rightleftharpoons C_2H_4(g) + H_2(g)$

$K_p^0$ is 0.05 and $\Delta_rG°$ is 22.384 kJ$mol^{-1}$ at 900 K. If an initial mixture comprising 20 mol of $C_2H_6$ and 80 mol of $N_2$(inert) is passed over a dehydrogenation catalyst at 900K, what is the equilibrium percentage composition of the effluent gas mixture? The total pressure is kept at 0.5 bar. Given : $\Delta_rS° = 135.143$ J $K^{-1}$ $mol^{-1}$ at 300K. Calculate $\Delta_rG°$ at 300K. (Assume $\Delta_rC_p = 0$)

Q.39(a) The equilibrium $H_2(g) + CO_2(g) \Leftrightarrow H_2O(g) + CO(g)$ is established in an evacuated vessel at 723 K starting with 0.1 mole of $H_2$ & 0.2 mole of $CO_2$. If the equilibrium mixture contains 10 mole per cent of water vapour, calculate $K_p$, given that the equilibrium pressure is 0.5 atm. Calculate the partial pressures of the component species & the volume of the container.

(b) If now, into the flask (mentioned in the preceding problem), solid CoO & solid Co are introduced two new equilibria are established.

$$CoO(s) + H_2(g) \text{ l } Co(s) + H_2O(g) \text{ ; } CoO(s) + CO(g) \text{ l } Co(s) + CO_2(g)$$

The new equilibrium mixture contains 30 mole precent of water vapour. Calculate the equilibrium constants for the new equilibria.

Q.40 Some iodine is dissolved in an aqueous solution of KI of concentration 0.102 mole/1, and the solution is then shaken with equal volume of $CCl_4$ until equilibrium is reached (at 15°C). The total amount of iodine (present as $I_3^-$ (aq) or as $I_2$ (aq) ) at equilibrium is found to be 0.048 mol/1 in the aqueous layer and 0.085 mol/1 in the $CCl_4$ layer. The distribution coefficient of iodine between $CCl_4$ and water is 85. Calculate the equilibrium constant at $15^0$C for the reaction:

$$I_3^- (aq) \text{ l } I_2 (aq) + I^- (aq)$$

## *EXERCISE III*

Q.1 Consider following reactions in equilibrium with equilibrium concentration 0.01 M of every species
(I) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$ (II) $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$
(III) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
Extent of the reactions taking place is:
(A) I > II > III (B) I < II < III (C) II < III < I (D) III < I < II

Q.2 For the reaction $3A(g) + B(g) \rightleftharpoons 2C(g)$ at a given temperature, $K_c = 9.0$. What must be the volume of the flask, if a mixture of 2.0 mol each of A, B and C exist in equilibrium?
(A) 6L (B) 9L (C) 36 L (D) None of these

Q.3 Sulfide ion in alkaline solution reacts with solid sulfur to form polysulfide ions having formulas $S_2^{2-}$, $S_3^{2-}$, $S_4^{2-}$ and so on. The equilibrium constant for the formation of $S_2^{2-}$ is 12 ( $K_1$) & for the formation of $S_3^{2-}$ is 132 ($K_2$), both from S and $S^{2-}$. What is the equilibrium constant for the formation of $S_3^{2-}$ from $S_2^{2-}$ and S?
(A) 11 (B) 12 (C) 132 (D) None of these

Q.4 For the following gases equilibrium. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
$K_p$ is found to be equal to $K_c$. This is attained when
(A) 0°C (B) 273 K (C) 1 K (D) 12.19 K

Q.5 1 mole $N_2$ and 3 mol $H_2$ are placed in a closed container at a pressure of 4 atm. The pressure falls to 3 atm at the same temperature when the following equilibrium is attained.
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$. The equilibrium constant $K_P$ for dissociation of $NH_3$ is:

(A) $\frac{1}{0.5} \times (1.5)^3$ atm$^{-2}$ (B) $0.5 \times (1.5)^3$ atm$^2$ (C) $\frac{0.5 \times (1.5)^3}{3 \times 3}$ atm$^2$ (D) $\frac{3 \times 3}{0.5 \times (1.5)^3}$ atm$^{-2}$

Q.6 One mole of $N_2O_4(g)$ at 300 K is left in a closed container under one atm. It is heated to 600 K when 20 % by mass of $N_2O_4(g)$ decomposes to $NO_2(g)$. The resultant pressure is :
(A) 1.2 atm (B) 2.4 atm (C) 2.0 atm (D) 1.0 atm

Q.7 For the reaction : $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$, the degree of dissociated ($\alpha$) of HI(g) is related to equilibrium constant $K_P$ by the expression

(A) $\frac{1 + 2\sqrt{K_p}}{2}$ (B) $\sqrt{\frac{1 + 2K_p}{2}}$ (C) $\sqrt{\frac{2K_p}{1 + 2K_p}}$ (D) $\frac{2\sqrt{K_p}}{1 + 2\sqrt{K_p}}$

Q.8 The vapour density of $N_2O_4$ at a certain temperature is 30. What is the % dissociation of $N_2O_4$ at this temperature?
(A) 53.3% (B) 106.6% (C) 26.7% (D) None

Q.9 For the reaction $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, the forward reaction at constant temperature is favoured by
(A) introducing an inert gas at constant volume
(B) introducing chlorine gas at constant volume
(C) introducing an inert gas at constant pressure
(D) increasing the volume of the container
(E) introducing $PCl_5$ at constant volume.

Q.10 When $N_2O_5$ is heated at temp. T, it dissociates as $N_2O_5 \rightleftharpoons N_2O_3 + O_2$, $K_c = 2.5$. At the same time $N_2O_3$ also decomposes as : $N_2O_3 \rightleftharpoons N_2O + O_2$. If initially 4.0 moles of $N_2O_5$ are taken in 1.0 litre flask and allowed to attain equilibrium, concentration of $O_2$ was formed to be 2.5 M. Equilibrium concentration of $N_2O$ is

(A) 1.0 (B) 1.5 (C) 2.166 (D) 0.334

Q.11 Densities of diamond and graphite are 3.5 and 2.3 gm/mL.

C (diamond) $\rightleftharpoons$ C (graphite) $\Delta_r H = -1.9$ kJ/mole

favourable conditions for formation of diamond are

(A) high pressure and low temperature (B) low pressure and high temperature

(C) high pressure and high temperature (D) low pressure and low temperature

Q.12 When $NaNO_3$ is heated in a closed vessel, oxygen is liberated and $NaNO_2$ is left behind. At equilibrium

(A) addition of $NaNO_2$ favours reverse reaction

(B) addition of $NaNO_3$ favours forward reaction

(C) increasing temperature favours forward reaction

(D) increasing pressure favours reverse reaction

Q.13 The equilibrium $SO_2Cl_2(g) \rightleftharpoons SO_2(g) + Cl_2(g)$ is attained at 25°C in a closed rigid container and an inert gas, helium is introduced. Which of the following statements is/are correct.

(A) concentrations of $SO_2$, $Cl_2$ and $SO_2Cl_2$ do not change

(B) more chlorine is formed

(C) concentration of $SO_2$ is reduced

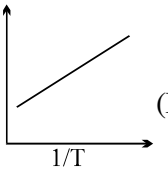
(D) more $SO_2Cl_2$ is formed

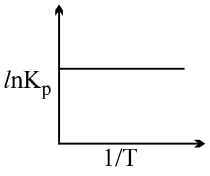
Q.14 For the gas phase reaction, $C_2H_4 + H_2 \rightleftharpoons C_2H_6$ ($\Delta H = -32.7$ kcal), carried out in a closed vessel, the equilibrium concentration of $C_2H_4$ can be increased by

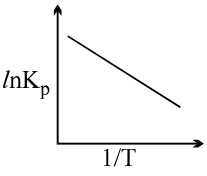
(A) increasing the temperature (B) decreasing the pressure

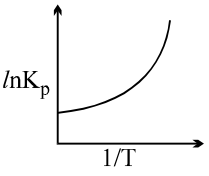
(C) removing some $H_2$ (D) adding some $C_2H_6$

Q.15 An exothermic reaction is represented by the graph :

(A) 


(B) 


(C) 


(D) 


Q.16 The correct relationship between free energy change in a reaction and the corresponding equilibrium constant K is

(A) $-\Delta G^\circ = RT \ln K$ (B) $\Delta G = RT \ln K$ (C) $-\Delta G = RT \ln K$ (D) $\Delta G^\circ = RT \ln K$

Q.17 The value of $\Delta G^\circ_f$ of gaseous mercury is 31 K J/mole. At what total external pressure mercury start boiling at 25°C. [R = 8.3]

(A) $10^{-5.44}$ (B) $10^{-12.5}$ (C) $10^{-6.52}$ (D) $10^{-3.12}$

Q.18 What is $\Delta_r G$ (KJ/mole) for synthesis of ammonia at 298 K at following sets of partial pressure:
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ ; $\Delta_r G^\circ = -33$ KJ/mole. [Take R = 8.3 J/K mole, log2 = 0.3; log3 = 0.48]

| Gas | $N_2$ | $H_2$ | $NH_3$ |
|---|---|---|---|
| Pressure (atm) | 1 | 3 | 0.02 |

(A) + 6.5 (B) – 6.5 (C) + 60.5 (D) – 60.5

Q.19 In a 7.0 L evacuated chamber, 0.50 mol $H_2$ and 0.50 mol $I_2$ react at 427°C .
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ . At the given temperature, $K_C$=49 for the reaction.

(i) What is the value of $K_p$ ?
(A) 7 (B) 49 (C) 24.5 (D) None

(ii) What is the total pressure (atm) in the chamber?
(A) 83.14 (B) 831.4 (C) 8.21 (D) None

(iii) How many moles of the iodine remain unreacted at equilibrium?
(A) 0.388 (B) 0.112 (C) 0.25 (D) 0.125

(iv) What is the partial pressure (atm) of HI in the equilibrium mixture?
(A) 6.385 (B) 12.77 (C) 40.768 (D) 646.58

Q.20 Equilibrium constants are given (in atm) for the following reactions at 0° C:

| | |
|---|---|
| $SrCl_2 \cdot 6H_2O(s) \rightleftharpoons SrCl_2 \cdot 2H_2O(s) + 4H_2O(g)$ | $K_p = 5 \times 10^{-12}$ |
| $Na_2HPO_4 \cdot 12 H_2O(s) \rightleftharpoons Na_2HPO_4 \cdot 7 H_2O(s) + 5H_2O(g)$ | $K_p = 2.43 \times 10^{-13}$ |
| $Na_2SO_4 \cdot 10 H_2O(s) \rightleftharpoons Na_2SO_4(s) + 10 H_2O(g)$ | $K_p = 1.024 \times 10^{-27}$ |

The vapor pressure of water at 0°C is 4.56 torr.

(i) Which is the most effective drying agent at 0°C?
(A) $SrCl_2 \cdot 2H_2O$ (B) $Na_2HPO_4 \cdot 7 H_2O$ (C) $Na_2SO_4$ (D) all equally

(ii) At what relative humidities will $Na_2SO_4 \cdot 10 H_2O$ be efflorescent when exposed to air at 0°C?
(A) above 33.33% (B) below 33.33 % (C) above 66.66% (D) below 66.66%

(iii) At what relative humidities will $Na_2SO_4$ be deliquescent (i.e. absorb moisture) when exposed to the air at 0°C?
(A) above 33.33% (B) below 33.33 % (C) above 66.66% (D) below 66.66%

## *EXERCISE IV*

Q.1 A sample of air consisting of $N_2$ and $O_2$ was heated to 2500K until the equilibrium $N_2(g) + O_2(g) \rightleftharpoons 2NO$ was established with an equilibrium constant $K_c = 2.1 \times 10^{-3}$. At equilibrium, the mol% of NO was 1.8. Estimate the initial composition of air in mol fraction of $N_2$ and $O_2$.

**[JEE 1997]**

Q.2 For the reaction $CO(g) + H_2O \rightleftharpoons CO_2(g) + H_2(g)$ at a given temperature the equilibrium amount of $CO_2(g)$ can be increased by :

(A) adding a suitable catalyst (B) adding an inert gas

(C) decreasing the volume of the container (D) increasing the amount of CO(g) .

**[JEE 1998]**

Q.3 For the reaction, $N_2O_5(g) = 2NO_2(g) + 0.5\ O_2(g)$, calculate the mole fraction of $N_2O_5(g)$ decomposed at a constant volume & temperature, if the initial pressure is 600 mm Hg & the pressure at any time is 960 mm Hg. Assume ideal gas behaviour . **[JEE 1998]**

Q.4 The degree of dissociation is 0.4 at 400K & 1.0 atm for the gasoeus reaction $PCl_5 \rightleftharpoons PCl_3 + Cl_2(g)$. Assuming ideal behaviour of all gases. Calculate the density of equilibrium mixture at 400K & 1.0 atm pressure. **[JEE 1999]**

Q.5 When 3.06g of solid $NH_4HS$ is introduced into a two litre evacuated flask at 27°C, 30% of the solid decomposes into gaseous ammonia and hydrogen sulphide.

(i) Calculate $K_C$ & $K_P$ for the reaction at 27°C.

(ii) What would happen to the equilibrium when more solid $NH_4HS$ is introduced into the flask?

**[JEE 2000]**

Q.6 When 1-pentyne (A) is treated with 4N alcoholic KOH at 175°C, it is converted slowly into an equilibrium mixture of 1.3% 1-pentyne (A), 95.2% 2–pentyne (B) & 3.5% of 1, 2,–pentadiene (C). The equilibrium was maintained at $175^0$C. Calculate $\Delta G°$ for the following equilibria.

$B = A \qquad \Delta G_1^0 = ?$

$B = C \qquad \Delta G_2^0 = ?$

From the calculated value of $\Delta G_1^0$ & $\Delta G_2^0$ indicate the order of stability of A, B & C. Write a reasonable reaction mechanism sharing all intermediate leading to A, B & C. **[JEE 2001]**

Q.7 $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

This reaction is carried out at 298 K and 20 bar. 5 mol each of $N_2O_4$ and $NO_2$ are taken initially.

Given: $\Delta G^o_{N_2O_4} = 100$ kJ mol$^{-1}$; $\Delta G^o_{NO_2} = 50$ kJ mol$^{-1}$

(i) Find $\Delta G$ for reaction at 298 K under given condition.

(ii) Find the direction in which the reaction proceeds to achieve equilibrium. **[JEE 2004]**

Q.8 $N_2 + 3H_2 \rightleftharpoons 2NH_3$

Which is correct statement if $N_2$ is added at equilibrium condition?

(A) The equilibrium will shift to forward direction because according to II law of thermodynamics the entropy must increases in the direction of spontaneous reaction.

(B) The condition for equilibrium is $G_{N_2} + 3G_{H_2} = 2G_{NH_3}$ where G is Gibbs free energy per mole of the gaseous species measured at that partial pressure. The condition of equilibrium is unaffected by the use of catalyst, which increases the rate of both the forward and backward reactions to the same extent.

(C) The catalyst will increase the rate of forward reaction by $\alpha$ and that of backward reaction by $\beta$.

(D) Catalyst will not alter the rate of either of the reaction. **[JEE 2006]**

# ***ANSWER KEY***
## ***EXERCISE I***

Q.1 (a) 25, shifts left, (b) 0.22, shifts right, (c) ∞, shifts left, (d) 1, shifts right, (e) 0, shift right, (f) 4, shifts left

Q.2 (a) $K = [Ag^+][Cl^-]$ is less than 1. AgCl is insoluble thus the concentration of ions are much less than 1 M
(b) $K = 1/[Pb^{2+}][Cl^-]^2$ is greater than one because $PbCl_2$ is insoluble and formation of the solid will reduce the concentration of ions to a low level

Q.4 K about 10 Q.6p (a) incomplete (b) almost complete Q.7 c Q.8 $\sim 9\times 10^{-32}$ mol/L

Q.9 The reaction is not an equilibrium because $Q_c > K_c$. The reaction will proceed from right to left to reach equilibrium

Q.11 $5.9 \times 10^{-3}$ M Q.12 [NO] = 0.056 M, $[N_2] = [O_2] = 1.37$ M

Q.13 $[PCl_3] = [Cl_2] = 0.071$ M, $[PCl_5] = 0.089$

Q.14 $P_{ClF} = P_{F_2} = 0.389$ atm, $P_{ClF_3} = 1.08$ atm

Q.15 $K_p = 0.4$, a ~ 0.1 Q.16 50%

Q.17 (a) $6.667 \times 10^{-3}$ mol $L^{-1}$; (b) n ($N_2O_4$) = 0.374 mol; n ($NO_2$) = 0.052 mol ; (c) 10.49 atm (d) 6.44 %

Q.18 0.97 atm Q.19 $K_P = 1.3 \times 10^{-3}$ $atm^{-2}$

Q.20 $K_p = 2.5$ atm, P = 15 atm Q.21 53.33%

Q.22 K = 4 Q.23 31/27 Q.24 22.4 mg

Q.25 $P_{H_2O} = 5 \times 10^{-15}$ atm Q.26 0.821 atm

Q.27 add $N_2$, add $H_2$, increase the pressure , heat the reaction

Q.28 (a) shift right, shift left, (b) shift right, no effect, (c) shift left, shift left, (d) shift left, shift right

Q.29 (a) $K = [CH_3OH]/[H_2]^2[CO]$ ,
(b) 1. $[H_2]$ increase, [CO] decrease, $[CH_3OH]$ increase ; 2. $[H_2]$ increase, [CO] decrease, $[CH_3OH]$ decrease ; 3. $[H_2]$ increase, [CO] increase, $[CH_3OH]$ increase ; 4. $[H_2]$ increase, [CO] increase, $[CH_3OH]$ increase ; 5. $[H_2]$ increase, [CO] increase, $[CH_3OH]$ decrease ; 6. no change

Q.30 (a) $K = [CO][H_2]/[H_2O]$ ;
(b) in each of the following cases the mass of carbon will change, but its concentration (activity) will not change. 1. $[H_2O]$ no change, [CO] no change, $[H_2]$ no change ; 2. $[H_2O]$ decrease, [CO] decrease, $[H_2]$ decrease ; 3. $[H_2O]$ increase, [CO] increase, $[H_2]$ decrease; 4. $[H_2O]$ increase, [CO] increase, $[H_2]$ increase ; 5. $[H_2O]$ decrease , [CO] increase , $[H_2]$ increase

Q.31 b

Q.32 Add NaCl or some other salt that produces $Cl^-$ in the solution. Cool the solution.

Q.33 a

Q.34 $k_f [A][B] = k_r [C]$ ; $\frac{k_f}{k_r} = \frac{[C]}{[A][B]} = k_c$ Q.36 216

Q.38 (i) 2; (ii) 1.2 mol/L; (iii) 0.1 moles/hr

Q.39 $k_r$ increase more than $k_f$, this means that $E_a$ (reverse) is greater than $E_a$ (forward). The reaction is exothermic when $E_a$ (reverse) > $E_a$ (forward).

Q.43 (a) –9.574 J/mol, (b) $A = 10^{10}$, (c) $9.96 \times 10^9$, (d) $9.98 \times 10^9$

Q.44 16.06 kJ Q.45 –810 J/mol ; – 5872 J/mol and 41.3 kJ / mol

Q.46 $1.3 \times 10^8$ Q.47 0.058

Q.48 29.0 Q.49 $K_p = 0.0313$ atm, $K_c = 1.28 \times 10^{-3}$

Q.50 (a) $K_c = \frac{[CO_2]^3}{[CO]^3}$ , $K_p = \frac{(P_{CO_2})^3}{(P_{CO})^3}$ , (b) $K_c = \frac{1}{[O_2]^3}$ , $K_p = \frac{1}{(P_{O_2})^3}$ , (c) $K_c = [SO_3]$, $K_p = P_{SO_3}$

$K_c = [Ba^{2+}]\,[SO_4^{2-}]$

Q.51 $K_c = 1.51k$ $K_p = 49.6$ Q.52 $1.5 \times 10^{-6}$ M
Q.53 $[CO] = [H_2] = 0.18$ M ; $[H_2O] = 1.02$ M
Q.54 (a) $K_c = 0.573$ and $K_p = 23.5$; (b) to the right, $[PCl_5] = 0.365$ M; $[PCl_3] = 0.285$ M, ; $[Cl_2] = 0.735$ M
Q.56 –1.005 kJ/ mol Q.57 $\Delta G° = 0$ ; $K = 1$
Q.58 $\Delta H° = 9.04$ kJ/mol; $\Delta S° = -8.64$ J/mol$^{-1}$K$^{-1}$

## ***PROFICIENCY TEST***

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | C | 2. | high | 3. | $\frac{1}{10}$ | 4. | $\frac{1}{\sqrt{K}}$ |
| 5. | exothermic | 6. | $K_P = K_C (RT)$ | 7. | backward | 8. | $\Delta G° = -RT \ln K$ |
| 9. | $\log \frac{K_2}{K_1} = \frac{\Delta H°}{2.303R}\left[\frac{T_2 - T_1}{T_2 T_1}\right]$ | | | 10. | zero | 11. | $mol^2L^{-2}$ |
| 12. | temperature | 13. | Guldberg and Waage | | | 14. | decreases |
| 15. | backward | 16. | high | 17. | $K_P = K_C (RT)^{\Delta n}$ | | |
| 18. | decreases | 19. | same amount of | | | 20. | larger value of |
| 21. | T | 22. | T | 23. | F | 24. | F |
| 25. | T | 26. | F | 27. | T | 28. | T |
| 29. | F | 30. | T | 31. | T | 32. | T |
| 33. | F | 34. | F | 35. | F | 36. | F |
| 37. | T | 38. | F | 39. | T | 40. | F |

## ***EXERCISE II***

Q.1 $K_p(atm) = 1.13$
Q.2 2.4 mole Q.3 $K_p = 0.01$ atm Q.4 0.379 atm
Q.5 $P_{CS_2} = 1.284$ atm, $P_{S2} = 0.1365$ atm
Q.6 300L
Q.7 (i) $x_{co} = 0.765$, $x_{co_2} = 0.235$; $p(CO_2) = 0.938$ atm (ii) $P_{Total} = 0.68$ atm
Q.8 ($K_c = 3$), $n_{SO_2} = 0.92$, $n_{SO_3} = 0.48$, $n_{NO} = 1.28$, $n_{NO_2} = 0.22$
Q.9 $P_{NO_2} = 0.64$ atm, $P_{N_2O_4} = 0.095$ atm Q.10 $\alpha = 0.415$ and 0.2
Q.11 2.7 g / lit Q.12 $K_c = 1.337$, $K_p = 0.0263$ Q.13 $K_p = 1.862 \times 10^{12}$ atm$^{-1/2}$
Q.14 $p_{Na} = 0.843$ M Pa; $p_{Na_2} = 0.170$ M Pa; $k_p = 0.239$ Q.15 V = 144 mL
Q.16 9.34 g Q.17 $K_c = 54$, $n_{HI} = 0.9$ mol, $n_{I2} = 0.05$ mol, $n_{H2} = 0.3$ mol
Q.18 48 atm Q.19 $n_{CO2} = 0.938$, $n_{H2} = 1.938$, $n_{CO} = 0.062$, $n_{H2Og} = 4.062$
Q.20 $6.71 \times 10^{-4}$ Q.21 (a) 400mm$^2$, 900mm$^2$ (b) 4: 9, (c) 72.15 mm Hg
Q.22 $\alpha = 0.5$
Q.23 $K_c = 1/12$, [R] = 4 (initial), = 1.5 (final) Q.24 dissociation = 48.5%, 80.05%
Q.25 $\Delta_r H = 75.5$ kJ mol$^{-1}$ Q.26 Total pressure = 84.34 atm
Q.27 K = 2.58 Q.28 K=707.2, backward reaction is favoured
Q.29 $K_c = 480$ Q.30 $1.32 \times 10^{-3}$
Q.31 $K_A = 779.4$, $K_B = 6.074 \times 10^5$; $K_c = 1.283 \times 10^{-3}$
Q.32 To be proved Q.33 $CuSO_4.5H_2O = 9.2 \times 10^{-4}$mol, $CuSO_4 = 8 \times 10^{-5}$ moles
Q.34 $k_{C_1} = 0.111$; $k_{C_2} = 0.14$ Q.35 (a) 1.05 atm, (b) 3.43 atm$^{-1}$ Q.36 314.1 atm

Q.37 $p_{CO_2}$ = 202.65 kPa; $p_{H_2O}$ = 3.16 kPa; $p_{CO}$ = 0.124 kPa

Q.38 103.47 kJ/mol

Q.39 (a) $K_p = 7.563 \times 10^{-2}$, v = 35.62, $p(H_2O) = p(CO) = 0.05$atm, $p(H_2) = 0.1167$atm, $p(CO_2) = 0.2833$atm
(b) $K_1 = 9$, $K_2 = 119$

Q.40 $K = 1.17 \times 10^{-3}$

## ***EXERCISE III***

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Q.1 | B | Q.2 | A | Q.3 | A | Q.4 | D |
| Q.5 | B | Q.6 | B | Q.7 | D | Q.8 | A |
| Q.9 | C,D,E | Q.10 | D | Q.11 | C | Q.12 | C,D |
| Q.13 | A | Q.14 | A,B,C,D | Q.15 | A | Q.16 | A |
| Q.17 | A | Q.18 | D | Q.19 | (i) B ,(ii) C,(iii) B,(iv) A | | |

Q.20 (i) A, (ii) B, (iii) A

## ***EXERCISE IV***

Q.1 $X_{N2} = 0.79$, $X_{O2} = 0.21$ Q.2 D

Q.3 Fraction decomposed = 0.4

Q.4 4.54 g $dm^{-3}$ Q.5 (i) $k_c = 8.1 \times 10^{-5}$ $mol^2 L^2$ ; $k_p = 4.91 \times 10^{-2}$ $atm^2$ (ii) Noeffect;

Q.6 15991 J $mol^{-1}$ , 12304 J $mol^{-1}$ ; B > C > A

Q.7 (i) $5.705 \times 10^3$ J $mol^{-1}$
(ii) Since initial Gibbs free energy change of the reaction is positive, so the reverse reaction will take place

Q.8 B

# STUDY PACKAGE

# Target: IIT-JEE (Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 8. Ionic Equilibrium

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

**4. Exercise III**

**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

# *THE KEY*

## Fundamentals of Acids, Bases & Ionic Equilibrium

**Acids & Bases**

When dissolved in water, acids release $H^+$ ions, base release $OH^-$ ions.

**Arrhenius Theory**

When dissolved in water, the substances which release

(i) $H^+$ ions are called acids (ii) $OH^-$ ions are called bases

**Bronsted & Lowry Concept**

Acids are proton donors, bases are proton acceptors

Note that as per this definition, water is not necessarily the solvent.

When a substance is dissolved in water, it is said to react with water e.g.

$HCl + H_2O \rightarrow H_3O^+ + Cl^-$ ; HCl donates $H^+$ to water, hence acid.

$NH_3 + H_2O \rightarrow NH_4^+ + OH^-$ ; $NH_3$ takes $H^+$ from water, hence base.

For the backward reaction, $NH_4^+$ donates $H^+$, hence it is an acid; $OH^-$ accepts $H^+$, hence it is base.

$NH_3$ (base) & $NH_4^+$ (acid) from conjugate acid base pair.

**Conjugate acid and bases**

To get *conjugate acid* of a given species add $H^+$ to it. e.g. conjugate acid of $N_2H_4$ is $N_2H_5^+$.

To get *conjugate base* of any species subtract $H^+$ from it. e.g. Conjugate base of $NH_3$ is $NH_2^-$.

***Note:*** Although $Cl^-$ is conjugate base of HCl, it is not a base as an independent species. In fact, anions of all strong acid like $Cl^-$, $NO_3^-$, $ClO_4^-$ etc. are neutral anions. Same is true for cations of strong bases like $K^+$, $Na^+$, $Ba^{++}$ etc. When they are dissolved in water, they do not react with water (i.e. they do not undergo hydrolysis) and *these ions* do not cause any change in pH of water (others like $CN^-$ do).

**Some examples of :**

Basic Anions : $CH_3COO^-$, $OH^-$, $CN^-$ (Conjugate bases of weak acids)

Acid Anions: $HSO_3^-$, $HS^-$ etc. Note that these ions are *amphoteric*, i.e. they can behave both as an acid and as a base. e.g. for $H_2PO_4^-$ :

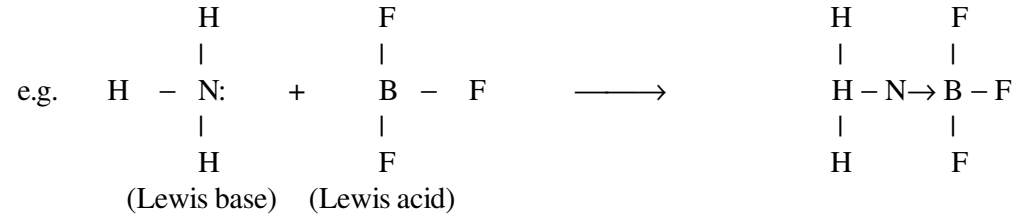
$HS^- + H_2O \rightleftharpoons S^{2-} + H_3O^+$ (functioning as an acid)

$HS^- + H_2O \rightleftharpoons H_2S + OH^-$ (functioning as a base)

Acid Cations : $NH_4^+$, $H_3O^+$ etc.(Conjugate acids of weak bases)

*Note :* Acid anions are rare.

**Lewis Concept :** Acids are substances which accept a pair of electrons to form a coordinate bond and bases are the substances which donate a pair of electrons to form a coordinate bond.



(Lewis base) (Lewis acid)

***Important :*** $Ca + S \rightarrow Ca^{2+} + S^{2-}$ is not a Lewis acid–base reaction since dative bond is not formed.

**Lewis Acids :** As per Lewis concept, following species can acts as Lewis Acids :

(i) Molecules in which central atom has *incomplete octet.* (e.g. $BF_3$, $AlCl_3$ etc.)

(ii) Molecules which have a central atom with empty d– orbitals (e.g. $SiX_4$, $GeX_4$, $PX_3$, $TiCl_4$ etc.)

(iii) *Simple Cations:* Though all cations can be expected to be Lewis acids, $Na^+$, $Ca^{++}$, $K^+$ etc. show no tendency to accept electrons. However $H^+$, $Ag^+$ etc. act as Lewis acids.

(iv) Molecules having multiple bond between atoms of dissimilar electronegativity.

e.g. $CO_2$, $SO_2$, $SO_3$ $\longrightarrow$ $(O=C=O + OH^- \longrightarrow \underset{\text{Lewis acid}}{} \quad \underset{\text{Lewis base}}{} \; \overset{-}{O}-C=O \text{ or } HCO_3^-)$

$\overset{|}{OH}$

**Lewis bases** are typically :

(i) Neutral species having at least one lone pair of electrons.

e.g. $\overset{\bullet\bullet}{N}H_2-\overset{\bullet\bullet}{N}H_2$, $R-\overset{\bullet\bullet}{\underset{\bullet\bullet}{O}}-H$

(ii) Negatively charged species (anions). e.g. $CN^-$, $OH^-$, $Cl^-$ etc.

**pH and pOH** $pH = -\log_{10}[H_3O^+]$, $pOH = -\log_{10}[OH^-]$

*Note :* * pH of very dilute (~ $10^{-8}$M or Lower) acids (or bases) is nearly 7 (not simply –log[acid] etc. due to ionization of water.

* pH of strong acids with concentration > 1M is never negative. It is zero only.

* At 25°C, if pH = 7, then solution is neutral, pH > 7 than solution is basic.

**Autoprotolysis of water (or any solvent)**

Autoprotolysis (or self–ionization) constant ($K_w$) = $[H_3O^+]\,[OH^-]$

Hence, $pH + pOH = pK_w$ at all temperatures

**Condition of neutrality** $[H_3O^+] = [OH^-]$ (for water as solvent)

At 25°C, $K_W = 10^{-14}$. $K_W$ increases with increase in temperature. Accordingly, the neutral point of water (pH = 7 at 25°C) also shifts to a value lower than 7 with increase in temperature.

***Important*:** $K_W = 10^{-14}$ is a value at (i) 25°C (ii) for water only. If the temperature changes or if some other solvent is used, autoprotolysis constant will not be same.

**Ionisation Constant**

* For dissociation of weak acids (eg. HCN), $HCN + H_2O \rightleftharpoons H_3O^+ + CN^-$ the equilibrium constant expression is written as $K_a = \dfrac{[H_3O^+][CN^-]}{[HCN]}$

* For the Polyprotic acids (e.g. $H_3PO_4$), sucessive ionisation constants are denoted by $K_1$, $K_2$, $K_3$ etc. For $H_3PO_4$,

$K_1 = \dfrac{[H_3O^+][H_2PO_4^-]}{[H_3PO_4]}$ $\qquad$ $K_2 = \dfrac{[H_3O^+][HPO_4^{2-}]}{[H_3PO_4^-]}$ $\qquad$ $K_3 = \dfrac{[H_3O^+][PO_4^{3-}]}{[HPO_4^{2-}]}$

Similarly, $K_b$ denotes basic dissociation constant for a base.

Also, $pK_a = -\log_{10}K_a$, $pK_b = -\log_{10}K_b$

***Some Important Results:*** $[H^+]$ concentration of

**<u>Case (i)</u> A weak acid in water**

(a) if $\alpha = \sqrt{\dfrac{K_a}{C}}$ is < 0.1, then $[H^+] \approx \sqrt{K_a c_0}$ .

(b) **General Expression :** $[H^+] = 0.5(-K_a + \sqrt{K_a^2 + 4K_a c_o})$

Similarly for a weak base, substitute $[OH^-]$ and $K_b$ instead of $[H^+]$ and $K_a$ respectively in these expressions.

**<u>Case (ii)</u>**(a) **A weak acid and a strong acid** $[H^+]$ is entirely due to dissociation of strong acid

(b) **A weak base and a strong base** $[H^+]$ is entirely due to dissociation of strong base
Neglect the contribution of weak acid/base usually.
***Condition for neglecting :*** If $c_0$ = concentration of strong acid, $c_1$ = concentration of weak acid then neglect the contribution of weak acid if $K_a \leq 0.01\ c_0^2/c_1$

**Case (iii) Two (or more) weak acids**

Proceed by the general method of applying two conditions
(i) of electroneutrality (ii) of equilibria.
The accurate treatement yields a cubic equation. Assuming that acids dissociate to a negligible extent [ i.e. $c_0 - x \approx c_0$] $[H^+] = (K_1c_1 + K_2c_2 + ... + K_w)^{1/2}$

**Case (iv) When dissociation of water becomes significant:**

Dissociation of water contributes significantly to $[H^+]$ or $[OH^-]$ only when for
*(i)* ***strong acids (or bases) :*** $10^{-8}M < c_0 < 10^{-6}M$. Neglecting ionisation of water at $10^{-6}M$ causes 1% error (approvable). Below $10^{-8}M$, contribution of acid (or base) can be neglected and pH can be taken to be practically 7.
***Weak acids (or bases) :*** When $K_ac_0 < 10^{-12}$, then consider dissociation of water as well.

**HYDROLYSIS**

* ***Salts of strong acids and strong bases*** do not undergo hydrolysis.
* ***Salts of a strong acids and weak bases*** give an acidic solution. e.g. $NH_4Cl$ when dissolved, it dissociates to give $NH_4^+$ ions and $NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$.
$K_h = [NH_3][H_3O^+] / [NH_4^+] = K_w/K_b$ of conjugate base of $NH_4^+$

**Important!** In general : $K_a$(of an acid)x$K_b$(of its conjugate base) = $K_w$

If the degree of hydrolysis(h) is small (<<1), $h = \sqrt{K_hc_0}$ .

Otherwise $h = \dfrac{-K_h + \sqrt{K_h^2 + 4K_hc_0}}{2c_0}$, $[H^+] = c_0h$

* ***Salts of strong base and weak acid*** give a basic solution (pH>7) when dissolved in water, e.g. $NaCN, CN^- + H_2O \rightleftharpoons HCN + OH^-$ $[OH^-] = c_0h$, $h = \sqrt{K_hc_0}$

* ***Salts of weak base and weak acid***

Assuming degree of hydrolysis to be same for the both the ions,
$K_h = K_w / (K_a.K_b)$, $[H^+] = [K_a K_w/K_b]^{1/2}$
***Note:*** Exact treatment of this case is difficult to solve. So use this assumption in general cases.
Also, degree of anion or cation will be much higher in the case of a salt of weak acid and weak base. This is because each of them gets hydrolysed, producing $H^+$ and $OH^-$ ions. These ions combine to form water and the hydrolysis equilibrium is shifted in the forward direaction.
**Buffer Solutions** are the solutions whose pH does not change significantly on adding a small quantity of strong base or on little dilution.

These are typically made by mixing a weak acid (or base) with its conjugate base (or acid). e.g. $CH_3COOH$ with
$CH_3COONa$, $NH_3$(aq) with $NH_4Cl$ etc.

If $K_a$ for acid (or $K_b$ for base) is not too high, we may write :
**Henderson's Equation**

$pH = pK_a + \log \{[salt] / [acid]\}$ for weak acid with its conjugate base.
or $pOH = pK_b + \log \{[salt] / [base]\}$ for weak base with its conjugate acid.

**Important** : For good buffer capacity, [salt] : [acid ratios should be as close to one as possible. In such a case, $pH = pK_a$. (This also is the case at midpoint of titration)
Buffer capacity = (no. of moles of acid (or base) added to 1L) / (change in pH)

**Indicators.** Indicator is a substance which indicates the point of equivalence in a titration by undergoing a change in its colour. They are weak acids or weak bases.

**Theory of Indicators.** The ionized and unionized forms of indicators have different colours. If 90 % or more of a particular form (ionised or unionised) is present, then its colour can be distinclty seen.In general, for an indicator which is weak acid, $HIn \rightleftharpoons H^+ + In^-$, the ratio of ionized to unionized form can be determined from

$$pH = pK_a + \log \frac{[In^-]}{[HIn]}$$

So, for detectable colour change, $pH = pK_a \pm 1$
This roughly gives the range of indicators. Ranges for some popular indicators are

**Table 1 : Indicators**

| Indicators | pH range | Colour | |
|---|---|---|---|
| | | acid medium | basic medium |
| Methyl Orange | 3.1-4.4 | pink | yellow |
| Methyl red | 4.2-6.3 | red | yellow |
| Litmus | 5.5-7.5 | red | blue |
| Phenol red | 6.8-8.4 | yellow | red |
| Phenolphathlene | 8.3-10 | colourless | pink |
| Thymol blue | 1.2-2.8 | red | yello |

**Equivalence point.** The point at which exactly equivalent amounts of acid and base have been mixed.

**Acid Base Titration.** For choosing a suitable indicator titration curves are of great help. In a titration curve, change in pH is plotted against the volume of alkali to a given acid. Four cases arise.

(a) **Strong acid vs strong base.** The curve is almost vertical over the pH range 3.5-10. This abrupt change corresponds to equivalence point. Any indicator suitable.

(b) **Weak acid vs strong base.** Final solution is basic 9 at equivalence point. Vertical region (not so sharp) lies in pH range 6.5-10. So, phenolphathlene is suitable.

(c) **Strong acid vs weak base.** Final solution acidic. Vertical point in pH range 3.8-7.2. Methyl red or methyl orange suitable.

(d) **Weak acid vs weak base.** No sharp change in pH. No suitable indicator.

*Note* : at midpoint of titration, $pH = pK_a$, thus by pH measurements, $K_a$ for weak acids (or $K_b$ for weak bases) can be determined.

**Polyprotic acids and bases.** Usually $K_2$, $K_3$ etc. can be safely neglected and only $K_1$ plays a significant role.

**Solubility product ($K_{sp}$).** For sparingly soluble salts (eg. $Ag_2C_2O_4$) an equilibrium which exists is

$$Ag_2C_2O_4 \rightleftharpoons 2Ag^+ (aq.)\ C_2O_4^{2-} (aq.)$$

Then $K_{sp} = [Ag^+]^2[C_2O_4^{2-}]$

**Precipitation.** Whenever the product of concentrations (raised to appropriate power) exceeds the solubility product, precipitation occurs.

**Common ion effects.** Suppression of dissociation by adding an ion common with dissociation products. e.g. $Ag^+$ or $C_2O_4^{2-}$ in the above example.

**Simultaneous solubility.** While solving these problems, go as per general method i.e.

(i) First apply condition of electroneutrality and
(ii) Apply the equilibria conditions.

# *THE ATLAS*

## The concept of ionic equilibria as equilibria involving ions in solution

(i) Arrhenius theory of electrolytes.
(ii) Ostwards dilution law for weak electrolyte $K_a = \left(\frac{\alpha^2}{1-\alpha}\right)\left(\frac{1}{V}\right)$

(iii) Acid & Bases
(a) Arrhenius $H^+/OH^-$ theory.
(b) Bronsted lowery - protonic concept.
(c) Lewis concept - electronic concept of acids and bases.

## Some basic concept

pH scale : $pH = -\log[H^+]$.
(i) Water as amphiprotic solvent.
(ii) Autoionization of water : $K_w = [H^+][OH^-]$.
(iii) $K_a[H_2O] = K_w/[H_2O]$
(iv) Change in pH of neutral $H_2O$ with temperature.

## Homogenons Ionic equilibria

(i) Acid/ base equilibrium.
(a) Strong acid/ base-$[H]^+ = \frac{c}{2} + \sqrt{\frac{c^2 + K_w}{4}}$ ;c=conc. of (acid)

(b) pH due to polyprotic weak acids
(c) Weak monobasic acid/base $[H]^+ = \sqrt{K_a.c}$ (if $\alpha < 0.1$)
(d) Mixture of S.A./W.A.
(e) Mixture of W.A./ W.A. $H^+ = \sqrt{K_1c_1 + K_2c_2}$

(f) Buffer solutions : $pH = pK_a + \log\left(\frac{salt}{acid}\right)$

$pOH = pK_b + \log\left(\frac{salt}{base}\right)$

(g) Salt hydrolysis – (W.A./S.B) $pH = \frac{1}{2}(pK_w + pK_a + \log c)$

(W.B/S.A.) $pH = \frac{1}{2}(pK_w - pK_b - \log c)$

(W.A./W.B.) $pH = \frac{1}{2}(pK_w + pK_a - pK_b)$

## Heterogenons equilibria

Solubility of sparingly soluble salt's
$(AB, AB_2, A_xB_y) K_{sp} = (S^{x+y})X^xY^y$
(i) Effect of pH on solubility.
(ii) Simultaneous solubility.

## Application of both heterogenous and homogenous equilibrium

(i) Extent of hydrolysis in buffer solution.
(ii) Change in solubility due to complex formation.
(iii) Solubility and hydrolysis.

## *GLOSSARY*

**Amphoteric substance.** A molecule which can act both as an acid and as a base.

**Autoprotolysis constant.** The equilibrium constant for the reaction in which one solvent molecule loses a proton to another, as $2H_2O \rightleftharpoons H_3O^+ + OH^-$.

**Amphiprotic solvent.** A solvent which possesses both acidic and basic properties.

**Aprotic solvent.** A solvent which is neither appreciably acidic or basic.

**Bronsted acid.** A substance which furnishes a proton.

**Bronsted base.** A substance which accepts a proton.

**Buffer capacity.** A measure of the effectiveness of a buffer in resisting changes in pH; the capacity is greater the concentrations of the conjugate acid-base pair.

**Buffer solution.** A solution which contains a conjugated acid-base pair. Such a solution resists large changes in pH when $H_3O^+$ or $OH^-$ ions are added and when the solution is diluted.

**Charge-balance equation.** The equation expressing the electroneutrality principle; i.e., the total concentration of positive charge must equal the total concentration of negative charge.

**Common-ion effect.** The effect produced by an ion, say from a salt, which is the same ion produced by the dissociation of a weak electrolyte. The "common" ion shifts the dissociation equilibrium in accordance with LeChatelier's principle.

**Central metal atom.** A cation which accepts electrons from a ligand to form a complex ion.

**Conjugate acid-base pair.** An acid-base pair which differ only by a proton, as HCl and $Cl^-$.

**Diprotic acid.** An acid which furnishes two protons.

**Electrolyte.** A compound which produces positive and negative ions in solution. Strong electrolytes are completely dissociated, whereas weak electrolytes are only partially dissociated.

**Hydrolysis.** An acid-base reaction of a cation or anion with water.

**Isoelectric point.** The pH at which there is an exact balance of positive and negative charge on an amino acid.

**Indicator.** A visual acid-base indicator is a weak organic acid or base which shows different colors in the molecular and ionic forms.

**Ligand.** An anion or neutral molecule which forms a complex ion with a cation by donating one or more pairs of electrons.

**Nonelectrolyte.** A substance which does not dissociate into ions in solution.

**pH.** The negative logarithm of the hydrogen ion concentration.

**pK.** The negative logarithm of an equilibrium constant.

**Polyprotic acid.** An acid which furnishes two or more protons.

**Range of an indicator.** That portion of the pH scale over which an indicator changes color, roughly the pK of the indicator ± 1 unit.

**Salt.** The product other than water which is formed when an acid reacts with a base; usually an ion solid.

**Simultaneous equilibria.** Equilibria established in the same solution in which one molecule or ions is a participant in more than one of the equilibria.

**Solubility product constant, $K_{sp}$.** The constant for the equilibrium established between a slightly soluble salt and its ions in solution.

**Stability constant.** The equilibrium constant for a reaction in which a complex is formed. Also called a formation constant.

# *EXERCISE I*

## IONIZATION CONSTANTS AND pH

Q.1.1 Calculate

(i) $K_a$ for $H_2O$ ($K_w = 10^{-14}$)

(ii) $K_b$ for $B(OH)_4^-$, $K_a$ $(B(OH)_3) = 6 \times 10^{-10}$

(iii) $K_a$ for HCN , $K_b$ $(CN^-) = 2.5 \times 10^{-5}$

Q.1.2 Calculate the ratio of degree of dissociation ($\alpha_2/\alpha_1$) when 1 M acetic acid solution is diluted to $\frac{1}{100}$ times. [Given $K_a = 1.8 \times 10^{-5}$]

Q.1.3 Calculate the ratio of degree of dissociation of acetic acid and hydrocyanic acid (HCN) in 1 M their respective solution of acids.[Given $K_{a(CH_3COOH)} = 1.8 \times 10^{-5}$; $K_{a(HCN)} = 6.2 \times 10^{-10}$]

Q.1.4 Calculate :

(a) $K_a$ for a monobasic acid whose 0.10 M solution has pH of 4.50.

(b) $K_b$ for a monoacidic base whose 0.10 M solution has a pH of 10.50.

Q.1.5 Calculate pH of following solutions :

(a) 0.1 M HCl

(b) 0.1 M $H_2SO_4$ (50 ml) + 0.4 M HCl 50 (ml)

(c) 0.1 M $CH_3COOH$ ($K_a = 1.8 \times 10^{-5}$)

(d) 0.1 M $NH_4OH$ ($K_b = 1.8 \times 10^{-5}$)

(e) $10^{-8}$ M HCl

(f) $10^{-10}$ M NaOH

(g) $10^{-6}$ M $CH_3COOH$

(h) $10^{-8}$ M $CH_3COOH$

(i) 0.1 M HA + 0.1 M HB [ $K_a$ (HA) = $2 \times 10^{-5}$ ; $K_a$ (HB) = $4 \times 10^{-5}$ ]

(j) Decimolar solution of Baryta ($Ba(OH)_2$), diluted 100 times.

(k) $10^{-3}$ mole of KOH dissolved in 100 L of water.

(l) 0.5 M HCl (25 ml) + 0.5 M NaOH (10 ml) + 40 ml $H_2O$

(m) equal volume of HCl solution (PH = 4) + 0.0019 N HCl solution

Q.1.6 The value of $K_w$ at the physiological temperature (37°C) is $2.56 \times 10^{-14}$. What is the pH at the neutral point of water at this temperature, where there are equal number of $H^+$ and $OH^-$?

Q.1.7 Calculate the number of $H^+$ present in one ml of solution whose pH is 13.

Q.1.8 Calculate change in concentration of $H^+$ ion in one litre of water, when temperature changes from 298 K to 310 K. Given $K_w(298) = 10^{-14}$ $K_w$ $(310) = 2.56 \times 10^{-14}$.

Q.1.9

(i) $K_w$ for $H_2O$ is $9.62 \times 10^{-14}$ at 60°C. What is pH of water at 60°C.

(ii) What is the nature of solution at 60°C whose

(a) pH = 6.7 (b) pH = 6.35

Q.1.10 pH of a dilute solution of HCl is 6.95. Calculate molarity of HCl solution.

Q.1.11 The pH of aqueous solution of ammonia is 11.5. Find molarity of solution. $K_b$ $(NH_4OH) = 1.8 \times 10^{-5}$.

Q.1.12 The solution of weak monoprotic acid which is 0.01 M has pH = 3. Calculate $K_a$ of weak acid.

Q.1.13 Boric acid is a weak monobasic acid. It ionizes in water as

$B(OH)_3 + H_2O \rightleftharpoons B(OH)_4^- + H^+ : K_a = 5.9 \times 10^{-10}$

Calculate pH of 0.3 M boric acid.

Q.1.14 Calculate $[H^+]$ and $[CHCl_2COO^-]$ in a solution that is 0.01 M in HCl and 0.01 M in $CHCl_2COOH$. Take ($K_a = 2.55 \times 10^{-2}$).

Q.1.15 Calculate the percent error in the $[H_3O^+]$ concentration made by neglecting the ionization of water in a $10^{-6}$M NaOH solution.

Q.1.16 Calculate $[H^+]$, $[CH_3COO^-]$ and $[C_7H_5O_2^-]$ in a solution that is 0.02 M in acetic acid and 0.01M in benzoic acid. $K_a$(acetic) = $1.8 \times 10^{-5}$ , $K_a$ (benzoic) = $6.4 \times 10^{-5}$.

Q.1.17 At 25°C , the dissociation constant of HCN and HF are $4 \times 10^{-10}$ and $6.7 \times 10^{-4}$. Calculate the pH of a mixture of 0.1 M HF and 0.1 M HCN.

## POLYPROTIC ACIDS & BASES

Q.2.1 Determine the $[S^{2-}]$ in a saturated (0.1M) $H_2S$ solution to which enough HCl has been added to produce a $[H^+]$ of $2 \times 10^{-4}$ . $K_1 = 10^{-7}$ , $K_2 = 10^{-14}$.

Q.2.2 Calculate $[H^+]$, $[H_2PO_4^-]$, $[HPO_4^{2-}]$ and $[PO_4^{3-}]$ in a 0.01M solution of $H_3PO_4$.
Take $K_1 = 7.225 \times 10^{-3}$, $K_2 = 6.8 \times 10^{-8}$ , $K_3 = 4.5 \times 10^{-13}$.

Q.2.3 Calculate the pH of a 0.1M solution of $H_2NCH_2CH_2NH_2$ ; ethylenediamine (en). Determine the en $H_2^{2+}$. Concentration in the solution. $K_{b_1}$ and $K_{b_2}$ values of ethylenediamine are $8.5 \times 10^{-5}$ and $7.1 \times 10^{-8}$ respectively.

Q.2.4 What are the concentrations of $H^+$, $HSO_4^-$ , $SO_4^{2-}$ and $H_2SO_4$ in a 0.20 M solution of sulphuric acid ?

Given : $H_2SO_4 \longrightarrow H^+ + HSO_4^-$ ; strong

$HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$ ; $K_2 = 1.3 \times 10^{-2}$ M

Q.2.5 What are the concentration of $H^+$, $H_2C_2O_4$, $HC_2O_4^-$ and $C_2O_4^{2-}$ in a 0.1 M solution of oxalic acid ?
[$K_1 = 5.9 \times 10^{-2}$ M and $K_2 = 6.4 \times 10^{-5}$ M ]

Q.2.6 Nicotine, $C_{10}H_{14}N_2$, has two basic nitrogen atoms and both can react with water to give a basic solution

$Nic\ (aq) + H_2O\ (l) \rightleftharpoons NicH^+\ (aq) + OH^-\ (aq)$

$NicH^+\ (aq) + H_2O\ (l) \rightleftharpoons NicH_2^{2+}\ (aq) + OH^-\ (aq)$

$K_{b1}$ is $7 \times 10^{-7}$ and $K_{b2}$ is $1.1 \times 10^{-10}$. Calculate the approximate pH of a 0.020 M solution.

Q.2.7 Ethylenediamine, $H_2N–C_2H_4–NH_2$, can interact with water in two steps, giving $OH^-$ in each step. Calculate the concentration of $OH^-$ and $[H_3N–C_2H_4–NH_3]^{2+}$ in a 0.15 M aqueous solution of the amine. $K_1 = 8.5 \times 10^{-5}$, $K_2 = 2.7 \times 10^{-8}$ for the base.

## BUFFER SOLUTION

Q.3.1 Determine $[OH^-]$ of a 0.050 M solution of ammonia to which has been added sufficient $NH_4Cl$ to make the total $[NH_4^+]$ equal to 0.100.[$K_{b(NH_3)} = 1.8 \times 10^{-5}$]

Q.3.2 Calculate the pH of a solution prepared by mixing 50.0 mL of 0.200 M $HC_2H_3O_2$ and 50.0 mL of 0.100 M NaOH.[$K_{a(CH_3COOH)} = 1.8 \times 10^{-5}$]

Q.3.3 A buffer of pH 9.26 is made by dissolving x moles of ammonium sulphate and 0.1 mole of ammonia into 100 mL solution. If $pK_b$ of ammonia is 4.74, calculate value of x.

Q.3.4 50 mL of 0.1 M NaOH is added to 75 mL of 0.1 M $NH_4Cl$ to make a basic buffer. If $pK_a$ of $NH_4^+$ is 9.26, calculate pH.

Q.3.5
(a) Determine the pH of a 0.2 M solution of pyridine $C_5H_5N$ . $K_b = 1.5 \times 10^{-9}$
(b) Predict the effect of addition of pyridinium ion $C_5H_5NH^+$ on the position of the equilibrium. Will the pH be raised or lowered ?
(c) Calculate the pH of 1.0 L of 0.10 M pyridine solution to which 0.3 mol of pyridinium chloride $C_5H_5NH^+Cl$, has been added, assuming no change in volume.

Q.3.6 Calculate the pH of a solution which results from the mixing of 50.0 ml of 0.3 M HCl with 50.0 ml of 0.4 M $NH_3$ . [$K_b$ ($NH_3$) = $1.8 \times 10^{-5}$]

Q.3.7 Calculate the pH of a solution made by mixing 50.0 ml of 0.2M $NH_4Cl$ & 75.0 ml of 0.1 M NaOH. [ $K_b$ ($NH_3$ ) = $1.8 \times 10^{-5}$ ]

Q.3.8 A buffer solution was prepared by dissolving 0.02 mol propionic acid & 0.015 mol sodium propionate in enough water to make 1.00 L of solution .($K_a$ for propionic acid is $1.34 \times 10^{-5}$)
(a) What is the pH of the buffer?
(b) What would be the pH if $1.0 \times 10^{-5}$ mol HCl were added to 10 ml of the buffer ?
(c) What would be the pH if $1.0 \times 10^{-5}$ mol NaOH were added to 10 ml of the buffer.
(d) Also report the percent change in pH of original buffer in cases (b) and (c).

Q.3.9 A solution was made up to be 0.01 M in chloroacetic acid, $ClCH_2COOH$ and also 0.002 M in sodium chloroacetate $ClCH_2COONa$ . What is [$H^+$] in the solution ? $K_a = 1.5 \times 10^{-3}$.

## INDICATORS

Q.4.1 A certain solution has a hydrogen ion concentration $4 \times 10^{-3}$ M. For the indicator thymol blue, pH is 2.0 when half the indicator is in unionised form. Find the % of indicator in unionised form in the solution with [$H^+$] = $4 \times 10^{-3}$ M.

Q.4.2 At what pH does an indicator change colour if the indicator is a weak acid with $K_{ind} = 4 \times 10^{-4}$. For which one(s) of the following neutralizations would the indicator be useful ? Explain.
(a) $NaOH + CH_3COOH$ (b) $HCl + NH_3$ (c) $HCl + NaOH$

Q.4.3 What indicator should be used for the titration of 0.10 M $KH_2BO_3$ with 0.10 M HCl ? $K_a$ ($H_3BO_3$) = $7.2 \times 10^{-10}$ .

Q.4.4 Bromophenol blue is an indicator with a $K_a$ value of $6 \times 10^{-5}$ . What % of this indicator is in its basic form at a pH of 5 ?

Q.4.5 An acid base indicator has a $K_a$ of $3 \times 10^{-5}$. The acid form of the indicator is red & the basic form is blue. By how much must the pH change in order to change the indicator form 75% red to 75 % blue?

## HYDROLYSIS

Q.5.1 What is the $OH^-$ concentration of a 0.08 M solution of $CH_3COONa$. [$K_a(CH_3COOH)=1.8 \times 10^{-5}$]

Q.5.2 Calculate the pH of a 2.0 M solution of $NH_4Cl$. [$K_b$ ($NH_3$) = $1.8 \times 10^{-5}$]

Q.5.3 0.25 M solution of pyridinium chloride $C_5H_6N^+Cl^-$ was found to have a pH of 2.699. What is $K_b$ for pyridine, $C_5H_5N$ ?

Q.5.4 Calculate the extent of hydrolysis & the pH of 0.02 M $CH_3COONH_4$. [$K_b$ ($NH_3$)= $1.8 \times 10^{-5}$, $K_a(CH_3COOH)=1.8 \times 10^{-5}$]

Q.5.5 Calculate the percent hydrolysis in a 0.06 M solution of KCN. [$K_a(HCN) = 6 \times 10^{-10}$]

Q.5.6 Calculate the extent of hydrolysis of 0.005 M $K_2CrO_4$. [$K_2 = 3.1 \times 10^{-7}$ for $H_2CrO_4$] (It is essentially strong for first ionization).

Q.5.7 Calculate the percent hydrolysis in a 0.0100 M solution of KCN.($K_a = 6.2 \times 10^{-10}$)

Q.5.8 A 0.010 M solution of $PuO_2(NO_3)_2$ was found to have a pH of 4.0. What is the hydrolysis constant, $K_h$, for $PuO_2^{2+}$ ,and what is $K_b$ for $PuO_2OH^+$ ?

Q.5.9 Calculate the pH of $1.0 \times 10^{-3}$ M sodium phenolate, $NaOC_6H_5$. $K_a$ for $HOC_6H_5$ is $1.05 \times 10^{-10}$.

Q.5.10 What is the pH of 0.1M $NaHCO_3$? $K_1 = 4.5 \times 10^{-7}$, $K_2 = 4.5 \times 10^{-11}$ for carbonic acids.

Q.5.11 Calculate pH of 0.05M potassium hydrogen phthalate, $KHC_8H_4O_4$.

$$H_2C_8H_4O_4 + H_2O \rightleftharpoons H_3O^+ + HC_8H_4O_4^- \qquad pK_1 = 2.94$$

$$HC_8H_4O_4^- + H_2O \rightleftharpoons H_3O^+ + C_8H_4O_4^{2-} \qquad pK_2 = 5.44$$

Q.5.12 Calculate $OH^-$ concentration at the equivalent point when a solution of 0.1 M acetic acid is titrated with a solution of 0.1 M NaOH. $K_a$ for the acid = $1.9 \times 10^{-5}$.

Q.5.13 The acid ionization (hydrolysis) constant of $Zn^{2+}$ is $1.0 \times 10^{-9}$

(a) Calculate the pH of a 0.001 M solution of $ZnCl_2$

(b) What is the basic dissociation constant of $Zn(OH)^+$?

## ACID BASE REACTIONS & TITRATIONS

Q.6.1 Calculate the hydronium ion concentration and pH at the equivalence point in the reaction of 22.0 mL of 0.10M acetic acid, $CH_3COOH$, with 22.0 mL of 0.10 M NaOH.

Q.6.2 Calculate the hydronium ion concentration and the pH at the equivalence point in a titration of 50.0 mL of 0.40 M $NH_3$ with 0.40M HCl.

Q.6.3 In the titration of a solution of a weak acid HX with NaOH, the pH is 5.8 after 10.0 mL of NaOH solution has been added and 6.402 after 20.0 mL of NaOH has been added. What is the ionization constant of HX?

Q.6.4 The equivalent point in a titration of 40.0 mL of a solution of a weak monoprotic acid occurs when 35.0 mL of a 0.10M NaOH solution has been added. The pH of the solution is 5.75 after the addition of 20.0 mL of NaOH solution. What is the dissociation constant of the acid?

Q.6.5 Phenol, $C_6H_5OH$, is a weak organic acid that has many uses, and more than 3 million ton are produced annually around the world. Assume you dissolve 0.515 g of the compound in exactly 100mL of water and then titrate the resulting solution with 0.123M NaOH.

$C_6H_5OH\,(aq) + OH^-\,(aq) \rightarrow C_6H_5O^-\,(aq) + H_2O(l)$

What are the concentrations of all of the following ions at the equivalence point: $Na^+$, $H_3O^+$, $OH^-$ and $C_6H_5O^-$ ? What is the pH of the solution ? [$K_a$ (phenol) = $1.3 \times 10^{-10}$]

Q.6.6 A weak base (50.0mL) was titrated with 0.1 M HCl. The pH of the solution after the addition of 10.0 mL and 25.0 mL were found to be 9.84 and 9.24, respectively. Calculate $K_b$ of the base and pH at the equivalence point.

Q.6.7 A weak acid (50.0mL) was titrated with 0.1 M NaOH. The pH values when 10.0 mL and 25.0 mL of base have been added are found to be 4.16 and 4.76, respectively. Calculate $K_a$ of the acid and pH at the equivalence point.

Q.6.8 $CH_3COOH$ (50 ml, 0.1 M) is titrated against 0.1 M NaOH solution. Calculate the pH at the addition of 0 ml, 10 ml 20 ml, 25 ml, 40 ml, 50 ml of NaOH. $K_a$ of $CH_3COOH$ is $2 \times 10^{-5}$.

## SOLUBILITY & SOLUBILITY PRODUCT'S

Q.7.1 The values of $K_{sp}$ for the slightly soluble salts MX and $QX_2$ are each equal to $4.0\times10^{-18}$. Which salt is more soluble? Explain your answer fully.

Q.7.2 The solubility of $PbSO_4$ water is 0.038 g/L. Calculate the solubility product constant of $PbSO_4$.

Q.7.3 Calculate the solubility of $Mg(OH)_2$ in water. $K_{sp} = 1.2 \times 10^{-11}$.

Q.7.4 How many mol CuI ($K_{sp} = 5 \times 10^{-12}$) will dissolve in 1.0 L of 0.10 M NaI solution ?

Q.7.5 A solution of saturated $CaF_2$ is found to contain $4.1 \times 10^{-4}$ M fluoride ion. Calculate the $K_{sp}$ of $CaF_2$. Neglect hydrolysis.

Q.7.6 The solubility of $ML_2$ (formula weight, 60 g/mol) in water is $2.4 \times 10^{-5}$ g/100 mL solution. Calculate the solubility product constant for $ML_2$.

Q.7.7 What is the solubility (in mol/L) of $Fe(OH)_3$ in a solution of pH = 8.0 ? [$K_{sp}$ for $Fe(OH)_3 = 1.0 \times 10^{-36}$]

Q.7.8 The solubility of $Ag_2CrO_4$ in water is 0.044 g/L. Determine the solubility product constant.

Q.7.9 Calculate the solubility of $A_2X_3$ in pure water, assuming that neither kind of ion reacts with water. For $A_2X_3$, [$K_{sp} = 1.1 \times 10^{-23}$]

Q.7.10 Determine the solubility of AgCl in 0.1 M $BaCl_2$. [$K_{sp}$ for $AgCl = 1 \times 10^{-10}$]

Q.7.11 What mass of $Pb^{2+}$ ion is left in solution when 50.0 mL of 0.20M $Pb(NO_3)_2$ is added to 50.0 mL of 1.5 M NaCl ?[Given $K_{sp}$ for $PbCl_2 = 1.7 \times 10^{-4}$]

Q.7.12 A solution has a $Mg^{2+}$ concentration of 0.0010 mol/L. Will $Mg(OH)_2$ precipitate if the $OH^-$ concentration of the solution is [$K_{sp} = 1.2 \times 10^{-11}$]
(a) $10^{-5}$ mol/L (b) $10^{-3}$ mol/L ?

Q.7.13 Calculate solubility of $PbI_2$ ($K_{sp} = 1.4 \times 10^{-8}$) in water at 25°, which is 90% dissociated.

Q.7.14 Calculate solubility of AgCN ($K_{sp} = 4 \times 10^{-16}$) in a buffer solution of PH = 3.

## SIMULTANEOUS SOLUBILITY

Q.8.1 Calculate the Simultaneous solubility of AgSCN and AgBr. $K_{sp}$ (AgSCN) = $1.1 \times 10^{-12}$, $K_{sp}$(AgBr) = $5 \times 10^{-13}$.

Q.8.2 Calculate $F^-$ in a solution saturated with respect of both $MgF_2$ and $SrF_2$. $K_{sp}(MgF_2) = 9.5 \times 10^{-9}$, $K_{sp}(SrF_2) = 4 \times 10^{-9}$.

Q.8.3 Equal volumes of 0.02M $AgNO_3$ and 0.02M HCN were mixed. Calculate [$Ag^+$] at equilibrium . Take $K_a(HCN) = 9 \times 10^{-10}$, $K_{sp}$ (AgCN) = $4 \times 10^{-16}$.

## COMPLEXATION EQUILIBRIA

Q.9.1 Assuming no change in volume, calculate the minimum mass of NaCl necessary to dissolve 0.010 mol AgCl in 100 L solution. [$K_f(AgCl_2^-) = 3 \times 10^5$, $K_{sp} = (AgCl) = 1 \times 10^{-10}$]

Q.9.2 A recent investigation of the complexation of $SCN^-$ with $Fe^{3+}$ led of 130, 16, and 1.0 for $K_1$, $K_2$, and $K_3$, respectively. What is the overall formation constant of $Fe(SCN)_3$ from its component ions, and what is the dissociation constant of $Fe(SCN)_3$ into its simplest ions on the basis of these data ?

Q.9.3 How much AgBr could dissolve in 1.0 L of 0.40 M $NH_3$ ? Assume that $Ag(NH_3)_2^+$ is the only complex formed. [$K_f(Ag(NH_3)_2^+) = 1 \times 10^8$ ; $K_{sp}$ (AgBr) = $5 \times 10^{-13}$]

## ***PROFICIENCY TEST***

Q.1 True / False. When a solution of a weak monoprotic acid is titrated against a strong base, at half-neutralization point, $pH = \frac{1}{2} pK_a$.

Q.2 True / False. A solution of sodium acetate and ammonium acetate can act as a buffer.

Q.3 True / False. If the solubility of the salt $Li_3Na_3(AlF_6)_2$ is x, then its solubility product would be $2916\ x^8$.

Q.4 True / False. A buffer has maximum buffer capacity when the ratio of salt to acid is 10.

Q.5 True / False. In the presence of a common ion (incapable of froming complex ion), the solubility of salt decreases.

Q.6 In a mixture of waek acid and its salt, the ratio of concentration of salt to acid is increased ten fold. The pH of the solution would _____ by _____ unit.

Q.7 The solubilty of $CH_3COOAg$ in water considering hydrolysis of $CH_3COO^-$ ions would be _____ than that ignoring the hydrolysis.

Q.8 From an equimolar solution of $Cl^-$ and $Br^-$ ions, the addition of $Ag^+$ will selectively precipitates _____ ($K_{sp}$ of AgCl & AgBr are $1 \times 10^{-10}$ & $1 \times 10^{-13}$ respectively).

Q.9 The solubility of AgCl in $NH_3$ is _____ than the solubility in pure water because of complex ion, $[Ag(NH_3)_2]^+$ formation.

Q.10 The hydrolytic constant $K_h$ for the hydrolytic equilibrium

$H_2PO_4^- + H_2O \longrightarrow H_3PO_4 + OH^-$ is $1.4 \times 10^{-12}$

What is the value of ionization constant for the $H_3PO_4 + H_2O \longrightarrow H_2PO_4^- + H_3O^+$ ?

Q.11 Given the equilibrium constants

$HgCl^+ + Cl^- \rightleftharpoons HgCl_2$ ; $K_1 = 3 \times 10^6$

$HgCl_2 + Cl^- \rightleftharpoons HgCl_3^-$ ; $K_2 = 8.9$

The equilibrium constant for the dispropotionation equilibrium.

$2HgCl_2 \rightleftharpoons HgCl^+ + HgCl_3^-$ is

Q.12 Under which set of conditions is the ionic product of water, $K_w$, constant at a given temperature in aqueous system?

Q.13 If the salts $M_2X$, $QY_2$ and $PZ_3$ have same solubilities (<<<1), their $K_{sp}$ values are related as _____.

Q.14 $K_a$ for an acid HA is $1 \times 10^{-6}$. $K_b$ for $A^-$ would be _____.

Q.15 An aqueous solution of $K_2SO_4$ has pH nearly equal to _____.

Q.16 The pH of a solution which is 0.1 M in sodium acetate and 0.01 M in acetic acid ($pK_a = 4.74$) would be _____.

Q.17 The conjugate acid of sulphate ($SO_4^{2-}$) is _____.

Q.18 The value of $K_w$ _____ with increase in temperature.

Q.19 AgCl is _____ soluble in aqueous sodium chloride solution than in pure water.

Q.20 The buffer HCOOH / HCOONa will have pH _____ than 7.

Q.21 In the reaction $I_2 + I^- \longrightarrow I_3^-$, $I_2$ acts as _____.

Q.22 An equimolar solution of $NaNO_2$ and $HNO_2$ can act as a _____ solution.

Q.23 Larger the value of $pK_a$,_____ is the acid.

Q.24 An aqueous solution of potash alum is _____ in nature.

Q.25 Salts of strong acids and weak bases undergo _____ hydrolysis.

Q.26 For salts of weak acid with weak bases, degree of hydrolysis is _____ of concentration of the salt in solution.

## ***BEWARE OF SNAKES***

1. ***General Mistake :*** pH of a neutral water solution is always equal to 7.

   ***Explanation :*** pH of neutral water depend on temperature. Since pH (neutral point) = $\frac{pK_w}{2}$; $pK_w$ decreases with temperature hence pH of neutral solution.

2. ***General Mistake :*** If a solution is diluted half times pH of solution becomes double.
   ***Explanation :*** Infact pH increases by 0.3010 unit. If it is diluted x times pH increases by log x.
   *e.g.* If solution is diluted 10 times pH increases by $\log_{10}10 = 1$ unit.

3. ***General Mistake :*** For calculation of pH of $10^{-6}$ M $CH_3COOH$ the formula $(H^+) = \sqrt{K_a c}$ will give

   $$pH = -\log\left(\sqrt{1.8\times10^{-5}\times10^{-6}}\right) = 5.37.$$

   ***Explanation :*** 5.37 is incorrect answer. pH should be calculated by taking $\alpha = \frac{-K_a + \sqrt{K_a^2 + 4K_a c}}{2c}$

4. ***General Mistake :*** If $10^3$ mole $CH_3COONa$ and 1 mole $CH_3COOH$ is added in $10^4$ litres water the pH of resulting solution is equal to $pH = pK_a + \log\frac{10^3}{1} = 7.74$.

   ***Explanation :*** 7.74 is incorrect answer. The $CH_3COOH$ concentration is too low to be taken as constituent of buffer solution. Use salt hydrolysis formula instead to calculate the pH.

5. ***General Mistake :*** The equilibrium concentration of anion and cation of a sparingly soluble salt ($A_2C_3$) are a and c moles $lit^{-1}$ respectively. The solubility product is $(2a)^2\,(3c)^3 = K_{sp}$
   ***Explanation :*** $K_{sp} = a^2c^3$.

6. ***General Mistake :*** pH of $10^{-8}$ M HCl is equal to 8.
   ***Explaination*** **:** pH = 8 means basic solution. Contribution of water can not be neglected in this case.

7. ***General Mistake :*** If NaOH is added to $NH_4Cl$ so that NaOH is limiting, the resulting solution is containing some remaining conc. of $NH_4Cl$. Now use salt hydrolysis condition to calculate pH of solution.
   ***Explanation :*** The addition of NaOH in $NH_4Cl$ results in a basic buffer solution.

8. ***General Mistake :*** Do not use the $K_1K_2$ form of equation unless you have an independent method of calculating $[H^+]$ or $[S^{2-}]$
   ***Explanation :*** Determine the $[S^{2-}]$ in a saturated $H_2S$ solution to which enough HCl has been added to produce a $[H^+]$ of $2 \times 10^{-4}$.

   ***Sol.*** : $K_1K_2 = \frac{[H^+]^2[S^{2-}]}{[H_2S]} = \frac{(2\times10^{-4})^2[S^{2-}]}{0.10} = 1.0 \times 10^{-21}$ or

   $$[S^{2-}] = \frac{1.0\times10^{-22}}{4\times10^{-8}} = 2.5 \times 10^{-15}.$$

## *EXERCISE II*

Q.1 At 25°C, the degree of dissociation of water was found to be $1.8 \times 10^{-9}$. Calculate the ionization constant and ionic product of water at this temperature.

Q.2 A solution contains HCl, $Cl_2HC$ COOH & $CH_3$ COOH at concentrations 0.09 M in HCl, 0.09 M in $Cl_2HC$ COOH & 0.1 M in $CH_3$ COOH. pH for the solution is 1. Ionization constant of $CH_3$ COOH = $10^{-5}$. What is the magnitude of K for dichloroacetic acid ?

Q.3 A solution of chloroacetic acid, $ClCH_2COOH$ containing 9.45 grams in 500 ml of the solution has a pH of 2.0. What is the degree of ionization of the acid.

Q.4 A solution of ammonia bought for cleaning the windows was found to be 10% ammonia by mass, having a density of 0.935 g . $ml^{-1}$. What is the pH of the solution. Take $K_b$ for protonation of ammonia = 5.5 x $10^{-6}$.

Q.5 The $K_w$ of water at two different temperatures is :

| T | 25°C | 50°C |
|---|---|---|
| $K_w$ | $1.08 \times 10^{-14}$ | $5.474 \times 10^{-14}$ |

Assuming that $\Delta H$ of any reaction is independent of temperature, calculate the enthalpy of neutralization of a strong acid and strong base.

Q.6 What is the pH of a 1.0 M solution of acetic acid ? To what volume must 1 litre of the solution be diluted so that the pH of the resulting solution will be twice the original value. Given $K_a = 1.8 \times 10^{-5}$.

Q.7 A handbook states that the solubility of methylamine $CH_3NH_2(g)$ in water at 1 atm pressure at 25°C is 959 volumes of $CH_3NH_2(g)$ per volume of water ( $pk_b$ =3.39)

(a) Estimate the max. pH that can be attained by dissolving methylamine in water.

(b) What molarity NaOH (aq.) would be required to yield the same pH ?

Q.8 The equilibrium constant of the reaction

$$2Ag(s) + 2I^- + 2H_2O \rightleftharpoons 2AgI(s) + H_2(g) + 2OH^-$$

is $1.2 \times 10^{-23}$ at 25°C. Calculate the pH of a solution at equilibrium with the iodine ion concentration = 0.10 and the pressure of $H_2$ gas = 0.60 atm.

Q.9 For the reaction

$$A + B \rightleftharpoons C + D$$

(all reactants in solution) calculate the value of the equilibrium constant for the following percentages of conversion of A and B into products. (Assume the initial concentrations of A and B are each 1.0 M)
(a) 67%; (b) 95%; (c) 99%.

Q.10 Mixtures of soutions. Calculate the pH of the following solutions. (Use data of Q.14)

(a) 50 ml of 0.12 M $H_3PO_4$ + 20 ml of 0.15 M NaOH;

(b) 50 ml of 0.12 M $H_3PO_4$ + 40 ml of 0.15 M NaOH;

(c) 40 ml of 0.12 M $H_3PO_4$ + 40 ml of 0.18 M NaOH;

(d) 40 ml of 0.10 M $H_3PO_4$ + 40 ml of 0.25 M NaOH.

Q.11 Mixtures of solution. Calculate the pH of the following solution.(Use data of Q.14)
(a) 40 ml of 0.050 M $Na_2CO_3$ + 50 ml of 0.040 M HCl;
(b) 40 ml of 0.020 M $Na_3PO_4$ + 40 ml of 0.040 M HCl;
(c) 50 ml of 0.10 M $Na_3PO_4$ + 50 ml of 0.10 M $NaH_2PO_4$ ;
(d) 40 ml of 0.10 M $H_3PO_4$ + 40 ml of 0.10 M $Na_3PO_4$.

Q.12 The electrolytic reduction of an organic nitro compound was carried out in a solution buffered by acetic acid and sodium acetate. The reaction was

$$RNO_2 + 4H_3O^+ + 4e \longrightarrow RNHOH + 5H_2O$$

300 ml of a 0.0100 M solution of $RNO_2$ buffered initially at pH 5.00 was reduced, with the reaction above going to completion. The total acetate concentration, [HOAc] + [$OAc^-$], was 0.50 M.Calculate the pH of the solution after the reduction is complete.

Q.13(a) It is desired to prepare 100 ml of a buffer of pH 5.00.Acetic, benzoic and formic acids and their salts are available for use. Which acid should be used for maximum effectiveness against increase in pH? What acid-salt ratio should be used ?$pK_a$ values of these acids are : acetic 4.74; benzoic 4.18 and formic 3.68.
(b) If it is desired that the change in pH of the buffer be no more than 0.10 unit for the addition of 1 m mol of either acid or base, what minimum concentrations of the acid and salt should be used ?

Q.14 Calculate the pH of 0.1 M solution of (i) $NaHCO_3$, (ii) $Na_2HPO_4$ and (iii) $NaH_2PO_4$. Given that:

$CO_2 + H_2O \rightleftharpoons H^+ + HCO_3^-$ ; $K_1 = 4.2 \times 10^{-7}$ M

$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$ ; $K_2 = 4.8 \times 10^{-11}$ M

$H_3PO_4 \rightleftharpoons H^+ + H_2PO_4^-$ ; $K_1 = 7.5 \times 10^{-3}$ M

$H_2PO_4^- \rightleftharpoons H^+ + HPO_4^{2-}$ ; $K_2 = 6.2 \times 10^{-8}$ M

$HPO_4^{2-} \rightleftharpoons H^+ + PO_4^{3-}$ ; $K_3 = 1.0 \times 10^{-12}$ M

Q.15 When a 40 mL of a 0.1 M weak base is titrated with 0.16M HCl, the pH of the solution at the end point is 5.23. What will be the pH if 15 mL of 0.12 M NaOH is added to the resulting solution.

Q.16 A buffer solution was prepared by dissolving 0.05 mol formic acid & 0.06 mol sodium formate in enough water to make 1.0 L of solution. $K_a$ for formic acid is $1.80 \times 10^{-4}$.
(a) Calculate the pH of the solution .
(b) If this solution were diluted to 10 times its volume, what would be the pH ?
(c) If the solution in (b) were diluted to 10 times its volume,what would be the pH?

Q.17 How many moles of sodium hydroxide can be added to 1.00 L of a solution 0.1 M in $NH_3$ & 0.1 M in $NH_4Cl$ without changing the pOH by more than 1.00 unit ? Assume no change in volume. $K_b(NH_3) = 1.8 \times 10^{-5}$.

Q.18 20 ml of a solution of 0.1 M $CH_3COOH$ solution is being titrated against 0.1 M NaOH solution. The pH values after the addition of 1 ml & 19 ml of NaOH are $(pH)_1$ & $(pH)_2$, what is $\Delta$pH ?

Q.19 Calculate the $OH^-$ concentration and the $H_3PO_4$ concentration of a solution prepared by dissolving 0.1 mol of $Na_3PO_4$ in sufficient water to make 1L of solution. $K_1 = 7.1 \times 10^{-3}$ , $K_2 = 6.3 \times 10^{-8}$ , $K_3 = 4.5 \times 10^{-13}$.

Q.20 Find the pH of 0.068M $Na_2HPO_4$ solution. Use K values from the above problem if required.

Q.21 Calculate the values of the equilibrium constants for the reactions with water of $H_2PO_4^-$, $HPO_4^{2-}$, and $PO_4^{3-}$ as bases. Comparing the relative values of the two equilibrium constants of $H_2PO_4^-$ with water, deduce whether solutions of this ion in water are acidic or basic. Deduce whether solutions of $HPO_4^{2-}$ are acidic or basic. Take $K_1 = 5 \times 10^{-3}$, $K_2 = 5 \times 10^{-8}$, $K_3 = 5 \times 10^{-13}$.

Q.22 Determine the equilibrium carbonate ion concentration after equal volumes of 1.0M sodium carbonate and 1.0M HCl are mixed. $K_1 = 5 \times 10^{-7}$, $K_2 = 5 \times 10^{-11}$.

Q.23 $K_1$ and $K_2$ for oxalic acid, $H_2C_2O_4$, are $5.6 \times 10^{-2}$ and $5.0 \times 10^{-5}$. What is $[OH^-]$ in a 0.4mM solution of $Na_2C_2O_4$?

Q.24 If 0.00050 mol $NaHCO_3$ is added to 1 litre of a buffered solution at pH 8.00, how much material will exist in each of the three forms $H_2CO_3$, $HCO_3^-$ and $CO_3^{2-}$? For $H_2CO_3$, $K_1 = 5 \times 10^{-7}$, $K_2 = 5 \times 10^{-13}$.

Q.25 Equilibrium constant for the acid ionization of $Fe^{3+}$ to $Fe(OH)^{+2}$ and $H^+$ is $6.5 \times 10^{-3}$. What is the max.pH, which could be used so that at least 95% of the total $Fe^{3+}$ in a dilute solution. exists as $Fe^{3+}$.

Q.26 Hydrazine, $N_2H_4$, can interact with water in two stages.

$$N_2H_4\ (aq) + H_2O\ (l) \rightleftharpoons N_2H_5^+\ (aq) + OH^-\ (aq.) \qquad K_{b1} = 8.5 \times 10^{-7}$$

$$N_2H_5^+\ (aq) + H_2O\ (l) \rightleftharpoons N_2H_6^{2+}\ (aq) + OH^-\ (aq.) \qquad K_{b2} = 8.9 \times 10^{-16}$$

(i) What are the concentration of $OH^-$, $N_2H_5^+$ and $N_2H_6^{2+}$ in a 0.010 M aqueous solution of hydrazine?

(ii) What is pH of the 0.010 M solution of hydrazine?

Q.27 How much $Na_2HPO_4$ must be added to one litre of 0.005M solution of $NaH_2PO_4$ in order to make a 1L of the solution of pH 6.7? $K_1 = 7.1 \times 10^{-3}$ , $K_2 = 6.3 \times 10^{-8}$, $K_3 = 4.5 \times 10^{-13}$ for $H_3PO_4$.

Q.28 A solution of volume V contains $n_1$ moles of QCl and $n_2$ moles of RCl where QOH and ROH are two weak bases of dissociation constants $K_1$ and $K_2$ respectively. Show that the pH of the solution is given

$$\text{by pH} = \frac{1}{2}\log\left[\left(\frac{K_1K_2}{K_W}\right)\frac{V}{(n_1K_2 + K_1n_2)}\right]$$

State assumptions, if any.

Q.29 The indicator phenol red is half in the ionic form when pH is 7.2. If the ratio of the undissociated form to the ionic form is 1 : 5, find the pH of the solution. With the same pH for solution, if indicator is altered such that the ratio of undissociated form to dissociated form becomes 1 : 4, find the pH when 50 % of the new indicator is in ionic form.

Q.30 A buffer solution, 0.080 M in $Na_2HPO_4$ and 0.020 M in $Na_3PO_4$, is prepared. The electrolytic oxidation of 1.00 mmol of the organic compound RNHOH is carried out in 100 ml of the buffer. The reaction is

$$RNHOH + H_2O \longrightarrow RNO_2 + 4H^+ + 4e$$

Calculate the approximate pH of the solution after the oxidation is complete.

Q.31 A solution of weak acid HA was titrated with base NaOH. The equivalence point was reached when 36.12 ml of 0.1 M NaOH has been added. Now 18.06 ml of 0.1 M HCl were added to titrated solution, the pH was found to be 4.92. What will be the pH of the solution obtained by mixing 10 ml of 0.2 M NaOH and 10 ml of 0.2 M HA.

Q.32 A weak base BOH was titrated against a strong acid . The pH at 1/4th equivalence point was 9.24. Enough strong base was now added (6m.eq.) to completely convert the salt. The total volume was 50ml. Find the pH at this point.

Q.33 An organic monoprotic acid [0.1M] is titrated against 0.1M NaOH. By how much does the pH change between one fourth and three fourth stages of neutralization? If at one third stage of neutralization, the pH is 4.45 what is the dissociation constant of the acid? Between what stages of neutralisation may the pH change by 2 units?

Q.34 50 ml of a solution which is 0.050 M in the acid HA, $pK_a = 3.80$ and 0.10 M in HB, $pK_a = 8.20$, is titrated with 0.2 M NaOH. Calculate the pH
(a) at the first equivalence point and
(b) at the second equivalence point.

Q.35 Calculate the solubility of solid zinc hydroxide at a pH of 5, 9 and 13. Given

| | | |
|---|---|---|
| $Zn(OH)_2(s) \rightleftharpoons Zn(OH)_2(aq)$ | $K_1 = 10^{-6}$ M | (1) |
| $Zn(OH)_2(aq) \rightleftharpoons Zn(OH)^+ + OH^-$ | $K_2 = 10^{-7}$ M | (2) |
| $Zn(OH)^+ \rightleftharpoons Zn^{2+} + OH^-$ | $K_3 = 10^{-4}$ M | (3) |
| $Zn(OH)_2(aq) + OH^- \rightleftharpoons Zn(OH)_3^-$ | $K_4 = 10^3$ $M^{-1}$ | (4) |
| $Zn(OH)_3^- + OH^- \rightleftharpoons Zn(OH)_4^{2-}$ | $K_5 = 10$ $M^{-1}$ | (5) |

Q.36 The salt $Zn(OH)_2$ is involved in the following two equilibria,
$Zn(OH)_2(s) \rightleftharpoons Zn^{2+}(aq) + 2OH^-(aq)$ ; $K_{sp} = 1.2 \times 10^{-17}$
$Zn(OH)_2(s) + 2OH^-(aq) \rightleftharpoons [Zn(OH)_4]^{2-}(aq.)$ ; $K_c = 0.13$
Calculate the pH of solution at which solubility is minimum.

Q.37 What is the solubility of AgCl in 0.20 M $NH_3$?
Given : $K_{sp}(AgCl) = 1.7 \times 10^{-10}$ $M^2$, $K_1 = [Ag(NH_3)^+] / [Ag^+][NH_3] = 2.33 \times 10^3$ $M^{-1}$ and $K_2 = [Ag(NH_3)_2^+] / [Ag(NH_3)^+][NH_3] = 7.14 \times 10^3$ $M^{-1}$.

Q.38 $H_2S$ is bubbled into a 0.2 M NaCN solution which is 0.02 M in each $Ag(CN)_2^-$ and $Cd(CN)_4^{2-}$. Determine which sulphide precipitates first.
Given : $K_{sp}(Ag_2S) = 1.0 \times 10^{-50}$ $M^3$
$K_{sp}(CdS) = 7.1 \times 10^{-28}$ $M^2$
$K_{inst}(Ag(CN)_2^-) = 1.0 \times 10^{-20}$ $M^2$
$K_{inst}(Cd(CN)_4^{2-}) = 7.8 \times 10^{-18}$ $M^4$

Q.39 Predict whether or not AgCl will be precipitated from a solution which is 0.02 M in NaCl and 0.05 M in $KAg(CN)_2$. Given $K_{inst}(Ag(CN)_2^-) = 4.0 \times 10^{-19}$ $M^2$ and $K_{sp}(AgCl) = 2.8 \times 10^{-10}$ $M^2$.

Q.40 Show that solubility of a sparingly soluble salt $M^{2+}A^{2-}$ in which $A^{2-}$ ions undergoes hydrolysis is given by : $S = \sqrt{K_{sp}\left(1 + \frac{[H^+]}{K_2} + \frac{[H^+]^2}{K_1K_2}\right)}$.
where $K_1$ and $K_2$ are the dissociation constant of acid $H_2A$. $K_{sp}$ is solubility product of MA.

## ***EXERCISE III***

Q.1 The conjugate acid of $NH_2^-$ is

(A) $NH_3$ (B) $NH_2OH$ (C) $NH_4^+$ (D) $N_2H_4$

Q.2 pH of an aqeous solution of NaCl at 85°C should be

(A) 7 (B) > 7 (C) < 7 (D) 0

Q.3 1 CC of 0.1 N HCl is added to 99 CC solution of NaCl. The pH of the resulting solution will be

(A) 7 (B) 3 (C) 4 (D) 1

Q.4 10 ml of $\frac{M}{200}H_2SO_4$ is mixed with 40 ml of $\frac{M}{200}H_2SO_4$. The pH of the resulting solution is

(A) 1 (B) 2 (C) 2.3 (D) none of these

Q.5 The pH of an aqueous solution of 1.0 M solution of a weak monoprotic acid which is 1% ionised is

(A) 1 (B) 2 (C) 3 (D) 11

Q.6 If $K_1$ & $K_2$ be first and second ionisation constant of $H_3PO_4$ and $K_1 >> K_2$ which is incorrect.

(A) $[H^+] = [H_2PO_4^-]$ (B) $[H^+] = \sqrt{K_1[H_3PO_4]}$

(C) $K_2 = [HPO_4^{--}]$ (D) $[H^+] = 3[PO_4^{3-}]$

Q.7 The degree of hydrolysis of a salt of weak acid and weak base in it's 0.1 M solution is found to be 50%. If the molarity of the solution is 0.2 M, the percentage hydrolysis of the salt should be

(A) 100% (B) 50% (C) 25% (D) none of these

Q.8 What is the percentage hydrolysis of NaCN in N/80 solution when the dissociation constant for HCN is $1.3 \times 10^{-9}$ and $K_w = 1.0 \times 10^{-14}$

(A) 2.48 (B) 5.26 (C) 8.2 (D) 9.6

Q.9 The compound whose 0.1 M solution is basic is

(A) Ammonium acetate (B) Ammonium chloride
(C) Ammonium sulphate (D) Sodium acetate

Q.10 Which of the following solution will have pH close to 1.0?

(A) 100 ml of M/100 HCl + 100 ml of M/10 NaOH
(B) 55 ml of M/10 HCl + 45 ml of M/10 NaOH
(C) 10 ml of M/10 HCl + 90 ml of M/10 NaOH
(D) 75 ml of M/5 HCl + 25 ml of M/5 NaOH

Q.11 The ≈ pH of the neutralisation point of 0.1 N ammonium hydroxide with 0.1 N HCl is

(A) 1 (B) 6 (C) 7 (D) 9

Q.12 If equilibrium constant of
$CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^-$
Is $1.8 \times 10^{-5}$, equilibrium constant for
$CH_3COOH + OH^- \rightleftharpoons CH_3COO^- + H_2O$ is

(A) $1.8 \times 10^{-9}$ (B) $1.8 \times 10^9$ (C) $5.55 \times 10^{-9}$ (D) $5.55 \times 10^{10}$

Q.13 If 40 ml of 0.2 M KOH is added to 160 ml of 0.1 M HCOOH [$K_a = 2\times10^{-4}$], the pOH of the resulting solution is

(A) 3.4 (B) 3.7 (C) 7 (D) 10.3

Q.14 The range of most suitable indicator which should be used for titration of $X^-$ $Na^+$ (0.1 M, 10 ml ) with 0.1 M HCl should be (Given: $k_{b(X^-)} = 10^{-6}$)

(A) 2–3 (B) 3–5 (C) 6–8 (D) 8–10

Q.15 When $NO_2$ is bubbled into water, it disproportionates completely into $HNO_2$ and $HNO_3$.

$$2NO_2 + H_2O\ (l) \longrightarrow NHO_2\ (aq.) + HNO_3\ (aq.)$$

The concentration of $NO_2^-$ in a solution prepared by dissolving 0.05 mole of $NO_2$ gas in 1 litre $H_2O$ is $\{K_a\ (HNO_2) = 5 \times 10^{-4}\}$ is

(A) ~ $5 \times 10^{-4}$ (B) ~ $4.8 \times 10^{-5}$ (C) ~ $4.8 \times 10^{-3}$ (D) ~ $2.55 \times 10^{-2}$

Q.16 Which of the following is most soluble in water?

(A) MnS ($K_{sp} = 8\times10^{-37}$) (B) ZnS ($K_{sp} = 7\times10^{-16}$)

(C) $Bi_2S_3$ ($K_{sp} = 1\times10^{-72}$) (D) $Ag_3(PO_4)$ ($K_{sp} = 1.8\times10^{-18}$)

Q.17 The precipitate of $CaF_2$($K_{sp} = 1.7 \times 10^{-10}$) is obtained when equal volumes of the following are mixed

(A) $10^{-4}$ M $Ca^{3+}$ + $10^{-4}$ M $F^-$ (B) $10^{-2}$ M $Ca^{2+}$ + $10^{-3}$ M $F^-$

(C) $10^{-5}$ M $Ca^{2+}$ + $10^{-3}$ M $F^-$ (D) $10^{-3}$ M $Ca^{2+}$ + $10^{-5}$ M $F^-$

Q.18 The solubility of AgCl in water, 0.01 M $CaCl_2$, 0.02 M NaCl and 0.05 M $AgNO_3$ are denoted by $S_1$, $S_2$, $S_3$ and $S_4$ respectively. Which of the following relationship is correct?

(A) $S_1 > S_2 > S_3 > S_4$ (B) $S_1 = S_2 = S_3 = S_4$

(C) $S_1 > S_3 > S_2 > S_1$ (D) $S_1 > S_2 = S_3 > S_4$

Q.19 How many moles $NH_3$ must be added to 2.0 litre of 0.80 M $AgNO_3$ in order to reduce the $Ag^+$ concentration to $5 \times 10^{-8}$ M. $K_f$ of $[Ag(NH_3)_2^+] = 10^8$

(A) 0.4 (B) 2 (C) 3.52 (D) 4

Q.20 The solubility of metal sulphides in saturated solution of $H_2S$ **{$[H_2S] = 0.1$ M}**can be represented by

$$MS + 2H^+ \rightleftharpoons M^{2+} + H_2S\ ;\ K_{eq} = \frac{[M^{2+}][H_2S]}{[H^+]^2}$$

The value of $K_{eq}$ is given for few metal sulphide. If conc. of each metal ion in solution is **0.01 M**, which metal sulphides are selectively ppt at total **$[H^+] = 1$M** in saturated $H_2S$ solution.

| **Metal sulphides** | MnS | ZnS | CoS | PbS |
|---|---|---|---|---|
| $\mathbf{K_{eq}} = \frac{[M^{2+}][H_2S]}{[H^+]^2}$ | $3 \times 10^{10}$ | $3 \times 10^{-2}$ | 3 | $3 \times 10^{-7}$ |

(A) MnS, ZnS, CoS (B) PbS, ZnS, CoS (C) PbS, ZnS (D) PbS

## *EXERCISE IV*

Q.1 In the reaction $I^- + I_2 \longrightarrow I_3^-$, the Lewis acid is _____ . **[ JEE '97, 1]**

Q.2 Between $Na^+$ & $Ag^+$ which is a stronger Lewis acid & why ? **[ JEE '97, 2]**

Q.3 Select the correct alternative . **[JEE'97,1+1]**
If $pK_b$ for fluoride ion at 25° C is 10.83, the ionisation constant of hydrofluoric acid in water at this temperature is :
(A) $1.74 \times 10^{-5}$ (B) $3.52 \times 10^{-3}$ (C) $6.75 \times 10^{-4}$ (D) $5.38 \times 10^{-2}$

Q.4 The solubility of $A_2X_3$ is y mol $dm^{-3}$. Its solubility product is **[JEE 97]**
(A) $6\ y^2$ (B) $64\ y^4$ (C) $36\ y^5$ (D) $108\ y^5$

Q.5 Which of the following statement(s) is/are correct ? **[ JEE '98, 2 ]**
(A) the pH of $1.0 \times 10^{-8}$ M solution of HCl is 8
(B) the conjugate base of $H_2PO_4^-$ is $HPO_4^{2-}$
(C) autoprotolysis constant of water increases with temperature
(D) when a solution of a weak monoprotic acid is titrated again a strong base, at half–neutralization point pH = (1/2) pKa .

Q.6 A buffer solution can be prepared from a mixture of **[JEE 99]**
(A) sodium acetate and acetic acid in water
(B) sodium acetate and hydrochloric acid in water
(C) ammonia and ammonium chloride in water
(D) ammonia and sodium hydroxide in water.

Q.7 The pH of 0.1 M solution of the following salts increases in the order **[JEE 99]**
(A) $NaCl < NH_4Cl < NaCN < HCl$ (B) $HCl < NH_4Cl < NaCl < NaCN$
(C) $NaCN < NH_4Cl < NaCl < HCl$ (D) $HCl < NaCl < NaCN < NH_4Cl$

Q.8 An aqueous solution of 6.3 g oxalic acid dihydrate is made up to 250 mL. The volume of 0.1 N NaOH required to completely neutralise 10 mL of this solution is **[JEE 2001]**
(A) 40 mL (B) 20 mL (C) 10 mL (D) 4 mL

Q.9 For sparingly soluble salt ApBq, the relationship of its solubility product (Ls) with its solubility (S) is
(A) $Ls = S^{p+q}, p^p . q^q$ (B) $Ls = S^{p+q}, p^p . q^p$ (C) $Ls = S^{pq}, p^p . q^q$ (D) $Ls = S^{pq}, (p.q)^{p+q}$
**[JEE 2001]**

Q.10 A solution which is $10^{-3}$ M each in $Mn^{2+}$, $Fe^{2+}$, $Zn^{2+}$ and $Hg^{2+}$ is treated with $10^{-16}$M sulphide ion. If $K_{sp}$, MnS, FeS, ZnS and HgS are $10^{-15}$, $10^{-23}$, $10^{-20}$ and $10^{-54}$ respectively, which one will precipitate first ?
(A) FeS (B) MnS (C) HgS (D) ZnS
**[JEE 2003]**

Q.11 HX is a weak acid ($K_a = 10^{-5}$). It forms a salt NaX (0.1 M) on reacting with caustic soda. The degree of hydrolysis of NaX is
(A) 0.01% (B) 0.0001% (C) 0.1% (D) 0.5% **[JEE 2004]**

Q.12 $CH_3NH_2$ (0.1 mole, $K_b = 5 \times 10^{-4}$) is added to 0.08 moles of HCl and the solution is diluted to one litre, resulting hydrogen ion concentration is
(A) $1.6 \times 10^{-11}$ (B) $8 \times 10^{-11}$ (C) $5 \times 10^{-5}$ (D) $2 \times 10^{-2}$ **[JEE 2005]**

## ***SUBJECTIVES***

Q.13 An acid type indicator, HIn differs in colour from its conjugate base ($In^-$) . The human eye is sensitive to colour differences only when the ratio $[In^-]/[HIn]$ is greater than 10 or smaller than 0.1 . What should be the minimum change in the pH of the solution to observe a complete colour change ($K_a = 1.0 \times 10^{-5}$) ? **[JEE '97, 2 ]**

Q.14 A sample of AgCl was treated with 5.00 ml of 1.5 M $Na_2CO_3$ solution to give $Ag_2CO_3$. The remaining solution contained 0.0026 g of $Cl^-$ per litre . Calculate the solubility product of AgCl. ($K_{sp}$ $Ag_2CO_3 = 8.2 \times 10^{-12}$) **[ JEE '97, 5 ]**

Q.15 Given : $Ag(NH_3)_2^+ \rightleftharpoons Ag^+ + 2\ NH_3$ , $K_c = 6.2 \times 10^{-8}$ & $K_{sp}$ of AgCl = $1.8 \times 10^{-10}$ at 298 K . Calculate the concentration of the complex in 1.0 M aqueous ammonia . **[JEE '98, 5 ]**

Q.16 What will be the resultant pH when 200 ml of an aqueous solution of HCl (pH = 2.0) is mixed with 300 ml of an aqueous solution of NaOH (pH = 12.0) ? **[ JEE '98, 2 ]**

Q.17 The solubility of $Pb(OH)_2$ in water is $6.7 \times 10^{-6}$M. Calculate the solubility of $Pb(OH)_2$ in a buffer solution of pH = 8. **[ JEE '99, 4 ]**

Q.18 The average concentration of $SO_2$ in the atmosphere over a city on a certain day is 10 ppm, when the average temperature is 298 K. Given that the solubility of $SO_2$ in water at 298 K is 1.3653 moles $litre^{-1}$ and the $pK_a$ of $H_2SO_3$ is 1.92, estimate the pH of rain on that day. **[JEE 2000]**

Q.19 500 ml of 0.2 M aqueous solution of acetic acid is mixed with 500 mL of 0.2 M HCl at 25°C.
(a) Calculate the degree of dissociation of acetic acid in the resulting solution and pH of the solution.
(b) If 6 g of NaOH is added to the above solution, determine final pH. Assume there is no change in volume on mixing. $K_a$ of acetic acid is $1.75 \times 10^{-5}$ M. **[JEE 2002]**

Q.20 Will the pH of water be same at 4°C and 25°C ? Explain. **[JEE 2003]**

Q.21 0.1 M of HA is titrated with 0.1 M NaOH, calculate the pH at end point. Given Ka(HA) = $5 \times 10^{-6}$ and $\alpha << 1$. **[JEE 2004]**

# ANSWER KEY

## *EXERCISE I*

### IONIZATION CONSTANTS AND pH

Q.1.1 (i) $1.8 \times 10^{-16}$, (ii) $1.66 \times 10^{-5}$, (iii) $4 \times 10^{-10}$ Q.1.2 10 Q.1.3 170.4

Q.1.4 (a) $K_a = 10^{-8}$, (b) $K_b = 10^{-6}$

Q.1.5 (a) +1, (b) 0.522, (c) 2.87, (d) 11.13 (e) 6.97, (f) 7, (g) 6.01, (h) 6.97, (i) 2.61, (j) 11.30 (k) 9 (l) 1 , (m) 3

Q.1.6 6.81 Q.1.7 $6.022 \times 10^{7}$ Q.1.8 $0.6 \times 10^{-7}$

Q.1.9 (i) 6.51 ; (ii) (a) Basic , (b) Acidic Q.1.10 $2.31 \times 10^{-8}$ M Q.1.11 0.556 M

Q.1.12 $1.11 \times 10^{-4}$ Q.1.13 4.87 Q.1.14 $[H^+] = 1.612 \times 10^{-2}$M, $[CHCl_2COO^-] = 6.126 \times 10^{-3}$M

Q.1.15 error = 1% Q.1.16 $[H^+] = 10^{-3}$M, $[CH_3COO^-] = 3.6 \times 10^{-4}$M, $[C_7H_5O_2^-] = 6.4 \times 10^{-4}$M

Q.1.17 2.08

### POLYPROTIC ACIDS & BASES

Q.2.1 $[S^{2-}] = 2.5 \times 10^{-15}$

Q.2.2 $[H^+] = [H_2PO_4^-] = 5.623 \times 10^{-3}$, $[HPO_4^{2-}] = 6.8 \times 10^{-8}$, $[PO_4^{3-}] = 5.441 \times 10^{-18}$

Q.2.3 pH = 11.46, $[enH_2^{2+}] = 7.1 \times 10^{-8}$ M Q.2.4 0.2116 M, 0.1884 M, 0.0116 M, 0

Q.2.5 0.0528 M, 0.0472 M, 0.0528 M, 0.000064 M Q.2.6 10.07

Q.2.7 $[OH^-] = 3.57 \times 10^{-3}$ M, $[H_2en]^{2+} = 2.7 \times 10^{-8}$ M

### BUFFER SOLUTION

Q.3.1 $[OH^-] = 9.0 \times 10^{-6}$ Q.3.2 4.74 Q.3.3 0.05 mol

Q.3.4 9.56 Q.3.5 (a) pH = 9.239 (b) lowered (c) pH = 4.699

Q.3.6 8.7782 Q.3.7 9.7324

Q.3.8 (a) 4.7525 (b) 4.697, (c) 4.798 (d) 1.134% on acid addition 0.96% on base addition.

Q.3.9 $[H^+] = 2.5 \times 10^{-3}$

### INDICATORS

Q.4.1 $[HI_n] = 28.57\%$ Q.4.2 (b), (c)

Q.4.3 (methyl red), one with pH = 5.22 as midpoint of colour range

Q.4.4 85.71% Q.4.5 $\Delta$pH = 0.954

### HYDROLYSIS

Q.5.1 $[OH^-] = 6.664 \times 10^{-6}$ Q.5.2 pH = 4.477 Q.5.3 $K_b = 6.25 \times 10^{-10}$

Q.5.4 0.56%, pH = 7 Q.5.5 1.667% Q.5.6 0.26%

Q.5.7 4.0% Q.5.8 $10^{-6}$ ; $10^{-8}$ Q.5.9 pH = 10.43

Q.5.10 8.34 Q.5.11 4.19 Q.5.12 $5.12 \times 10^{-6}$ M

Q.5.13 (a) 6, (b) $1 \times 10^{-5}$

### ACID BASE REACTIONS & TITRATIONS

Q.6.1 8.71 Q.6.2 4.98 Q.6.3 6.1

Q.6.4 $2.37 \times 10^{-6}$ Q.6.5 pH = 8.73, $[Na^+] = 0.0379$, $[C_6H_5O^-] = 0.0373$

Q.6.6 $K_b = 1.8 \times 10^{-5}$, 5.27 Q.6.7 8.73

Q.6.8 (i) 2.85, (ii) 4.0969, (iii) 4.5229, (iv) 4.699, (v) 5.301, (vi) 8.699

### SOLUBILITY & SOLUBILITY PRODUCT'S

Q.7.1 $QX_2$ is more soluble Q.7.2 $1.6 \times 10^{-8}$ Q.7.3 $1.4 \times 10^{-4}$

Q.7.4 $[Cu^+] = 5 \times 10^{-11}$ M Q.7.5 $3.4 \times 10^{-11}$ Q.7.6 $2.6 \times 10^{-16}$

Q.7.7 $1.0 \times 10^{-18}$ M Q.7.8 $8.8 \times 10^{-12}$ Q.7.9 $1.0 \times 10^{-5}$ mol/lit

Q.7.10 $5 \times 10^{-10}$ M Q.7.11 12 mg

Q.7.12 (a) no precipitation will occur, (b) a precipitate will form Q.7.13 $1.6 \times 10^{-3}$

Q.7.14 $2.1 \times 10^{-5}$

## SIMULTANEOUS SOLUBILITY

Q.8.1 $4 \times 10^{-7}$ mol/L AgBr, $9 \times 10^{-7}$ mol/L AgSCN Q.8.2 $[F^-] = 3 \times 10^{-3}$M
Q.8.3 $[Ag^+] = 6.667 \times 10^{-5}$M

## COMPLEXATION EQUILIBRIA

Q.9.1 19.3 kg Q.9.2 $K_d = 1/K_f = 4.8 \times 10^{-4}$ Q.9.3 $2.8 \times 10^{-3}$ M

## PROFICIENCY TEST

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Q.1 | False | Q.2 | False | Q.3 | True | Q.4 | False | Q.5 | True |
| Q.6 | Increase, one | Q.7 | Greater | Q.8 | $Br^-$ ion | Q.9 | Greater | | |
| Q.10 | $7.14 \times 10^{-3}$ | Q.11 | $3 \times 10^{-6}$ | Q.12 | in both dil acidic and alkaline solution | | | | |
| Q.13 | $M_2X = QY_2 > PZ_3$ | | | | | | | | |
| Q.14 | $10^{-8}$ | Q.15 | 7 | Q.16 | 5.74 | Q.17 | $HSO_4^-$ | | |
| Q.18 | increases | Q.19 | less | Q.20 | less | Q.21 | Lewis acid | | |
| Q.22 | Buffer | Q.23 | Weaker | Q.24 | acidic | Q.25 | cationic | Q.26 | independent |

## *EXERCISE II*

Q.1 $1.8 \times 10^{-16}$, $10^{-14}$ Q.2 $K_a = 1.25 \times 10^{-2}$ Q.3 $\alpha = 0.05$
Q.4 11.74 Q.5 $\Delta H_{neut} = -51.963$ kJ mol$^{-1}$
Q.6 $V = 2.77 \times 10^4$ litre Q.7 (a) 13.1, (b) 0.13 M Q.8 1.650
Q.9 (a) 4.1, (b) $3.6 \times 10^2$, (c) $9.8 \times 10^3$ Q.10 (a) 2.12 (b) 4.66 (c) 7.2 (d) 12
Q.11 (a) 8.34 (b) 4.66 (c) 9.6 (d) 7.20 Q.12 5.158
Q.13 (a) acetic acid, salt-acid molar ratio 1.8 :1 ;
(b) [HOAc] = 0.066 mmol/ml and $[OAc^-]$ = 0.119 mmol/ml
Q.14 8.35, 9.60, 4.66 Q.15 9.168 Q.16 (a) pH = 3.83 (b) pH = 3.85 , (c) = 3.99
Q.17 0.0818 moles Q.18 2.558 Q.19 $[OH^-] = 3.73 \times 10^{-2}$M, $[H_3PO_4] = 6 \times 10^{-18}$M
Q.20 9.7736
Q.21 $K_h(H_2PO_4^-) = 2 \times 10^{-12}$; $K_h(HPO_4^{2-}) = 2 \times 10^{-7}$, $K_h(PO_4^{3-}) = 2 \times 10^{-2}$ ; acidic, basic
Q.22 $[CO_3^{2-}] = 4.9 \times 10^{-3}$M Q.23 $[OH^-] = 3 \times 10^{-7}$M
Q.24 $[H_2CO_3] = 9.85 \times 10^{-6}$M ; $[HCO_3^-] = 4.9 \times 10^{-4}$ $[CO_3^{2-}] = 2.45 \times 10^{-8}$
Q.25 0.908 Q.26 (a) $9.21 \times 10^{-5}$ M, $9.21 \times 10^{-5}$ , $8.9 \times 10^{-16}$ (b) 9.96
Q.27 1.6 mmol Q.29 pH = 7.3 Q.30 7.81 Q.31 8.96
Q.32 11.22 Q.33 0.9542, $pK_a = 4.751$, $\frac{1}{11}$th & $\frac{10}{11}$th stages of neutralisation
Q.34 (a) 5.85 (b) 10.48 Q.35 10 M, $1.12 \times 10^{-6}$ M, $2 \times 10^{-4}$ M
Q.36 9.99, $s = 2.5 \times 10^{-5}$ M Q.37 $9.66 \times 10^{-3}$ Q.38 $[Cd^{2+}]$
Q.39 Precipitation will occur

## *EXERCISE III*

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Q.1 | A | Q.2 | C | Q.3 | B | Q.4 | B | Q.5 | C | Q.6 | D | Q.7 | B |
| Q.8 | A | Q.9 | D | Q.10 | D | Q.11 | B | Q.12 | B | Q.13 | D | Q.14 | B |
| Q.15 | A | Q.16 | D | Q.17 | B | Q.18 | D | Q.19 | D | Q.20 | D | | |

## *EXERCISE IV*

Q.1 $I_2$ Q.2 $Ag^+$, $Na^+$ has no tendency to accept $e^-$ Q.3 C Q.4 D
Q.5 B, C Q.6 A, B, C Q.7 B Q.8 A Q.9 A
Q.10 C Q.11 A Q.12 B

***SUBJECTIVES***

Q.13 $\Delta pH = 2$ Q.14 $K_{sp} = 1.71 \times 10^{-10}$ Q.15 $[Ag(NH_3)_2^+] = 0.0539$
Q.16 pH = 11.3010 Q.17 $s = 1.203 \times 10^{-3}$M Q.18 think ?
Q.19 (a) 0.0175% , (b) 4.757 Q.20 No it will be > 7 Q.21 pH = 9

# STUDY PACKAGE

# Target: IIT-JEE (Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 12. Thermodynamics

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

**4. Exercise III**

**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

# THE KEY

The subject of ***Thermodynamics*** deals basically with the interaction of one body with another in terms of quantities of heat & work. It may be *defined* as the branch of science which deals with energy changes associated with various physical & chemical processes. The entire formulation of thermodynamics is based on a few (Three) fundamental laws which have been established on the basis of human experience of the experimental behaviour of macroscopic aggregates of matter collected over a long period of time. ***Thermochemistry*** is the branch of physical chemistry which deals with the thermal or heat changes caused by chemical reactions. It is based on first law of thermodynamics. There are two laws of thermochemistry:

(a) Lavoisier & Laplace law (b) Hess's law.

**TERMS & CONVENTIONS**

A number of terms & conventions are used in thermodynamics.

A ***System*** is defined as that part of the universe which is at the moment under investigation.

Universe less the system is defined as ***Surroundings***. The actual or imaginary surface that separates the system from the surroundings is called the ***Boundary***.

**TYPES OF SYSTEMS:**

A system is said to be ***Isolated*** if it cannot exchange matter and energy with the surroundings (coffee in a thermos flask).

A system is said to be ***Closed*** if it can exchange energy but not matter. Coffee in a closed stainless steel flask is an example. A system is said to be ***Open*** if it can exchange matter and energy both. A thermo flask or a steel flask if not closed is an example. A system is said to be homogeneous when it is completely uniform throughout, made up of one phase only, pure liquid. solid, gas.

A system is said to be heterogeneous if it consists of two or more phases, liquid in contact with vapour.

**STATE OF A SYSTEM :**

The state of a system is defined by a particular set of its measurable properties. For example, we can describe the state of a gas by quoting its pressure (P) , volume (V) and temperature (T) etc.

Variables like P, V , T are *State Functions* OR *State Variables* because their values depend only on the state of a system and not on how the state was reached.

State variables can be intensive or extensive. An intensive variable (eg. temperature, pressure, concentration) is one whose value is independent of the size of the system. An extensive variable (eg. volume , mass, surface area is one whose value is proportional to the size of the system.

| **Extensive Properties (Depend upon quantity of Matter present and are additive)** | **Intensive Properties (Do not depend upon quantity of Matter present and are non additive)** |
|---|---|
| Volume | Molar volume |
| Number of moles | Density |
| Mass | Refractive index |
| Free Energy G | Surface tension |
| Entropy S | Viscosity |
| Enthalpy H | Free energy per mole |
| Internal energy E | Specific heat |
| Heat capacity | Pressure |
| | Temperature |
| | Boiling point, freezing point etc |

**THERMODYNAMIC PROCESS :**

A thermodynamic process involves change of a system from one state to another state.

**TYPES:**

A process is called ***Isothermal***, if the temperature of the system remains constant during the change. It is carried out in a thermostat and in such a process the exchange of energy between the system and surroundings takes place. In such a process $dT = 0$ & $dE = 0$.

A process is called ***Adiabatic***, if the system does not exchange energy with surroundings. Such a process is carried out in perfectly insulated containers. During it the temperature of the system may change. In such a process $dQ = 0$.

A process carried out at a constant pressure is called an **isobaric** process. In such a process $dP = 0$.

A process in which the volume of the system remains constant is called an **isochoric** process, whereby $dV = 0$.

**CYCLIC PROCESS :**

When a system undergoes a number of different processes and finally returns to its initial state.

$\Delta E = 0$ & $\Delta H = 0$.

**REVERSIBLE PROCESS : (QUASI-STATIC)**

A process which is carried out so slowly that the system and the surroundings are always in equilibrium is known as a ***Reversible Process***. If this condition does not hold good, the process is said to be, ***Irreversible***.

In a reversible process the driving force is infinitesimally larger than the opposing force. If the driving force is made infinitely smaller than opposing force, the system can be brought back without producing any permanent change.

A process which proceeds of its own i.e. without any external help, is called as ***Spontaneous Process*** (or a natural process).

Internal Energy (Intrinsic Energy) E – Every system having some quantity of matter is associated with a definite amount of energy, called internal energy.

$$E = E_{Translational} + E_{Rotational} + E_{vibrational} + E_{Bonding} + E_{Electronic} + .....$$

It is a state function & is an extensive property.

$\Delta E = E_{final} - E_{initial}$ ; $\Delta E = q_v$

**ZEROTH LAW OF THERMODYNAMICS**

It states that, two systems in thermal equilibrium with a third system, are also in thermal equilibrium with each other.

**FIRST LAW OF THERMODYNAMICS**

It is law of conservation of energy. Mathematically, this law is written as :

$\Delta E = q + w$, where $\Delta E$ is change in internal energy of the system and is a state function, q is the transfer of heat from / to the system and w is the work involved (either done on the system or by the system) .

According to IUPAC , heat, added to the system and work done on the system are assigned positive values as both these *Modes* increase the internal energy of the system.

**TYPES OF WORK :**

Two TYPES of work normally come across in chemistry. These are *Electrical Work* in system involving ions, while the *Mechanical Work* is involved when a system changes its volume in the, presence of an externally applied pressure (i.e. pressure volume work). It is especially important in system containing gases.

If a system expands from a volume $V_1$ to $V_2$ at constant pressure P, then the first law equation becomes,
$\Delta E = q - P\,\Delta V$ $(\Delta V = V_2 - V_1)$ ....(1)

For a process carried at constant volume A B = q (heat absorbed at constant volume)

Work = Intensity factor X capacity factor
w – by the system (Expansion) negative
w – on the system (compression) positive
q → absorbed by the system positive
q → given out by the system negative

Work done in irreversible process (Expansion) $w = -P_{Ext}\,\Delta V$

Work done in isothermal reversible process $w = -2.303\ nRT \log \frac{V_2}{V_1}$

(maximum work) (Expansion) $= -2.303\ nRT \log \frac{P_1}{P_2}$

work done is adiabatic reversible process $w = \frac{nR}{\gamma - 1}[T_2 - T_1]$

$\gamma = \frac{C_P}{C_V}$ = Poisson's ratio

**ENTHALPY :**

Chemical reactions are generally carried out at constant pressure (atmospheric pressure) so it has been found useful to define a new state function ***Enthalpy*** (H) as :

$H = E + PV$ (By definition) or $\Delta H = \Delta E + P\,\Delta V + V\Delta P$

or $\Delta H = \Delta E + P\,\Delta V$ (at constant pressure) combining with first law. Equation (1) becomes

$\Delta H = q_p$

Hence transfer of heat at constant volume brings about a change in the internal energy of the system whereas that at constant pressure brings about a change in the enthalpy of the system.

The difference between $\Delta H$ & $\Delta E$ becomes significant only when gases are involved (insignificant in solids and liquids) and is given by: $\Delta H = \Delta E + (\Delta n)\,RT$, where $\Delta n$ is the INCREASE in the number of moles of the gases involved (i.e. Total number of moles of product gases less the total number of moles of reactant gases).

**FACTORS AFFECTING ΔH OF THE REACTIONS ARE :**

(i) Temperature (ii) Physical states of reactants & products
(iii) Allotropic forms of elements & (iv) Pressure & volume (in case of gases)

**Two Types of Reactions may be distinguished :**

(i) **Exothermic Reactions :** For these $\Delta H$ is negative, which implies that

$\sum H$ (products) $< \sum H$ (reactants)

(ii) **Endothermic Reactions :** For these $\Delta H$ is positive, which implies that

$\sum H$ (products) $> \sum H$ (reactants)

**HESS'S LAW OF CONSTANT HEAT SUMMATION :**

According to Hess's law (a consequence of first law), if a set of reactants is converted into a set of product by more than one sequence of reactions, the total enthalpy change will be the same for every sequence.

As such, the chemical equations can be treated ordinary algebraic expressions and can be added or subtracted to yield the required equation. The corresponding enthalpy changes are also manipulated in the same way to obtain the enthalpy change for the desired equation.

**APPLICATIONS OF HESS'S LAW :**

It helps us in Calculation of :

(i) Heat of formation ($\Delta H_f$) of many substances which cannot be synthesised directly from their elements.

(ii) Bond energies.

(iii) Enthalpy changes of slow reactions and

(iv) Enthalpy of transformation, say from one allotropic form to the other.

**CONVENTIONAL VALUES OF MOLAR ENTHALPIES :**

It is not possible to determine the absolute value of enthalpy of a substance and further it also depends on the conditions under which its determination is carried out. It is therefore necessary to choose some standard conditions for reporting the enthalpy data.

Conventionally, the enthalpy of every element in its most stable state of aggregation at 1 atm. (101.325 k Pa) and 298 K is assigned a zero value.

Based on the above convention, the relative values of "Standard molar enthalpies" ($\Delta H°$) of other substances are obtained and it is obvious that in terms of $\Delta H_f^o$ values, the enthalpy change of any reaction is given as :

$$\Delta H° = \sum \Delta H_f^o \text{ (products)} - \sum \Delta H_f^o \text{ (reactants)}$$

i.e. $$\Delta H° = \begin{pmatrix} sum\ of\ standard\ enthalpies \\ of\ formation\ of\ product \end{pmatrix} - \begin{pmatrix} sum\ of\ standard\ enthalpies \\ of\ formation\ of\ reactants \end{pmatrix}$$

Reactions are frequently classified according to type of thermochemical purpose and the enthalpies of reactions are given different names.

A balanced chemical equation which expresses the heat changes taking place in a reaction as well as the physical states of various reactants and products is known as a thermochemical equation.

Heat capacity $C = \frac{dq}{dT}$ ; $C_V = \left(\frac{\partial E}{\partial T}\right)_V$ ; $Cp = \left(\frac{\partial H}{\partial T}\right)_P$

$q = C (T_2 - T_1)$ for 1 mole; $q = nC (T_2 - T_1)$ for n moles

Kirchoffs Equation: $\frac{\Delta H_2 - \Delta H_1}{T_2 - T_1} = \Delta C_P$ ; $\frac{\Delta E_2 - \Delta E_1}{T_2 - T_1} = \Delta C_V$

**Trouton's Rule :** Entropy of vaporization of non-associated or non-dissociated liquid is constant & may be taken as about 87.3 $J\ k^{-1} mol^{-1}$.

**Thermochemical Equations :** An equation which indicates the amount of heat change in the reaction. These can be added, subtracted or multiplied whenever required.

**The various named Enthalpies are defined as the Enthalpy change when .....**

**Enthalpy of reaction :** "quantities of substances indicated in the balanced equation react completely to form the product."

**Enthalpy of formation :** "one mole of the substance is formed directly from its constituent elements."

**Enthalpy of combustion** : "one mole of the substance undergoes complete combustion" (it is always negative)

**Calorific Value :** "it is the amount of heat given out by complete combustion of unit weight of a solid or liquid or unit volume of a gas".

**Enthalpy of solution :** "one mole of the substance is completely dissolved in a large excess of the given solvent under given conditions of temperature and pressure".

**Enthalpy of neutralisation :** "one gram equivalent of an acid is neutralised by one gram equivalent of a base in fairly dilute solution" .

**Enthalpy of hydration :** "one mole of an anhydrous (or a partly hydrated salt) combines with the required number of mole of water to form a specific hydrate".

**Enthalpy of sublimation :** "one mole of a solid is directly converted into its vapour at a given temperature below its melting point" .

**Enthalpy of fusion :** "one mole of the solid substance is completely converted into the liquid state at its melting point" .

**Enthalpy of vaporisation :** "one mole of a substance is converted from the liquid state to its vapour state at its boiling point".

**Resonance Energy = Observed heat of formation - Calculated heat of formation**

**BOND ENTHALPIES' (BOND ENERGIES) :**

The bond enthalpy of a diatomic molecule ($H_2$, $C1_2$, $O_2$) is equal to its dissociation energy and is defined as "the enthalpy change involved in breaking the bond between atoms of a gaseous molecule" (Bond breaking is an endothermic process). Average bond enthalpy (energy) is the average value of bond energy obtained from molecules that contain more than one bond of that type.

$$\text{Av. BE} = \frac{\Delta H_f \text{ of molecules}}{\text{no. of bonds}}$$

BE is an additive property.

## SECOND LAW OF THERMODYNAMICS

The essence of first law is that all physical and chemical processes take place in such a manner that the total energy of the universe remain constant.

However, it is observed that all processes have a natural direction ,i.e. a direction in which they take place spontaneously. First law fails to answer this. Another feature of the spontaneous processes is that they proceed only until an equilibrium is achieved.

The direction of a spontaneous process and that it eventually reaches equilibrium, can be understood on the basis of entropy concept introduced through the second law of thermodynamics.

**ENTROPY AND SPONTANEITY:**

Entropy (denoted by S) in s state function. When the state of a system changes, entropy also changes.

The change of entrops $\Delta S$ is defined by, $\Delta S = \frac{q_{rev}}{T}$, where $q_{rev}$ means that the heat is being supplied "Isothermally" and "Reversibly" ($JK^{-1}$).

One can think entropy as a measure of the degree of randomness or disorder in a system. The greater the disorder, in a system, the higher is the entropy.

A useful form of 2nd law of thermodynamics is :

"The entropy of the universe increases in the course of every spontaneous (natural) change".

**OR**

"For a spontaneous process in an isolated system, the change in entropy is positive".

When a system is in equilibrium the entropy is maximum. So mathematically

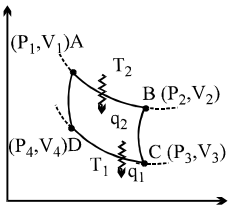
$\Delta S = 0$ (at equilibrium)

**SECOND LAW** :

***Statements :***

(i) No cyclic engine is possible which take heat from one single source and in a cycle completely convert it into work without producing any change in surrounding.

(ii) Efficiency of Carnot engine working reversibly is maximum.




Carnot cycle

AB – Iso. Rev.Exp. $\quad w_{AB} = -nRT_2 \ln \frac{V_2}{V_1}$

BC – Ada. Rev. Exp. $\quad w_{BC} = C_V (T_1 - T_2)$

CD – Iso. Rev. Comp. $\quad w_{CD} = -nRT_1 \ln \left(\frac{V_4}{V_3}\right)$

DA – Iso. Rev. Comp. $\quad w_{DA} = C_V (T_2 - T_1)$

Carnot efficiency $\eta = \frac{-w_{Total}}{q_2} = \frac{T_1 - T_2}{T_2} = \frac{q_1 + q_2}{q_2}$

**CARNOT CYCLE :**

$\frac{q_1}{T_1} + \frac{q_2}{T_2} = 0$ for rev. cycle

Irreversible engine $\frac{q_1}{T_1} + \frac{q_2}{T_2} < 0$

$\oint \frac{q_{rev}}{T} = 0 \Rightarrow \frac{q_{rev}}{T}$ is a state function.

$\Delta S = \int \frac{dq_{rev}}{T}$

Also $\quad \Delta S_{syt} + \Delta S_{surr} = 0 \quad$ for rev. process

$\quad \Delta S_{syt} + \Delta S_{surr} > 0 \quad$ for irrev. process

$\Rightarrow \quad \Delta S_{syt} + \Delta S_{surr} \geq 0 \quad$ ( In general )

**ENTROPY CHANGE (General Expression ):**

$$\Delta S = nC_V \, ln \, \frac{T_2}{T_1} + nR \, ln \, \frac{V_2}{V_1}$$

Change in state function for various processes.
Reversible irreversible isothermal expansion and contraction : (ideal gas )

$$\Delta E = 0; \Delta H = 0; \Delta S = nR \, ln \, \frac{V_2}{V_1}$$

Isobaric heating or cooling :

$$\Delta E = C_V \Delta T$$
$$\Delta H = C_P \Delta T = q_P$$
$$\Delta S = nC_P \, ln\left(\frac{T_2}{T_1}\right)$$

Isochoric heating or cooling :

$$\Delta E = C_V \Delta T = q_V$$
$$\Delta H = C_P \Delta T$$
$$\Delta S = nC_V \, ln\left(\frac{T_2}{T_1}\right)$$

Adiabatic process :

$$\Delta E = C_V \Delta T$$
$$\Delta H = C_P \Delta T$$
$$\Delta S = nC_V \, ln\frac{T_2}{T_1} + nR \, ln \, \frac{V_2}{V_1} \text{ for irreversible process}$$

$\Delta S = 0$ for reversible adiabatic compression and expansion.

***Gibb's function :***

$$G = H - TS$$

at constant T and pressure

$$\Delta G = \Delta H - T\Delta S$$

$\Delta G = (\Delta H - T\Delta S) \leq 0$ for rev. process.
$(-\Delta G)_{T, P}$ = work done by system max. non P – V

$$\frac{\Delta G}{T} = -(\Delta S_{syst} + \Delta S_{surr.})$$

$\Delta G < 0$ for spontaneous process
$\Delta G = 0$ for equilibrium.

**GIBBS FREE ENERGY (G) AND SPONTANEITY:**

A new thermodynamic (state) function G , the Gibbs free energy is defined as :

$G = H - TS$ or $\Delta G = \Delta H - T\,\Delta S$ (at constant temperature and pressure)

For a spontaneous reaction $\Delta G$ must be negative. The use of Gibbs free energy has the advantage that it refers to the system only (and not surroundings also as in entropy).

To summaries, the spontaneity of a chemical reaction is decided by two factors taken together:

(i) the enthalpy factor and (ii) the entropy factor.

The equation $\Delta G = \Delta H - T\,\Delta S$ takes both the factors into consideration.

The most favorable situation for a negative value of $\Delta G$ is a negative value of $\Delta H$ and a positive value of $\Delta S$. However a large negative value of $\Delta H$ may outweigh an unfavorable $\Delta S$ value and a large value of $\Delta S$ may outweigh an unfavorable value of $\Delta H$.

**STANDARD FREE ENERGY CHANGE ($\Delta G^\circ$) :**

The standard free energy change $\Delta G^\circ$ is defined as the free energy change for a process at a specified temperature in which the reactants in their standard state are converted to the products in their standard state. It is denoted by $\Delta G^\circ$.

Like the standard enthalpy of formation of an element "the standard free energy of formation of an element in its standard state is zero". And so ;

$$\Delta G_r^o = \sum \Delta G_f^o \text{ (products)} - \sum \Delta G_f^o \text{ (reactants)}$$

The standard free energy change. $\Delta G^\circ$ is related to the equilibrium constant $k_{eq}$ by the relation;

$$\Delta G^\circ = -2.303\ RT \log k_{eq}.$$

It can be shown that free energy change for a process is equal to the maximum possible work that can be derived from the process i.e.

$\Delta G^\circ = W_{max}$ (for a reversible change at constant pressure and temperature)

In case of a galvanic cell, free energy change, $\Delta G$ is related to the electrical work done in the cell.

$\Delta G = -nFE_{cell}$ , where $E_{cell}$ = e.m.f. of the cell ; F = Faraday constant and

n = number of electrons being transferred in the chemical process

So $\Delta G = -nF E_{cell}^o$ , where $E_{cell}^o$ is the standard cell potential.

Clausius Claperyon's Equation : $\log \dfrac{p_2}{p_1} = \dfrac{\Delta H}{2.303R}\left[\dfrac{1}{T_1} - \dfrac{1}{T_2}\right]$ (For liquid $\Leftrightarrow$ gas equilibrium)

$p_1$ & $p_2$ are vapour pressure at $T_I$ & $T_2$

**THIRD LAW OF THERMODYNAMICS**

"At absolute zero, the entropy of a perfectly crystalline substance is taken as zero", which means that at absolute zero every crystalline solid is in a state of perfect order and its entropy should be zero.

By virtue of the third law, the absolute value of entropy (unlike absolute value of enthalpy) for any pure substance can be calculated at room temperature.

The standard absolute entropy of, a substance" So, is the entropy of the substance in its standard at 298K and 1 atm.

Absolute entropies of various substances have been tabulated and these value are used to calculate entropy changes for the reactions by the formula;

$$\Delta S^\circ = \sum S^\circ \text{ (products)} - \sum S^\circ \text{ (reactants)}$$

# EXERCISE-I

***First law : Heat (q), work (w) and ΔU, ΔH***

Q.1 In which of the following changes at constant pressure is work done by system on surrounding? By the surrounding on system?

| | Initial state | | Final state |
|---|---|---|---|
| (i) | $H_2O$ (g) | $\longrightarrow$ | $H_2O$ (*l*) |
| (ii) | $H_2O$ (s) | $\longrightarrow$ | $H_2O$ (g) |
| (iii) | $H_2O$ (*l*) | $\longrightarrow$ | $H_2O$ (s) |
| (iv) | $2H_2(g) + N_2(g)$ | $\longrightarrow$ | $2NH_3$ (g) |
| (v) | $CaCO_3$ (s) | $\longrightarrow$ | CaO (s) + $CO_2$ (g) |

Q.2 The gas is cooled and loses 65 J of heat. The gas contracts as it cools and work done on the system equal to 20 J is exchanged with the surroundings. What are q, w and ΔE ?

Q.3 The enthalpy change for the reaction of 50 ml of ethylene with 50.0 ml of $H_2$ at 1.5 atm pressure is ΔH = – 0.31 KJ. What is the ΔE.

Q.4 The enthalpy of combustion of glucose is – 2808 $KJmol^{-1}$ at 25°C. How many grams of glucose do you need to consume [Assume wt = 62.5 Kg].

(a) to climb a flight of stairs rising through 3M.

(b) to climb a mountain of altitude 3000 M?

Assume that 25% of enthalpy can be converted to useful work.

Q.5 What is ΔE when 2.0 mole of liquid water vaporises at 100°C ? The heat of vaporisation , ΔH vap. of water at 100°C is 40.66 $KJmol^{-1}$.

Q.6 If 1.0 k cal of heat is added to 1.2 L of $O_2$ in a cylinder of constant pressure of 1 atm, the volume increases to 1.5 L. Calculate ΔE and ΔH of the process.

Q.7 When the following reaction was carried out in a bomb calorimeter, ΔE is found to be – 742.7 kJ/mol of $NH_2CN$ (s) at 298 K.

$$NH_2CN\ (s) + \frac{3}{2} O_2\ (g) \rightarrow N_2\ (g) + CO_2\ (g) + H_2O\ (l)$$

Calculate $\Delta H_{298}$ for the reaction.

Q.8 When 1 mole of ice melt at 0°C and at constant pressure of 1 atm. 1440 calories of heat are absorbed by the system. The molar volumes of ice and water are 0.0196 and 0.0180 litre respectively. Calculate ΔH and ΔE for the reaction.

Q.9 Water expands when it freezes. Determine amount of work in joules, done when a system consisting of 1.0 L of liquid water freezes under a constant pressure of 1.0 atm and forms 1.1 L of ice.

Q.10 Lime is made commercially by decomposition of limestone $CaCO_3$. What is the change in internal energy when 1.00 mole of solid $CaCO_3$ (V = 34.2 ml) absorbs 177.9 kJ of heat and decomposes at 25°C against a pressure of 1.0 atm to give solid CaO. (Volume = 16.9 ml) and $CO_2$ (g) (V = 24.4 L).

Q.11 One mole of solid Zn is placed in excess of dilute $H_2SO_4$ at 27 °C in a cylinder fitted with a piston. Find the value of ΔE, q and w for the process if the area of piston is 500 $cm^2$ and it moves out by 50 cm against a pressure of 1 atm during the reaction. The heat given to surrounding is 36.5 KJ.

$$Zn(s) + 2H^+\ (aq) \text{ l } Zn^{2+}\ (aq) + H_2(g)$$

Q.12 Two mole of ideal diatomic gas ($C_{V,m}$ = 5/2 R) at 300 K and 5 atm expanded irreversly & adiabatically to a final pressure of 2 atm against a constant pressure of 1 atm. Calculate q, w, ΔH & ΔV.

Q.13 Five moles of an ideal gas at 300 K, expanded isothermally from an initial pressure of 4 atm to a final pressure of 1 atm against a cont. ext. pressure of 1 atm. Calculate q, w, $\Delta U$ & $\Delta H$. Calculate the corresponding value of all if the above process is carried out reversibly.

Q.14 Calculate the max. work done by system in an irreversible (single step) adiabatic expansion of 1 mole of a polyatomic gas from 300K and pressure 10 atm to 1 atm.($\gamma = 1.33$).

Q.15 1 mole of $CO_2$ gas at 300 K is expanded under reversible adiabatic condition such that its volume becomes 27 times.

(a) What is the final temperature. (b) What is work done.

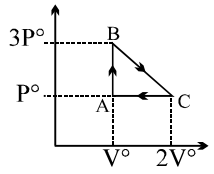
Given $\gamma = 1.33$ and $C_V = 25.08$ J $mol^{-1}K^{-1}$ for $CO_2$.

Q.16 Three moles of a ideal gas at 200 K and 2.0 atm pressure undergo reversible adiabatic compression until the temperature becomes 250 K for the gas $C_V$ is 27.5 $JK^{-1}$ $mol^{-1}$ in this temperature range. Calculate q, w, $\Delta U$, $\Delta H$ and final V and final P.

Q.17 A sample of a fluorocarbon was allowed to expand reversibly and adiabatically to twice its volume. In the expansion the temperature dropped from 298.15 K to 248.44 K. Assume the gas behaves perfectly. Estimate the value of $C_{V,m}$.

Q.18 Find the work done when one mole of the gas is expanded reversibly and isothermally from 5 atm to 1 atm at 25°C.

Q.19 One mole of ideal monoatomic gas is carried through the reversible cyclic process as shown in figure. Calculate.



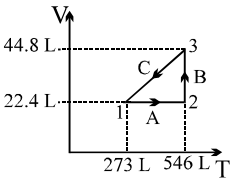

(a) work done by the gas

(b) The heat exchanged by the gas in path CA and AB.

(c) Net heat absorbed by the gas in the path BC.

(d) The max. temperature attained by the gas during the cycle.

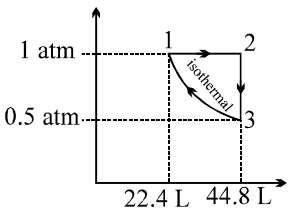
Q.20 One mole of an ideal monoatomic gas is carried through the cycle of the given figure consisting of step A, B and C and involving state 1,2 and 3. Fill in the blank space in the table given below assuming reversible steps.




| Table-1 | | | |
|---|---|---|---|
| **State** | **P** | **V** | **T** |
| 1 | | | |
| 2 | | | |
| 3 | | | |

| **Step** | **Name of process** | **q** | **w** | **$\Delta E$** | **$\Delta H$** |
|---|---|---|---|---|---|
| **A** | | | | | |
| B | | | | | |
| C | | | | | |
| | | | | | |

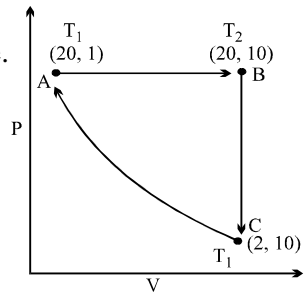
Q.21 One mole of an ideal monoatomic gas is put through rev path as shown in figure. Fill in the balnk in the table given below:




| State | P | V | T |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |

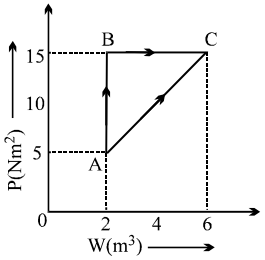
| Step | Name of process | q | w | ΔE | ΔH |
|---|---|---|---|---|---|
| A | | | | | |
| B | | | | | |
| C | | | | | |
| | | | | | |

Q.22 One mole of a perfect monoatomic gas is put through a cycle consisting of the following three reversible steps :
(CA) Isothermal compression from 2 atm and 10 litres to 20 atm and 1 litre.
(AB) Isobaric expansion to return the gas to the original volume of 10 litres with T going from $T_1$ to $T_2$.
(BC) Cooling at constant volume to bring the gas to the original pressure and temperature.
The steps are shown schematically in the figure shown.

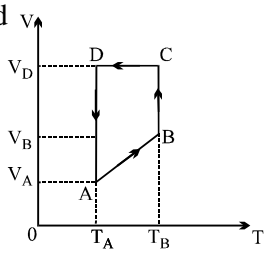



(a) Calculate $T_1$ and $T_2$.
(b) Calculate ΔE, q and W in calories, for each step and for the cycle.

Q.23 The given figure shows a change of state A to state C by two paths ABC and AC for an ideal gas. Calculate the:
(a) Path along which work done is least.
(b) Internal energy at C if the internal energy of gas at A is 10 J and amount of heat supplied to change its state to C through the path AC is 200 J.
(c) Amount of heat supplied to the gas to go from A to B, if internal energy change of gas is 10 J.




Q.24 A monoatomic ideal gas of two moles is taken through a reversible cyclic process starting from A as shown in figure. The volume ratios are $\frac{V_B}{V_A} = 2$ and $\frac{V_D}{V_A} = 4$. If the temperature $T_A$ at A is 27°C, calculate:




(a) The temperature of the gas at point B.
(b) Heat absorbed or released by the gas in each process.
(c) The total work done by the gas during complete cycle.

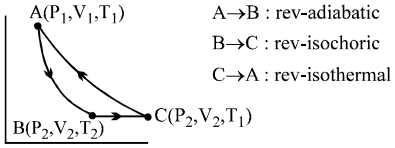
***Kirchoff's Rule : Variation of Enthalpy with Temperature***

Q.25 The standard enthalpy of formation of water liquid is – 285.76 kJ at 298 K. Calculate the value of 373K. The molar heat capacities at constant pressure ($C_P$) in the given temperature range of $H_2$ (g), $O_2$(g) and $H_2O$ (*l*) are respectively 38.83, 29.16 and 75.312 $JK^{-1}mol^{-1}$.

Q.26 Methan (Considered to be an ideal gas) initially at 25°C and 1 bar pressure is heated at constant pressure until the volume has doubled. The variation of the molart heat capacity with ansolute temperature is given by $C_P = 22.34 + 48.1 \times 10^{-3}\ T$
where $C_P$ is in $JK^{-1}\ mol^{-1}$. Calculate molar (a) ΔH (b) ΔU.

***Second law & Entropy change in thermodynamic processes***

Q.27 One mole of monoatomic gas was taken through a cylic process as shown in figure.




Calculate $\sum_{ABCA} \frac{dq_{rev}}{T}$.

Q.28 One mole of NaCl (s) on melting absorved 30.5 KJ of heat and its entropy is increased by 28.8 $JK^{-1}$. What is the melting point of sodium chloride?

Q.29 Oxygen is heated from 300 to 600 at a constant pressure of 1 bar. What is the increases in molar entropy? The molar heat capacity in $JK^{-1}$ $mol^{-1}$ for the $O_2$ is

$$C_P = 25.5 + 13.6 \times 10^{-3} T - 42.5 \times 10^{-7} T^2$$

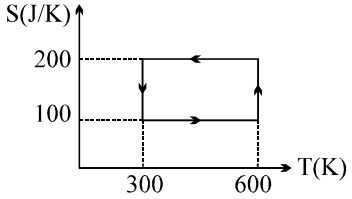
Q.30 A heat engine absorbs 760 kJ heat from a source at 380K. It rejects (1) 650 kJ, (ii) 560 kJ, (iii) 504 kJ of heat to sink at 280 K. State which of these represent a reversible, an irreversible and an impossible cycle.

Q.31 From the given T-S diagram of a reversible carnot engine, find

(i) work delivered by engine in each cycle

(ii) heat taken from the source in each cycle.

(iii) $\Delta S_{sink}$ in each cycle.




***Gibb's Function***

Q.32 Calculate the free energy change at 298 K for the reaction ;
$Br_2(l) + Cl_2(g) \longrightarrow 2\,BrCl(g)$ . For the reaction $\Delta H^\circ = 29.3$ kJ & the entropies of $Br_2(l)$, $Cl_2(g)$ & BrCl(g) at the 298 K are 152.3, 223.0, 239.7 J $mol^{-1}$ $K^{-1}$ respectively.

Q.33 Using the date given below , establish that the vaporization of $CCl_4(l)$ at 298 K to produce $CCl_4(g)$ at 1 atm pressure does not occur spontaneously .

Given : $CCl_4(l, 1\text{ atm}) \longrightarrow CCl_4(g, 1\text{ atm})$ ; $\Delta S^\circ = 94.98$ $JK^{-1}$ $mol^{-1}$;

$\Delta H^\circ_f(CCl_4, g) = -106.7$ kJ $mol^{-1}$ & $\Delta H^\circ_f(CCl_4, l) = -139.3$ kJ $mol^{-1}$

Q.34 Animals operate under conditions of constant pressure and most of the processes that maintain life are electrical (in a broad sense). How much energy is available for sustaining this type of muscular and nervous activity from the combustion of 1 mol of glucose molecules under standard conditions at 37°C (blood temperature)? The entropy change is + 182.4 $JK^{-1}$ for the reaction as stated.

$$\Delta H_{combustion\ [glucose]} = -2808 \text{ KJ}$$

Q.35 From the given table answer the following questions:

| | **CO(g)** | **$CO_2$(g)** | **$H_2O$(g)** | **$H_2$(g)** |
|---|---|---|---|---|
| $\Delta H^\circ_{298}$ (-KCal/mole) | –26.42 | –94.05 | –57.8 | 0 |
| $\Delta G^\circ_{298}$ (-KCal/mole) | –32.79 | –94.24 | –54.64 | 0 |
| $S^\circ_{298}$ (-Cal/Kmole) | 47.3 | 51.1 | ? | 31.2 |

Reaction: $H_2O(g) + CO(g) \rightleftharpoons H_2(g) + CO_2(g)$

(i) Calculate $\Delta_r H^\circ_{298}$ (ii) Calculate $\Delta_r G^\circ_{298}$ (iii) Calculate $\Delta_r S^\circ_{298}$

(iv) Calculate $\Delta_r E^\circ_{298}$ (v) Calculate $S^\circ_{298}$ $[H_2O(g)]$

# PROFICIENCY TEST

***Q.1 Mark the following statement as True or False.***

1. Pressure is an intensive property.
2. Like U and H, S is also a state function.
3. When a system undergoes a change at constant pressure, it is referred to an isothermal process.
4. A reversible process is always quasi-static.
5. The workdone by a gas during free expansion is equal to zero.
6. First law of T.D. is applicable to all processes irrespective to whether they are reversible or irreversible.
7. All spontaneous processes proceed in one direction only.
8. Whenever a system undergoes a cyclic change

$$\oint \frac{dQ}{T} \leq 0$$

9. Positive value of $\Delta S_{system}$ during the process can be taken as sole criterion of spontaneity.
10. A real crystal has higher entropy than the ideal crystal.

***Q.2 Fill in the blank with appropriate items:***

1. According to IUPAC conventions work done on the surroundings is_____ .
2. A system is said to be _____ if it can neither exchange matter nor energy with surrounding.
3. A carnot cycle uses only _____ thermal reservoir.
4. A carnot cycle consists of only _____ processes.
5. The efficiency of a carnot engine can be increased by _____ sink temperature when the source temperature is held constant.
6. For a reversible adiabatic process, S = constnat and hence it is called as an _____ process.
7. Entropy change of a system is determined by the _____ and _____ states only, irrespective of how the system has changed its states.
8. Solidification of liquid shows _____ in entropy.
9. When Fe(s) is dissolved in a aqueous HCl in a closed rigid vessel the work done is _____.
10. For Non-spontaneous process at constant T & P $\Delta G$ is _____.

## EXERCISE-II

Q.1 Calculate workdone in adiabatic compression of one mole of an ideal gas (monoatomic) from an initial ressure of 1 atm to final pressure of 2 atm. Initial temperature = 300 K.

(a) If process is carried out reversible

(b) if process is carried out irreversible against 2 atm external pressure.

Compute the final volume reached by gas in two cases and describe the work graphically.

Q.2 1 mole of ice at 0°C and 4.6 mm Hg pressure is converted to water vapour at a constant temperature and pressure. Find $\Delta H$ and $\Delta E$ if the latent heat of fusion of ice is 80 cal/gm and latent heat of vaporisation of liquid water at 0°C is 596 cal per gram and the volume of ice in comparison of that water (vapour) is neglected.

Q.3 Two moles of an ideal gas ($\gamma = 5/3$) are initially at a temperature of 27°C and occupy a volume of 20 litre. The gas is first expanded at constant pressure until the volume is doubled. It then undergoes adiabatic change until the temperature returns to its initial value.

(a) Sketch the process on P – V diagram.

(b) What are final pressure and final volume of gas.

(c) What is the work done by the gas.

Q.4 20.0 $dm^3$ of an ideal gas (diatomic $C_{v,m} = 5R/2$) at 673 K and 0.7 MPa expands until pressure of the gas is 0.2 MPa. Calculate q, w, $\Delta U$ and $\Delta H$ for the process if the expansion is :

(i) Isothermal and reversible (ii) Adiabatic and reversible

(iii) Isothermal and adiabatic (iv) Against 0.2 MPa and adiabatic

(v) Against 0.2 MPa and isothermal.

Q.5 One mole of an ideal monoatomic gas ($C_{V,m} = 1.5\ R$) is subjected to the following sequence of steps:

(a) The gas is heated reversibly at constant pressure of 101.325 kPa from 298 K to 373 K.

(b) Next, the gas is expanded reversibly and isothermally to double its volume.

(c) Finally, the gas is cooled reversibly and adiabatically to 308K.

Calculate q , w , $\Delta U$ and $\Delta H$ for the overall process.

Q.6 Calculate $\Delta S_f^\circ$ at 298 K of ; (i) NaCl(s) , (ii) $NH_4Cl$(s) & (iii) diamond.

The values of $S^\circ$ of Na, $Cl_2$, NaCl, $NH_4Cl$, $N_2$, $H_2$, diamond & graphite are 51, 223, 72, 95, 192, 131, 2.43 & 5.69 $JK^{-1}\ mol^{-1}$ respectively.

Q.7 One mole of an ideal gas is expanded isothermally at 298 K until its volume is tripled. Find the values of $\Delta S_{gas}$ and $\Delta S_{total}$ under the following conditions.

(i) Expansion is carried out reversibly.

(ii) Expansion is carried out irreversibly where 836.8J of heat is less absorbed than in (i)

(iii) Expansion is free.

Q.8 10 g of neon initially at a pressure of 506.625 kPa and temperature of 473 K expand adiabatically to a pressure of 202.65 kPa. Calculate entropy change of the system and total entropy change for the following ways of carrying out this expansion.

(i) Expansion is carried out reversibly.

(ii) Expansion occurs against a constant external pressure of 202.65 kPa.

(iii) Expansion is a free expansion.

Q.9 One mole of an ideal gas (not necessarily monoatomic) is subjected to the following sequence of steps.

(a) It is heated at constant volume from 298 K to 373 K

(b) It is expanded freely into a vacuum to double volume.

(c) It is cooled reversibly at constant pressure to 298 K.

Calculate q, w , $\Delta U$ and $\Delta H$ for the overall process.

Q.10 Calculate the heat of vaporisation of water per gm at 25°C and 1 atm.

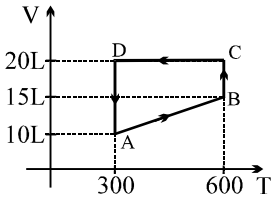
Given $\Delta H_f^\circ[H_2O(l)] = -285.57$ kJ/mol, $\Delta H_f^\circ[H_2O(g)] = -241.6$ kJ/mol.

Comment why $\Delta H_{vap}$ (25°C) > $\Delta H_{vap}$ (100°C).

Use data of **Q.20**

Q.11 Pressure over 1000 ml of a liquid is gradually increases from 1 bar to 1001 bar under adiabatic conditions. If the final volume of the liquid is 990 ml, calculate ΔU and ΔH of the process, assuming linear variation of volume with pressure.

Q.12 One mole monoatomic ideal gas was taken through process ABCD as shown in figure. Calculate




(i) $w_{AB}$, $w_{BC}$ , $w_{CD}$ , $w_{DA}$

(ii) $q_{AB}$, $q_{BC}$, $q_{CD}$, $q_{DA}$

(iii) $\Delta H_{AB}$, $\Delta H_{BC}$, $\Delta H_{CD}$, $\Delta H_{DA}$

[Use : $l$n (3/2) = 0.40 ; $l$n (4/3) = 0.29]

Q.13 One mole of ideal monoatomic gas was taken through reversible isochoric heating from 100 K to 1000 K. Calculate $\Delta S_{system}$ , $\Delta S_{surr}$, and $\Delta S_{total}$ in

(i) when the process carried out reversibly

(ii) when the process carried out irreversibly (one step)

Q.14 Calculate the entropy of a substance at 600 K using the following data.

(i) Heat capacity of solid from 0 K to normal melting point 200 K

$C_{P,m}(s) = 0.035\ T$ $JK^{-1}mol^{-1}$.

(ii) Enthalpy of fusion = 7.5 KJ $mol^{-1}$, (iii)Enthalpy of vaporisation = 30 KJ $mol^{-1}$.

(iv) Heat capacity of liquid from 200 K to normal boiling point 300 K

$C_{P,m}(l) = 60 + 0.016\ T$ $JK^{-1}mol^{-1}$.

(v) Heat capacity of gas from 300 K to 600 K at 1 atm

$C_{P,m}(g) = 50.0$ $JK^{-1}mol^{-1}$.

Q.15(a) An ideal gas undergoes a single stage expansion against a constant opposing pressure from ($P_1$, $V_1$, T) to ($P_2$, $V_2$, T). What is the largest mass m which can be lifted through a height h in this expansion?

(b) The system in (a) restored to its initial state by a single stage compression. What is the smallest mass m' which must fall through the height h to restore the system?

(c) What is the net mass lowered through height h in the cyclic transformation in (a) and (b)?

Q.16 Calculate the free energy change in the freezing of 18 gm of water at 263.15 K, given that the vapour pressure of water and ice at 263.15 K are 0.287 Pa and 0.260 Pa, respectively.

Q.17 A 32 g sample of $CH_4$ gas initially at 101.325 kPa and 300K is heated to 550 K. $C_{P,m}/JK^{-1}mol^{-1} = 12.552 + 8.368 \times 10^{-2}\ T/K$. Assuming $CH_4$ behaves ideally, compute w, q . ΔU and ΔH for (a) an isobaric reversible process, and (b) an isochoric reversible process.

Q.18 Derive a mathematical expression for the work done on the surrounding when a gas that has the equation of state $PV = nRT - \frac{n^2a}{V}$ expands reversibly from $V_i$ to $V_f$ at constant temperature.

Q.19 For the reaction

$FeCO_3(s) = FeO(g) + CO_2(g)$

$\Delta_r G^\circ/J\ mol^{-1} = 78073.4 - 60.33\ (T/K) \log (T/K) - 25.397\ (T/K) + 34.476 \times 10^{-3}\ (T/K)^2$

find $\Delta_r H^\circ$ and $\Delta_r S^\circ$ for the reaction at 25°C.

Q.20 Compute $\Delta_r G$ for the reaction $H_2O$ ($l$, 1 atm, 323 K) → $H_2O$ (g, 1 atm, 323 K)

Given that : $\Delta_{vap}H$ at 373 K = 40.639 $kJmol^{-1}$, $C_P(H_2O, l) = 75.312\ J\ K^{-1}mol^{-1}$,

$C_p(H_2O, g) = 33.305\ J\ K^{-1}mol^{-1}$.

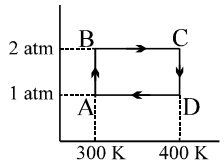
## EXERCISE-III

Q.1 Out of boiling point (I), entropy (II), pH (III) and e.m.f. of a cell (IV) Intensive properties are:
(A) I, II (B) I,II, III (C) I, III, IV (D) All of these

Q.2 Which has maximum internal energy at 298 K?
(A) helium gas (B) oxygen gas (C) ozone gas (D) equal

Q.3 Ethyl chloride ($C_2H_5Cl$), is prepared by reaction of ethylene with hydrogen chloride:
$C_2H_4(g) + HCl(g) \longrightarrow C_2H_5Cl(g)$ $\Delta H = -72.3$ kJ
What is the value of $\Delta E$ (in kJ), if 98 g of ethylene and 109.5 g of HCl are allowed to react at 300 K.
(A) – 64.81 (B) –190.71 (C) –209.41 (D) – 224.38

Q.4 Two moles of Helium gas undergo a reversible cyclic process as shown in figure. Assuming gas to be ideal, what is the net work involved in the cyclic process?




(A) –100 R*l*n4 (B) +100R*l*n4
(C) +200R*l*n4 (D) –200R*l*n4

Q.5 Benzene burns according to the following equation
$2C_6H_6(l) + 15\ O_2(g) \longrightarrow 12\ CO_2(g) + 6H_2O(l)$ $\Delta H^\circ = -6542$ kJ
What is the $\Delta E^\circ$ for the combustion of 1.5 mol of benzene
(A) –3271 kJ (B) –9813 kJ (C) – 4906.5 kJ (D) None of these

Q.6 One mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of 27°C. If the work done by the gas in the process is 3 kJ, the final temperature will be equal to ($C_V$=20 J/K mol)
(A) 100 K (B) 450 K (C) 150 K (D) 400 K

Q.7 Two moles of an ideal gas ($C_V = \frac{5}{2}R$) was compressed adiabatically against constant pressure of 2 atm. Which was initially at 350 K and 1 atm pressure. The work involve in the process is equal to
(A) 250 R (B) 300 R (C) 400 R (D) 500 R

Q.8 The maximum efficiency of a heat engine operating between 100°C and 25°C is
(A) 20% (B) 22.2% (C) 25% (D) none

Q.9 A heat engine operating between 227°C and 27°C absorbs 2 Kcal of heat from the 227°C reservoir reversibly per cycle. The amount of work done in one cycle is
(A) 0.4 Kcal (B) 0.8 Kcal (C) 4 Kcal (D) 8 Kcal

Q.10 A reversible heat engine A (based on carnot cycle) absorbs heat from a reservoir at 1000K and rejects heat to a reservoir at $T_2$. A second reversible engine B absorbs, the same amount of heat as rejected by the engine A, from the reservoir at $T_2$ and rejects energy to a reservoir at 360K.
If the efficiencies of engines A and B are the same then the temperature $T_2$ is
(A) 680 K (B) 640 K (C) 600 K (D) none

Q.11 For the reaction at 300 K

$A(g) + B(g) \longrightarrow C(g)$

$\Delta E = -3.0$ kcal ; $\Delta S = -10.0$ cal/K

value of $\Delta G$ is

(A) –600 cal (B) –6600 cal (C) –6000 cal (D) None

Q.12 The entropy change when two moles of ideal monoatomic gas is heat from 200 to 300°C reversibly and isochorically

(A) $\frac{3}{2} R \ln\left(\frac{300}{200}\right)$ (B) $\frac{5}{2} R \ln\left(\frac{573}{273}\right)$ (C) $3R \ln\left(\frac{573}{473}\right)$ (D) $\frac{3}{2} R \ln\left(\frac{573}{473}\right)$

Q.13 What is the free energy change ($\Delta G$) when 1.0 mole of water at 100°C and 1 atm pressure is converted into steam at 100°C and 1 atm pressure?

(A) 80 cal (B) 540 cal (C) 620 cal (D) zero

Q.14 What is the free energy change ($\Delta G$) when 1.0 mole of water at 100°C and 1 atm pressure is converted into steam at 100°C and 2 atm pressure?

(A) zero cal (B) 540 cal (C) 515.4 cal (D) none

Q.15 When two equal sized pieces of the same metal at different temperatures $T_h$ (hot piece) and $T_c$(cold piece) are brought into contact into thermal contact and isolated from it's surrounding. The total change in entropy of system is given by

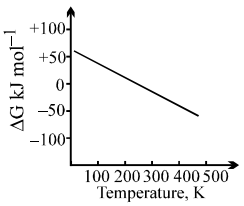
(A) $C_v \ln \frac{T_c + T_h}{2T_c}$ (B) $C_v \ln \frac{T_2}{T_1}$ (C) $C_v \ln \frac{(T_c + T_h)^2}{2T_h . T_c}$ (D) $C_v \ln \frac{(T_c + T_h)^2}{4T_h . T_c}$

Q.16 Pick out the correct statement among the following.

(A) $\Delta S_f^o \{He(g)\} > 0$ at 298 K (B) $\Delta S_f^o \{H_2O(g)\} > 0$ at 298 K

(C) S° of $H_2$ gas > 0 at 298 K (D) $\Delta G_f^o \{H_2(g)\} > 0$ at 298 K

Q.17 What can be concluded about the values of $\Delta H$ and $\Delta S$ from this graph?




(A) $\Delta H > 0, \Delta S > 0$ (B) $\Delta H > 0, \Delta S < 0$

(C) $\Delta H < 0, \Delta S > 0$ (D) $\Delta H < 0, \Delta S < 0$

Q.18 Which of the following statement(s) is/are incorrect:

Statement (a) : Reversible isothermal compression of an ideal gas represents the limiting minimum value of the workdone (|w|) by the surrounding on the system.

Statement (b) : In an irreversible process, the cyclic integral of work is not zero.

Statement (c) : For thermodynamic changes in adiabatic process $T^{\left(\frac{C_{p,m}}{R}\right)} . P = \text{constant}$

Statement (d) : $\Delta S_{system}$ is zero for reversible adiabatic expansion of an ideal gas.

(A) Statement c (B) Statement a, b, c (C) Statement a, b, d (D) All

Q.19 9.0 gm ice at 0°C is mixed with 36 gm of water at 50°C in a thermally insulated container.
Using the following data, answer the question that follow
$C_P\ (H_2O) = 4.18\ Jg^{-1}K^{-1}$ ; $\Delta H_{fusion}$ (ice) = 335 J $g^{-1}$

(i) final temperature of water is
(A) 304.43 K (B) 296.97 K (C) 303.93 K (D) 287 K

(ii) $\Delta S_{ice}$ is
(A) 11.04 $JK^{-1}$ (B) 3.16 $JK^{-1}$ (C) 14.2 $JK^{-1}$ (D) 7.84 $JK^{-1}$

(iii) $\Delta S_{water}$ is
(A) –12.64 $JK^{-1}$ (B) –0.34 $JK^{-1}$ (C) –5.42 $JK^{-1}$ (D) 12.64 $JK^{-1}$

(iv) What is the total entropy change in the process?
(A) –1.56 $JK^{-1}$ (B) –1.60 $JK^{-1}$ (C) 1.56 $JK^{-1}$ (D) 1.60 $JK^{-1}$

Q.20 Liquid water freezes at 273 K under external pressure of 1 atm. The process is at equilibrium
$H_2O\ (l)$ ⥮ $H_2O$ (s) at 273 K & 1 atm.
However it was required to calculate the thermodynamic parameters of the fusion process occuring at same pressure & different temperature. Using the following data, answer the question that follow.

$d_{ice}$ = 0.9 gm/cc ; $d_{H_2O(l)}$ = 1 gm/cc ; $C_P\ [H_2O\ (s)] = 36.4\ JK^{-1}mol^{-1}$ ;
$C_P\ [H_2O\ (l)] = 75.3\ JK^{-1}mol^{-1}$ ; $\Delta H_{fusion} = 6008.2\ Jmol^{-1}$.

(i) The value of "$\Delta H_{fusion}$" at 263 K & 1 atm will be
(A) +6008.2 J $mole^{-1}$ (B) 5619.2 J $mole^{-1}$ (C) –5619.2 J $mole^{-1}$ (D) 6619.2 J $mole^{-1}$

(ii) "$\Delta S_{fusion}$" at 263 K & 1 atm will be
(A) 22.01 $JK^{-1}\ mol^{-1}$ (B) 22.84 $JK^{-1}\ mol^{-1}$ (C) 21.36 $JK^{-1}\ mol^{-1}$ (D) 20.557 $KJ^{-1}\ mol^{-1}$

(iii) At 1 atm & at differnt temperature given below. Match the conditions & the temperature for the "fusion" process

| | **Condition** | | **Temperature** |
|---|---|---|---|
| (1) | Spontaneous | (a) | 273 K |
| (2) | At equilibrium | (b) | 260 K |
| (3) | Not feasible | (c) | 280 K |

(A) (1–c), (2–a), (3–b) (B) (1–b), (2–a), (3–c)
(C) (1–c), (2–b), (3–a) (D) (1–a), (2–b), (3–c)

(iv) For the fusion process at 263 K, match the conditions with the pressure

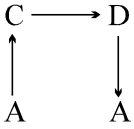
| | **Conditions** | | **Pressure** |
|---|---|---|---|
| (1) | Spontaneous | (a) | 1 atm |
| (2) | At equilibrium | (b) | 1054 atm |
| (3) | Not feasible | (c) | 2000 atm |

(A) (1–b), (2–c), (3–a) (B) (1–a), (2–b), (3–c)
(C) (1–c), (2–b), (3–a) (D) (1–a), (2–c), (3–b)

# EXERCISE-IV

## OBJECTIVE

Q.1 A process A ⟶ B is difficult to occur directly instead it takes place in three successive steps.




$\Delta S$ (A ⟶ C ) = 50 e.u.
$\Delta S$ (C ⟶ D ) = 30 e.u.
$\Delta S$ (B ⟶ D ) = 20 e.u.

where e.u. is entropy unit.
Then the entropy change for the process $\Delta S$ (A⟶ B) is
(A) +100 e.u. (B) –60 e.u. (C) –100 e.u. (D) + 60 e.u. **[JEE 2006]**

Q.2 The molar heat capacity of a monoatomic gas for which the ratio of pressure and volume is one.
(A) 4/2 R (B) 3/2 R (C) 5/2 R (D) zero **[JEE 2006]**

Q.3 One mole of monoatomic ideal gas expands adiabatically at initial temp. T against a constant external pressure of 1 atm from one litre to two litre. Find out the final temp. (R = 0.0821 litre. atm $K^{-1}$ $mol^{-1}$)

(A) T (B) $\frac{T}{(2)^{\frac{5}{3}-1}}$

(C) $T - \frac{2}{3\times0.0821}$ (D) $T + \frac{2}{3\times0.0821}$ **[JEE 2005]**

Q.4 Two mole of an ideal gas is expanded isothermally and reversibly from 1 litre to 10 litre at 300 K. The enthalpy change (in kJ) for the process is
(A) 11.4 kJ (B) –11.4 kJ
(C) 0 kJ (D) 4.8 kJ **[JEE 2004]**

Q.5 The enthalpy of vapourization of a liquid is 30 kJ $mol^{-1}$ and entropy of vapourization is 75 J $mol^{-1}$ K. The boiling point of the liquid at 1 atm is
(A) 250 K (B) 400 K
(C) 450 K (D) 600 K **[JEE 2004]**

Q.6 One mol of non-ideal gas undergoes a change of state (2.0 atm, 3.0 L, 95 K) to (4.0 atm, 5.0 L, 245 K) with a change in internal energy ($\Delta U$) = 30.0 L-atm. The change in enthalpy ($\Delta H$) of the process in L-atm.
(A) 40.0
(B) 42.3
(C) 44.0
(D) not defined, because pressure is not constant **[JEE 2002]**

Q.7 Which of the following statement is false?
(A) Work is a state function
(B) Temperature is a state function
(C) Change of state is completely defined when initial and final states are specified.
(D) Work appears at the boundary of the solution. **[JEE 2001]**

Q.8 Molar heat capacity of water in equilibrium with ice at constant pressure is
(A) zero (B) ∞
(C) 40.45 kJ $K^{-1}$ $mol^{-1}$ (D) 75.48 $JK^{-1}$ $mol^{-1}$ **[JEE 1997]**

## *SUBJECTIVE*

Q.9 For the reaction,
$2CO(g) + O_2(g) \longrightarrow 2CO_2(g)$; $\Delta H = -560$ kJ $mol^{-1}$
In one litre vessel at 500 K the initial pressure is 70 atm and after the reaction it becomes 40 atm at constant volume of one litre. Calculate change is internal energy. All the above gases show significant deviation from ideal behaviour. (1 L atm = 0.1 kJ) **[JEE 2006]**

Q.10 One mole of a liquid (1 bar, 100 ml) is taken in an adiabatic container and the pressure increases steeply to 100 bar. Then at a constant pressure of 100 bar, volume decreases by 1 ml. Find $\Delta U$ and $\Delta H$ **[JEE 2004]**

Q.11 Two moles of a perfect gas undergoes the following processes : **[JEE 2002]**
(a) a reversible isobaric expansion from (1.0 atm, 20.0 L) to (1.0 atm, 40.0 L);
(b) a reversible isochoric change of state from (1.0 atm, 40.0 L) to (0.5 atm, 40.0 L);
(c) a reversible isothermal compression from (0.5 atm, 40.0 L) to (1.0 atm, 20.0 L);
(i) Sketch with labels each of the processes on the same P-V diagram.
(ii) Calculate the total work (w) and the total heat change (q) involved in the above processes.
(iii) What will be the values of $\Delta U$, $\Delta H$ and $\Delta S$ for the overall process ?

Q.12 Show that the reaction $CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$ at 300 K is spontaneous and exothermic, when the standard entropy is $-0.094$ kJ $mol^{-1}$ $K^{-1}$. The standard Gibbs free energies of formation for $CO_2$ and CO are $-394.4$ and $-137.2$ kJ $mol^{-1}$, respectively. **[JEE 2001]**

Q.13 A sample of argon gas at 1 atm pressure and 27°C expands reversibly and adiabatically from 1.25 $dm^3$ to 2.50 $dm^3$. Calculate the enthalpy change in this process. $C_{v.m.}$ for argon is 12.48 $JK^{-1}$ $mol^{-1}$. **[JEE 2000]**

# *ANSWER KEY*
# *EXERCISE-I*

***Heat (q), work (w) and ΔU, ΔH (First Law)***

Q.1 (i) w, (ii) – w, (iii) – w, (iv) w, (v) – w Q.2 q = – 65 J ; w = 20 J; ΔE = – 45 J

Q.3 – 0.3024 kJ Q.4 (a) 0.47 gm, (b) 0.47 kg Q.5 ΔE =75.11 kJ

Q.6 ΔE = 0.993 k cal, ΔH = 1 k cal Q.7 – 741.5 kJ

Q.8 ΔH ≅ ΔE = 1440 calories Q.9 – 10 J

Q.10 q = 177.9 kJ, w = – 2.5 kJ ; ΔE = 175.4 kJ

Q.11 DE = – 39.03 KJ/mole; q = – 36.5 KJ;w = – 2.53 KJ

Q.12 ΔU = w = –1247.1 ; ΔH = – 1745.94 J

Q.13 $w_{irr}$ = – 9353.25, $w_{rev}$ = –17288.47 J, ΔU = ΔH =0 Q.14 – 1.683 kJ

Q.15 $T_2$ = 100 K ; W = 5.016 KJ

Q.16 q = 0; w =ΔU= 4.12 KJ; ΔH = 5.37 KJ; $V_f$ = 11.8 $dm^3$; P = 5.21 atm

Q.17 $C_{V,m}$ = 31.6 $JK^{-1}mol^{-1}$ Q.18 w = – 3.988 kJ

Q.19 $W = P_0V_0$ ; $q_{CA} = -\frac{5}{2}P°V°$ ; $q_{AB} = 3P°V°$ ; $q_{BC} = \frac{1}{2}P°V°$ $T_{max} = \frac{25}{8}\left(\frac{P°V°}{R}\right)$

Q.20

| Table-1 | | | |
|---|---|---|---|
| **State** | **P** | **V** | **T** |
| 1 | 1 atm | 22.4 | 273 |
| 2 | 2 atm | 22.4 | 546 |
| 3 | 1 atm | 44.8 | 546 |

| **Step** | **Name of process** | **q** | **w** | **ΔE** | **ΔH** |
|---|---|---|---|---|---|
| **A** | Isochoric | 3/2 R(273) | 0 | 3/2 R(273) | 5/2 R(273) |
| B | Isothermal | R ln 2 | R ln 2 | 0 | 0 |
| C | Isobaric | –5/2 R (273) | R (273) | –3/2 R (273) | –5/2 R (273) |
| | Cyclic | –R(273) + R ln 2 | R(273)-R ln 2 | 0 | 0 |

Q.21

| **State** | **P** | **V** | **T** |
|---|---|---|---|
| 1 | 1 atm | 22.4 | 273 |
| 2 | 1 | 44.8 | 546 |
| 3 | 0.5 | 44.8 | 273 |

| **Step** | **Name of process** | **q** | **w** | **ΔE** | **ΔH** |
|---|---|---|---|---|---|
| **A** | Isobaric | 5/2 R(273) | – R (273) | 3/2 R(273) | 5/2 R(273) |
| B | Isochoric | –3/2 R (273) | 0 | –3/2 R (273) | –5/2 R(273) |
| C | Isothermal | –R ln 2 | R ln 2 | 0 | 0 |
| | Cyclic | –R(273) + R ln 2 | –R(273)–R ln 2 | 0 | 0 |

]

Q.22 (a) $T_1$ = 243.60 K; $T_2$ = 2436.0 K, (b) ΔE = 0; q = –w = +3262.88 cal

Q.23 (a) AC, (b) 170 J, (c) 10 J

Q.24 (a) 600 K, (b) $q_{AB}$ = 3000 cal; $q_{BC}$=1663 cal; $q_{CD}$= –1800 cal; $q_{DA}$=–1663 cal; Total Q = 1200 cal, (c) W = –1200 cal

***Kirchoff's Rule : Variation of Enthalpy with Temperature***

Q.25 $\Delta H^o_{373}$ ($H_2O$ (*l*) )= – 284.11 kJ Q.26 (a) 13.064 kJ $mol^{-1}$, (b) 10.587 kJ $mol^{-1}$

***Second law & Entropy change in thermodynamic processes***

Q.28 T = 1059 K Q.29 21.18 $JK^{-1}$ $mol^{-1}$ Q.30 (i) irreversible, (ii) reversible, (iii) impossible

Q.31 (i) 30 kJ, (ii) + 60 kJ, (iii) 100 J/K

***Gibb's Function***

Q.32 – 1721.8 J  Q.33 $\Delta G^\circ = 4.3$ kJ $mol^{-1} > 0$  Q.34 – 2864.5 KJ

Q.35 (i) – 9.83 Kcal/mole; (ii) – 6.81 Kcal/mole, (iii) – 10.13 Cal / Kmole, (iv) –9.83 Kcal/mole, (v) + 45.13 Cal/ K mole

## PROFICIENCY TEST

***Q.1***

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1. | T | 2. | T | 3. | F | 4. | T | 5. | T |
| 6. | T | 7. | T | 8. | T | 9. | F | 10. | T |

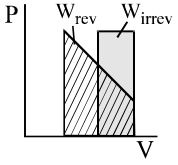
***Q.2***

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | negative | 2. | isolated | 3. | two | 4. | reversible |
| 5. | decresing | 6. | isentropic | 7. | Initial, final | 8. | decrease |
| 9. | zero | 10. | positive | | | | |

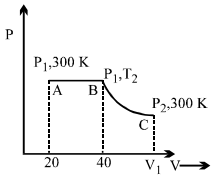
## EXERCISE II

Q.1 (a) $T_2 = 395.8$ ; $V_2 = 16.24$ L; $W_{rev} = 1194.72$ J,

(b) $V_2^1 = 17.24$ L ; $T_2^1 = 420$ K , $W_{irrev} = 1496.52$ J




Q.2 $\Delta H = 12168$ calories; $\Delta E = 11623$ calories

Q.3 (a) 

, (b) $P_2 = 0.435$ atm $V_1 = 113.13$ litre, (c) $W_T = -3000$ cal

Q.4 (i) $q = -w = 17.54$ kJ , $\Delta U = 0$ and $\Delta H = 0$ ; (ii) $q = 0$ , $w = \Delta U = -10.536$ kJ and $\Delta H = -14.75$kJ (iii) $q = 0$ , $w = 0$ , $\Delta U = 0$ and $\Delta H = 0$ (iv) $q = 0$; $\Delta U = w = -7.14$ KJ; $\Delta H = -9.996$ KJ, (v) $q = -w = 10.0$ KJ, $\Delta U = \Delta H = 0$

Q.5 (a) $q = \Delta H = 1558.88$, $\Delta U = 935.33$ ; $w = -P(\Delta U) = -623.55$ J $mol^{-1}$
(b) $w = -2149.7$ ; $\Delta U$ & $\Delta H = 0$, $q = -w$
(c) $q = 0$, $w = -810.62$ , $\Delta H = -1351.03$ $Jmol^{-1}$
for overall process $q = 3708.59$ ; $w = -3583.88$, $\Delta U = 124.71$ ; $\Delta H = 207.85$

Q.6 (i) −90.5 (ii) −374.5 (iii) −3.26 (all in J $mol^{-1}$ $K^{-1}$)

Q.7 (i) $\Delta S_{gas} = -\Delta S_{surr}$ and $\Delta S_{total} = 0$, (ii) $\Delta S_{total} = 2.808$ J $K^{-1}$ (iii) $\Delta S_{total} = \Delta S_{sys} = 9.134$ J $K^{-1}$

Q.8 (i) $\Delta S_{sys} = 0$ ; $\Delta S_{surr} = 0$ and $\Delta S_{total} = 0$, (ii) $\Delta S_{surr} = 0$ ; $\Delta S_{total} = \Delta S_{sys} = 0.957$ $JK^{-1}$
(iii) $\Delta S_{sys} = \Delta S_{total} = 3.81$ $JK^{-1}$

Q.9 $\Delta U$ & $\Delta H = 0$ ; $w = 623.55$ ; $q = -623.55$ J $mol^{-1}$

Q.10 $\Delta H_{vap}(25°C) = 43.97$ kJ/mol $= 2.433$ kJ/gm, $\Delta H_{vap}(100°C) = 40.62$ kJ/mol (given)

Q.11 $\Delta U = 501$ J ; $\Delta H = 99.5$ kJ

Q.12 (i) $w_{AB} = -1496.52$ J, $w_{BC} = -1446.63$ Joule, $w_{CD} = 0$, $w_{DA} = 1728.84$ Joule ;
(ii) $q_{AB} = 5237.82$ Joule, $q_{BC} = 1446.63$ Joule, $q_{CD} = -3741.3$ Joule, $q_{DA} = 1728.84$ Joule;
(iii) $\Delta H_{AB} = 6235.5$ Joule, $\Delta H_{BC} = 0$ , $\Delta H_{CD} = -6235.5$ Joule, $\Delta H_{DA} = 0$ Joule

Q.13 (i) Rev. Process $\Delta S_{syst} = \frac{3}{2} R \ln 10$ ; $\Delta S_{surr} = -\frac{3}{2} R \ln 10$

(ii) Irr Process $\Delta S_{sys} = \frac{3}{2} R \ln 10$ ; $\Delta S_{surr} = \frac{3}{2} R (0 - 9)$ ; $\Delta S_{total} = \frac{3}{2} R (1.403)$

Q.14 205.08 $JK^{-1} mol^{-1}$

Q.15 (a) $m = \frac{nRT}{gh}\left(1 - \frac{P_2}{P_1}\right)$, (b) $m' = \frac{nRT}{gh}\left(\frac{P_1}{P_2} - 1\right)$, (c) $m' - m = \frac{nRT}{gh}\left(\frac{(P_1 - P_2)^2}{P_1 P_2}\right)$

Q.16 – 216.198 J $mol^{-1}$

Q.17 (a) $q_P = \Delta H = 24.058$ kJ , w = – 4.157 kJ , $\Delta U$ = 19.90 kJ ; (b) $\Delta U$=19.90 kJ ; $\Delta H$=24.058 ; w = 0

Q.18 $w = -nRT\, ln \frac{V_f}{V_i} - n^2 a\left(\frac{1}{V_f} - \frac{1}{V_i}\right)$

Q.19 $\Delta_r H° = 82.801$ kJ $mol^{-1}$, $\Delta_r S° = 180.33$ J $K^{-1}$ $mol^{-1}$ Q.20 $\Delta_r G = 5.59$ kJ $mol^{-1}$

## EXERCISE III

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Q.1 | C | Q.2 | C | Q.3 | C | Q.4 | A | Q.5 | D | Q.6 | C | Q.7 | D |
| Q.8 | A | Q.9 | B | Q.10 | C | Q.11 | A | Q.12 | C | Q.13 | D | Q.14 | C |
| Q.15 | D | Q.16 | C | Q.17 | A | Q.18 | A | Q.19 | (i) B (ii) C (iii) A (iv) C | | | | |

Q.20 (i) B (ii) D (iii) A (iv) C

## EXERCISE IV

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Q.1 | D | Q.2 | A | Q.3 | C | Q.4 | C |
| Q.5 | B | Q.6 | C | Q.7 | A | Q.8 | B |

Q.9 –557 kJ/mol

Q.10 $\Delta U$ = 0.1 litre atm, $\Delta H$ = 9.9 litre atm

Q.11 (ii) –W = q = 620.77 J, (iii) $\Delta H = 0$, $\Delta U = 0$, $\Delta S = 0$

Q.12 $\Delta H° = -285.4$ kJ/mol, $\Delta G° = -257.2$ kJ/mol Q.13 $\Delta H = -114.52$ J

# STUDY PACKAGE

# Target: IIT-JEE (Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 17 IUPAC Nomenclature

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

**4. Exercise III**

**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

# ***IUPAC***

Nomenclature according to IUPAC system involves use of following terms:

(i) Word root : The word root represents the number of C atoms in parent chain.

| No. of C atom | W.R. | No. of C atom | W.R. |
|---|---|---|---|
| 1 | Meth | 8 | Oct |
| 2 | Eth | 9 | Non |
| 3 | Prop. | 10 | Dec |
| 4 | But | 11 | Undec |
| 5 | Pent | 12 | Dodec |
| 6 | Hex | 13 | Tridec |
| 7 | Hep | | |

(ii) Primary suffix : Primary suffix is used to indicate saturation or unsaturation in carbon chain.
While writing name, primary suffix is added to word root.

| | Nature of C chain | Primary suffix |
|---|---|---|
| (1) | Saturated C chain | ane |
| (2) | Unsaturated C chain | |
| | C=C | ene |
| | C≡C | yne |
| | 2 C = C | diene |
| | 2 C≡C | diyne |

(iii) Secondary suffix : Secondary suffix is used to indicate functional group in organic comp. It is added primary suffix by dropping its terminal "e".

Prefix : The part of the name C appears before the word root is called prefix. Different prefixes are used for dif categories of group as:

(a) **Alkyl groups:**

$CH_3–CH_2–CH_2$ ⟶ 1-propyl

$CH_3–CH(–CH_3)$ ⟶ 2-propyl

$CH_3–CH_2–CH_2–CH_2$ ⟶ 1-Butyl

$CH_3–CH_2–CH(–CH_3)$ ⟶ 2-Butyl

$CH_3–CH(CH_3)–CH_2$ ⟶ 2-Methyl-1-propyl

(b) In IUPAC system,for nomenclature some groups are not considered as functional group but treated as substituent. These functional group are always indicated by prefixes instead of secondary suffixes.

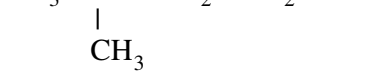
| | |
|---|---|
| $–NO_2$ | Nitro |
| –OR | Alkoxy |
| –Cl | Chloro |
| –Br | Bromo |
| –I | Iodo |
| –F | Fluoro |
| –N=O | Nitroso |
| $–NO_2$ | Nitro |
| =N | Diazo |

(c) In polyfunctional compound, one of the functional group is treated as principal functional group & indicated by secondary suffix while other functional groups are treated as substituents & indicated by prefixes.

| G.F | O.Co. | Suffix | IUPAC Name | Prefix |
|---|---|---|---|---|
| R–OH | Alcohols | ol | alkanol | Hydroxy |
| R–SH | Thioalcohols | thiol | alkanethiol | Mercapto |
| $R–NH_2$ | Amines | amine | alkanamine | Amino |
| R–CHO | Aldehyde | al | alkanal | formyl |
| RCOR | Ketone | one | alkanone | Keto or oxo |
| RCOOH | carboxylic acid | -oic acid | alkanoic acid | Carboxy |
| Amides | $RCONH_2$ | amide | alkanamide | Carbamoyl |
| Acid halide | RCOX | oyl halide | alkanoyl halide | haloformyl |
| Ester | RCOOR | oate | alkylalkanoate | Carbalkoxy |
| Nitriles | R–C≡N | nitrile | alkanenitrile | Cyano |
| Isonitrile | R–NC | isonitrile | alkane isonitrle | Carbylamino |

**Arrangement :**

Prefix(es) + word root + p. suffix + sec. suffix



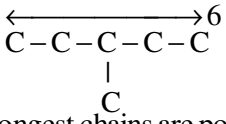
Methyl + but + an + ol

**Rules**:

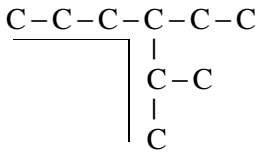
For saturated compounds:

(i) **Selection of longest chain** :

(a) Longest continuous chain of carbon atoms is selected. This is called parent chain while all other C atoms C are not included in parent chain is called side chain.

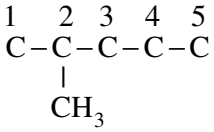


(b) If more than one set of longest chains are possible, the chain C max. no. of substituant acts as parent chain.

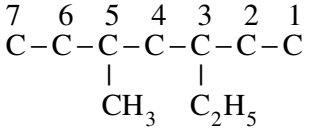


(ii) **Numbering of selected chain** :

(a) The selected chain is numbered from one end to other. The number are called locants. Numbering is done in such a way that lowest no. is assigned to side chain or substituent



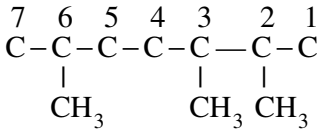
(b) If two different alkyl groups or substituents are at same position from opposite ends, lowest no. is given in alphabetical order.

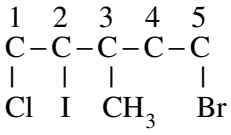


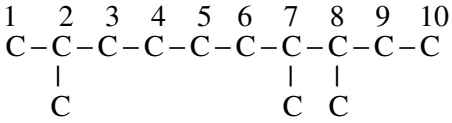
(c) If two different substituents are at same position from opp. ends, lowest no. is assigned in order of their alphabets.



(d) The numbering is done is such a way that the substituted carbon atoms have the lowest possible numbers. Where series of locants conatining the same no. of terms are compared term by term, the chosen series should contain the number on the occasion of first difference.

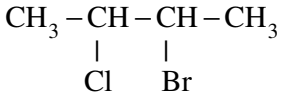




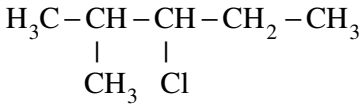


(iii) **Arrangement of prefixes** :

(a) Side chain or substituent group are added as prefix $\overline{C}$ its locant in alphabetical order.

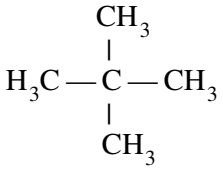


2-bromo-3-chlorobutane



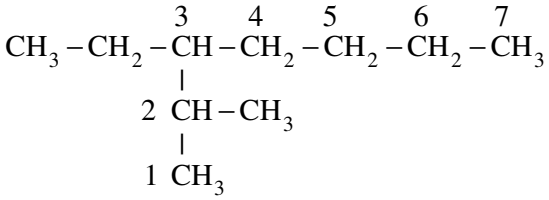
3 chloro- 2-methyl butane

(b) If more than one similar alkyl group or substituents are present then di, tri, tetra are used.

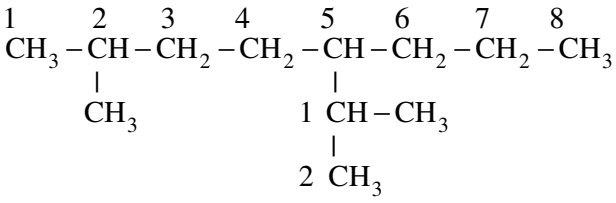


2,2-dimethyl propane

(c) In case side chain is also branched, it is also numbered form carbon atom attached to main chain & is generally written in brackets.

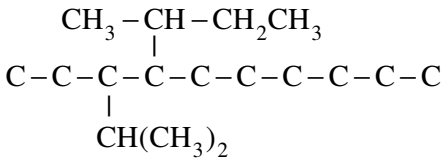


**3-ethyl-2-methyl heptane**



**2-methyl-5-(1-methyl ethyl) octane**

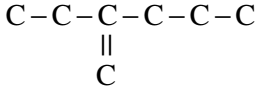
The use of iso & related common prefixes for describing alkyl group as long as these are not further substituted are also allowed by IUPAC nomenclature. While writing name in alphabetical order prefixes iso & neo are considered to be part of fundamental name of alkyl group. However sec. & tert are not considered to be part of fundamental name.



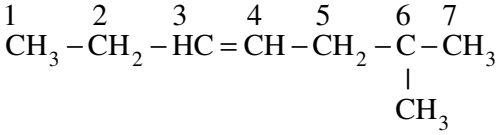
**3-Ethyl-2-methyl-4-(1-methylpropyl) decane**

***For unsaturated hydrocarbon :***

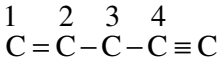
(i) Select the longest possible carbon chain having maximum no. of unsaturated carbon atoms or max. no. of double or triple bonds, even if prior rule is violated.



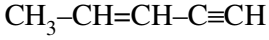
(ii) Lowest no. is assigned to first unsaturated carbon even if prior rule is violated.



(iii) If double & triple bonds are at same position from either ends, lowest no is assigned to double bond



(iv) If both alkene & alkyne group are present, the org. compound is named as derivative of alkyne rather than alkene.



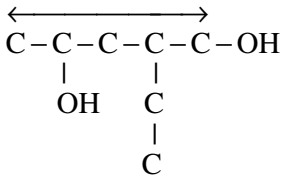
**Pent-3-en-1-yne**

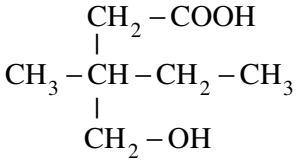
In some cases all the double & triple bonds present in molecule can't be included in longest chain. In such cases following prefixes.

| | |
|---|---|
| $CH_2=$ | $CH_3–CH=$ |
| Methylene | Ethylidene |
| $CH\equiv C–$ | $CH_2=CH–$ |
| Ethynyl | Ethenyl |

***For functional groups*** :

(i) Select the longest possible carbon chain having senior functional group.



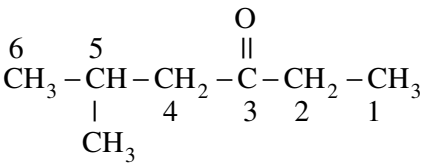


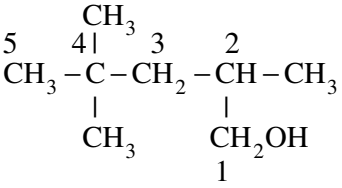
(ii) The carbon atom of functional group is to be included in deciding the longest carbon chain.

C–C–CN 3 carbon chain

C–C–C–CHO 4 carbon chain

(iii) The lowest no is assigned to functional group even if prior rules are violated.



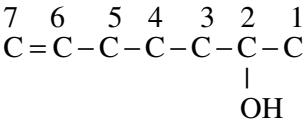


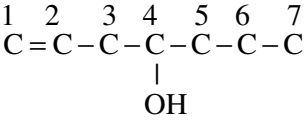
(iv) The order of numbering a carbon chain, thus follows the order:

(a) Functional group

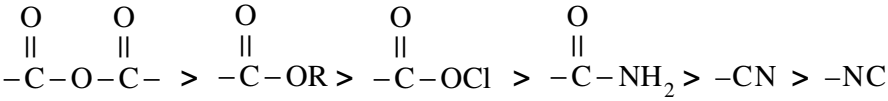
(b) Unsaturation

(c) Substituents & side chains





(v) If more than 1 functional group; then choice of principal functional group is made on the basis following order of preference:



> aldehyde > Ketone > alcohol > thiols > amines > ene > yne

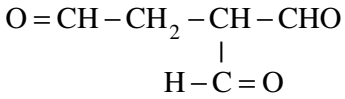
## ***EXERCISE - I***

Q.1 How many 1°carbon atom will be present in a simplest hydrocarbon having two 3° and one 2° carbon atom?
(A) 3 (B) 4 (C) 5 (D) 6

Q.2 $C_3H_6Br_2$ can shows:
(A) Two gem dibromide (B) Two vic dibromide
(C) Two tert. dibromo alkane (D) Two sec. dibromo alkane

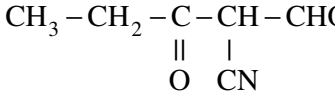
Q.3 The IUPAC name of the compound $CH_3CH = CHCH{=}CHC{\equiv}CCH_3$ is:
(A) 4,6-octadiene-2-yne (B) 2,4-octadiene-6-yne
(C) 2-octyn-4,6-diene (D) 6-octyn-2,4-diene

Q.4 The correct IUPAC name of the following compound is:



(A) 1,1-diformyl propanal (B) 3-formyl butanedial
(C) 2-formyl butanedial (D) 1,1,3-ethane tricarbaldehyde

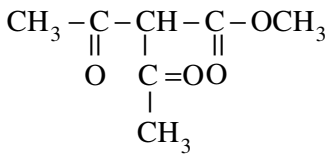
Q.5 The correct IUPAC name of compound:



is:
(A) 2-cyano-3-oxopentanal (B) 2-formyl-3-oxopentanenitrile
(C) 2-cyano-1,3-pentanedione (D) 1,3-dioxo-2-cyanopentane

Q.6 All the following IUPAC names are correct except:
(A) 1-chloro-1-ethoxy propane (B) 1-amino-1-ethoxypropane
(C) 1-ethoxy-2-propanol (D) 1-ethoxy-1-propanamine

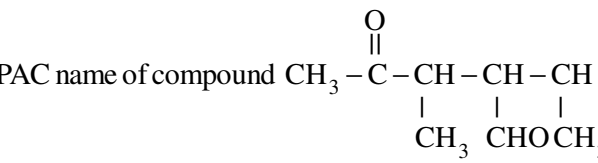
Q.7 IUPAC name of:



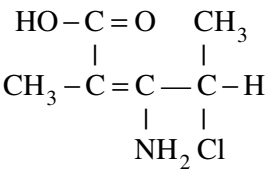
(A) Methyl-2,2 acetyl ethanoate (B) 2,2 acetyl-1-methoxy ethanone
(C) Methyl-2-acetyl-3-oxobutanoate (D) None

Q.8 The IUPAC name of β-ethoxy-α-hydroxy propionic acid (trivial name) is:
(A) 1,2-dihydroxy-1-oxo-3-ethoxy propane (B) 1-carboxy-2-ethoxy ethanol
(C) 3-Ethoxy-2-hydroxy propanoic acid (D) All above

Q.9 The IUPAC name of compound

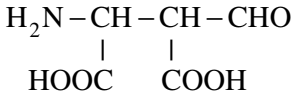


is:
(A) 3,5-Dimethyl-4-Formyl pentanone (B) 1-Isopropyl-2-methyl-4-oxo butanal
(C) 2-Isopropyl-3-methyl-4-oxo pentanal (D) None of the above

Q.10 The IUPAC name of compound  is:

(A) 2-amino-3-chloro-2-methyl-2-pentenoic acid
(B) 3-amino-4-chloro-2-methyl-2-pentenoic acid
(C) 4-amino-3-chloro-2-methyl-2-pentenoic acid
(D) All of the above

Q.11 The IUPAC name of the structure is:



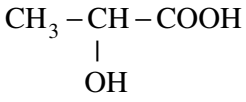
(A) 3-amino-2-formyl butane-1, 4-dioic acid (B) 3-amino-2, 3-dicarboxy propanal
(C) 2-amino-3-formyl butane-1, 4-dioic acid (D) 1-amino-2-formyl succinic acid

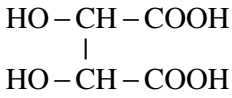
Q.12 $C_4H_6O_2$ does not represent:
(A) A diketone (B) A compound with two aldehyde
(C) An alkenoic acid (D) An alkanoic acid

Q.13 Esters are fiunctional isomers of:
(A) Hydroxy aldehyde (B) Ketone (C) Diketone (D) Diols

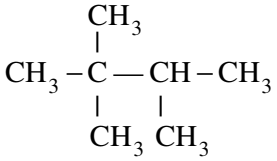
Q.14 How many carbons are in simplest alkyne having two side chains?
(A) 5 (B) 6 (C) 7 (D) 8

Q.15 Which of the following is not correctly matched:

(A) Lactice acid 

(B) Tartaric acid 

(C) Pyvaldehyde $CH_3C(CH_3)_2CHO$

(D) Iso-octane 

Q.16 Which of the following pairs have absence of carbocyclic ring in both compounds?
(A) Pyridine, Benzene (B) Benzene, Cyclohexane
(C) Cyclohexane, Furane (D) Furane, Pyridine

Q.17 The commerical name of trichloroethene is:
(A) Westron (B) Perclene (C) Westrosol (D) Orlone

Q.18 The compound which has one isopropyl group is:
(A) 2,2,3,3-tetramethyl pentane (B) 2,2-dimethyl pentane
(C) 2,2,3-trimethyl pentane (D) 2-methyl pentane

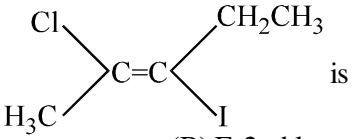
Q.19 How many secondary carbon atoms does methyl cyclopropane have?
(A) None (B) One (C) Two (D) Three

Q.20 The IUPAC name of the compound $\underset{OH}{CH_2}-\underset{OH}{CH}-\underset{OH}{CH_2}$ is:
(A) 1,2,3-tri hydroxy propane (B) 3-hydroxy pentane-1,5-diol
(C) 1,2,3-hydroxy propane (D) Propane-1,2,3-triol

Q.21 As per IUPAC rules, which one of the following groups, will be regarded as the principal functional group?
(A) –C≡C– (B) –OH (C) $-\overset{\overset{O}{\|}}{C}-$ (D) $-\overset{\overset{O}{\|}}{C}-H$

Q.22 Which of the following is the first member of ester homologous series?
(A) Ethyl ethanoate (B) Methyl ethanoate (C) Methyl methanoate (D) Ethyl methanoate

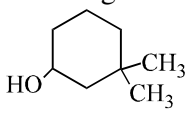
Q.23 The correct IUPAC name of 2-ethyl-3-pentyne is:
(A) 3-methyl hexyne-4 (B) 4-ethyl pentyne-2 (C) 4-methyl hexyne-2 (D) None of these

Q.24 IUPAC name for the compound  is
(A) E-3-iodo-4-chloro-3-pentene (B) E-2-chloro-3-iodo-2-pentene
(C) Z-2-chloro-3-iodo-2-pentene (D) Z-3-iodo-4-chloro-3-pentene

Q.25 The IUPAC name of the compound is $CH_3-\underset{CH_3}{\underset{|}{CH}}-\overset{Ph}{\overset{|}{CH}}-NH_2$
(A) 1-amino-1-phenyl-2-methyl propane (B) 2-methyl-1-phenyl propane-1-amine
(C) 2-methyl-1-amino-1-phenyl propane (D) 1-isopropyl-1-phenyl methyl amine

Q.26 Which of the following compound is wrongly named?
(A) $CH_3CH_2CH_2\underset{Cl}{\underset{|}{CH}}COOH$ ; 2-Chloro pentanoic acid
(B) $CH_3C\equiv\underset{CH_3}{\underset{|}{C}}CHCOOH$ ; 2-Methyl hex-3-enoic acid
(C) $CH_3CH_2CH=CHCOCH_3$ ; Hex-3-en-2-one
(D) $CH_3-\underset{CH_3}{\underset{|}{CH}}CH_2CH_2CHO$ ; 4-Methyl pentanal

Q.27 The IUPAC name of the given compound is:



(A) 1,1-dimethyl-3-hydroxy cyclohexane (B) 3,3-dimethyl-1-hydroxy cyclohexane
(C) 3,3-dimethyl-1-cyclohexanol (D) 1,1-dimethyl-3-cyclohexanol

Q.28 The IUPAC name of $(C_2H_5)_2\,NCH_2\underset{\underset{Cl}{|}}{CH}.COOH$ is:

(A) 2-chloro-4-N-ethylpentanoic acid (B) 2-chloro-3-(N,N-diethyl amino)-propanoic acid
(C) 2-chloro-2-oxo diethylamine (D) 2-chloro-2-carboxy-N-ethyl ethane

Q.29 The IUPAC name of the compound Br(Cl) CH. $CF_3$ is:
(A) haloethane (B) 1,1,1-trifluoro-2-bromo-2-chloroethane
(C) 2-bromo-2-chloro-1,1,1-trifluoroethane (D) 1-bromo-1-chloro-2,2,2-trifloro ethane

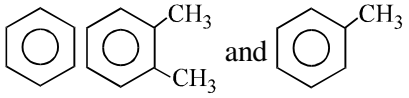
Q.30 The group of hetrocylic compounds is:
(A) Phenol, Furane (B) Furane, Thiophene (C) Thiophene, Phenol (D) Furane, Aniline

Q.31 The correct IUPAC name of $CH_3-CH_2-\underset{\underset{CH_2}{||}}{C}-COOH$ is:

(A) 2-methyl butanoic acid (B) 2-ethyl-2-propenoic acid
(C) 2-carboxy-1-butene (D) None of the above

Q.32 The IUPAC name of the following structure $(CH_3)C.C.C.(CH_3)CH(CH_3)$ is:
(A) 3-methyl-4-hexynene-2 (B) 3-methyl-2-hexenyne-4
(C) 4-methyl-4-hexenyne-4 (D) all are correct

Q.33 The IUPAC name of the following structure is $[CH_3CH(CH_3)]_2\,C(CH_2CH_3)C(CH_3)\,C(CH_2CH_3)_2$
(A) 3,5-diethyl-4,6-dimethyl-5-[1-methylethyl]-3-heptene
(B) 3,5-diethyl-5-isopropyl-4,6-dimethyl-2-heptene
(C) 3,5-diethyl-5-propyl-4,6-dimethyl-3-heptene
(D) None of these

Q.34  and 

Number of secondary carbon atoms present in the above compounds are respectively:
(A) 6,4,5 (B) 4,5,6 (C) 5,4,6 (D) 6,2,1

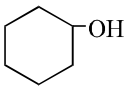
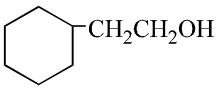
Q.35 The IUPAC name of acetyl acetone is:
(A) 2,5-Pentane dione (B) 2,4-Pentane dione (C) 2,4-Hexane dione (D) 2,4-butane dione

Q.36 When vinyl & allyl are joined each other, we get
(A) Conjugated alkadiene (B) comulative alkadiene
(C) Isolated alkadiene (D) Allenes

Q.37 Glycerine is:

(A) Propane triol-1,2,3 (B) Propylene trialcohol

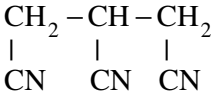
(C) Propyl glycol (D) Hydroxy methyl glycol

Q.38 (a)  and (b) 

True statement for the above compounds is:

(A) (a) is phenol while (b) is alcohol (B) Both (a) and (b) are primary alcohol

(C) (a) is primary and (b) is secondary alcohol (D) (a) is secondary and (b) is primary alcohol

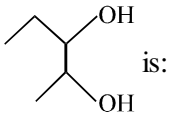
Q.39 IUPAC name will be 

(A) 1,2,3-Tricyano propane (B) Propane trinitrile-1,2,3

(C) 1,2,3-cyano propane (D) 3-cyano pentane-1,3-dinitrile

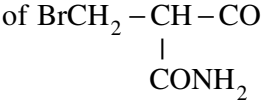
Q.40 A substance containing an equal number of primary, secondary and tertiary carbon atoms is:

(A) Mesityl Oxide (B) Mesitylene (C) Maleic acid (D) Malonic acid

Q.41 IUPAC name of  is:

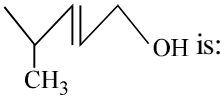
(A) But-2-ene-2,3-diol (B)Pent-2-ene-2,3-diol

(C) 2-methylbut-2-ene-2,3-diol (D) Pent-3-ene-3,4-diol

Q.42 The IUPAC name of  is:

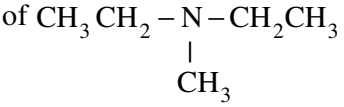
(A) 2-bromo methyl-3-oxo hexanamide (B) 1-bromo-2-amino-3-oxo hexane

(C) 1-bromo-2-amino-n-propyl ketone (D) 3-bromo-2-propyl propanamide

Q.43 IUPAC name of  is:

(A) 5-methyl hexanol (B) 2-methyl hexanol

(C) 2-methyl hex-3-enol (D) 4-methyl pent-2-en-1-ol

Q.44 The IUPAC name of  is:

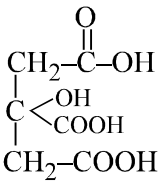
(A) N-methyl-N-ethylethyl amine (B) diethyl methyl amine

(C) N-ethyl-N-methyl ethyl amine (D) methyl diethyl amine

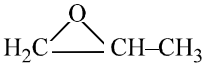
Q.45 The molecular formula of the first member of the family of alkenynes and its name is given by the set

(A) $C_3H_2$, alkene (B) $C_5H_6$, 1-penten-3-yne

(C) $C_6H_8$, 1-hexen-5-yne (D) $C_4H_4$, butenyne

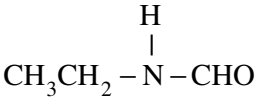
Q.46 The IUPAC name of compound 

(A) 1,2,3-tricarboxy-2,1-propane (B) 3-Carbox-3-hydroxy-1,5-pentane dioic acid
(C) 3-hydroxy-3-Carboxy-1,5-pentane dioic acid (D) None

Q.47 The IUPAC name of the compound: 

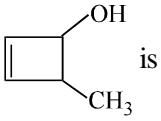
(A) Propylene Oxide (B) 1,2-Oxo propane
(C) 1,2-Epoxy propane (D) 1,2-Propoxide

Q.48 One among the following is the correct IUPAC name of the compound



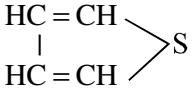
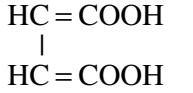
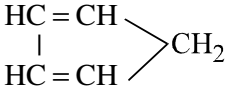
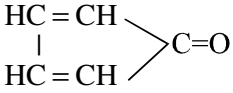
(A) N-Formyl aminoethane (B) N-Ethyl formyl amine
(C) N-Ethyl methanamide (D) Ethylamino methanal

Q.49 Which among the following is the correct IUPAC name of isoamylene:
(A) 1-Pentene (B) 2-Methyl-2-butene (C) 3-Methyl-1-butene (D) 2-Methyl-1-butene

Q.50 The IUPAC name of  is

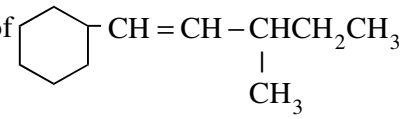
(A) 3-Methyl cyclo-1-butene-2-ol (B) 4-Methyl cyclo-2-butene-1-ol
(C) 4-Methyl cyclo-1-butene-3-ol (D) 2-Methyl cyclo-3-butene-1-ol

Q.51 Which of the following is a heterocyclic compound

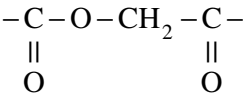
(A)  (B)  (C)  (D) 

Q.52 The number of primary, secondary and tertiary amines possible with the molecular fomula $C_3H_9N$ is:
(A) 1,2,2 (B) 1,2,1 (C) 2,1,1 (D) 3,0,1

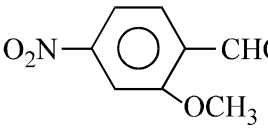
Q.53 The IUPAC name of $C_6H_5CH{=}CH{-}COOH$ is:
(A) cinnamic acid (B) 1-phenyl-2-carboxy ethane
(C) 3-phenyl prop-2-enoic acid (D) dihydroxy-3-phenyl propionic acid

Q.54 The IUPAC name of  is:

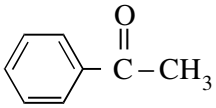
(A) 1-cyclohexyl-3-methyl-1-pentene (B) 3-methyl-5-cyclohexyl-pent-ene
(C) 1-cyclohexyl-3-ethyl-but-1-ene (D) 1-cyclohexyl-3,4-dimethyl-but-1-ene

Q.55 The IUPAC name of  is:

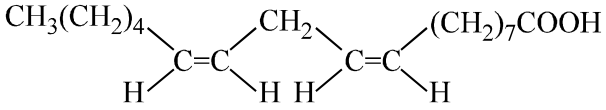
(A) 1-acetoxy acetic acid
(B) 2-acetoxy ethanoic acid
(C) 2-ethanoyl oxyacetic acid
(D) 2-ethanoyl oxyethanoic acid

Q.56 The IUPAC name of  is:

(A) 2-methoxy-4-nitro benzaldehyde
(B) 4-nitro anisaldehyde
(C) 3-methoxy-4-formyl nitro benzene
(D) 2-formyl-4-nitro anisole

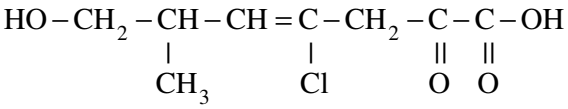
Q.57 The IUPAC name of  is:

(A) phenyl ethanone
(B) methyl phenyl ketone
(C) acetophenone
(D) phenyl emethyl ketone

Q.58 The IUPAC name of  is:

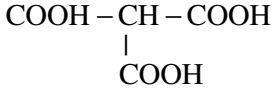
(A) cis-cis-9, 12-octadecan dienoic acid
(B) cis-trans-9,12-octadecan dienoic acid
(C) 9,10-octa decadienoic acid
(D) 9,14-octa decadienoic acid

Q.59 The suffix of the principal group, the prefixes for the other groups and the name of the parent in the structure

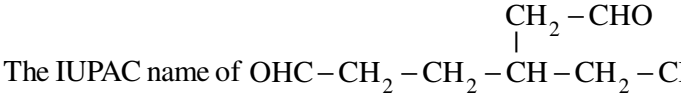


(A) -oic acid, chloro, hydroxy, oxo, methyl, 4-heptene
(B) -oic acid, chloro, hydroxy, methyl, oxo, 4-heptene
(C) -one, carboxy, chloro. methyl, hydroxy, 4-heptene
(D) -one, carboxy, chloro, methyl, hydroxy, 4-heptene

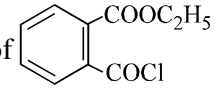
Q.60 The IUPAC name of compound



(A) Tricarboxy methane
(B) Propane trioic acid
(C) Tributanoic acid
(D) 2-carboxy propanedioic acid

Q.61 The IUPAC name of  is:

(A) 4,4-di(formylmethyl) butanal
(B) 2-(formylmethyl) butane-1, 4-dicarbaldehyde
(C) hexane-3-acetal-1, 6-dial
(D) 3-(formylmethyl) hexane-1, 6-dial

Q.62 The IUPAC name of  is:

(A) 2-chlorocarbonyl ethylbenzoate
(B) 2-carboxyethyl benzoyl chloride
(C) ethyl-2-(chlorocarbonyl) benzoate
(D) ethyl-1-(chlorocarbonyl) benzoate

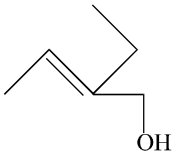
Q.63 Which of the following is crotonic acid:

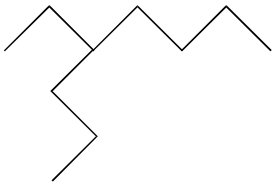
(A) $CH_2$= CH–COOH
(B) $C_6H_5$–CH=CH–COOH
(C) $CH_3$–CH=CH–COOH
(D) $\begin{matrix} CH-COOH \\ \| \\ CH-COOH \end{matrix}$

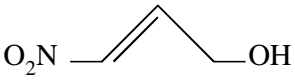
Q.64 $\begin{matrix} CH_3-O-C-CH_2-COOH \\ \quad\quad\quad\

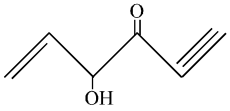
## ***EXERCISE - II***

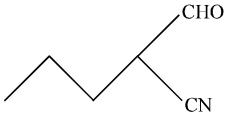
***Give the IUPAC names for each of the following :***

Q.1 

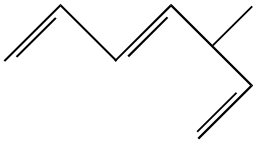

Q.2 

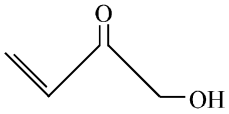
Q.3 

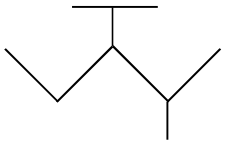

Q.4 

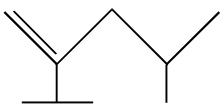

Q.5 

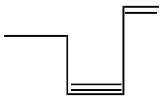

Q.6 

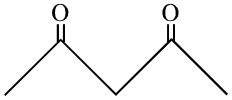

Q.7 

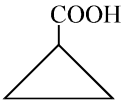
Q.8 

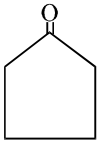

Q.9 

Q.10 

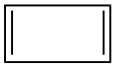
Q.11 

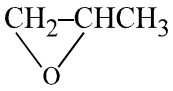
Q.12 


Q.13 

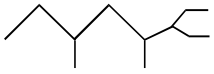

Q.14 

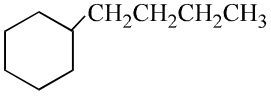

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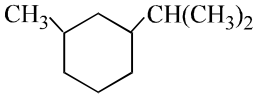
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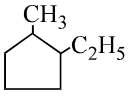
Q.17 

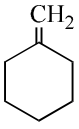

Q.18 

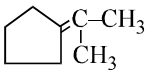
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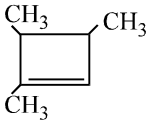
Q.20 

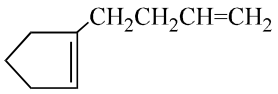

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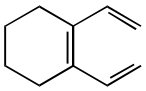

Q.22 

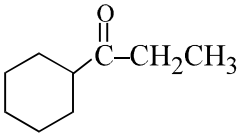

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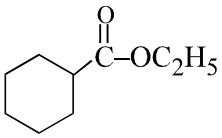
Q.24 

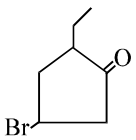
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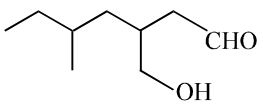
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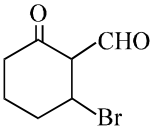
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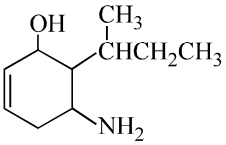
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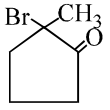
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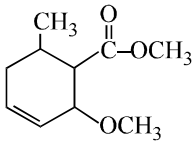
Q.30 

Q.31 

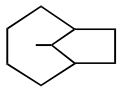
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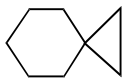
Q.33 

Q.34 

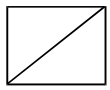
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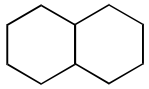
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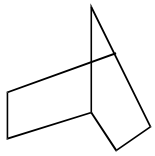
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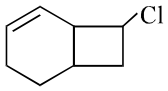
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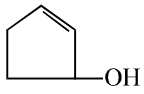
Q.39 

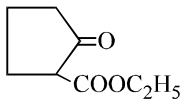
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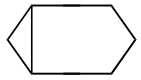
Q.41 

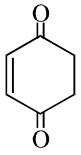
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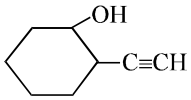
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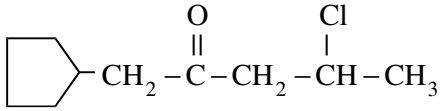
Q.44 

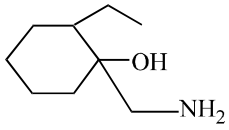
Q.45 

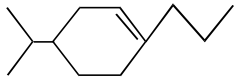
Q.46 

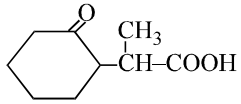
Q.47 

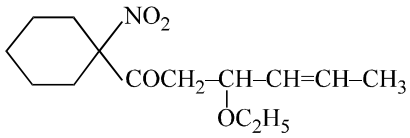
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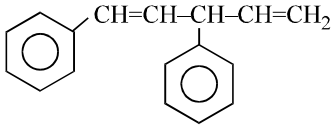
Q.49 

Q.50 

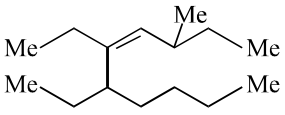
Q.51 

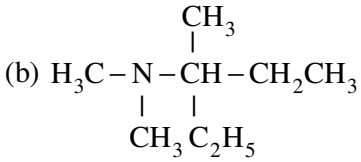
Q.52 

Q.53 

Q.54 

Q.55 Write IUPAC Name of following :

(a)  Me = methyl group

(b) 

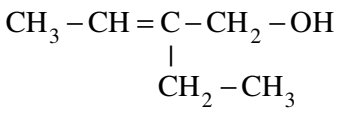
Q.56 Write IUPAC name of succinic acid

# ANSWER KEY

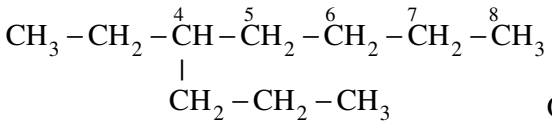
## EXERCISE - I

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Q.1 | B | Q.2 | A | Q.3 | B | Q.4 | C | Q.5 | B | Q.6 | B | Q.7 | C |
| Q.8 | C | Q.9 | C | Q.10 | B | Q.11 | C | Q.12 | D | Q.13 | A | Q.14 | B |
| Q.15 | D | Q.16 | D | Q.17 | C | Q.18 | D | Q.19 | C | Q.20 | D | Q.21 | D |
| Q.22 | C | Q.23 | C | Q.24 | C | Q.25 | B | Q.26 | B | Q.27 | C | Q.28 | B |
| Q.29 | D | Q.30 | B | Q.31 | B | Q.32 | B | Q.33 | A | Q.34 | A | Q.35 | B |
| Q.36 | C | Q.37 | A | Q.38 | D | Q.39 | A | Q.40 | B | Q.41 | B | Q.42 | A |
| Q.43 | D | Q.44 | C | Q.45 | D | Q.46 | B | Q.47 | C | Q.48 | C | Q.49 | C |
| Q.50 | B | Q.51 | A | Q.52 | C | Q.53 | C | Q.54 | A | Q.55 | D | Q.56 | A |
| Q.57 | A | Q.58 | A | Q.59 | B | Q.60 | D | Q.61 | D | Q.62 | C | Q.63 | C |
| Q.64 | B | Q.65 | D | Q.66 | A | Q.67 | A | | | | | | |

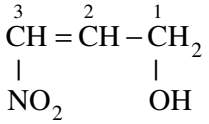
## EXERCISE - II

Q.1 

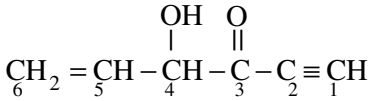
2-ethyl-2-butene-1-ol

Q.2 

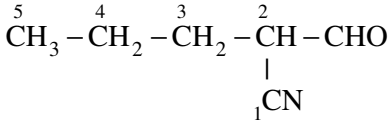
4-Ethyl octane

Q.3 

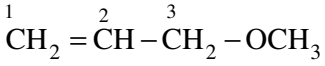
3-nitro-2-propene-1-ol

Q.4 

4-hydroxy-5-hexene-1yne-3-one

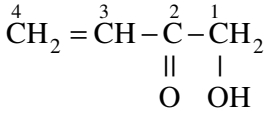
Q.5 

2-Formyl pentane nitrile

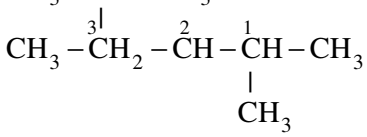
Q.6 

3-Methoxy-1-propene

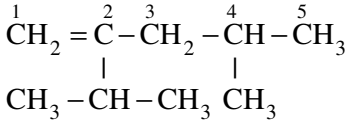
Q.7 3-methyl-1,4,6-Heptatriene

Q.8 

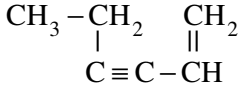
1-Hydroxy-3-Butene-2-one

Q.9 

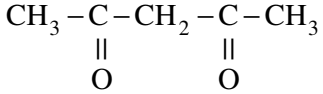
3-Ethyl-2,4-dimethyl pentane

Q.10 

2-isopropyl-4-methyl-1-pentene

Q.11 

1-Hexene-3-yne

Q.12 

2-4, pentane dione

Q.13 Cyclopropanecarboxylic acid

Q.14 Cyclopropane carboxylic acid

Q.15 1,3,5-cyclohexatriene

Q.16 1,3-cyclobutadiene

Q.17 1,2-epoxy propane

Q.18 2,2,6,7-tetramethylocatane

Q.19 3-ethyl-4,6-dimethyloctane

Q.20 Butyl cyclohexane

Q.21 3-isopropyl-1-methylcyclohexane

Q.22 2-ethyl-1-methylcyclopentane

Q.23 Methyl enyl cyclohexane

Q.24 Isopylidenecyclopentane

Q.25 1,3,4-dimethyl-1-cyclobutene

Q.26 1-(3-butenyl) cyclopentene

Q.27 1,2-diethenyl cyclohexene

Q.28 1-cyclohexyl-1-propanone

Q.29 Ethyl cyclohexanecarboxylate

Q.30 4-Bromo-2-ethyl cyclopentaneone

Q.31 3-(1-hydroxyethyl)-5-methylheptanal

Q.32 6-Bromo-2-oxocyclohexanecarbaldehyde

Q.33 3-amino-2-sec-butyl-5-cyclohexen-1-ol

Q.34 2-bromo-2-methyl cyclopentanone

Q.35 Methyl-2-methoxy-6-methyl-3- cyclohexene carboxylate

Q.36 Bicylo(2.2.1)heptane

Q.37 9-methyl bicyclo(4.2.1) nonane

Q.38 spiro (2.5) octane

Q.39 spiro(4.5) decane

Q.40 Bicyclo (2.2.1) heptane

Q.41 Bicyclo(4.4.0) decane

Q.42 Bicyclo(2.2.1) heptane

Q.43 8-chloro bicyclo(4.2.0) oct-2-ene

Q.44 2-cyclepenten-1-ol

Q.45 2-carbethoxy cyclopentanone

Q.46 Bicyclo (3.1.0) hexane

Q.47 Cyclohex-2-en-1,4-dione

Q.48 2-ethynyl cyclohexanol

Q.49 4-chloro-1-cyclopentyl pentane-2-one

Q.50 1-Amino methyl-2-ethyl cyclohexanol

Q.51 1-propyl-4-isopropyl-1-cyclohexene

Q.52 2-(β-keto cyclohexyl) propanoic acid

Q.53 3-ethoxyl-1(1-nitrocyclohexyl)-hexe-4-one-1

Q.54 1,3-diphenyl-1,4-pentadiene

Q.55 (a) 5,6-diethyl-3-methyl-dec-4-ene
(b) N,N,3-trimethyl-3-pentanamine

Q.56 Butane-1,4-dioic acid

# STUDY PACKAGE

# Target: IIT-JEE (Advanced)

# SUBJECT: CHEMISTRY-XI

# TOPIC: 18. Hydrocarbons

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

**4. Exercise III**

**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

# REACTION CHART FOR ALKANES

## GMP

(1) $R-C\equiv CH$ or $R-CH=CH_2$ $\xrightarrow[200-300^\circ C]{H_2, Ni}$ Sabatier senderens reaction

(2) $R-X$ $\xrightarrow[Red\ P-Hi,\ LiAlH_4]{Zn-Cu+HCl}$

(3) $R-Mg-X$ $\xrightarrow[or\ NH_3\ or\ RNH_2]{+HOH\ or\ ROH}$

(4) RX $\xrightarrow[Wurtz\ reaction]{Na,\ dry\ ether}$

(5) RX $\xrightarrow[Frankland's\ reaction]{Zn}$

(6) $R-\underset{\underset{O}{\|}}{C}-Cl$ or ROH $\xrightarrow{Red\ P/Hi}$ or $R-\underset{\underset{O}{\|}}{C}-R$ or RCHO

(7) $R-\underset{\underset{O}{\|}}{C}-R$ $\xrightarrow[Clemension's\ reduction]{Zn-Hg/Conc.HCl}$

(8) $R-\underset{\underset{R}{|}}{C}=O$ or $(RCH_2CH_2)_3B$ $\xrightarrow[Wolf/Kishner\ reduction]{H_2N-NH_2}$ $\xrightarrow{+H_2O}$

(9) RCOONa $\xrightarrow{NaOH+CaO}$

(10) RCOONa $\xrightarrow{Kolbe's\ electrolytic\ synthesis}$

R–H or R–R or $C_nH_{2n+2}$

## GR

(1) $\xrightarrow{X_2, h\nu \text{ or UV light or } 400^\circ C}$ RX

(2) $\xrightarrow{Nitration}$ $R-N\begin{matrix}O\\ \searrow\\ O\end{matrix}$

(3) $\xrightarrow{Sulphonation\ H_2S_2O_7}$ Alkyl Sulphonic acid

(4) $SO_2 + Cl_2 \xrightarrow[h\nu]{Reed\ reaction} RSO_2Cl$

(5) $\xrightarrow[Isomerisation]{AlCl_3/HCl}$ branched alkanes

(6) $\xrightarrow[500-700^\circ C]{Pyrolysis}$ Alkenes + $CH_4$ or $C_2H_6$

(7) $\xrightarrow[+Al_2O_3\ 500^\circ C]{Cr\ or\ Mo\ or\ V\ oxide}$ Aromatic compound

(8) $\xrightarrow[step\ up\ reaction]{CH_2N_2}$ Higher alkane

(9) $\xrightarrow[\Delta]{O_2}$ $CO_2 + H_2O$

Combustion

# REACTION CHART FOR ALKENES

## GMP

(1) $R–CH_2–CH_2–OH \xrightarrow[-H_2O]{conc.\ H_2SO_4}$

(2) $R–CH_2–CH_2–X \xrightarrow[-HX]{alc.\ KOH}$

(3) $R–CH_2–CH\langle^{X}_{X} \xrightarrow[\text{for higher alkene}\ -X_2]{Zn\ dust}$

(4) $R-\underset{\underset{X}{|}}{CH}-\underset{\underset{X}{|}}{CH_2} \xrightarrow{Zn\ dust}$

(5) $R–C\equiv CH \xrightarrow[200–300°C]{Ni,\ H_2}$

(6) $\begin{matrix}RCH-COOK\\|\\RCH-COOK\end{matrix} \xrightarrow{Kolbe's\ electrolytic\ synthesis}$

(7) $(C_2H_5)_4N^+OH \xrightarrow{\Delta}$

(8) $R-\underset{\underset{O}{\|}}{C}-O-CH_2-CH_2-R \xrightarrow{Pyrolysis}$

(9) $R–H \xrightarrow{Pyrolysis}$

(10) $CH_2=CHCl \xrightarrow{CuR_2}$

$R–CH=CH_2$ or $C_nH_{2n}$

## GR

(1) $\xrightarrow[200–300°C]{H_2,\ Ni} R–CH_2–CH_3$

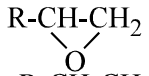
(2) $\xrightarrow{X_2} R–CHX–CH_2X$

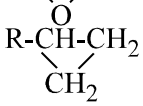
(3) $\xrightarrow{HX} R–CHX–CH_3$

(4) $\xrightarrow{HBr,\ Peroxide} R–CH_2–CH_2Br$

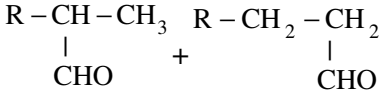
(5) $\xrightarrow{HOCl} R–CH(OH)–CH_2Cl$

(6) $\xrightarrow[+H_2O]{dil.\ H_2SO_4} R–CH_2(OH)–CH_3$

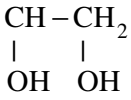
(7) $\xrightarrow[Ag\ 300°C]{1/2\ O_2}$ 

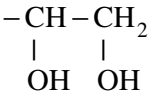
(8) $\xrightarrow{+CH_2N_2}$ 

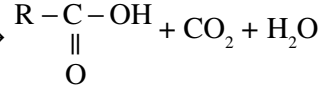
(9) $\xrightarrow{BH_3} (RCH_2CH_2)_3B$

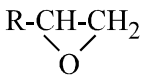
(10) $\xrightarrow[HCo(CO)_4]{CO+H_2}$ 

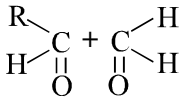
(11) $\xrightarrow[\Delta]{O_2} CO_2 + H_2O$

(12) $\xrightarrow{O_S\ O_4}$ 

(13) $\xrightarrow[1\%\ alkaline\ KMnO_4]{Bayer\ reagent}$ 

(14) $\xrightarrow{strong\ oxidant}$ 

(15) $\xrightarrow[Pr\ iles–chalev's\ reaction]{Per\ acid}$ 

(16) $\xrightarrow{O_3+H_2O}$ Ozonolysis 

(17) $\xrightarrow[200°C\ high\ P]{O_2}$ Polyalkene

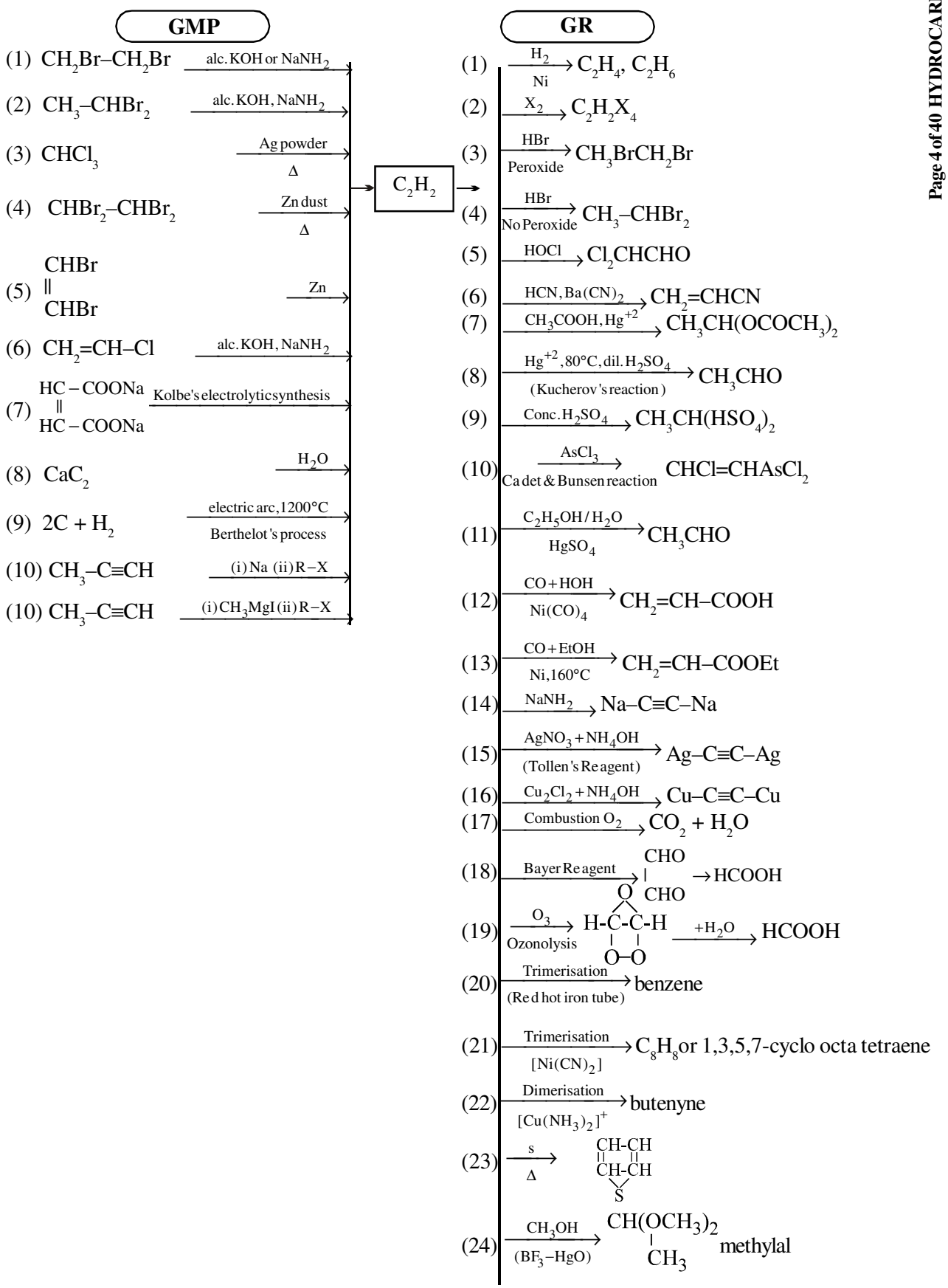
(18) $\xrightarrow[500°C]{Cl_2}$ Substitution product

(19) $\xrightarrow[200–300°C]{Al_2(SO_4)_3}$ Isomerisation

(20) $\xrightarrow{acetic\ anhydride} R–CH_2=CH–COCH_3$
Methyl alkenyl ketone

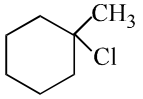
(21) $\xrightarrow{Alkane}$ Higher alkane

# REACTION CHART FOR ALKYNES

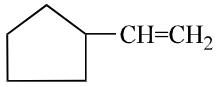
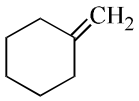
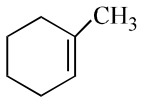


## *EXERCISE–I (A)*

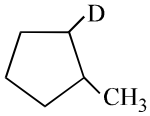
Q.1 In the given reaction

$C_7H_{12}$ (A) $\xrightarrow{HCl}$  as major product

(A) is:

(A)  (B)  (C)  (D) All of these

Q.2 1-Methylcyclopentene can be converted into the given compound

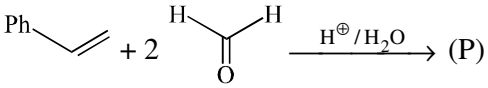


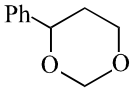
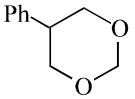
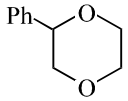
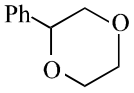
by the use of which of the following reagents?

(A) $BD_3$ followed by HCOOH (B) $BH_3$ followed HCOOD

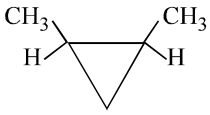
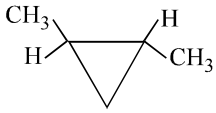
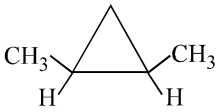
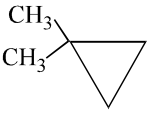
(C) $BD_3$ followed by HCOOD (D) $BH_3$ followed by $D-\overset{\overset{O}{\|}}{C}-O-H$

Q.3 Identify (P) in the following reaction:



(A)  (B)  (C)  (D) 

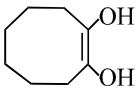
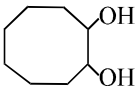
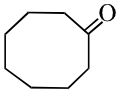
Q.4 The reaction of E-2-butene with $CH_2I_2$ and Zn–Cu Couple in either medium leads to formation of

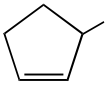
(A)  (B)  (C)  (D) 

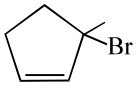
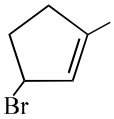
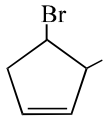
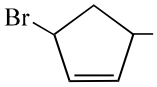
Q.5 (E)-3-bromo-3-hexene when treated with $CH_3O^r$ in $CH_3OH$ gives

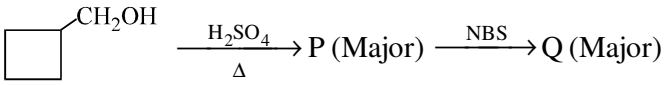
(A) 3-hexyne (B) 2-hexyne (C) 2,3-hexadiene (D) 2,4-hexadiene

Q.6 The reaction of cyclooctyne with $HgSO_4$ in the presence of aq. $H_2SO_4$ gives

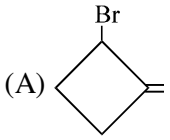
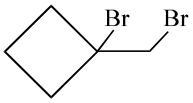
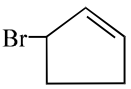
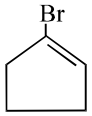
(A)  (B)  (C)  (D) 

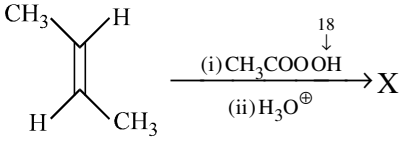
Q.7  + $Br_2$ $\xrightarrow{h\nu}$ mixture of product. Among the following which product will formed minimum amount.

(A)  (B)  (C)  (D) 

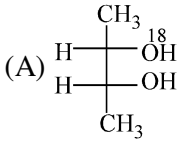
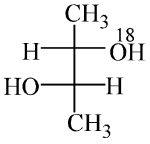
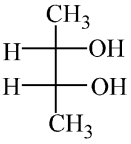
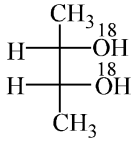
Q.8 

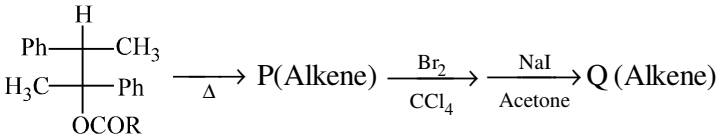
The structure of Q is

(A)  (B)  (C)  (D) 

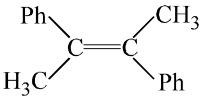
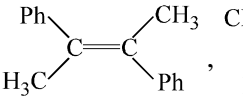
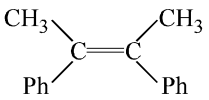
Q.9 

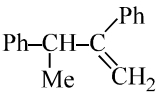
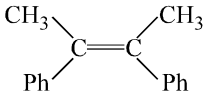
The probable structure of 'X' is

(A)  (B)  (C)  (D) 

Q.10 

Alkene (P) & (Q) respectively are

(A) Both  (B)  , 

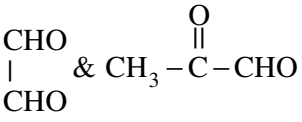
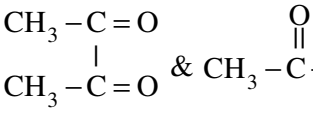
(C) Both  (D) Both 

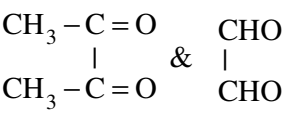
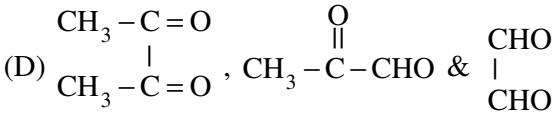
Q.11 Ozonolysis of $CH_3$–CH=C=$CH_2$ will give

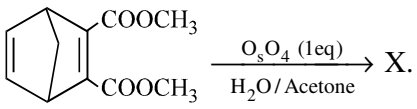
(A) Only $CH_3CHO$ (B) Only HCHO

(C) Only $CO_2$ (D) Mixture of $CH_3CHO$, HCHO & $CO_2$

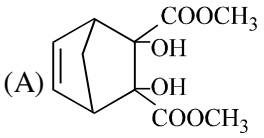
Q.12 O-xylene on ozonolysis will give

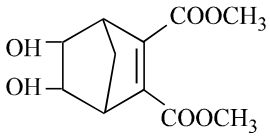
(A)  (B) 

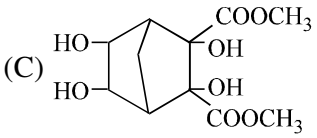
(C)  (D) 

Q.13  X.

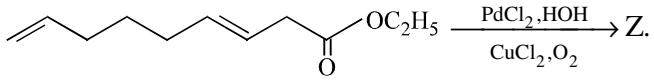
Identify 'X'.

(A) 

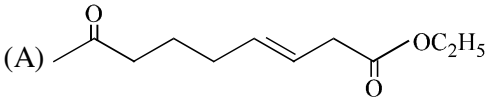
(B) 

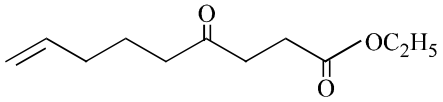
(C) 

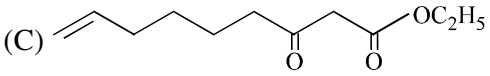
(D) Reaction will not occur

Q.14  Z.

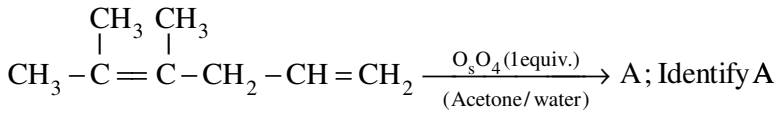
Identify Z.

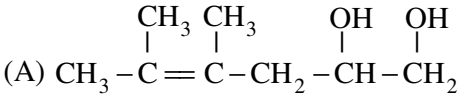
(A) 

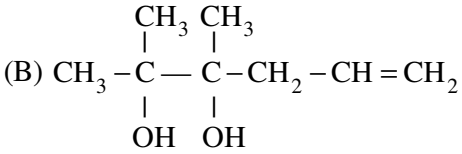
(B) 

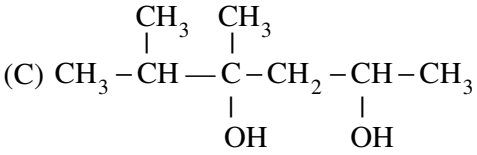
(C) 

(D) All are correct

Q.15 

(A) 

(B) 

(C) 

(D) Reaction will not occur

Q.16 1-Penten-4-yne reacts with bromine at – 80°C to produce:

(A) 4,4,5,5-Tetrabromopentene (B) 1,2-Dibromo-1,4-pentadiene

(C) 1,1,2,2,4,5-hexabromopentane (D) 4,5-dibromopentyne

Q.17 Compound (A) on oxidation with hot $KMnO_4$ / $OH^r$ gives two compound

$$CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-COOH \;\&\; CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-CH_2CH_2CH_3$$

compound A will have structure.

(A) $CH_3CH_2-\underset{\displaystyle CH_3}{\underset{|}{C}}=\underset{\displaystyle CH_3}{\underset{|}{C}}-CH_2CH_3$

(B) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH=\underset{\displaystyle CH_3}{\underset{|}{C}}-CH_2CH_2CH_3$

(C) $CH_3\underset{\displaystyle CH_3}{\underset{|}{CH}}-C\equiv C-CH_3$

(D) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-C\equiv C-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$

Q.18 Consider the following reaction

(A) $C_6H_{12} \xrightarrow{KMnO_4/OH^-/\Delta} C_5H_{10}O$

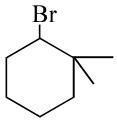
In the above reaction (A) will be

(A) $CH_3–CH_2–CH_2–CH_2–CH=CH_2$

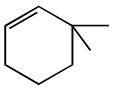
(B) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-CH=CH_2$

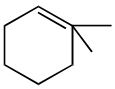
(C) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH=CH_2$

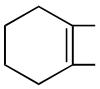
(D) $CH_3CH_2CH_2-\underset{\displaystyle CH_3}{\underset{|}{C}}=CH_2$

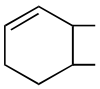
Q.19  $\xrightarrow{\text{alcoholic KOH}}$ product

Major product is:

(A) 

(B) 

(C) 

(D) 

Q.20 Number of required $O_2$ mole for complete combustion of one mole of propane –
(A) 7 (B) 5 (C) 16 (D) 10

Q.21 How much volume of air will be needed for complete combustion of 10 lit. of ethane –
(A) 135 lit. (B) 35 lit. (C) 175 lit. (D) 205 lit.

Q.22 When n-butane is heated in the presence of $AlCl_3$/HCl it will be converted into –
(A) Ethane (B) Propane (C) Butene (D) Isobutane

Q.23 The reacting species of alc. KOH is –
(A) $OH^-$ (B) $OR^+$ (C) $OK^+$ (D) $RO^-$

Q.24 The product of reaction between one mole of acetylene and two mole of HCHO in the presence of $Cu_2Cl_2$ –
(A) $HOCH_2 – C\equiv C – CH_2OH$ (B) $H_2C = CH – C\equiv C – CH_2OH$
(C) $HC \equiv C – CH_2OH$ (D) None of these

Q.25 PMA polymer is formed by methyl acrylate, which is prepared as follows –

(A) $R - C \equiv CH \xrightarrow{CO+ROH}$ (B) $HC \equiv CH \xrightarrow[Ni(CO)_4]{CO+CH_3OH}$

(C) $HC \equiv CH \xrightarrow[Ni(CO)_4]{CO+H_2O}$ (D) None of these

Q.26 During the preparation of ethane by Kolbe's electrolytic method using inert electrodes the pH of the electrolyte –

(A) Increases progressively as the reaction proceeds

(B) Decreases progressively as the reaction proceeds

(C) Remains constant throughout the reaction

(D) May decrease of the the concentration of the electrolyte is not very high

Q.27 Ethylene forms ethylene chlorohydrin by the action of –

(A) Dry HCl gas (B) Dry chlorine gas

(C) Solution of chlorine gas in water (D) Dilute hydrochloric acid

Q.28 Anti–Markownikoff's addition of HBr is not observed in –

(A) Propene (B) But–2–ene (C) Butene (D) Pent–2–ene

Q.29 Which alkene on heating with alkaline $KMnO_4$ solution gives acetone and a gas, which turns lime water milky –

(A) 2–Methyl–2–butene (B) Isobutylene

(C) 1–Butene (D) 2–Butene

Q.30 Acetylene may be prepared using Kolbe's electrolytic method employing –

(A) Pot. acetate (B) Pot. succinate (C) Pot. fumarate (D) None of these

Q.31 $\xleftarrow{\text{Lindlar}}$ R–C≡C–R $\xrightarrow{Na/NH_3}$ A

A and B are geometrical isomers (R–CH=CH–R) –

(A) A is trans, B is cis (B) A and B both are cis

(C) A and B both are trans (D) A is cis, B is trans

Q.32 Which is expected to react most readily with bromine –

(A) $CH_3CH_2CH_3$ (B) $CH_2=CH_2$ (C) CH≡CH (D) $CH_3–CH=CH_2$

Q.33 By the addition of CO and $H_2O$ on ethene, the following is obtained –

(A) Propanoic acid (B) Propanal (C) 2–Propenoic acid (D) None of the above

Q.34 An alkyne $C_7H_{12}$ on reaction with alk. $KMnO_4$ and subsequent acidification with HCl yields a mixture of $\begin{matrix} CH_3-CHCOOH \\ \quad\quad\quad | \\ \quad\quad\quad CH_3 \end{matrix}$ + $CH_3CH_2COOH$. The alkyne is –

(A) 3–Hexyne (B) 2–Methyl–3–hexyne

(C) 2–Methyl–2–hexyne (D) 2–Methyl–2–hexene

Q.35 A compound ($C_5H_8$) reacts with ammonical $AgNO_3$ to give a white precipitate and reacts with excess of $KMnO_4$ solution to give $(CH_3)_2CH–COOH$. The compound is –

(A) $CH_2=CH–CH=CH–CH_3$ (B) $(CH_3)_2CH–C\equiv CH$

(C) $CH_3(CH_2)_2C\equiv CH$ (D) $(CH_3)_2C=C=CH_2$

Q.36 Which of the following reagents cannot be used to locate the position of triple bond in $CH_3–C\equiv C–CH_3$

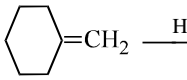
(A) $Br_2$ (B) $O_3$ (C) $Cu_2^{2+}$ (D) $KMnO_4$

Q.37 $CH_3–CH_2–C\equiv CH \underset{B}{\overset{A}{\rightleftharpoons}} CH_3C\equiv C–CH_3$

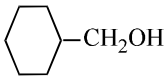
A and B are –

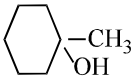
(A) alcoholic KOH and $NaNH_2$ (B) $NaNH_2$ and alcoholic KOH

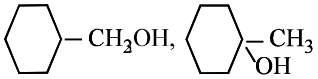
(C) $NaNH_2$ and Lindlar (D) Lindlar and $NaNH_2$

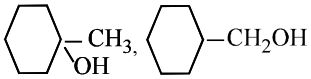
Q.38 B $\xleftarrow[H_2O_2/OH^-]{BH_3/THF}$  $\xrightarrow{H_3O^+}$ A

A and B are –

(A) Both 

(B) Both 

(C) 

(D) 

Q.39 B $\xleftarrow[H_2O_2,\ OH^-]{BH_3\ THF}$ $CH_3–C\equiv CH \xrightarrow{HgSO_4/H_2SO_4}$ A

A and B are –

(A) $CH_3CH_2CHO, CH_3-\overset{\overset{O}{\|}}{C}-CH_3$

(B) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3\ CH_3CH_2CHO$

(C) $CH_3CH_2CHO$ (both)

(D) $CH_3-\underset{\underset{O}{\|}}{C}-CH_3$ (both)

Q.40 $CH_3CH=CH_2 \xrightarrow[H_2O_2/OH^-]{B_2D_6}$ product X

X is –

(A) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2D$

(B) $CH_3-\underset{\underset{D}{|}}{CH}-CH_2OH$

(C) $CH_3-\underset{\underset{OD}{|}}{CH}-CH_3$

(D) none is correct

Q.41 $CH_2=CH–CH=CH_2 \xrightarrow{CCl_3Br}$ product. The major product is –

(A) $Br–CH_2–CH=CH–CH_2–CCl_3$

(B) $CH_2=CH-\underset{\underset{Br}{|}}{CH}-CH_2-CCl_3$

(C) $CH_2=CH-\underset{\underset{CCl_3}{|}}{CH}-CH_2-Br$

(D) None is correct

Q.42 Mixture of one mole each of ethene and propyne on reaction with Na will form $H_2$ gas at S.T.P. –
(A) 22.4 L (B) 11.2 L (C) 33.6 L (D) 44.8 L

Q.43 Dehydration of 2, 2, 3, 4, 4–pentamethyl–3–pentanol gave two alkenes A and B. The ozonolysis products of A and B are -

(A) A gives $(CH_3)_3C-\overset{\overset{O}{\|}}{C}-C(CH_3)_3$ and HCHO

B gives $CH_3-\overset{\overset{O}{\|}}{C}-CH_2-C(CH_3)_3$ and HCHO

(B) A gives $(CH_3)_3C-\overset{\overset{O}{\|}}{C}-C(CH_3)_3$ and HCHO

B gives $CH_3-\overset{\overset{O}{\|}}{C}-\overset{\overset{CH_3}{|}}{C}-C(CH_3)_3$ and HCHO

(C) A gives $(CH_3)_3C-\overset{\overset{O}{\|}}{C}-CH(CH_3)_2$ and HCHO

B gives $(CH_3)-CH_2-\overset{\overset{O}{\|}}{C}-C(CH_3)_3$ and $CH_3CH_2CHO$

(D) None of these

Q.44 $CH\equiv CH \xrightarrow[Cu_2Cl_2]{NH_4Cl}$ product

Product is –
(A) Cu–C≡C–Cu (B) $CH_2$=CH–C≡CH (C) CH≡C–Cu (D) Cu–C≡C–$NH_4$

Q.45 Alkene A $\xrightarrow{O_3/H_2O}$ $CH_3-\overset{\overset{O}{\|}}{C}-CH_3 + CH_3COOH + CH_3-\underset{\underset{O}{\|}}{C}-COOH$

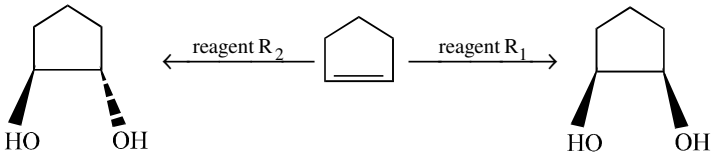
A can be –

(A) $CH_3\text{-}\overset{\overset{CHCH_3}{\|}}{C}\text{-}CH{=}C\begin{matrix}\nearrow CH_3\\ \searrow CH_3\end{matrix}$

(B) $CH_3-\overset{\overset{C(CH_3)_2}{\|}}{C}-CH=HC-CH_3$

(C) Both correct

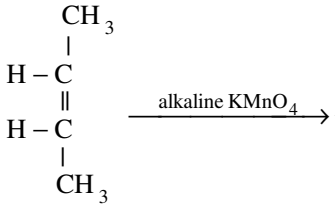
(D) None is correct

Q.46 


$R_1$ and $R_2$ are –

(A) Cold alkaline $KMnO_4$, $OsO_4/H_2O_2$ (B) Cold alkaline $KMnO_4$, $HCO_3H$

(C) Cold alkaline $KMnO_4$, $CH_3–O–O–CH_3$ (D) $C_6H_5CO_3H$, $HCO_3H$

Q.47 

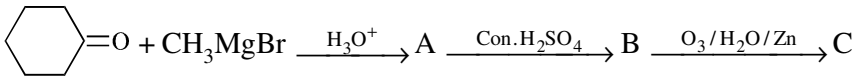
A, which is true about this reaction?

(A) A is meso 2, 3–butan–di–ol formed by *syn* addition

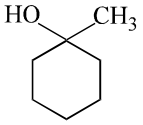
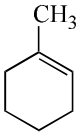
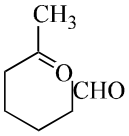
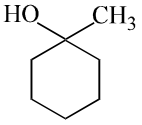
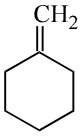
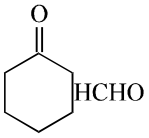
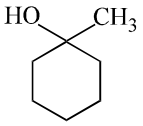
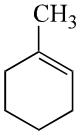
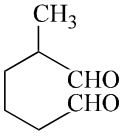
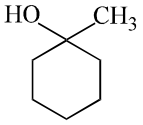
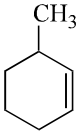
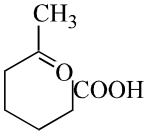
(B) A is meso 2, 3–butan–di–ol formed by *anti* addition

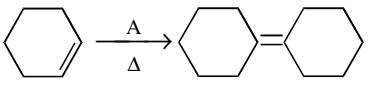
(C) A is a racemic mixture of d and *l*, 2, 3–butan–di–ol formed by anti addition

(D) A is a racemic mixture of d and *l* 2,3–butan–di–ol formed by syn addition

Q.48 


A, B and C are –

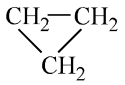
| | A | B | C |
|---|---|---|---|
| (A) |  |  |  |
| (B) |  |  |  |
| (C) |  |  |  |
| (D) |  |  |  |

Q.49 

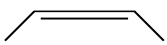
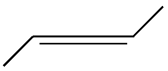
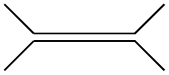
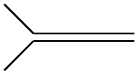

A can be –

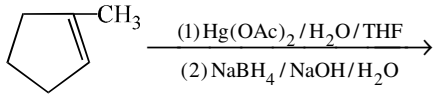
(A) Conc. $H_2SO_4$ (B) alcoholic KOH (C) $Et_3N$ (D) t-BuOK

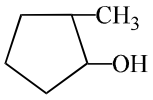
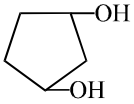
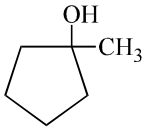
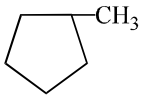
Q.50 $BrCH_2–CH_2–CH_2Br$ reacts with Na in the presence of ether at 100 °C to produce –

(A) $BrCH_2–CH=CH_2$ (B) $CH_2=C=CH_2$ (C)  (D) All of these

Q.51 Which has least heat of hydrogenation –

(A)  (B)  (C)  (D) 

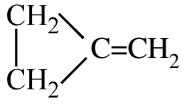
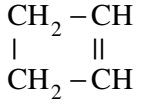
Q.52  A. A is –

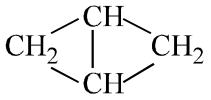
(A)  (B)  (C)  (D) 

Q.53 An organic compound of molecular formula $C_4H_6$, (A), forms precipitates with ammoniacal silver nitrate and ammoniacal cuprous chloride. 'A' has an isomer 'B', one mol of which reacts with one mol of $Br_2$ to form 1, 4-dibromo-2-butene. Another isomer of A is 'C', one mole of C reacts with only 1 mol. of $Br_2$ to give vicinal dibromide. A, B & C are

(A) $CH_3–CH_2–C\equiv CH$ and $CH_2=CH–CH=CH_2$ ; 

(B) $CH_3–C\equiv C–CH_3$ and $CH_3–CH=C=CH_2$ ; $CH_3–C\equiv C–CH_3$

(C)  and  ; $CH_2 = CH–CH=CH_2$

(D) $CH_3–C\equiv C–CH_3$ and  ; $CH_2 = CH–CH = CH_2$

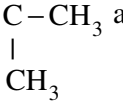
Q.54 $CH_3–CH=CH–CH_3 \xrightarrow{x}$ product is Y (non–resolvable) then X can be –

cis

(A) $Br_2$ water (B) $HCO_3H$

(C) Cold alkaline $KMnO_4$ (D) all of the above

Q.55 Electrophilic addition reaction is not shown by

(A)  and $Br_2$ (B) $CH\equiv CH_2$ and HO–Cl

(C) $CH_3–C\equiv CH$ and $CH_3MgBr$ (D) $CH_2=CH_2$and dil. $H_2SO_4$ solution

Q.56 A mixture of $CH_4$, $C_2H_4$ and $C_2H_2$ gaseous are passed through a Wolf bottle containing ammonical cuprous chloride. The gas coming out is

(A) Methane (B) Acetylene

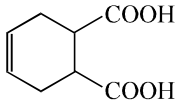
(C) Mixture of methane and ethylene (D) original mixture

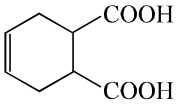
Q.57 In the presence of strong bases, triple bonds will migrate within carbon skeletons by the

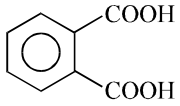
(A) removal of protons (B) addition of protons

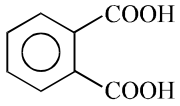
(C) removal and readdition of protons (D) addition and removal of protons.

Q.58 $CH_2=CH-CH=CH_2 + \begin{matrix}CHCOOH\\ ||\\ CHCOOH\end{matrix} \xrightarrow{\Delta}$ product X by reaction R. X and R are

(A)  Diels Alder

(B)  Friedel-Crafts

(C)  Diels Alder

(D)  Friedel-Crafts

Q.59 For the ionic reaction of hydrochloric acid with the following alkenes, predict the correct sequence of reactivity as measured by reaction rates:

(I) $ClCH=CH_2$ (II) $(CH_3)_2.C=CH_2$ (III) $OHC.CH=CH_2$ (IV) $(NC)_2C=C(CN)_2$

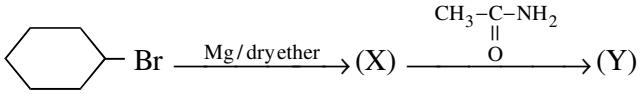
(A) IV > I > III > II (B) I > IV > II > III (C) III > II > IV > I (D) II > I > III > IV

Q.60 The addition of bromine to 2-cyclohexenyl benzoate in 1,2-dichloroethane produces ____ dibromo derivatives:

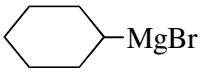
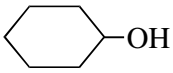
(A) 2 (B) 3 (C) 4 (D) 6

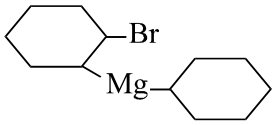
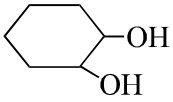
Q.61 How many products will be formed when methylenecyclohexane reacts with NBS?

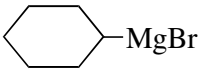
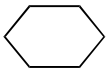
(A) 3 (B) 1 (C) 2 (D) 4

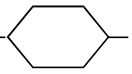
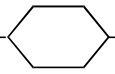
Q.62 

The structures of (X) and (Y) respectively are

(A) X =  ; Y = 

(B) X =  ; Y = 

(C) X =  ; Y = 

(D) X = BrMg  MgBr ; Y = HO  OH

## *EXERCISE–I (B)*

Q.1 $A \xrightarrow[\xrightarrow[H_2O_2/OH^-]{BH_3/THF}]{HgSO_4/H_2SO_4} B$

B is identical when A is –

(A) H–≡–H (B) –≡–H (C)  (D) 

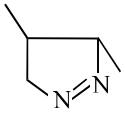
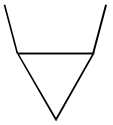
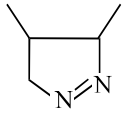
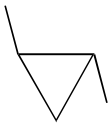
Q.2 An alkene on ozonolysis yields only ethanal. There is an isomer of this which on ozonolysis yields:

(A) propanone (B) ethanal

(C) methanal (D) only propanal

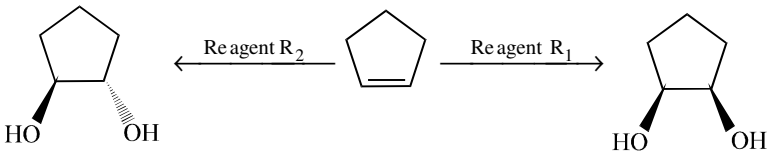
Q.3 $CH_3-CH=CH-CH_3 + CH_2N_2 \xrightarrow{\Delta} A$

A can be

(A)  (B)  (C)  (D) 

Q.4 Aqueous solution of potassium propanoate is electrolysed. Possible organic products are:

(A) n-Butane (B) $C_2H_5COOC_2H_5$ (C) $CH_3-CH_3$ (D) $CH_2=CH_2$

Q.5 

$R_1$ and $R_2$ are:

(A) cold alkaline $KMnO_4$, $OsO_4$ / $H_2O_2$ (B) $OsO_4$ / $NaHSO_3$ ; $Ag_2O$ , $H_3O^{\oplus}$

(C) cold alkaline $KMnO_4$, $C_6H_5CO_3H$ / $H_3O^{\oplus}$ (D) $C_6H_5CO_3H$ ; $OsO_4$ / $NaHSO_3$

Q.6 (A) $C_4H_6 \xrightarrow{H_2/Pt}$ (B) $C_4H_8 \xrightarrow{O_3/H_2O} CH_3COOH$

Hence A and B are

(A) $CH_3C \equiv CCH_3$, $CH_3CH=CHCH_3$ (B) $CH_2=CHCH_3=CH_2$, $CH_3CH=CHCH_3$

(C)  , $CH_3CH=CHCH_3$ (D) None

Q.7 Which is / are true statements/ reactions?

(A) $Al_4C_3 + H_2O \longrightarrow CH_4$ (B) $CaC_2 + H_2O \longrightarrow C_2H_2$

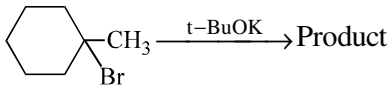
(C) $Mg_2C_3 + H_2O \longrightarrow CH_3C \equiv CH$ (D) $Me_3C-H + KMnO_4 \longrightarrow Me_3C-OH$

Q.8 $Ph-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow{A} Ph-CH_2-CH_3$

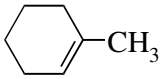
A could be:

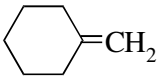
(A) $NH_2NH_2$, glycol/$OH^-$ (B) Na(Hg)/conc. HCl

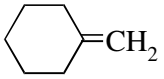
(C) Red P/HI (D) $\begin{matrix} CH_2 & - & CH_2 \\ | & & | \\ SH & & SH \end{matrix}$ ; Raney Ni – $H_2$

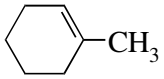
Q.9  Product

which is / are correct statements about the product:

(A)  is an endocyclic Saytzeff product

(B)  is an exocyclic Saytzeff product

(C)  is an exocyclic Hoffmann product

(D)  is an endocyclic Hoffmann product

Q.10 $CH_2 = CHCH_2CH = CH_2 \xrightarrow{NBS}$ A, A can be

(A) $CH_2 = CH\underset{\underset{Br}{|}}{CH}CH = CH_2$

(B) $CH_2$=CHCH=CH–$CH_2$Br

(C) $CH_2 = CH\,CH_2\,CH = CHBr$

(D) $CH_2 = CH\,CH_2\,\underset{\underset{Br}{|}}{C} = CH_2$

Q.11 Which are correct statements?

(A) meso-2, 3-dibromo-butane on reaction with NaI / acetone gives trans-2-butene

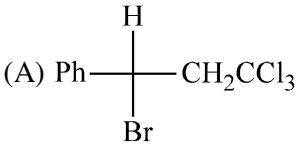
(B) d-or *l*- 2, 3-dibromobutane on reaction with NaI/acetone gives cis-2-butene

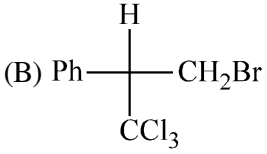
(C) meso-2, 3-dibromo-butane on reaction with NaI / acetone gives cis-2-butene

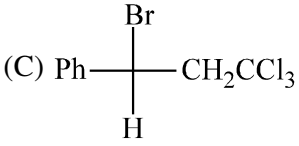
(D) d-or *l*- 2, 3-dibromobutane on reaction with NaI/acetone gives trans-2-butene

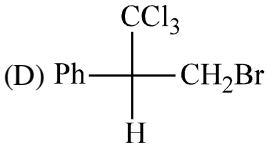
Q.12 $Ph–CH=CH_2 + BrCCl_3 \xrightarrow{peroxide}$

Product is:

(A) 

(B) 

(C) 

(D) 

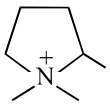
Q.13 Which of the following elimination reactions will occur to give but-1-ene as the major product?

(A) $CH_3.CHCl.CH_2.CH_3 + KOH \xrightarrow{EtOH}$

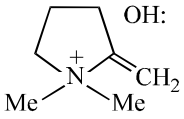
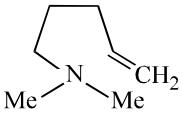
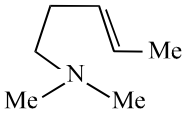
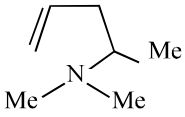
(B) $CH_3.\underset{\underset{\overset{+}{N}Me_3}{|}}{CH_2}.CH.CH_3 + NaOEt \xrightarrow[\Delta]{EtOH}$

(C) $CH_3.CH_2.CHCl.CH_3 + Me_3CoK \xrightarrow{\Delta}$

(D) $CH_3.CH_2.CH(OH).CH_3 + \text{conc. } H_2SO_4 \xrightarrow[\Delta]{}$

Q.14 

The above compound undergoes ready elimination on heating to yield which of the following products?

(A)  (B)  (C)  (D) 

Q.15 Which of the following will give same product with HBr in presence or absence of peroxide.

(A) Cyclohexene (B) 1-methylcyclohexene

(C) 1,2-dimethylcyclohexene (D) 1-butene

Q.16 The ionic addition of HCl to which of the following compounds will produces a compound having Cl on carbon next to terminal.

(A) $CF_3.(CH_2)_3.CH=CH_2$ (B) $CH_3.CH=CH_2$

(C) $CF_3.CH=CH_2$ (D) $CH_3.CH_2CH=CH.CH_3$

Q.17 Select true statement(s):

(A) $I_2$ does not react with ethane at room temperature even though $I_2$ is more easily cleaved homolytically than the other halogens.

(B) Stereochemical outcome of a radical substitution and a radical addition reaction is identical.

(C) The rate of bromination of methane is decreased if HBr is added to the reaction mixture.

(D) Allylic chloride adds halogens faster than the corresponding vinylic chloride.

Q.18 Select true statement(s):

(A) Instead of radical substitution, cyclopropane undergoes electrophilic addition reactions in sun light.

(B) In general, bromination is more selective than chlorination.

(C) The 2,4,6-tri-tert, butylphenoxy radical is resistant to dimerization.

(D) The radical-catalysed chlorination, $ArCH_3 \rightarrow ArCH_2Cl$, occurs faster when Ar = phenyl than when Ar = p-nitrophenyl.

Q.19 Nitrene is an intermediate in one of the following reactions:

(A) Schmidt rearrangement (B) Beckmann rearrangement

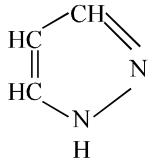
(C) Baeyer-Villiger oxidation (D) Curtius reaction

Q.20 Which reagent is the most useful for distinguishing compound I from the rest of the compounds

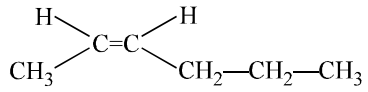
| $CH_3CH_2C\equiv CH$ | $CH_3C\equiv CCH_3$ | $CH_3CH_2CH_2CH_3$ | $CH_3CH=CH_2$ |
|---|---|---|---|
| I | II | III | IV |
| (A) alk. $KMnO_4$ | (B) $Br_2/CCl_4$ | (C) $Br_2/CH_3COOH$ | (D) Ammonical $AgNO_3$ |

Q.21 Indicate among the following the reaction **not correctly** formulated.

(A) $CH_2=CH-CH_3 \xrightarrow[\text{UV light}]{+SO_2Cl_2} CH_2Cl-CHCl-CH_3$

(B) $HC\equiv CH + CH_2N_2 \longrightarrow$ 

(C) $(CH_3)_3CH + Cl_2 \xrightarrow[\text{halogenation}]{\text{photo}-} (CH_3)_3C-Cl$ as major product

(D) $CH_3-C\equiv C-CH_2-CH_2-CH_3 \xrightarrow[\text{in } NH_3\,(liq)]{Na}$ 

Q.22

| List I | List II |
|---|---|
| (A) n-Hexane $\xrightarrow{Cr_2O_3-Al_2O_3,\Delta}$ | (1) Substitution reaction |
| (B) $CH\equiv CH \xrightarrow[\text{Red hot Fe tube}]{}$ | (2) Elimination reaction |
| (C) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-X \longrightarrow$ aq. | (3) Aromatisation |
| (D) $CH_3-CH_2-X \longrightarrow$ alc. KOH | (4) Cyclization |

Q.23

| List I | List II |
|---|---|
| (A) $CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}=CH_2 \xrightarrow[(ii)\,H_2O_2/\overset{\oplus}{O}H]{(i)\,BH_3}$ | (1) $CH_3-CH_2-CH=CH_2$ |
| (B) $CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}=CH_2 \xrightarrow[(ii)\,NaBH_4]{(i)\,Hg(OAc)_2/HOH}$ | (2) $CH_3-CH=CH-CH_3$ |
| (C) $CH_3-CH_2-\underset{|}{\overset{Cl}{\overset{|}{C}}}H-CH_3 \xrightarrow{CH_3ONa/\Delta}$ | (3) $CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}H-CH_2OH$ |
| (D) $CH_3-CH_2-\underset{|}{\overset{Cl}{\overset{|}{C}}}H-CH_3 \xrightarrow[\Delta]{(CH_3)_3CONa}$ | (4) $CH_3-\underset{\displaystyle CH_3}{\overset{|}{\overset{OH}{\overset{|}{C}}}}-CH_3$ |

**Codes:**

| | A | B | C | D |
|---|---|---|---|---|
| (a) | 4 | 3 | 1 | 2 |
| (b) | 4 | 3 | 2 | 1 |
| (c) | 3 | 4 | 2 | 1 |
| (d) | 3 | 4 | 1 | 2 |

Q.24

| List I | List II |
|---|---|
| (A) Walden Inversion | (1) Cis addition |
| (B) Racemic mixture | (2) Trans addition |
| (C) Alkene $\xrightarrow[\text{Reagent}]{\text{Baeyer}}$ | (3) $SN_1$ reaction |
| (D) Alkene $\xrightarrow{Br_2}$ | (4) $SN_2$ reaction |

| Codes: | A | B | C | D |
|---|---|---|---|---|
| (a) | 3 | 4 | 2 | 1 |
| (b) | 3 | 4 | 1 | 2 |
| (c) | 4 | 3 | 1 | 2 |
| (d) | 4 | 3 | 2 | 1 |

Q.25

| List I | List II |
|---|---|
| (A) $CH_3–C\equiv C–CH_3 \longrightarrow$ cis-2-butene | [illegible]$_3(l)$ |
| (B) $CH_3-C\equiv C-CH_3 \longrightarrow$ trans-2-butene | (2) $H_2$/Pd/$BaSO_4$ |
| (C) $CH_3C\equiv C–CH_3 \longrightarrow$ 1-Butyne | (3) alc. KOH, $\Delta$ |
| (D) $CH_3–CH_3–C\equiv CH \longrightarrow$ 2-Butyne | (4) $NaNH_2$ , $\Delta$ |

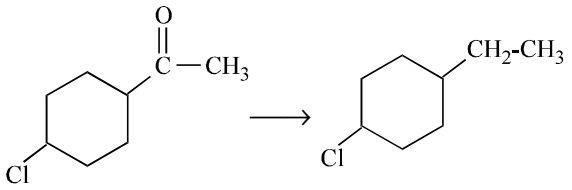
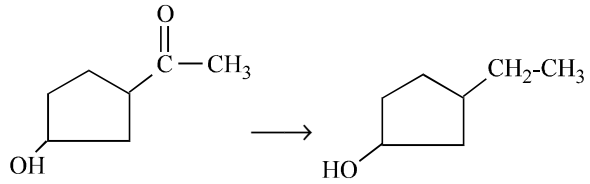
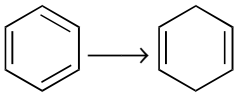
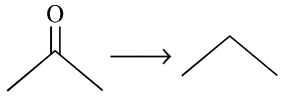
| Codes: | A | B | C | D |
|---|---|---|---|---|
| (a) | 2 | 1 | 3 | 4 |
| (b) | 1 | 2 | 4 | 3 |
| (c) | 1 | 2 | 3 | 4 |
| (d) | 2 | 1 | 4 | 3 |

Q.26

| List I | List II |
|---|---|
| (A) $RCOONa \xrightarrow{\text{electrolysis}} R–R$ | (1) Corey-Housh reaction |
| (B) $R–CH_2–COOH \xrightarrow[\Delta]{\text{Soda lime}} R–CH_3$ | (2) Kolbe electrolysis |
| (C) $RCOOH \xrightarrow[(ii)\,Cl_2/\Delta]{(i)\,AgNO_3} R–Cl$ | (3) Oakwood degradration |
| (D) $R'–X + R_2CuLi \longrightarrow R–R'$ | (4) Hunsdiecker reaction |

| Codes: | A | B | C | D |
|---|---|---|---|---|
| (a) | 2 | 3 | 4 | 1 |
| (b) | 1 | 3 | 4 | 2 |
| (c) | 2 | 4 | 3 | 2 |
| (d) | 2 | 4 | 1 | 3 |

Q.27

| List I | List II |
|---|---|
| (A)  | (1) Birch reduction |
| (B)  | (2) Stephen's reduction |
| (C)  | (3) Wolf-Kishner reduction |
| (D)  | (4) Clemmensen reduction |

Q.28

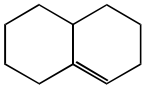
| List I | List II |
|---|---|
| (A) n-Hexane $\longrightarrow$ Benzene | (1) Wurtz reaction |
| (B) $CH \equiv CH \longrightarrow$ Benzene | (2) Coupling of reactants is taking place |
| (C) $CH_3-(CH_2)_6-CH_3 \longrightarrow$ 2,2,3,3 tetramethyl butane | (3) $AlCl_3$ + HCl at 300°C |
| (D) $CH_3-CH_2-X \longrightarrow$ n-Butane | (4) Polymerisation |
| | (5) Aromatic procducts is formed |
| | (6) Zn + $\Delta$ used as reagent |
| | (7) $Al_2O_3$ at high temperature |

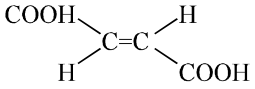
Q.29 Match List-I with List-II and select the correct answer using the codes given below the lists:

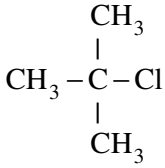
| | **List-I (Reaction)** | | **List-II (Reagents)** |
|---|---|---|---|
| (A) | $CH_3-CH=CH_2 \rightarrow CH_3-CHBr-CH_3$ | (P) | HBr |
| (B) | $CH_3-CH=CH_2 \rightarrow CH_3-CH_2-CH_2Br$ | (Q) | $Br_2$ |
| (C) | $CH_3-CH=CH_2 \rightarrow BrCH_2-CH=CH_2$ | (R) | HBr / Peroxide |
| (D) | $CH_3-CH=CH_2 \rightarrow CH_3-CHBr-CH_2Br$ | (S) | NBS |

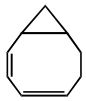
## *EXERCISE–II*

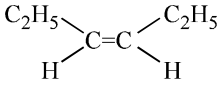
Q.1 Give the product of

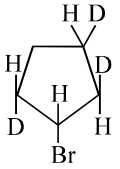
(a)  $\xrightarrow[THF]{BH_3}$ A $\xrightarrow{H_2O_2}$ B

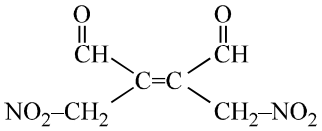
(b)  $\xrightarrow[2.\ D_2O_2,\ DO^-]{1.\ BD_3.THF}$

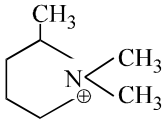
Q.2  $\xrightarrow{LiAlH_4}$ ?

Q.3 What are the ozonolysis products of 

Q.4  $\xrightarrow[2.\ H_2O/H^+]{1.\ CF_3CO_3H}$ A + B. What are A and B?

Q.5  $\xrightarrow[Alc.]{NaOH}$ A. Write the structure of **A**.

Q.6  $\xrightarrow{NH_2-NH_2/H_2O_2}$ A. Write the structure of **A**

Q.7  $\xrightarrow[\Delta]{^-OH}$ A. Write the structure of major product **A**.

Q.8 Give the structure of the alkene that yields on ozonolysis

(i) $CH_3CH_2CH_2CHO$ & HCHO

(ii) $C_2H_5COCH_3$ & $CH_3CH(CH_3)CHO$

(iii) Only $CH_3CO.CH_3$

(iv) $CH_3.CHO$ & HCHO & $OHC.CH_2.CHO$

(v) Only $OHC-CH_2CH_2CH_2-CHO$.

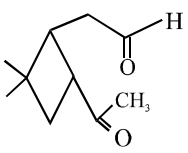
Q.9 One of the constituent of turpentine is α-pinene having molecular formula $C_{10}H_{16}$. The following scheme give reaction of α-pinene. Determine the structure of α–pinene & of the reaction products A through E.

$E(C_{10}H_{18}O_2)$ $\quad$ $A(C_{10}H_{16}Br_2)$

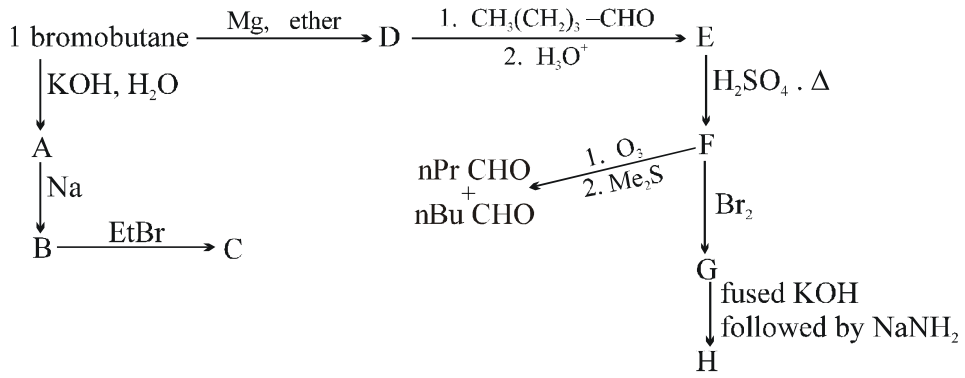
$\uparrow H_2O$ $\quad$ $\uparrow Br_2/CCl_4$

$C_{10}H_{16}O(D) \xleftarrow{PhCO_3H}$ α–pinene $\xrightarrow{Br_2+H_2O}$ B $\quad (C_{10}H_{17}OBr)$

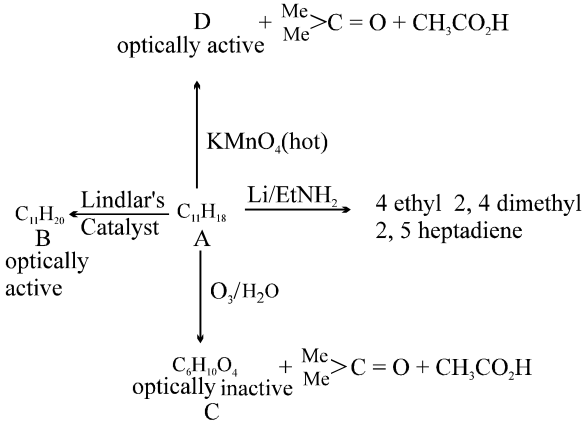
$\xrightarrow[H_2O]{OsO_4}$ F $\quad$ $\downarrow$ (i) $O_3$, (ii) $Me_2S$ $\quad$ $\downarrow$ (i) $H_2SO_4$, (ii) $\Delta$ $\rightarrow$ $C(C_{10}H_{15}Br)$



Q.10 Propose structures for intermediates & products A to K



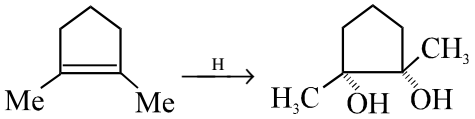

Q.11 Identify the following (A to D).

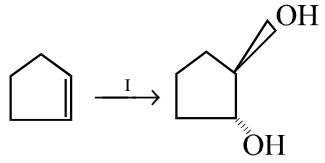



Q.12 What are A to K for the following reactions

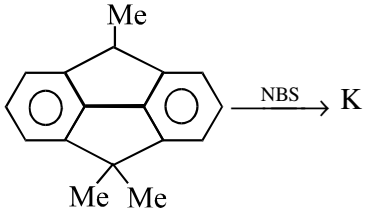
(i) $PhC \equiv CH + CH_3MgX \rightarrow A \xrightarrow{ArCH_2Cl} B \xrightarrow{Li/NH_3} C.$

(ii) $PhCH_2CH_2CH_3 + Br_2 \xrightarrow{h\gamma} D \xrightarrow[KOH]{alcoholic} E \xrightarrow{\text{cold dil. } KMnO_4} F \xrightarrow{\text{hot } KMnO_4} G$

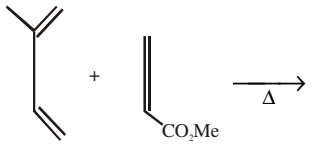
(iii) 

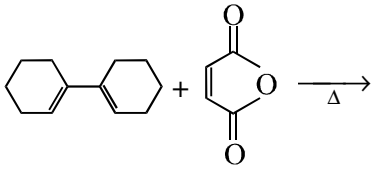

(iv) 

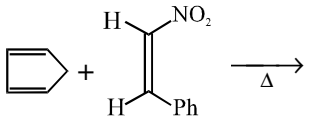

(v) $CF_3 - CH = CH_2 \xrightarrow{HBr} J$

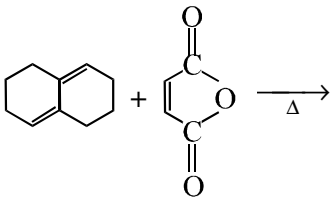
(vi) 

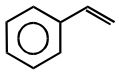
Q.13 What will be the product in the following reaction

(i) 

(ii) 

(iii) 

(iv) 

(v)  + $Br_2$ ⟶ A $\xrightarrow[2.\ CH_3I]{1.\ NaNH_2\ (3equiv.)\ NH_3}$ B

Q.14 (i) Compare the reaction of $CH_2 = CH_2$ & $CF_2 = CF_2$ with NaOEt in EtOH
(ii) $CCl_2 = CCl_2$ does not decolourise $Br_2$ solution - explain.

Q.15 Account for the collowing facts
(i) Ozonolysis if carried out in MeOH solvent a hydroxy peroxy ether is formed as unexpected product.
(ii) When 2, 3 dimethyl 2 butene is treated with $O_3$ in presence of HCHO in $CH_2Cl_2$ medium, an ozonide other than that expected of the starting alkene is formed. Identify the unexpected ozonide.

Q.16 Explain the following:
(i) 1, 2 shift does not take place during oxymericuration demercuration. Why?
(ii) Halogneation of alkene is anti addition but not syn addition. Why?
(iii) Anti markovnikov addition is not applicable for HCl. Why?
(iv) 1,4–addition takes place in butadi-ene. Why?
(v) C–H bond is stronger than C–C bond but in chlorination C–H bonds get cleaved but not C–C bond. Why?

Q.17 Conversion:
(i) $C_2H_2$ ⟶ racemic 2, 3 dibromobutane (ii) 2 butyne ⟶ 2 pentyne
(iii) Ethyne ⟶ Acetone (iv) Methane ⟶ n Butane (v) Ethene ⟶ Propionic Acid

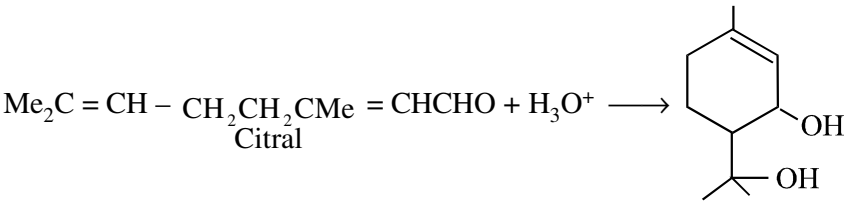
Q.18 Conversion:
(i) $C_2H_2$ ⟶ ethylidene diacetate (ii) $C_2H_2$ ⟶ Butyne diol
(iii) $C_2H_2$ ⟶ m nitroaniline (iv) cis but 2 ene ⟶ Trans but 2 ene

Q.19 Outline a stereospecific synthesis of meso 3, 4 dibromohexane from ethyne.

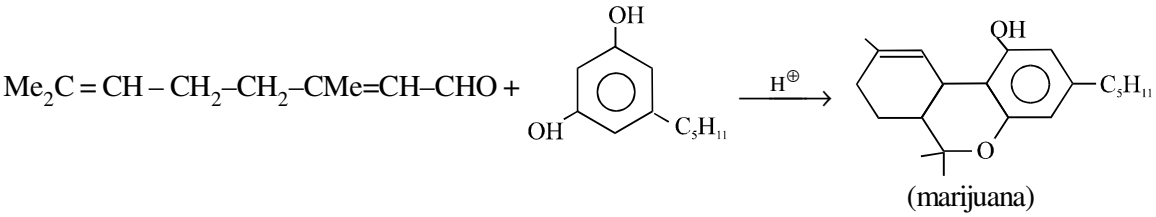
Q.20 How can you convert
(a) Ethane in to meso 2, 3 dimethyl oxiran
(b) $CaC_2$ into 1, 3, 5 hexatriene
(c) Trimethylsecbutyl amonium hydroxide into 1,4-butan-dial
(d) Cyclo hexanol into trans cyclo hexane-1, 2-diol

Q.21 How will you conver
(a) Hexane dial in to 1,3,5 hexatriene
(b) 1-methyl propyl ethanoate into 1,4-dichloro-2-butene

Q.22 Explain the mechanism of following conversion.



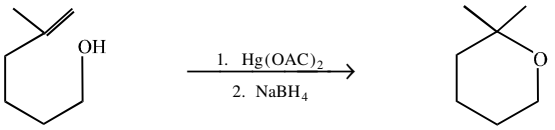

Q.23 When citral is allowed to react in presence of dilute acid with olivetol, there is obtained a mixture of products, one of which is drug marijuana. Reaction is as follows.




(marijuana)

Explain the mechanism.

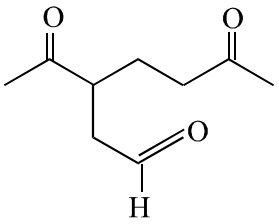
Q.24 The following cyclisation has been observed in the oxymercuration & demercuration of this unsaturated alcohol. Propose a mechanism for this reaction.




Q.25 Write the structural formula of limonene from the following observation:

(a) Limonene when treated with excess $H_2$ & Pt catalyst, the product formed is 1 isopropyl · 4 methyl cyclohexane

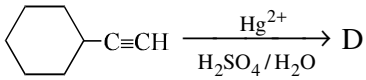
(b) When it is treated with $O_3$ & then $Zn/H_2O$ the products of the reaction are HCHO & following compound



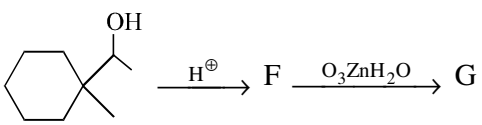

Q.26 (a) $MeCH_2–C\equiv CBr + CH\equiv CMe \xrightarrow{Cu^+} A$

(b) $CH_2Cl-CHCl_2 \xrightarrow{\bar{O}H} B$

(c) $CH_2=CH-\underset{\displaystyle OH}{\underset{|}{CH}}-CH=CH-CH_3 \xrightarrow[H_2SO_4]{MeOH} C$

(d) 


(e) $Cl_3C–CH=CH_2 \xrightarrow{HOBr} E$

(f) 

Q.27 Acetylene is acidic but it does not react with NaOH or KOH. Why?

Q.28 $CH\equiv C-CH_2-CH=CH_2$, adds up HBr to give $CH\equiv C-CH_2-CHBr-CH_3$ while $CH\equiv C-CH=CH_2$ adds up HBr to give $CH_2=C\,.\,Br\,.\,CH=CH_2$.

Q.29 Chlorination of ethane to ethyl chloride is more practicable than the chlorination of n-pentane to 1-chloropentane.

Q.30 Why n-pentane has higher boiling point than neopentane?

## *EXERCISE–III*

Q.1 0.37 gm of ROH was added to $CH_3MgI$ and the gas evolved measured 112 cc at STP. What is the molecular wt. of ROH ? On dehydration ROH gives an alkene which on ozonolysis gives acetone as one of the products. ROH on oxidation easily gives an acid containing the same number of carbon atoms. Gives the structures of ROH and the acid with proper reasoning.

Q.2 An alkane A ($C_5H_{12}$) on chlorination at $300^0$ gives a mixture four different mono chlorinated derivatives B, C, D and E. Two of these derivatives give the same stable alkene F on dehydrohalogenation, On oxidation with hot alkaline $KMnO_4$ followed by acidification of F gives two products G and H. Give structures of A to H with proper reasoning.

Q.3 There are six different alkene A, B, C, D, E and F. Each on addition of one mole of hydrogen gives G which has the lowest molecular wt hydrocarbon containing only one asymmetric carbon atom. None of the above alkene give acetone as a product on ozonolysis. Give the structures of A to F. Identify the alkenes that is likely to give a ketone containing more than five carbon atoms on treatment with a warm conc. solution of alkaline $KMnO_4$.

Q.4 3, 3–dimethyl–1–butene and HI react to give two products, $C_6H_{13}I$. On reaction with alc. KOH one isomer, (I) gives back 3,3–dimethyl–1–butene the other (J) gives an alkene that is reductively ozonized to $Me_2C=O$. Give the structures of (I) and (J) and explain the formation of the later.

Q.5 Three isomeric alkenes A, B and C, $C_5H_{10}$ are hydrogenated to yield 2–methylbutane A and B gave the same $3^0$ ROH on oxymercuration – demercuration. B and C give different $1^0$ ROH's on hydroboration -oxidation. Supply the structures of A, B & C.

Q.6 Two isomeric alkyl bromides A and B ($C_5H_{11}Br$) yield the following results in the laboratory. A on treatment with alcoholic KOH gives C and D ($C_5H_{10}$). C on ozonolysis gives formaldehyde and 2 methyl propanal. B on treatment with alcoholic KOH gives only C ($C_5H_{10}$). Deduce the structures of A, B, C and D. Ignore the possibility of geometrical and optical isomerism.

Q.7 Give the structure of A, B and C.

(a) A ($C_4H_8$) which adds on HBr in the presence and in the absence of peroxide to give the same product $C_4H_9Br$.

(b) B ($C_4H_8$) which when treated with $H_2SO_4$ / $H_2O$ give ($C_4H_{10}O$) which cannot be resoslved into optical isomers.

(c) C ($C_6H_{12}$), an optically active hydrocarbon on catalytic hydrogenation gives an optically inactive compound $C_6H_{14}$.

Q.8 An alkylhalide, X, of formula $C_6H_{13}Cl$ on treatment with potassium tertiary butoxide gives two isomeric alkenes Y and Z ($C_6H_{12}$). Both alkenes on hydrogenation gives 2, 3–dimethylbutane predict the structures of X, Y and Z.

Q.9 Identify a chiral compound C, $C_{10}H_{14}$, that is oxidized with hot $KMnO_4$ to Ph COOH, and an achiral compound D, $C_{10}H_{14}$, inert to oxidation under the same conditions.

Q.10 $C_7H_{14}$ (A) decolorises $Br_2$ in $CCl_4$ and reacts with $Hg(OAC)_2$ in THF. $H_2O$ followed by reduction with $NaBH_4$ to produced a resolvable compound B. (A) undergoes reductive ozonolysis to give the same compound (C) obtained by oxidation of 3–hexanol with $KMnO_4$. Identity A, B and C compound D, and isomers of A reacts with $BH_3$. THF and then $H_2O_2$/OH to give chiral E. Oxidation of E with $KMnO_4$ or acid dichromate affords a chiral carboxylic acid F. Reductive Ozonolysis of D, gives the same compound G which is obtained by oxidation of 2–methyl–3pentanol with $KMnO_4$ identify D, E, F and G.

Q.11 Three compounds A, B and C are isomers of the formula $C_5H_8$. All of them decolorises bromine in $CCl_4$ and gives a positive test with Baeyer's reagent. All the three compounds dissoslve in conc. $H_2SO_4$. Compound A gives a white ppt. with ammonical silver nitrate whereas B and C do not. On hydrogenation in presence of Pt catalyst, A and B both yield n–pentane whereas C gives a product of formula $C_5H_{10}$. On oxidation with hot alkaline $KMnO_4$ (B) gives acetic acid and $CH_3CH_2COOH$. Identify A, B & C.

Q.12 An unsaturated hydrocarbon (A) $C_6H_{10}$ readily gives (B) on treatment with $NaNH_2$ in liquid $NH_3$. When (B) is allowed to react with 1–chloropropane a compound (C) is obtained. On partial hydrogenation in the presence of lindlar's catalyst, (C) gives (D), $C_9H_{18}$. On ozonolysis, (D) gives 2, 2–dimethylpropanal and 1–butanal with proper reasoning give the structures of (A) (B), (C) and (D).

Q.13 A hydrocarbon A, of the formula $C_8H_{10}$, on ozonolysis gives compound B ($C_4H_6O_2$) only. The compound B can also be obtained from the alkylbromide ($C_3H_5Br$) upon treatment with magnesium in dry ether, followed by carbondioxide and acidification. Identify A, B and C and also give equations for the reactions.

Q.14 An organic compound (A), $C_6H_{10}$ on reduction first gives (B), $C_6H_{12}$ and finally (C), $C_6H_{14}$. (A) on ozonolysis followed by hydrolysis gives two aldehydes (D), $C_2H_4O$ and (E) $C_2H_2O_2$. Oxidation of (B) with acidified $KMnO_4$ gives the acid (F), $C_4H_8O_2$. Determine the structures of the compounds (A) to (F) with proper reasoning.

Q.15 Compound A ($C_6H_{12}$) is treated with $Br_2$ to form compound B ($C_6H_{12}Br_2$). On treating B with alcoholic KOH followed by $NaNH_2$ the compound C ($C_6H_{10}$) is formed. C on treatment with $H_2$/Pt forms 2-methylpentane. The compound 'C' does not react with ammonical $Cu_2Cl_2$ or $AgNO_3$. When A is treated with cold $KMnO_4$ solution, a diol D is formed which gives two acids E and F when heated with $KMnO_4$ solution. Compound E is found to be ethanoic acid. Deduce the structures from A to F.

Q.16 An optically active hydrocarbon (A), $C_8H_{12}$ gives an optically inactive compound (B) after hydrogenation. (A) gives no ppt. with $Ag(NH_3)_2^+$ and gives optically inactive (C), $C_8H_{14}$ with $H_2$ in presence of Pd / $BaSO_4$. Determine the structures, give suitable names for A, B, C & give your reasoning.

Q.17 A organic compound A having carbon and hydrogen, adds one mole of $H_2$ in presence of Pt catalyst to form normal hexane. On vigorous oxidation with $KMnO_4$, it gives a simple carboxylic acid containing 3 carbon atoms. Assign the structure to A.

Q.18 An organic compound A, $C_6H_{10}$ , on catalytic reduction first gives B, $C_6H_{12}$ , and finally C, $C_6H_{14}$. A on ozonolysis followed by hydrolysis gives two aldehydes D, $C_2H_4O$ and E, $C_2H_2O_2$. Oxidation of B with acidified $KMnO_4$ gives acid F.

Q.19 A hydrocarbon has 88.89% carbon and 11.11% hydrogen. 0.405 g sample of the hydrocarbon occupies 229.54 ml at 100°C and 1 atm pressure. It decolourises potassium permanganate solution and bromine water without evolving hydrobromic acid. It gave no precipitate with either ammoniacal silver nitrate or cuprous chloride solution. When it reacts with dilute $H_2SO_4$ in presence of mercuric sulphate, under appropriate conditions, methyl ethyl ketone is formed. What is the hydrocarbon. Write the structural formulae of the eight possible isomer of this compound.

Q.20 6g sample of a natural gas consisting of methane ($CH_4$) and ethylene ($C_2H_4$) was burned with excess of oxygen and 17.2g of carbon dioxide and some water was obtained as products. What percent by weight of the sample was ethylene.

## *EXERCISE–IV (A)*

Q.1 Alcoholic solution of KOH is a specific reagent for – **[IIT '90]**
(A) Dehydration (B) Dehydrogenation
(C) Dehydro halogenation (D) Dehalogenation

Q.2 Of the following, unsaturated hydrocarbons are – **[IIT '90]**
(A) ethyne (B) cyclohexane (C) n–propane (D) ethene

Q.3 1-chlorobutane on reaction with alcoholic potash gives – **[IIT '91]**
(A) 1–butene (B) 1–butanol (C) 2–butene (D) 2–butanol

Q.4 The hybridisation of carbon atoms in C–C single bond of $HC\equiv C-CH=CH_2$ is – **[IIT '91]**
(A) $sp^3–sp^3$ (B) $sp^2–sp^3$ (C) $sp–sp^2$ (D) $sp^2–sp^2$

Q.5 The product(s) obtained via oxymercuation ($HgSO_4 + H_2SO_4$) of 1–butyne would be –

(A) $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-CH_3$ (B) $CH_3–CH_2–CH_2–CHO$

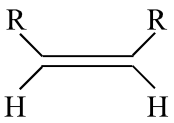
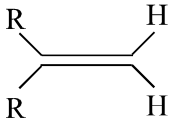
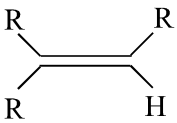
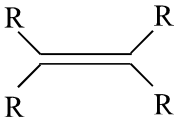
(C) $CH_3–CH_2–CHO + HCHO$ (D) $CH_3–CH_2–COOH + HCOOH$

Q.6 When cyclohexane is poured on water, it floats, because – **[IIT '97]**
(A) Cyclohexane is in 'boat' form (B) Cyclohexane is in 'chair' form
(C) Cyclohexane is in 'crown' form (D) Cyclohexane is less dense than water

Q.7 Which of the following compounds will show geometrical isomerism? **[IIT '98]**
(A) 2–butene (B) Propene (C) 1–phenylpropene (D) 2–methyl–2–butene

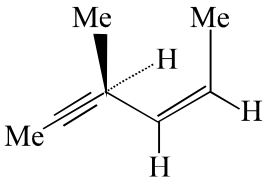
Q.8 In the compound $CH_2=CH–CH_2–CH_2–C\equiv CH$, the $C_2–C_3$ bond is of the type – **[IIT '99]**
(A) $sp–sp^2$ (B) $sp^3–sp^3$ (C) $sp–sp^3$ (D) $sp^2–sp^3$

Q.9 Which one of the following alkenes will react fastest with $H_2$ under catalytic hydrogenation condition – **[IIT '2000]**

(A)  (B)  (C)  (D) 

Q.10 Propyne and propene can be distinguished by – **[IIT '2000]**
(A) conc. $H_2SO_4$ (B) $Br_2$ in $CCl_4$ (C) dil. $KMnO_4$ (D) $AgNO_3$ in ammonia

Q.11 In the presence of peroxide, hydrogen chloride and hydrogen iodide do not give anti–Markovnikov addition to alkene because – **[IIT S'2001]**
(A) both are highly ionic
(B) one is oxidising and the other is reducing
(C) one of the step is endothermic in both the cases
(D) All the steps are exothermic in both cases

Q.12 

Hydrogenation of the above compound in the presence of poisoned paladium catalyst gives –

(A) An optically active compound (B) An optically inactive compound **[IIT '2001]**

(C) A racemic mixture (D) A diastereomeric mixture

Q.13 The reaction of propene with HOCl proceeds via the addition of – **[IIT '2001]**

(A) $H^+$ in first step (B) $Cl^+$ in first step

(C) $OH^-$ in first step (D) $Cl^+$ and $OH^-$ in single step

Q.14 The nodal plane in the π–bond of ethene is located in – **[IIT '2002]**

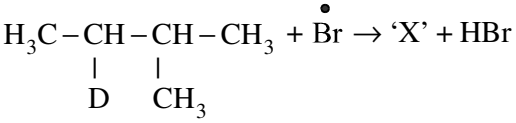
(A) the molecular plane

(B) a plane parallel to the molecular plane

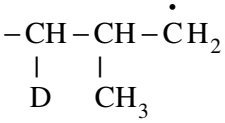
(C) a plane perpendicular to the molecular plane which contains the carbon–carbon σ–bond at right angle

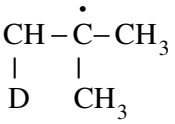
(D) a plane perpendicular to the molecular plane which contains the carbon–carbon σ–bond

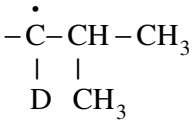
Q.15 Consider the following reactions – **[IIT '2002]**

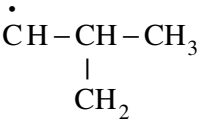


Identify the structure of the major product 'X'

(A) 

(B) 

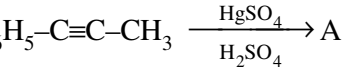
(C) 

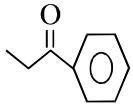
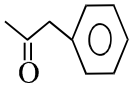
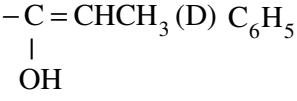
(D) 

Q.16 Identify a reagent from the following list which can easily distinguish between 1–butyne and 2-butyne- **[IIT '2002]**

(A) bromine, $CCl_4$ (B) $H_2$, Lindlar catalyst

(C) dilute $H_2SO_4$, $HgSO_4$ (D) ammonical $Cu_2Cl_2$ solution

Q.17  **[IIT '2003]**

(A)  (B)  (C)  (D) 

Q.18  5 compounds of molecular formula $C_4H_8Br_2$

Number of compounds in X will be: **[IIT '2003]**

(A) 2 (B) 3 (C) 4 (D) 5

Q.19 2–hexyne can be converted into trans–2–hexene by the action of : **[IIT '2004]**

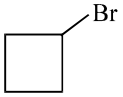
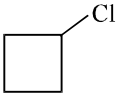
(A) $H_2$–Pd-$BaSO_4$ (B) Li in liq. $NH_3$ (C) $H_2$–$PtO_2$ (D) $NaBH_4$

Q.20 When Phenyl Magnesium Bromide reacts with tert. butanol, which of the following is formed?

(A) Tert. butyl methyl ether (B) Benzene

(C) Tert. butyl benzene (D) Phenol **[IIT '2005]**

Q.21 1–bromo–3–chlorocyclobutane when treated with two equivalents of Na, in the presence of ether which of the following will be formed? **[IIT '2005]**

(A)  (B)  (C)  (D) 

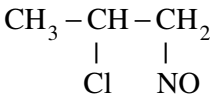
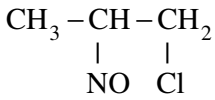
Q.22 Cyclohexene is best prepared from cyclohexanol by which of the following:

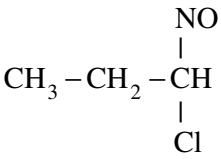
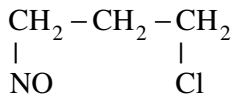
(A) conc. $H_3PO_4$ (B) conc. HCl/$ZnCl_2$ (C) conc. HCl (D) conc. HBr

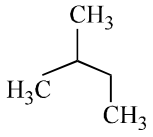
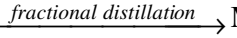
**[IIT '2005]**

Q.23 $CH_3$–CH=$CH_2$ + NOCl $\rightarrow$ P **[IIT 2006]**

Identify the adduct.

(A)  (B) 

(C)  (D) 

Q.24  N(isomeric products) $C_5H_{11}Cl$  M(isomeric products)

What are N and M?

(A) 6, 6 (B) 6, 4 (C) 4, 4 (D) 3, 3 **[IIT 2006]**

## *EXERCISE–IV (B)*

Q.1 $(CH_3)_2\overset{Cl}{\overset{|}{C}}-CH_2CH_3 \xrightarrow{\text{alc.KOH}} ?$ **[IIT 1992]**

Q.2 $C_6H_5CH_2\underset{Br}{\underset{|}{C}}HCH_3 \xrightarrow[\text{heat}]{\text{alc.KOH}} ? \xrightarrow{HBr} ?$ **[IIT 1993]**

Q.3 $C(C_6H_{12})$, an optically active hydrocarbon which on catalytic hydrogenation gives an optically inactive compound, $C_6H_{14}$. **[IIT 1993]**

Q.4 Draw the stereochemical structure of the product in the following reactions. **[IIT 1994]**

$$R–C\equiv C–R \xrightarrow[\text{Lindlar catalyst}]{H_2}$$

Q.5 Write down the structures of the stereoisomers formed when cis–2–butene is reacted with bromine. **[IIT 1995]**

Q.6 An organic compound $E(C_5H_8)$ on hydrogenation gives compound $F(C_5H_{12})$. Compound E on ozonolysis gives formaldehyde and 2–ketopropanal. Deduce the structure of compound E. **[IIT 1995]**

Q.7 Give the structures of the major organic products from 3–ethyl–2–pentene under each of the following reaction conditions. **[IIT 1996]**

(a) HBr in the presence of peroxide (b) $Br_2/H_2O$

(c) $Hg(OAc)_2/H_2O$; $NaBH_4$

Q.8 An alkyl halide, (X) of formula $C_6H_{13}Cl$ on treatment with potassium tertiary butoxide gives two isomeric alkenes (Y) and (Z) $(C_6H_{12})$. Both alkenes on hydrogenation give 2, 3–dimethylbutane. Predict the structures of (X), (Y) and (Z) **[IIT 1996]**

Q.9 3,3–Dimethyl–butan–2–ol loses a molecule of water in the presence of concentrated sulphuric acid to give tetramethylethylene as a major product. Suggest a suitable mechanism. **[IIT 1996]**

Q.10 One mole of the compound A (molecular formula $C_8H_{12}$), incapable of showing stereoisomerism, reacts with only one mole of $H_2$ on hydrogenation over Pd. A undergoes ozonolysis to give a symmetrical diketone B $(C_8H_{12}O_2)$. What are the structure of A and B? **[IIT 1997]**

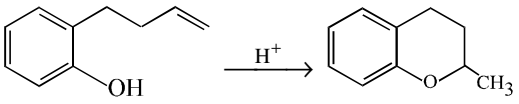
Q.11 Compound (A) $C_6H_{12}$ gives a positive test with bromine in carbon tetrachloride. Reaction of (A) with alkaline $KMnO_4$ yields only (B) which is the potassium salt of an acid. Write structure formulae and IUPAC name of (A) and (B). **[IIT 1997]**

Q.12 The central carbon–carbon bond in 1,3–butadiene is shorter than that of n–butane. Why? **[IIT 1998]**

Q.13 Write the intermediate steps for each of the following reaction **[IIT 1998]**

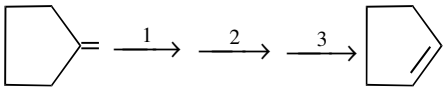
$C_6H_5CH(OH)C\equiv CH \rightarrow C_6H_5CH=CHCHO$

Q.14 Write the intermediate steps for each of the following reaction. **[IIT 1998]**

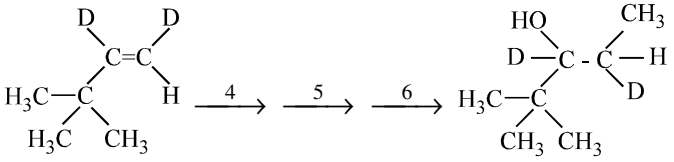


Q.15 Discuss the hybridisation of carbon atoms in allene ($C_3H_4$) and show the π–orbital overlaps. **[IIT 1999]**

Q.16 Complete the following – **[IIT 1999]**

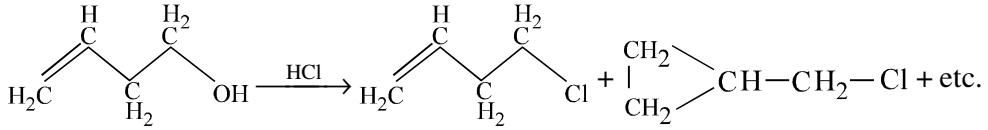


Q.17 Complete the following – **[IIT 1999]**



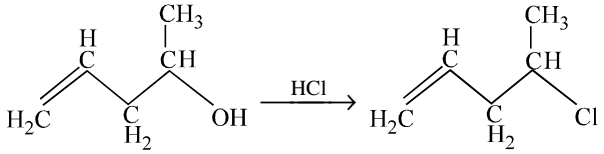
Q.18 Explain briefly the formation on the products giving the structures of the intermediates. **[IIT 1999]**

(i)



But

(ii)



Explain the non formation of cyclic product in (ii)

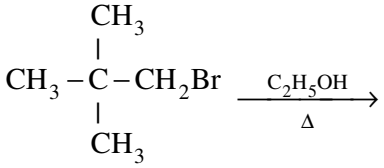
Q.19 Carry out the following transformation in not more than three steps. **[IIT 1999]**

$$CH_3–CH_2–C\equiv C–H \rightarrow CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-CH_3$$

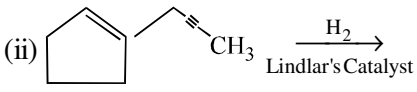
Q.20 $CH_2{=}CH^-$ is more basic than $HC\equiv C^-$ **[IIT 2000]**

Q.21 What would be the major product in each of the following reactions? **[IIT 2000]**

(i)



(ii)



Q.22 On reaction with 4N alcoholic KOH at 175 °C 1–pentyne is slowly converted into equilibrium mixture of 1.3% 1–pentyne (A), 95.2% 2–pentyne (B) and 3.5% 1,2–pentadiene (C). Give the suitable mechanism of formation of A, B and C with all intermediates. **[IIT 2001]**

Q.23 Identify X, Y and Z in the following synthetic scheme and write their structures. Is the compound Z optically active? Justify your answer. **[IIT 2002]**

$$CH_3CH_2C\equiv C–H \xrightarrow[(ii)\,CH_3CH_2Br]{(i)\,NaNH_2} X \xrightarrow{H_2/Pd-BaSO_4} Y \xrightarrow{\text{alkaline } KMnO_4} Z$$

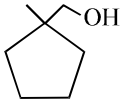
Q.24 A biologically active compound, Bombykol ($C_{16}H_{30}O$) is obtained from a natural source. The structure of the compound is determined by the following reactions.

(a) On hydrogenation, Bombykol gives a compound A, $C_{16}H_{34}O$, which reacts with acetic anhydride to give an ester.

(b) Bombykol also reacts with acetic anhydride to give another ester, which on oxidative ozonolysis ($O_3/H_2O_2$) gives a mixture of butanoic acid, oxalic acid and 10-acetoxy decanoic acid.

Determine the number of double bonds in Bombykol. Write the structures of compound A and Bombykol. How many geometrical isomers are possible for Bombykol? **[IIT 2002]**

Q.25 If after complete ozonolysis of one mole of monomer of natural polymer gives two moles of $CH_2O$ and one mole of $O=\overset{\overset{CH_3}{|}}{C}-CH=O$. Identify the monomer and draw the all-cis structure of natural polymer. **[IIT 2005]**

Q.26  $\xrightarrow{H^+,\Delta}$ X $\xrightarrow[(ii)\ Zn/Cl_3COOH]{(i)\ O_3}$ Y.

Identify X and Y. **[IIT 2005]**

# ANSWER KEY

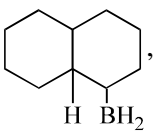
## *EXERCISE–I (A)*

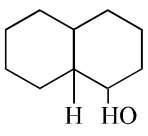
| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Q.1 | D | Q.2 | B | Q.3 | A | Q.4 | B | Q.5 | A | Q.6 | D | Q.7 | C |
| Q.8 | C | Q.9 | A | Q.10 | C | Q.11 | D | Q.12 | A | Q.13 | B | Q.14 | A |
| Q.15 | B | Q.16 | D | Q.17 | B | Q.18 | D | Q.19 | A | Q.20 | B | Q.21 | C |
| Q.22 | D | Q.23 | A | Q.24 | A | Q.25 | B | Q.26 | A | Q.27 | C | Q.28 | B |
| Q.29 | B | Q.30 | C | Q.31 | A | Q.32 | D | Q.33 | A | Q.34 | B | Q.35 | B |
| Q.36 | A | Q.37 | A | Q.38 | D | Q.39 | B | Q.40 | B | Q.41 | A | Q.42 | B |
| Q.43 | B | Q.44 | B | Q.45 | C | Q.46 | B | Q.47 | A | Q.48 | A | Q.49 | A |
| Q.50 | C | Q.51 | C | Q.52 | C | Q.53 | A | Q.54 | C | Q.55 | C | Q.56 | C |
| Q.57 | C | Q.58 | A | Q.59 | D | Q.60 | A | Q.61 | A | Q.62 | C | | |

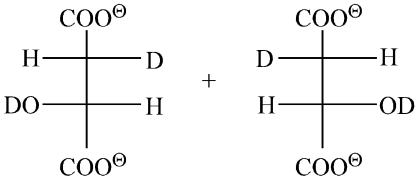
## *EXERCISE–I (B)*

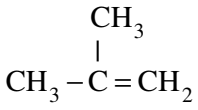
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Q.1 | A,C | Q.2 | A,C | Q.3 | A,B,C,D | Q.4 | A,B,C,D |
| Q.5 | B,C | Q.6 | A,B | Q.7 | A,B,C | Q.8 | A,B,C,D |
| Q.9 | A,C | Q.10 | A,B | Q.11 | A,B | Q.12 | A,C |
| Q.13 | B,C | Q.14 | B,C,D | Q.15 | A,C | Q.16 | A,B,D |
| Q.17 | A,C,D | Q.18 | B,C,D | Q.19 | A,D | Q.20 | D |
| Q.21 | A,C,D | Q.22 | (A) 3,4 ; (B) 3,4 ; (C) 1,2 ; (D) 2 | | | Q.23 | (c) |
| Q.24 | (c) | Q.25 | (d) | Q.26 | (a) | | |
| Q.27 | (A) 4; (B) 3; (C) 1; (D) 3,4 | | | Q.28 | (A)5,7 ; (B) 4,5 ;(C) 3 ; (D) 1,2,6 | | |
| Q.29 | (A) P; (B) R; (C) S; (D) Q | | | | | | |

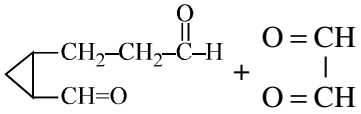
## *EXERCISE–II*

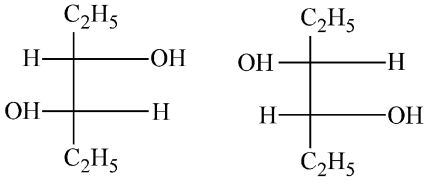
Q.1 (a) (A) = 

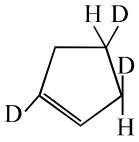
, (B) = 

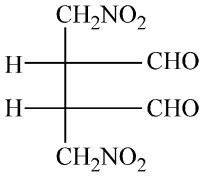

(b) 

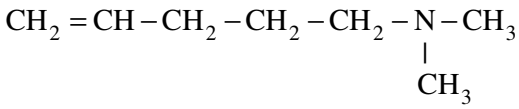
Q.2 

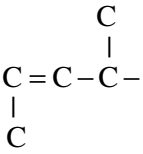
Q.3 

Q.4 A + B are two enatiomers 

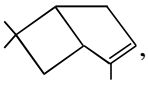
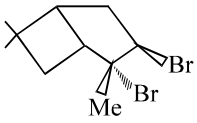
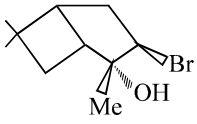
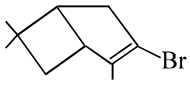
Q.5 

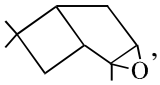
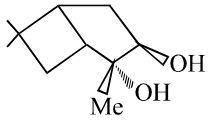
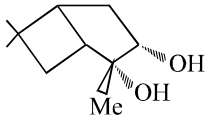
Q.6 

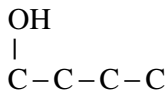
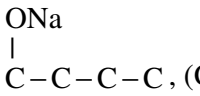
Q.7 

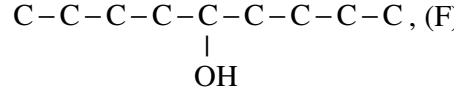
Q.8 (i) C–C–C–C=C, (ii)  , (iii)  , (iv) C–C=C–C–C=C,

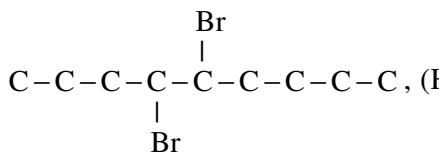
(v) 

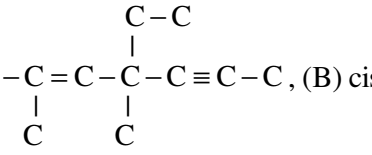
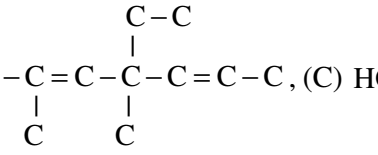
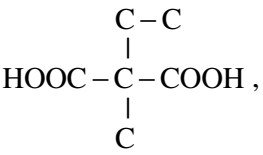
Q.9 α-pinene →  , (A) =  , (B) =  , (C) = 

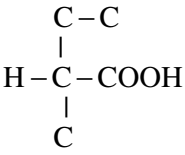
(D) =  , (E) =  , (F) = 

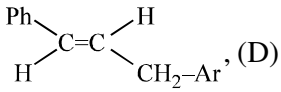
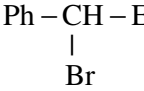
Q.10 (A)  , (B)  , (C) C–C–C–C–O–C–C, (D) BrMg–C–C–C–C

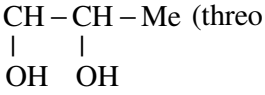
(E)  , (F) C–C–C–C=C–C–C–C–C ,

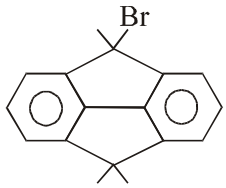
 , (H) C–C–C–C≡C–C–C–C–C

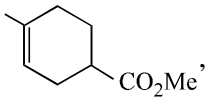
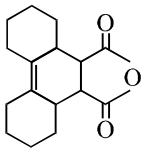
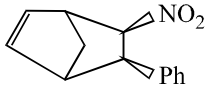
Q.11 (A)  , (B) cis  , (C)  ,

(D) 

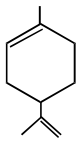
Q.12 (A) PhC≡CMgx, (B) Ph–C≡C–$CH_2$Ar, (C)  , (D)  ,

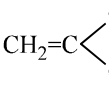
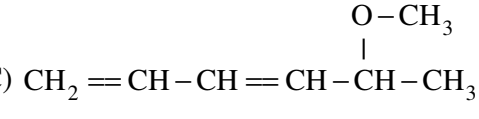
(E) Ph–CH=CH–Me trans, (F)  (threo mix.), (G) Ph–COOH

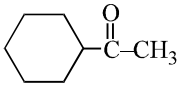
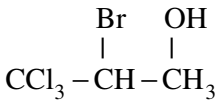
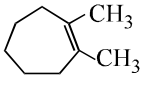
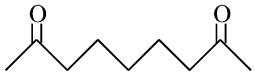
(H) cold dil. $KMnO_4$, (I) $HCO_3H$, (J) $CF_3CH_2CH_2Br$, (K) 

Q.13 (i)  , (ii)  , (iii)  , (iv) no reaction

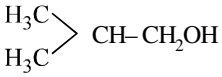
(v) (A)  , (B) Ph–C≡C–$CH_3$

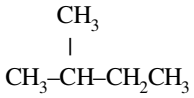
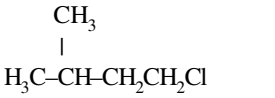
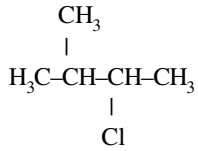
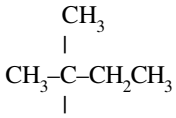
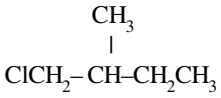
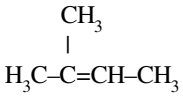
Q.14 (i) II is faster , (ii) unstable intermediate Q.25 

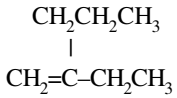
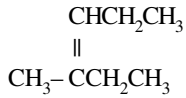
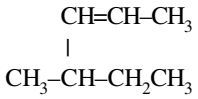
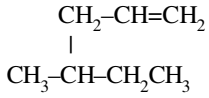
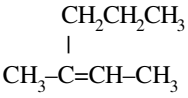
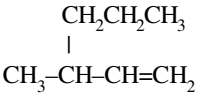
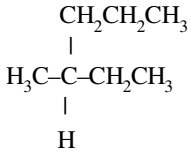
Q.26 (A) $MeCH_2$–C≡C–C≡CMe, (B)  , (C) 

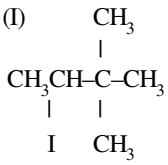
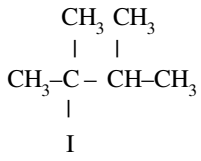
(D)  , (E)  , (F)  , (G) 

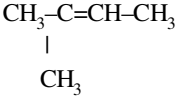
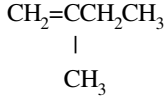
## *EXERCISE–III*

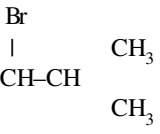
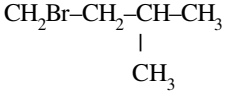
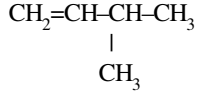
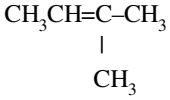
Q.1 

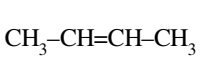
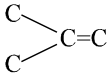
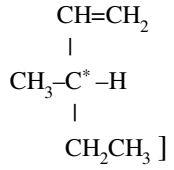
Q.2 (A) 
(B) 
(C) 
(D) 
(E) 
(F) 
(G) 
(H) 

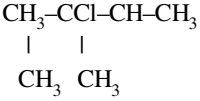
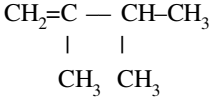
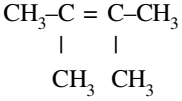
Q.3 (A) 
(B) 
(C) 
(D) 
(E) 
(F) 
(G) 

Q.4 (I) 
(J) 

Q.5 (A) 
(B) 
(C) 

Q.6 
(B) 
(C) 
(D) 

Q.7 (A) 
(B) 
(C) 

Q.8 (X) 
(Y) 
(Z) 

Q.9 (A) $PhCH(CH_3)CH_2CH_3$ (B) $PhC(CH_3)_3$

Q.10 (A) $CH_3–CH_2–C(=CH_2)–CH_2–CH_2CH_3$ (B) $CH_3CH_2–C(OH)(CH_3)–CH_2CH_2CH_3$ (C) $CH_3CH_2–C(=O)–CH_2CH_3$

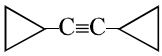
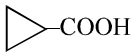
(D) $CH_3–CH(CH_3)–C(=CH_2)–CH_2CH_3$ (E) $CH_3CH(CH_3)–CH(CH_2OH)–CH_2CH_3$ (F) $CH_3–CH(CH_3)–CH(CO_2H)–CH_2–CH_3$

(G) $CH_3–CH(CH_3)–C(=O)–CH_2–CH_3$

Q.11 (A) $CH_3CH_2CH_2–C\equiv CH_3$ (B) $CH_3CH_2C\equiv C–CH_3$ (C) Cyclopentene

Q.12 (A) $CH_3–C(CH_3)_2–C\equiv CH$ (B) $H_3C–C(CH_3)_2–C\equiv C^- Na^+$

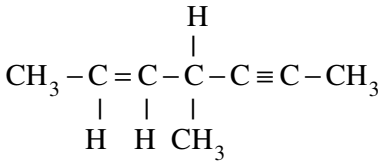
(C) $H_3C–C(CH_3)_2–C\equiv C–CH_2CH_2CH_3$ (D) $H_3C–C(CH_3)_2–C=CH–CH_2CH_2CH_3$

Q.13 (A)  (B)  (C) 

Q.14 (A) $CH_3–CH=CH–CH=CH–CH_3$ (B) $CH_3CH_2CH_2CH=CHCH_3$ (C) $CH_3CH_2CH_2CH_2–CH_2CH_3$
(D) $CH_3CHO$ (E) $CHO–CHO$ (F) $CH_3CH_2CH_2COOH$

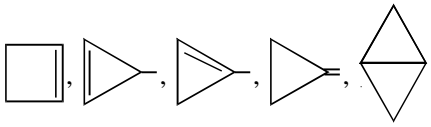
Q.15 (A) $CH_3–CH=CH–CH(CH_3)–CH_3$ (B) $CH_3–CH(Br)–CH(Br)–CH(CH_3)–CH_3$ (C) $CH_3–C\equiv C–CH(CH_3)–CH_3$

(D) $CH_3–CH(OH)–CH(OH)–CH(CH_3)–CH_3$ (E) $CH_3COOH$ (F) $H_3C–CH(CH_3)–COOH$

Q.16 

Q.17 $CH_3CH_2CH=CHCH_2CH_3$

Q.18 (A) $CH_3–CH=CH–CH=CH–CH_3$, (B) $CH_3–CH_2–CH=CH–CH_2–CH_3$,
(C) $CH_3–CH_2–CH_2–CH_2–CH_2–CH_3$, (D) $CH_3CHO$, (E) $CHO–CHO$, (F) $CH_3CH_2COOH$

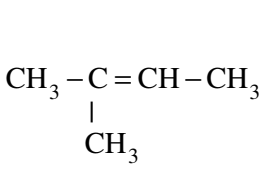
Q.19 Isomer are : C≡C–C–C, C=C–C=C, C=C=C–C, 

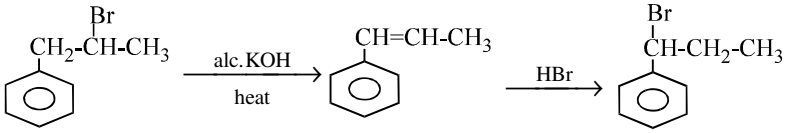
Q.20 23.7

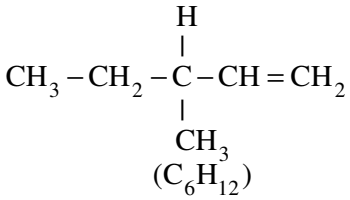
## *EXERCISE–IV (A)*

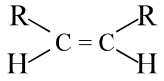
| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Q.1 | C | Q.2 | A,D | Q.3 | A | Q.4 | C | Q.5 | A |
| Q.6 | D | Q.7 | A,C | Q.8 | D | Q.9 | A | Q.10 | D |
| Q.11 | C | Q.12 | B | Q.13 | B | Q.14 | A | Q.15 | B |
| Q.16 | D | Q.17 | A | Q.18 | B | Q.19 | B | Q.20 | B |
| Q.21 | D | Q.22 | A | Q.23 | A | Q.24 | B | | |

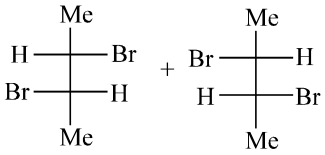
## *EXERCISE–IV (B)*

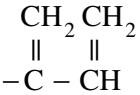
Q.1 

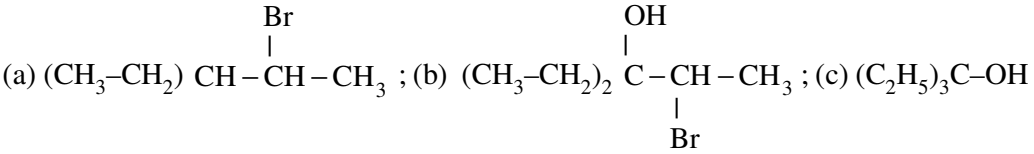
Q.2 

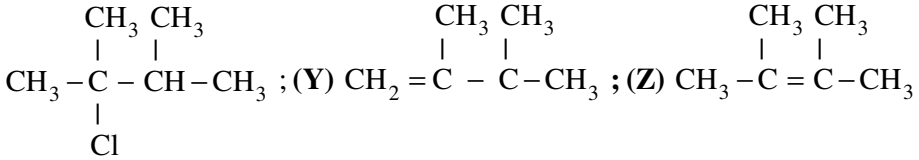
Q.3 

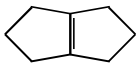
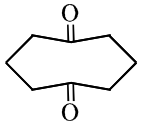
Q.4 

Q.5 

Q.6 

Q.7 

Q.8 

Q.10 **(A)**  **(B)** 

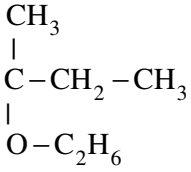
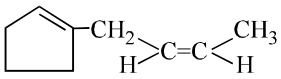
Q.11 **(A)** $CH_3–CH_2–CH=CH–CH_2–CH_3$ **(B)** $CH_3CH_2COOK$

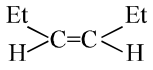
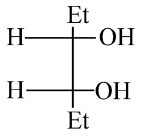
Q.16 $1 \rightarrow$ ozonolysis ; $2 \rightarrow LiAlH_4$ ; $3 \rightarrow H_2SO_4$

Q.17 $(4) \rightarrow HO–Cl$ ; $(5) \rightarrow CH_3MgCl$ ; $(6) \rightarrow H_2O/H^+$

Q.19 (1) $NaNH_2$, (2) Me–I, (3) $HgSO_4$ dil $H_2SO_4$

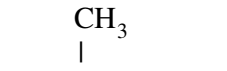
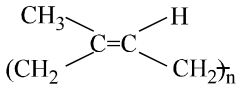
Q.20 higher electronegativity of sp carbon

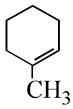
Q.21 (i)  (ii) 

Q.23 **(X)** $\rightarrow$ Et–C≡C–Et **(Y)** $\rightarrow$  **(Z)** $\rightarrow$  Z is meso so optically inactive.

Q.24 Bombykol :- HO–C–C–C–C–C–C–C–C–C–C–C=C–C=C–C–C–C

(A) :- $HO\text{-}(CH_3)_{15}CH_3$ 4 geometrical isomers are possible

Q.25 (a)  (b) 

Q.26 (X)  , (Y) 

# STUDY PACKAGE

# Target: IIT-JEE (Advanced)

# SUBJECT: CHEMISTRY-XII

# TOPIC: 16. Nitrogen Family

**Index:**


**1. Key Concepts**

**2. Exercise I**

**3. Exercise II**

**4. Exercise III**

**5. Exercise IV**

**6. Answer Key**

**7. 34 Yrs. Que. from IIT-JEE**

**8. 10 Yrs. Que. from AIEEE**

# NITROGEN FAMILY

## *Nitrogen($N_2$)*

***Occurence:*** Nitrogen is widely distributed in nature both in free as well as in the combined state. Air is the most abundant source of free nitrogen. It forms 75% by mass and 78% by volume of the air. In combined state, it is found as nitrates such as Chile saltpetre ($NaNO_3$), Indian saltpetre ($KNO_3$) and ammonium comopunds.

***Preparation:*** Nitrogen can be obtained from the following two sources:
(i) Nitrogen Compounds (ii) Air

**(i) Nitrogen from nitrogen compounds:** (a) Nitrogen in the laboratory can be obtained by heating ammonium nitrite or ammonium dichromate.

$$NH_4Cl + NaNO_2 \longrightarrow \underset{\text{Ammonium nitrite}}{NH_4NO_2} + NaCl$$

$$NH_4NO_2 \longrightarrow N_2 + 2H_2O$$

$$(NH_4)_2Cr_2O_7 \longrightarrow N_2 + Cr_2O_3 + 4H_2O$$

Nitrogen is collected by downward dispalcement of water.

(b) Pure nitrogen can be obtained by passing the ammonia vapours over heated CuO.

$$2NH_3 + 3CuO \longrightarrow N_2 + 3Cu + 3H_2O$$

$NH_3$ can also be oxidised to nitrogen by $Cl_2$, $Br_2$, a hypochlorite, a hypobromite or bleaching powder.

(c) It can be obtained by the action of nitrous acid (or $NaNO_2$ and dil. $H_2SO_4$) on urea.

$$\underset{\text{Urea}}{NH_2CONH_2} + 2HNO_2 \longrightarrow 2N_2 + CO_2 + 3H_2O$$

(d) Pure nitrogen is obtained in small amounts by heating sodium or barium azides in vacuum.

$$\underset{\text{Barium azide}}{Ba(N_3)_2} \longrightarrow 3N_2 + Ba$$

**(ii) From air:** (a) Commercially nitrogen is obtained by liquefaction of air. The resultant liquid isfractionally distilled in *Claude's apparatus*.

(b) By removing oxygen of the air with the use of chemical substances.

$$\text{Purified air} \longrightarrow \text{Hot Cu} \longrightarrow \text{Nitrogen}$$

$$2Cu + O_2 \longrightarrow 2CuO$$

$$\text{Purified air} \longrightarrow \text{Hot Cake} \longrightarrow CO_2, CO, N_2$$

$CO_2$ and CO are removed by usual methods.

$$\text{Purified air} \longrightarrow \text{Phosphorus} \longrightarrow P_2O_5 + N_2$$

$$P_2O_5 \xrightarrow{H_2O} H_3PO_4$$

***Properties:***

(i) It is a colourless, tasteless and odourless gas. It is slightly lighter than air as its vapour density is 14.0. It is sparingly soluble in water.

(ii) It can be liquefied to a colourless liquid (b. pt. $-195.8^0$C).

(iii) It does not help in combustion. Nitrogen itself is non-combustible.

(iv) It is chemically inert under ordinary conditions. However, it shows chemical activity under high temperatures.

(a) Nitrogen combines with oxygen under the influence of very high temperature like electric spark.

$$N_2 + O_2 \underset{}{\overset{3000^0C}{\rightleftharpoons}} 2NO \text{ (Nitric oxide)}$$

(b) Nitrogen combines with hydrogen in the presence of a catalyst (finely divided iron) at 200 atmospheres and 400-500$^0$C temperature.

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 \text{ (Ammonia)}$$

(c) Nitrogen combines with metals at red heat to form nitrides.

$$6Li + N_2 \xrightarrow{450^0C} 2Li_3N \text{ (Lithium nitride)}$$

$$3Mg + N_2 \xrightarrow{450^0C} Mg_3N_2 \text{ (Magnesium nitride)}$$

$$2Al + N_2 \xrightarrow{800^0C} 2AlN \text{ (Aluminium nitride)}$$

Non-metals like boron, silicon at bright red heat also combine with nitrogen.

$$2B + N_2 \longrightarrow 2BN \text{ (Boron nitride)}$$

$$3Si + 2N_2 \longrightarrow Si_3N_4 \text{ (Silicon nitride)}$$

(d) Nitrogen combines with calcium carbide to form calcium cyanamide at 1000$^0$C.

$$CaC_2 + N_2 \longrightarrow CaCN_2 + C$$

The mixture of calcium cyanamide and carbon is technically known as nitrolinm.

**Uses:** It is used in the manufacture of $NH_3$, $HNO_3$, $CaCN_2$ and other nitrogen compounds.

**Active nitrogen:** When an electric discharge is allowed to pass through nitrogen under very low pressure (about 2 mm), a brialliant luminiscence is observed which persists for sometime after the stoppage of the discharge. It is observed that nitrogen after the discharge is more active. This nitrogen is termed **active nitrogen.**

The exact nature of active nitrogen is not yet known.

## *Important compounds of Nitrogen*

### *1. Ammonia*

Nitrogen forms three well known hydrides with hydrogen:
(i) Ammonia, $NH_3$, (ii) Hydrazine, $NH_2 \cdot NH_2$ ($N_2H_4$); (iii) Hydrazoic acid, $N_3H$. Ammonia is the most important of these hydrides.

**Occurence:** $NH_3$ is found in traces in atmopshere. Ammonium salts such as ammonium chloride and ammonium sulphate are found in small amounts in the soil.

Discovery: It was first isolated by Priestly by the action of ammonium chloride and lime. It was named alkaline air.

**Preparation:** (i) Ammonium is obtained on a small scale from ammonium salts which evolve it when heated with caustic soda or lime.

$$NH_4Cl + NaOH \longrightarrow NH_3 + NaCl + H_2O$$

(Laboratory Preparation) $2NH_4Cl + \underset{\text{(slaked lime)}}{Ca(OH)_2} \longrightarrow 2NH_3 + CaCl_2 + 2H_2O$

(ii) Ammonia is formed when ammonium chloride is heated with litharge.

$$2NH_4Cl + PbO \longrightarrow 2NH_3 + PbCl_2 + H_2O$$

(iii) By reacting nitrides with water, ammonia is obtained.

$$AlN + 3H_2O \longrightarrow Al(OH)_3 + NH_3$$

$$Mg_3N_2 + 6H_2O \longrightarrow 3Mg(OH)_2 + 2NH_3$$

(iv) Ammonium can also be formed by doing reduction of nitrates and nitrites with zinc and caustic soda. Zinc and caustic soda produce nascent hydrogen which reacts with nitrates and nitrites to form ammonia.

$$NaNO_3 + 8H \xrightarrow{Zn/NaOH} NaOH + NH_3 + 2H_2O$$

$$NaNO_2 + 6H \xrightarrow{Zn/NaOH} NaOH + NH_3 + H_2O$$

(v) Calcium cyanamide is also obtained by heating ammonium compounds.

$$\underset{\text{Ammonium sulphate}}{(NH_4)_2SO_4} \xrightarrow{\text{Heat}} NH_3 + \underset{\text{Ammnoium hydrogen sulphate}}{NH_4HSO_4}$$

$$\underset{\text{Ammonium dihydrogen phosphate}}{NH_4H_2\cdot PO_4} \xrightarrow{\text{Heat}} NH_3 + \underset{\text{Metaphosphoric acid}}{HPO_3} + H_2O$$

(vi) Urea on treatment with caustic soda forms ammonia.

$$\underset{\text{urea}}{NH_2CONH_2} + 2NaOH \longrightarrow Na_2CO_3 + 2NH_3$$

**Drying of Ammonia gas:**

The common dehydrating agents like sulphuric acid or $CaCl_2$ or $P_2O_5$ cannot be used as these react with ammonia.

$$2NH_3 + H_2SO_4 \longrightarrow \underset{\text{Ammonium sulphate}}{(NH_4)_2SO_4}$$

$$CaCl_2 + 8NH_3 \longrightarrow \underset{\text{Addition product}}{CaCl_2\cdot 8NH_3}$$

$$P_2O_5 + 6NH_3 + 3H_2O \longrightarrow \underset{\text{Ammonium phosphate}}{2(NH_4)_3PO_4}$$

Fro drying, quick lime is used as it does not react with ammonia but reacts readily with moisture.

$$\underset{\text{quick lime}}{CaO} + H_2O \longrightarrow Ca(OH)_2$$

## *Manufacture of Ammonia:*

**(i) Haber's process:** The method involves the direct combination of nitrogen and hydrogen according to the following reaction:

$$N_2 + 3H_2 \rightleftharpoons 2NH_3 + 24.0 \text{ kcal}$$

Raw materials: Nitrogen and hydrogen are the chief raw materials. Nitrogen is obtained from air by liquefaction followed by fractional evaporation of liquid air. Hydrogen is obtained by electrolysis of water.

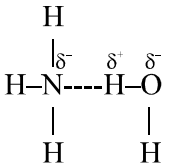
**(ii) Bosch Process:** From Powder gas & water gas

**(iii) Cyanamide process:** $CaCN_2 + \underset{\text{(steam)}}{3H_2O} \xrightarrow[\text{3-4 atm}]{180^0C} CaCO_3 + 2NH_3$

**(iv) From ammoniacal liquor obtained during coal distillation:** Large quantities of ammonia are obtained as a by-product in the manufacture of coal gas.

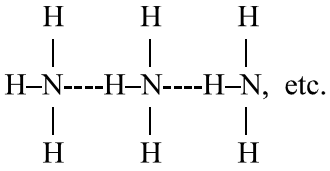
**Physical properties:** (i) Ammonia is a colourless gas with a characterstic pungent odour. it brings tears into the eyes.

(ii) It is highly soluble in water. This high solubility is due to the hydrogen bonding. The solubility of ammonia increases with increase of pressure and decreases with increase of temperature.



(iii) It can be easily liquefied at room temperature by the application of pressure.

(iv) Ammonia molecules link together to form associated molecules through hydrogen bonding.



Higher melting point and boiling point in comparison to other hydrides of V group are due to hydrogen bonding.

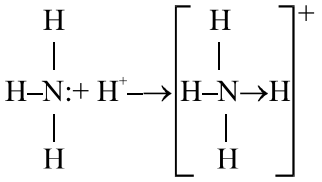
**Chemical Properties:(i) Stability :** It is highly stable. It decomposes into nitrogen and hydrogen at red heat or when electric sparks are passed through it.

$$2NH_3 \rightleftharpoons N_2 + 3H_2$$

**(ii) Combustion:** Ordinary, ammonia is neither combustible nor a supporter of combustion. However, it burns in the presence of oxygen to form nitrogen and water.

$$4NH_3 + 3O_2 \longrightarrow 2N_2 + 6H_2O$$

**(iii) Basic nature:** Ammonia is a Lewis base, accepting proton to form ammonium ion as it has tendency to donate an electron pair.



It forms salts with acids.

$$NH_3 + HCl \longrightarrow NH_4Cl \qquad \text{(Ammonium chloride)}$$

Thick white fumes

$$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4 \qquad \text{(Ammonium sulphate)}$$

It's solution is a weak base. the solution is described as aqueous ammonia. It's ionisation in water is represented as:

$$NH_3 + H_2O \longrightarrow NH_4OH \rightleftharpoons NH_4^+ + OH^-$$

The solution turns red litmus to blue and phenolphthalein pink.

**(iv) Oxidation:** It is oxidised to nitrogen when passed over heated CuO or PbO

$$3CuO + 2NH_3 \longrightarrow 3Cu + N_2 + 3H_2O$$
$$3PbO + 2NH_3 \longrightarrow 3Pb + N_2 + 3H_2O$$

Both chlorine and bromine oxidise ammonia.

$$2NH_3 + 3Cl_2 \longrightarrow N_2 + 6HCl$$
$$6NH_3 + 6HCl \longrightarrow 6NH_4Cl$$

---

$$8NH_3 + 3Cl_2 \longrightarrow N_2 + 6NH_4Cl$$

(excess)

When chlorine is in excess an explosive substance nitrogen trichloride is formed.

$$NH_3 + 3Cl_2 \longrightarrow NCl_3 + 3HCl$$

Iodine flakes when rubbed with liquor ammonia form a dark brown precipitate of ammoniated nitrogen iodide which explodes readily on drying.

$$2NH_3 + 3I_2 \longrightarrow NH_3 \cdot NI_3 + 3HI$$

Hypochlorites and hypobromites oxidise ammonia to nitrogen.

$$2NH_3 + 3NaClO \longrightarrow N_2 + 3NaCl + 3_2O$$

The oxidation of ammonia with bleaching powder occurs on warming.

$$3CaOCl_2 + 2NH_3 \longrightarrow 3CaCl_2 + N_2 + 3H_2O$$

Thus, ammonia acts as a reducing agent.

The restricted oxidation of $NH_3$ can be done with air, when the mixture is passed over heated platinum gauze at 700-800$^0$C.

$$4NH_3 + 5O_2 \longrightarrow 4NO + 6H_2O$$

This is the Ostwald's process and used for the manufacture of $HNO_3$.

**(v) Formation of amides:** When dry ammonia is passed over heated sodium or potassium, amides are formed with evolution of hydrogen.

$$2Na + 2NH_3 \longrightarrow \underset{\text{Sodamide}}{2NaNH_2} + H_2$$

**(vi) Reactions of aqueous ammonia:** Many metal hydroxides are formed which may be precipitated or remain dissolved in the form of complex compound in excess of $NH_4OH$.

$$FeCl_3 + 3NH_4OH \longrightarrow \underset{\text{ppt.}}{Fe(OH)_3} + 3NH_4Cl$$

$$AlCl_3 + 3NH_4OH \longrightarrow \underset{\text{ppt.}}{Al(OH)_3} + 3NH_4Cl$$

$$CrCl_3 + 3NH_4OH \longrightarrow \underset{\text{ppt.}}{Cr(OH)_3} + 3NH_4Cl$$

$$CuSO_4 + 2NH_4OH \longrightarrow \underset{\text{Blue ppt.}}{Cu(OH)_2} + (NH_4)_2SO_4$$

$$Cu(OH)_2 + (NH_4)_2CO_4 + 2NH_4OH \longrightarrow \underset{\substack{\text{Tetramine copper sulphate} \\ \text{(colourless solution)}}}{[Cu(NH_3)_4]SO_4} + 4H_2O$$

$$CdSO_4 + 4NH_4OH \longrightarrow \underset{\substack{\text{Cadmium tetramine sulphate} \\ \text{(Colourless solution)}}}{[Cd(NH_3)_4]SO_4} + 4H_2O$$

$$AgNO_3 + NH_4OH \longrightarrow \underset{\text{White ppt.}}{AgOH} + NH_4NO_3$$

$$AgOH + 2NH_4OH \longrightarrow \underset{\text{soluble}}{[Ag(NH_3)_2](OH)} + 2H_2O$$

AgCl also dissolve in $NH_4OH$ solution

$$AgCl + 2NH_4OH \longrightarrow \underset{\text{Diamine silver chloride}}{[Ag(NH_3)_2]Cl} + 2H_2O$$

$$ZnSO_4 + 2NH_4OH \longrightarrow \underset{\text{ppt.}}{Zn(OH)_2} + (NH_4)_2SO_4$$

$$Zn(OH)_2 + (NH_4)_2SO_4 + 2NH_4OH \longrightarrow \underset{\substack{\text{Tetramine zinc sulphate} \\ \text{(soluble) colourless}}}{[Zn(NH_3)_4]SO_4} + 4H_2O$$

Nickel salt first gives a green precipitate which dissolves in excess of $NH_4OH$.

$$NiCl_2 + 2NH_4OH \longrightarrow Ni(OH)_2 + 2NH_4Cl$$

$$Ni(OH)_2 + 2NH_4Cl + 4NH_4OH \longrightarrow [Ni(NH_3)_6]Cl_2 + 6H_2O$$

It forms a white precipitate with mercuric chloride.

$$HgCl_2 + 2NH_4OH \longrightarrow \underset{\substack{\text{Amido mercuric} \\ \text{chloride}}}{HgNH_2Cl} + NH_4Cl + H_2O$$

It forms a grey precipitate with mercurous chloride.

$$Hg_2Cl_2 + 2NH_4OH \longrightarrow \underbrace{Hg + HgNH_2Cl}_{\text{Grey}} + NH_4Cl + H_2O$$

**(vii) Reaction with Nessler's reagent:** A reddish brown ppt. is formed.

$$2KI + HgCl_2 \longrightarrow HgI_2 + 2KCl$$
$$2KI + HgI_2 \longrightarrow K_2HgI_4$$

Alkaline solution of $K_2HgI_{44}$ is called Nessler's reagent.
This gives brown ppt. with $NH_3$ called iodide of Million's base.

$$2K_2HgI_4 + NH_3 + 3KOH \longrightarrow \underset{\text{Brown ppt.}}{H_2NHgOHgI} + 7KI + 2H_2O$$

**Uses:** (i) Liquid hydrogen is not safe to transport in cylinders. Ammonia can be easily liquefied and transported safely in cylinders. Ammonia can be decomposed into hydrogen and nitrogen by passing over heated metallic catalyst. Thus, ammonia is the source for the production of hydrogen at any destination.
(ii) Ammonia is also used in the manufacture of urea which is an excellent fertilizer of nitrogen.

## *Hydrazine or Diamide $NH_2NH_2$ or $N_2H_4$*

This is another hydride of nitrogen. It is prepared by following methods:

**(i) Raschig's method:** A strogn aqueous solution of ammonia is boiled with sodium hypochlorite in presence of a little glue.

$$NH_3 + NaOCl \longrightarrow NH_2Cl + NaOH$$
$$\underset{\text{Chloramine}}{NH_2Cl} + NH_3 \longrightarrow \underset{\text{Hydrazine}}{NH_2{\cdot}NH_2} + HCl$$

$$2NH_3 + NaOCl \longrightarrow NH_2NH_2 + NaCl + H_2O$$

It burns in air liberating huge amount of energy. The alkyl derivatives of hydrazine are used these days as potential rocket fuels. It reacts with nitrous acid to give hydrazoic acid, $N_3H$.

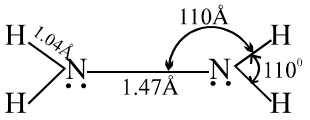
$$N_2H_4 + HNO_2 \longrightarrow N_3H + 2H_2O$$

Hydrazine and its salts act as powerful reducing agents.

$$PtCl_4 + N_2H_4 \longrightarrow Pt + N_2 + 4HCl$$
$$4AgNO_3 + N_2H_4 \longrightarrow 4Ag + N_2 + 4HNO_3$$
$$4AuCl_3 + 3N_2H_4 \longrightarrow 4Au + 3N_2 + 12\ HCl$$

It reduces Fehling's solution to red cuprous oxide, iodates to iodides and decolourises acidified $KMnO_4$ solution. It is used as a fuel for rockets, reducing agent and a reagent in organic chemistry.

Structure




## *Hydrazoic Acid, $N_3H$*

It is the third hydride of nitrogen. It is an acid while other hydrides, $NH_3$ and $N_2H_4$ are bases. It is prepared by the action of nitrous acid on hydrazine.

$$NH_2{\cdot}NH_2 + HNO_2 \longrightarrow N_3H + 2H_2O$$

It is also formed in the form of sodium salt by passing nitrous oxide on sodamide.

$$\underset{\text{Sodium}}{Na} \xrightarrow{\text{Dry } NH_3} \underset{\text{Sodamide}}{NaNH_2} \xrightarrow{N_2O} \underset{\text{Sodium azide}}{NaN_3} + H_2O$$

$$NaN_3 \xrightarrow{H_2SO_4} N_3H$$

It reduces acidified $KMnO_4$, nitrous acid, etc.

$$2N_3H + O \longrightarrow 3N_2 + H_2O$$

$$N_3H + HNO_2 \longrightarrow N_2 + N_2O + H_2O$$

It oxidises HCl into $Cl_2$

$$N_3H + 2HCl \longrightarrow N_2 + NH_3 + Cl_2$$

## *Oxides of Nitrogen*

## $N_2O_3$ and $N_2O_5$ monomeric other are dimeric

Nitrogen forms a number of oxides. The well known oxides of nitrogen are:

(i) Nitrogen oxide, $N_2O$
(ii) Nitric oxide, NO
(iii) Nitrogen trioxide, $N_2O_3$
(iv) Nitrogen dioxide or Di-nitrogen tetroxide, $NO_2$ or $N_2O$
(v) Nitrogen pentoxide, $N_2O_5$

### (I) Nitrogen Oxide, $N_2O$ or Laughing Gas (Neutral)

***Preparation:***

It can be prepared by heating ammonium nitrate or a mixture of sodium nitrate and ammonium sulphate.

$$NH_4NO_3 \rightarrow N_2O + 2H_2O$$

$$2NaNO_3 + (NH_4)_2SO_4 \rightarrow 2NH_4NO_3 + Na_2SO_4$$

$$\downarrow$$

$$2N_2O + 4H_2O$$

$$FeSO_4 + NO \rightarrow FeSO_4.NO$$

Addition product

$$H_2SO_4 + 2NH_3 \rightarrow (NH_4)_2SO_4$$

Ammonium Sulphate

The following reactions can also be used to prepare nitrous oxide.

$$NaNH_2 + N_2O \rightarrow NaN_3 + NaOH + NH_3$$

(a) By the action of cold and dilute nitric acid on zinc metal.

$$Fe/4Zn + 10HNO_3 \rightarrow 4Zn(NO_3)_2 + N_2O + 5H_2O$$

(b) By reducing nitric acid with stannous chloride and hydrochloric acid.

$$4SnCl_2 + 8HCl + 2HNO_3 \rightarrow 4SnCl_4 + N_2O + 5H_2O$$

(c) By reducing nitric oxide with sulphur dioxide.

$$2NO + SO_2 + H_2O \rightarrow H_2SO_4 + N_2O$$

(d) By heating the mixture of hydroxylamine hydrochloride and sodium nitrite (1 : 1)

$$NH_2OH.HCl + NaNO_2 \rightarrow N_2O + NaCl + 2H_2O$$

***Properties:***

(a) It is a colourless gas with pleasant odour and sweet taste.

(b) When inhaled in moderate quantity, it produces hysterical laughter, hence named as laughing gas. However, when inhaled for long, it produces insensibility and may prove fatal too.

(c) It is heavier than air.

(d) It is fairly soluble in cold water but not in hot water.

(e) It is neutral to litmus.

(f) It does not burn but support combustion. The burning material decompose nitrous oxide into nitrogen and oxygen. The oxygen then helps in the buring.

$$2N_2O \xrightarrow{520-900^oC} 2N_2 + O_2$$

It supports combustion of sulphur, phosphorus, magnesium, sodium, candle and a splineter.

$$S + 2N_2O \rightarrow SO_2 + 2N_2$$
$$4P + 10N_2O \rightarrow 2P_2O_5 + 10N_2$$
$$Mg + N_2O \rightarrow MgO + N_2$$

(g) It is decomposed by red hot copper.

$$Cu + N_2O \rightarrow CuO + N_2$$

(h) A mixture of hydrogen and nitrous oxide (equal volumes) explodes with violence.

$$N_2O + H_2 \rightarrow N_2 + H_2O$$

(i) $N_2O + NaNH_2 \rightarrow NaH_3 + NH_3 + NaOH$

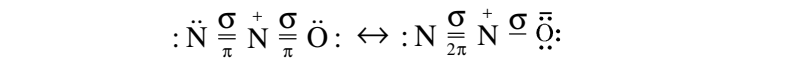
***Uses:***

(i) It is used as the propellant gas for whipped ice-cream.

(ii)A mixture of nitrous oxide and oxygen is used as an ananesthetic in dental and other minor surgical operations.

***Structure:***

$N_2O$ is linear and unsymmetrical molecule. It is considered as a resonance hybrid of the following two structures:



It has a very small value of dipole moment (0.116D)

***Tests:***

(i) It has sweet smell.

(ii) It supports the combustion of glowing splinter.

(iii) It does not form brown fumes with nitric oxide.

(iv) $N_2O$ does not form $H_2N_2O_2$ with $H_2O$ nor hyponitrites with alkali.

## (II) Nitric oxide, NO Neutral

***Preparation:***

(a) By the action of dilute nitric acid on copper (Lab. Method). the nitric oxide liberated is collected over water.

$$Ag/Hg/Pb/3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

The liberated gas may contain $NO_2$ and $N_2O$. These are separated by passing the mixture through ferrous sulphate solution. NO forms a dark nitroso-ferrous sulphate. When this solution is heated, pure nitric oxide is liberated.

$$FeSO_4 + NO \rightarrow FeSO_4.NO \xrightarrow{heate} FeSO_4 + NO$$

(impure gas) (Dark Brown) (Pure gas)

(b) A pure sample of nitric oxide is obtained when a mixture of $KNO_3$, $FeSO_4$ and dilute $H_2SO_4$ is heated. This is also a laboratory method.

$$2KNO_3 + H_2SO_4 \longrightarrow K_2SO_4 + 2HNO_3$$
$$2HNO_3 \longrightarrow H_2O + 2NO + 3O$$
$$[2FeSO_4 + H_2SO_4 + O \longrightarrow Fe_2(SO_4)_3 + H_2O] \times 3$$

$$2KNO_3 + 6FeSO_4 + 4H_2SO_4 \longrightarrow K_2SO_4 + 3Fe_2(SO_4)_3 + 2NO + 4H_2O$$

(c) Nitric oxide is the first product obtained from the following two processes during the manufacture of nitric acid.

### *(i) Electric arc process:*

By passing air through an electric arc, nitrogen and oxygen of the air combine together to form nitric oxide.

$$N_2 + O_2 \underset{}{\overset{\text{Electric arc}}{\rightleftharpoons}} 2NO$$

### *(ii) Ostwald's process:*

By restricted oxidation of ammonia with air in presence of platinum gauze catalyst at $750^0C$, nitric oxide is formed.

$$4NH_3 + 5O_2 \xrightarrow[750^0C,\ 6atm]{\text{Pt. gauze}} 4NO + 6H_2O$$

**Properties:**

(a) It is a colourless gas, slightly heavier than air.
(b) It is sparingly soluble in water.
(c) It is paramagnetic indicating the presence of unpaired electron in the molecule.
(d) It is neutral to litmus.
(e) It at once reacts with oxygen to give brown fumes of nitrogen dioxide.

$$2NO + O_2 \longrightarrow 2NO_2$$

(f) It is stable oxide. It decomposes into nitrogen and oxygen when heated at $800^0C$.

$$2NO \xrightarrow{800^0C} N_2 + O_2$$

(g) It is combustible and supports combustion of boiling sulphur and burning phosphorus.

$$S + 2NO \longrightarrow SO_2 + N_2$$

(h) It dissolves in cold ferrous sulphate solution by forming a hydrated nitrosyl complex.

$$[Fe(H_2O)_6]SO_4 + NO \longrightarrow [Fe(H_2O)_5NO]SO_4 + H_2O$$

Ferrous sulphate — Hydrated nitrosyl complex (Brown colour)

$$\downarrow \text{ Heat}$$
$$FeSO_4 + NO + 5H_2O$$

(i) It is oxidised to nitric acid by oxidising agents like acidified $KMnO_4$ or hypochlorous acid. Thus, it acts as a reducing agent.

$$6KMnO_4 + 9H_2SO_4 + 10NO \longrightarrow 3K_2SO_4 + 6MnSO_4 + 10HNO_3 + 4H_2O$$
$$[HClO \longrightarrow HCl + O] \times 3$$
$$[NO + O \longrightarrow NO_2] \times 3$$
$$3HClO + 2NO + H_2O \longrightarrow 2NO_3 + 3HCl$$

$HNO_3$ oxidises nitric oxide into $NO_2$.

$$2HNO_3 + NO \longrightarrow H_2O + 3NO_2$$

(j) It acts as an oxidising agent. It oxidises $SO_2$ to $H_2SO_4$ ande $H_2S$ to S.

$$SO_2 + 2NO + H_2O \longrightarrow H_2SO_4 + N_2O$$
$$H_2S + 2NO \longrightarrow H_2O + S + N_2O$$

When exploded with hydrogen it liberates nitrogen.

$$2H_2 + 2NO \longrightarrow 2H_2O + N_2$$

However, when a mixture of hydrogen and nitric oxide is passed over platinum black, ammonia is formed.

$$2NO + 5H_2 \longrightarrow 2NH_3 + 2H_2O$$

Stannous chloride reduces nitric oxide to hydroxylamine.

$$[SnCl_2 + 2HCl \longrightarrow SnCl_4 + 2H] \times 3$$
$$2NO + 6H \longrightarrow 2NH_2OH$$

---

$$3SnCl_2 + 6HCl + 2NO \longrightarrow 3SnCl_4\ 2NH_2OH$$

(k) Nitric oxide directly combines with halogen (fluorine, chlorine, bromine) to form corresponding nitrosyl halides.

$$2NO + X_2 \longrightarrow 2NOX$$

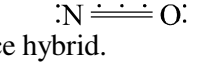
($X_2$ = $F_2$, $Cl_2$ or $Br_2$; 2NOX = Nitrosyl halide)

***Uses:***

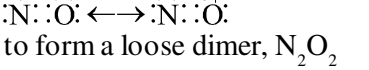
(i) In the manufacture of nitric acid.
(ii) As a catalyst in lead chamber process for the manufacture of sulphuric acid.
(iii) In the detection of oxygen to ditinguish it from nitrous oxide.

***Structure:***

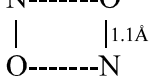
The molecule NO has eleven valnecy electrons and it is impossible for all of them to be paired.Hence, the molecule contains an odd electron which makes the gaseous nitric oxide as paramagnetic.



The structure is represented as a resonance hybrid.



In the liquid and solid states NO is known to form a loose dimer, $N_2O_2$




## *(III) Dinitrogen Trioxide, $N_2O_3$*

This oxide is also called nitrogen sesquioxide or nitrous anhydride.

***Preparation:*** It is obtained by the reduction of nitric acid with arsenious oxide.

$$2HNO_3 \rightarrow H_2O + N_2O_3 + 2O$$
$$As_2O_3 + 2O \rightarrow As_2O_5$$
$$As_2O_5 + 3H_2O \rightarrow 2H_3AsO_4$$

---

$$As_2O_3 + 2HNO_3 + 2H_2O \rightarrow 2H_3AsO_4 + N_2O_3$$

It is known in pure state in solid form at very very low temperature. In the vapour state, it is present as an equimolar mixture of NO and $NO_2$. The mixture of NO and $NO_2$ may be obntained by the action of 6N nitric acid on copper.

$$2Cu + 6HNO_3 \longrightarrow 2Cu(NO_3)_2 + \underbrace{NH + NO_2}_{N_2O_3} + 3H_2O$$

(a) It condenses to a bluw coloured liquid at $-30^0$C. The liquid when warmed at room temperature, decomposes to a mixture of NO and $NO_2$ (Brown coloured)

$$\underset{\text{Blue coloured liquid}}{N_2O_3} \xrightarrow{\text{Room Temperature}} \underbrace{NO + O_2}_{\text{Brown coloured gas}}$$

(b) It is an acidic oxide. If forms nitrous acid and water and hence the name nitrous anhydride.

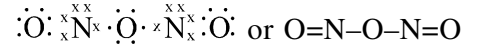
$$N_2O_3 + H_2O \longrightarrow 2HNO_2$$

The oxide combines with caustic alkali forming corresponding nitrite.

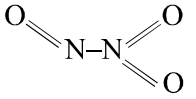
$$2NaOH + N_2O_3 \longrightarrow 2NaNO_2 + H_2O$$

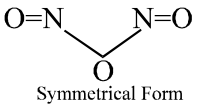
Structure:

Since the oxide is unstable in liquid and gaseous states and decomposes into NO and $NO_2$, it may be assumed that it has the following electronic structure:

 or O=N–O–N=O

The structure is supported by its diamagnetic behaviour strucutre of $N_2O_3$ is of two forms:


Assymmetrical Form


Symmetrical Form

## *(IV) Nitrogen Dioxide, $NO_2$ or DInitrogen Tetroxide, $N_2O_4$*

This oxide exists as $NO_2$ in gaseous state while at low temperature, it exists as a dimer $N_2O_4$ (Solid state)

$$\underset{\text{Brown gas}}{2NO_2} \overset{-11^0C}{\rightleftharpoons} \underset{\text{Colourless solid}}{N_2O_4}$$

**Preparation:** It is prespared in the laboratory either by heating nitrates of heavy metal or by the action of concentrated nitric acid on metals like copper, silver, lead etc.

$$2Pb(NO_3)_2 \longrightarrow 2PbO + 4NO_2 + O_2$$

The mixture of nitrogen dioxide and oxygen is passed through a U-tube cooled by freezing mixture. Nitrogen dioxide condenses to a pale yellow liquid while oxygen escapes.

$$Zn(NO_3)_2 \longrightarrow ZnO + NO_2 + O_2$$

$$AgNO_3 \longrightarrow Ag + NO_2 + O_2$$

$$Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

It is also obtained by air oxidation of nitric oxide.

$$2NO + O_2 \longrightarrow 2NO_2$$

**Properties:**

(a) It is brown colored gas wit pungent odour. Above $140^0$C, it is 100%$NO_2$. The liquid as well as solid is entirely $N_2O_4$ (dimer) at low temperature. The liquid boils at $22^0$C and solid melts at $-11^0$C.

(b) It decomposes completely into nitric oxide and oxygen at $620^0$C.

$$NO_2 + HCl \longrightarrow NOCl + Cl_2 + H_2O$$

$$2NO_2 \underset{}{\overset{620^oC}{\rightleftharpoons}} \underbrace{2NO + O_2}_{\text{Gaseous mixture colourless}}$$

(c) When reacted with cold water, it forms a mixture of nitrous acid and nitric acid.

$$2NO_2 + H_2O \longrightarrow HNO_2 + HNO_3$$

On account of this, it is known as mixed anhydride of these two acids. However, with an excess of warm water it forms nitric acid and nitric oxide.

$$3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$$

(d) When absorbed by alkalies, nitrites and nitrates are formed.

$$2NO_2 + 2NaOH \longrightarrow NaNO_2 + NaNO_3 + H_2O$$

(e) It acts as an oxidising agent. It oxidises metals like sodium, potassium, mercury, tin copper, etc.

$$NO_2 + 2Na \longrightarrow Na_2O + NO$$
$$NO_2 + 2Cu \longrightarrow Cu_2O + NO$$

None metals like carbon, sulphur, phosphorus when burnt in its stmosphere, are converted into corresponding oxides.

$$5NO_2 + 2P \longrightarrow P_2O_5 + 5NO$$
$$2NO_2 + S \longrightarrow SO_2 + 2NO$$
$$2NO_2 + C \longrightarrow CO_2 + 2NO$$

It liberates iodine from KI and turns starch-iodide paper blue.

$$2KI + 2NO_2 \longrightarrow 2KNO_2 + I_2$$

In aqueous solution, it oxidises $SO_2$ to sulphuric acid

$$SO_2 + H_2O + NO_2 \longrightarrow H_2SO_4 + NO$$

This reaction is used for the manufacture of $H_2SO_4$ by lead chamber process.
$H_2S$ is oxidised to S and CO to $CO_2$.

$$H_2S + NO_2 \longrightarrow H_2O + S + NO$$
$$CO + NO_2 \longrightarrow CO_2 + NO$$

(f) It behaves also as a reducing agent. It redduces ozone to oxygen.

$$2NO_2 + O_3 \longrightarrow N_2O_5 + O_2$$

It decolourises acidified $KMnO_4$ solution.

$$2KMnO_4 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2MnSO_4 + 3O_2 + 5O$$
$$10NO_2 + 5H_2O + 5O \longrightarrow 10HNO_3$$

---

$$2KMnO_4 + 3H_2SO_4 + 10NO_2 + 2H_2O \longrightarrow K_2SO_4 + 2MnSO_4 + 10\ HNO_3$$

**Uses**

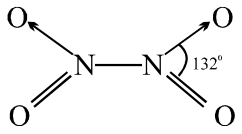
(i) It is used for the manufacture of nitric acid.
(ii) It is employed as a catalyst in the lead chamber process for the manufacture of sulphuric acid.

**Structure**

$NO_2$ molecule has V-shaped structure with O-N-O bond angle 132° and N-O bond length of about 1.19Å which is intermediate between a single and a double bond. Hence, $NO_2$ is regarded as a resonance hybrid of the following two streuctures.



The molecule is an odd electron molecule. The paramagnetic behaviour of $NO_2$ confirms this view. Due to prossession of odd eledtron, it is colored and has a tendency to polymerize to form a colour less dimer, $N_2O_4$. the dimer is planar in structure with N-N bond length 1.75Å.




## (v) Nitrogen Pentoxide, $N_2O_5$

This oxide is also known as nitric anhydride.

**Preparation**

It is prepared by distilling concentrated nitric acid with phosphorus pentoxide in a glass apparatus.

$$2NHO_3 \longrightarrow H_2O + N_2O_5$$
$$P_2O_5 + H_2O \longrightarrow 2HPO_3$$
$$P_2O_5 + 2HNO_3 \longrightarrow 2HPO_3 + N_2O_5$$

It is also prepared by the action of dry chlorine on solid silver nitrate at 95°C.

$$4AgNO_3 + 2Cl_2 \longrightarrow 4AgCl + 2N_2O_5 + O_2$$

**Properties**

(a) It is a white crystalline solid. The crystals melt at 30°C giving a yellow liquid which decomposes at 40°C to give brown $NO_2$. The decompostion occurs with explosion.

$$2N_2O_5 \longrightarrow 4NO_2 + O_2$$

(b) It is an acidic oxide. It reacts with water with hissing sound forming nitric acid.

$$N_2O_5 + H_2O \longrightarrow 2HNO_3$$

On account of this, it is known as nitric anhydride. With alkalies if forms nitrates.

$$2NaOH + N_2O_5 \longrightarrow 2NaNO_3 + H_2O$$

(c) It acts as a strong oxidising agent. It affect organic substances such as cork, rubber, etc. It oxidises iodine readily into iodine pentoxide.

$$I_2 + 5N_2O_5 \longrightarrow I_2O_5 + 10NO_2$$

(d) With aqueous NaCl, the ionic reaction takes place.

$$N_2O_5 + NaCl \longrightarrow naNO_3 + NO_2Cl$$

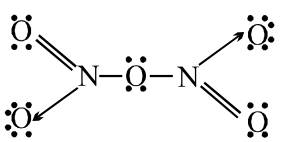
The reaction proves that $N_2O_5$ exists as ionic nitronium nitrate ($NO_2$, $NO_3$)

(e) $N_2O_5$ is decomposed by alkali metals.

$$N_2O_5 + Na \longrightarrow naNO_3 + NO_2$$

**Structure**

In the gaseous state, it exists as a symmetrical molecule having the structure $O_2N - O - NO_2$, N - O - N bond is almost linear. It may be represented as:



X- ray studies suggest that solid $N_2O_5$ is ionic in nature, i.e. nitronium nitrate, $NO_2$, $NO_3^-$.

## 5. Oxyacids of Nitrogen

Nitrogen forms a number of oxyacids. The most common and important oxyacids are:

### (i) Hyponitrous acid, $H_2N_2O_2$

**Preparation**

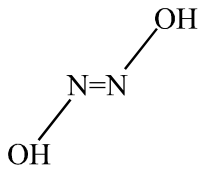
$$NaNO_2 \xrightarrow[\text{reduction}]{Na\ \ Hg} Na_2N_2O_2 \xrightarrow{AgNO_3} AgN_2O_2 \xrightarrow[AgCl]{HCl} H_2N_2O_2$$

**Properties**

It is colourless, hygroscopic. It is very weak dibasic acid but a strong reducing agent.

$$H_2N_2O_2 \longrightarrow H_2O + N_2O$$

It has zero dipole moment which is based on its trans structure.




### (ii) Nitrous Acid, $HNO_2$

The free acid is unknown. It is known only in solution.

**Preparation**

A solution of nitrous acid can be prepared by acidifying solutions of nitrites with mineral acids.

$$2NaNO_2 + H_2SO_4 \longrightarrow Na_2SO_4 + 2HNO_2$$
$$KNO_2 + HCl \longrightarrow KCl + HNO_2$$

Nitrates on heating with lead decompose to give nitrite.

$$NaNO_3 + Pb \xrightarrow{\Delta} NaNO_2 + 2HNO_2$$

An aqueous solution of the acid, free from any salt, can be prepared by treating barium nitrite with calculated amount of dilute sulphuric acid. Singce the acid is very unstable, the reaction is carried out at low temperature (freezing mixture temperature). The insoluble barium sulphate is filtered off.

$$Ba(NO_2)_2 + H_2SO_4 \longrightarrow \underset{\text{insoluble}}{BaSO_4} + 2HNO_2$$

A solution of nitrous acid may also the prepared by dissolving $N_2O_3$ in water.

$$HNO_2 + NaOH \longrightarrow NaNO_2 + H_2O$$
$$NH_3 + H_2O_2 \longrightarrow HNO_2 + H_2O$$

**Properties**

(a) Aqueous solution of nitrous acid is pale blue. This is due to the presence of nitrogen trioxide, $N_2O_3$ the colour fades on standing for sometime.

(b) It is weak acid and reacts with alkalies to form salts known as nitrites.

$$HNO_2 + NaOH \longrightarrow NaNO_2 + H_2O$$

(c) **Auto-oxidation**

The acid is unstable and even in cold solution, it undergoes auto-oxidation.

$$2HNO_2 \longrightarrow 2NO + H_2O + O$$
$$HNO_2 + O \longrightarrow HNO_3$$
---
$$2HNO_2 \longrightarrow 2NO + HNO_3 + H_2O$$
---

On heating, it decomposes into nitric oxide and nitrogen dioxide.

$$2HNO_2 \longrightarrow H_2O + N_2O_3 \rightleftharpoons NO + NO_2$$
(brown)

**(d) Oxidising nature**

It acts as an oxidising agent due to ease with which it decomposes to give nascent oxygen. the potential equation when it acts as an oxidising agent is:

$$2HNO_2 \longrightarrow H_2O + 2NO + O$$

(i) Iodine is liberated from potassium iodide.

$$2KI + H_2SO_4 + 2HNO_2 \longrightarrow K_2SO_4 + 2NO + I_2 + 2H_2O$$

(ii) Stannous chloride is oxidised to stannic chloride.

$$SnCL_2 + 2HCL + 2HNO_2 \longrightarrow SnCl_4 + 2NO + 2H_2O$$

(iii) Sulphur chloride is oxidised to sulphuric acid.

$$SO_2 + 2HNO_2 \longrightarrow H_2SO_4 + 2NO$$

(iv) Sulphur is formed by oxidation of hydrogen sulphide.

$$H_2S + 2HNO_2 \longrightarrow S + 2H_2O + 2NO$$

(v) Acidified ferrous sulphate is oxidised to ferric sulphate.

$$2FeSO_4 + H_2SO_4 + 2HNO_2 \longrightarrow Fe_2(SO_4)_3 + 2NO + 2H_2O$$

(vi) Sodium arsentie is oxidised to sodium arsenate.

$$Na_3AsO_3 + 2HNO_2 \longrightarrow Na_3AsO_4 + 2NO + H_2O$$

**(e) Reducing nature**

Nitrous acid acts as a reducing agent as it can be oxidised into nitric acid.

$$HNO_2 + O \longrightarrow HNO_3$$

(i) It reduces bromine to hydrobromic acid.

$$Br_2 + H_2O + HNO_2 \longrightarrow 2HBr + HNO_3$$

(ii) Acidified potassium permanganate is decolourised.

$$2KMnO_4 + 3H_2SO_4 + 5HNO_2 \longrightarrow K_2SO_4 + 2MnSO_4 + 5HNO_3 + 3H_2O$$

(iii) Acidified potassium dichromate is reduced to chromic sulphate (green)

$$K_2Cr_2O_7 + 4H_2SO_4 + 3HNO_2 \longrightarrow K_2SO_4 + Cr_2(SO_4)_3 + 3HNO_3 + 4H_2O$$

(iv) Hydrogen peroxide is reduced to water

$$H_2O_2 + HNO_2 \longrightarrow HNO_3 + H_2O$$

**(f) Reaction with ammonia**

It reacts with ammonia to form nitrogen and water.

$$NH_3 + HNO_2 \longrightarrow \underset{\text{Intermediate}}{[NH_4NO_2]} \longrightarrow N_2 + 2H_2O$$

**(g) Reaction with urea**

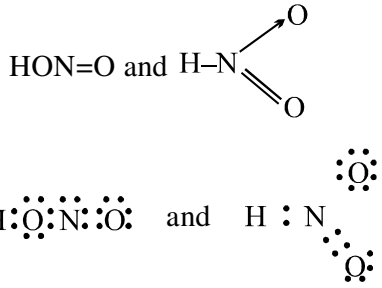
It decomposes urea and aliphatic primary amines to nitrogen

$$\underset{\text{Urea}}{NH_2CONH_2} + 2HNO_2 \longrightarrow 2N_2 + CO_2 + 3H_2O$$

$$\underset{\text{Ethylamine}}{C_2H_5NH_2} + HO \cdot NO \longrightarrow \underset{\text{Ethyl alcohol}}{C_2H_5OH} + N_2 + H_2O$$

**Structure**

Since nitrous acid forms two types of organic derivatives, the nitrites (R-ONO) and nitro compounds (R-$NO_2$), it is considered to be a automeric mixture ot two forms.




## *(iii) Nitric acid, $HNO_3$*

It was named aqua fortis (means strong water) by alchemists.

**Preparation**

**(i) Laboratory Method**

$$KNO_3 + \text{conc. } H_2SO_4 \longrightarrow KHSO_4 + HNO_3(\text{vap})$$

vapours of nitric acid evolved are condensed in a glass receiver.

**(ii) Industrial Preparation**

**(A) Birkeland Eyde Process or arc process**

step 1 $N_2 + O_2 \xrightarrow[\text{Electric Arc}]{3000^0\,C} 2NO - \text{heat}$

step 2 $NO + O_2 \longrightarrow NO_2$

step 3 $NO_2 + H_2O \longrightarrow HNO_2 + HNO_3$

step 4 $HNO_2 \longrightarrow HNO_3 + NO + H_2O$

**(B) Ostwald's Process**

step 1 $NH_3 + O_2 \xrightarrow[700-800^0\,C]{\text{Pt. gauze}} NO + H_2O + \text{heat}$

step 2 $NO + O_2 \longrightarrow NO_2$

step 3 $NO_2 + H_2O \longrightarrow HNO_2 + HNO_3$

step 4 $HNO_2 \longrightarrow HNO_3 + NO + H_2O$

**Properties**

**Physical**

Nitric acid usually acquires yellow colour due to its decomposition by sunlight into $NO_2$.

$$4HNO_3 \xrightarrow{\text{Sunlight}} 4NO_2 + 2H_2O + O_2$$

The yellow colour of the acid can be removed by warming it to 60-80°C and bubbling dry air through it. It has extremely corrosive action on the skin and causes painful sores.

**Chemical**

**(a)** It is very strong acid. It exhibits usual properties of acids. It reacts with basic oxides, carbonates, bicarbonates and hydroxides forming corresponding salts.

$$CaO + 2HNO_3 \longrightarrow Ca(NO_3)_2 + H_2O$$
$$Na_2CO_3 + 2HNO_3 \longrightarrow 2NaNO_3 + H_2O + CO_2$$
$$NaOH + HNO_3 \longrightarrow NaNO_3 + H_2O$$

**(b)Oxidising nature :** Nitric acid acts as a strong oxidising agent as it decomposes to give nascent oxygen easily.

$$2HNO_3 \longrightarrow H_2O + 2NO_2 + O$$
or $$2HNO_3 \longrightarrow H_2O + 2NO + 3O$$

**(i) Oxidation of non-metals:** The nascent oxygen oxidises various non-metals to their corresponding highest oxyacids.

(1) Sulphur is oxides to sulphuric acid

$$S + 6HNO_3 \xrightarrow[\text{conc. and hot}]{} H_2SO_4 + 6NO_2 + 2H_2O$$

(2) Carbon is oxidised to carbonic acid

$$C + 4HNO_3 \rightarrow H_2CO_3 + 4NO_2 + 2H_2O$$

(3) Phosphorus is oxidised to orthophosphoric acid.

$$2P + 10HNO_3 \xrightarrow[\text{conc. and hot}]{} 2H_3PO_4 + 10NO_2 + 2H_2O$$

(4) Iodine is oxidised to iodic acid

$$I_2 + 10HNO_3 \xrightarrow[\text{conc. and hot}]{} 2HIO_3 + 10NO_2 + 4H_2O$$

***(ii) Oxidation of metalloids***

Metalloids like non-metals also form highest oxyacids

(1) Arsenic is oxidised to arsenic acid

$$2As + 10HNO_3 \rightarrow 2H_3AsO_4 + 10NO_2 + 2H_2O$$
or $$As + 5HNO_3 \xrightarrow[\text{conc. and hot}]{} H_3AsO_4 + 5NO_2 + H_2O$$

(2) Antimony is oxidised to antimonic acid

$$Sb + 5HNO_3 \xrightarrow[\text{conc. and hot}]{} H_3SbO_4 + 5NO_2 + H_2O$$

(3) Tin is oxidised to meta-stannic acid.

$$Sn + 2HNO_3 \rightarrow H_2SnO_3 + 4NO_2 + H_2O$$

***(iii) Oxidation of Compounds:***

(1) Sulphur dioxide is oxidised to sulphuric acid

$$SO_2 + 2HNO_3 \rightarrow H_2SO_4 + 2NO_2$$

(2) Hydrogen sulphiode is oxidised to sulphur

$$H_2S + 2HNO_3 \rightarrow 2NO_2 + 2H_2O + S$$

(3) Ferrous sulphate is oxidised to ferric sulphate in presence of $H_2SO_4$

$$6FeSO_4 + 3H_2SO_4 + 2HNO_3 \rightarrow 3Fe_2(SO_4)_3 + 2NO + 4H_2O$$

(4) Iodine is liberated from KI.

$$6KI + 8HNO_3 \rightarrow 6KNO_3 + 2NO + 3I_2 + 4H_2O$$

(5) HBr, HI are oxidised to $Br_2$ and $I_2$, respectively.

$$2HBr + 2HNO_3 \rightarrow Br_2 + 2NO_2 + 2H_2O$$

Similarly, $2HI + 2HNO_3 \rightarrow I_2 + 2NO_2 + 2H_2O$

(6) Ferrous sulphide is oxidised to ferrous sulphate

$$FeS + HNO_3 \rightarrow FeSO_4 + 8NO_2 + 4H_2O$$

(7) Stannous chloride is oxidised to stannic chloride is presence of HCl.

$$2HNO_3 + 14H \rightarrow NH_2OH + NH_3 + 5H_2O$$

Hydroxylamine

$$NH_3 + HNO_3 \rightarrow NH_4NO_3$$

---

$$7SnCl_2 + 14HCl + 3HNO_3 \rightarrow 7SnCl_4 + Nh_2OH + NH_4NO_3 + 5H_2O$$

(8) Cane sugar is oxidised to oxalic acid.

$$C_{12}H_{22}O_{11} + 36HNO_3 \rightarrow 6(COOH)_2 + 36NO_2 + 23H_2O$$

**(c) Action on Metals:** Most of the metals will the exveption of noble metals like gold and platinum are attacked by Nitric acid plays a double role in the action of metals, i,e, it acts as an acid as well as an oxidising agent. ARmstrong postulated that primary action of nitric acid is to produce hydrogen in the nascent form. Before this hydrogen is allowed to escape, it reduces the nitric acid into number of products like $NO_2$, NO, $N_2O$, $N_2$ or $NH_3$ according to the following reactions:

$$\text{Metal} + HNO_3 \longrightarrow \text{Nitrate} + H$$

$$2HNO_3 + 2H \longrightarrow 2NO + 2H_2O$$

$$2HNO_3 + 6H \longrightarrow 2NO + 4H_2O$$

$$2HNO_3 + 10H \longrightarrow N_2 + 6H_2O$$

$$2HNO_3 + 16\ H \longrightarrow 2NH_3 + 6H_2O$$

The progress of the reaction is controlled by a number of factors:

(a) the nature of the metal,
(b) the concentration of the acid,
(c) the temperature of the reaction,
(d) the presence of other impurities.

| Concentration of nitric acid | Metal | Main Products |
|---|---|---|
| Very dilute $HNO_3$ (6%) | Mg, Mn | $H_2$ + Metal nitrate |
| | Fe, Zn, Sn | $NH_4NO_3$ + metal nitrate + $H_2O$ |
| Dilute $HNO_3$ (20%) | Pb, Cu, Ag, Hg | NO + metal nitrate + $H_2O$ |
| | Fe, Zn | $N_2O$ + metal nitrate + $H_2O$ |
| | Sn | $NH_4NO_3$ + $Sn(NO_3)_2$ |
| Conc. $HNO_3$(70%) | Zn, Fe, Pb, Cu, Ag | $NO_2$ + metal nitrate + $H_2O$ |
| | Sn | $NO_2$ + $H_2SnO_3$ Metastannic acid |

**Action on Proteins**

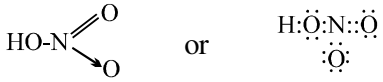
(i) Nitric acid attacks proteins forming a yellow nitro compound called xanthoprotein. It, therefore, stains skin and renders wool yellow. This property is utilized for the test of proteins.

(ii) **Oxidation** A number of organic compounds are oxidised.

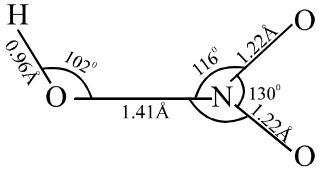
Sawdust catches fire when nitric acid is poured on it. Turpentine oil bursts into flames when treated with fuming nitric acid. Cane sugar is oxidised to oxalic acid. Toluene is oxidised to benzoic acid with dil. $HNO_3$.

**Structure**

Nitric acid is a monobasic acid, i.e., the molecule consist of one hydroxyl group as it is formed by the hydrolysis of nitryl chloride, $NO_2Cl$. It may be structurally represented as below:




Gaseous nitric aicd is a planar molecule. The bond lengths and bond angles as present in the molecule are represented in the figure:




It is supposed to exist in two resonting forms.



# PHOSPHORUS

It glows in the dark and was, therefore, called phosphorus (Greek work, phos=light, and phero=1 carry)

**Occurrence**

Since phosphorus is an active element, it is not found free in nature. It is widely distributed in nature in the combined state. It occurs as phosphates in the rocks and in the soil and as phosphoproteins in all living beings. It is an essential constituents of bones, teeth, blood and nervous tissure. It is necessary for the growth of plants. Bone ash contains about 80% calcium phosphate. it is present in milk, eggs and guano (excreta of seabirds). The principal minerals of phosphorous are

(i) Phosphorite $Ca_3(FO_4)_2$
(ii) Fluorapatite $3Ca_3(PO_4)_2.CaF_2$
(iii) Chlorapatite $3Ca_3(PO_4)_2.CaCl_2$

**Extraction**

Phosphorus is extracted either from phosphorite or bone ash by the application of following two processes.

**(i) Retort process or old process.**
**(ii) Electrothermal process or modern process.**

**(i) Retort process or old process**

The phosphorite mineral or bone ash is digested with concentrated sulphuric acid (about 60%). Insoluble calcium sulphate and orthophosphoric acid are formed.

$$Ca(PO_4)_2 + 3H_2SO_4 \rightarrow 3CaSO_4 + 3H_3PO_4$$

The syrupy liquid is separated from insoluble residue by filtration. The liquid is evaporated when it changes into metaphosphoric acid with evolution of water.

$$H_3PO_4 \rightarrow \underset{\text{Metaphosphoric acid}}{HPO_3} + H_2O$$

The metaphosphoric acid is mixed with powdered coke and distilled in fireclay retorts at a bright red het. The acid is reduced to phosphorus by carbon which comes in vaporized form. The vapours are condensed below water.

$$4HPO_3 + 10C \rightarrow P_4 + 10CO + 2H_2O$$

**(ii) Electrothermal process or modern process.**

The mixture of phosphorite, carbon and silica is dried and then introduced into the electric furnace. The furnace is an iron tank lined inside with refractory bricks. Carbon electrodes are fitted on either side of the furnace. The furnace has two exits, one for removal of vapours in the upper part of the furnace and the other for removal of slag in the lower part of furnace. The charge is introduced through the closed hopper arrangement.

The mixture is heated at 1400-1500°C by the discharge of an alternating current between carbon electrodes. Silica combines with calcium phosphate and forms phosphorus pentaxide which is reduced by carbon into phosphorus.

$$[Ca_3(PO_4)_2 + 2SiO_2 \rightarrow 3CaSiO_3 + P_2O_5] \times 2$$
$$2P_2O_5 + 10C \rightarrow P_4 + 10CO$$

---

$$2Ca_3(PO_4)_2 + 6SiO_2 + 10C \rightarrow 6CaSiO_3 + P_4 + 10COs$$

Vapours of phosphorus and carbon monoxide leave the furnace through the upper exit and are condensed under water. The liquid slag is tapped out periodically through an exit in the base.s

**Purification**

Phosphorus obtained is further purified by melting under acidified potassium dichromate solution when the impurities are oxidised. It is redistilled.

**Allotropic modifications of phosphorus**

Phosphorus exists in a number of allotropic forms. These forms are:

(i) Yellow or white phosphorus
(ii) Red phosphorus
(iii) Scarlet phosphorus
(iv) $\alpha$ -black phosphorus
(v) $\beta$ -black phosphorus
(vi) violet phosphorus

The main allotropic forms, however, are white and red.

**White or yellow phosphorus**

This is the common variety and is obtained by the methods described above. This form is chemically very active.

**Properties**

(a) The pure form is white but attains yellow colour on long standing due to the formation of a thin film of the red variety on the surface.
(b) It is a transparent waxy solid (sp. gr. 1.8) and can be easily cut with knife.
(c) It has characteristic garlic smell and is poisonous in nature. 0.15 g is the fatal dose. Vapours are also injurious. Persons working with phosphorus develop a disease in which the jaw bones decay. This disease is known as **phossy jaw.**
(d) It is insoluble in water but readily soluble in carbon disulphide.
(e) It melts at 44°C into a yellow liquid. It boils at 280°C.
(f) In contact with air, it undergoes slow combustion and glows dark. ths property is called phosphorescence.
(g)Its ignition temperature is low (about 30°C). It readily catches fire giving dense fumes of phosphorus pentoxide. it si, therefore, kept in water.

$$P_4 + 5O_2 \rightarrow P_4O_{10} \text{ or } 2O_2O_5$$

(h) It dissolves in caustic alkalies on boiling in an inert atmosphere and forms phosphite.

$$P_4 + \underset{\text{Caustic soda}}{3NaOH} + 3H_2O \rightarrow \underset{\text{sodium hypophosphite}}{3NaH_2PO_2} + PH_3 \uparrow$$

(i) It directly combines with halogens forming first trihalides and then pentahalides.

$$P_4 + 6Cl_2 \rightarrow 4PCl_3; \quad P_4 + 10Cl_2 \rightarrow 4PCl_5$$

(j) It combines with a number of metals forming phosphides.

$$6Mg + P_4 \rightarrow 2Mg_3P_2 \text{(Magnesium phosphide)}$$
$$6Ca + P_4 \rightarrow 2Ca_3P_2 \text{ (Calcium phosphide)}$$

(k) It combines with sulphur with explosive violence forming a number of sulphides such as $P_2S_3$, $P_2S_5$, $P_4S_3$ and $P_4S_7$.
(l) It acts as a strong reducing agent. It reduces nitric acid and sulphuric acid.

$$P_4 + 20HNO_3 \rightarrow 4H_3PO_4 + 20NO_2 + 4H_2O$$
$$P_4 + 10H_2SO_4 \rightarrow 4H_3PO_4 + 10SO_2 + 4H_2O$$

It reduces solutions of copper, silver and gold salts to corresponding metals,

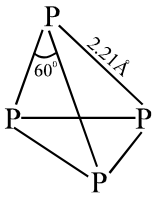
$$P_4 + 10CuSO_4 + 16H_2O \rightarrow 10Cu + 4H_3PO_4 + 10H_2SO_4$$

When the solution is heated, cuprous phosphide is formed.

$$3P_4 + 12\ CuSO_4 + 24\ H_2O \xrightarrow{\text{heat}} 4Cu_3P + 8H_3PO_3 + 12H_2SO_4$$
$$P_4 + 20AgNO_3 + 16H_2O \rightarrow 20Ag + 4H_3PO_4 + 20HNO_3$$

(m) when heated in inert atmosphere at 240°C, it changes into red variety.

$$\text{Yellow P} \xrightarrow[\text{Inert atm.}]{240-250^{\circ}\text{C}} \text{Red P}$$




**(n) Structure**

The vapour density of white phosphorus between 500-700°C, is 62 which corresponds to the molecular formula $P_4$. The four phosphorus atoms lie at the corners of a regular tetrahedron. Each phosphorus atom is linked to each of the other three atoms by covalent bonds. The P-P bond length is equal to 2.21Å. The bond angle is equal to 60° which suggests that he molecule is under strain and hence active in nature.

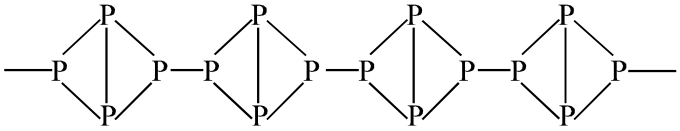
**Red Phosphorus**

**Preparation**

Red phosphorus is formed by heating yellow phosphorus, between 240-250°C, in presence of an inert gas. The heating is done in an egg shaped iron vessel provided with a upright tube closed by safety value. the thermometers placed in iron tubes help to regulate the temperature.

**Structure of red phosphorus**

The exact structure of red phosphorus is not yet known. It is regarded as a polymer consisting of chains of $P_4$ tetrahedral linked together possible in the manner as shown in the figure.



***Proposed molecular structure of red phophorus***

| ***Comparison between White and Red Phosphorus*** | | | |
|---|---|---|---|
| **S.No.** | **Property** | **White Phosphorus** | **Red Phosphorus** |
| 1. | Physical State | Soft waxy solid | Brittle powder |
| 2. | Colour | White when pure. Attains yellow colour on standing | Red |
| 3. | Odour | Garlic | Odourless |
| 4. | Specific gravity | 1.8 | 2.1 |
| 5. | Melting point | 44°C | Sublimes in absence of air at 290°C |
| 6. | Ignition temperature | Low, 30°C | High, 260°C |
| 7. | Solubility in water | Insoluble | Insoluble |
| 8. | Solubility in $CS_2$ | Soluble | Insoluble |
| 9. | Physiological action | Poisonous | Non-poisonous |
| 10. | Chemical activity | Very active | Less active |
| 11. | Stability | Unstable | Stable |

| | | | |
|---|---|---|---|
| 12. | Phosphorescence | Glows in dark | Does not glow in dark |
| 13. | Burning in air | Forms $P_4O_{10}$ | Forms $P_4O_{10}$ |
| 14. | Reaction with NaOH | Evolves phosphine | No action |
| 15. | Action of $Cl_2$ | Combines spontaneously to form $PCl_3$ & $PCl_5$ | Reacts on heating to form $PCl_3$ & $PCl_5$ |
| 16. | Reaction with hot $HNO_3$ | Forms $H_3PO_4$ | Forms $H_3PO_4$ |
| 17. | Molecular formula | $P_4$ | Complex Polymer |

**Uses of phosphorus**

(i) It is largely used in the match industry. Red phosphorus or scarlet phosphorus is preferred to yellow variety.
(ii) Yellow phosphorus is used as a rat poison.
(iii) Red phosphorus is used for the preparation of HBr and HI.
(iv) Radioactive phosphorus ($P^{32}$) is used in th treatement of leukemia and other blood disorders.
(v) It is used for making incendiary bombs and smoke screens.
(vi) It is used in the manufacture of phosphor bronze, an alloy of phosphorus, copper and tin.
(vii)It is used in the manufacture of compounds like hypophosphites (medicine), phosphorus chlorides in industry, calcium phosphide used in making Holme's signals and orthophosphoric acid.

**Compouns of Phosphorus**

**1. Phosphine, $PH_3$**

It is analogous to ammonia.

**Laboratory preparation**

It is prepared by blining yellow phosphorus with a concentrated solution of solution of sodium hydroxide in an inert atmosphere.

$$P_4 + 3NaOH + 3H_2O \rightarrow \underset{\text{sodium hypophosphite}}{3NaH_2PO_2} + PH_3$$

Besides $PH_3$, small amounts of hydrogen and phosphorus dihydride ($P_2H_4$) are also formed.

$$P_4 + 4NaOH + 4H_2O \rightarrow 3NaH_2PO_2 + 2H_2$$
$$3P_4 + 8NaOH + 8H_2O \rightarrow 8NaH_2PO_2 + 2P_2H_4$$

As soon as the bubbles of the gas come in contact with air, they catch fire spontaneously forming rings of smoke known as vortex rings. This combustion is due to the presence of highly inflammable phosphorus dihydride ($P_2H_4$).

$$2P_2H_4 + 7O_2 \rightarrow \underset{\text{Metaphosphoric acid}}{4HPO_3} + 2H_2O$$

$P_2H_4$ can be removed from phosphine by the following methods:
(i) By passing the evolved gas through a freezing mixture which condenses $P_2H_4$.
(ii) By passing the gas through HI. $PH_3$ is absorbed forming phosphonium iodide. This on treatment with casutic potash gives pure phosphine.

$$PH_4I + KOH \rightarrow KI + H_2O + PH_3$$

**Physical properties**

It is a colourless gas having unpleasant garlic like odour or rotten fish odour.

**(a) Decomposition**

When heated out of contact of air to 440°C or when electric sparks are passed through, phosphine decomposes into red phosphorus and hydrogen.

$$PH_3 \rightarrow P_4 + 6H_2$$

**(b) Combustibility**

A pure sample of phosphine is not spontaneously inflammable. It burns in air or oxygen when heated at 150°C.

$$2PH_3 + 4O_2 \rightarrow P_2O_5 + 3H_2O$$

The spontaneous in flammability of phosphine at the time of preparation is due to the presence of highly inflammable phosphorus dihydride, $P_2H_4$. This property is used in making Holme's signal. A mixture of calcium carbide and calcium phosphide is placed in metallic containers. Two holes are made and the container is thrown into the sea. Water enters and produces acetylene and phosphine respectively. The gaeous mixture catches fire spontaneously due to the presence of $P_2H_4$. The acetylene produces a bright luminous flame which serves as a signal to the approaching ship.

**(c) Action of chlorine**

Phosphine burns in the atmosphere of chlorine and forms phosphorus pentachloride.

$$PH_3 + 4Cl_2 \rightarrow PCl_5 + 3HCl$$

**(d) Basic nature**

Phosphine is neutral to litmus. However, it si a weak base, even weaker than ammonia. It reacts with HCl, HBr or HI to form phosphonium compounds.

$PH_3 + HCl \rightarrow PH_4Cl$ (Phosphonium chloride)

$PH_3 + HBr \rightarrow PH_4Br$ (Phosphonium bromide)

$PH_3 + HI \rightarrow PH_4I$ (Phosphonium iodide)

**(e) Action of nitric acid**

In contact with nitric acid phosphine begins to burn.

$$2PH_3 + 16\ HNO_3 \rightarrow P_2O_5 + 16NO_2 + 11H_2O$$

**(f) Addition compounds**

In forms addition compounds with anhydrous $AlCl_3$ and $SnCl_4$

$$AlCl_3 + 2PH_3 \rightarrow AlCl_3.2PH_3$$

$$SnCl_4 + 2PH_3 \rightarrow SnCl_4.2PH_3$$

When $PH_3$ is passed through cuprous chloride solution in HCl, it forms an addition compound.

$$Cu_2Cl_2 + 2PH_3 \rightarrow 2CuCl.\ PH_3$$

**(g) Formation of phosphides**

(i) When phosphine is passed through copper sulphate solution, a black precipitate of cupric phosphide is formed.

$$3CuSO_4 + 2PH_3 \rightarrow Cu_3P_2 + 3H_2SO_4$$

(ii) A black precipitate of silver phosphide is formed when phosphine is circulated through silver nitrate solution.

$$3AgNO_3 + PH_3 \rightarrow Ag_3P + 3HNO_3$$

**(h)** The mixture of $PH_3$ and $N_2O$ or $PH_3$ and NO explodes in presence of electric spark.

$$PH_3 + 4N_2O \rightarrow H_3PO_4 + 4N_2$$

**2. Oxides of Phosporus**

Phosphorus forms three important oxides. These exist in dimeric forms.

(i) Phosphorus trioxide, $P_2O_3$ or $P_4O_6$

(ii) Phosphorus tetroxide, $P_2O_4$ or $P_4O_8$

(iii) Phosphorus pentoxide, $P_2O_5$ or $P_4O_{10}$

**(i) Phosphorus trioxide $P_2O_3$ or $P_4O_6$**

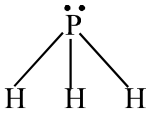
It is formed by burning phosphorus in limited supply of air.

$$P_4 + 3O_2 \rightarrow P_4O_6$$

The pentoxide, formed in small amount, is removed by passing through glass wool. The vapours of trioxide pass through the glass wool and are condensed in a reveiver cooled by a freezing mixture.

**Structure of Phosphine**

Phosphine is a covalent molecule. It has pyramidal structure like ammonia.




The bond angle H-P-H is 93°.

**Uses:**

(i) For making Holme's signals.

(ii) For making smoke screens.

(iii) For making metallic phosphides.

Vapours of phosphorus at low pressure react with $N_2O$ at 600°C to form $P_2O_3$.

$$P_4 + 6N_2O \xrightarrow{600^\circ C} 2P_2O_3 + 6N_2$$

**Properties**

(a) It is a waxy solid having garlic odour.

(b) It is poisonous in nature.

(c) It is soluble in benzene or chloroform.

(d) When heated above 210°C, it forms red phosphorus and another oxide, $P_4O_8$.

$$4P_4O_6 \rightarrow \underset{\substack{\text{Phosphorus}\\\text{tetroxide}}}{3P_4O_8} + \underset{\text{Red}}{4P}$$

When heated at about 440°C, it dissociates to phosphorus dioxide.

$$2P_2O_3 \rightarrow 3PO_2 + P$$

(e) In contact with air, it is oxidised to phosphorus pentoxide.

$$P_4O_6 + 2O_2 \rightarrow P_4O_{10}$$

(f) It burns in chlorine forming oxy-chlorides.

$$P_4O_6 + 4Cl_2 \rightarrow \underset{\substack{\text{Phosphorus}\\\text{Oxy-chloride}}}{2POCl_3} + \underset{\substack{\text{Metaphosphorus}\\\text{Oxy-chloride}}}{2PO_2Cl}$$

(g) In cold water it dissolves slowly forming phosphorus acid.

$$P_4O_6 + 6H_2O \text{ (cold)} \rightarrow 4H_3PO_3$$

With hot water, a violent reaction occurs forming orthophosphoric acid and phosphine.

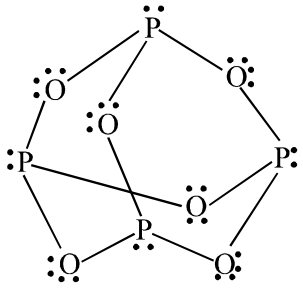
$$P_4O_6 + 6H_2O \text{ (hot)} \rightarrow 3H_3PO_4 + PH_3$$

The above reaction is actually the conversion of phosphorus acid into orthophosphoric acid and phosphine.

$$4H_3PO_3 \rightarrow 3H_3PO_4 + PH_3$$

This oxide is known as acid anhydride of phosphorus acid.

**Structure of phosphorus trioxide**



**(ii) Phosphorus pentoxide, $P_4O_{10}$**

It is prepared by heating phosphorus in a free supply of air or oxygen.

$$P_4 + 5O_2 \rightarrow P_4O_{10}$$

It is further purified by sublimation.

Phosphorus burns in $CO_2$ at 100°C to form $P_2P_5$.

$$2P + 5CO_2 \rightarrow P_2O_5 + 5CO$$

**Properties**

(a) It is a white crystalline compounds

(b) It is odourless when pure. The usual garlic odour is due to presence of small amount of $P_4O_6$ as impurity.

(c) It sublimes on heating.

(d) It has great affinity for water. The final product is orthophosphoric acid. It is therefore, termed phosphoric anhydride.

$$P_4O_{10} \xrightarrow{2H_2O} \underset{\text{Metaphosphoric acid}}{4HPO_3} \xrightarrow{2H_2O} \underset{\text{Pyrophosphoric acid}}{2H_4P_2O_7} \xrightarrow{2H_2O} \underset{\text{Orthophosphoric acid}}{4H_3P_2O_7}$$

$$\text{i.e.} P_4O_{10} + 6H_2O \longrightarrow 4H_3PO_4$$

It is, thus used as a powerful dehydrating or drying agent. It removes water from inorganic and organic compounds like $H_2SO_4$, $HNO_3$, RCOOH, $RCONH_2$, etc.

$$2H_2PO_4 + P_4O_{10} \rightarrow 2SO_3 + 4HPO_3$$
$$4HNO_3\ P_4O_{10} \rightarrow 2N_2O_5 + 4HPO_3$$
$$4CH_3COOH + P_4O_{10} \rightarrow 2(CH_3CO)_2O + 4HPO_3$$
$$2CH_3CONH_2 + P_4O_{10} \rightarrow 2CH_3Cn + 4HPO_3$$

It also chars wood, paper, sugar etc.

(e) when heated strongly with carbon, it forms red phosphorus.

$$P_4O_{10} + 10C \rightarrow 10CO + 4P(\text{red phosphorus})$$

(f) when fused with basic oxides, it forms phosphates.

$$6CaO + P_4O_{10} \rightarrow 2Ca_3(PO_4)_2$$

(g) 30% $H_2O_2$ react on $P_2O_5$ in acetonitrile solution at low temperature to form peroxy monophosphoric acid.

$$P_2O_5 + 2H_2O_2 + H_2O \rightarrow 2H_3PO_5$$

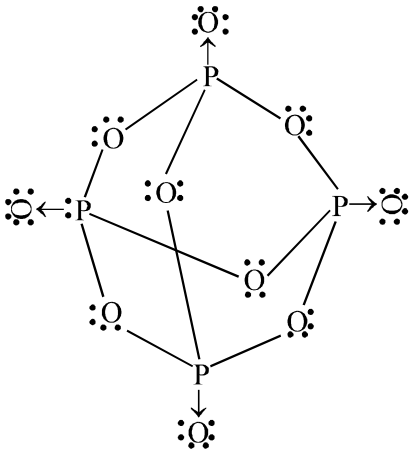
(h) Mixture of $P_2O_5$ and $O_2$ in vapour state combine in presence of electric discharge to form $P_2O_6$ called phosphorus peroxide.

$$P_2O_5 + 1/2O_2 \rightarrow P_2O_6$$

**Uses**

It is most effective drying or dehydrating agent below 100°C

## *Structure of phosphorus pentoxide*



**3. Oxyacids of phosphorus**

Phosphorus forms a number of oxyacids. Common oxyacids are given below.

| Name of Oxyacid | Formula | Basicity | Oxidation state of P |
|---|---|---|---|
| Hydrophosphorus acid | $H_3PO_2$ | 1 | +1 |
| Phosphorus acid | $H_3PO_3$ | 2 | +3 |
| Orthophosphoric acid | $H_3PO_4$ | 3 | +5 |
| Metaphosphoric acid | $HPO_3$ | 1 | +5 |
| Hypophosphoric acid | $H_2P_2O_6$ | 4 | +4 |
| Pyrophosphoric acid | $H_4P_2O_7$ | 4 | +5 |

**(i) Phosphorus acid, $H_3PO_3$, Prepartion**

(i) It is obtained by dissolving phosphorus trioxide in water.

$$P_4O_6 + 6H_2O \rightarrow H_3PO_3$$

(ii) It is also obtained by hydrolysis of phosphorus trichloride.

$$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$$

Chlorine is passed over molten white phosphorus under water when phosphorus trichloride formed undergoes hydrolysis.

$$2P + 3Cl_2 \rightarrow 2PCl_3$$

The solution is heated until the temperature becomes 180°C. On cooling crystals of phosphorus acid are obtained.

**Properties**

(a) It is colourless crystalline compound. It melts at 73°C. It is highly soluble in water.

(b) When heated, it forms orthophosphoric acid and phosphine.

$$4H_3PO_3 \xrightarrow{200^oC} 3H_3PO_4 + PH_3$$

(c) It si strong acid. It is diabasic in nature.

$$H_3PO_3 \Leftrightarrow H^- + H_2PO_3^- \Leftrightarrow 2H^- + HPO_3^-$$

$K_1 = 10^{-1}$; $K_2 = 2 \times 10^{-7}$

It thus forms two series of salts such as $NaH_2PO_3$ adn $Na_2HPO_3$ known as primary phosphites and secondary phosphites respectively.

(d) It acts as a strong reducing agent. The potential equation is:

$$H_3PO_3 + H_2O \rightarrow H_3PO_4 + 2H$$

It reduces $CuSO_4$ to Cu, $AgNO_3$ to Ag, $HgCl_2$ to $Hg_2Cl_2$, $I_2$ to HI and acidfied $KMnO_4$ solution.

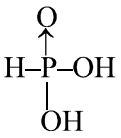
$$CuSO_4 + 2H \rightarrow Cu + H_2SO_4$$
$$AgNO_3 + H \rightarrow Ag + HNO_3$$
$$2HgCl_2 + 2H \rightarrow Hg_2Cl_2 + 2HCl$$
$$I_2 + 2H \rightarrow 2HI$$
$$2KMnO_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 3H_2O + 5O$$
$$[H_3PO_3 + O \rightarrow K_3PO_4] \times 5$$

---

$$2KMnO_4 + 3H_2SO_4 + 5H_3PO_3 \rightarrow K_2SO_4 + 2MnSO_4 + 5H_3PO_4 + 3H_2O$$

**Structure of Phosphorus acid**

It is diabasic acid, i.e. two hydrogen atoms are insoluble or two hydroxyl groups are present. Thus, the structure is:



The phosphorus lies in $sp^3$ hybrid state.

**(ii) Orthophosphoric aicd, $H_3PO_4$**

This acid is commonly called as phosphoric acid.

**Preparation**

(i) It is formed when phosphorus pentoxide is boiled with hot water.

(ii) ORthophosphoric acid is also formed when $PCl_5$ (Phosphorus pentachloride) is boiled with water.

$$PCl_5 + 4H_2O \rightarrow H_3PO_4 + 5HCl$$

**(iii) Laboratory preparation**

The best method for its preparation in the laboratory is to heat red phosphorus with concentrated nitric acid in a flask with a reflux condenser.

$$P + 5HNO_3 \rightarrow H_3PO_4 + H_2O + 5NO_2$$

The reaction is usually carried out is presence of a crystal of iodine. The iodine acts as a catalyst.

$$P + 3I \rightarrow PI_3$$
$$PI_3 + 3H_2O \rightarrow H_3PO_3 + 3HI$$
$$3HI + 3HNO_3 \rightarrow 3H_2O + 3NO_2 + 3I$$
$$H_3PO_3 + 2HNO_3 \rightarrow H_3PO_4 + H_2O + 2NO_2$$

---

$$P + 5HNO_3 \rightarrow H_3PO_4 + 5NO_2 + H_2O$$

The solution is concentrated till it becomes syrupy about 170°C. It is cooled over conc. $H_2SO_4$ in vacuum dessicator when crystals of orthophosphoric acid are formed.

**Manufacture**

It is prepared on large scale from bone ash or phosphoric mineral.

(a) By decomposing calcium phosphate present is bone ash or phosphoric meneral with conc. $H_2SO_4$.

$$Ca_3(PO_4)_2 + 3H_2SO_4 \rightarrow 3CaSO_4 + 2H_3PO_4$$

On standing calcium sulphate settles down and the clear supernatant liquid separates out. The liquid is concentrated when about 85% concentrated orthophosphoric acid is obtained.

(b) The bone ash is dissolved in minimum amount of nitric acid and lead acetate is added as to precipitate lead phosphate. The lead salt is then decomposed by passing $H_2S$.

$$Pb_3(PO_4)_2 + 3H_2S \rightarrow \underset{\text{insoluble}}{3PbS} + \underset{\text{solution}}{2H_3PO_4}$$

(c) Bone ash or calcium phosphate is converted into phosphorous pentoxide when heated with silica in electric furnace.

$$2Ca_3(PO_4)_2 + 6SiO_2 \rightarrow 6CaSiO_3 + P_4O_{10}$$

$P_4O_{10}$ is then dissolved in hot water.

$$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$$

**properties**

(a) It is transparent deliquescent solid. It melts at 42.3°C. It absorbs water and forms colourless syrupy mass. It is highly soluble in water.

**(b) Heating effect**

When heated at 250°C, it is converted into pyrophosphoric acid.

$$2H_3PO_4 \xrightarrow{250^\circ C} H_4P_2O_7 + H_2O$$

On further heating, it is converted into metaphosphoric acid.

$$H_3PO_4 \xrightarrow[600^\circ C]{\text{heat}} HPO_3 + H_2O$$

$$H_4P_2O_7 \xrightarrow{\text{heat}} 2HPO_3 + H_2O$$

When strongly heated at red heat, it forms $P_4O_{10}$.

**(c) Acidic mature**

It is tribasic acid, i.e. all the three hydrogen atoms are ionizable. It forms three series of salts.

$$H_3PO_4 \Leftrightarrow H^+ + H_2PO_4^- \Leftrightarrow 2H^+ + HPO_4^{2-} \Leftrightarrow 3H^+ + PO_4^{3-}$$

$$\underset{\text{(primary salt)}}{NaH_2PO_4} \qquad \underset{\text{(secondary salt)}}{Na_2HPO_4} \qquad \underset{\text{(normal salt)}}{Na_3PO_4}$$

Primary salt on heating forms a slat of metaphosphoric acid.

$$\underset{\substack{\text{Sodium dihydrogen}\\ \text{phosphate}}}{NaH_2PO_4} \xrightarrow{\text{heat}} \underset{\substack{\text{Sodium}\\ \text{metaphosphate}}}{NaPO_3} + H_2O$$

Secondary salt on heating forms a slat of pyrophosphoric acid.

$$\underset{\substack{\text{Disod.Hydrogen}\\ \text{phosphate}}}{2Na_2HPO_3} \xrightarrow{\text{heat}} \underset{\substack{\text{Sodium}\\ \text{pyrophosphate}}}{Na_4P_2O_7} + H_2O$$

Normal salt is not affected by heating.

In case, ammonium ion is present in the slat it behaves as hydrogen.

$$2MgNH_4PO_4 \xrightarrow{\text{heat}} Mg_2P_2O_7 + 2NH_3 + H_2O$$

$$NaNH_4HPO_4 \xrightarrow{\text{heat}} naPO_3 + NH_3 + H_2O$$

**(d) Reaction with $AgNO_3$**

A yellow precipitate of silver phosphate is formed.

$$3AgNO_4 + H_3PO_4 \rightarrow \underset{\text{yellow ppt.}}{Ag_3PO_4} + 3HNO_3$$

**(e) Reaction with $BaCl_2$**

A white ppt. of barium phosphate is formed in neutral or alkaline solution.

$$3BaCl_2 + 2H_3PO_4 \rightarrow \underset{\text{white ppt.}}{Ba_3(PO_4)_2} + 6HCl$$

**(f) Reaction with bromides and iodides**

Hydrobromic and hydroiodic acids are liberated from bromides and iodides respectively. This is te laboratory preparation of HBr and HI.

$$3NaBr + H_3PO_4 \rightarrow Na_3PO_4 + 3HBr$$
$$3NaI + H_3PO_4 \rightarrow Na_3PO_4 + 3HI$$

**(g) Reaction with magnesium salt**

Magnesium slats combine with orthophosphoric acid in presence o ammonium chloride nd ammonium hydroxide to form a white precipitate of magnesium ammonium phosphate.

$$MgSO_4 + NH_4Cl + H_3O_4 \rightarrow Mg(NH_4)PO_4 + H_2SO_4 + HCl$$

This reaction is used to test $Mg^{2+}$ ion.

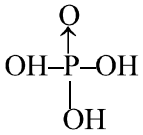
**(h)** On heating orthophosphoric acid is presence of nitric acid with ammonium molybdate a canary yellow ppt. of ammonium phosphomolybdate is formed.

$$H_3PO_4 + 21\ HNO_3 + 12\ (NH_4)_2\ MoO_4 \rightarrow \underset{\text{Ammoniumphosphomolybdate}}{(NH_4)_3PO_4 12MoO_3} + 21\ NH_4NO_3 + 12\ H_2O$$

This reaction is used to test $PO_4^{3-}$ ion.

## *Structure of orthophosphoric acid*

Orthophosphoric acid is a tribasic. i.e. 3 hydroxyl groups are present. The structure of the acid is thus represented as:




Phosphorus atom lies in $sp^3$ hybrid state.

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